MATCH LINE SEE 1 / EX-131SP

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EX-136SP
ENLARGED PLAZA
POOL LAYOUT PLAN

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CENTER OF DRAIN
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FFE = 191.00

FFE = 190.00

EX. OLMS
WALK (BRIC

EX-134SP / 1 MATCH LINE SEE

1
EX-133SP
Layout Plan C
1/4" = 1'-0"

0' 2' 4' 8'

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TRUE NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

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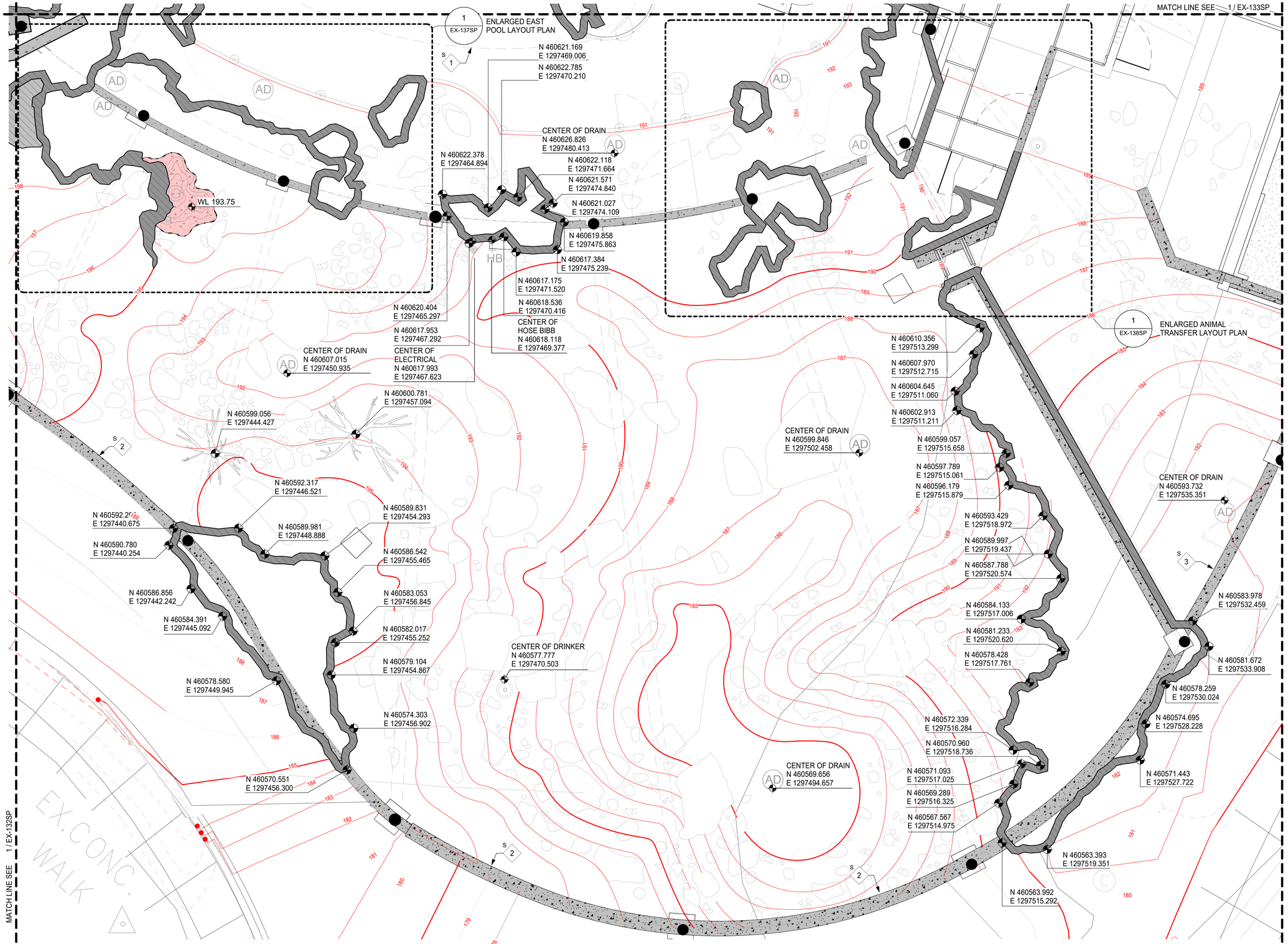
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LAYOUT PLAN C |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 201 OF 378

EX-133SP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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TRUE NORTH

KEY PLAN

0 2' 4' 8'
1/4" = 1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
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| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LAYOUT PLAN D |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

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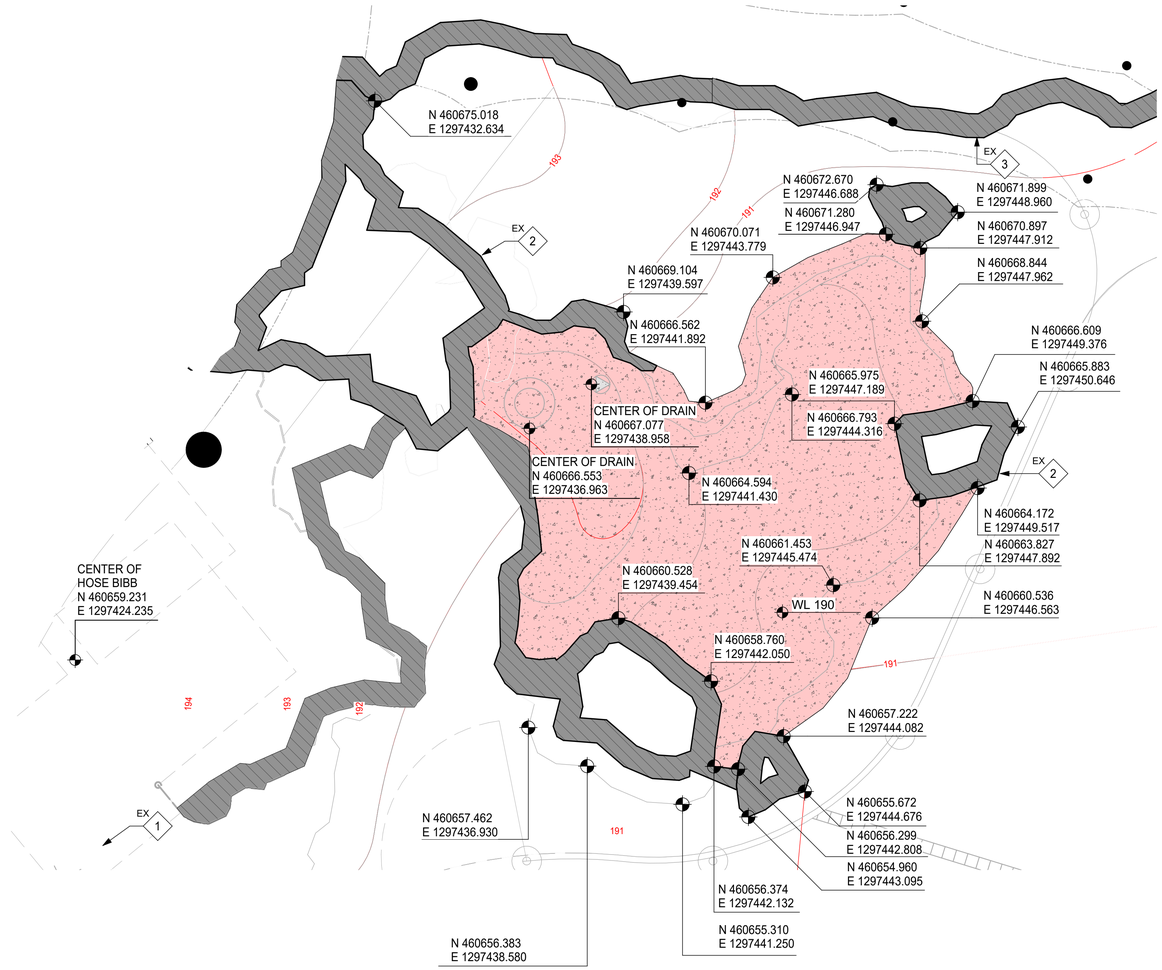
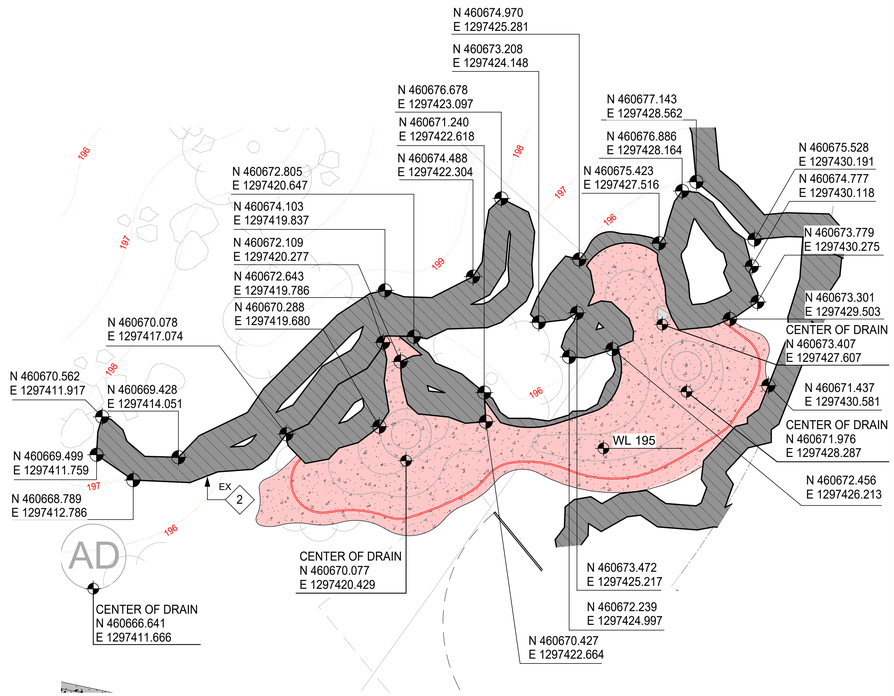



1 EX-136SP **Enlarged Plaza Pool Layout Plan**
1/2" = 1'-0"




2 EX-136SP **Enlarged Habitat West Pool Layout Plan**
1/2" = 1'-0"

0' 1' 2' 4'

FOR REFERENCE ONLY

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TRUE NORTH

KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
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| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ENLARGED LAYOUT PLANS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 203 OF 378

EX-136SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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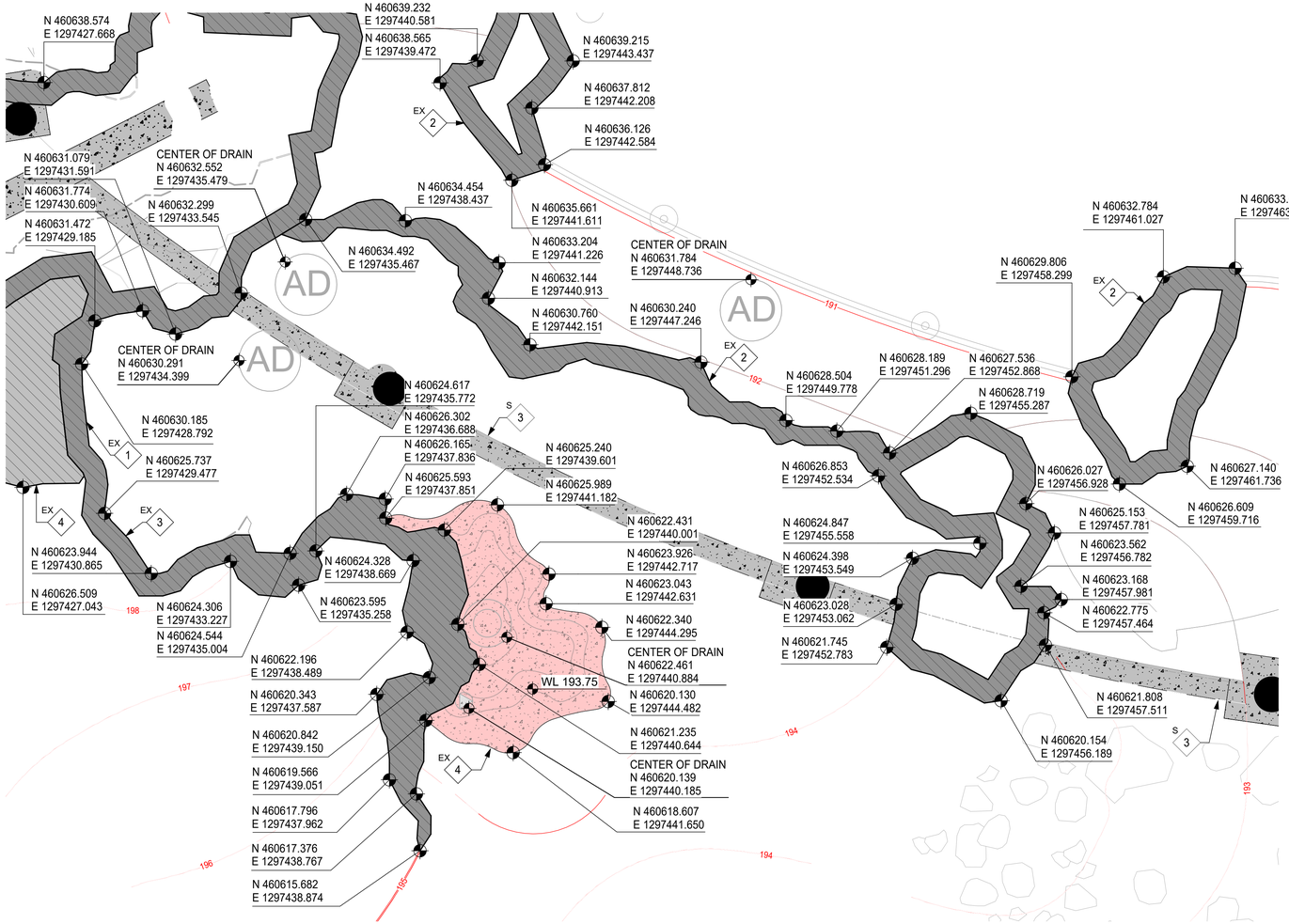


1 EX-137SP Enlarged Habitat East Pool Layout Plan

1/2" = 1'-0"

0' 1' 2' 4'

# ALF

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TRUE NORTH

KEY PLAN

0 1' 2' 4'

1/2"=1'-0"

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|---|---|
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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ENLARGED LAYOUT PLAN |
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EX-137SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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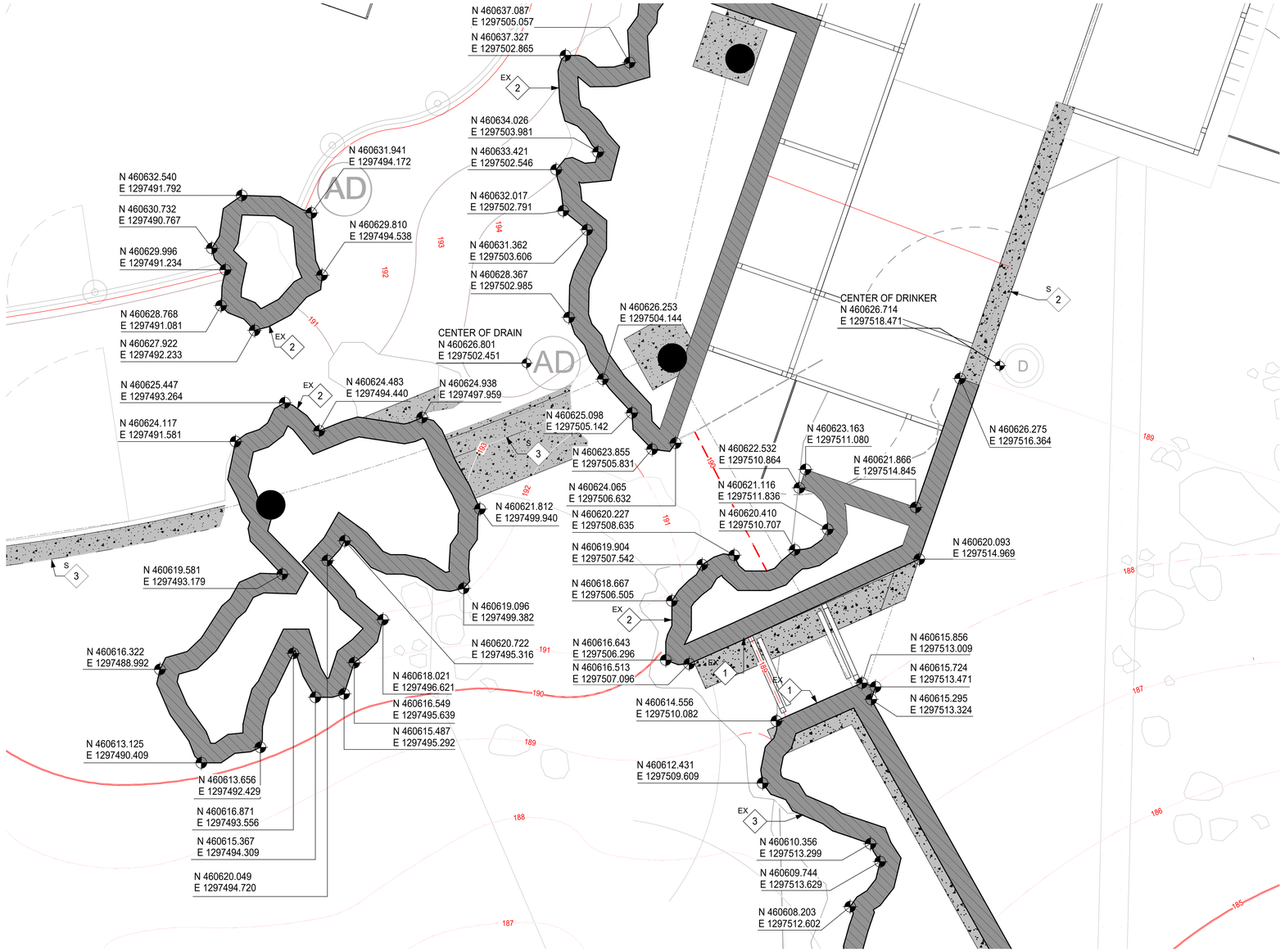


1 EX-138SP **Enlarged Animal Animal Transfer Layout Plan**

1/2" = 1'-0"

0' 1' 2' 4'

# ALF

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TRUE NORTH

KEY PLAN

0 1' 2' 4'

1/2"=1'-0"

**GRAPHIC SCALE(S)**

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ENLARGED LAYOUT PLAN |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 205 OF 378 |
| | EX-138SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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EX-138SP BY: Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

# PLANTING SYMBOLS & SCHEDULE

| SYMBOL | BOTANICAL NAME | COMMON NAME | QUANTITY | NATIVE | CONTAINER SIZE | HEIGHT | SPREAD | USDA ZONE | EVERGREEN | WATER NEEDS | TOXIC? | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **TREES** | | | | | | | | | | | | |
| | *ARCTOSTAPHYLOS DENSIFLORA* 'HOWARD McMINN' | 'HOWARD McMINN' MANZANITA | 9 | N | 15 GAL | 4-8' | 6-10' | 6-7 | Y | LOW | N | ATTRACTS POLLINATORS |
| | *PINUS THUNBERGII* | JAPANESE BLACK PINE | | N | 5" CAL MIN. 15' HT MIN. 8' W MIN. | 20-80' | 20-35' | 5-8 | Y | MEDIUM | Y | SPECIMEN TREES, VERIFY WITH LANDSCAPE ARCHITECT BEFORE PLANTING. SEE REFERENCE PHOTOS BELOW. |
| | ALTERNATES: | | | | | | | | | | | |
| | *JUNIPERUS VIRGINIANA VAR. SILICOLA* | SOUTHERN RED CEDAR | 7 | N | 5" CAL MIN. 15' HT MIN. 8' W MIN. | 30-40' | 25' | 7-11 | Y | LOW | N | |
| | *JUNIPERUS VIRGINIANA* | EASTERN RED CEDAR | | Y | 5" CAL MIN. 15' HT MIN. 8' W MIN. | 30-40' | 10-20' | 2-9 | Y | LOW | Y | |
| | *MORELLA CERIFERA* | SOUTHERN WAX MYRTLE | 9 | Y | 5 GAL | 20' | 6-12' | 7-10 | Y | HIGH | N | DROUGHT TOLERANT, DEER & RABBIT RESISTANT, ATTRACTS POLLINATORS |
| | *SALIX HUMILIS* | PRAIRIE WILLOW | 15 | Y | 15 GAL | 6-10' | 6-10' | 3-8 | Y | HIGH | N | |
| **SHRUBS** | | | | | | | | | | | | |
| | *ARTEMISIA FRIGIDA* | SAND SAGEBRUSH | 80 | N | 1 GAL | 9-16" | 12-18" | 3-10 | N | LOW | N | DROUGHT TOLERANT, DEER & RABBIT RESISTANT |
| | *SARCOCCOCA HOOKERIANA* | SWEETBOX | 10 | N | 5 GAL | 4-5' | 5-6' | 6-8 | Y | MEDIUM | N | FRAGRANT FLOWERS |
| **PERENNIALS** | | | | | | | | | | | | |
| | *CISTUS LADANIFER* | GUM ROCKROSE | 15 | N | 5 GAL | 3-5' | 3-5' | 8 | Y | LOW | N | SEASONAL INTEREST, ATTRACTS POLLINATORS |
| | *SOLIDAGO 'DANSOLITLEM'* | 'LITTLE LEMON' GOLDEN ROD | 35 | Y | 1 GAL | 18-24" | 12-18" | 6-9 | N | LOW | N | SEASONAL INTEREST, ATTRACTS POLLINATORS, DEER RESISTANT |
| | *KNIPHOFIA UVARIA* | RED HOT POKER | 20 | N | 5 GAL | 4-8' | 2-3' | 5-10 | Y | LOW | N | SEASONAL INTEREST, ATTRACTS POLLINATORS, DROUGHT TOLERANT, DEER & RABBIT RESISTANT |
| **GRASSES (1 GAL OR LARGER)** | | | | | | | | | | | | |
| | *CAREX LEAVENWORTHII* | LEAVENWORTH'S SEDGE | 250 | N | 1 GAL | 6-12" | 6-12" | 6-9 | Y | LOW | N | DROUGHT TOLERANT |
| | *CAREX PLANTAGINEA* | PLANTAINLEAF SEDGE | 115 | N | 1 GAL | 1-2' | 1-2' | 4-8 | Y | MEDIUM | N | |
| | *DESCHAMPSIA CESPITOSA* | TUFTED HAIRGRASS | 143 | N | 1 GAL | 2-3' | 1-2"' | 4-9 | N | MEDIUM | N | SEASONAL INTEREST |
| | *ELYMUS CANADENSIS* | CANADA WILD RYE | 25 | Y | 1 GAL | 2-4' | 1-2' | 3-8 | N | HIGH | N | SEASONAL INTEREST, ATTRACTS POLLINATORS, DEER RESISTANT, DROUGHT TOLERANT |
| | *SPOROBOLUS HETEROLEPIS 'TARA'* | 'TARA' PRAIRIE DROPSEED | 75 | N | 1 GAL | 1-2' | 1-2' | 3-9 | N | LOW | N | DROUGHT TOLERANT |

**ABBREVIATIONS**

B&B BALLED & BURLAPPED

NOTE: USDA HARDINESS ZONE FOR WASHINGTON, D.C. IS 7B.

## SYMBOLS

EXISTING TREE

## NOTES

1. DRAWINGS FROM ALL DISCIPLINES MUST BE REVIEWED IN TANDEM WITH EXHIBIT DRAWINGS. THIS IS REQUIRED FOR A FULL UNDERSTANDING OF THE DESIGN INTENT FOR THE COMPLETED PROJECT. DISCREPANCIES MUST BE BROUGHT TO THE ATTENTION OF EXHIBITS AND THE PROJECT PRIME AND A RESOLUTION AGREED UPON BEFORE WORK MAY PROCEED.

## SPECIMEN TREE REFERENCE PHOTOS

*PINUS THUNBERGII*

*JUNIPERUS VIRGINIANA VAR. SILICOLA*

JUNIPERUS VIRGINIANA

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ADDRESS: 3001 Connecticut Ave NW Washington, DC 20008 USA

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

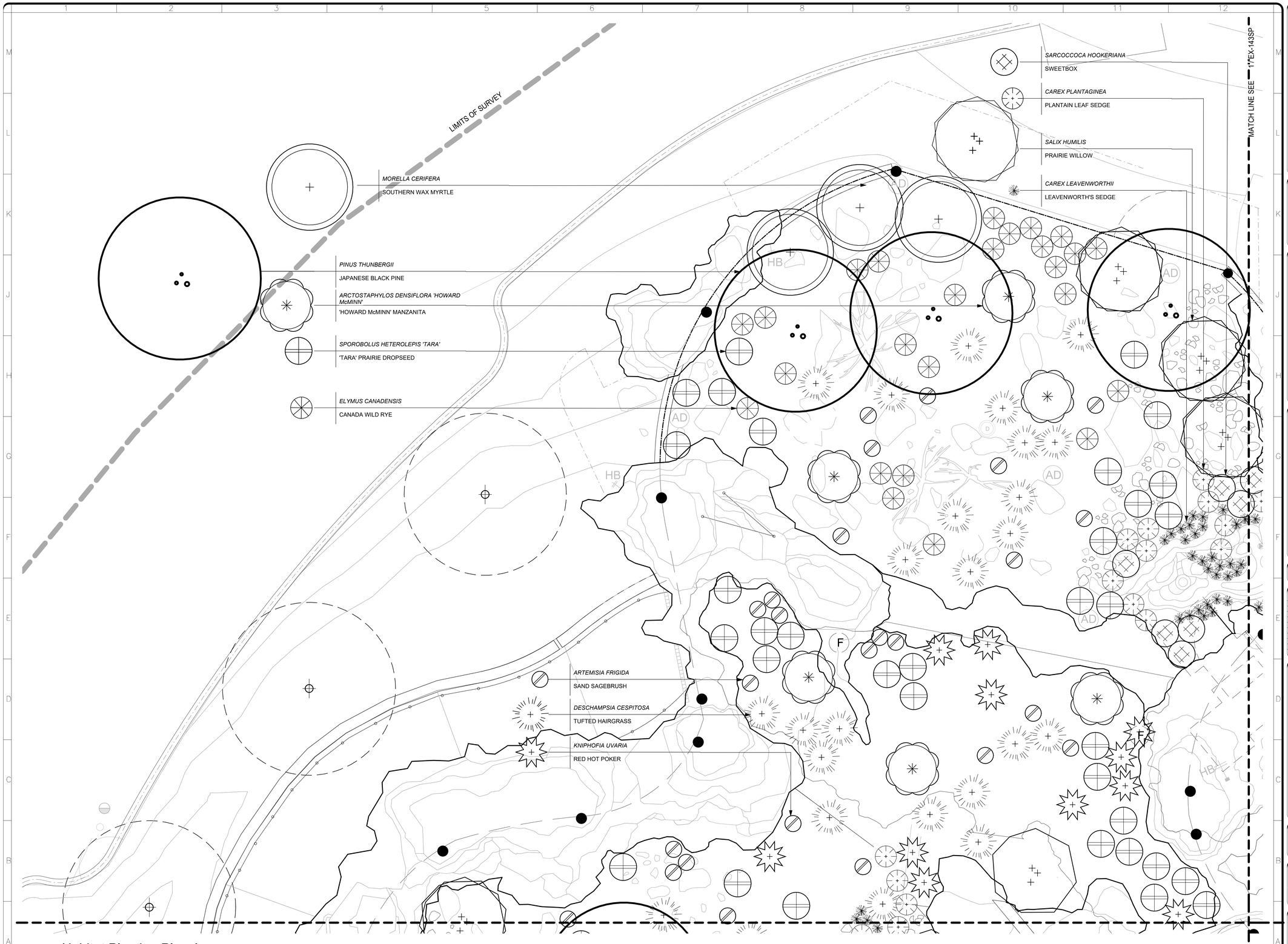
DRAWING TITLE: HABITAT PLANTING SYMBOLS AND SCHEDULE

WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| DF | SHR | JH |

SHEET NO. 206 OF 378

EX-140SP (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)




1 EX-141SP

## Habitat Planting Plan A

1/4" = 1'-0"

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TRUE NORTH

0 2' 4' 8'

1/4"=1'-0"

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT PLANTING PLAN A |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 207 OF 378

EX-141SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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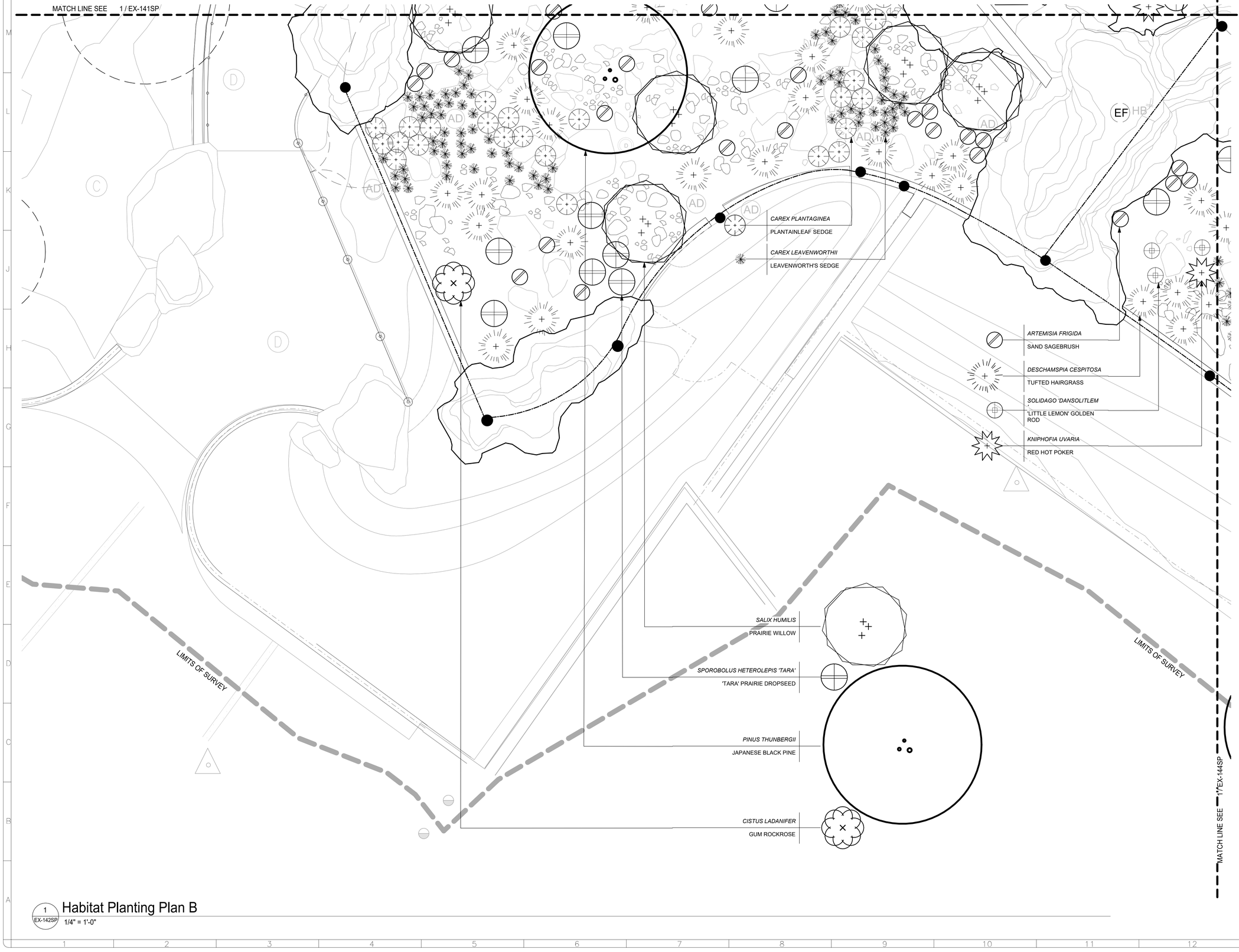
MATCH LINE SEE 1 / EX-141SP




1 EX-142SP Habitat Planting Plan B

1/4" = 1'-0"

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TRUE NORTH

0 2' 4' 8'
1/4"=1'-0"

**GRAPHIC SCALE(S)**

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| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT PLANTING PLAN B |

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| DF | SHR | JH |

WORKING STAFF

SHEET NO. 208 OF 378

EX-142SP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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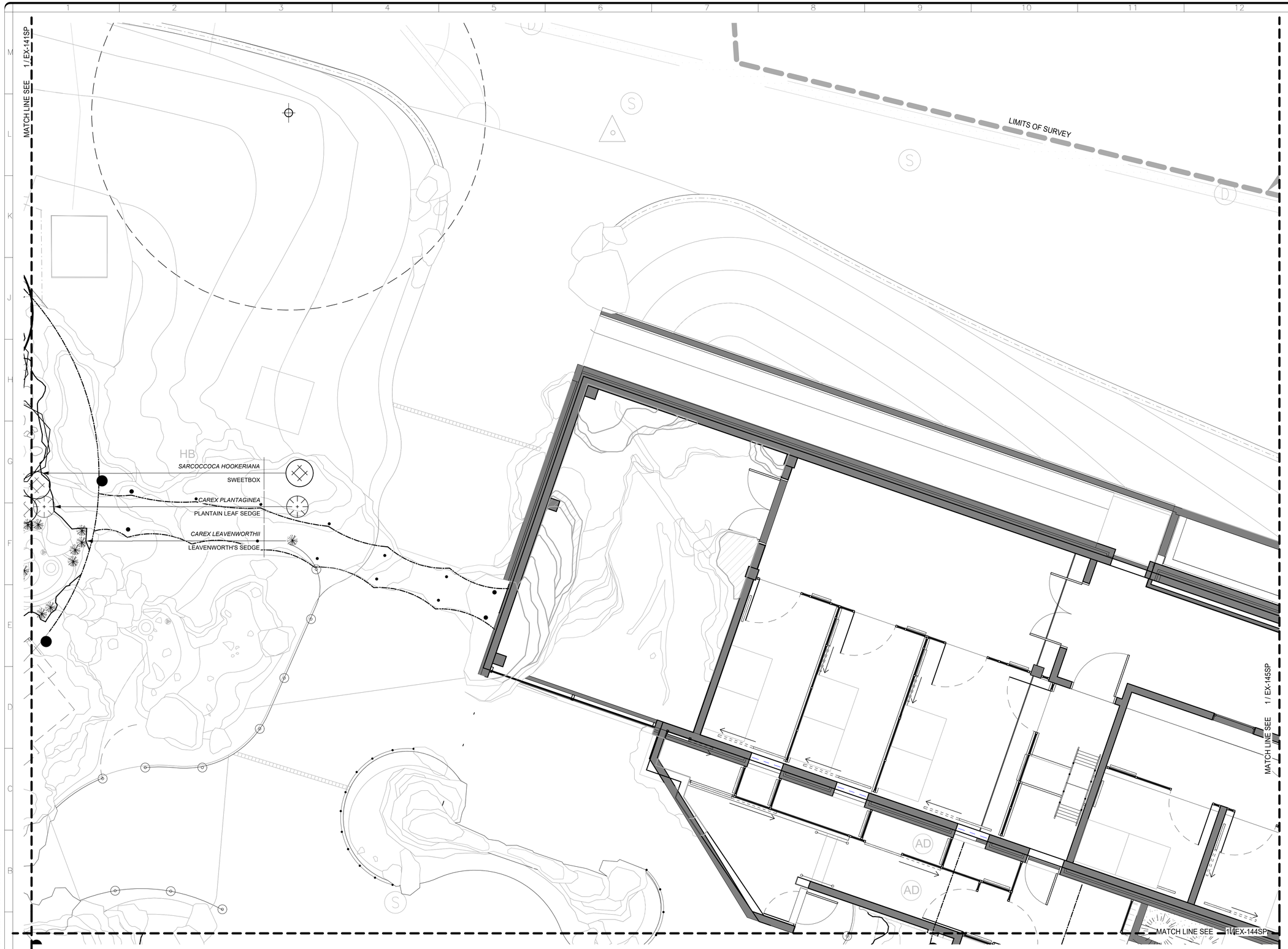


1 EX-143SP **Habitat Planting Plan C**

1/4" = 1'-0"

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1/4"=1'-0"

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| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT PLANTING PLAN C |
| WORKING STAFF | DESIGNED BY: DF; DRAWN BY: SHR; CHECKED BY: JH |
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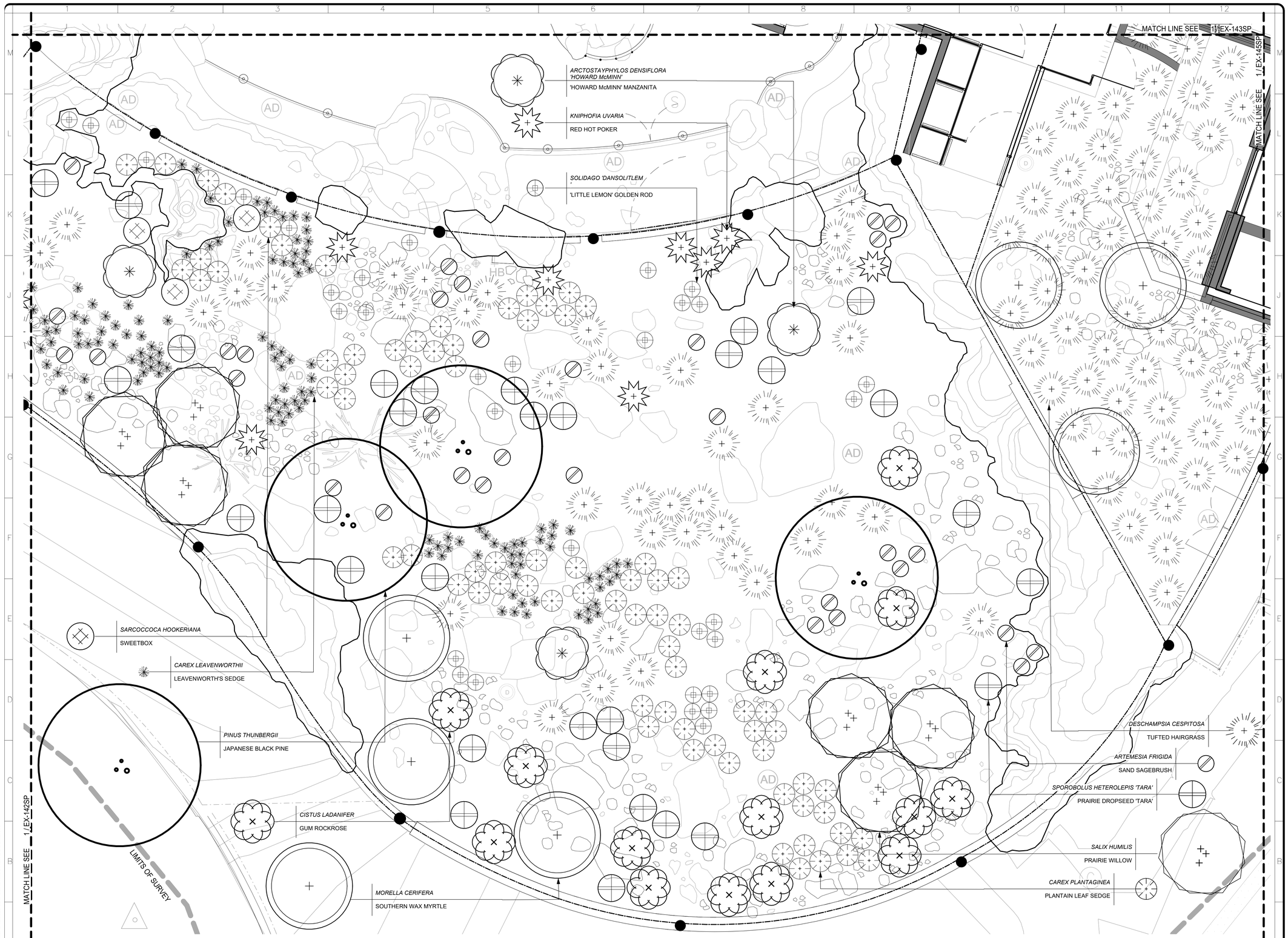


1 EX-144SP Habitat Planting Plan D
1/4" = 1'-0"

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0 2' 4' 8'
1/4"=1'-0"

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT PLANTING PLAN D |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
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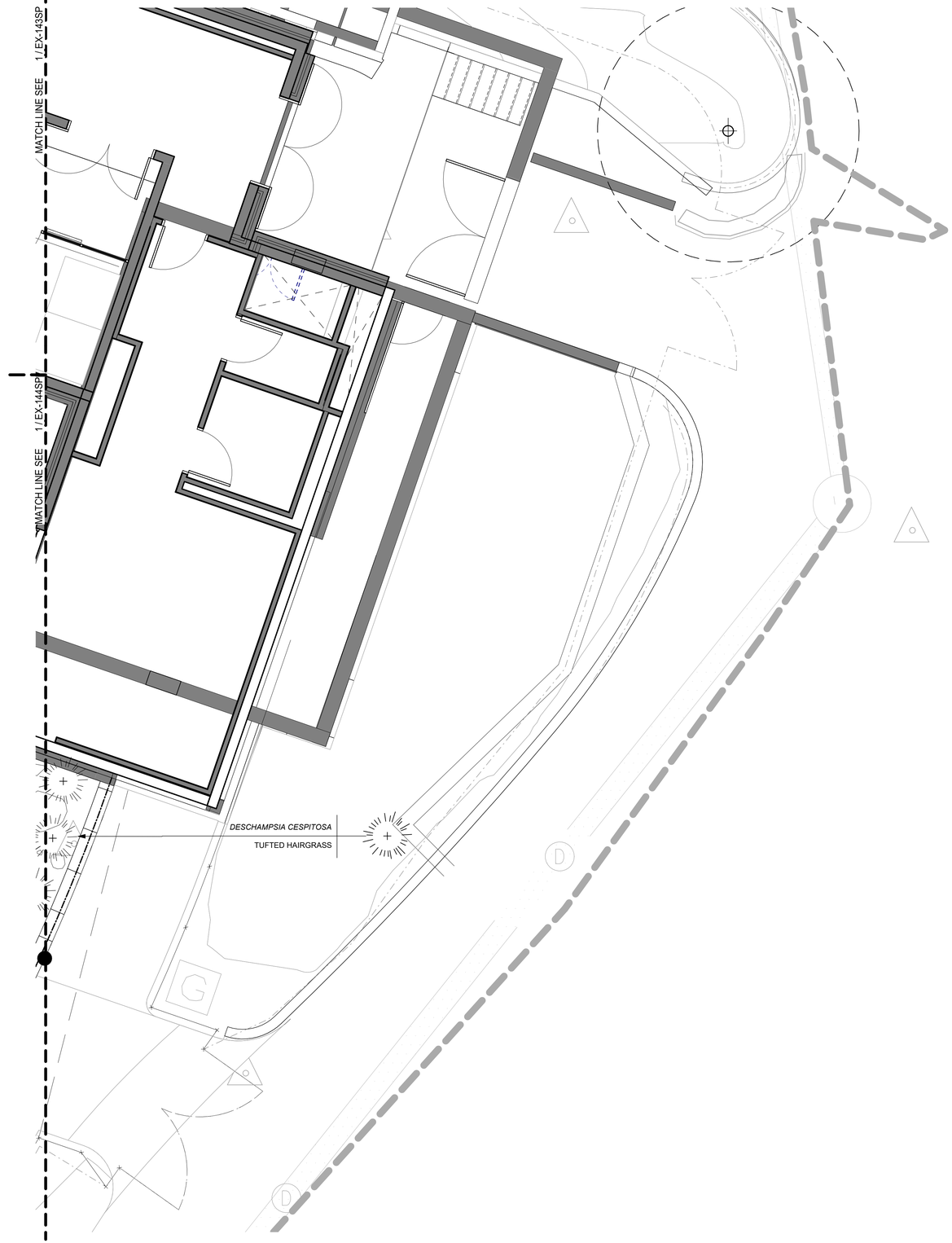
EX-144SP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-145SP Habitat Planting Plan E

1/4" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

TRUE NORTH

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
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| REVISION # DATE | REVISION DESCRIPTION |
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| BUILDING NAME / ADDRESS | ARABIAN LEOPARD FACILITY (ALF)<br>3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT PLANTING PLAN E |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 211 OF 378 |
| | EX-145SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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## LINES

- LIMIT OF WORK
- CENTERLINE
- WATERLINE
- 100 — CONTOUR LINE - PROPOSED
- (100) — CONTOUR LINE - EXISTING
- GRADE BREAK
- FOOTING DRAIN
- LANDSCAPE DRAIN

## SYMBOLS

- DATUM / DIMENSION / CONTROL POINT
- WL 100.50 — WATER LEVEL / WATER SURFACE
- 110.50 — SPOT ELEVATION
- (110.50) — EXISTING SPOT ELEVATION
- (D) — DRINKER - PEDESTAL
- AD — AREA DRAIN
- POOL DRAIN
- POOL OVERFLOW
- HB — HOSE BIB
- ELECTRICAL RECEPTACLE

## MATERIALS

- 6" GRADED AGGREGATE WITH CLEAN SAND SURFACE
  SLOPES LESS THAN 5% — 6 / EX-630DT
- 6" GRADED AGGREGATE WITH STONE DUST SURFACE
  SLOPES 5-20% — 7 / EX-630DT
- CRUSHER RUN AGGREGATE
  SLOPES GREATER THAN 20% — 8 / EX-630DT

## NOTES

1. DRAWINGS FROM ALL DISCIPLINES MUST BE REVIEWED IN TANDEM WITH EXHIBIT DRAWINGS. THIS IS REQUIRED FOR A FULL UNDERSTANDING OF THE DESIGN INTENT FOR THE COMPLETED PROJECT. DISCREPANCIES MUST BE BROUGHT TO THE ATTENTION OF EXHIBITS AND THE PROJECT PRIME AND A RESOLUTION AGREED UPON BEFORE WORK MAY PROCEED.

# ALF

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KEY PLAN

**GRAPHIC SCALE(S)**

DATE: **6/15/2026**

SUBMISSION: **65% CONSTRUCTION DOCUMENTS**

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BUILDING NAME: **ARABIAN LEOPARD FACILITY (ALF)**

ADDRESS: **3001 Connecticut Ave NW Washington, DC 20008 USA**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**

OPDC PROJECT NUMBER: **2533105**

A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **HABITAT STONE MULCH LEGEND**

WORKING STAFF: DESIGNED BY **DF** | DRAWN BY **SHR** | CHECKED BY **JH**

SHEET NO. 212 OF 378

**EX-150SP** (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

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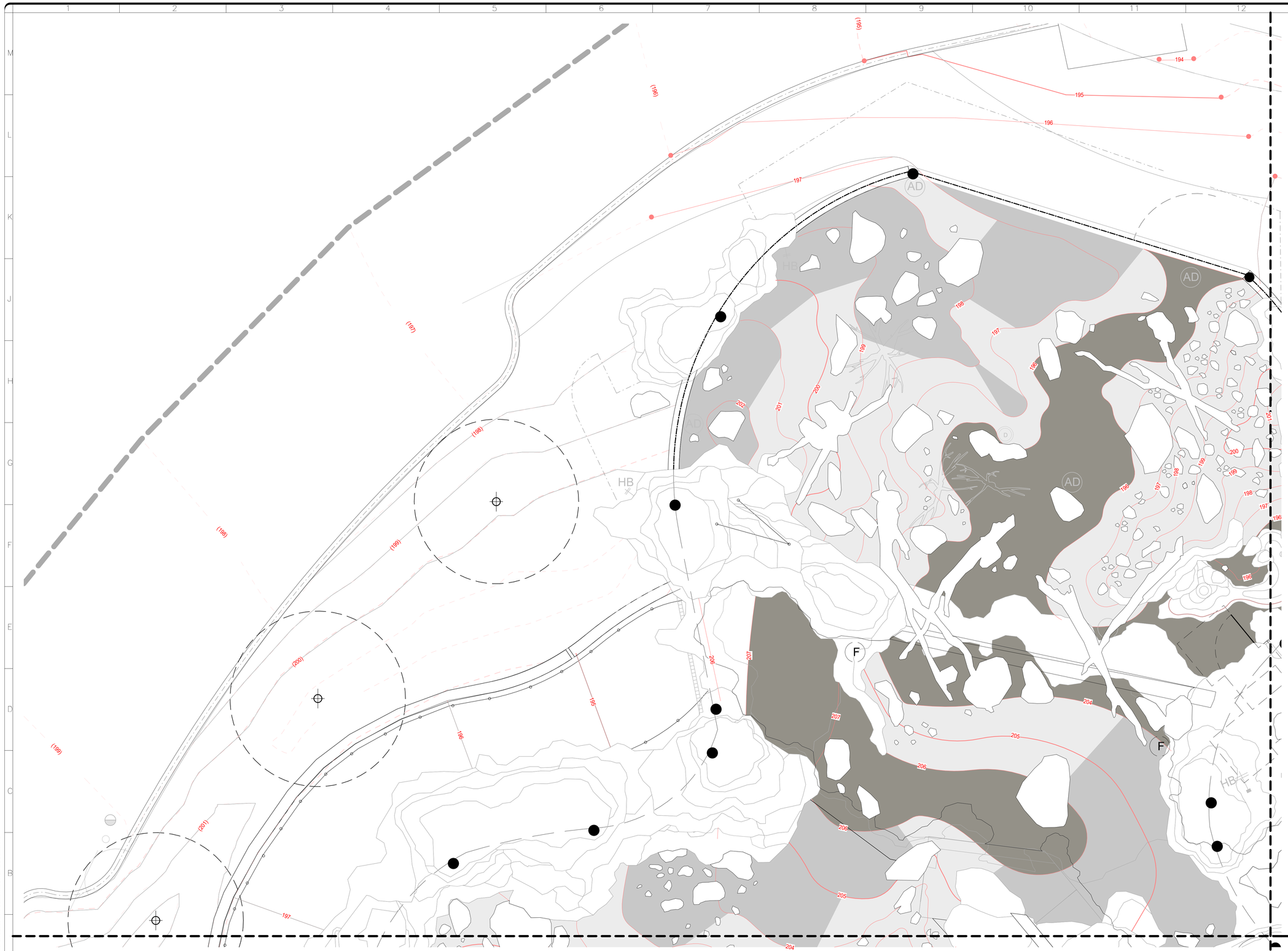
6/12/2026 1:32:07 AM
EX-150SP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-151SP Habitat Stone Mulch Plan A

1/4" = 1'-0"

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EX-151SP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

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TRUE NORTH

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

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| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT STONE MULCH PLAN A |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 213 OF 378

EX-151SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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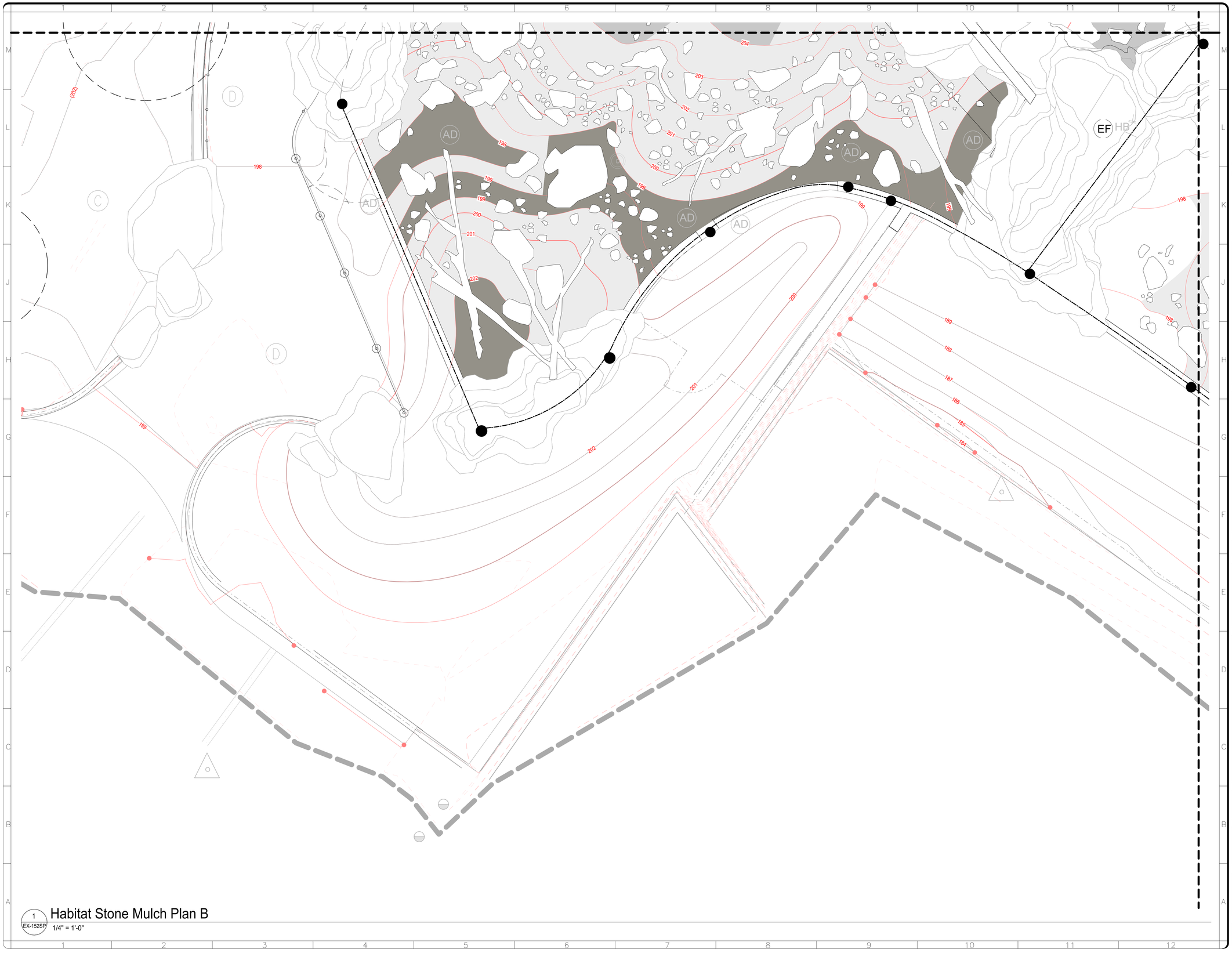


1 EX-152SP
Habitat Stone Mulch Plan B
1/4" = 1'-0"

# ALF

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TRUE NORTH

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

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| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT STONE MULCH PLAN B |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 214 OF 378 |
| | EX-152SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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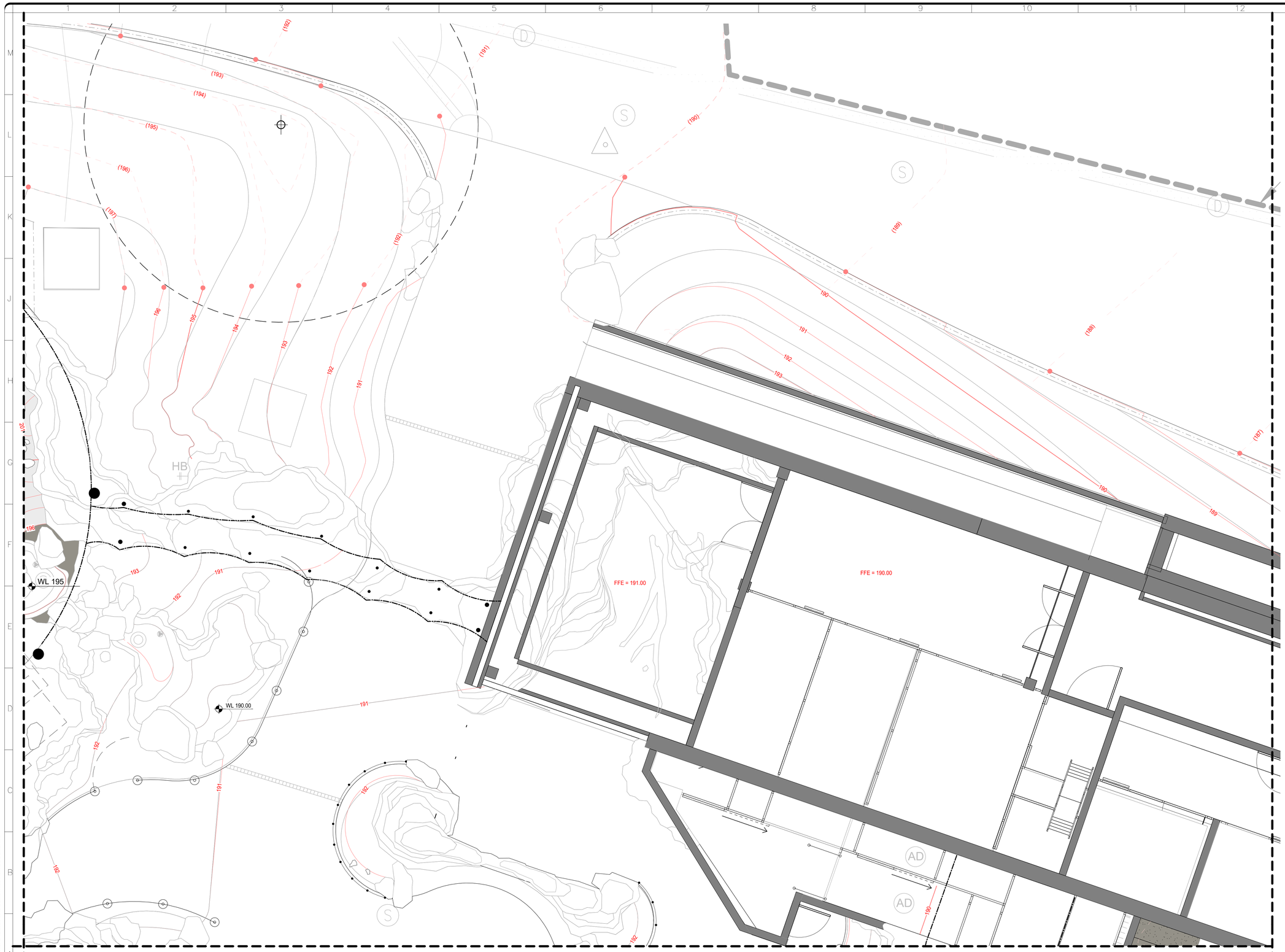
6/12/2026 1:32:08 AM
EX-152SP BY Apicorner 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-153SP
Habitat Stone Mulch Plan C
1/4" = 1'-0"

6/12/2026 1:32:08 AM
EX-153SP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

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STAMP

TRUE NORTH

0 2' 4' 8'
1/4"=1'-0"

**GRAPHIC SCALE(S)**

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT STONE MULCH PLAN C |

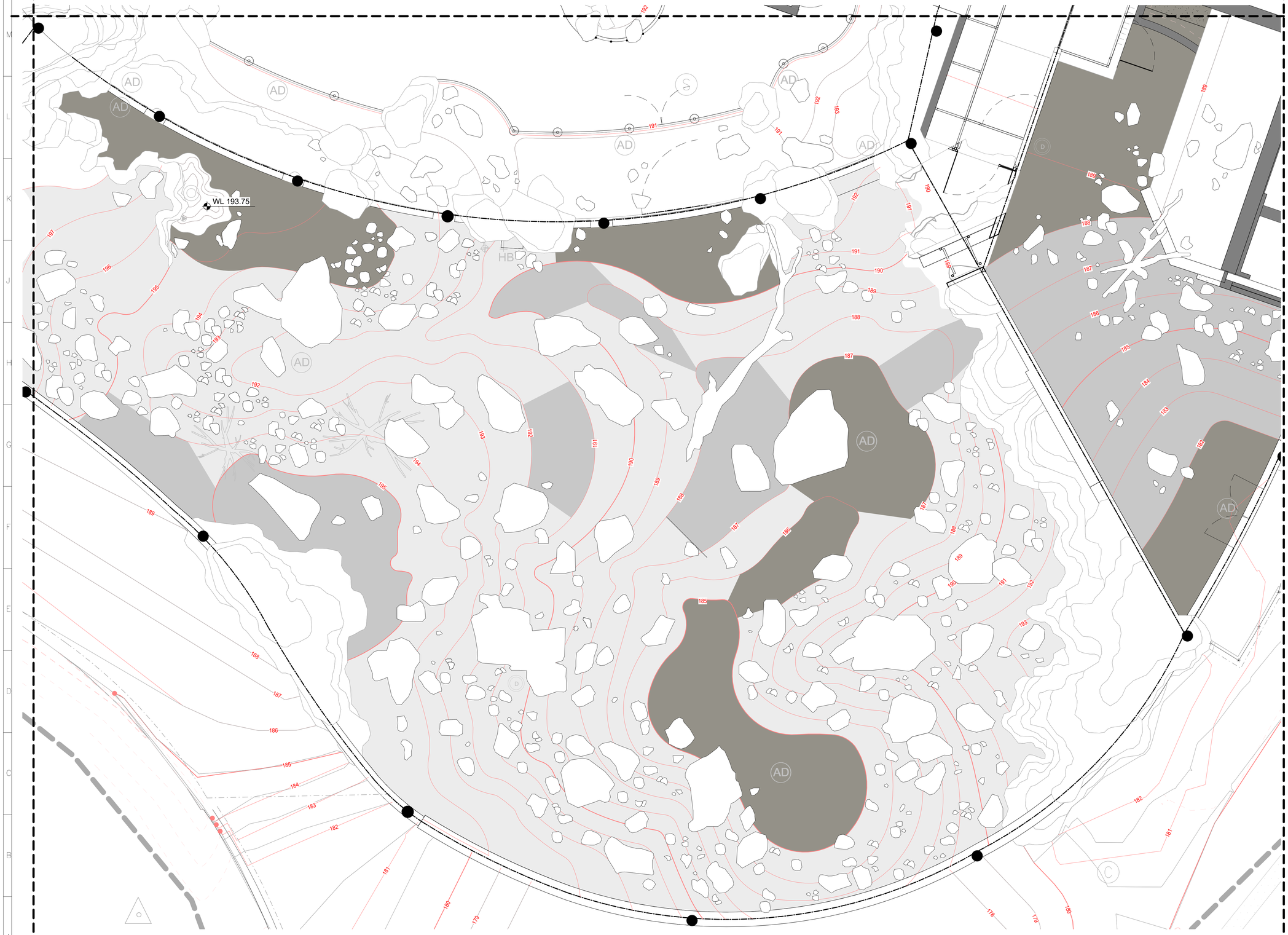
| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| DF | SHR | JH |

WORKING STAFF

SHEET NO. 215 OF 378

EX-153SP
DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

FOR REFERENCE ONLY




1 EX-154SP Habitat Stone Mulch Plan D

1/4" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

NOT FOR CONSTRUCTION

STAMP

TRUE NORTH

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT STONE MULCH PLAN D |

| WORKING STAFF | DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|---|
| | DF | SHR | JH |

SHEET NO. 216 OF 378

EX-154SP

DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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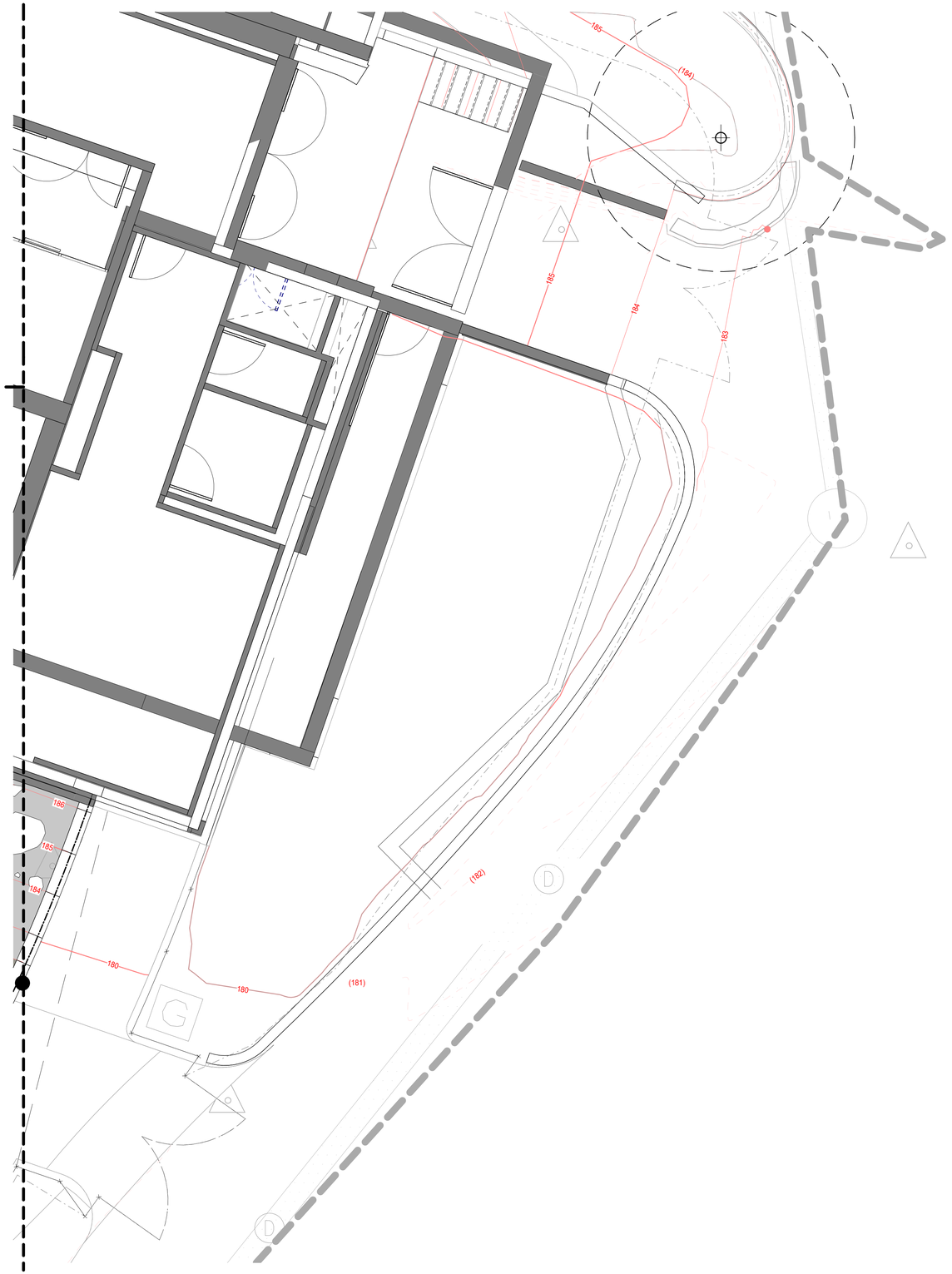
6/12/2026 1:32:09 AM
EX-154SP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-155SP Habitat Stone Mulch Plan E

1/4" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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TRUE NORTH

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT STONE MULCH PLAN E |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 217 OF 378 |
| DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE | EX-155SP |

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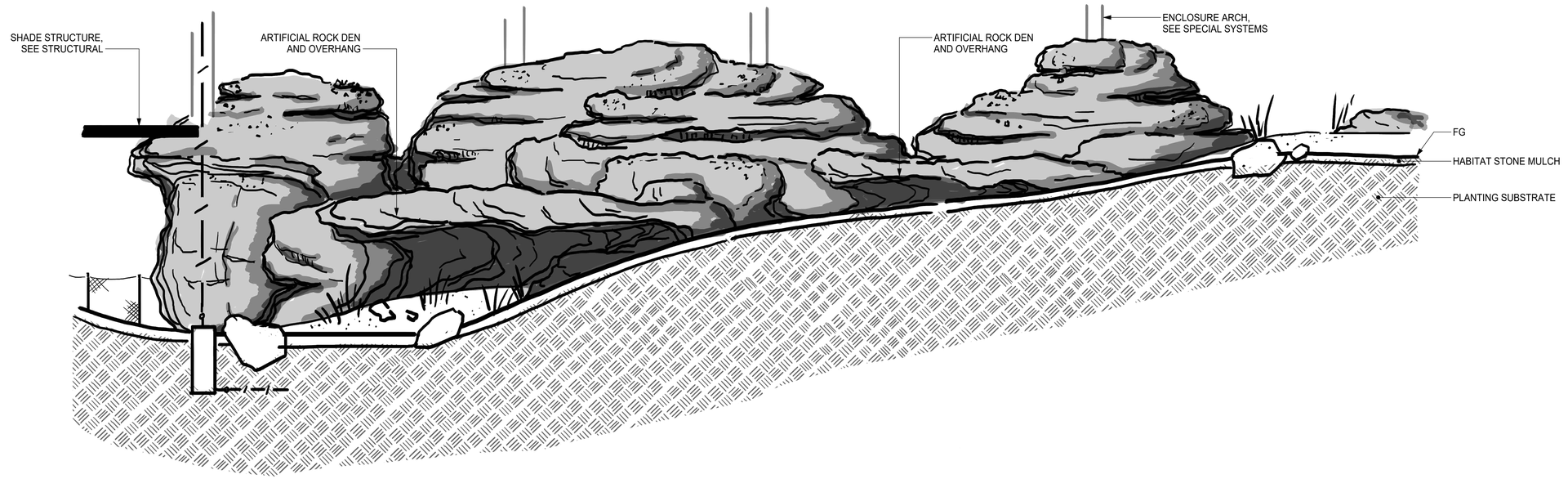
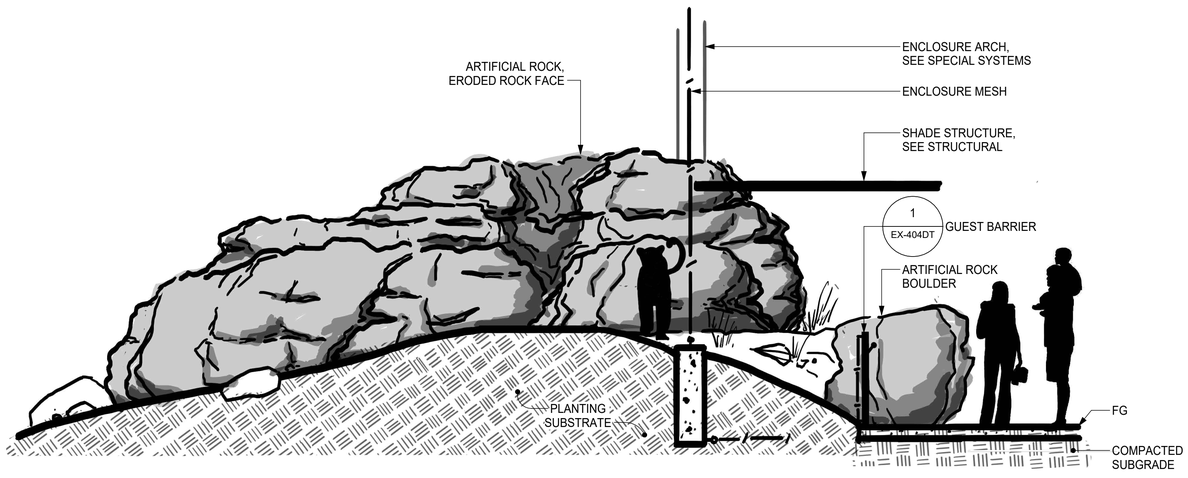
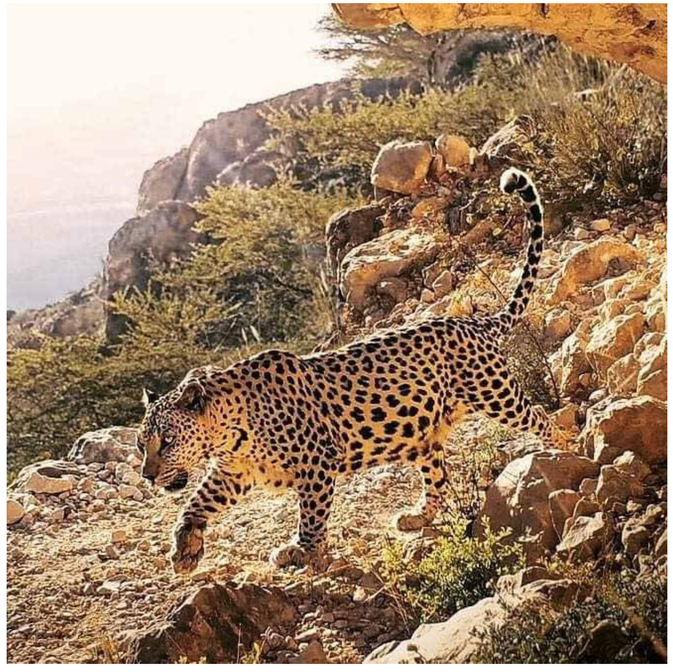
EX-155SP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-201SC
## Section Elevation B
3/8" = 1'-0"




2 EX-201SC
## Section Elevation A
3/8" = 1'-0"



3 EX-201SC
## Arabian Rock Reference Image
1/4" = 1'-0"

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

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STAMP

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SITE ELEVATIONS AND SCENES |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 218 OF 378 |
| | EX-201SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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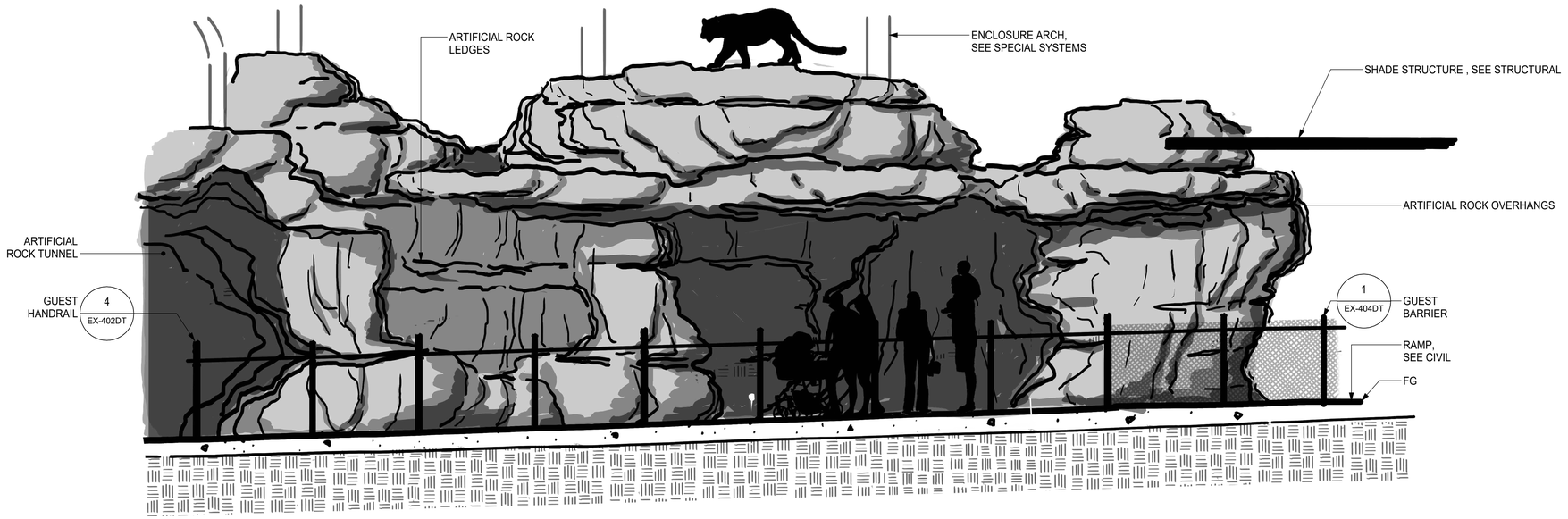
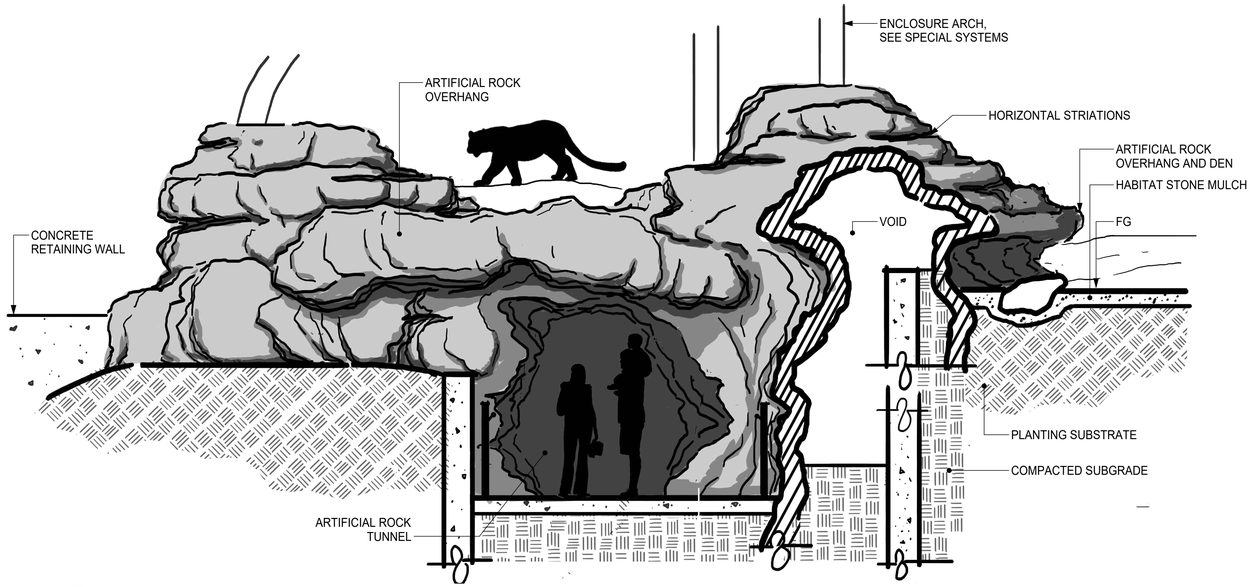



1 EX-202SC **Section Elevation C**
3/8" = 1'-0"




2 EX-202SC **Section Elevation D**
3/8" = 1'-0"

3 **Arabian Rock Overhang Reference Image**
NTS

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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STAMP

KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SITE ELEVATIONS AND SCENES |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 219 OF 378 |
| | EX-202SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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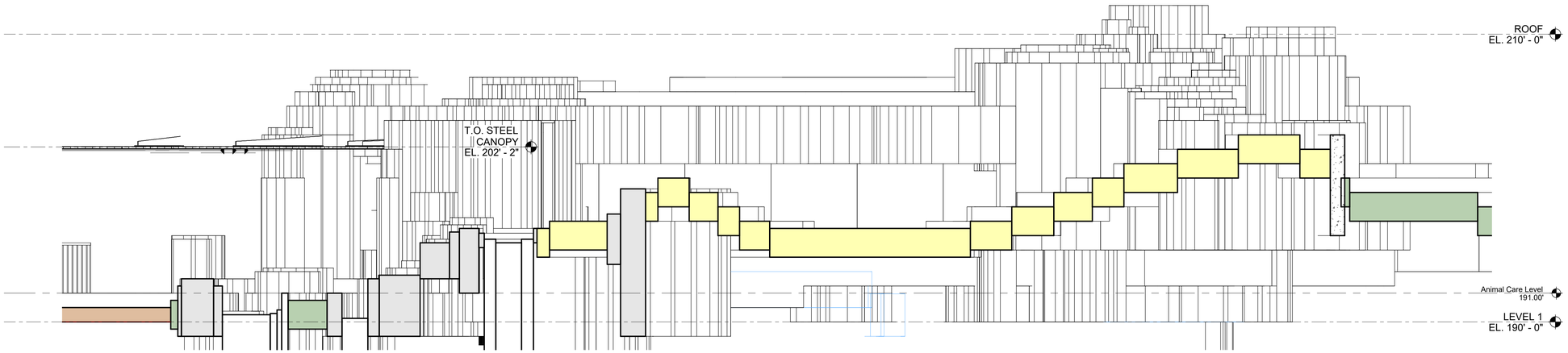
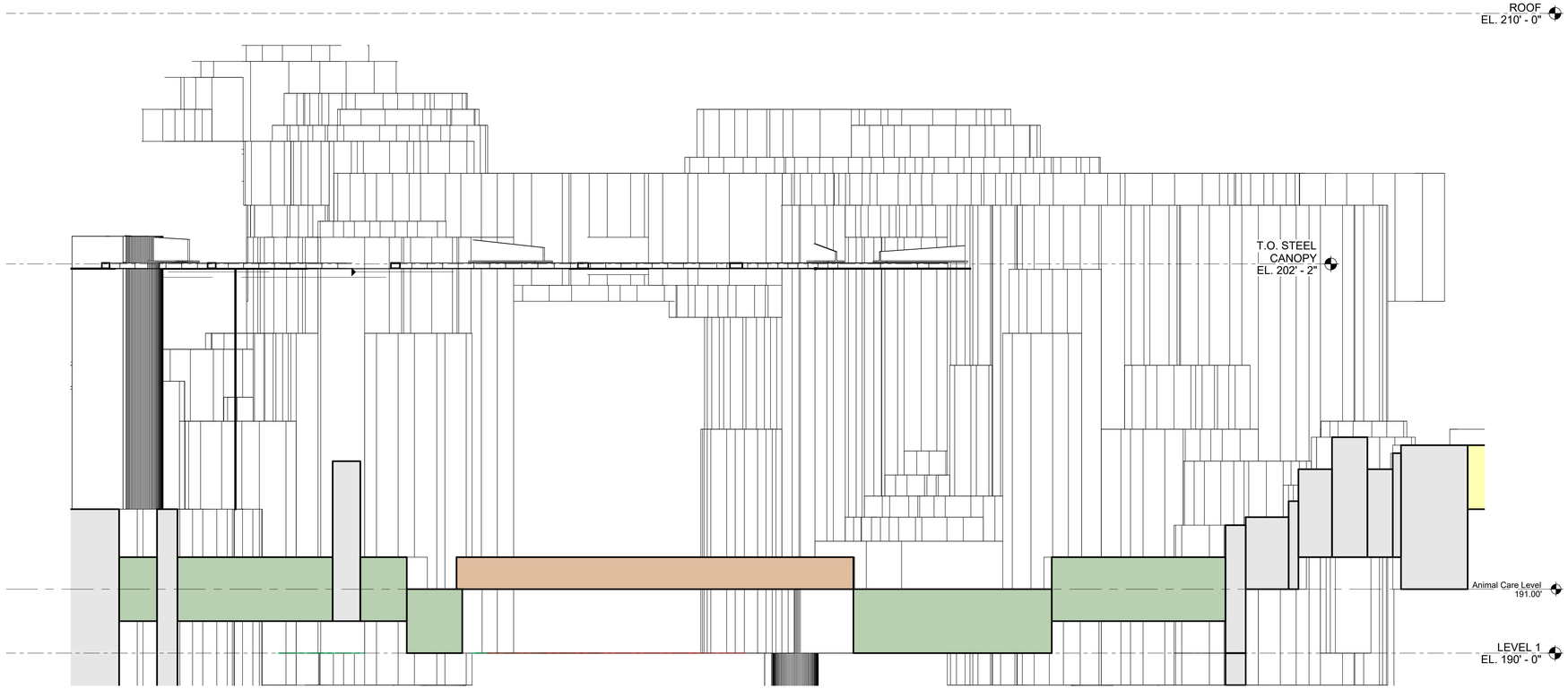



1 EX-203SC **Section E**
1/4" = 1'-0"




2 EX-203SC **Section F**
1/2" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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STAMP

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
0 1' 2' 4'
1/2"=1'-0"

**GRAPHIC SCALE(S)**

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| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SITE ELEVATIONS AND SCENES |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 220 OF 378

EX-203SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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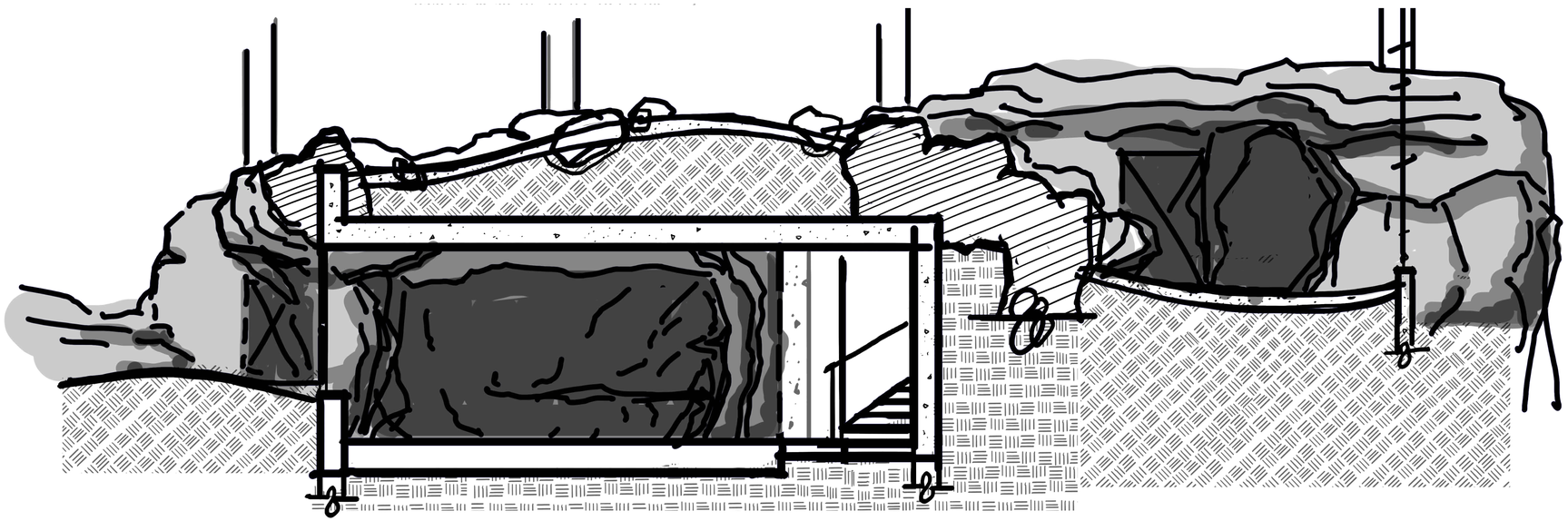
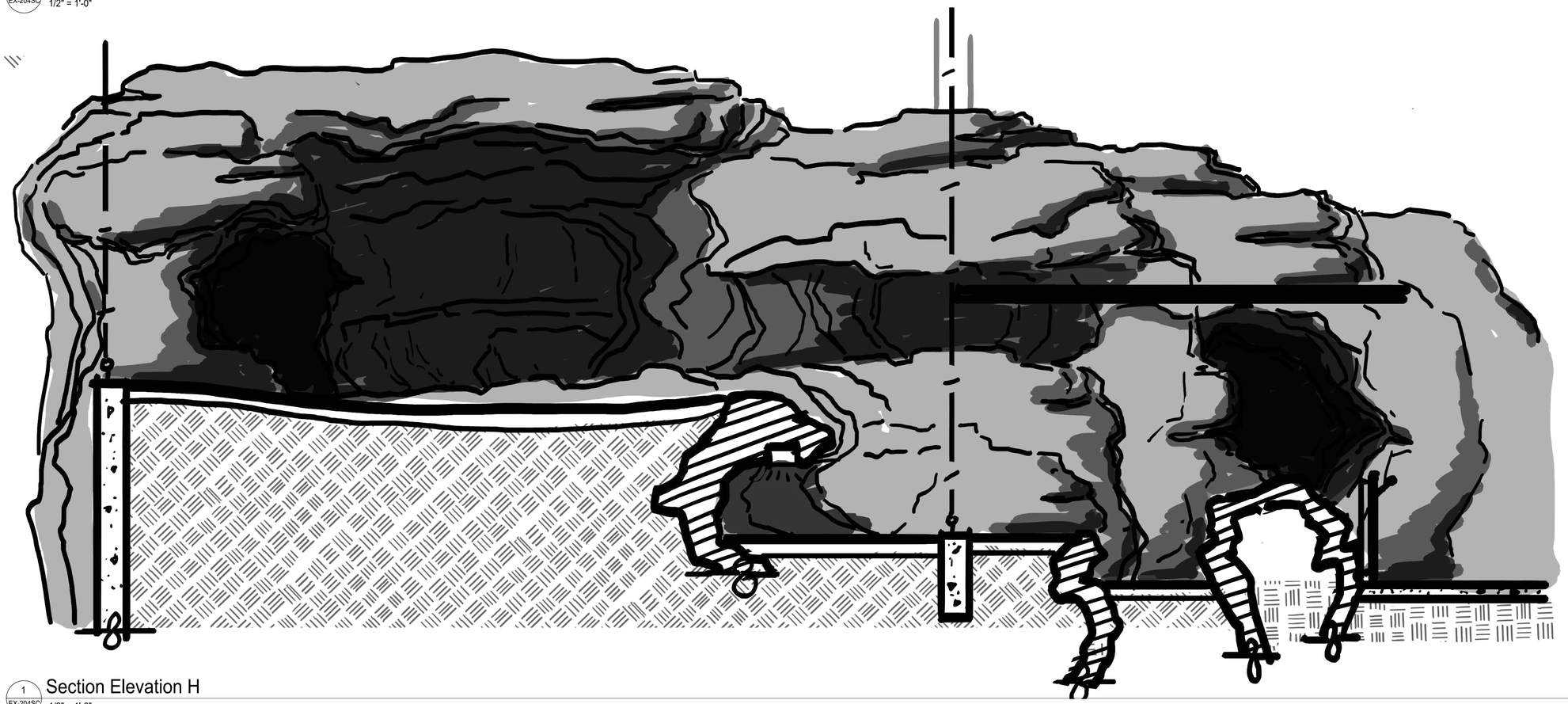
EX-203SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



2 EX-204SC
Section Elevation G
1/2" = 1'-0"



1 EX-204SC
Section Elevation H
1/2" = 1'-0"

# ALF

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STAMP

KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

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| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SITE ELEVATIONS AND SCENES |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 221 OF 378

EX-204SC
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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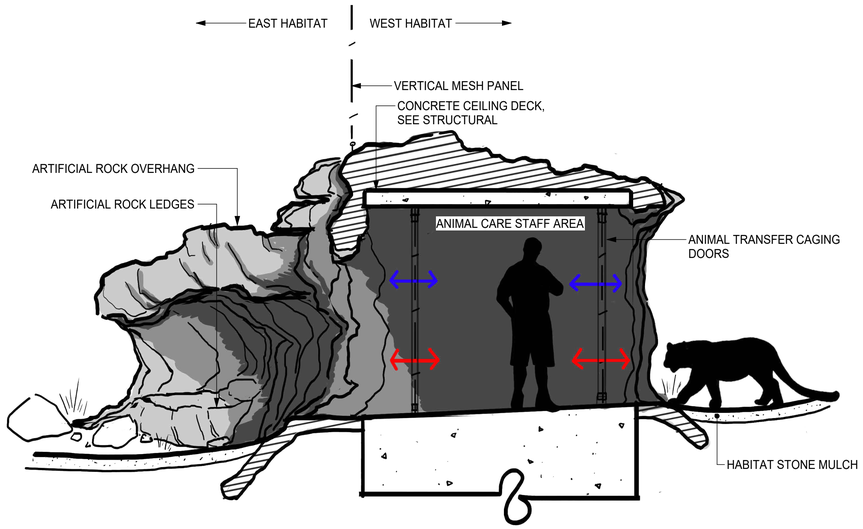
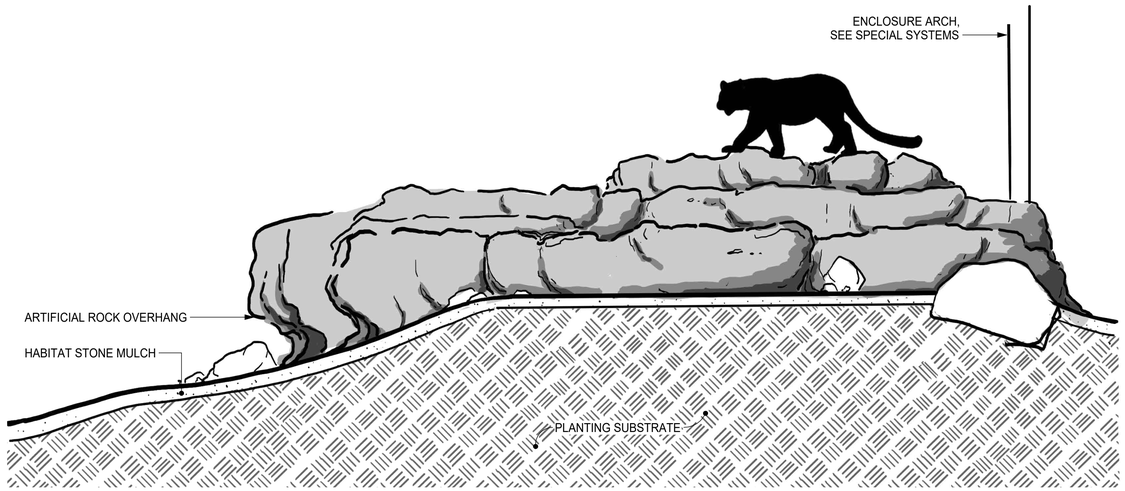
EX-204SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



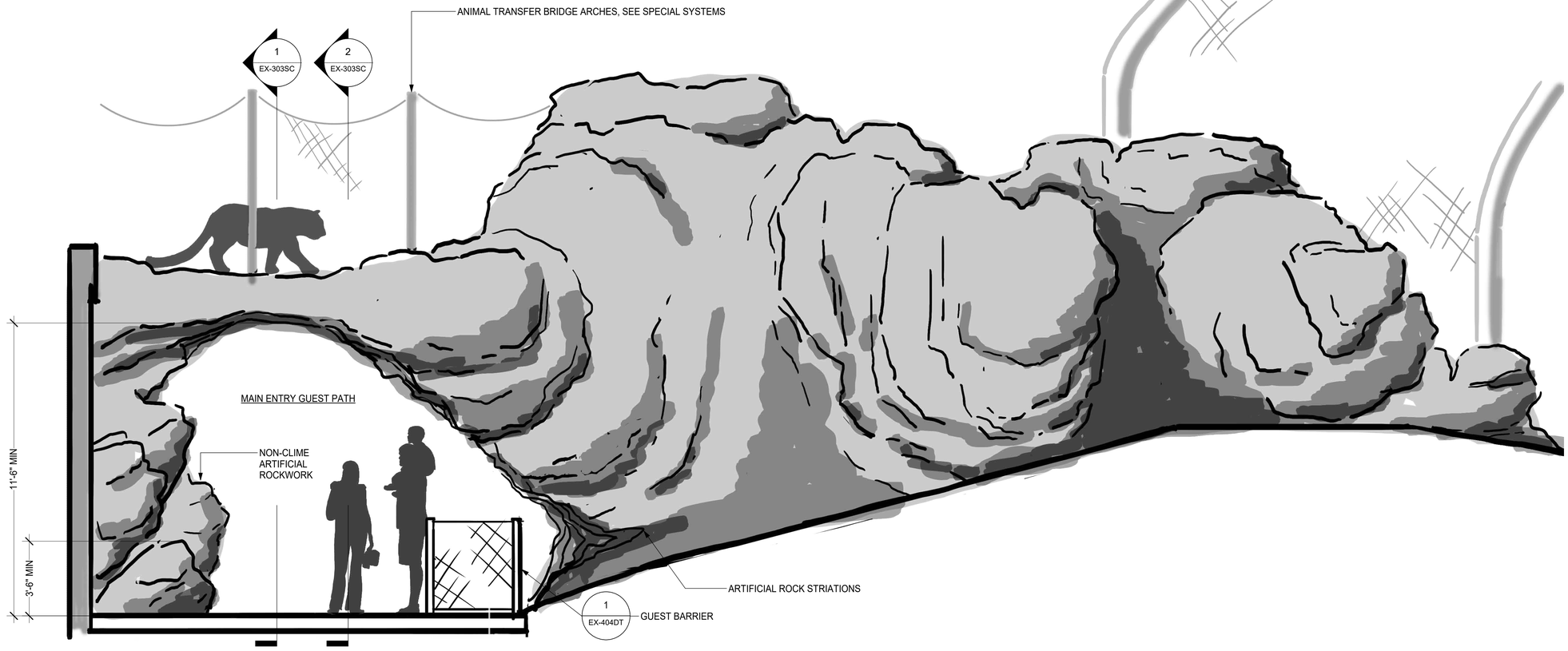

2 EX-205SC Section Elevation I

1/2" = 1'-0"




1 EX-205SC Section Elevation J

1/2" = 1'-0"




3 EX-205SC Section Elevation M

1/2" = 1'-0"

FOR REFERENCE ONLY

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STAMP

KEY PLAN

0 1' 2' 4'

1/2"=1'-0"

GRAPHIC SCALE(S)

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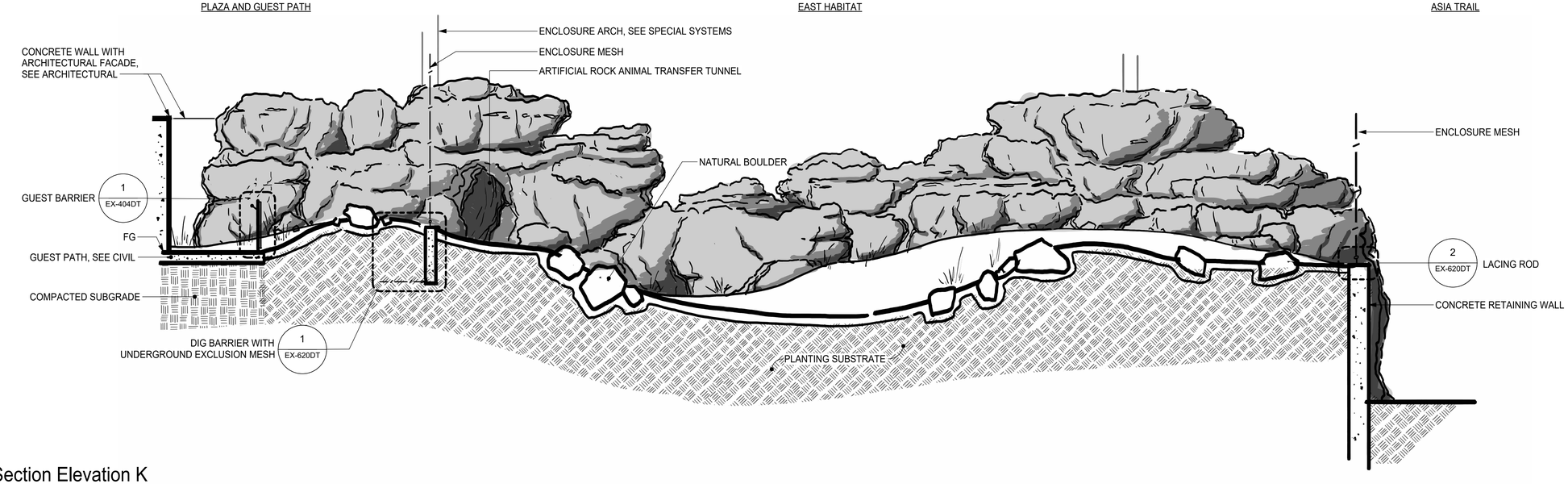
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SITE ELEVATIONS AND SCENES |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 222 OF 378 |
| | EX-205SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

6/11/2026 10:21:24 PM
EX-205SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-206SC **Section Elevation K**

1/4" = 1'-0"




2 EX-206SC **Section Elevation N**

1/2" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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STAMP

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
0 1' 2' 4'
1/2"=1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SITE ELEVATIONS AND SCENES |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 223 OF 378

EX-206SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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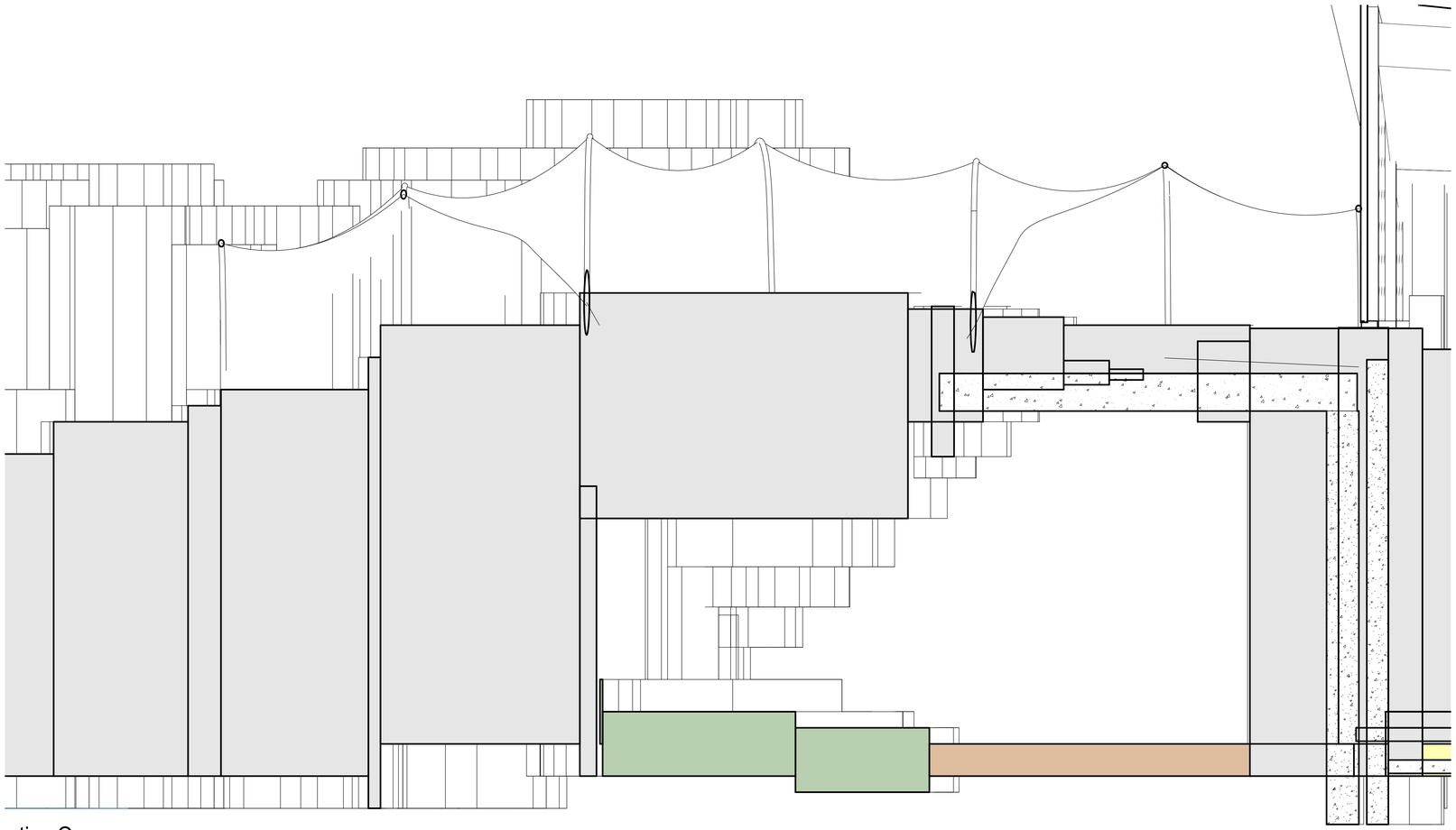
6/11/2026 10:22:18 PM
EX-206SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



1 EX-207SC
## Section O
1/2" = 1'-0"

2
## Arabian Arch Reference Image
NTS

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**CONSTRUCT ARABIAN LEOPARD FACILITY**

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STAMP

KEY PLAN

0 1' 2' 4'
1/2" = 1'-0"
**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SITE ELEVATIONS AND SCENES |

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| DF | SHR | JH |

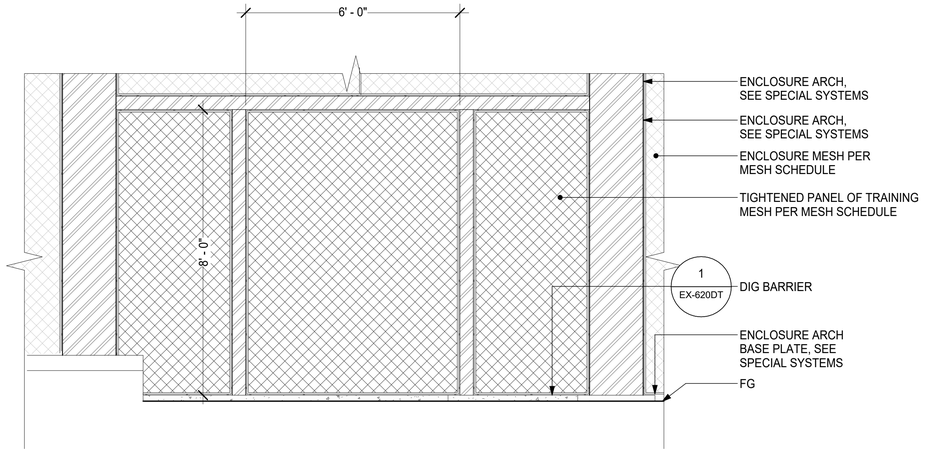
WORKING STAFF

SHEET NO. 224 OF 378

EX-207SC
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

*FOR REFERENCE ONLY*

6/11/2026 10:34:21 PM
EX-207SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-220DT Leopard Training Panel

1/2" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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STAMP

TRUE NORTH

KEY PLAN

0 1' 2' 4'

1/2"=1'-0"

GRAPHIC SCALE(S)

DATE: 6/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

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BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)

ADDRESS: 3001 Connecticut Ave NW Washington, DC 20008 USA

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: ENCLOSURE ELEVATIONS

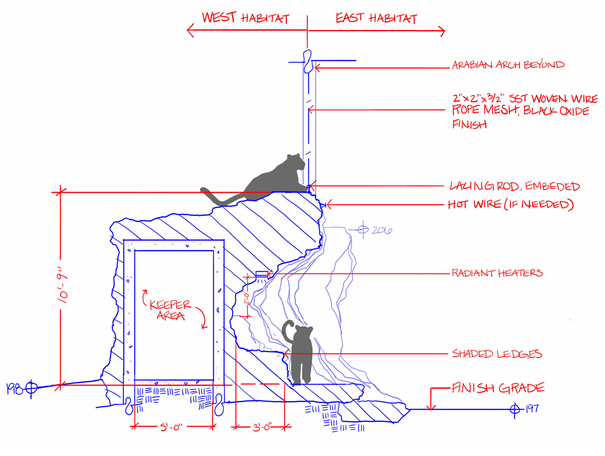
WORKING STAFF: DF (DESIGNED BY) | SHR (DRAWN BY) | JH (CHECKED BY)

SHEET NO. 225 OF 378

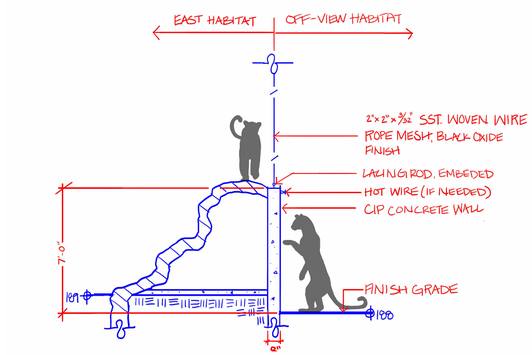
EX-220DT (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

FOR REFERENCE ONLY

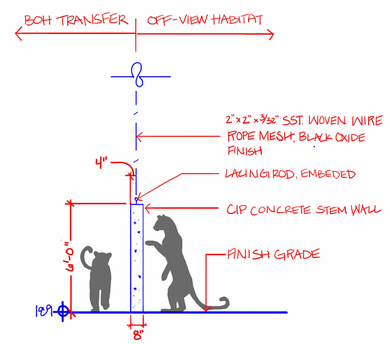
6/11/2026 10:54:46 PM
EX-220DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



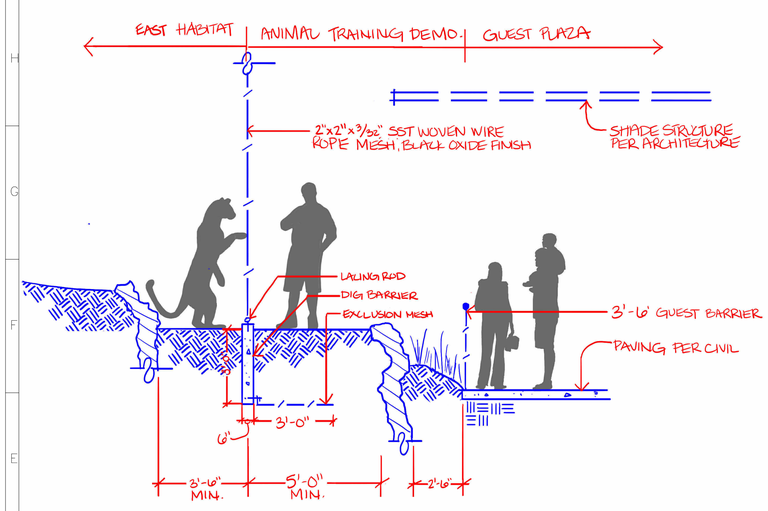

1 EX-300SC **Cat to Cat Condition Section - Habitat A & Habitat B**
1/4" = 1'-0"



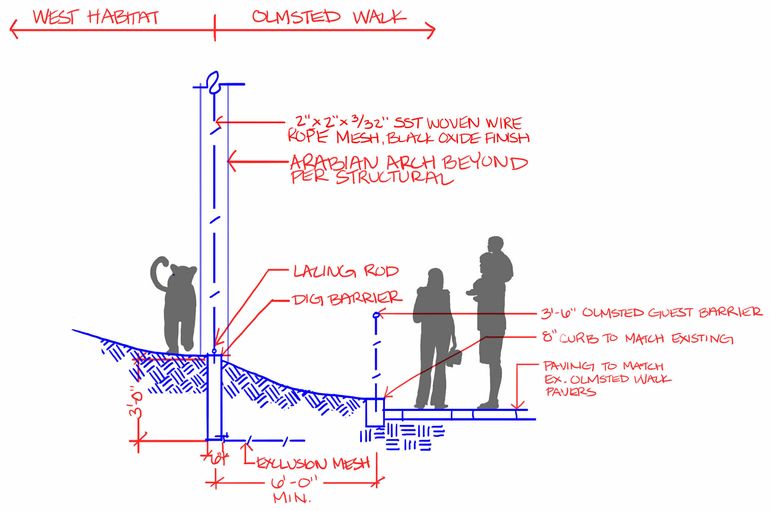

2 EX-300SC **Cat to Cat Condition Section - Habitat B & Off-View Habitat**
1/4" = 1'-0"



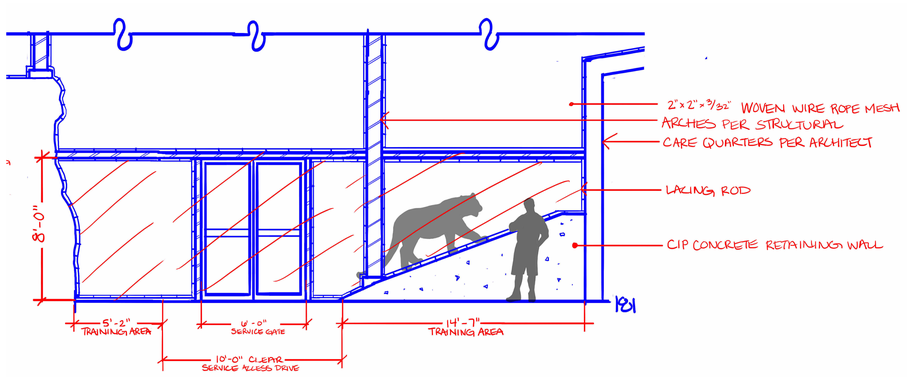

3 EX-300SC **Cat to Cat Condition Section - Off-View Habitat at Transfer**
1/4" = 1'-0"




4 EX-300SC **Cat to Staff Condition Section - Plaza Training**
3/8" = 1'-0"




5 EX-300SC **Cat to Guest Condition Section - Olmstead Walk Viewing**
3/8" = 1'-0"




6 EX-300SC **Cat to Staff Condition Section - VIP Mesh Training**
1/4" = 1'-0"

FOR REFERENCE ONLY

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

NOT FOR CONSTRUCTION

STAMP

TRUE NORTH

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
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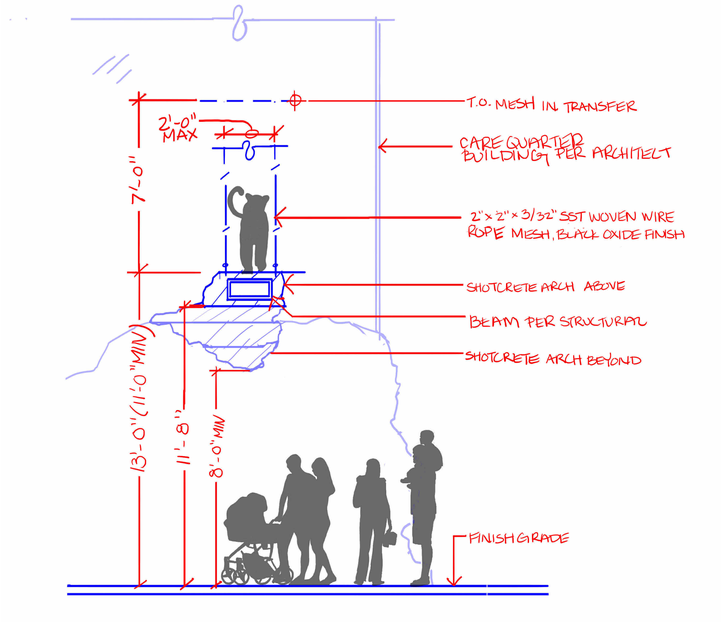
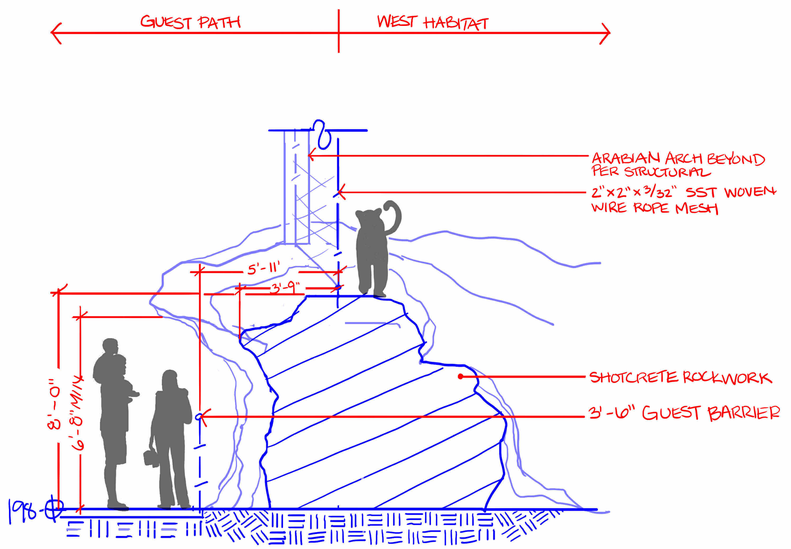
| BUILDING NAME / ADDRESS | ARABIAN LEOPARD FACILITY (ALF)<br>3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CRITICAL ANIMAL SECTIONS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 226 OF 378 |
| | EX-300SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

6/11/2026 11:00:03 PM
EX-300SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



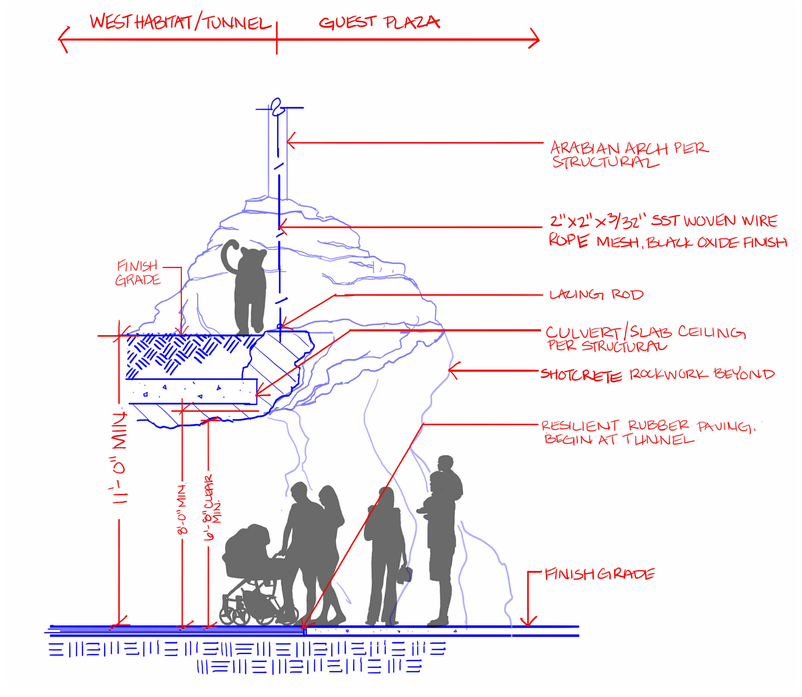

1 EX-301SC Cat to Guest Condition Section - Entry Arch Animal Overhead Transfer

3/8" = 1'-0"



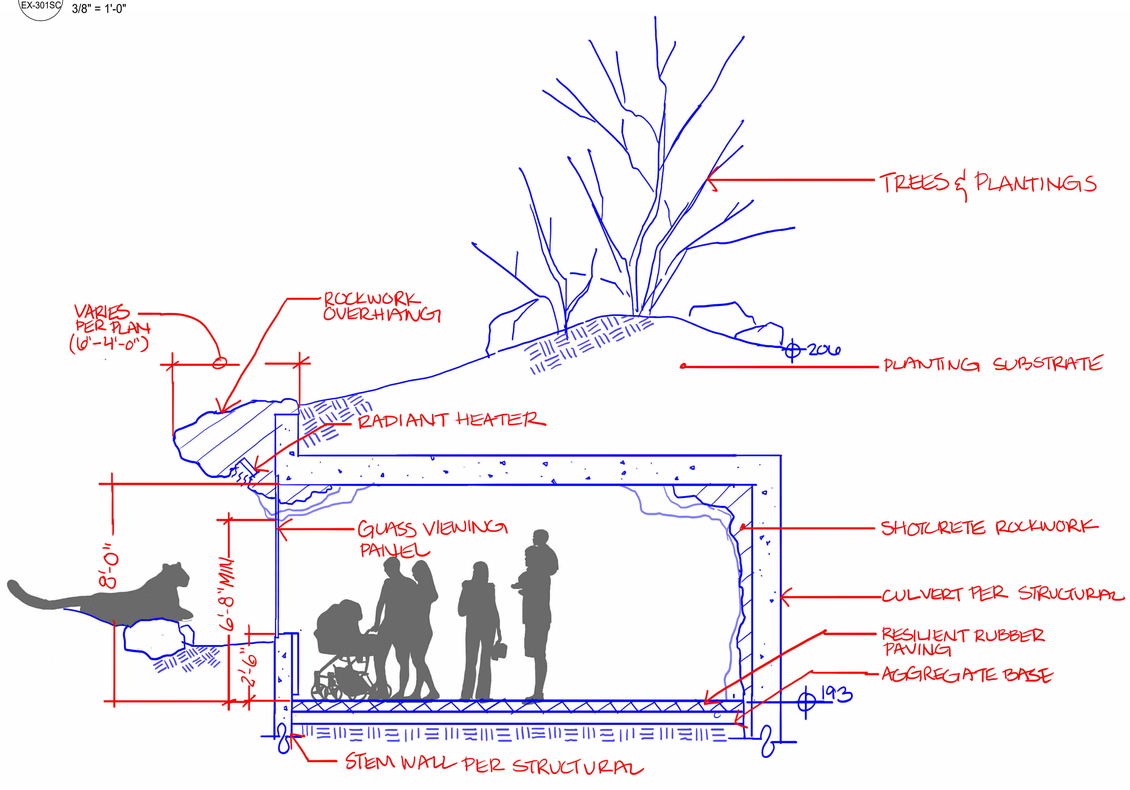

2 EX-301SC Cat to Guest Condition Section - Habitat A Rock Overhang

3/8" = 1'-0"




3 EX-301SC Cat to Guest Condition Section - Tunnel Entrance

3/8" = 1'-0"




4 EX-301SC Cat to Guest Condition Section - Tunnel Viewing

3/8" = 1'-0"

6/11/2026 11:00:05 PM
EX-301SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

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**CONSTRUCT ARABIAN LEOPARD FACILITY**

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STAMP

TRUE NORTH

KEY PLAN

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
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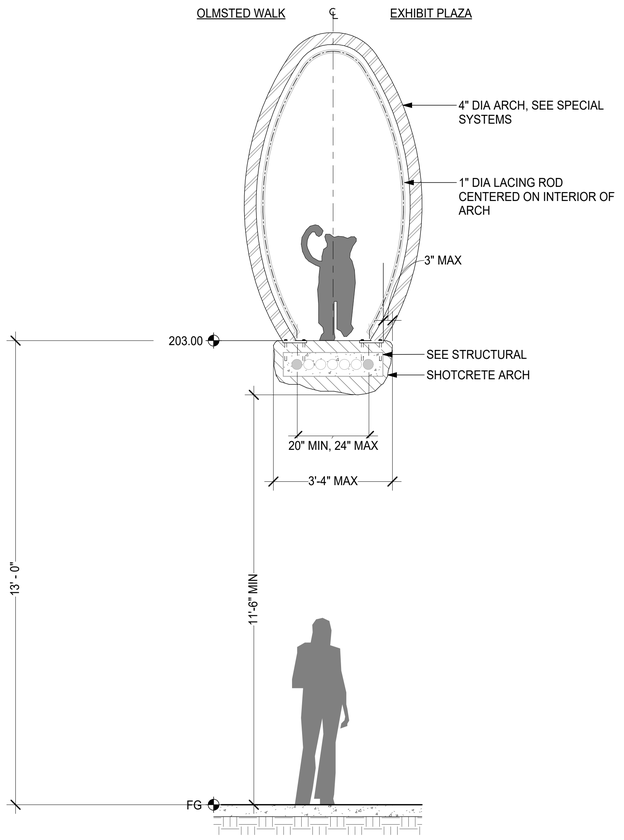
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CRITICAL ANIMAL SECTIONS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 227 OF 378

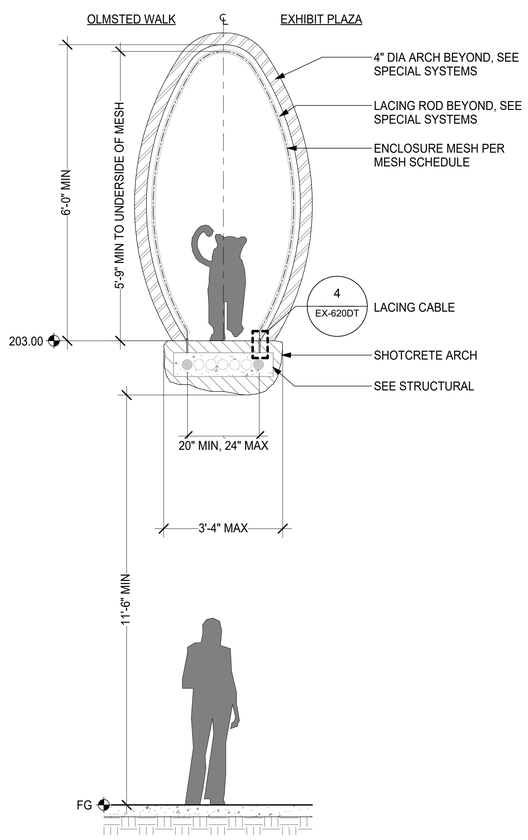
EX-301SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

FOR REFERENCE ONLY



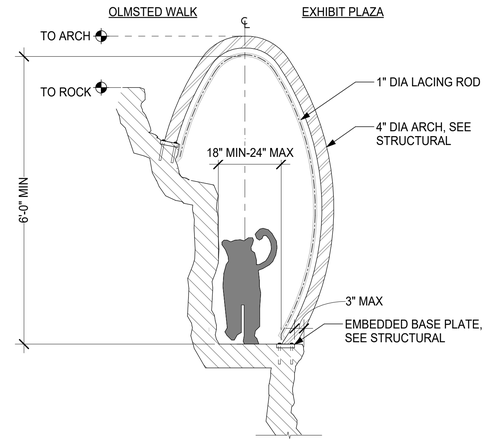

## 1 EX-302SC Overhead Transfer - End Condition Arch Section

1/2" = 1'-0"



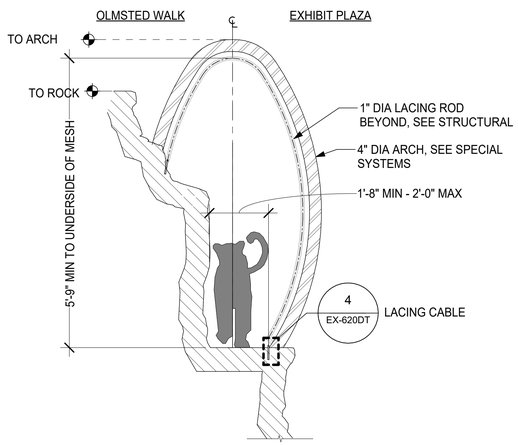

## 2 EX-302SC Overhead Transfer - End Condition Mesh Section

1/2" = 1'-0"



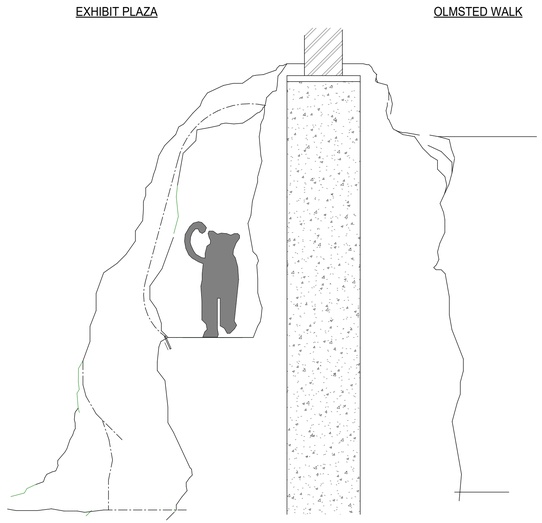

## 3 EX-302SC Overhead Transfer - Arch at Artificial Rock Section

1/2" = 1'-0"




## 4 EX-302SC Overhead Transfer - Mesh at Artificial Rock

1/2" = 1'-0"




## 5 EX-302SC Overhead Transfer to Enclosure Section

1/2" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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TRUE NORTH

0 1' 2' 4'

1/2"=1'-0"

**GRAPHIC SCALE(S)**

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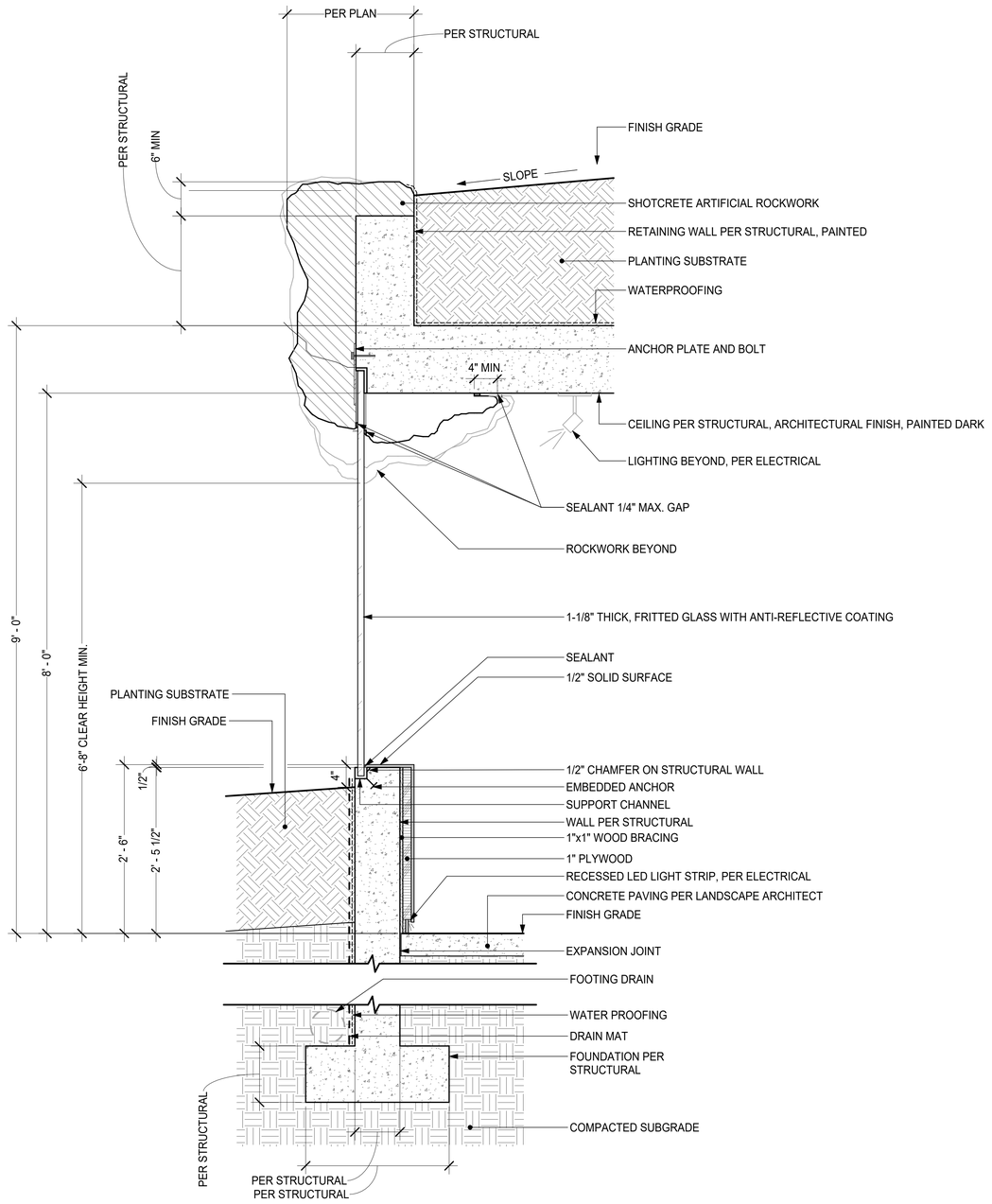
| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | OVERHEAD LEOPARD TRANSFER SECTIONS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 228 OF 378

EX-302SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

6/11/2026 11:00:07 PM
EX-302SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-303DT
## Artificial Rockwork At Glass Condition
1" = 1'-0"

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EX-303DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

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KEY PLAN

0 1" 2"
1" = 1"

GRAPHIC SCALE(S)

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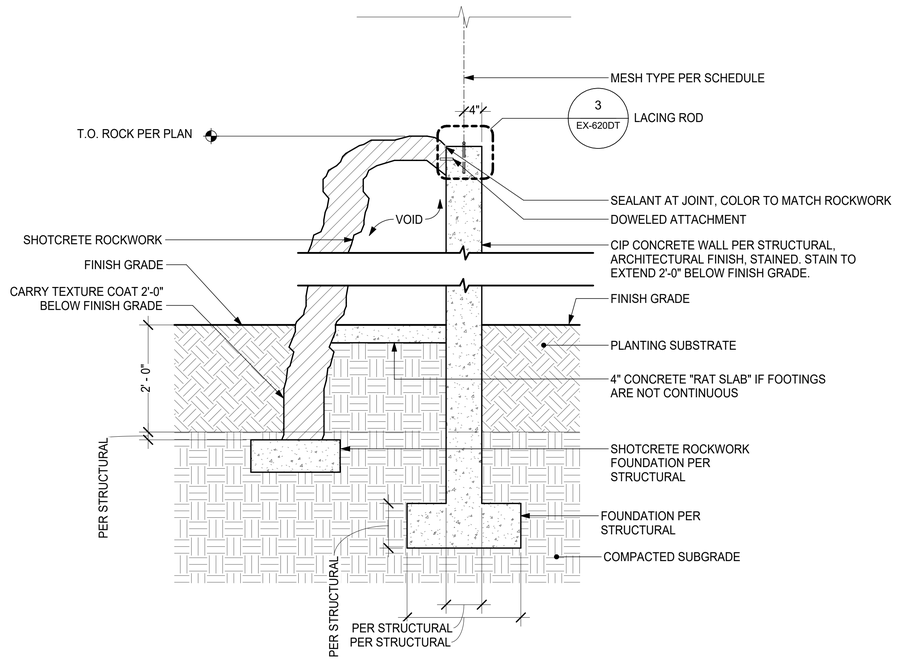
| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ARTIFICIAL ROCKWORK SECTION |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

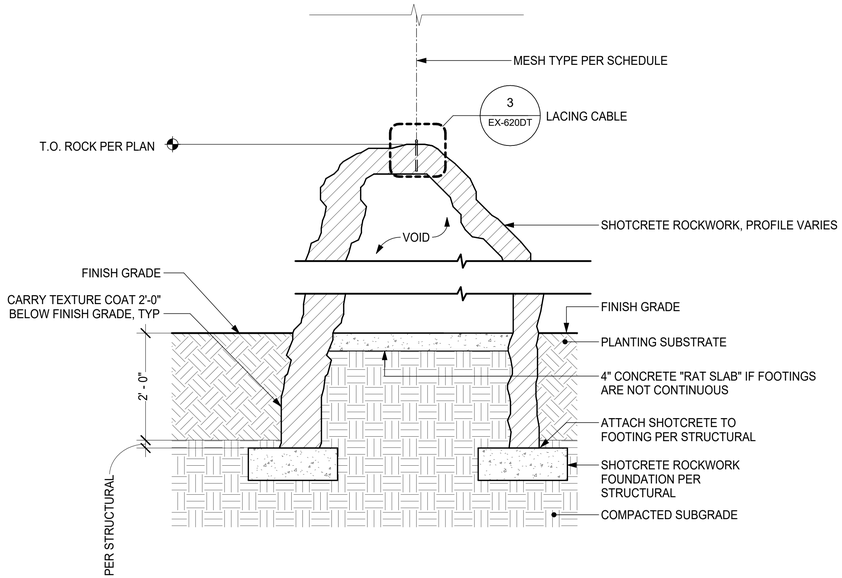
SHEET NO. 229 OF 378

EX-303DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)




1 EX-304DT **Shotcrete Rockwork At Off-View Habitat**

3/4" = 1'-0"




2 EX-304DT **Shotcrete Rockwork Section**

3/4" = 1'-0"

# ALF

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STAMP

TRUE NORTH

KEY PLAN

**GRAPHIC SCALE(S)**

DATE: 6/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

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Washington, DC 20024-2520

BUILDING NAME:
ADDRESS:
**ARABIAN LEOPARD FACILITY (ALF)**
**3001 Connecticut Ave NW**
**Washington, DC 20008 USA**

PROJECT TITLE:
**CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**

OPDC PROJECT NUMBER: **2533105**

A/E PROJECT NUMBER: **00100084**

DRAWING TITLE:
**ARTIFICIAL ROCKWORK SECTIONS**

WORKING STAFF:

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| DF | SHR | JH |

SHEET NO. 230 OF 378

EX-304DT

DISCIPLINE TYPE SEQUENCE DRAWING TYPE

*FOR REFERENCE ONLY*

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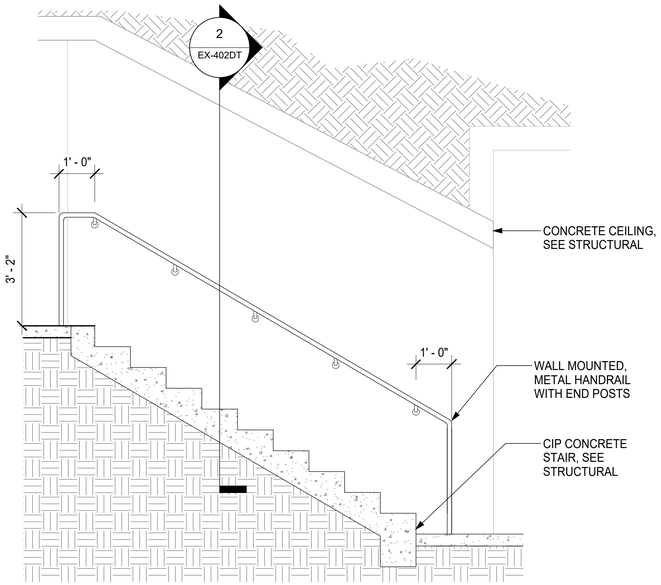
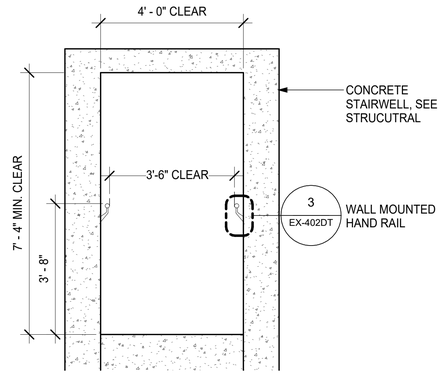
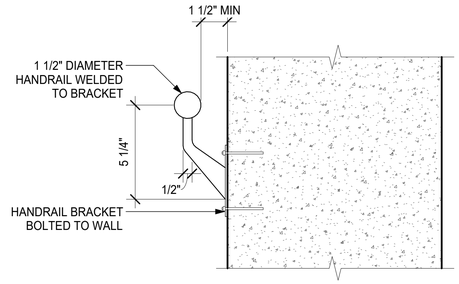
EX-304DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



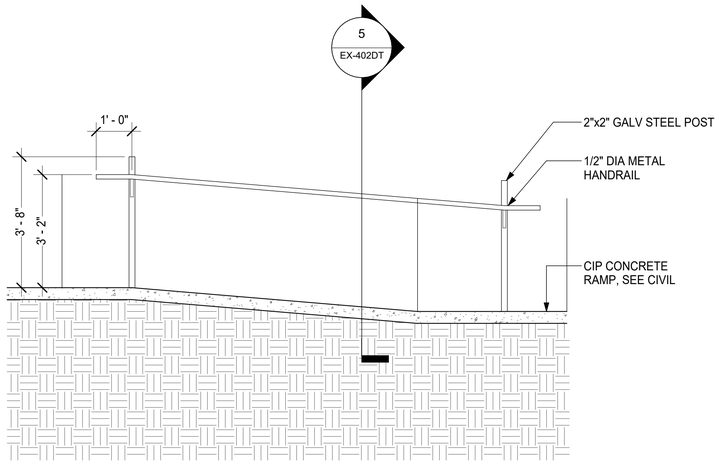

1 EX-402DT — Handrail at Tunnel Service Stairs - Section
1/2" = 1'-0"



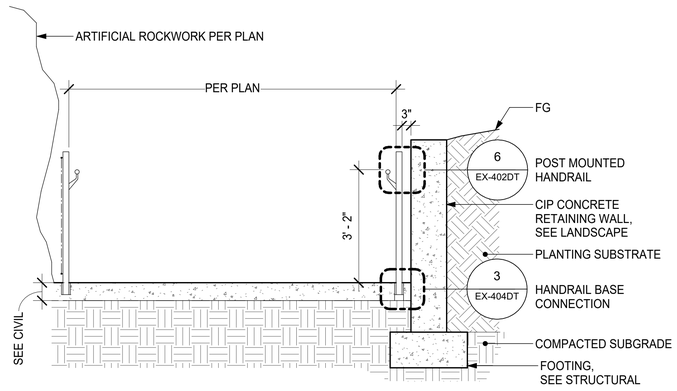

2 EX-402DT — Handrail at CIP Stairs - Section
1/2" = 1'-0"



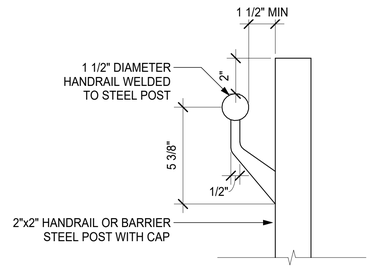

3 EX-402DT — Wall Mounted Handrail - Detail
3" = 1'-0"




4 EX-402DT — Handrail at Ramp - Section
1/2" = 1'-0"




5 EX-402DT — Handrail at CIP Ramp - Section
1/2" = 1'-0"




6 EX-402DT — Post Mounted Handrail - Detail
3" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

FOR REFERENCE ONLY

STAMP

TRUE NORTH

KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
| 6/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
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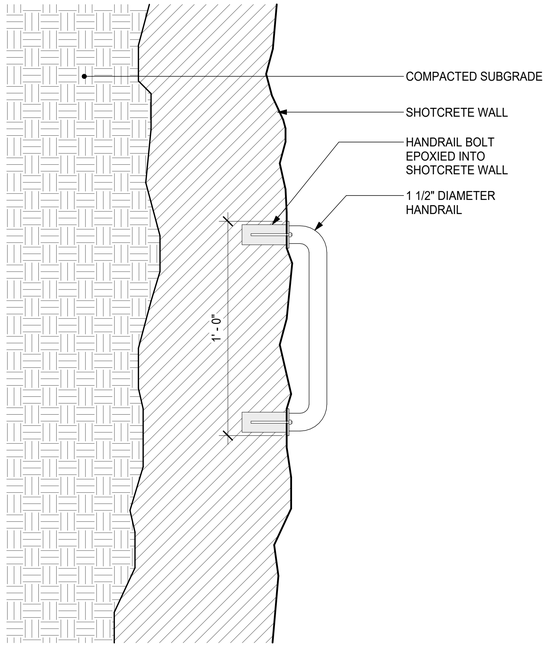
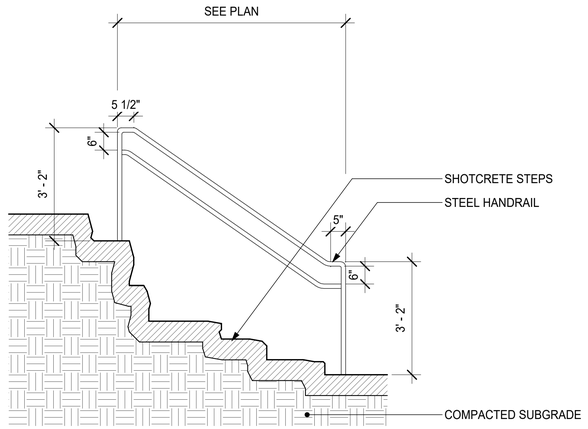
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | STAIRS, RAMPS , AND HANDRAILS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 231 OF 378 |
| | EX-402DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

6/11/2026 10:59:22 PM
EX-402DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



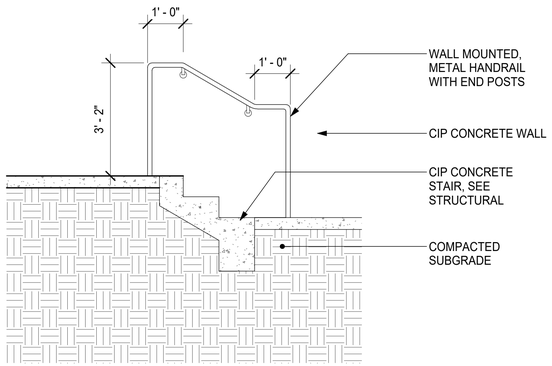

1 EX-403DT — Mounted Hand Grip - Detail

3" = 1'-0"



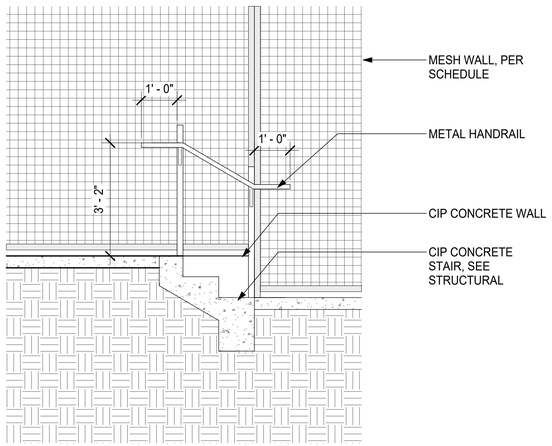

2 EX-403DT — Mounted Handrail on Shotcrete - Elevation

1/2" = 1'-0"




3 EX-403DT — Handrail on Wall at Care Staff Exterior Area - Section

1/2" = 1'-0"




4 EX-403DT — Handrail on Mesh at Care Staff Exterior Area - Section

1/2" = 1'-0"

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

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FOR REFERENCE ONLY

STAMP

TRUE NORTH

KEY PLAN

0 1' 2' 4'

1/2"=1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 6/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|
| | |

Smithsonian Institution

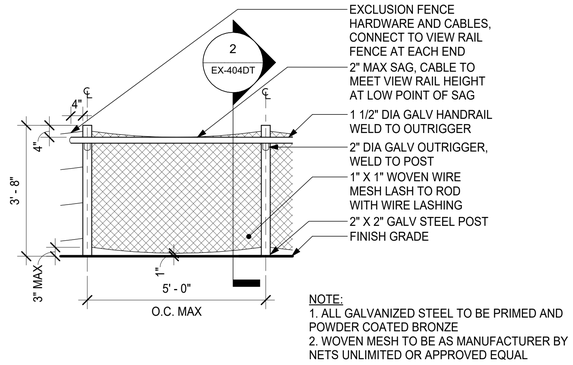
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | STAIRS, RAMPS , AND HANDRAILS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 232 OF 378

EX-403DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

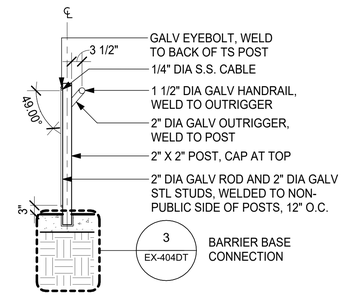
6/11/2026 10:59:23 PM
EX-403DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




NOTE:
1. ALL GALVANIZED STEEL TO BE PRIMED AND POWDER COATED BRONZE
2. WOVEN MESH TO BE AS MANUFACTURER BY NETS UNLIMITED OR APPROVED EQUAL

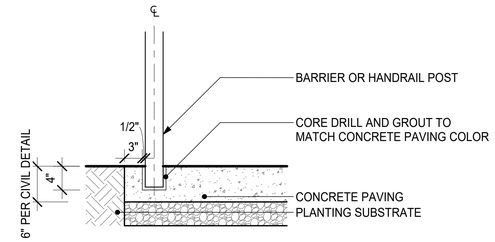
## 1 EX-404DT Guest Barrier - Elevation

1/2" = 1'-0"



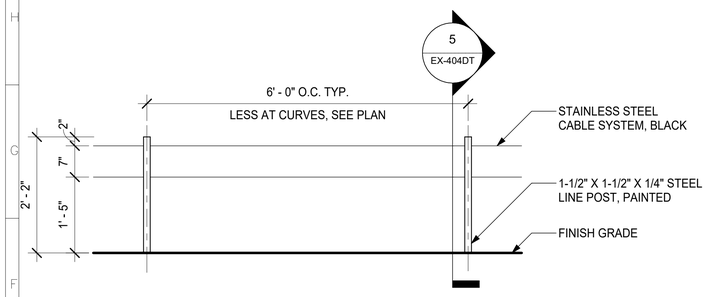

## 2 EX-404DT Guest Barrier - Section

1/2" = 1'-0"



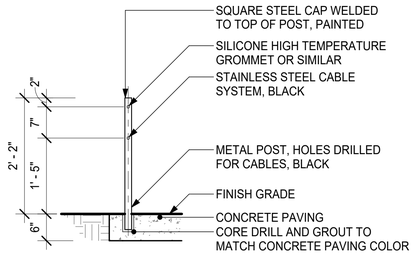

## 3 EX-404DT Barrier Base Connection - Section

1" = 1'-0"




## 4 EX-404DT Plant Protection Barrier - Elevation

3/4" = 1'-0"




## 5 EX-404DT Plant Protection Barrier - Section

3/4" = 1'-0"

# ALF

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TRUE NORTH

KEY PLAN

0 1" 2"
1"=1"
0 1' 2' 4'
1/2"=1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 6/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

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600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

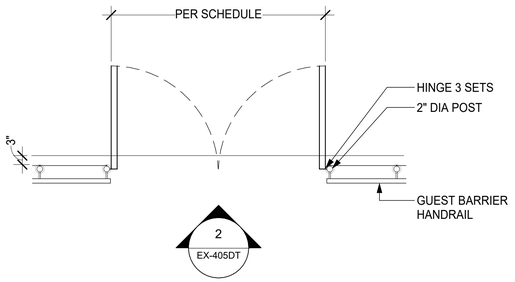
| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BARRIERS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 233 OF 378

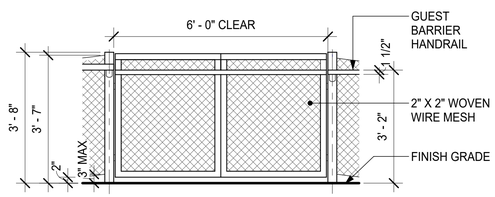
EX-404DT
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

FOR REFERENCE ONLY

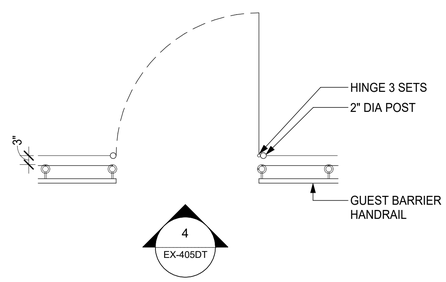
6/11/2026 10:59:24 PM
EX-404DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



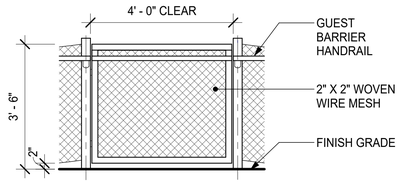

1 EX-405DT **Double Service Gate - Plan**
1/2" = 1'-0"




2 EX-405DT **Double Service Gate - Elevation**
1/2" = 1'-0"




3 EX-405DT **Single Service Gate - Plan**
1/2" = 1'-0"




4 EX-405DT **Single Service Gate - Elevation**
1/2" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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STAMP

TRUE NORTH

KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

DATE: **6/15/2026**

SUBMISSION: **65% CONSTRUCTION DOCUMENTS**

| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: **ARABIAN LEOPARD FACILITY (ALF)**

ADDRESS: **3001 Connecticut Ave NW Washington, DC 20008 USA**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**

OPDC PROJECT NUMBER: **2533105**

A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **BARRIERS**

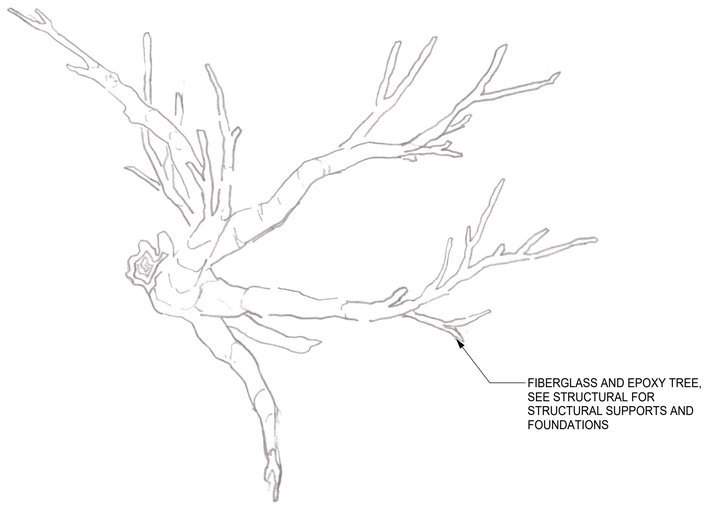
WORKING STAFF: DESIGNED BY **DF** | DRAWN BY **SHR** | CHECKED BY **JH**

SHEET NO. 234 OF 378

**EX-405DT**
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

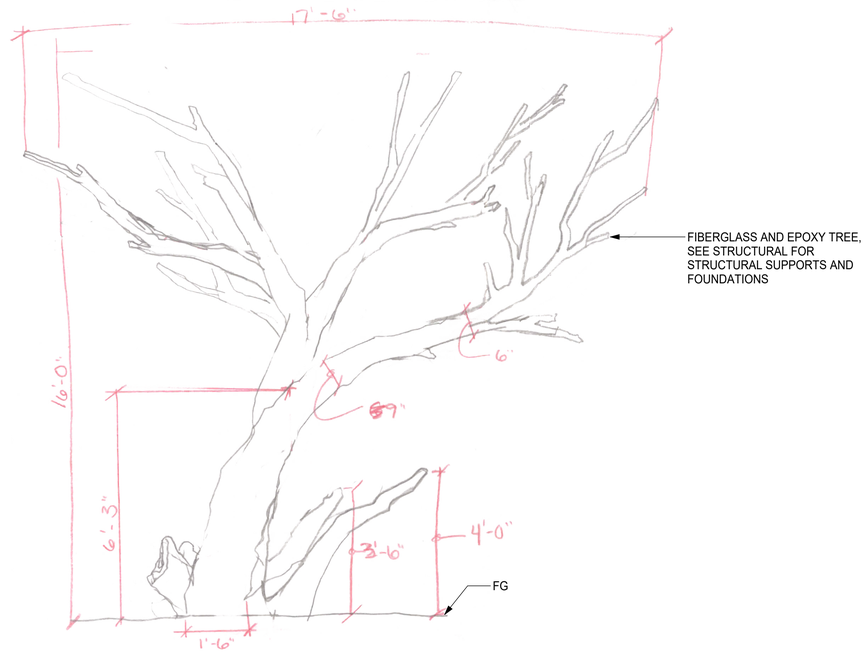
*FOR REFERENCE ONLY*

6/11/2026 10:59:25 PM
EX-405DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



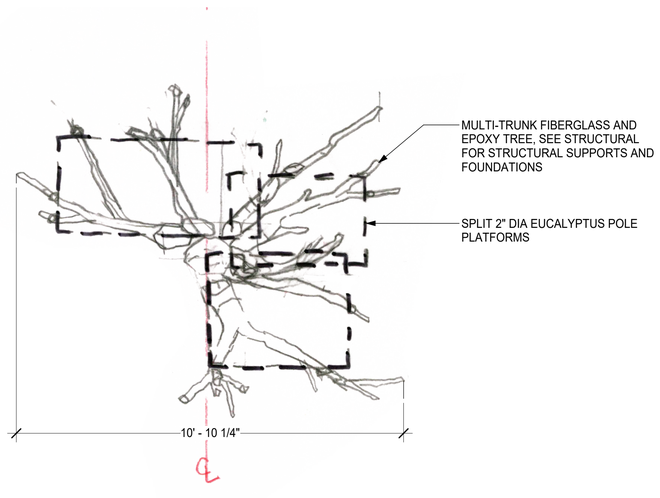

## 1 EX-510DT Artificial Tree AL2.1 - Plan

1/2" = 1'-0"



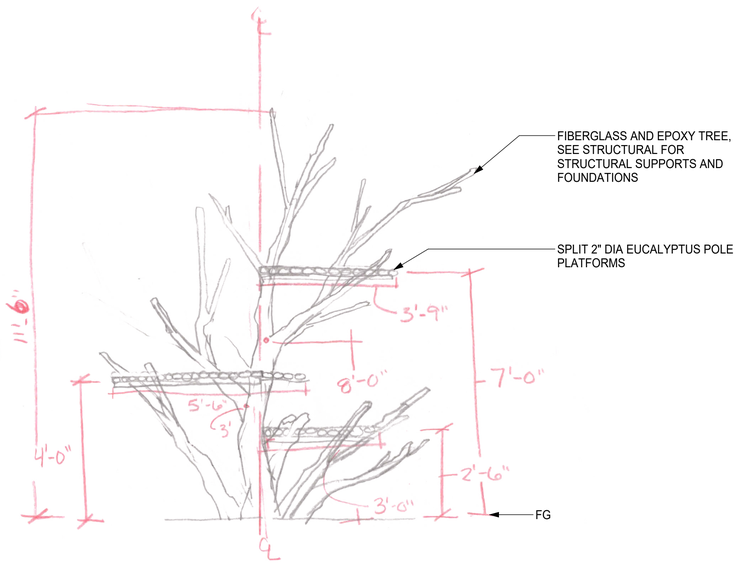

## 4 EX-510DT Artificial Tree AL2.1 - Elevation

1/2" = 1'-0"




## 3 EX-510DT Artificial Tree With Platforms AL2.2 - Plan

1/2" = 1'-0"




## 2 EX-510DT Artificial Tree With Platforms AL2.2 - Elevation

1/2" = 1'-0"

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

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STAMP

TRUE NORTH

KEY PLAN

0 1' 2' 4'

1/2"=1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 6/15/2026 | 65% CONSTRUCTION DOCUMENTS |

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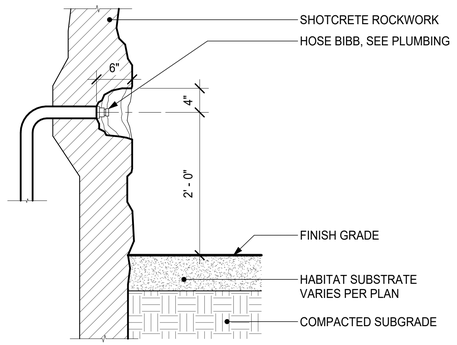
| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ARTIFICIAL TREES |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

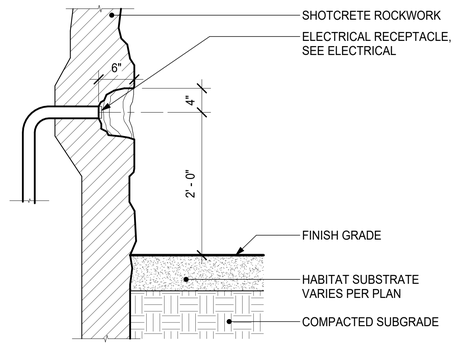
SHEET NO. 235 OF 378

EX-510DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

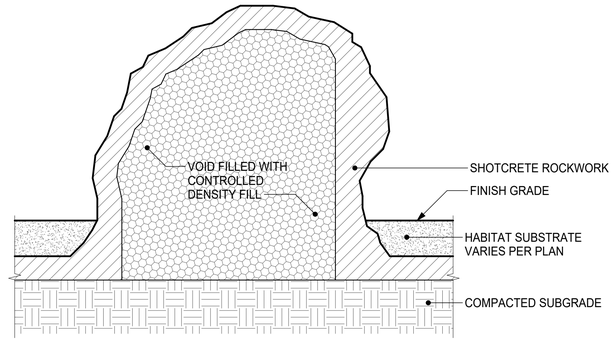

6/11/2026 9:52:17 PM
EX-510DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



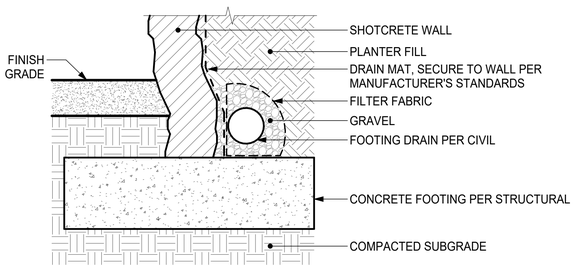

1 EX-601DT **Hose Bibb in Shotcrete - Section**
1" = 1'-0"



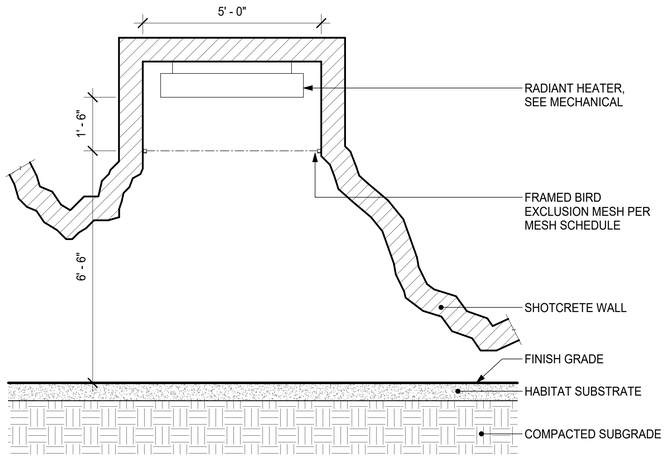

2 EX-601DT **Electrical Receptacle in Shotcrete - Section**
1" = 1'-0"



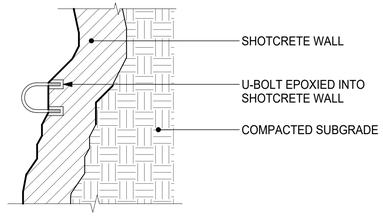

3 EX-601DT **Shotcrete Boulder - Section**
1/2" = 1'-0"

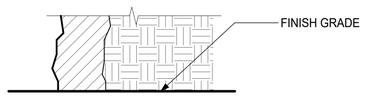



4 EX-601DT **Footing Drain - Section**
1" = 1'-0"

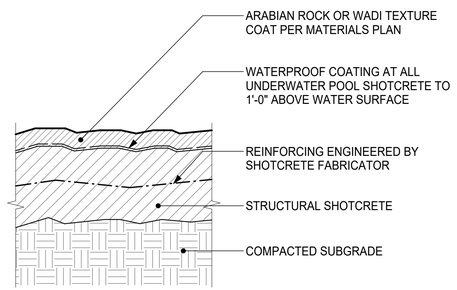



5 EX-601DT **Recessed Heater in Rockwork - Section**
1/2" = 1'-0"

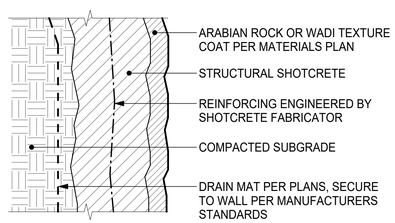



6 EX-601DT **Enrichment Connection in Shotcrete Wall - Section**
1" = 1'-0"




7 EX-601DT **Shotcrete Rockwork Under Water - Section**
1 1/2" = 1'-0"




8 EX-601DT **Shotcrete Rockwork Above Water - Section**
1 1/2" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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TRUE NORTH

KEY PLAN

0 1/2' 1' 1 1/2'
1 1/2"=1'-0"
0 1" 2"
1"=1"
0 1' 2' 4'
1/2"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
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Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME / ADDRESS | ARABIAN LEOPARD FACILITY (ALF)<br>3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SHOTCRETE DETAILS |

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| DF | SHR | JH |

WORKING STAFF

SHEET NO. 236 OF 378

EX-601DT
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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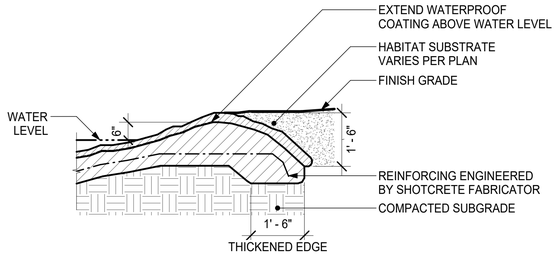
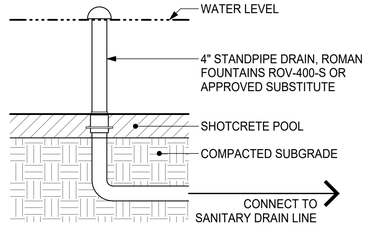
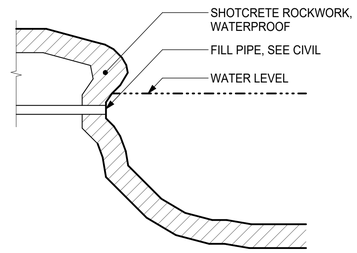
6/11/2026 11:04:11 PM
EX-601DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-602DT **Shotcrete Pool Edge**
1/2" = 1'-0"




2 EX-602DT **Pool Stand Pipe**
1/2" = 1'-0"




3 EX-602DT **Pool Fill Pipe**
1/2" = 1'-0"

# ALF

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TRUE NORTH

KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 6/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

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| | |
|---|---|
| BUILDING NAME / ADDRESS | **ARABIAN LEOPARD FACILITY (ALF)**<br>**3001 Connecticut Ave NW**<br>**Washington, DC 20008 USA** |
| PROJECT TITLE | **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)** |
| OPDC PROJECT NUMBER | **2533105** |
| A/E PROJECT NUMBER | **00100084** |
| DRAWING TITLE | **SHOTCRETE DETAILS** |
| WORKING STAFF | DESIGNED BY: **DF** · DRAWN BY: **SHR** · CHECKED BY: **JH** |
| SHEET NO. | 237 OF 378 |
| | **EX-602DT** (DISCIPLINE · TYPE · SEQUENCE · DRAWING TYPE) |

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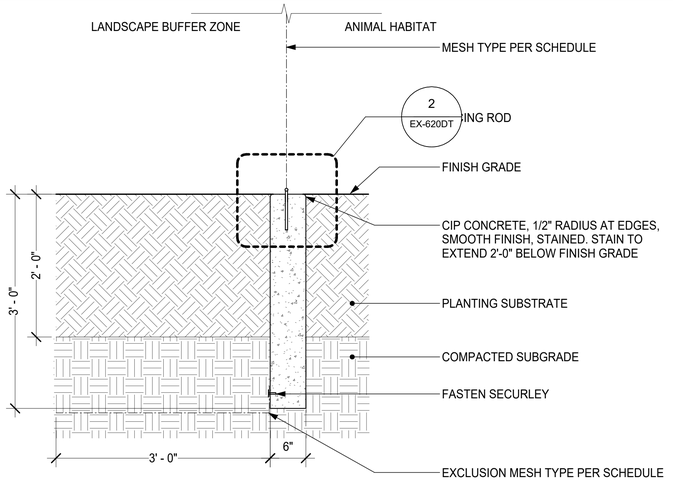
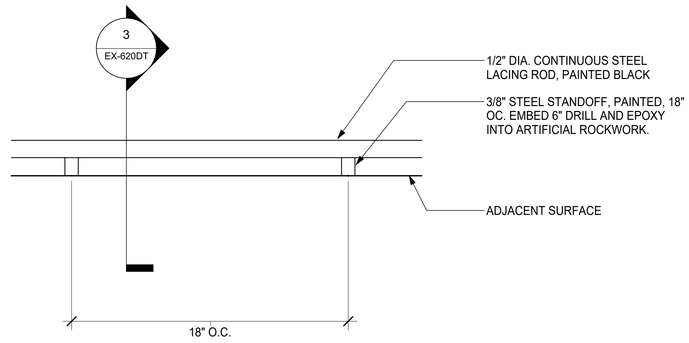
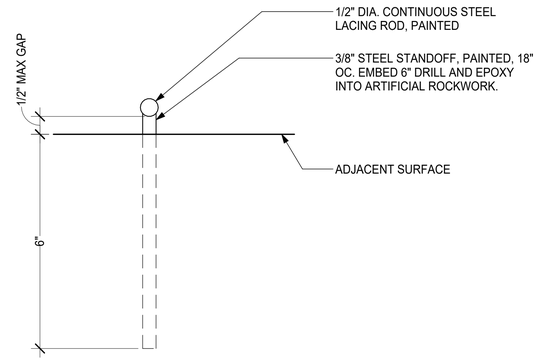
6/11/2026 11:04:43 PM
EX-602DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS



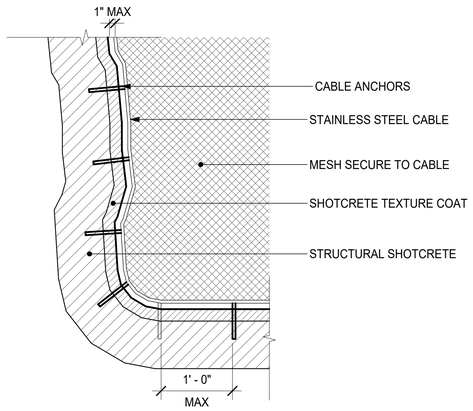

1 EX-620DT **Dig Barrier with Underground Exclusion Mesh**
1" = 1'-0"



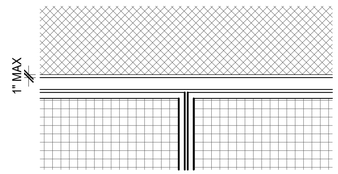

2 EX-620DT **Lacing Rod - Elevation**
6" = 1'-0"



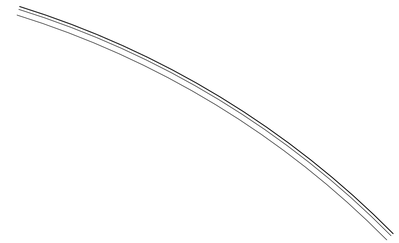

3 EX-620DT **Lacing Rod - Section**
6" = 1'-0"



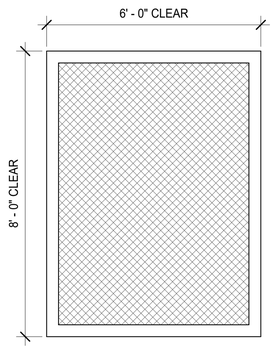

4 EX-620DT **Lacing Cable - Section**
1" = 1'-0"



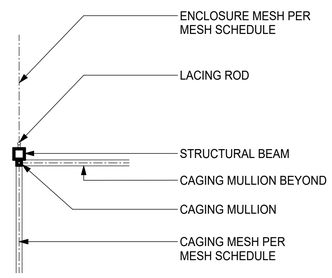

5 EX-620DT **Lacing Rod Attachment at Vestible**
1/2" = 1'-0"



6 EX-620DT **Lacing Rod at Structural Arch**
1/2" = 1'-0"




7 EX-620DT **Enclosure Service Access Panel - Elevation**
1/2" = 1'-0"




8 EX-620DT **Lacing Rod Above Caging**
1" = 1'-0"

# ALF

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TRUE NORTH

KEY PLAN

0 1" 2"
1" = 1"
0 1' 2' 4'
1/2" = 1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MESH DETAILS |
| WORKING STAFF | DF (DESIGNED BY) · SHR (DRAWN BY) · JH (CHECKED BY) |

SHEET NO. 238 OF 378

EX-620DT (DISCIPLINE · TYPE · SEQUENCE · DRAWING TYPE)

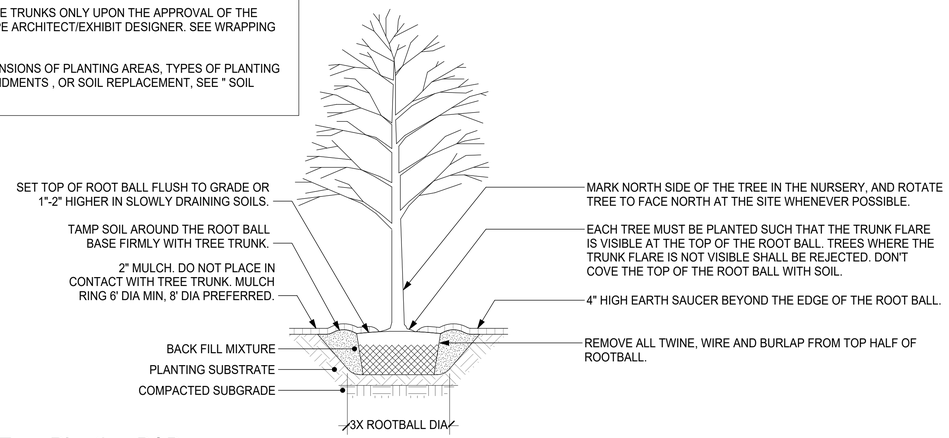
FOR REFERENCE ONLY

6/11/2026 11:05:07 PM
EX-620DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

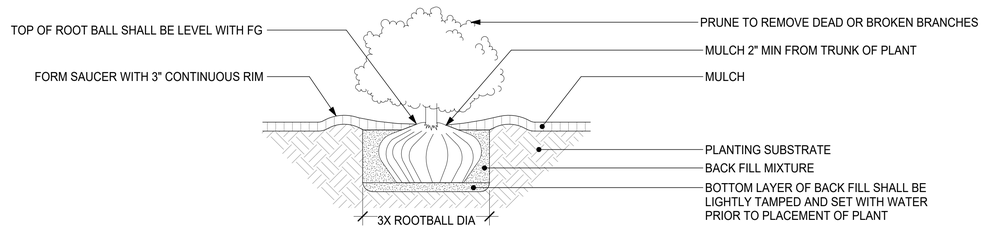
NOTE:
DO NOT HEAVILY PRUNE THE TREE AT PLANTING. PRUNE ONLY CROSSOVER LIMBS, CO-DOMINATE LEADERS, AND BROKEN OR DEAD BRANCHES, SOME INTERIOR TWIGS AND LATERAL BRANCHES MAY BE PRUNED; HOWEVER DO NOT REMOVE THE TERMINAL BUDS OF BRANCHES THAT EXTEND TO THE EDGE OF THE CROWN.

WRAP TREE TRUNKS ONLY UPON THE APPROVAL OF THE LANDSCAPE ARCHITECT/EXHIBIT DESIGNER. SEE WRAPPING SPEC.

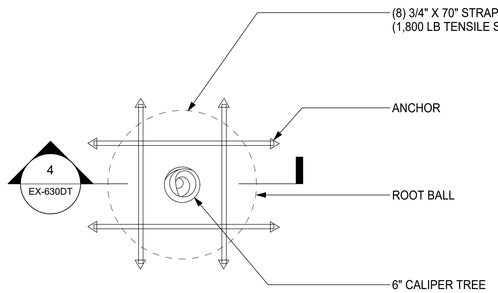
FOR DIMENSIONS OF PLANTING AREAS, TYPES OF PLANTING SOIL AMENDMENTS , OR SOIL REPLACEMENT, SEE " SOIL SPECS".




1 EX-630DT **Tree Planting B&B**
1/2" = 1'-0"

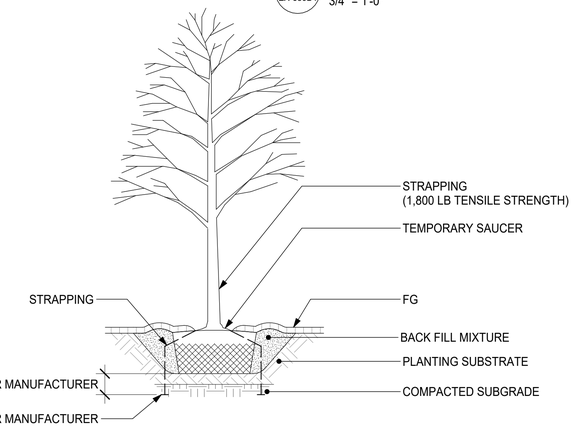



2 EX-630DT **Shrub Planting**
3/4" = 1'-0"

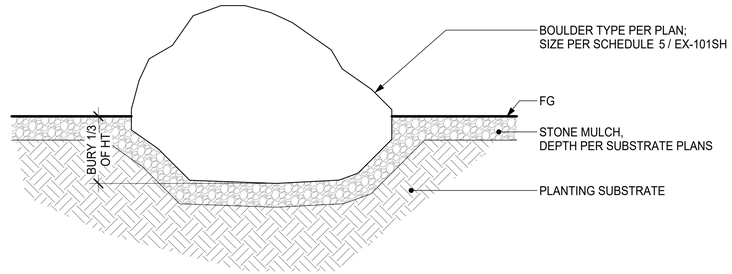



4 EX-630DT

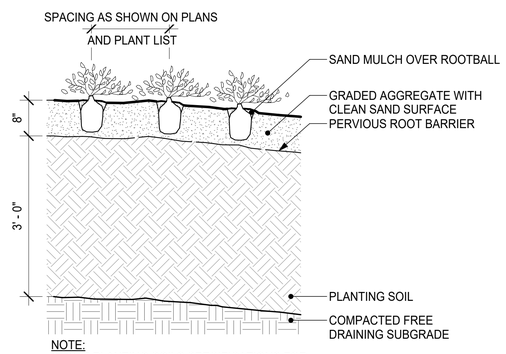
3 EX-630DT **Under Ground Guying - Plan**
1" = 1'-0"




4 EX-630DT **Under Ground Guying - Section**
1/2" = 1'-0"

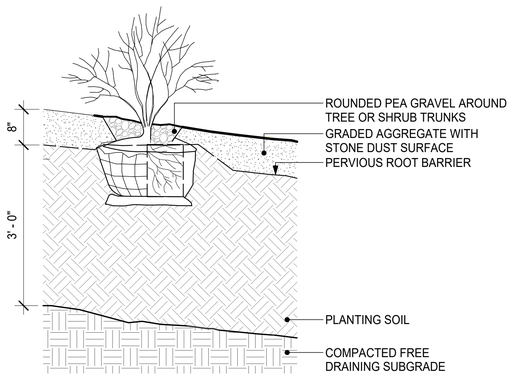



5 EX-630DT **Stand Alone Boulder**
1/2" = 1'-0"



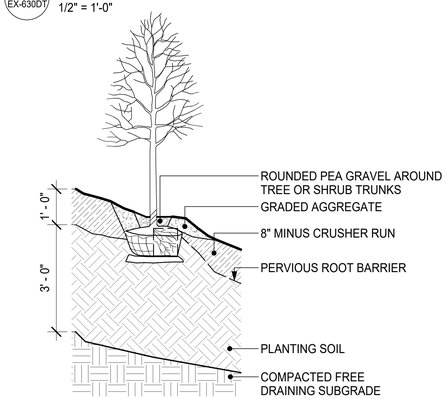

NOTE:
1. SEE PLANTING SOILS SPECIFICATIONS FOR SOIL MIXES AND PROPERTIES
2. HABITAT PLANTING SOILS FOR <5% SLOPES

6 EX-630DT **Shrub Planting on Slopes**
3/4" = 1'-0"




NOTE:
1. SEE PLANTING SOILS SPECIFICATIONS FOR SOIL MIXES AND PROPERTIES
2. HABITAT PLANTING SOILS FOR 5% - 20% SLOPES

7 EX-630DT **Tree or Shrub Planting on Slopes**
3/4" = 1'-0"




NOTE:
1. SEE PLANTING SOILS SPECIFICATIONS FOR SOIL MIXES AND PROPERTIES
2. HABITAT PLANTING SOILS FOR SLOPES GREATER THAN 20%

8 EX-630DT **Tree Planting on Slopes**
1/2" = 1'-0"

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TRUE NORTH

KEY PLAN

1" = 1"
1/2" = 1'-0"

GRAPHIC SCALE(S)

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| ADDRESS | 3001 Connecticut Ave NW Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT PLANTING DETAILS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 239 OF 378

EX-630DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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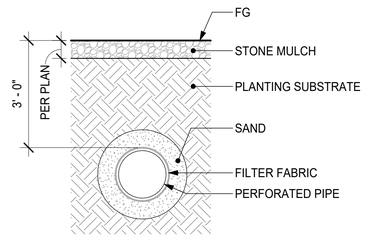
6/11/2026 11:05:31 PM
EX-630DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-631DT **Landscape Drain**
1/2" = 1'-0"

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KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

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| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HABITAT PLANTING DETAILS |
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SHEET NO. 240 OF 378

EX-631DT
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

*FOR REFERENCE ONLY*

6/11/2026 11:05:56 PM
EX-631DT BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

## LINES

DOOR / GATE OPERATOR OVERHEAD

## MATERIALS

ANIMAL SUBSTRATE

## SYMBOLS

K1.01 — DOOR / GATE ID - SEE SCHEDULE

DOOR / GATE OPERATOR

DRINKER - WALL-MOUNTED

DRINKER - PEDESTAL

HOSE BIBB

ELECTRICAL RECEPTACLE

HB — HOSE BIBB

## KEYNOTES

EX 1 — KEYNOTE TEXT

EX 2 — KEYNOTE TEXT

EX 3 — KEYNOTE TEXT

EX 4 — KEYNOTE TEXT

EX 5 — KEYNOTE TEXT

## NOTES

1. DRAWINGS FROM ALL DISCIPLINES MUST BE REVIEWED IN TANDEM WITH EXHIBIT DRAWINGS TO FULLY UNDERSTAND THE DESIGN INTENT. IF THERE IS A DISCREPANCY IT MUST BE BROUGHT TO THE ATTENTION OF EXHIBITS AND A RESOLUTION AGREED UPON BEFORE WORK MAY PROCEED.
2. ENSURE MAXIMUM GAP SIZE IS NOT EXCEEDED ALONG ENTIRE EXTENT OF ADJOINING SURFACES, ESPECIALLY IF THEY ARE NOT PARALLEL, E.G., A SLOPED FLOOR.

# ALF

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KEY PLAN

**GRAPHIC SCALE(S)**

DATE: 6/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

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PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**

OPDC PROJECT NUMBER: **2533105**

A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **CAGING LEGEND**

WORKING STAFF: DESIGNED BY **DF** | DRAWN BY **SHR** | CHECKED BY **JH**

SHEET NO. 241 OF 378

EX-700FP — DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

*FOR REFERENCE ONLY*

6/11/2026 5:41:03 PM
EX-700FP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

ANIMAL CARE KITCHEN

DEN 1 8' - 0"

DEN 2 8' - 0"

DEN 3 11' - 9"

DEN 4

DEN 5

KEEPER WORK SPACE

15' - 2 7/8"

FFE = 191.00

FFE = 190.00

HB

AD

1 EX-701FP Caging Plan 1/4" = 1'-0"

0' 2' 4' 8'

# ALF

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KEY PLAN

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
| 6/15/2026 | 65% CONSTRUCTION DOCUMENTS |

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| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CAGING PLANS |

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| DF | SHR | JH |

WORKING STAFF

SHEET NO. 242 OF 378

EX-701FP

DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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6/11/2026 5:41:45 PM
EX-701FP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

# ALF

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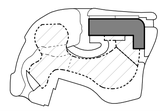

KEY PLAN

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CAGING CEILING PLANS |

| WORKING STAFF | DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|---|
| | DF | SHR | JH |

SHEET NO. 243 OF 378

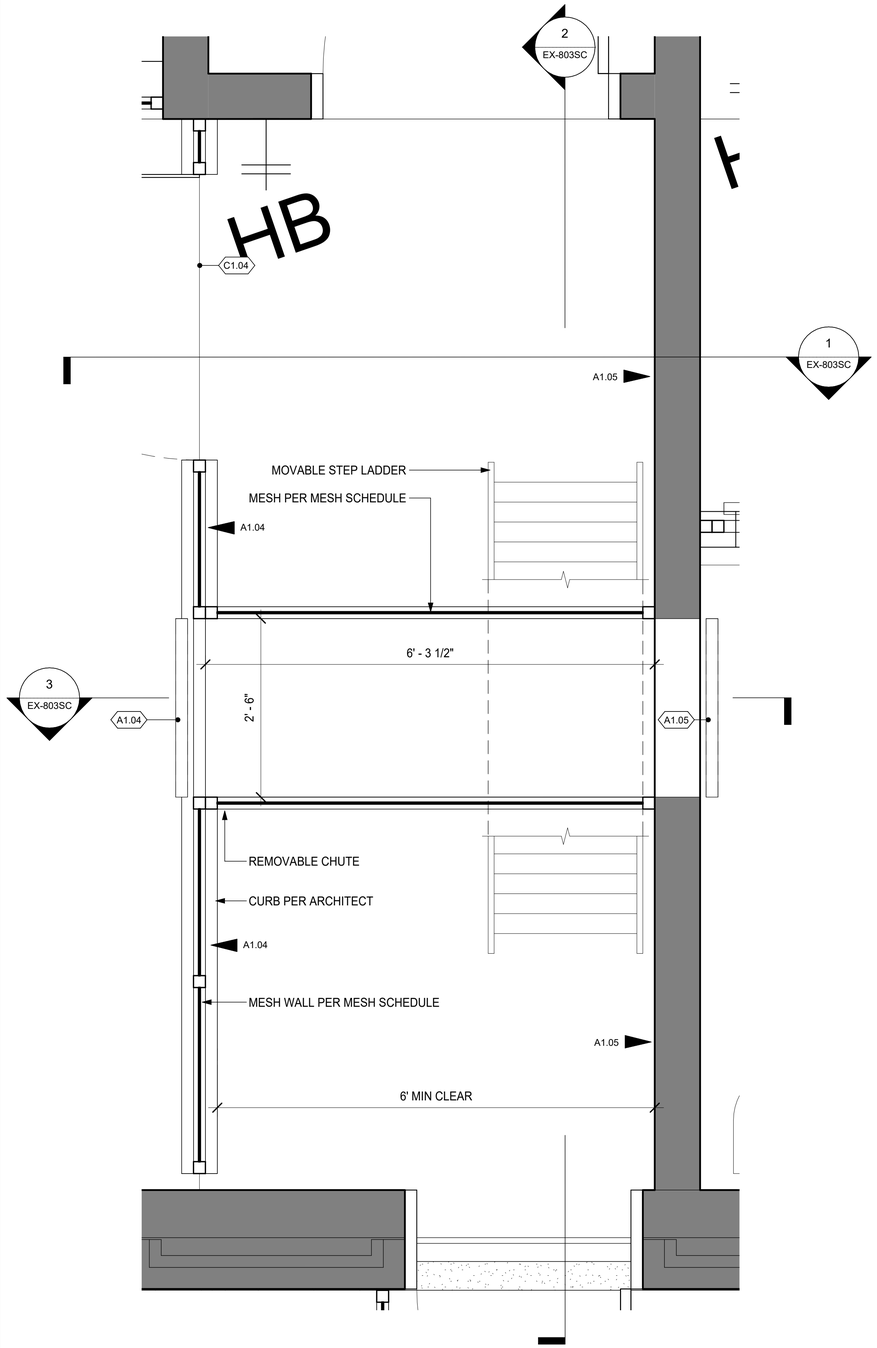
EX-702FP

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

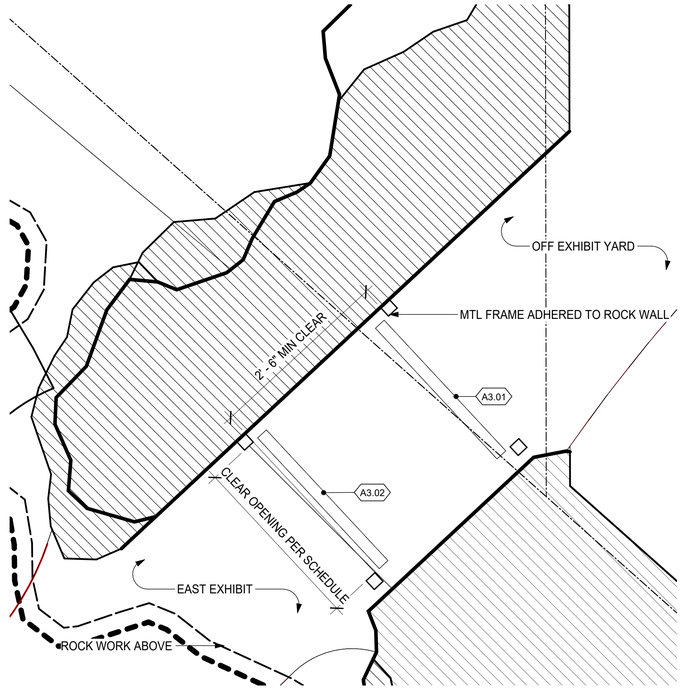
6/11/2026 5:42:27 PM
EX-702FP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

1 EX-702FP — Care Quarters Caging Ceiling Plan

1/4" = 1'-0"




3 EX-703FP **Interior Animal Chute - Plan**
1" = 1'-0"




2 EX-703FP **Off View to East Habitat Shift - Plan**
1" = 1'-0"

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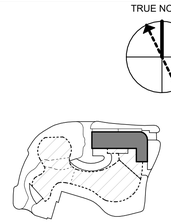

KEY PLAN

0 1" 2"
1" = 1"

GRAPHIC SCALE(S)

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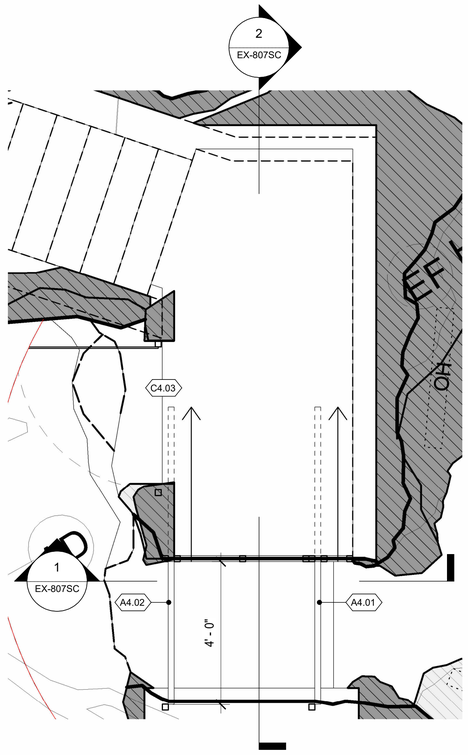
| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CAGING ENLARGED PLANS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 244 OF 378

EX-703FP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

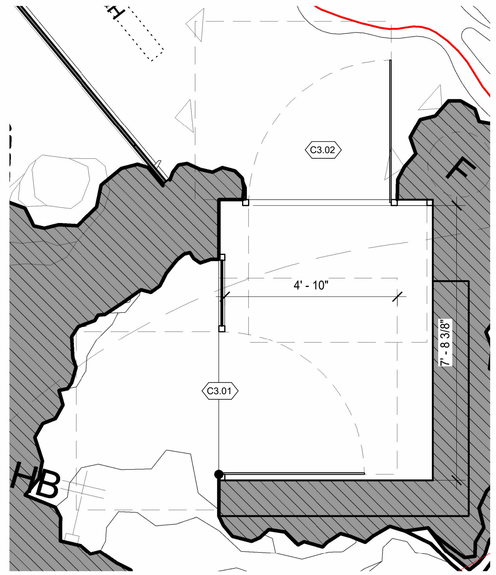
FOR REFERENCE ONLY

6/11/2026 5:43:54 PM
EX-703FP BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-704FP — East Habitat to West Habitat Shift

1/2" = 1'-0"




2 EX-704FP — Care Staff Vestibule - Plan

1/2" = 1'-0"

# ALF

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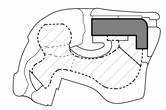

KEY PLAN

0 1' 2' 4'

1/2"=1'-0"

**GRAPHIC SCALE(S)**

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|---|---|
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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CAGING ENLARGED PLANS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 245 OF 378

EX-704FP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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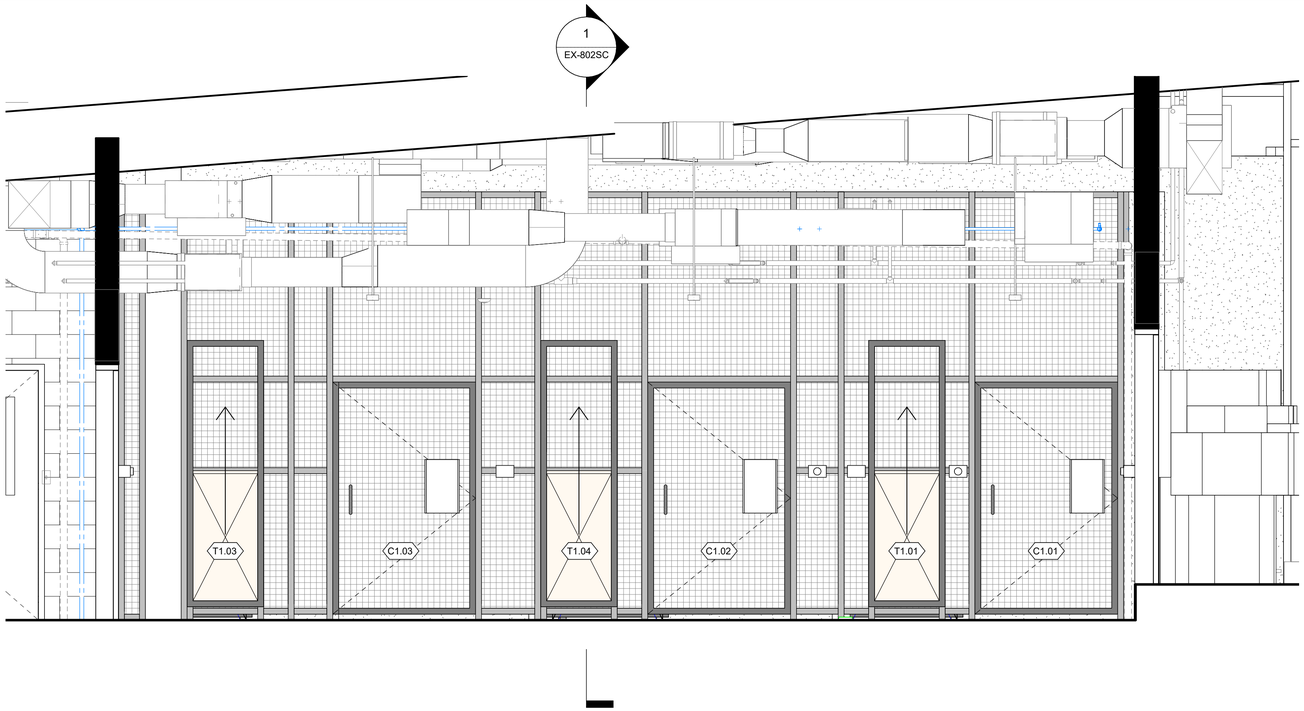
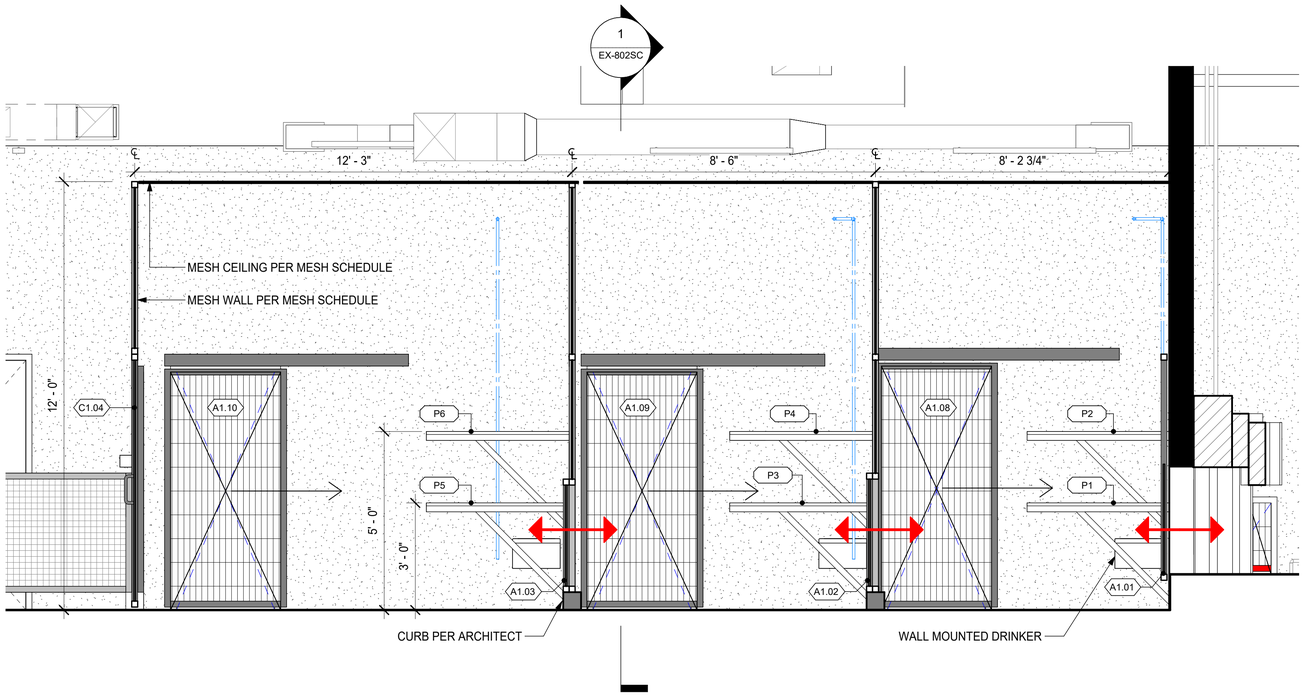
6/11/2026 5:44:03 PM
EX-704FP BY: Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS




1 EX-801SC — Keeper Hallway Den 1-3 - Elevation

1/2" = 1'-0"




3 EX-801SC — Interior Den 1-3 - Elevation

1/2" = 1'-0"

6/11/2026 5:44:04 PM
EX-801SC BY Approver 6/15/2026 65% CONSTRUCTION DOCUMENTS

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KEY PLAN

0 1' 2' 4'

1/2"=1'-0"

**GRAPHIC SCALE(S)**

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| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CAGING ELEVATIONS AND SECTIONS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |
| SHEET NO. | 246 OF 378 |
| | EX-801SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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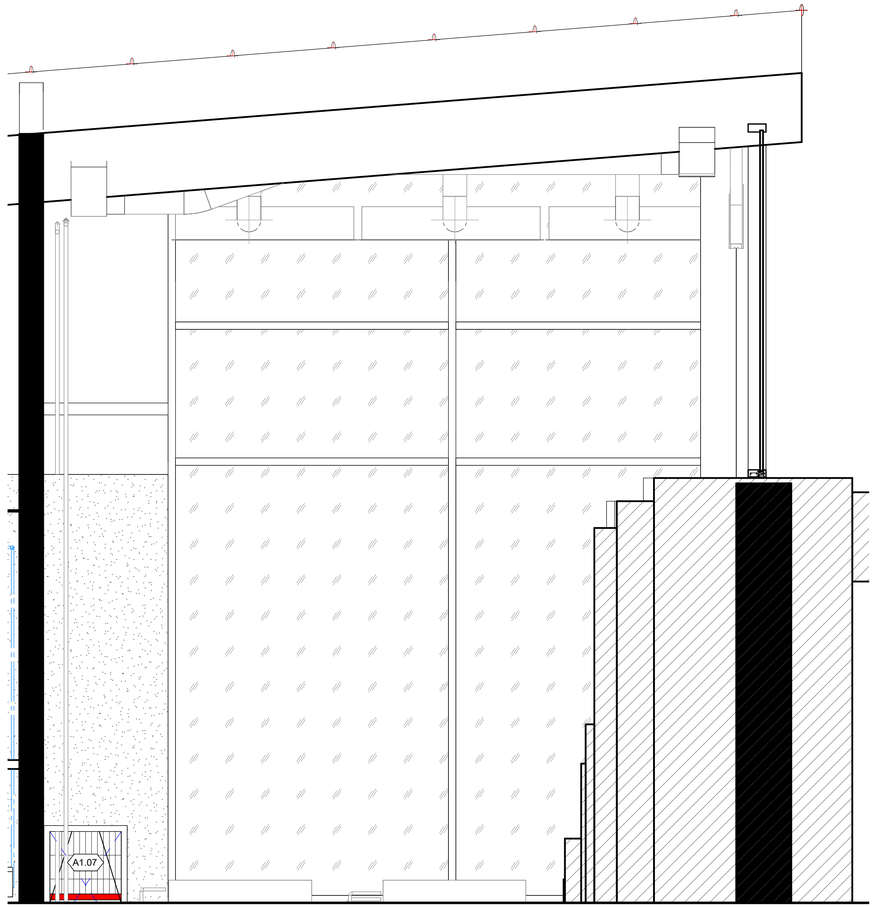
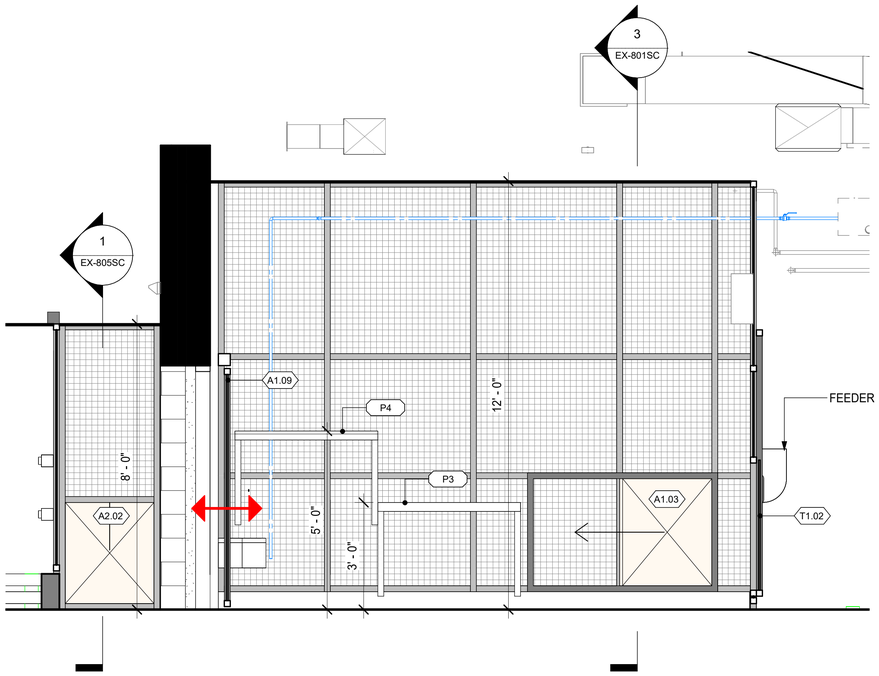



2 EX-802SC **Dayroom - Section**
1/2" = 1'-0"




1 EX-802SC **Interior Den 2 - Elevation**
1/2" = 1'-0"

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0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

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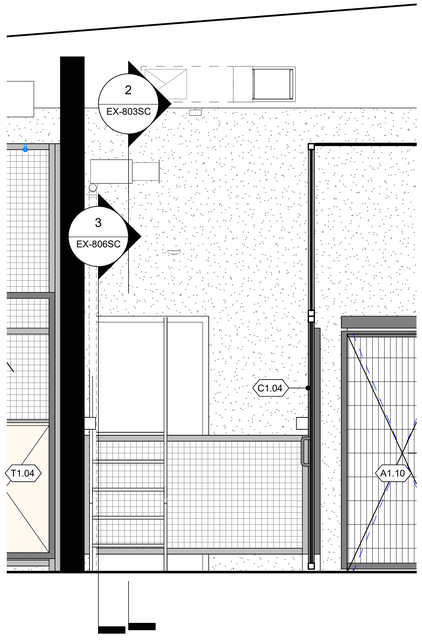
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CAGING ELEVATIONS AND SECTIONS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

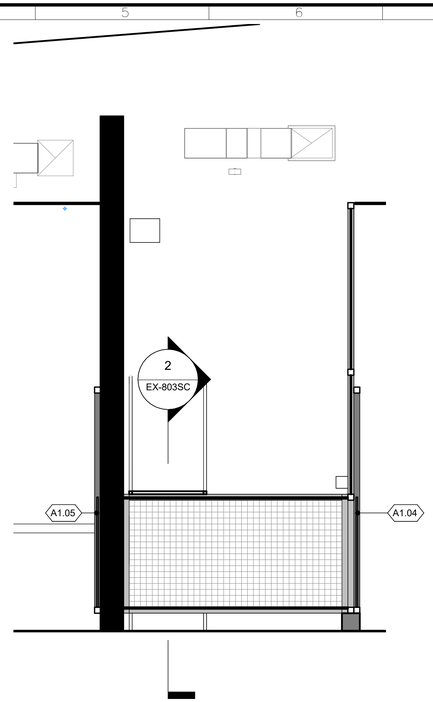
SHEET NO. 247 OF 378

EX-802SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

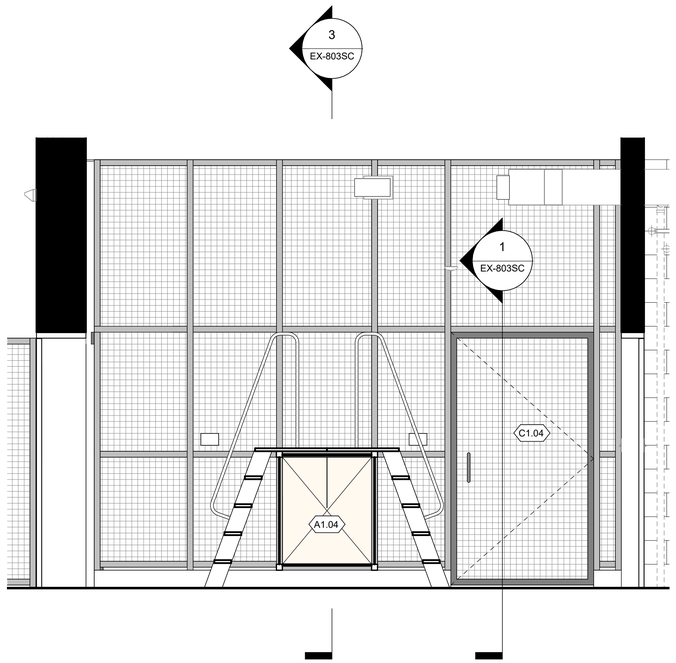
6/11/2026 5:44:05 PM
EX-802SC BY Aprowar 6/15/2026 65% CONSTRUCTION DOCUMENTS



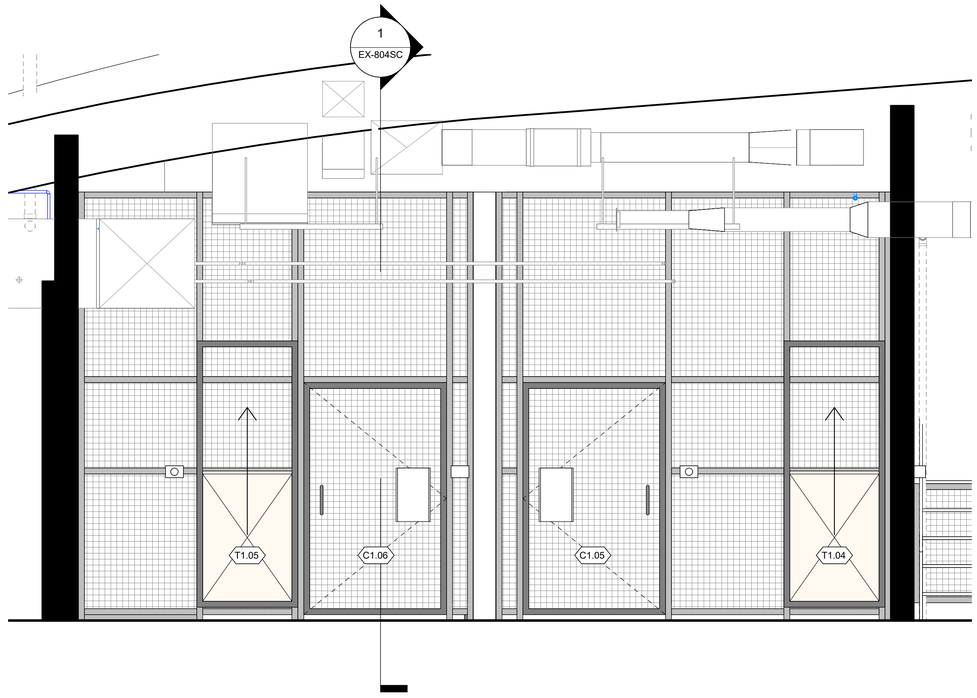

1 EX-803SC — Transfer Chute Elevation
1/2" = 1'-0"




3 EX-803SC — Transfer Chute Longitudinal Section
1/2" = 1'-0"




2 EX-803SC — Transfer Hallway - Section
1/2" = 1'-0"




4 EX-803SC — Keeper Hallway Den 4-5 - Elevation
1/2" = 1'-0"

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KEY PLAN

1/2"=1'-0" 0 1' 2' 4'
1/8"=1'-0" 0 4' 8' 16'
1"=100'-0" 0 100' 200'
1"=100'-0" 0 8' 16' 32'
1/16"=1'-0" 0 1' 2' 4'
1/2"=1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
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Washington, DC 20024-2520


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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Ave NW<br>Washington, DC 20008 USA |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | CAGING ELEVATIONS AND SECTIONS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 248 OF 378

EX-803SC (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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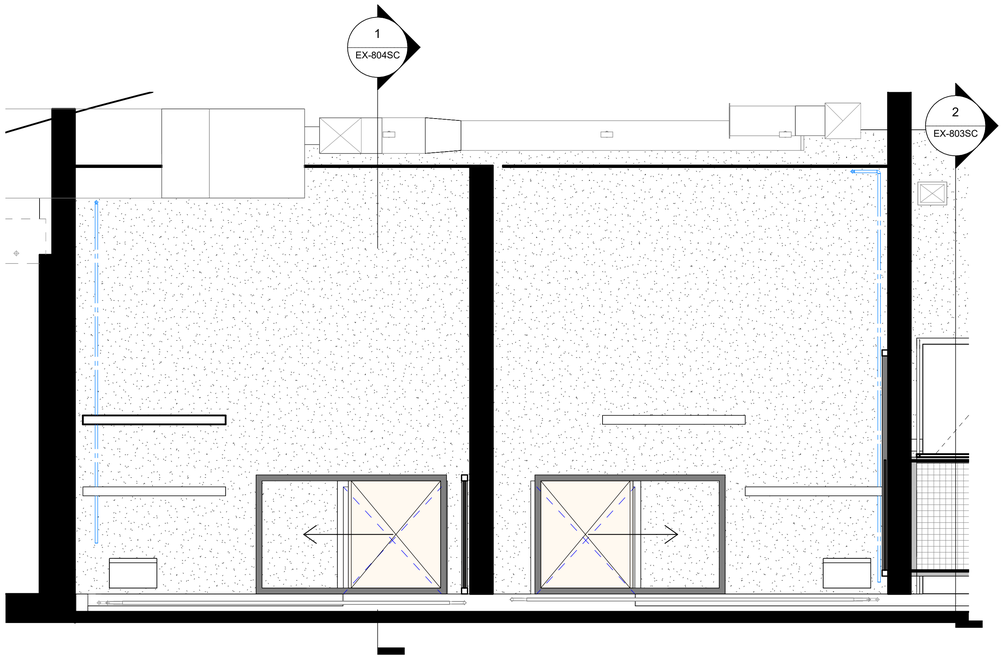
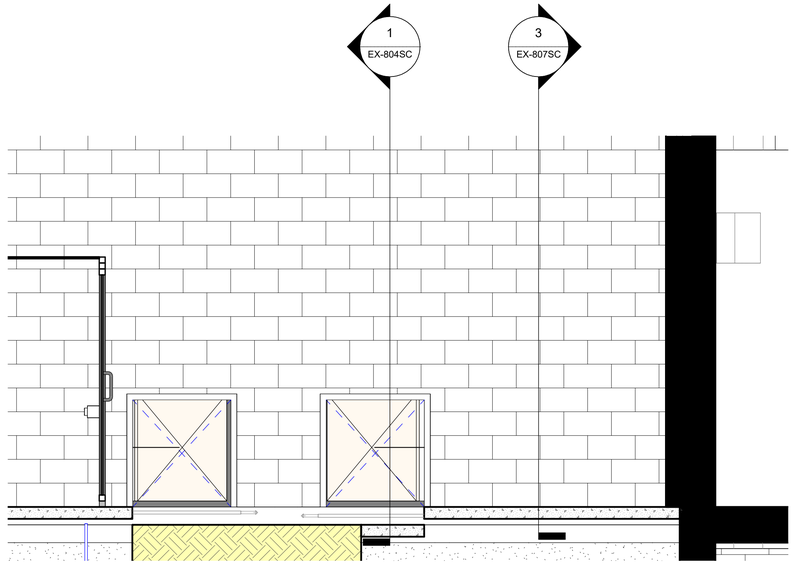
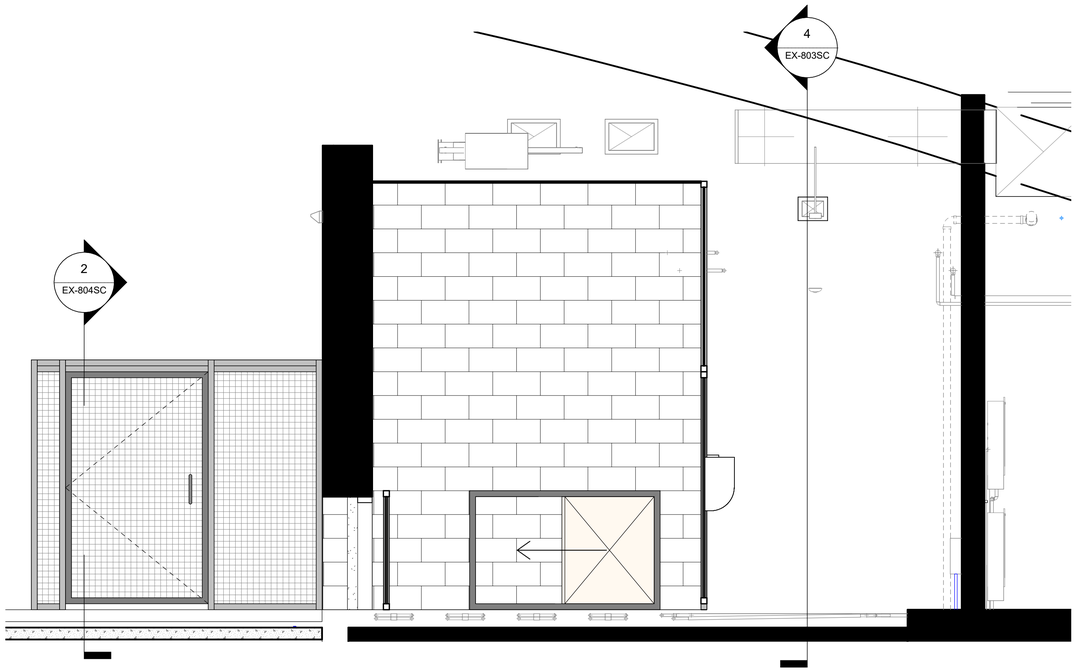



3 EX-804SC — Interior Den 4 -5 - Section
1/2" = 1'-0"




2 EX-804SC — Exterior Keeper Hallway - Elevation2
1/2" = 1'-0"




1 EX-804SC — Interior Den 5 - Section
1/2" = 1'-0"

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KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

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| DRAWING TITLE | CAGING ELEVATIONS AND SECTIONS |
| WORKING STAFF | DF (DESIGNED BY) / SHR (DRAWN BY) / JH (CHECKED BY) |

SHEET NO. 249 OF 378

EX-804SC
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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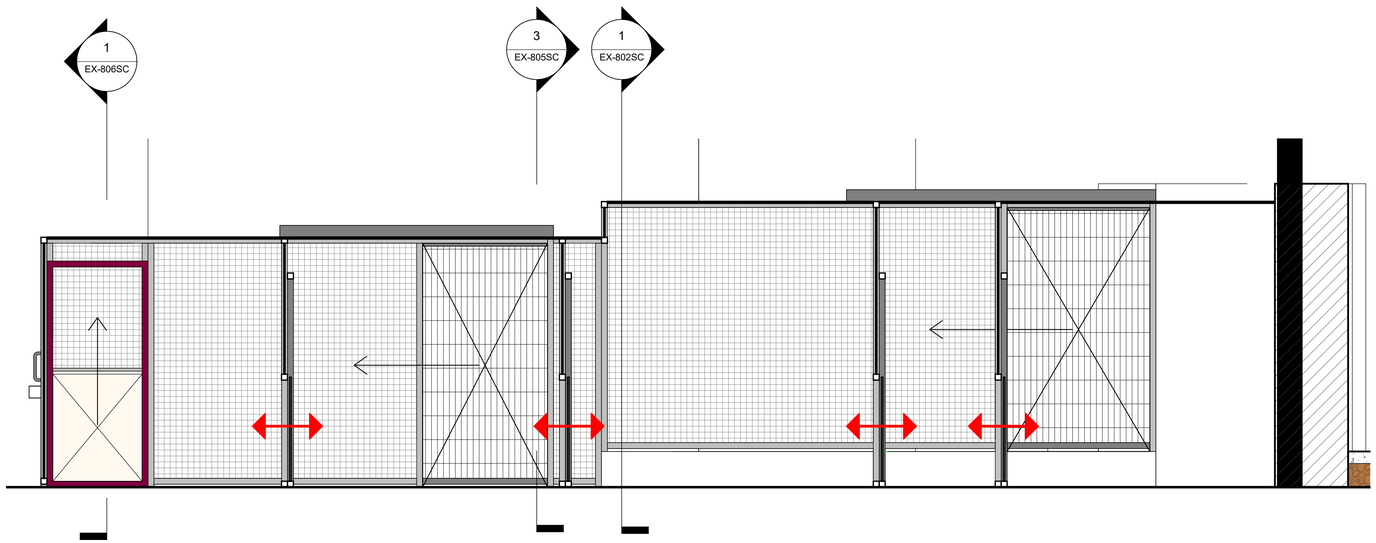
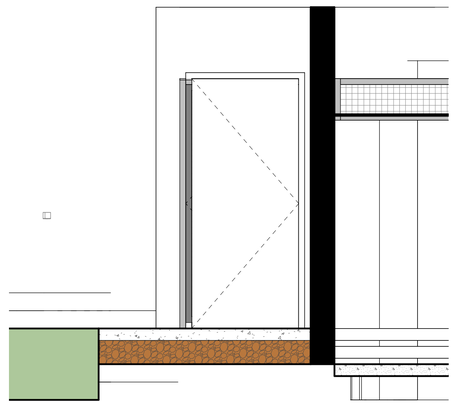
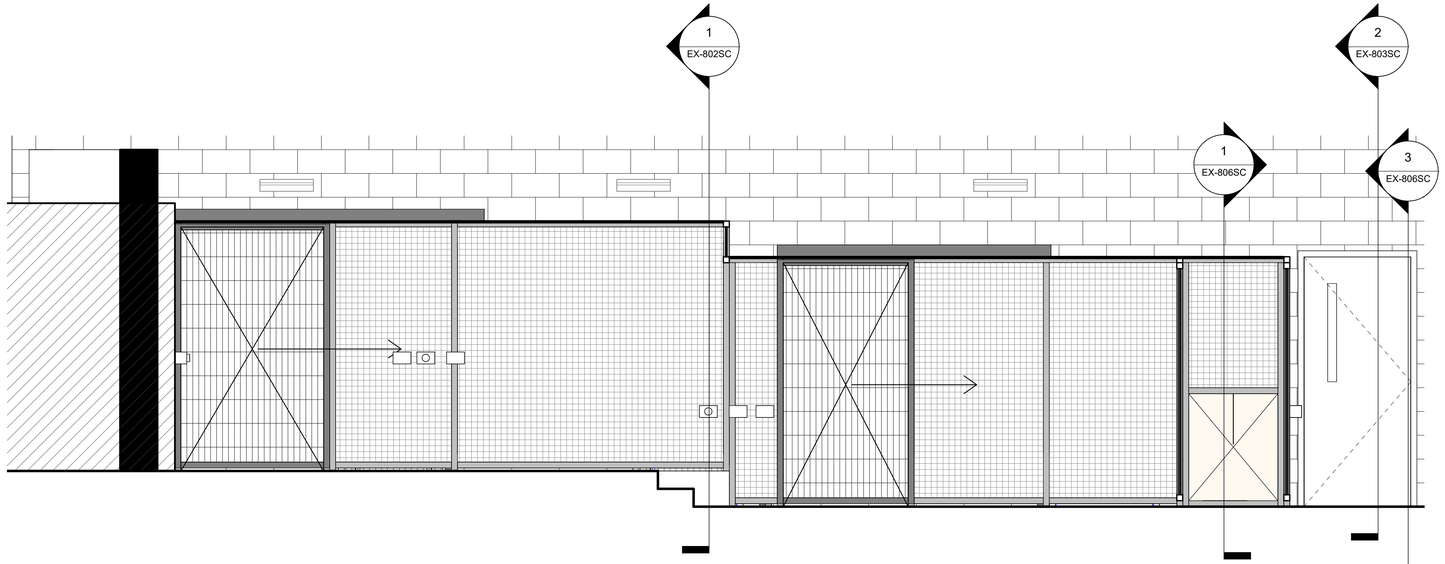



1 EX-805SC — Exterior Animal Transfer 1
1/2" = 1'-0"



3 EX-805SC — Exterior Keeper Access - Elevation
1/2" = 1'-0"




2 EX-805SC — Exterior Keeper Hallway - Elevation
1/2" = 1'-0"

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KEY PLAN

0 1' 2' 4'
1/2"=1'-0"
**GRAPHIC SCALE(S)**

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|---|---|---|---|
| | DF | SHR | JH |

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EX-805SC
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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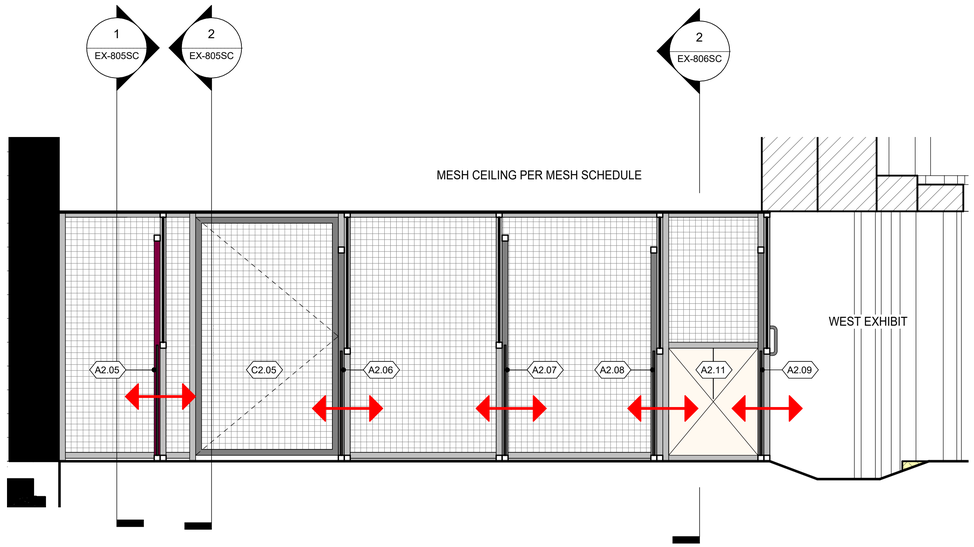
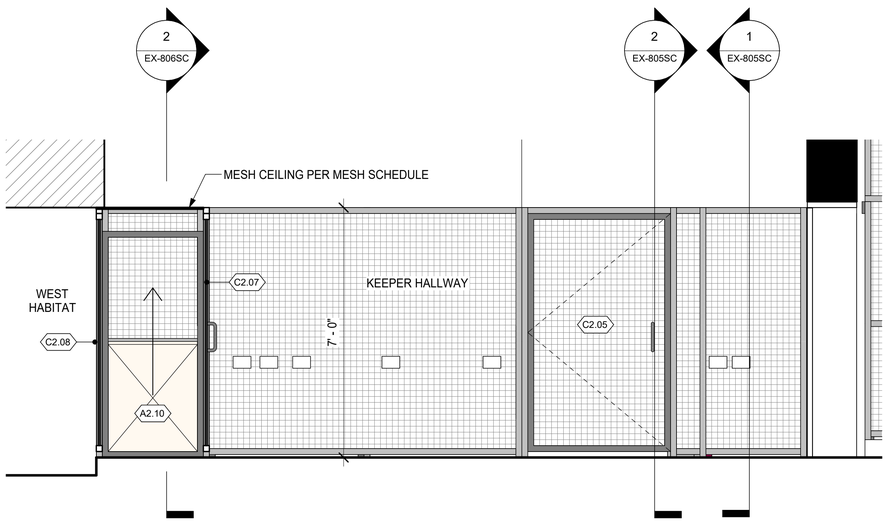
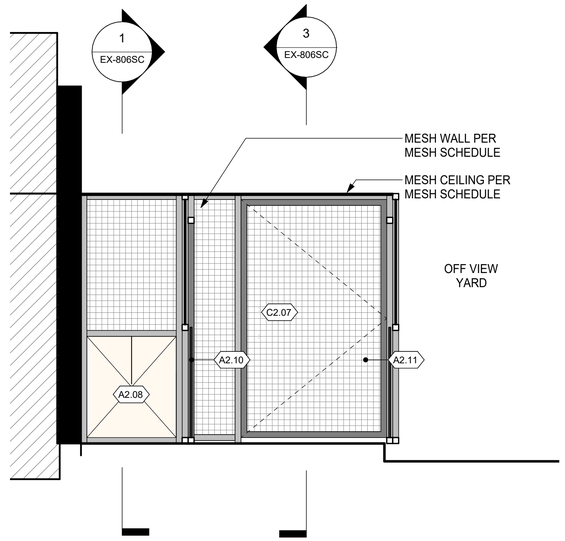



1 EX-806SC **Exterior Animal Transfer 2**
1/2" = 1'-0"




3 EX-806SC **Keeper Hallway**
1/2" = 1'-0"




2 EX-806SC **Exterior Keeper Hallway - Elevation1**
1/2" = 1'-0"

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KEY PLAN

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1/2"=1'-0"
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| OPDC PROJECT NUMBER | **2533105** |
| A/E PROJECT NUMBER | **00100084** |
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**EX-806SC**
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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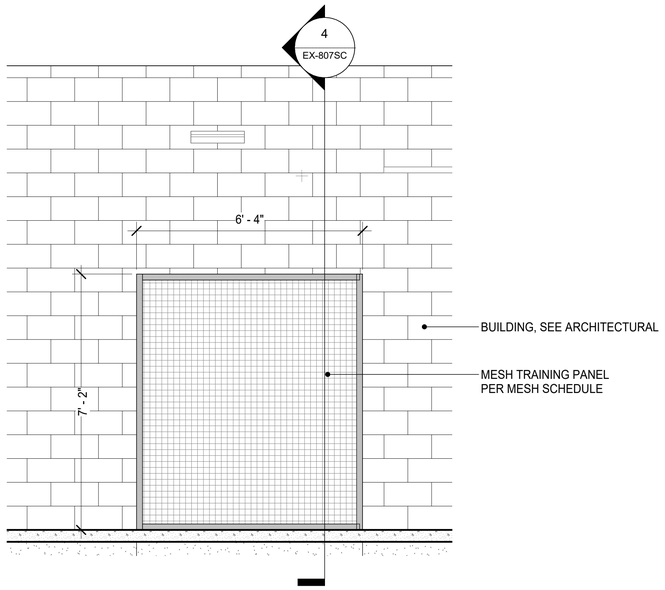
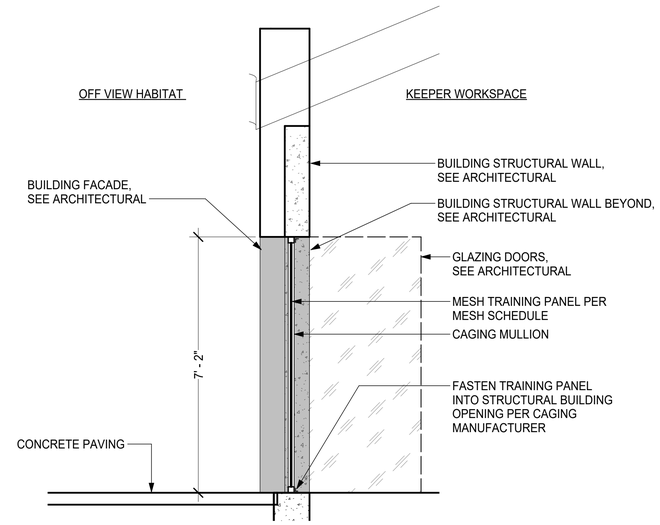
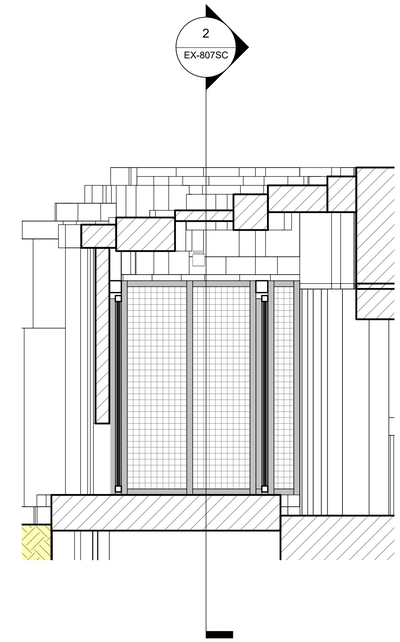


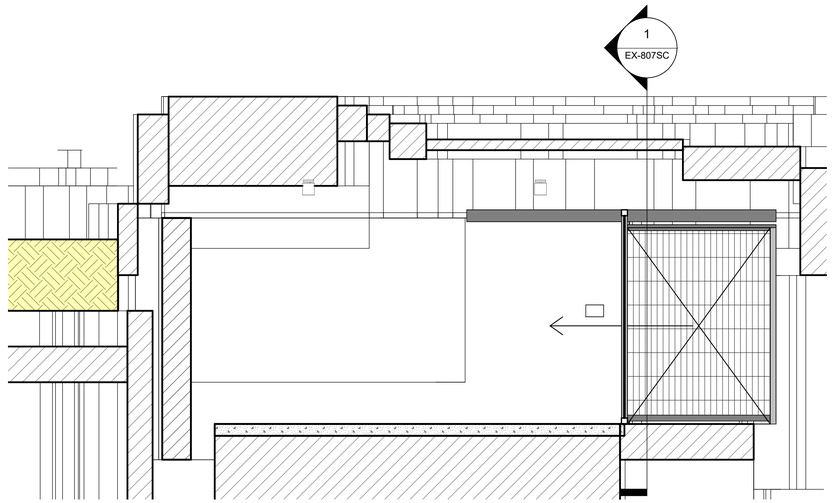

3 EX-807SC **Off-View Training Panel - Elevation**
1/2" = 1'-0"




4 EX-807SC **Off-View Habitat Training Panel - Section**
1/2" = 1'-0"




1 EX-807SC **East West Exhibit Section 1**
1/2" = 1'-0"




2 EX-807SC **East West Exhibit Section 2**
1/2" = 1'-0"

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1/2"=1'-0"
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EX-807SC
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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# MECHANICAL ABBREVIATIONS

| | | |
|---|---|---|
| A | AC | AIR CONDITIONING |
| | ACB | ACTIVE CHILLED BEAM |
| | ACC | AIR COOLED CHILLER |
| | ACCU | AIR COOLED CONDENSING UNIT |
| | ACFM | ABSOLUTE CUBIC FEET PER MINUTE |
| | ACU | AIR CONDITIONING UNIT |
| | AD | ACCESS DOOR |
| | AFF | ABOVE FINISHED FLOOR |
| | AFM | AIRFLOW MEASURING DEVICE |
| | AFMS | AIRFLOW MEASURING STATION |
| | AHU | AIR HANDLING UNIT |
| | AMB | AMBIENT |
| | AP | ACCESS PANEL |
| | APPROX | APPROXIMATE |
| | AS | AIR SEPARATOR |
| B | B | BOILER |
| | BHP | BRAKE HORSEPOWER |
| | BOD | BOTTOM OF DUCT |
| | BOP | BOTTOM OF PIPE |
| | BTU(H) | BRITISH THERMAL UNIT (PER HOUR) |
| C | CAP | CAPACITY |
| | CC | COOLING COIL |
| | CDSR | CONDENSER |
| | CFH | CUBIC FEET PER HOUR |
| | CFM | CUBIC FEET PER MINUTE |
| | CH | CHILLER |
| | CI | CAST IRON |
| | COMP | COMPRESSOR |
| | COND | CONDENSATE |
| | CONT | CONTINUATION |
| | CP | CONDENSATE PUMP |
| | CS | CLEAN STEAM |
| | CT | COOLING TOWER |
| | CUH | CABINET UNIT HEATER |
| | CV | CONSTANT VOLUME/CONTROL VALVE |
| | CW | COLD WATER |
| | D&T | DRIP & TRAP |
| D | DB | DECIBELS |
| | DBA | DECIBELS (A-WEIGHTED SCALE) |
| | DBT | DRY BULB TEMPERATURE °F |
| | DDC | DIRECT DIGITAL CONTROL |
| | DIA | DIAMETER |
| | DN | DOWN |
| | DP | DIFFERENTIAL PRESSURE |
| | DPS | DIFFERENTIAL PRESSURE SENSOR |
| | DPT | DEW POINT TEMPERATURE °F |
| | DWG | DRAWING |
| | DX | DIRECT EXPANSION |
| | (E) | EXISTING |
| E | EA | EACHEXHAUST AIR |
| | EAT | ENTERING AIR TEMPERATURE |
| | EER | ENERGY EFFICIENCY RATIO |
| | EF | EXHAUST FAN |
| | EFF | EFFICIENCY |
| | EG | ETHYLENE GLYCOL WATER SOLUTION |
| | EL | ELEVATION |
| | ELEC | ELECTRICAL |
| | ENT | ENTERING |
| | EQ | EQUIVALENT |
| | ERU | ENERGY RECOVERY UNIT |
| | ESP | EXTERNAL STATIC PRESSURE |
| | ET | EXPANSION TANK |
| | EV | EXHAUST VALVE |
| | EWC | ELECTRIC WATER COOLER |
| | EWT | ENTERING WATER TEMPERATURE |
| | EXH | EXHAUST |
| | EXP | EXPANSION |
| | EXT | EXTERAL |
| F | F | FAHRENHEIT |
| | FCU | FAN COIL UNIT |
| | FD | FIRE DAMPER |
| | FH | FUME HOOD |
| | FLA | FULL LOAD AMPERES |
| | FLEX | FLEXIBLE CONNECTION |
| | FLR | FLOOR |
| | FM | FLOW METER |
| | FPM | FEET PER MINUTE |
| | FS | FLOW SWITCH |
| | FSD | FIRE & SMOKE DAMPER |
| | FT | FEET or FOOT |
| | FTR | FIN TUBE RADIATION |
| G | G | GLYCOL |
| | GAL | GALLONS |
| | GPH | GALLONS PER HOUR |
| | GPM | GALLONS PER MINUTE |
| | GR | GRAINS |
| | H | HEIGHT |
| H | HB | HOSE BIBB |
| | HC | HEATING COIL |
| | HEX | HEAT EXCHANGER |
| | HG | MERCURY |
| | HO | HUB OUTLET |
| | HP | HORSE POWER |
| | HR | HOUR |
| | HUM | HUMIDIFEIR |
| | HZ | HERTZ |
| I | ID | INSIDE DIAMETER |
| | IE | INVERT ELEVATION |
| | IFB | INTEGRAL FACE AND BY-PASS |
| | IN | INCHES |
| | IPLV | INTEGRATED PART LOAD VALUE |
| K | KW | KILOWATTS |
| | KWH | KILOWATT HOURS |
| L | L | LENGTH |
| | LAT | LEAVING AIR TEMPERATURE |
| | LBS | POUNDS |
| | LBS/HR | POUNDS PER HOUR |
| | LEF | LAB EXHAUST FAN |
| | LP | LOW POINT |
| | LRA | LOCKED ROTOR AMPERES |
| | LVG | LEAVING |
| | LWT | LEAVING WATER TEMPERATURE |
| M | MAX | MAXIMUM |
| | MBH | 1000 BTU/HR |
| | MCA | MAXIMUM CURRENT AMPACITY |
| | MCC | MOTOR CONTROL CENTER |
| | MD | MOTORIZED DAMPER |
| | MECH | MECHANICAL |
| | MER | MECHANICAL EQUIPMENT ROOM |
| | MIN | MINIMUM |
| | MISC | MISCELLANEOUS |
| | MND | MANUAL DAMPER |
| | MOCP | MAXIMUM OVER CURRENT PROTECTION |
| | MTG | MOUNTING |
| | (NRL) | NEW LOCATION OF RELOCATED MATERIAL OR EQUIPMENT |
| | N.C | NORMALLY CLOSED |
| | N.O. | NORMALLY OPEN |
| N | NC | NOISE CRITERIA |
| | NIC | NOT IN CONTRACT |
| | NO. | NUMBER |
| | NOM | NOMINAL |
| | NPLV | NON STANDARD PART LOAD VALUE |
| | NPSH | NET POSITIVE SUCTION HEAD |
| | NTS | NOT TO SCALE |
| O | OA | OUTSIDE AIR |
| | OAD | OUTSIDE AIR DAMPER |
| | OD | OUTSIDE DIAMETER |
| | OPR | OPERATING |
| | OV | OUTLET VELOCITY |
| P | P | PUMP |
| | PCB | PASSIVE CHILLED BEAM |
| | PD | PRESSURE DROP (FEET OF WATER) |
| | PG | PROPYLENE GLYCOL WATER SOLUTION |
| | PH | PHASE |
| | PRESS | PRESSURE |
| | PRV | PRESSURE REGULATING VALVE |
| | PSI | POUNDS PER SQUARE INCH |
| | PSIG | POUNDS PER SQUARE INCH - GAUGE |
| Q | QTY | QUANTITY |
| | (RL) | RELOCATED MATERIAL OR EQUIPMENT |
| R | RA | RETURN/RELIEF AIR |
| | RC | ROOF CONDUCTOR |
| | RE | REFERENCE |
| | RET | RETURN |
| | RF | RETURN FAN |
| | RH | RELATIVE HUMIDITY |
| | RHC | REHEAT COIL |
| | RLA | RUNNING LOAD AMPS |
| | RPM | REVOLUTIONS PER MINUTE |
| | RTU | ROOFTOP UNIT |
| S | SA | SUPPLY AIR |
| | SA | SHOCK ABSORBER |
| | SA | SOUND ATTENUATOR |
| | SAN | SANITARY WASTE |
| | SCFM | STANDARD CUBIC FEET PER MINUTE |
| | SD | SMOKE DAMPER/SMOKE DETECTOR |
| | SD | SMOKE DAMPER |
| | SEER | SEASONAL ENERGY EFFICIENCY RATIO |
| | SENS | SENSIBLE |
| | SF | SQUARE FEET |
| | SG | SPECIFIC GRAVITY |
| | SL | SOUND LINING |
| | SP | STATIC PRESSUR (INCHES OF WATER) |
| | SPD | SPEED |
| | SPL | SOUND POWER LEVEL |
| | SPS | STATIC PRESSURE SENSOR |
| | SS | STAINLESS STEEL |
| | STM | STEAM |
| | SV | SUPPLY VALVE |
| T | T | TEMPERATURE |
| | TAB | TESTING, ADJUSTING AND BALANCING |
| | TEF | TOILET EXHAUST FAN |
| | TOD | TOP OF DUCT |
| | TOP | TOP OF PIPE |
| | TP | TOTAL PRESSURE |
| | TYP | TYPICAL |
| U | UH | UNIT HEATER |
| | UON | UNLESS OTHERWISE NOTED |
| V | V | VOLTAGE/VENT |
| | VAC | VACUUM |
| | VAV | VARIABLE AIR VOLUME |
| | VEL | VELOCITY |
| | VFD | VARIABLE FREQUENCY DRIVE |
| | VIFB | VERTICAL INTEGRAL FACE AND BYPASS |
| | VTR | VENT THRU ROOF |
| W | W | WATTS/WIDTH |
| | W/ | WITH |
| | W/O | WITHOUT |
| | WB | WET BULB TEMPERATURE |
| | WG | WATER GAUGE |
| | WMS | WIRE MESH SCREEN |

# MECHANICAL SYMBOLS

- ISOLATION VALVE
- THROTTLING VALVE
- ANGLE VALVE
- CHECK VALVE
- 2-WAY PNEUMATIC CONTROL VALVE
- 3-WAY PNEUMATIC CONTROL VALVE
- 2-WAY ELECTRIC CONTROL VALVE
- 3-WAY ELECTRIC CONTROL VALVE
- SOLENOID VALVE
- AUTOMATIC FLOW CONTROL VALVE
- PLUG VALVE (ISOLATION OR THROTTLING)
- SELF ACTUATING TYPE DIFFERENTIAL PRESSURE CONTROL VALVE (DPCV)
- AUTOMATIC DPCV
- PRESSURE REGULATING VALVE, FLANGED
- PRESSURE REGULATING VALVE, THREADED
- PRESSURE REDUCING VALVE
- FLOW INDICATING BALANCE VALVE
- SAFETY RELIEF VALVE
- AIR SEPARATOR
- BASKET STRAINER
- STRAINER
- MECHANICAL EXPANSION JOINT
- FLEXIBLE CONNECTION
- ORIFICE PLATE
- REDUCER - CONCENTRIC
- REDUCER - ECCENTRIC
- PIPE UNION
- PIPE ANCHOR
- ALIGNMENT GUIDE
- PRESSURE GAUGE WITH COCK
- THERMOMETER
- CONNECT OUT OF TOP
- DROP OR RISE
- DIRECTION OF FLOW
- DIRECTION OF PIPE PITCH
- REDUCED PRESSURE BACKFLOW PREVENTER
- QUICK COUPLING
- CAP OR PLUG
- ELBOW - TURNED UP/RISER CONNECTION
- ELBOW - TURNED DOWN
- TEE OUTLET - DOWN
- TEE OUTLET - UP
- MANUAL AIR VENT WITH VALVE
- AUTOMATIC AIR VENT
- FLOW METER

- ROOM THERMOSTAT/SENSOR
- DUCT MOUNTED THERMOSTAT/SENSOR
- CARBON DIOXIDE SENSOR
- CARBON MONOXIDE SENSOR
- HUMIDITY SENSOR/HUMIDISTAT
- SMOKE DETECTOR
- FLOW SWITCH
- STATIC PRESSURE SENSOR
- HOSE CONNECTION
- STEAM TRAP
- TEST PORT (PRESSURE / TEMPERATURE)
- FLEXIBLE DUCT OF SIZE AS SHOWN
- FLEXIBLE DUCT CONNECTION
- AUTOMATIC DAMPER (MOTORIZED)
- VOLUME DAMPER (MANUAL)
- FIRE DAMPER
- SMOKE DAMPER
- COMBINATION FIRE & SMOKE DAMPER
- RFI SHIELD DAMPER
- SECURITY BARS
- DIRECTION OF FLOW
- ACOUSTICALLY LINED DUCT

- SUPPLY
- RETURN
- EXHAUST

(AIR SYSTEMS)

- SIDE WALL SUPPLY DIFFUSER
- SIDE WALL RETURN or EXHAUST GRILLE/REGISTER
- MATERIAL OR EQUIPMENT EXISTING TO REMAIN
- MATERIAL OR EQUIPMENT TO BE DEMOLISHED
- RETURN OR EXHAUST AIRFLOW
- SUPPLY OR OUTSIDE AIRFLOW
- POINT OF CONNECTION
- POINT OF DISCONNECT
- PUMP (RISER)
- FAN (RISER)
- HUMIDIFIER

(VERTICAL DUCT RISE)

(VERTICAL DUCT DROP)

| | | |
|---|---|---|
| 18x6 | 18x6 | INDICATES RECTANGULAR DUCT WITH DUCT SIZE 18 INCHES WIDE (IN PLANE OF DRAWING) AND 6 INCHES DEEP. SIZE PERTAINS TO THE ENTIRE RUN OF DUCT UNLESS OTHERWISE NOTED. |
| 18x6~ | 18x6~ | INDICATES FLAT OVAL DUCT WITH DUCT SIZE 18 INCHES WIDE (IN PLANE OF DRAWING) AND 6 INCHES DEEP. SIZE PERTAINS TO THE ENTIRE RUN OF DUCT UNLESS OTHERWISE NOTED. |
| 18Ø | 18Ø | INDICATES ROUND DUCT WITH DUCT SIZE OF 18 INCHES IN DIAMETER. SIZE PERTAINS TO THE ENTIRE RUN OF DUCT (FROM DUCT ORIGIN AT TAP TO END OF DUCT) UNLESS OTHERWISE NOTED. |

# DUCT AND PIPING LEGEND

| | |
|---|---|
| CHWS | CHILLED WATER SUPPLY |
| CHWR | CHILLED WATER RETURN |
| RL | REFRIGERANT LIQUID |
| RS | REFRIGERANT SUCTION |
| RHG | REFRIGERANT HOT GAS |
| HHWS | HEATING HOT WATER SUPPLY |
| HHWR | HEATING HOT WATER RETURN |
| D | CONDENSATE DRAIN |
| SA | SUPPLY AIR |
| RA | RETURN AIR |
| TE | TOILET EXHAUST |
| GE | GENERAL EXHAUST |
| OA | OUTSIDE AIR |

# REFERENCE SYMBOLS

- SECTION NUMBER (1 / E0.1)
- DRAWING NUMBER WHERE DRAWN
- HHWS / 4 — RISER DESIGNATION (TYPE AS INDICATED)
- MXXX — KEY NOTE
- 4 — DEMO KEY NOTE

# CODES AND STANDARDS

THE FOLLOWING CODES, REGULATIONS AND REFERENCE STANDARDS WILL DIRECT THE MECHANICAL DESIGN FOR THIS PROJECT, WHERE APPLICABLE. REFER TO THE PROJECT MANUAL COVER SHEET FOR EXTENSIVE LIST OF ALL THE CODES THAT APPLY TO THIS PROJECT. THIS LIST IS NOT COMPREHENSIVE. ANY ADDITIONAL CODES AND REGULATIONS ENCOUNTERED DURING DESIGN WILL BE ADDED THEREAFTER. A FULL LIST OF REFERENCE CODES, STANDARDS, AND GUIDELINES INCLUDED IN THE 'SMITHSONIAN DESIGN STANDARDS VOL. 1 – DESIGN GUIDELINES AND TECHNICAL SECTIONS' WILL BE CONTINUALLY REFERENCED THROUGHOUT THE DESIGN PHASE.

1. INTERNATIONAL CODE COUNCIL (ICC), 2024 EDITIONS
   a. INTERNATIONAL BUILDING CODE (IBC), SUBSTITUTING NFPA 101 FOR IBC CHAPTER 10
   b. INTERNATIONAL EXISTING BUILDING CODE (IEBC)
   c. INTERNATIONAL ENERGY CONSERVATION CODE (IECC)
   d. INTERNATIONAL FIRE CODE (IFC)
   e. INTERNATIONAL MECHANICAL CODE (IMC)
   f. INTERNATIONAL PLUMBING CODE (IPC)
2. NFPA 70-2023 – NATIONAL ELECTRIC CODE (NEC)
3. NFPA 72-2022 – NATIONAL FIRE ALARM AND SIGNAL CODES
4. LEED VERSION 4.1: BUILDING DESIGN AND CONSTRUCTION
5. ASHRAE STD. 15/34-2022, SAFETY STANDARDS FOR REFRIGERATION SYSTEMS.
6. ASHRAE STD. 90.1-2022, ENERGY STANDARD FOR BUILDINGS EXCEPT LOW-RISE RESIDENTIAL (ENERGY CODE COMPLIANCE)
7. AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS (ASHRAE) GUIDELINE 36: HIGH PERFORMANCE SEQUENCES FOR OPERATION OF HVAC SYSTEMS
8. ASSOCIATED AIR BALANCE COUNCIL (AABC) NATIONAL STANDARDS FOR TOTAL SYSTEM BALANCE
9. ASHRAE HANDBOOKS (2019-2025)
10. 2023 SMITHSONIAN DESIGN STANDARDS (SDS)
11. OSHA STANDARDS AND REGULATIONS

# GENERAL NOTES

1. INSTALL ALL WORK IN ACCORDANCE WITH THE 2024 INTERNATIONAL MECHANICAL CODE AND 2024 INTERNATIONAL ENERGY CONSERVATION CODE ALONG WITH THE REGULATIONS OF SMITHSONIAN AND ALL AUTHORITIES HAVING JURISDICTION.
2. OBTAIN ALL CONSTRUCTION INFORMATION, ALL SPECIFICATION AND DRAWINGS FROM THE CONTRACTOR(S). WHEN SPECIFICATION REQUIRE DETAILS (STRUCTURAL, CIVIL, MECHANICAL, ELECTRICAL, PLUMBING) NOT PROVIDED, CONSULT THE ARCHITECT/ENGINEER.
3. BRING TO THE ATTENTION OF THE ARCHITECT/ENGINEER ANY CONDITIONS CONTRARY TO THE SPECIFICATIONS AND DRAWINGS. THE CONTRACTOR(S) SHALL NOT PROCEED WITH ANY WORK EXCEPT AT ITS OWN RISK UNTIL ALL CONFLICTS ARE RESOLVED AND WRITTEN INSTRUCTIONS ARE GIVEN TO THE CONTRACTOR(S) BY THE ARCHITECT/ENGINEER.
4. PROVIDE OFFSETS AND TRANSITIONS IN DUCTWORK AND PIPING AS REQUIRED TO MAINTAIN COORDINATION WITH THE WORK OF OTHERS.
5. SHEET METAL FABRICATION IN THIS PROJECT ACCORDING TO THE LATEST ASHRAE STANDARDS. THE CONTRACTOR(S) SHALL PROVIDE SHOP DRAWINGS TO SHOW ALL PIPING & SHEETMETAL OFFSETS & TRANSITIONS REQUIRED.
6. FULLY COORDINATE THE MECHANICAL WORK WITH ARCHITECTURAL AND STRUCTURAL FEATURES TO PROVIDE COMPLETE AND OPERABLE SYSTEMS WITHOUT INTERFERENCES.
7. INSTALL ALL EQUIPMENT LEVEL AND PLUMB; PROVIDE BLOCKING/SUPPORTING ACCORDING TO MANUFACTURER REQUIREMENTS.
8. CONTRACTOR(S) TO FURNISH MANUFACTURER'S CERTIFIED DRAWINGS FOR ALL EQUIPMENT TO BE INCORPORATED INTO WORK; PROVIDE TRANSITIONS AND MAKE FINAL CONNECTIONS TO ALL EQUIPMENT.
9. INSTALL EQUIPMENT FURNISHED PRE-PIPED SERVICE RIGHT TO THE FIRST FITTING OR EDGE OF EQUIPMENT UNLESS PROTECTIVE RAILING OR PARAPET OF 42 INCHES AFF HAS BEEN PROVIDED. VERIFY WITH OTHER TRADES TO ASSURE NO CONFLICT WITH THE PROVIDED EQUIPMENT.
10. GENERAL NOTES DO NOT RELIEVE THE CONTRACTOR/ENGINEER FROM COMPLYING WITH APPLICABLE CODES; DUCT CONSTRUCTION MUST COMPLY UNLESS PROTECTIVE CONDITIONS OCCUR AT OTHER LOCATIONS. PROVIDE ADDITIONAL SUPPORT BRACING AND ANGLE BRACING.
11. DUCT CONSTRUCTION SHALL COMPLY WITH THE LATEST VERSION OF SMACNA CONSTRUCTION STANDARDS FOR THE SPECIFIC PRESSURE CLASSIFICATIONS INDICATED. ALL 90-DEGREE ELBOWS SHALL BE RADIUS TYPE UNLESS OTHERWISE NOTED; MITERED ROUND ELBOWS REQUIRE TURNING VANES. USE TURNING VANES UNLESS OTHERWISE NOTED. PROVIDE OVAL TAPS FOR METAL DUCTS AND 90-DEGREE TAPS FOR RECTANGULAR DUCTS.
12. FLEXIBLE DUCTWORK SHALL NOT BE USED IN RETURN, EXHAUST OR OUTSIDE-AIR DUCT SYSTEMS. FLEXIBLE DUCT MAY BE USED IN SUPPLY SYSTEMS ONLY IF INDICATED ON THE DRAWING. FLEXIBLE DUCT SHALL MATCH THE STATIC PRESSURE CLASS RATING OF DUCT SYSTEM. FLEXIBLE DUCT ROUTING SHALL BE STRAIGHT WITHOUT ELBOWS AND/OR KINKS, AND MAXIMUM LENGTH SHALL NOT EXCEED 5 FEET.
13. INSTALL DUCTWORK AS CLOSE AS POSSIBLE TO LOCATION ON DRAWINGS; DO NOT DEVIATE WITHOUT ARCHITECT/ENGINEER APPROVAL.
14. ALL DUCTWORK RUNS LOCATED INSIDE CLEAR DIMENSIONS SHOWN ON DRAWINGS.
15. PROVIDE DUCTWORK TO TERMINAL AIR DEVICES AT SCHEDULED CFM UNLESS OTHERWISE INDICATED. PROVIDE TRANSITIONS AS REQUIRED.
16. DUCTWORK PENETRATIONS OF ALL FIRE-RATED WALLS REQUIRE CONSTRUCTION, COORDINATION, AND FIRE PROTECTION PER LOCAL CODES. PENETRATION ASSEMBLY TO BE UL LISTED.
17. PAINT FLAT BLACK THE INSIDE OF ALL DUCTWORK VISIBLE THROUGH LINEAR DIFFUSERS OR EGGCRATE GRILLES.
18. PROVIDE OPEN END DUCTWORK (OED) WITH 22-GAUGE METAL UL-LISTED; NO GYPSUM FRAMES.
19. FURNISH AND INSTALL ALL MANUALLY OPERATED VOLUME CONTROL (BALANCING) DAMPERS NECESSARY TO PERFORM BALANCING WORK AT EACH BRANCH TAKEOFF, EACH AIR TERMINAL DEVICE, AND WHERE INDICATED.
20. DUCT SYSTEMS SHALL BE BALANCED TO AIRFLOWS INDICATED ON DRAWINGS.
21. TEST AND BALANCE SYSTEMS TO ENSURE COMPLIANCE WITH SCHEDULED FAN PERFORMANCE AND SCHEDULED AIRFLOW AT DESIGN STATIC PRESSURES.
22. PROVIDE FLEXIBLE CONNECTIONS BETWEEN DUCTWORK AND ALL FANS.
23. PROVIDE EXACT LOCATION OF DIFFUSERS, GRILLES, AND REGISTERS WITH THE REFLECTED CEILING PLAN BEFORE INSTALLATION; PROVIDE SYMMETRY WITH ROOM LIGHTING SYSTEMS.
24. VERIFY DIFFUSER GRILLE AND REGISTER MOUNTING/FACE TYPE MATCHES CEILING CONFIGURATION.
25. PROVIDE CAPTURE HOOD/EXTENDED PROBE FOR MOUNTED MECHANICAL SYSTEMS.
26. PROVIDE ACCESS DOORS FOR HIDDEN PANELS FOR ROOF-MOUNTED MECHANICAL SYSTEMS; ALIGN WITH ROOF STRUCTURE FRAMING AND LOCATE WITH ALL TRADES.
27. PROVIDE ACCESS DOORS AND PANELS IN HARD CEILINGS ON THE JOB SITE FOR VALVES, DAMPERS, AND MECHANICAL EQUIPMENT.
28. REFER TO ARCHITECTURAL GENERAL INFORMATION SHEET FOR MECHANICAL PENETRATIONS; IF ROOF ACCESS/SLEEVES NOT PROVIDED, MOUNT ROOM THERMOSTAT AT 42 INCHES AFF AND WIRE PER ELECTRICAL SHEET.
29. SQUARE OR RECTANGULAR DUCTWORK USES THE AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA) GUIDELINES.
30. COORDINATE ALL WORK UNDER UNITS INDICATED AND AS REQUIRED BY MANUFACTURER; DRAWINGS DO NOT PROVIDE COMPLETE DETAILS.
31. WHERE PER ALL PROVIDED COPPER DAMPERS EXCEED THE LIMITS OF A SINGLE OPERATION HANDLE, PROVIDE MULTIPLE DAMPERS PER MANUFACTURER RECOMMENDATIONS.
32. PIPE ROUTING MUST PROVIDE SLOPE FOR EQUIPMENT.
33. ALL SHUTOFF VALVES SHALL BE 1/4 TURN; ORIENT VALVES SO HANDLE POSITION COMMUNICATES STATUS TO THE OPERATOR. COORDINATE VALVE HEIGHTS SO HANDLE OPERATION DOES NOT CONFLICT WITH CEILING.
34. CONTRACTOR SHALL COORDINATE EQUIPMENT TAGS WITH SMITHSONIAN TO ENSURE THAT THE FORMATTING AND TAGS ARE CONSISTENT WITH THE SMITHSONIAN SERVER STANDARDS. CONTRACTOR SHALL INCLUDE BOTH THE SERVER TAG AND THE PHYSICAL TAG IN THE BAS AS REFERENCE INFORMATION (IF DIFFERENT).

# STATIC PRESSURE CLASSIFICATION FOR DUCT CONSTRUCTION

| DUCT SYSTEM | LOCATION | | PRESSURE CLASSIFICATION (INCH WATER GAUGE) | DUCTWORK SEAL/LEAKAGE CLASS | REMARKS |
|---|---|---|---|---|---|
| | FROM | TO | | | |
| SUPPLY | AHU SUPPLY FAN DISCHARGE | SUPPLY RISER | +6 | A | |
| SUPPLY | SUPPLY RISER | TERMINAL UNIT | +6 | A | |
| SUPPLY | TERMINAL UNIT | AIR TERMINAL UNIT | +2 | A | |
| EXHAUST | AIR TERMINAL UNIT | TERMINAL UNIT | -2 | A | |
| EXHAUST | TERMINAL UNIT | EXHAUST FAN | -4 | A | |
| EXHAUST | EXHAUST FAN | EXHAUST PLENUM | -2 | A | |
| OUTSIDE AIR DUCT | ALL | | -2 | A | |

GENERAL NOTES:
A. ALL DUCTWORK WITH PRESSURE CLASSIFICATIONS IN EXCESS OF NEGATIVE 3 INCH WATER GAUGE (I.E. -4,-5,-6, -10 ETC) SHALL BE CONSTRUCTED IN ACCORDANCE WITH THE SMACNA INDUSTRIAL DUCT STANDARDS (ROUND & RECTANGULAR)
B. ALL SUPPLY DUCTWORK NOT INDICATED ABOVE SHALL BE 4 INCH WATER GAUGE CONSTRUCTION.
C. ALL RETURN/EXHAUST DUCTWORK NOT INDICATED ABOVE SHALL BE NEGATIVE 4 INCH WATER GAUGE CONSTRUCTION.

ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

**NOT FOR CONSTRUCTION**

STAMP

**GRAPHIC SCALE(S)**

DATE: 06/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF); 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL ABBREVIATIONS AND SYMBOLS

WORKING STAFF: OP (DESIGNED BY) | YW (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 253 OF 378

M-001CS (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

FOR REFERENCE ONLY

6/15/2026 2:56:43 PM
M-001CS BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# VENTILATION CALCULATION

| | |
|---|---|
| System name and number | AHU-1 |
| Condition analyzed | Heating |
| Calculation Methodology | IMC-2024 |
| | Appendix A |

Notes:

1. Zone population is based on the larger of the design occupancy and the occupancy calculated using the IMC-2024 default occupancy rates unless otherwise noted.

2. Zone primary airflow is the larger of: (1) the calculated airflow based on winter heat loss and full occupancy, (2) the airflow provided at the associated terminal unit's minimum airflow set point, and (3) make-up air required to meet code required exhaust.

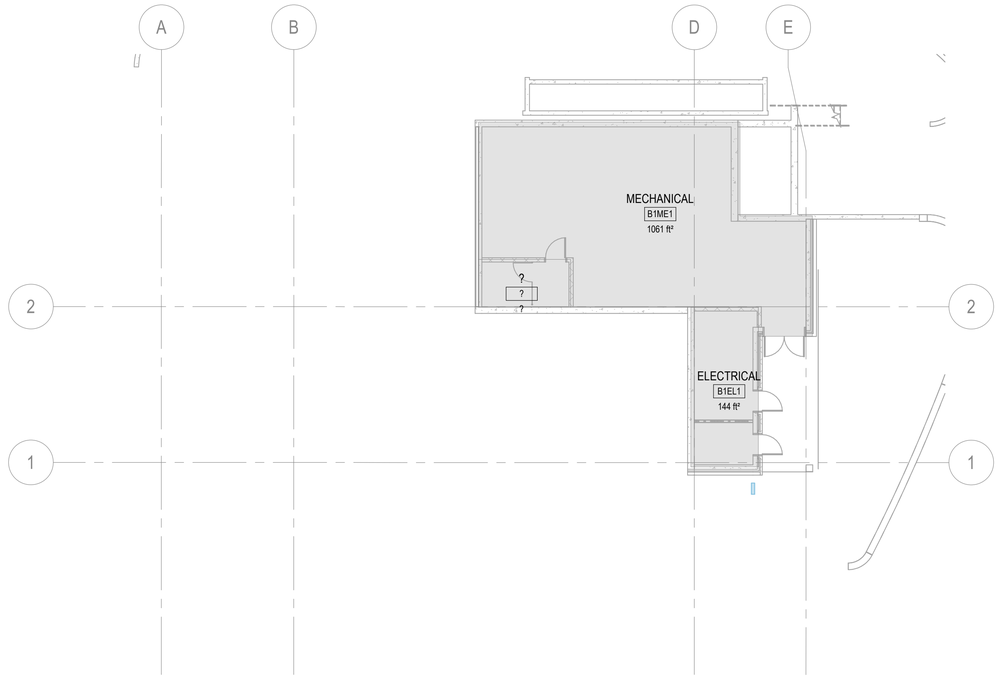
3. Corridors that directly connect to and are adjacent to spaces designated as Class 3 Air per ASHRAE 62.1-2022 shall also be classified as Class 3. Consequently, they are exhausted at the same rate as the adjoining Class 3 spaces.

Note 1

| Zone Name and Number | Occupancy Category | Zone Floor Area | Zone Population | Area Outdoor Air Rate | People Outdoor Air Rate | Breathing Zone Outdoor Airflow | Zone Air Distribution Effectiveness | Zone Outdoor Airflow | Zone Exhaust Airflow Rate | Zone Exhaust Airflow | Zone Airflow |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | *Az* | *Pz* | *Ra* | *Rp* | *Vbz* | *Ez* | *Voz* | *Rex* | *Vex* | *V* |
| | | (sq ft) | (people) | (cfm per sq ft) | (cfm per person) | (cfm) | | (cfm) | (cfm per sq ft) | (cfm) | (cfm) Designed or measured |
| 1CR3 CORRIDOR | Corridors | 161 | 0 | 0.06 | 0.0 | 10 | 0.8 | 12 | 2.25 | 362 | 400 |
| 104 CUBBING | Large-animal holding room | 108 | 3 | 0.18 | 10.0 | 49 | 0.8 | 62 | 2.25 | 243 | 300 |
| 103 QUARANTINE | Large-animal holding room | 99 | 3 | 0.18 | 10.0 | 48 | 0.8 | 60 | 2.25 | 223 | 300 |
| 107 DEN1 | Large-animal holding room | 144 | 3 | 0.18 | 10.0 | 55 | 0.8 | 68 | 2.25 | 324 | 350 |
| 1CR1 KEEPER CORRIDOR | Corridors | 348 | 0 | 0.06 | 0.0 | 21 | 0.8 | 26 | 2.25 | 783 | 800 |
| 106 DEN2 | Large-animal holding room | 128 | 3 | 0.18 | 10.0 | 49 | 0.8 | 61 | 2.25 | 288 | 300 |
| 105 DEN3 | Large-animal holding room | 192 | 4 | 0.18 | 10.0 | 73 | 0.8 | 91 | 2.25 | 432 | 450 |
| 1CR2 KEEPER CORRIDOR | Corridors | 112 | 0 | 0.06 | 0.0 | 7 | 0.8 | 8 | 2.25 | 252 | 300 |
| 100 KITCHEN PREP | Kitchens (Cooking) | 472 | 10 | 0.12 | 7.5 | 127 | 0.8 | 159 | 1.40 | 661 | 700 |
| 101 VIP | Office Spaces | 522 | 4 | 0.06 | 5.0 | 51 | 0.8 | 64 | 0.00 | 0 | 700 |
| 1WC1 RESTROOM | Toilet Rooms - Public | 81 | 1 | 0.00 | 0.0 | 0 | 0.8 | 0 | 0.00 | 100 | 100 |
| 1CS1 CUSTODIAL | Storage Rooms | 28 | 0 | 0.12 | 0.0 | 3 | 0.8 | 4 | 0.00 | 100 | 100 |
| 108 DAYROOM | Large-animal holding room | 583 | 12 | 0.18 | 10.0 | 221 | 0.8 | 277 | 2.25 | 1311 | 1,700 |
| | | | | | | | | | | | |

| | |
|---|---|
| System area (As, sq ft) | 2,978 |
| System population (Ps, people) | 43 |
| Sum of zone population (people) | 43 |
| Occupant diversity (D) | 1.00 |
| Uncorrected outdoor air intake (Vou, CFM) | 717 |
| LEED WITH - Additional 30% Uncorrected outdoor air intake (Vou, CFM) | 931 |

| | |
|---|---|
| Outdoor air intake flow (Required) | 931 |
| Code Required Exhaust (Required) | 5,079 |
| Outdoor air intake flow (Design) (includes tolerance for air leakage and building pressurization) | 6,645 |




1 M-002 BASEMENT LEVEL VENTILATION PLAN ZONING REFERENCE
SCALE: NOT TO SCALE

# VENTILATION CALCULATION (METRIC UNITS - FOR REFERENCE ONLY)

| | |
|---|---|
| System name and number | AHU-1 |
| Condition analyzed | Heating |
| Calculation Methodology | IMC-2024 |
| | Appendix A |

Notes:

1. Zone population is based on the larger of the design occupancy and the occupancy calculated using the IMC-2024 default occupancy rates unless otherwise noted.

2. Zone primary airflow is the larger of: (1) the calculated airflow based on winter heat loss and full occupancy, (2) the airflow provided at the associated terminal unit's minimum airflow set point, and (3) make-up air required to meet code required exhaust.

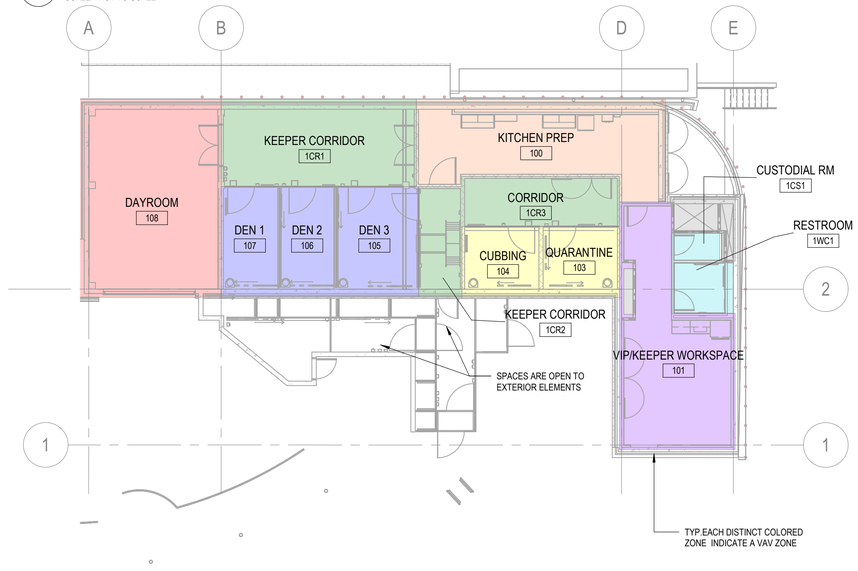
3. Corridors that directly connect to and are adjacent to spaces designated as Class 3 Air per ASHRAE 62.1-2022 shall also be classified as Class 3. Consequently, they are exhausted at the same rate as the adjoining Class 3 spaces.

Note 1

| Zone Name and Number | Occupancy Category | Zone Floor Area | Zone Population | Area Outdoor Air Rate | People Outdoor Air Rate | Breathing Zone Outdoor Airflow | Zone Air Distribution Effectiveness | Zone Outdoor Airflow | Zone Exhaust Airflow Rate | Zone Exhaust Airflow | Zone Airflow |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | *Az* | *Pz* | *Ra* | *Rp* | *Vbz* | *Ez* | *Voz* | *Rex* | *Vex* | *V* |
| | | (m²) | (people) | (m³/hr per m²) | (m³/hr per person) | (m³/hr) | | (m³/hr) | (m³/hr per m²) | (m³/hr) | (m³/hr) Designed or measured |
| 1CR3 CORRIDOR | Corridors | 15 | 0 | 1.10 | 0.0 | 16 | 0.8 | 21 | 41.15 | 616 | 680 |
| 104 CUBBING | Large-animal holding room | 10 | 3 | 3.29 | 17.0 | 84 | 0.8 | 105 | 41.15 | 413 | 510 |
| 103 QUARANTINE | Large-animal holding room | 9 | 3 | 3.29 | 17.0 | 81 | 0.8 | 102 | 41.15 | 378 | 510 |
| 107 DEN1 | Large-animal holding room | 13 | 3 | 3.29 | 17.0 | 93 | 0.8 | 116 | 41.15 | 550 | 595 |
| 1CR1 KEEPER CORRIDOR | Corridors | 32 | 0 | 1.10 | 0.0 | 35 | 0.8 | 44 | 41.15 | 1,330 | 1,359 |
| 106 DEN2 | Large-animal holding room | 12 | 3 | 3.29 | 17.0 | 83 | 0.8 | 103 | 41.15 | 489 | 510 |
| 105 DEN3 | Large-animal holding room | 18 | 4 | 3.29 | 17.0 | 124 | 0.8 | 155 | 41.15 | 734 | 765 |
| 1CR2 KEEPER CORRIDOR | Corridors | 10 | 0 | 1.10 | 0.0 | 11 | 0.8 | 14 | 41.15 | 428 | 510 |
| 100 KITCHEN PREP | Kitchens (Cooking) | 44 | 10 | 2.19 | 12.7 | 217 | 0.8 | 271 | 25.60 | 1,123 | 1,189 |
| 101 VIP | Office Spaces | 48 | 4 | 1.10 | 8.5 | 87 | 0.8 | 109 | 0.00 | 0 | 1,189 |
| 1WC1 RESTROOM | Toilet Rooms - Public | 8 | 1 | 0.00 | 0.0 | 0 | 0.8 | 0 | 0.00 | 170 | 170 |
| 1CS1 CUSTODIAL | Storage Rooms | 3 | 0 | 2.19 | 0.0 | 6 | 0.8 | 7 | 0.00 | 170 | 170 |
| 108 DAYROOM | Large-animal holding room | 54 | 12 | 3.29 | 17.0 | 376 | 0.8 | 470 | 41.15 | 2,228 | 2,888 |
| | | | | | | | | | | | |

| | |
|---|---|
| System area (As, m²) | 277 |
| System population (Ps, people) | 41.6 |
| Sum of zone population (people) | 41.6 |
| Occupant diversity (D) | 1.00 |
| Uncorrected outdoor air intake (Vou, m³/hr) | 1,217 |
| LEED - Additional 30% Uncorrected outdoor air intake (Vou, m³/hr) | 1,583 |

| | |
|---|---|
| Outdoor air intake flow (Required) | 1,583 |
| Code Required Exhaust (Required) | 8,629 |
| Outdoor air intake flow (Design) (includes tolerance for air leakage and building pressurization) | 11,290 |




2 M-002 LEVEL 1 VENTILATION PLAN ZONING REFERENCE
SCALE: NOT TO SCALE

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | VENTILATION CALCULATIONS - AHU-1 |
| WORKING STAFF | DESIGNED BY: OP; DRAWN BY: OP; CHECKED BY: IP |

SHEET NO. 254 OF 378

M-002 (DISCIPLINE: M; TYPE: -; SEQUENCE: 002; DRAWING TYPE: -)

FOR REFERENCE ONLY

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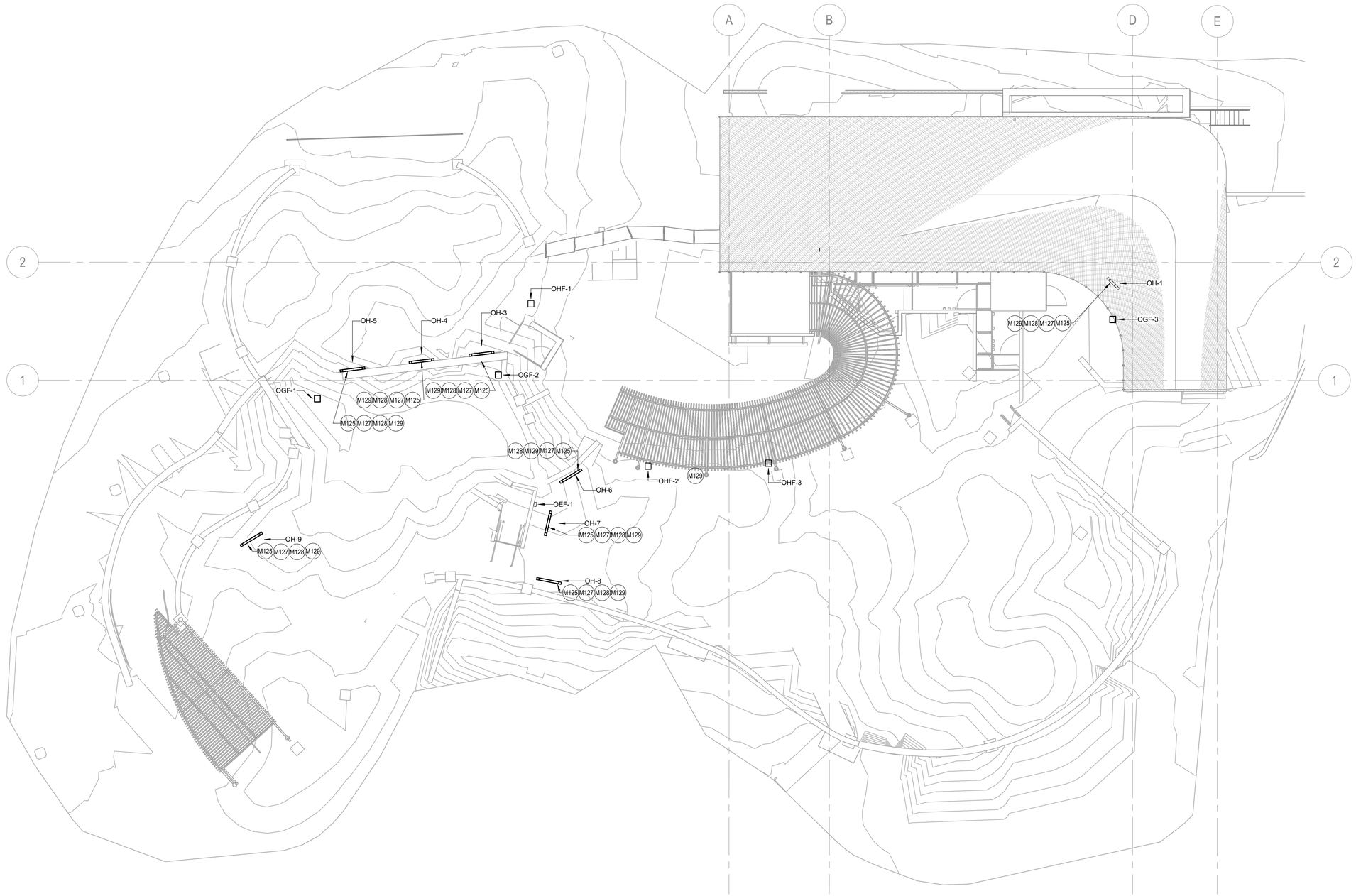


1 MH-100FP LEVEL 1 MECHANICAL SITE PLAN
SCALE: 1" = 10'-0"

## GENERAL SHEET NOTES

A. INSTALLATION INSTRUCTIONS FROM THE MANUFACTURER OF PURCHASED PRODUCT(S) MUST BE REVIEWED ALONG WITH THE CONTRACT DOCUMENTS. THE MOST STRINGENT INSTALLATION/CLEARANCE REQUIREMENT SHALL APPLY.
B. DUE TO UNIQUE INSTALLATION REQUIREMENTS FOR EACH ELECTRIC HEATING ELEMENT, THE CONTRACTOR SHALL FORMALLY CONFIRM THE INSTALLATION INTENT FOR EACH INSTANCE OF HEATER WITH THE MANUFACTURER OF THE PURCHASED PRODUCT. SUBMIT WRITTEN CONFIRMATION FROM THE MANUFACTURER OF PURCHASED EQUIPMENT TO THE E.O.R. AND OWNER. SPECIFICALLY CONFIRM ANY HAZARDS ASSOCIATED WITH FIRE, OVERHEATING, INADEQUATE CLEARANCES, PROXIMITY TO COMBUSTIBLE MATERIALS, EXPOSURE TO ELEMENTS, AND ELECTRICAL CONNECTIONS. CONFIRM REQUIREMENTS FOR GROUNDING, WEATHERPROOFING, AND ENVIRONMENTAL RATINGS.
C. CONTRACTOR SHALL NOT PROCEED WITH INSTALLATION UNTIL ALL IDENTIFIED RISKS ARE REVIEWED, MITIGATED, AND ACCEPTED BY THE MANUFACTURER, E.O.R., AND OWNER.
D. DOCUMENT SERVICING INSTRUCTIONS FOR EACH INSTANCE OF PURCHASED OUTDOOR EQUIPMENT BASED ON INSTALLED LOCATION, AND SUBMIT TO THE E.O.R. AND OWNER FOR RECORD.
E. PROTECT CONDUCTORS IN ACCORDANCE WITH DIVISION 26 DOCUMENTS (DRAWINGS AND SPECIFICATIONS) FROM DAMAGE DUE TO EXPOSURE TO ELEMENTS.

## SHEET KEYNOTES

M125 COORDINATE WITH DIVISION 26. PROVIDE INTERMEDIATE CONTROL ENCLOSURE WITH HEAVY-DUTY CONTACTOR ADJACENT TO HEATER RECEPTACLE. FIELD-WIRE LOW-VOLTAGE CONTROL SIDE TO BACNET RELAY MODULE. RATED FOR (AT LEAST) CONTINUOUS 5KW LOAD.

M127 TYP. FOR OUTDOOR HEATER. SEE DRAWINGS ISSUED BY OTHER DISCIPLINES FOR ADDITIONAL INFORMATION. HEATER MOUNTING SHALL BE AS SHOWN ON DETAILS CALLED OUT ON EX-300 SERIES AND ACCORDING TO MANUFACTURER RECOMMENDATIONS ONLY. HEATER SHALL BE DRIVEN BY A THERMOSTAT WITHIN THE MECHANICAL ROOM (SEE MI-800CP FOR MORE INFORMATION). SEE MECHANICAL SCHEDULES FOR SELECTION INFORMATION.

M128 PROVIDE A MANUAL DISCONNECT FOR THE HEATER AT THE LOCATIONS IDENTIFIED IN DIVISION 26 DOCUMENTS. COORDINATE WITH THE DIVISION 26 CONTRACTOR IF THE MANUFACTURER-PROVIDED CABLE LENGTH IS INSUFFICIENT FOR ELECTRICAL CONNECTION.

M129 ROUTE, INSTALL AND PROTECT CONTROL CONDUCTOR PER SPECIFICATIONS ISSUED BY DIVISION 26 FOR LOW VOLTAGE OUTDOOR CONDUCTORS, AND ALONGSIDE CONDUCTORS OF DIVISION 26 SERVING THIS EQUIPMENT.

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# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

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STAMP

TRUE NORTH
TRUE NORTH
PROJECT NORTH

KEY PLAN

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
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| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

Smithsonian Institution

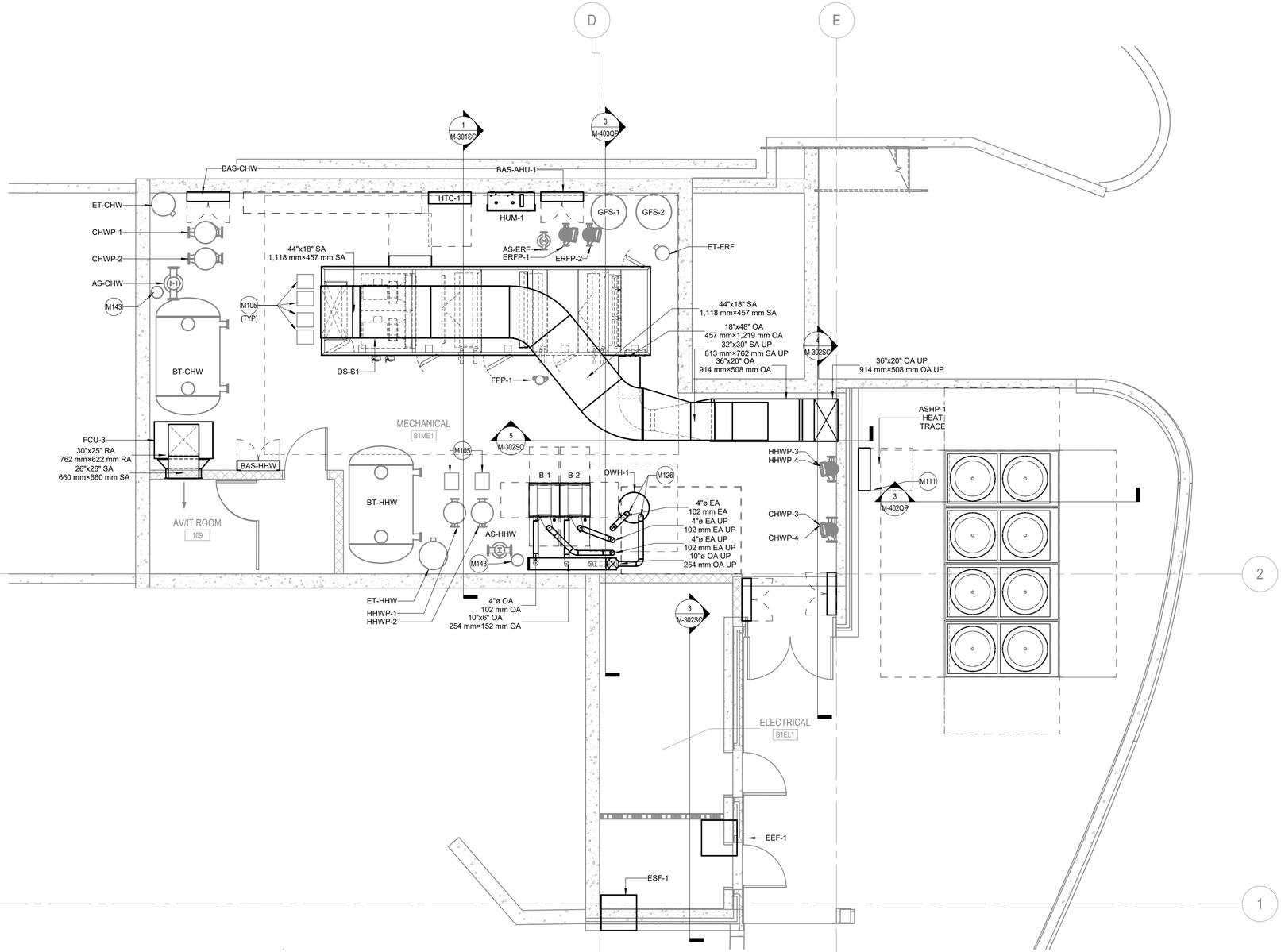
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL SITE PLAN - EXTERIOR ELEMENTS |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 255 OF 378

MH-100FP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

*FOR REFERENCE ONLY*




1 MH-100HP BASEMENT LEVEL DUCTWORK
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ALL GENERAL EXHAUST DUCTS TERMINATING IN EXPOSED AREAS SHALL TERMINATE IN EXHAUST GRILLE TO BE COORDINATED WITH ARCHITECTURE.
B. ALL DUCT DIMENSIONS SHOWN ON PLAN ARE CLEAR INTERIOR DIMENSIONS. CONTRACTOR TO COORDINATE EXTERIOR DIMENSIONS.
C. PROVIDE NO MEDIA SOUND ATTENUATORS FOR TERMINAL UNITS CALLED OUT IN SILENCER SCHEDULE. SILENCER MAY BE INTEGRAL TO TERMINAL UNIT.
D. COORDINATE ALL DIFFUSERS, REGISTERS, AND GRILLES WITH REFLECTED CEILING PLAN.
E. ALL LIFE SAFETY DAMPERS/ASSEMBLIES SHOWN ON PLANS SHALL CONTAIN RATINGS THAT ARE COORDINATED WITH THE RESPECTIVE WALL/SLAB RATING. COORDINATE WITH ARCHITECTURAL OR LIFE SAFETY PLANS FOR WALLS/SLAB RATING.
F. ALL DUCT PENETRATIONS THROUGH 1-HOUR RATED PARTITIONS SHALL BE FIRE STOPPED PER UL DETAILS TO ACHIEVE 1-HR RATING.
G. DUCT SIZE TO AIR TERMINAL SHALL BE SAME AS AIR TERMINAL INLET SIZE UNLESS OTHERWISE NOTED.
H. PROVIDE BALANCING DAMPER AT EACH DIFFUSER, GRILLE, AND BRANCH TAP-OFF IN ALL SUPPLY, RETURN AND EXHAUST DUCTWORK. LOCATE BALANCING DAMPER AS CLOSE TO BRANCH TAKE-OFF AS POSSIBLE.
I. PROVIDE CABLE OPERATED REMOTE DAMPERS TO ALL DIFFUSERS LOCATED IN DRY WALL CEILING. COORDINATE LOCATION OF REMOTE DAMPER ADJUSTMENT PANEL WITH ALL DISCIPLINES' CEILING ELEMENTS.
J. USE OF FLEXIBLE DUCTWORK IN EXHAUST AIR SYSTEM IS NOT PERMITTED.
K. USE OF FLEXIBLE DUCTWORK TO CONNECT AN AIR TERMINAL LOCATED ABOVE AN INACCESSIBLE CEILING IS NOT PERMITTED.

## SHEET KEYNOTES

M105 DIVISION 26 EQUIPMENT/FIXTURE/CONDUCTOR ONLY SHOWN FOR COORDINATION. REFER TO DIVISION 26 DOCUMENT FOR EXACT LOCATION/DIMENSION.

M106 SEE DOCUMENTS ISSUED BY OTHER DISCIPLINES TO VIEW EQUIPMENT AROUND THIS GENERAL REGION.

M111 MANUFACTURER FURNISHED HEAT TRACE CONTROL PANEL FOR AIR SOURCE HEAT PUMP. THE HEAT TRACE PANEL MAY BE INTEGRAL TO THE UNIT. COORDINATE LOCATION OF PANEL WITH PURCHASED PRODUCT. COORDINATE WITH ELECTRICAL CONTRACTOR ON EXACT LOCATION.

M126 PROVIDE A 4" (102MM) EXHAUST AND 4" (102MM) OUTSIDE AIR CONNECTION TO PLUMBING EQUIPMENT. COORDINATE WITH DIVISION 22 CONTRACTOR ON PURCHASED PRODUCT. PROVIDE END FITTINGS (REDUCERS/COUPLINGS) AS REQUIRED BY MANUFACTURER TO COMPLETE CONNECTION TO EQUIPMENT.

M143 CHEMICAL SHOT FEEDER

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STAMP

TRUE NORTH
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PROJECT NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

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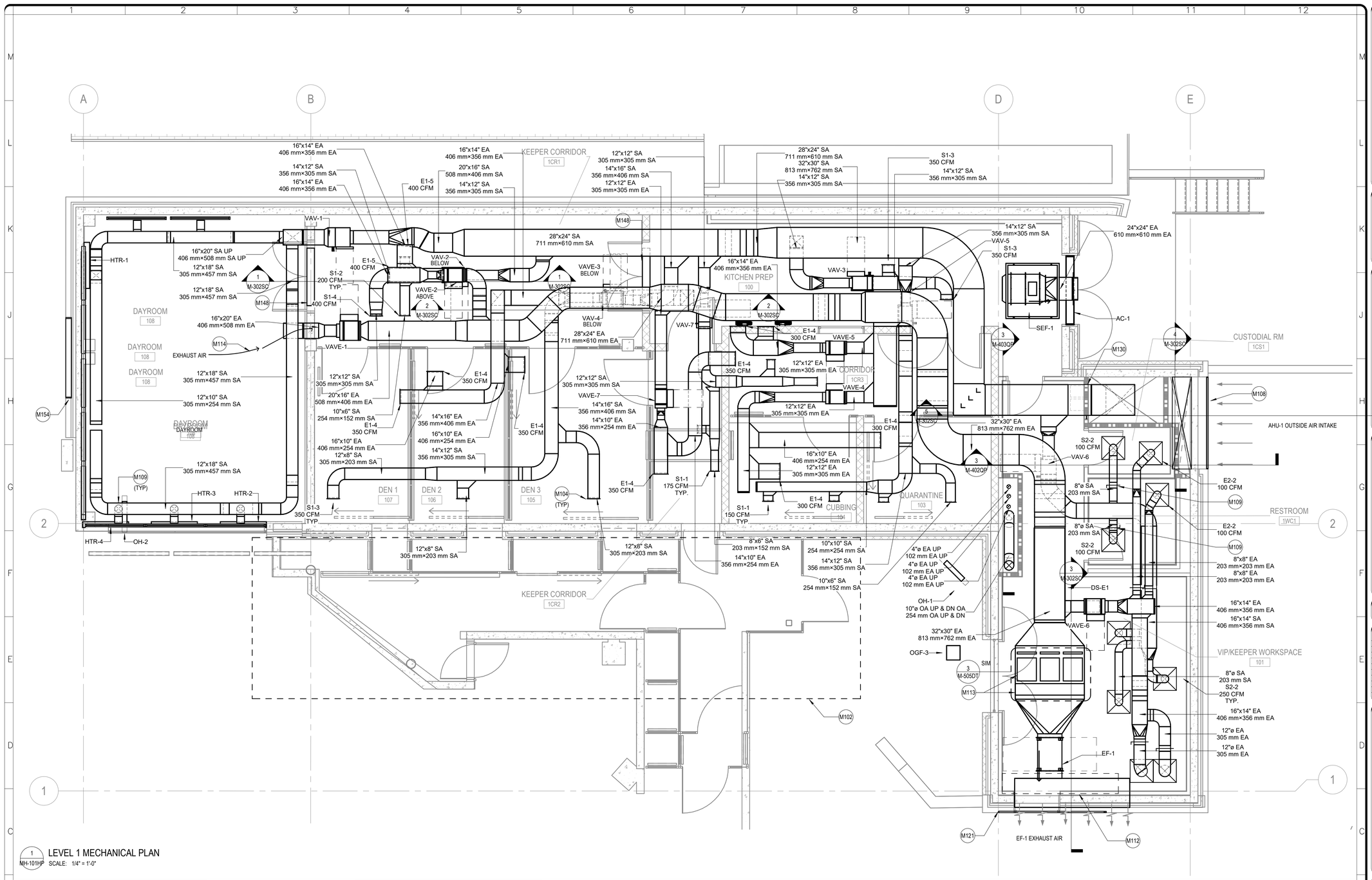
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BASEMENT MECHANICAL PLAN |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |
| SHEET NO. | 256 OF 378 |
| | MH-100HP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |




1 MH-101HP LEVEL 1 MECHANICAL PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ALL GENERAL EXHAUST DUCTS TERMINATING IN EXPOSED AREAS SHALL TERMINATE IN EXHAUST GRILLE TO BE COORDINATED WITH ARCHITECTURE.
B. ALL DUCT DIMENSIONS SHOWN ON PLAN ARE CLEAR INTERIOR DIMENSIONS. CONTRACTOR TO COORDINATE EXTERIOR DIMENSIONS.
C. PROVIDE NO MEDIA SOUND ATTENUATORS FOR TERMINAL UNITS CALLED OUT IN SILENCER SCHEDULE. SILENCER MAY BE INTEGRAL TO TERMINAL UNIT.
D. COORDINATE ALL DIFFUSERS, REGISTERS, AND GRILLES WITH REFLECTED CEILING PLAN.
E. ALL LIFE SAFETY DAMPERS/ASSEMBLIES SHOWN ON PLANS SHALL CONTAIN RATINGS THAT ARE COORDINATED WITH THE RESPECTIVE WALL/SLAB RATING. COORDINATE WITH ARCHITECTURAL OR LIFE SAFETY PLANS FOR WALLS/SLAB RATING.
F. ALL DUCT PENETRATIONS THROUGH 1-HOUR RATED PARTITIONS SHALL BE FIRE STOPPED PER UL DETAILS TO ACHIEVE 1-HR RATING.
G. DUCT SIZE TO AIR TERMINAL SHALL BE SAME AS AIR TERMINAL INLET SIZE UNLESS OTHERWISE NOTED.
H. PROVIDE BALANCING DAMPER AT EACH DIFFUSER, GRILLE, AND BRANCH TAP-OFF IN ALL SUPPLY, RETURN AND EXHAUST DUCTWORK. LOCATE BALANCING DAMPER AS CLOSE TO BRANCH TAKE-OFF AS POSSIBLE.
I. PROVIDE CABLE OPERATED REMOTE DAMPERS TO ALL DIFFUSERS LOCATED IN DRY WALL CEILING. COORDINATE LOCATION OF REMOTE DAMPER ADJUSTMENT PANEL WITH ALL DISCIPLINES' CEILING ELEMENTS.
J. USE OF FLEXIBLE DUCTWORK IN EXHAUST AIR SYSTEM IS NOT PERMITTED.
K. USE OF FLEXIBLE DUCTWORK TO CONNECT AN AIR TERMINAL LOCATED ABOVE AN INACCESSIBLE CEILING IS NOT PERMITTED.

## SHEET KEYNOTES

M102 SPACES SHOWN IN DASHED BOUNDARY ARE NOT A PART OF CONDITIONED BUILDING.
M104 TERMINATE IN A GRILLE. GRILLE SHALL MATCH FACE SIZE.
M108 SEE ARCHITECTURAL DOCUMENTS FOR MORE INFORMATION ON LOUVER/FINISH. PROVIDE AT LEAST 20SF IN FREE AREA, ACROSS THE ENTIRE INTAKE ASSEMBLY. INTAKE ASSEMBLY CONSISTS OF LOUVER, AND/OR DECORATIVE SCREENS, MESHES. PURCHASED PRODUCT MUST DOCUMENT THIS INFORMATION.
M109 DO NOT PROVIDE FLEXIBLE DUCT END CONNECTION TO DIFFUSER. PROVIDE CABLE OPERATED DAMPER. CONSOLIDATE AND LOCATE THE DAMPER ADJUSTMENT DEVICE(S) AT AN ACCESSIBLE LOCATION. DOCUMENT THE LOCATION.
M112 PROVIDE A EXHAUST PLENUM WITH A LOUVER ASSEMBLY.
M113 ENERGY RECOVERY COIL AND AIR FILTER UPSTREAM OF EXHAUST FAN. SEE MECHANICAL SCHEDULES FOR ADDITIONAL INFORMATION.
M114 OPEN CEILING EXHAUST FROM DAY ROOM INTO THE CHASE THROUGH A GRILLE. THE CHASE SHALL HAVE THE EXHAUST AIR DUCT TAP (16 IN X 20IN) AS SHOWN TO VAVE-1.
M121 SEE ARCHITECTURAL DOCUMENTS FOR MORE INFORMATION ON LOUVER/FINISH. PROVIDE AT LEAST 20SF IN FREE AREA, ACROSS THE ENTIRE EXHAUST ASSEMBLY. EXHAUST ASSEMBLY CONSISTS OF LOUVER, AND/OR DECORATIVE SCREENS, MESHES. PURCHASED PRODUCT MUST DOCUMENT THIS INFORMATION.
M130 FIRE SMOKE DAMPER; BASIS OF DESIGN POTTORFF FD-350 AIRFOIL BLADE. DAMPER SHALL COMPLY WITH AIR DUCT ACCESSORIES SPECIFICATION SECTION AND SHALL BE DYNAMIC RATED FOR 3HR.
M148 TRANSFER OPENING FOR SMOKE EXHAUST SYSTEM. SEE SHEET M-822CP FOR MORE INFORMATION.
M154 INTAKE LOUVER AS A PART OF SMOKE EXHAUST SYSTEM. LOUVER IS SHOWN ONLY FOR REFERENCE. ALL SCOPE IS DEFINED BY OTHER TRADES. PRECISE LOCATION AND OPENING DIMENSIONS SHALL BE PER ARCHITECTURAL AND FIRE ALARM DRAWINGS. SEE M-821CP AND M-822CP FOR ADDITIONAL INFORMATION.

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# ALF

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TRUE NORTH
PROJECT NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

FOR REFERENCE ONLY

DATE: 06/15/2026
SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE
REVISION DESCRIPTION

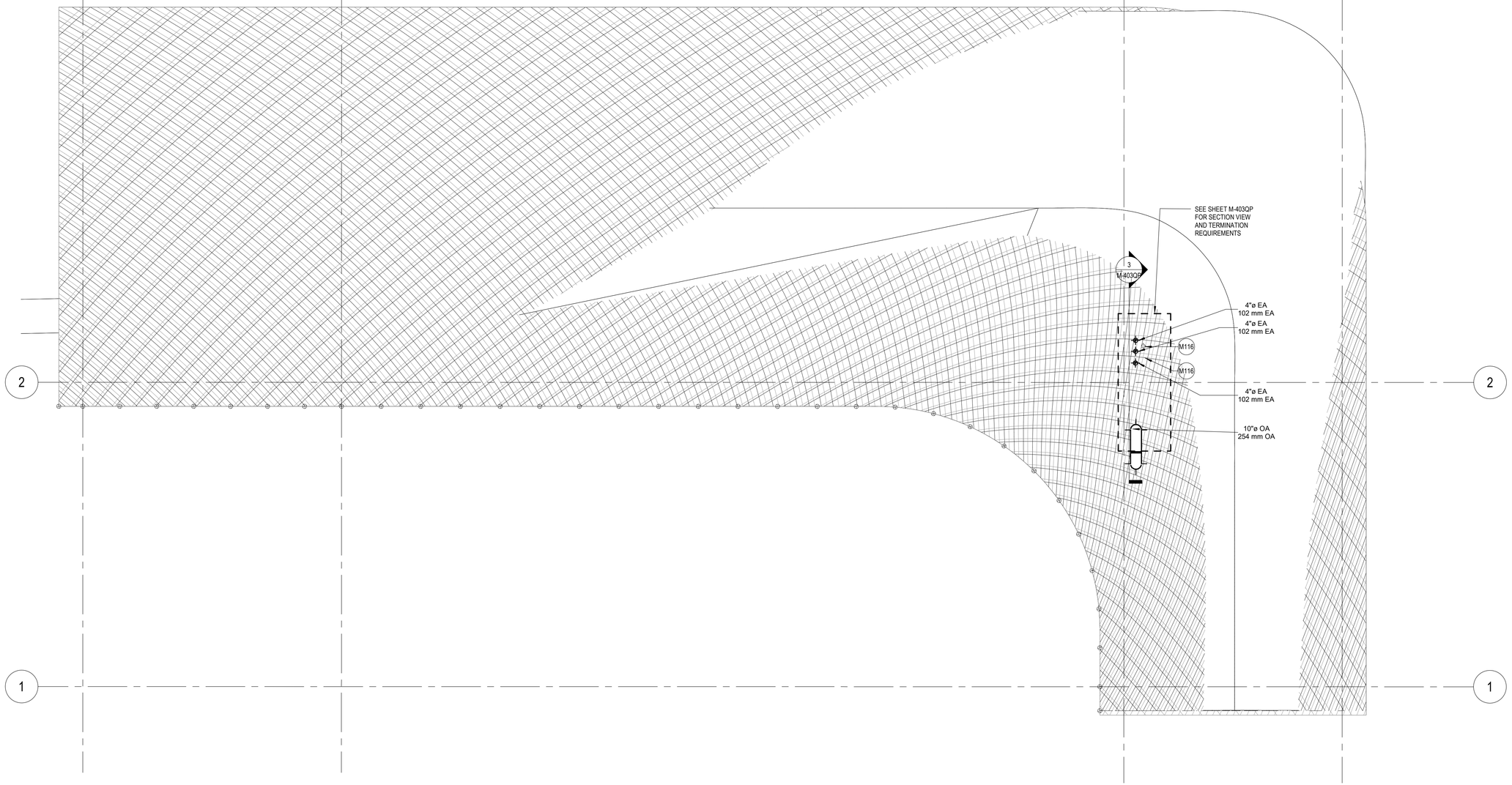
Smithsonian Institution
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: LEVEL 1 MECHANICAL PLAN
WORKING STAFF: OP (DESIGNED BY) | YW (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 257 OF 378
MH-101HP (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)




1 MH-102RO ROOF MECHANICAL PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ALL GENERAL EXHAUST DUCTS TERMINATING IN EXPOSED AREAS SHALL TERMINATE IN EXHAUST GRILLE TO BE COORDINATED WITH ARCHITECTURE.
B. ALL DUCT DIMENSIONS SHOWN ON PLAN ARE CLEAR INTERIOR DIMENSIONS. CONTRACTOR TO COORDINATE EXTERIOR DIMENSIONS.
C. PROVIDE NO MEDIA SOUND ATTENUATORS FOR TERMINAL UNITS CALLED OUT IN SILENCER SCHEDULE. SILENCER MAY BE INTEGRAL TO TERMINAL UNIT.
D. COORDINATE ALL DIFFUSERS, REGISTERS, AND GRILLES WITH REFLECTED CEILING PLAN.
E. ALL LIFE SAFETY DAMPERS/ASSEMBLIES SHOWN ON PLANS SHALL CONTAIN RATINGS THAT ARE COORDINATED WITH THE RESPECTIVE WALL/SLAB RATING. COORDINATE WITH ARCHITECTURAL OR LIFE SAFETY PLANS FOR WALL/SLAB RATING.
F. ALL DUCT PENETRATIONS THROUGH 1-HOUR RATED PARTITIONS SHALL BE FIRE STOPPED PER UL DETAILS TO ACHIEVE 1-HR RATING.
G. DUCT SIZE TO AIR TERMINAL SHALL BE SAME AS AIR TERMINAL INLET SIZE UNLESS OTHERWISE NOTED.
H. PROVIDE BALANCING DAMPER AT EACH DIFFUSER, GRILLE, AND BRANCH TAP-OFF IN ALL SUPPLY, RETURN AND EXHAUST DUCTWORK. LOCATE BALANCING DAMPER AS CLOSE TO BRANCH TAKE-OFF AS POSSIBLE.
I. PROVIDE CABLE OPERATED REMOTE DAMPERS TO ALL DIFFUSERS LOCATED IN DRY WALL CEILING. COORDINATE LOCATION OF REMOTE DAMPER ADJUSTMENT PANEL WITH ALL DISCIPLINES' CEILING ELEMENTS.
J. USE OF FLEXIBLE DUCTWORK IN EXHAUST AIR SYSTEM IS NOT PERMITTED.
K. USE OF FLEXIBLE DUCTWORK TO CONNECT AN AIR TERMINAL LOCATED ABOVE AN INACCESSIBLE CEILING IS NOT PERMITTED.

## SHEET KEYNOTES

M116 UTILITY/FIXTURE/EQUIPMENT BY DIVISION 22. SEE PLUMBING DOCUMENTS FOR MORE INFORMATION.

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**CONSTRUCT ARABIAN LEOPARD FACILITY**

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STAMP

TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

**GRAPHIC SCALE(S)**

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ROOF MECHANICAL PLAN |

| WORKING STAFF | OP | OP | IP |
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| | DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 258 OF 378

MH-102RO
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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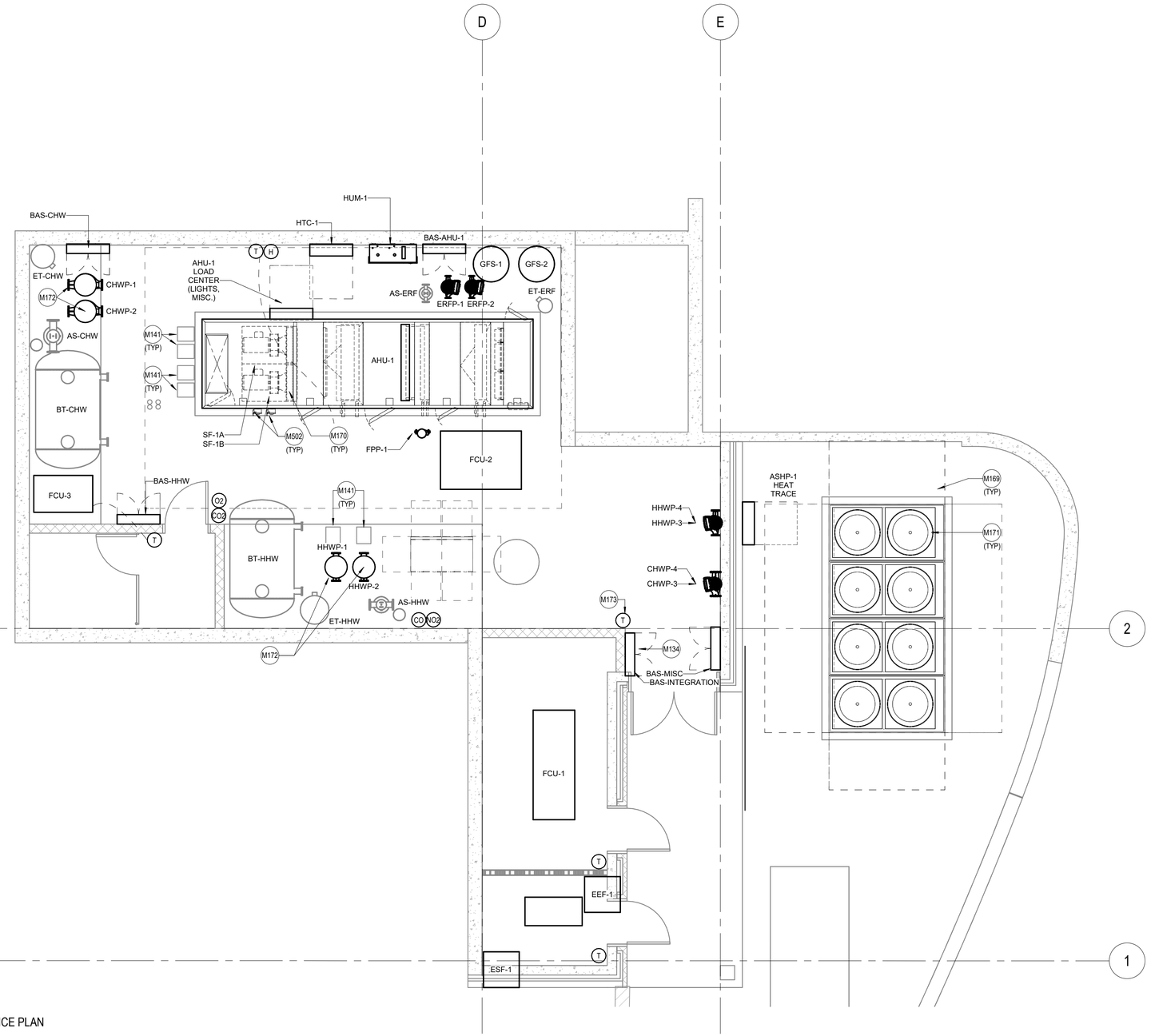


1 MI-100CP MECHANICAL CONTROLS - BASEMENT CONTROL DEVICE PLAN

SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. COORDINATE FINAL THERMOSTAT LOCATIONS WITH ARCHITECTURE.
B. COORDINATE THERMOSTAT CONDUIT, WHERE EXPOSED, TO MATCH ADJACENT COLUMNS OR WALLS.
C. MOUNT TEMPERATURE/HUMIDITY SENSORS WITHIN ANIMAL OCCUPIED SPACES AT AN ELEVATION OF 6FT OR ABOVE.
D. CONTRACTOR SHALL COORDINATE EQUIPMENT TAGS WITH THE SMITHSONIAN TO ENSURE FORMATTING IS CONSISTENT WITH THE SMITHSONIAN SERVER. CONTRACTOR SHALL INCLUDE BOTH THE SERVER TAG AND THE PHYSICAL TAG IN THE BAS AS REFERENCE INFORMATION.

## SHEET KEYNOTES

M134 PROVIDE LOCAL STANDALONE PROGRAMMABLE CONTROLLER WITH SUFFICIENT INPUT/OUTPUT CAPACITY TO SERVE ALL OUTDOOR ELECTRIC RADIANT HEATERS, PROVIDING ON/OFF OVERRIDE FOR ALL HEATERS; MOUNT CONTROLLER WITHIN BAS-INTEGRATION PANELBOARD LOCATED ADJACENT TO THE THERMOSTAT. SYSTEM SHALL OPERATE WITHOUT BAS DEPENDENCY, WITH BAS MONITORING ONLY.

M141 TYP. VARIABLE FREQUENCY DRIVE FOR A MOTOR/FAN. VFD IS SHOWN ON PLAN FOR REFERENCE ONLY. SEE ELECTRICAL DOCUMENTS FOR MORE INFORMATION.

M169 SMITHSONIAN DESIGN STANDARD SDS VOL.1 23.2.14.8. - PROVIDE AND INTEGRATE HIGH-PRECISION TRIAXIAL ACCELEROMETERS FOR PREVENTATIVE MAINTENANCE OF HEAT PUMP COMPRESSORS (8 TOTAL SENSORS FOR 8 ON/OFF SCROLL COMPRESSORS ACROSS 4 MODULES). SENSORS SHALL BE STUD MOUNTED ON COMPRESSOR HOUSINGS OR APPROVED LOCATIONS. COORDINATE WITH MANUFACTURER TO CONFIRM FINAL MOUNTING POSITIONS. INTEGRATE VIA BACNET MSTP TO BAS FOR VIBRATION MONITORING (X, Y, Z AXES). COORDINATE WITH SMITHSONIAN PREVENTATIVE MAINTENANCE TEAM. TREND DATA SHALL BE FULLY COMPATIBLE WITH OWNER'S PREVENTATIVE MAINTENANCE SYSTEM AS WELL. OBTAIN BASELINE VIBRATION DATA REPRESENTING HEALTHY OPERATION FOR TRENDING AND ALARM SETPOINTS.

M170 SMITHSONIAN DESIGN STANDARD SDS VOL.1 23.2.14.8 - PROVIDE AND INSTALL VIBRATION MONITORING FOR VFD DRIVEN AHU FANS FOR PREVENTATIVE MAINTENANCE. SENSORS SHALL BE DUAL AXIS, FIELD MOUNTED ON FAN BEARINGS OR SHAFT HOUSING. COORDINATE FINAL MOUNTING WITH EQUIPMENT MANUFACTURER. INTEGRATE SIGNALS TO BAS FOR TRENDING AND ALARMING. TREND DATA SHALL BE FULLY COMPATIBLE WITH OWNER'S PREVENTATIVE MAINTENANCE SYSTEM AS WELL. COORDINATE WITH OWNER TO ESTABLISH BASELINE VIBRATION LEVELS REPRESENTING NORMAL OPERATION AND DEVELOP ALERT THRESHOLDS FOR EARLY FAULT DETECTION.

M171 SMITHSONIAN DESIGN STANDARD SDS VOL.1 23.2.14.8 - PROVIDE AND INSTALL VIBRATION MONITORING FOR ECM FANS ASSOCIATED WITH HEAT PUMPS FOR PREVENTATIVE MAINTENANCE. PROVIDE ONE (1) SENSOR PER FAN (SINGLE AXIS PIEZOELECTRIC ACCELEROMETER OR EQUIVALENT) TO MEASURE VIBRATION. FAN SPEED (RPM) SHALL BE OBTAINED FROM THE INTEGRAL CONTROLLER, AND VIBRATION DATA SHALL BE CORRELATED WITH FAN SPEED FOR CONDITION MONITORING. INSTALL SENSOR ALONG THE AXIS OF GREATEST VIBRATION AND COORDINATE FINAL MOUNTING LOCATION WITH EQUIPMENT MANUFACTURER. INTEGRATE ALL POINTS TO BAS FOR TRENDING AND ALARMING, AND COORDINATE WITH OWNER TO ESTABLISH BASELINE VIBRATION VS. SPEED PROFILES REPRESENTING NORMAL OPERATION. TREND DATA SHALL BE FULLY COMPATIBLE WITH OWNER'S PREVENTATIVE MAINTENANCE SYSTEM AS WELL.

M172 SMITHSONIAN DESIGN STANDARD SDS VOL.1 23.2.14.8. - PROVIDE AND INSTALL VIBRATION MONITORING FOR HYDRONIC PUMPS FOR PREVENTATIVE MAINTENANCE. PROVIDE ONE (1) SENSOR PER PUMP (SINGLE AXIS PIEZOELECTRIC ACCELEROMETER OR EQUIVALENT) TO MEASURE VIBRATION AT BEARINGS OR OTHER REPRESENTATIVE LOCATIONS. PUMP SPEED (IF VARIABLE) SHALL BE OBTAINED FROM THE VFD OR CONTROLLER. AND VIBRATION DATA SHALL BE CORRELATED WITH SPEED FOR CONDITION MONITORING. INSTALL SENSOR ALONG THE AXIS OF GREATEST VIBRATION AND COORDINATE FINAL MOUNTING LOCATION WITH EQUIPMENT MANUFACTURER. INTEGRATE ALL POINTS TO BAS FOR TRENDING AND ALARMING, AND COORDINATE WITH OWNER TO ESTABLISH BASELINE VIBRATION VS. SPEED PROFILES REPRESENTING NORMAL OPERATION.

M173 TOUCH SCREEN GRAPHIC INTERFACE THERMOSTAT. INCLUDE READ ONLY OUTDOOR AIR TEMPERATURE AND HUMIDITY READING FROM WEATHER STATION. THERMOSTAT SHALL BE USED TO ACTIVATE/DEACTIVATE OUTDOOR SITE HEATERS TOGETHER AND SIMULTANEOUSLY BY OWNER. SEE M-820CP FOR MORE INFORMATION.

M502 PROVIDE CASING MOUNTED FAN DISCONNECT. COORDINATE EXACT LOCATION WITH ELECTRICAL CONTRACTOR WITH PURCHASED PRODUCT'S SHOP DRAWINGS. SEE MECHANICAL SPECIFICATIONS FOR ADDITIONAL INFORMATION.

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# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - BASEMENT CONTROL DEVICE PLAN |
| WORKING STAFF | OP / OP / IP (DESIGNED BY / DRAWN BY / CHECKED BY) |

SHEET NO. 259 of 378

MI-100CP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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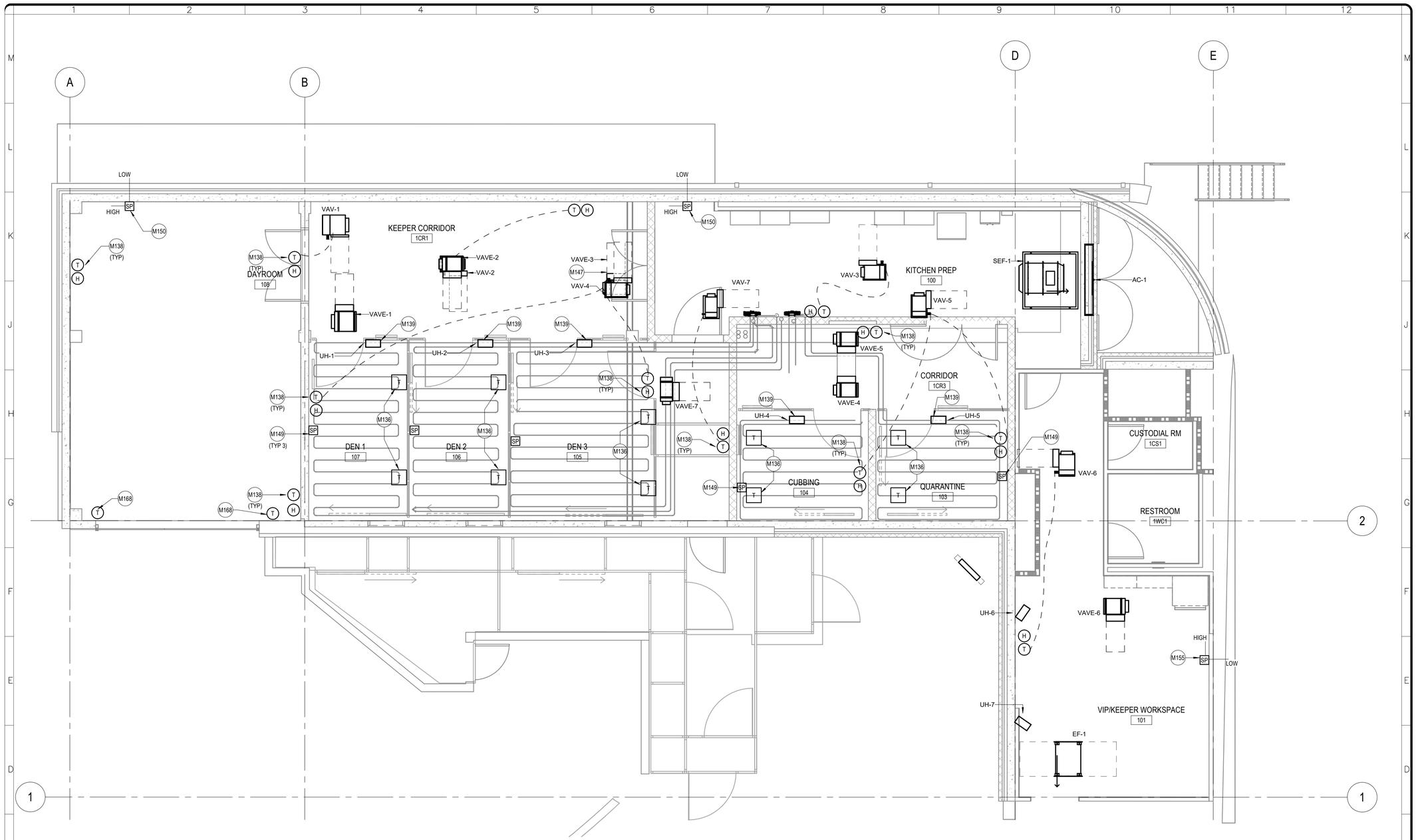


1 MI-101CP LEVEL 1 MECHANICAL CONTROL DEVICE PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. COORDINATE FINAL THERMOSTAT LOCATIONS WITH ARCHITECTURE.
B. COORDINATE THERMOSTAT CONDUIT, WHERE EXPOSED, TO MATCH ADJACENT COLUMNS OR WALLS.
C. MOUNT TEMPERATURE/HUMIDITY SENSORS WITHIN ANIMAL OCCUPIED SPACES AT AN ELEVATION OF 6FT OR ABOVE.
D. CONTRACTOR SHALL COORDINATE EQUIPMENT TAGS WITH THE SMITHSONIAN TO ENSURE FORMATTING IS CONSISTENT WITH THE SMITHSONIAN SERVER. CONTRACTOR SHALL INCLUDE BOTH THE SERVER TAG AND THE PHYSICAL TAG IN THE BAS AS REFERENCE INFORMATION.

## SHEET KEYNOTES

M136 INSTALL TEMPERATURE SENSORS AS EMBEDDED IN THE CONCRETE SLAB. SEE UNDERFLOOR PIPING FOR RADIANT PIPING DETAIL ON SHEET M-505DT. THE TEMPERATURE SLAB SENSORS TO CONTROL PUMP, MIXING VALVE, AND ISOLATION VALVE.

M138 PROVIDE CUSTOM FABRICATED WELDED STEEL WIRE CAGE ENCLOSURE AROUND ALL EXPOSED SENSORS WITHIN THE SPACE TO PROTECT EQUIPMENT FROM DAMAGE DUE TO ANIMAL. PROVIDE REMOVABLE OR HINGED ACCESS PANEL AS REQUIRED FOR MAINTENANCE. CAGE MATERIAL SHALL NOT INTERFERE WITH OPERATION OF SENSING ELEMENT. NO SENSOR/TRANSMITTER/ACCESSORY THAT RELY ON ELECTROMAGNETIC/RADIO WAVES IS PERMITTED.

M139 PROVIDE CUSTOM FABRICATED WELDED STEEL WIRE CAGE ENCLOSURE AROUND TEMPERATURE SENSOR TO PROTECT EQUIPMENT FROM DAMAGE DUE TO ANIMAL. CAGE SHALL BE RIGIDLY MOUNTED TO STRUCTURE AND SHALL ALLOW ADEQUATE AIRFLOW, SERVICE ACCESS, AND CLEARANCES REQUIRED FOR PROPER OPERATION. CAGE DIMENSIONS AND SPACING SHALL BE COORDINATED WITH AND INSTALLED PER MANUFACTURER'S RECOMMENDED CLEARANCES AND APPROVED SUBMITTALS. PROVIDE REMOVABLE OR HINGED ACCESS PANEL AS REQUIRED FOR MAINTENANCE.

M147 INSTALL HUMIDITY SENSOR IN DUCTWORK DOWNSTREAM OF EXHAUST TERMINAL UNIT. SEE MECHANICAL CONTROLS FOR ADDITIONAL INFORMATION.

M149 ROUTE ONE END OF PRESSURE TUBING TO BUILDING EXTERIOR. ENSURE THAT THE TUBING AND TRANSDUCER ARE MOUNTED ON SOLID WALL. LOCATE THE OTHER PRESSURE TUBING PORT AS SHOWN IN OPEN CEILING OF THE DEN. ENSURE THAT THE PRESSURE SENSING ELEMENT IS MOUNTED AT LEAST 5 FEET AWAY FROM AN OPERABLE OPENING. CONTRACTOR SHALL FIELD COORDINATE INSTALLATION.

M150 BUILDING PRESSURIZATION SYSTEM. ROUTE ONE END OF PRESSURE TUBING TO BUILDING EXTERIOR. ENSURE THAT THE TUBING AND TRANSDUCER ARE MOUNTED ON SOLID WALL. LOCATE THE OTHER PRESSURE TUBING PORT AS SHOWN IN OPEN CEILING OF THE DAYROOM/KITCHEN PREP AS SHOWN. ENSURE THAT THE PRESSURE SENSING ELEMENT IS MOUNTED AT LEAST 5 FEET AWAY FROM AN OPERABLE OPENING. CONTRACTOR SHALL FIELD COORDINATE INSTALLATION.

M155 BUILDING PRESSURIZATION SYSTEM. ROUTE ONE END OF PRESSURE TUBING TO BUILDING EXTERIOR. ENSURE THAT THE TUBING AND TRANSDUCER ARE MOUNTED ON SOLID WALL. LOCATE THE OTHER PRESSURE TUBING PORT WITHIN OCCUPIED SPACE (DO NOT TERMINATE OVER THE CEILING GRID). ENSURE THAT THE PRESSURE SENSING ELEMENT IS MOUNTED AT LEAST 5 FEET AWAY FROM AN OPERABLE OPENING. CONTRACTOR SHALL FIELD COORDINATE INSTALLATION.

M168 MULLION TEMPERATURE SENSOR. LOCATE SENSOR CLOSE TO GLASS MULLION MOUNTED AT FLOOR ELEVATION.

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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TRUE NORTH
PROJECT NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|


Smithsonian Institution
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520


| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - LEVEL 1 CONTROL DEVICE PLAN |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 260 OF 378

MI-101CP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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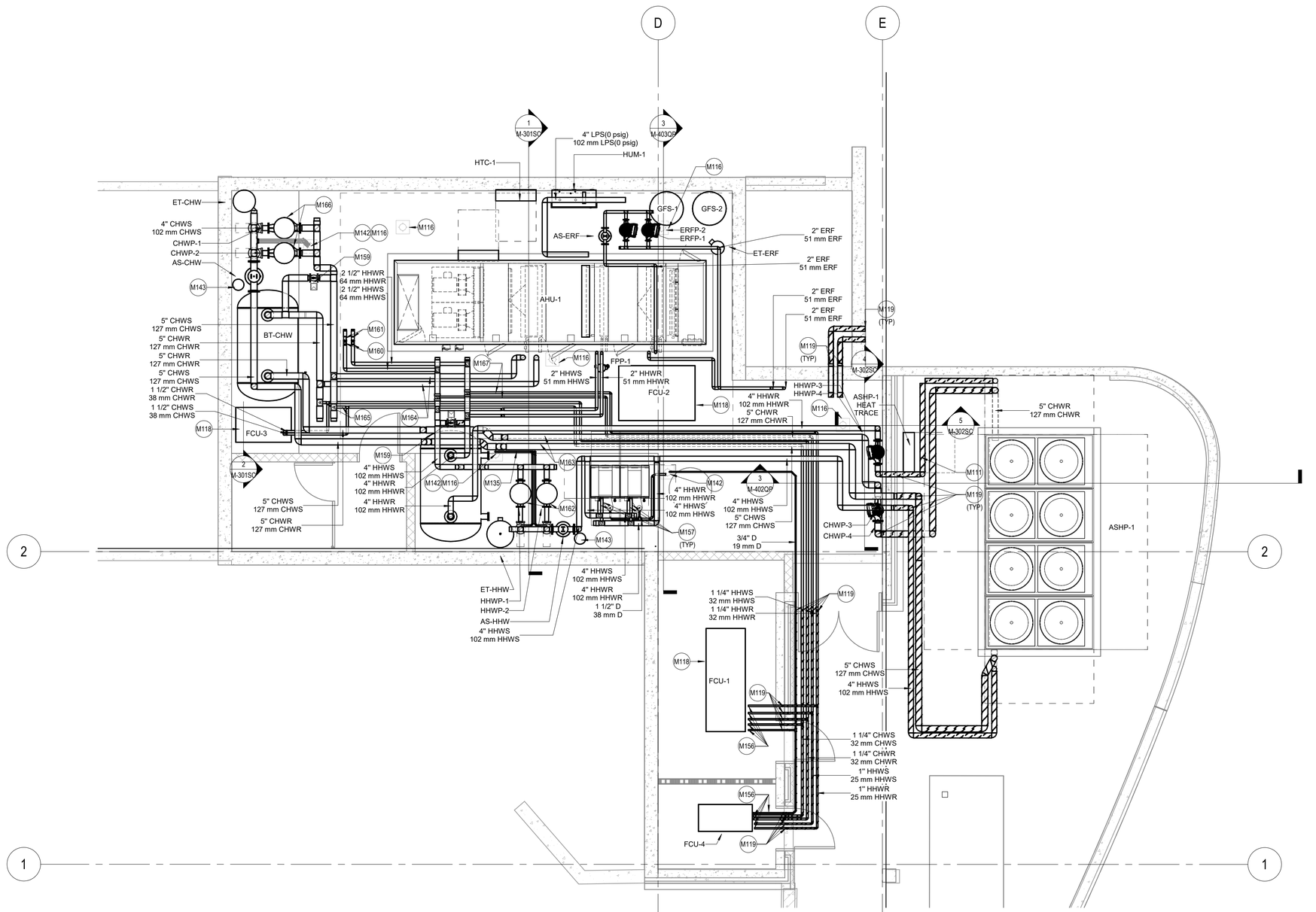
MI-101CP BY Approved 06/15/2026 65% CONSTRUCTION DOCUMENTS




1 MP-100PP BASEMENT - MECHANICAL PIPING PLAN

SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ENSURE ALL TERMINAL UNIT ISOLATION VALVES ARE INSTALLED WITHIN 2'X2' MAINTENANCE ACCESS.
B. PROVIDE ISOLATION VALVES FOR ALL EQUIPMENT AND PIPING TAKEOFFS.
C. REFER TO PLUMBING DRAWINGS ALONG WITH MECHANICAL DRAWINGS FOR CONDENSATE PIPING.
D. ALL CHILLED WATER AND HEATING HOT WATER SHALL BE MINIMUM 0.75 INCH DIAMETER, UNLESS OTHERWISE EXPLICITLY NOTED ON THE PLANS.
E. IRRESPECTIVE OF PIPE MATERIAL, ALL PIPE ACCESSORIES INCLUDING BUT NOT LIMITED TO STRAINERS, FLEXIBLE PIPES AND COIL CONNECTION KITS SHALL BE A MINIMUM OF 0.75 INCH DIAMETER.
F. PROVIDE REDUCERS AS NEEDED FOR FINAL CONNECTION TO TERMINAL UNIT OR EQUIPMENT.
G. INSTALL PIPING AND EQUIPMENT ACCESSORIES ACCORDING TO MECHANICAL DETAILS.
H. ALL EXTERIOR PIPING, VALVES AND ACCESSORIES SHALL BE HEAT TRACED AND INSTALLED IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS.

## SHEET KEYNOTES

M111 MANUFACTURER FURNISHED HEAT TRACE CONTROL PANEL FOR AIR SOURCE HEAT PUMP. THE HEAT TRACE PANEL MAY BE INTEGRAL TO THE UNIT. COORDINATE LOCATION OF PANEL WITH PURCHASED PRODUCT. COORDINATE WITH ELECTRICAL CONTRACTOR ON EXACT LOCATION.

M116 UTILITY/FIXTURE/EQUIPMENT BY DIVISION 22. SEE PLUMBING DOCUMENTS FOR MORE INFORMATION.

M118 PROVIDE 1IN CHILLED WATER SUPPLY AND RETURN PIPE FOR SHOWN FAN COIL UNIT. PROVIDE CONDENSATE PIPING TO NEAREST DRAIN. PIPING SHALL BE SLOPED TOWARDS THE DRAIN AT 1% OR MORE.

M119 DASHED LINE PATTERN ON A PIPE INDICATES HEAT TRACED PIPING.

M135 COORDINATE DRAIN PIPE ROUTING TO PREVENT TRIP HAZARD.

M142 REFERENCE PLUMBING DRAWINGS FOR DRAIN LOCATION. TERMINATE OVER DRAIN PER MECHANICAL DRAWINGS AND SPECIFICATIONS.

M143 CHEMICAL SHOT FEEDER

M156 1.25 INCH (32 MM) CHILLED WATER SUPPLY/RETURN PIPE TO FAN COIL. 1 INCH (26 MM) HEATING HOT WATER SUPPLY/RETURN PIPE TO SERVICE FAN COIL UNIT. 0.75 INCH (19 MM) CONDENSATE PIPING FROM FAN COIL UNIT.

M157 2 INCH (51MM) HEATING HOT WATER SUPPLY/RETURN PIPE TO BOILER B-1 AND B-2. SHOWN PIPING LAYOUT CORRESPONDS TO BASIS OF DESIGN PRODUCT. COORDINATE EXACT LOCATION AND FINAL FITTINGS WITH PURCHASED PRODUCT. PIPING ARRANGEMENT SHALL STILL COMPLY WITH DESIGN DOCUMENTS - REVERSE RETURN.

M159 PROVIDE HIGH PERFORMANCE SEGMENTED BALL VALVE WITH A TURNDOWN RATIO OF 300:1; BASIS OF DESIGN: VSI, SERIES V. PROVIDE VALVE WITH AN ELECTRIC ACTUATOR WITH A RESOLUTION OF 200 STEPS, 0.04VDC/0.5 DEGREES. BASIS OF DESIGN: VSI, SERIES 1000-X

M160 1.5 INCH (38 MM) HEATING HOT WATER SUPPLY AND RETURN PIPING SERVICING THE RADIANT SLAB LOOPS. PROVIDE ISOLATION VALVE IN MECHANICAL ROOM AT INDICATED LOCATION. SEE MP-101PP AND MP-102PP FOR CONTINUATION.

M161 2.5 INCH (64 MM) HEATING HOT WATER SUPPLY AND RETURN PIPING SERVICING THE REHEAT COILS, UNIT HEATERS, DAY ROOM HEATERS AND AIR CURTAINS. PROVIDE ISOLATION VALVE IN MECHANICAL ROOM AT INDICATED LOCATION. SEE MP-101PP AND MP-102PP FOR CONTINUATION.

M162 PIPING TO HHWP-1 AND HHWP-2. SEE MECHANICAL SCHEDULES FOR ADDITIONAL INFORMATION. SUCTION DIFFUSERS AND PUMPS TO BE ISOLATED FROM BUILDING STRUCTURE/SLAB WITH SPECIFIED VIBRATION ISOLATION. PROVIDE 8INCH (76MM) HEATING HOT WATER SUPPLY PIPE CONNECTION TO PUMP SUCTION FROM SUCTION DIFFUSER. PROVIDE 4 INCH (102MM) HEATING HOT WATER SUPPLY PIPE CONNECTION OUT OF PUMP DISCHARGE. PROVIDE REDUCERS/INCREASERS AS NEEDED.

M163 5 INCH (127 MM) CHILLED WATER SUPPLY AND RETURN PIPE.

M164 4 INCH (102 MM) CHILLED WATER SUPPLY AND RETURN PIPE TO/FROM CHILLED WATER COIL WITHIN AHU-1.

M165 1.5 INCH (38 MM) CHILLED WATER SUPPLY AND RETURN PIPE TO/FROM CHILLED WATER COILS WITHIN FOUR (4) FAN COIL UNITS.

M166 PIPING TO CHWP-1 AND CHWP-2. SEE MECHANICAL SCHEDULES FOR ADDITIONAL INFORMATION. SUCTION DIFFUSERS AND PUMPS TO BE ISOLATED FROM BUILDING STRUCTURE/SLAB WITH SPECIFIED VIBRATION ISOLATION. PROVIDE 4INCH (76MM) CHILLED WATER SUPPLY PIPE CONNECTION TO PUMP SUCTION FROM SUCTION DIFFUSER. PROVIDE 5 INCH (102MM) CHILLED WATER SUPPLY PIPE CONNECTION OUT OF PUMP DISCHARGE. PROVIDE REDUCERS/INCREASERS AS NEEDED.

M167 1.25 INCH (32 MM) HEATING HOT WATER SUPPLY AND RETURN PIPE TO/FROM FCU-1, FCU-2 AND FCU-4.

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PROJECT NORTH

KEY PLAN

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

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|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

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600 Maryland Avenue S.W. Suite 5001
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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BASEMENT MECHANICAL PIPING PLAN |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 261 OF 378

MP-100PP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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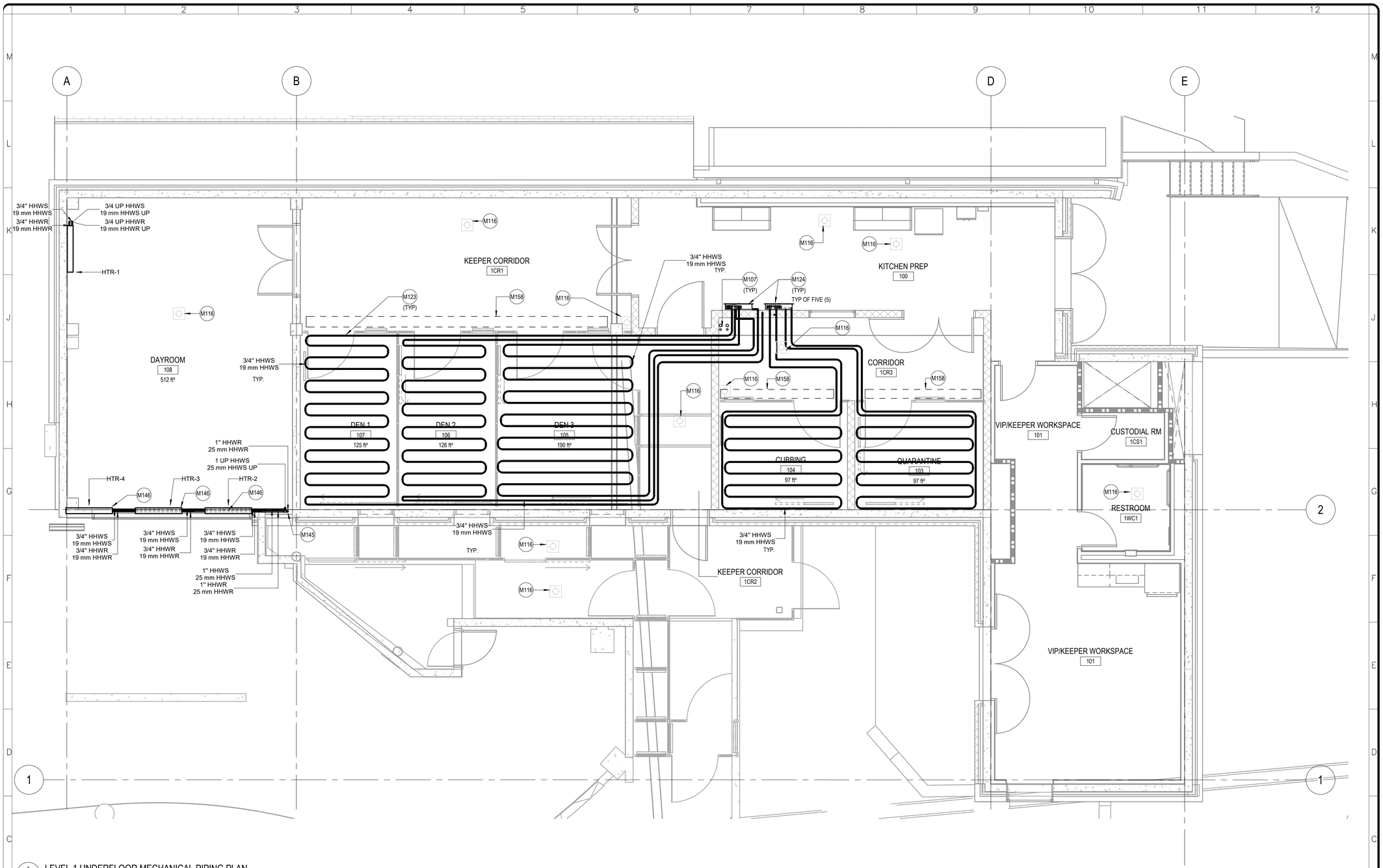
MP-100PP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




1 LEVEL 1 UNDERFLOOR MECHANICAL PIPING PLAN
MP-101PP SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ENSURE ALL TERMINAL UNIT ISOLATION VALVES ARE INSTALLED WITHIN 2'X2' MAINTENANCE ACCESS.
B. PROVIDE ISOLATION VALVES FOR ALL EQUIPMENT AND PIPING TAKEOFFS.
C. REFER TO PLUMBING DRAWINGS ALONG WITH MECHANICAL DRAWINGS FOR CONDENSATE PIPING.
D. ALL CHILLED WATER AND HEATING HOT WATER SHALL BE MINIMUM 0.75 INCH DIAMETER, UNLESS OTHERWISE EXPLICITLY NOTED ON THE PLANS.
E. IRRESPECTIVE OF PIPE MATERIAL, ALL PIPE ACCESSORIES INCLUDING BUT NOT LIMITED TO STRAINERS, FLEXIBLE PIPES AND COIL CONNECTION KITS SHALL BE A MINIMUM OF 0.75 INCH DIAMETER.
F. PROVIDE REDUCERS AS NEEDED FOR FINAL CONNECTION TO TERMINAL UNIT OR EQUIPMENT.
G. INSTALL PIPING AND EQUIPMENT ACCESSORIES ACCORDING TO MECHANICAL DETAILS.
H. ALL EXTERIOR PIPING, VALVES AND ACCESSORIES SHALL BE HEAT TRACED AND INSTALLED IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS.

## ◯ SHEET KEYNOTES

M107 RADIANT PIPING SHOWN FOR DESIGN INTENT. CONTRACTOR SHALL COORDINATE WITH THE MANUFACTURER AND PROVIDE A COORDINATED LAYOUT.
M116 UTILITY/FIXTURE/EQUIPMENT BY DIVISION 22. SEE PLUMBING DOCUMENTS FOR MORE INFORMATION.
M123 SEE M-505DT FOR ADDITIONAL INFORMATION ON SLAB SENSOR INSTALLATION AND SYSTEM LAYOUT CONSTRUCTION/SCOPE.
M124 EACH RADIANT MANIFOLD SYSTEM SHALL CONSIST OF A PUMP, THREE WAY FLOW CONTROL MODULATING VALVE AND ACCESSORIES AS SHOWN ON M-505DT. IT IS ACCEPTABLE TO CONSOLIDATE ALL ACCESSORIES IN A SINGULAR PANEL IF PERFORMANCE CRITERIA IS MET AND IF COST EFFECTIVE.
M145 SEE SHEET M101PP FOR CONTINUATION
M146 TYP. FLOOR CONVECTOR. SEE MECHANICAL SCHEDULES FOR MORE INFORMATION ON SELECTION AND CAGED ENCLOSURE AROUND EACH INSTANCE. ANGLE THE DISCHARGE BLADE TOWARDS THE GLAZING
M158 SEE DRAWINGS ISSUED BY OTHER DISCIPLINES. COORDINATE RADIANT PEX PIPING WITH TRENCH DRAIN.

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KEY PLAN

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Smithsonian Institution

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600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

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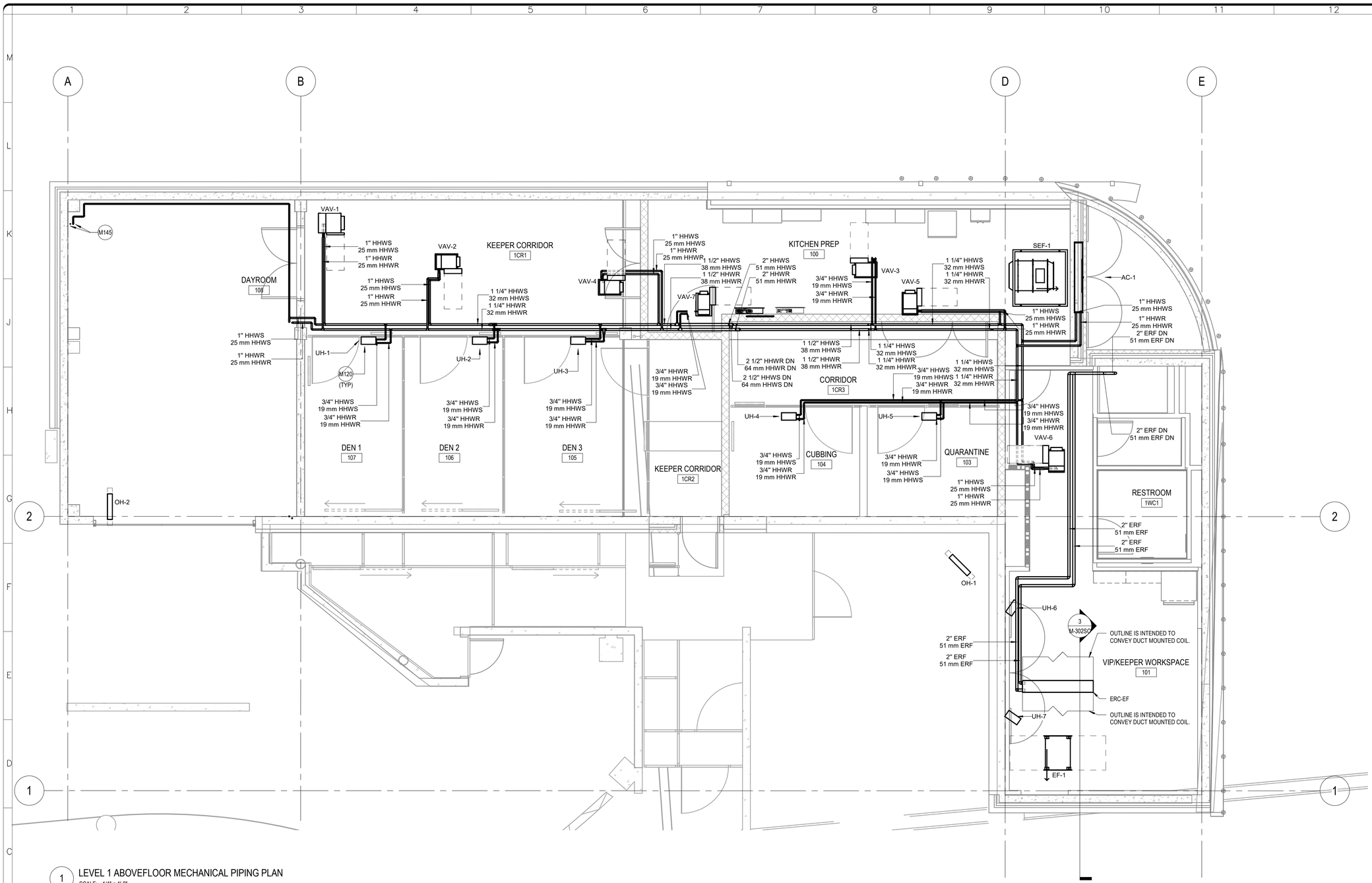
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LEVEL 1 UNDERFLOOR MECHANICAL PIPING PLAN |
| WORKING STAFF | OP (DESIGNED BY) / YW (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 262 OF 378

MP-101PP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE




1 LEVEL 1 ABOVEFLOOR MECHANICAL PIPING PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ENSURE ALL TERMINAL UNIT ISOLATION VALVES ARE INSTALLED WITHIN 2'X2' MAINTENANCE ACCESS.
B. PROVIDE ISOLATION VALVES FOR ALL EQUIPMENT AND PIPING TAKEOFFS.
C. REFER TO PLUMBING DRAWINGS ALONG WITH MECHANICAL DRAWINGS FOR CONDENSATE PIPING.
D. ALL CHILLED WATER AND HEATING HOT WATER SHALL BE MINIMUM 0.75 INCH DIAMETER, UNLESS OTHERWISE EXPLICITLY NOTED ON THE PLANS.
E. IRRESPECTIVE OF PIPE MATERIAL, ALL PIPE ACCESSORIES INCLUDING BUT NOT LIMITED TO STRAINERS, FLEXIBLE PIPES AND COIL CONNECTION KITS SHALL BE A MINIMUM OF 0.75 INCH DIAMETER.
F. PROVIDE REDUCERS AS NEEDED FOR FINAL CONNECTION TO TERMINAL UNIT OR EQUIPMENT.
G. INSTALL PIPING AND EQUIPMENT ACCESSORIES ACCORDING TO MECHANICAL DETAILS.
H. ALL EXTERIOR PIPING, VALVES AND ACCESSORIES SHALL BE HEAT TRACED AND INSTALLED IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS.

## SHEET KEYNOTES

M120 PROVIDE A CAGE ENCLOSURE AROUND THE UNIT HEATER TO PREVENT DAMAGE TO EQUIPMENT FROM ANIMAL.
M145 SEE SHEET M101PP FOR CONTINUATION

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TRUE NORTH
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KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

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| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

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600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

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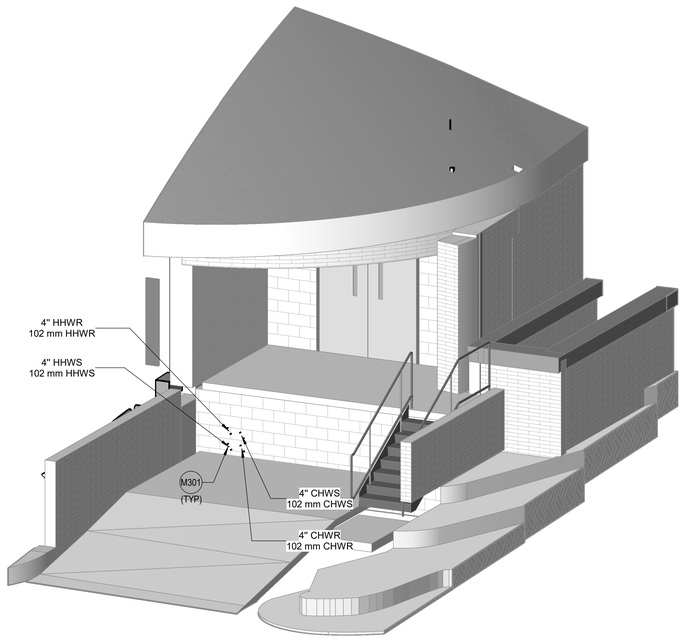
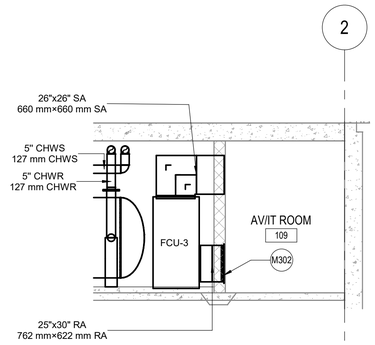
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LEVEL 1 ABOVE FLOOR MECHANICAL PIPING PLAN |
| WORKING STAFF | OP (DESIGNED BY) / YW (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 263 OF 378

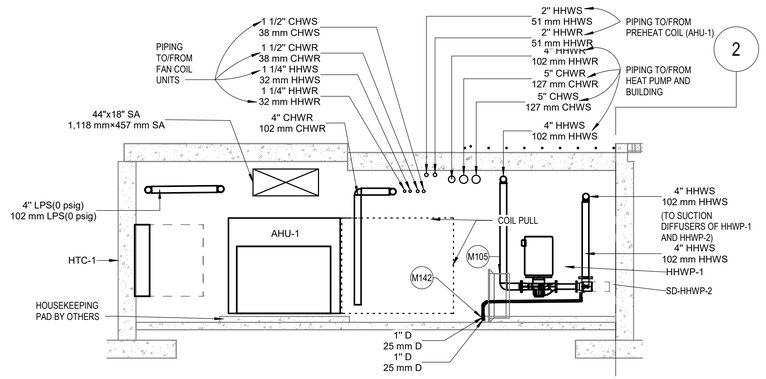
MP-102PP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE




4 M-301SC 3D - LOADING DOCK TIE IN SPOTS
SCALE: NOT TO SCALE




2 M-301SC VERTICAL FAN COIL UNIT FOR THE AV/IT ROOM
SCALE: 1/4" = 1'-0"




1 M-301SC AHU COIL PULL SECTION VIEW
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ALL GENERAL EXHAUST DUCTS TERMINATING IN EXPOSED AREAS SHALL TERMINATE IN EXHAUST GRILLE TO BE COORDINATED WITH ARCHITECTURE.
B. ALL DUCT DIMENSIONS SHOWN ON PLAN ARE CLEAR INTERIOR DIMENSIONS. CONTRACTOR TO COORDINATE EXTERIOR DIMENSIONS.
C. PROVIDE NO MEDIA SOUND ATTENUATORS FOR TERMINAL UNITS CALLED OUT IN SILENCER SCHEDULE. SILENCER MAY BE INTEGRAL TO TERMINAL UNIT.
D. COORDINATE ALL DIFFUSERS, REGISTERS, AND GRILLES WITH REFLECTED CEILING PLAN.
E. ALL LIFE SAFETY DAMPERS/ASSEMBLIES SHOWN ON PLANS SHALL CONTAIN RATINGS THAT ARE COORDINATED WITH THE RESPECTIVE WALL/SLAB RATING. COORDINATE WITH ARCHITECTURAL OR LIFE SAFETY PLANS FOR WALLS/SLAB RATING.
F. ALL DUCT PENETRATIONS THROUGH 1-HOUR RATED PARTITIONS SHALL BE FIRE STOPPED PER UL DETAILS TO ACHIEVE 1-HR RATING.
G. DUCT SIZE TO AIR TERMINAL SHALL BE SAME AS AIR TERMINAL INLET SIZE UNLESS OTHERWISE NOTED.
H. PROVIDE BALANCING DAMPER AT EACH DIFFUSER, GRILLE, AND BRANCH TAP-OFF IN ALL SUPPLY, RETURN AND EXHAUST DUCTWORK. LOCATE BALANCING DAMPER AS CLOSE TO BRANCH TAKE-OFF AS POSSIBLE.
I. PROVIDE CABLE OPERATED REMOTE DAMPERS TO ALL DIFFUSERS LOCATED IN DRY WALL CEILING. COORDINATE LOCATION OF REMOTE DAMPER ADJUSTMENT PANEL WITH ALL DISCIPLINES' CEILING ELEMENTS.
J. USE OF FLEXIBLE DUCTWORK IN EXHAUST AIR SYSTEM IS NOT PERMITTED.
K. USE OF FLEXIBLE DUCTWORK TO CONNECT AN AIR TERMINAL LOCATED ABOVE AN INACCESSIBLE CEILING IS NOT PERMITTED.

## SHEET KEYNOTES

M105 DIVISION 26 EQUIPMENT/FIXTURE/CONDUCTOR ONLY SHOWN FOR COORDINATION. REFER TO DIVISION 26 DOCUMENT FOR EXACT LOCATION/DIMENSION.
M142 REFERENCE PLUMBING DRAWINGS FOR DRAIN LOCATION. TERMINATE OVER DRAIN PER MECHANICAL DRAWINGS AND SPECIFICATIONS.
M301 PROVIDE ENDCAP FOR EMERGENCY CHILLER AND BOILER HOOKUP. SEE SYSTEM DIAGRAM FOR ADDITIONAL INFORMATION.
M302 TERMINATE IN GRILLE TO MATCH DUCT SIZE.

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M-301SC BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

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## CONSTRUCT ARABIAN LEOPARD FACILITY

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DATE: 06/15/2026
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REVISION # DATE
REVISION DESCRIPTION

Smithsonian Institution
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL SECTIONS - BASEMENT MECHANICAL ROOM
WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 264 OF 378
M-301SC (DISCIPLINE M | TYPE 3 | SEQUENCE 01 | DRAWING TYPE SC)

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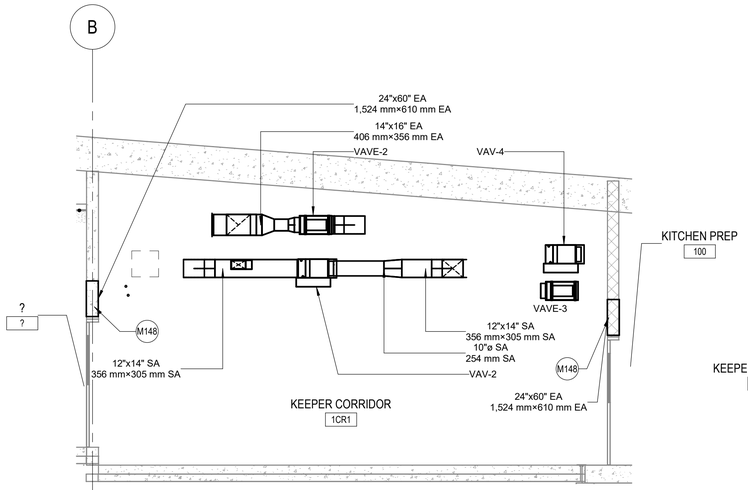
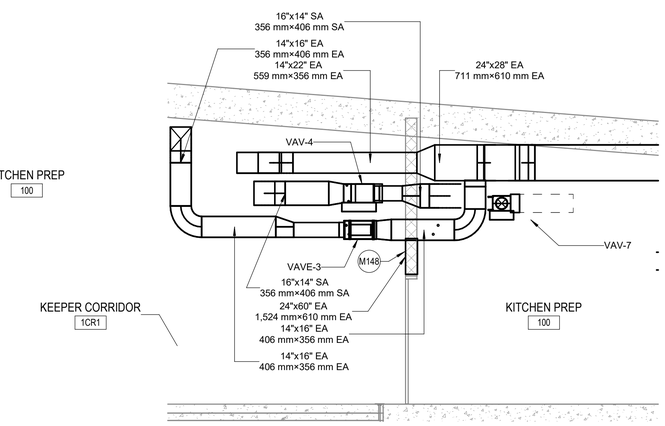
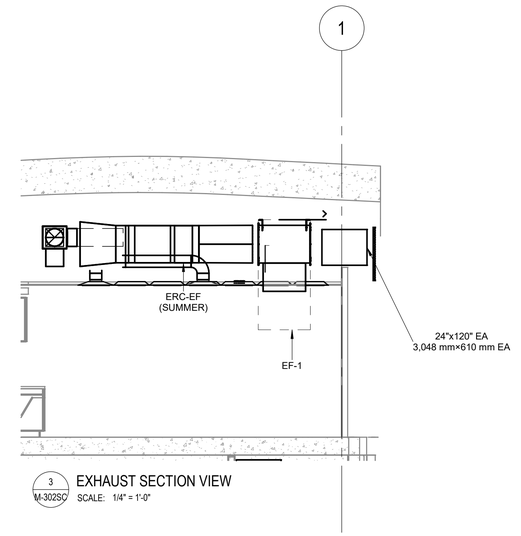



1 M-302SC STACKED VAV-2 AND VAVE-2
SCALE: 1/4" = 1'-0"




2 M-302SC DUCT SECTION VIEW - 2
SCALE: 1/4" = 1'-0"




3 M-302SC EXHAUST SECTION VIEW
SCALE: 1/4" = 1'-0"




4 M-302SC PRIMARY PUMPS
SCALE: 1/4" = 1'-0"




5 M-302SC INTAKE AND SUPPLY AIR DUCTWORK
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ALL GENERAL EXHAUST DUCTS TERMINATING IN EXPOSED AREAS SHALL TERMINATE IN EXHAUST GRILLE TO BE COORDINATED WITH ARCHITECTURE.
B. ALL DUCT DIMENSIONS SHOWN ON PLAN ARE CLEAR INTERIOR DIMENSIONS. CONTRACTOR TO COORDINATE EXTERIOR DIMENSIONS.
C. PROVIDE NO MEDIA SOUND ATTENUATORS FOR TERMINAL UNITS CALLED OUT IN SILENCER SCHEDULE. SILENCER MAY BE INTEGRAL TO TERMINAL UNIT.
D. COORDINATE ALL DIFFUSERS, REGISTERS, AND GRILLES WITH REFLECTED CEILING PLAN.
E. ALL LIFE SAFETY DAMPERS/ASSEMBLIES SHOWN ON PLANS SHALL CONTAIN RATINGS THAT ARE COORDINATED WITH THE RESPECTIVE WALL/SLAB RATING. COORDINATE WITH ARCHITECTURAL OR LIFE SAFETY PLANS FOR WALL/SLAB RATING.
F. ALL DUCT PENETRATIONS THROUGH 1-HOUR RATED PARTITIONS SHALL BE FIRE STOPPED PER UL DETAILS TO ACHIEVE 1-HR RATING.
G. DUCT SIZE TO AIR TERMINAL SHALL BE SAME AS AIR TERMINAL INLET SIZE UNLESS OTHERWISE NOTED.
H. PROVIDE BALANCING DAMPER AT EACH DIFFUSER, GRILLE, AND BRANCH TAP-OFF IN ALL SUPPLY, RETURN AND EXHAUST DUCTWORK. LOCATE BALANCING DAMPER AS CLOSE TO BRANCH TAKE-OFF AS POSSIBLE.
I. PROVIDE CABLE OPERATED REMOTE DAMPERS TO ALL DIFFUSERS LOCATED IN DRY WALL CEILING. COORDINATE LOCATION OF REMOTE DAMPER ADJUSTMENT PANEL WITH ALL DISCIPLINES' CEILING ELEMENTS.
J. USE OF FLEXIBLE DUCTWORK IN EXHAUST AIR SYSTEM IS NOT PERMITTED.
K. USE OF FLEXIBLE DUCTWORK TO CONNECT AN AIR TERMINAL LOCATED ABOVE AN INACCESSIBLE CEILING IS NOT PERMITTED.

## SHEET KEYNOTES

M108 SEE ARCHITECTURAL DOCUMENTS FOR MORE INFORMATION ON LOUVER/FINISH. PROVIDE AT LEAST 20SF IN FREE AREA, ACROSS THE ENTIRE INTAKE ASSEMBLY. INTAKE ASSEMBLY CONSISTS OF LOUVER, AND/OR DECORATIVE SCREENS, MESHES. PURCHASED PRODUCT MUST DOCUMENT THIS INFORMATION.

M130 FIRE SMOKE DAMPER; BASIS OF DESIGN POTTORFF FD-350 AIRFOIL BLADE. DAMPER SHALL COMPLY WITH AIR DUCT ACCESSORIES SPECIFICATION SECTION AND SHALL BE DYNAMIC RATED FOR 3HR.

M148 TRANSFER OPENING FOR SMOKE EXHAUST SYSTEM. SEE SHEET M-822CP FOR MORE INFORMATION.

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M-302SC BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

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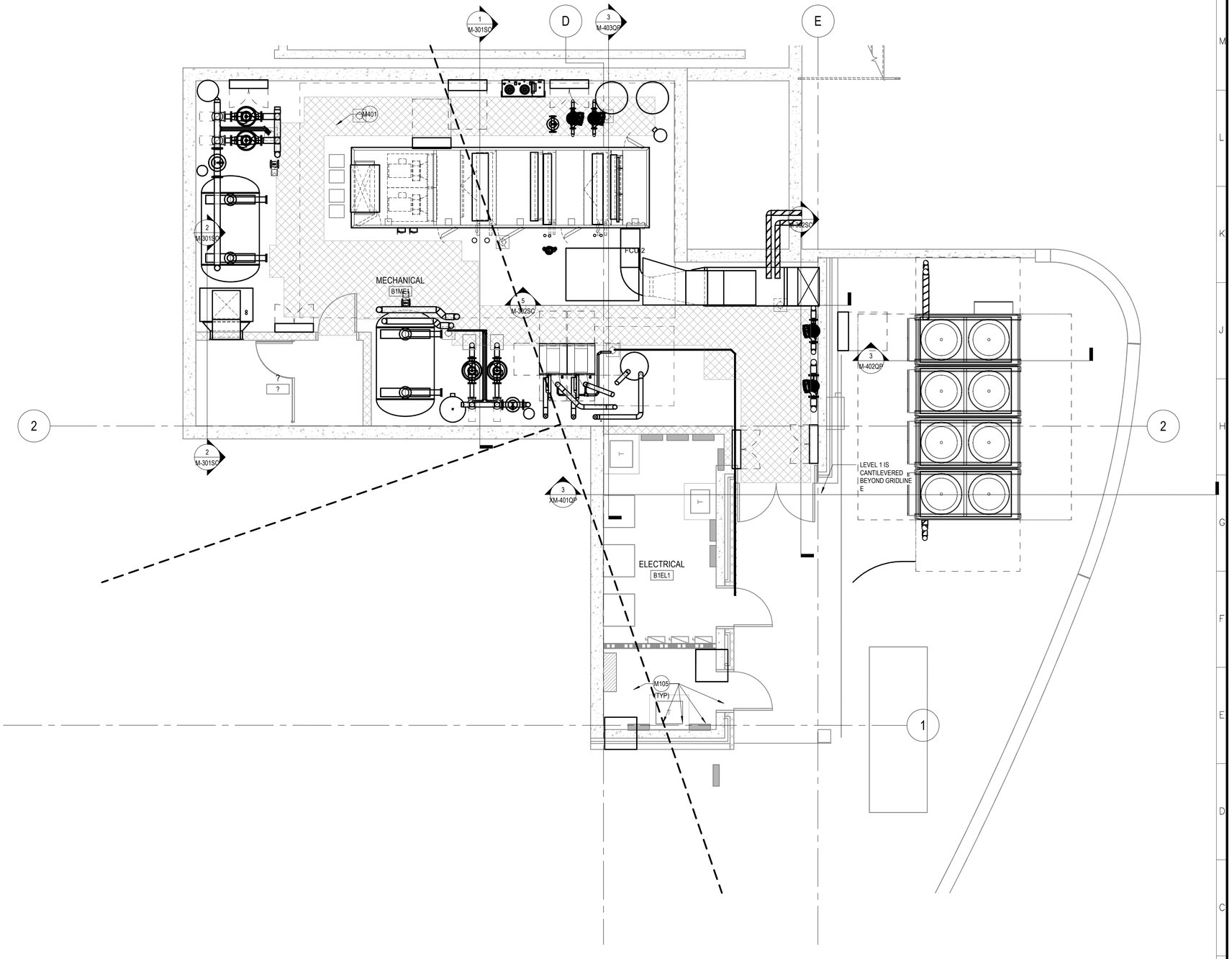
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL SECTIONS |
| WORKING STAFF | OP (DESIGNED BY) / YW (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 265 OF 378

M-302SC
DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE




1 M-401QP BASEMENT MECHANICAL ROOM WALKING ACCESS
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. ALL GENERAL EXHAUST DUCTS TERMINATING IN EXPOSED AREAS SHALL TERMINATE IN EXHAUST GRILLE TO BE COORDINATED WITH ARCHITECTURE.
B. ALL DUCT DIMENSIONS SHOWN ON PLAN ARE CLEAR INTERIOR DIMENSIONS. CONTRACTOR TO COORDINATE EXTERIOR DIMENSIONS.
C. PROVIDE NO MEDIA SOUND ATTENUATORS FOR TERMINAL UNITS CALLED OUT IN SILENCER SCHEDULE. SILENCER MAY BE INTEGRAL TO TERMINAL UNIT.
D. COORDINATE ALL DIFFUSERS, REGISTERS, AND GRILLES WITH REFLECTED CEILING PLAN.
E. ALL LIFE SAFETY DAMPERS/ASSEMBLIES SHOWN ON PLANS SHALL CONTAIN RATINGS THAT ARE COORDINATED WITH THE RESPECTIVE WALL/SLAB RATING. COORDINATE WITH ARCHITECTURAL OR LIFE SAFETY PLANS FOR WALL/SLAB RATING.
F. ALL DUCT PENETRATIONS THROUGH 1-HOUR RATED PARTITIONS SHALL BE FIRE STOPPED PER UL DETAILS TO ACHIEVE 1-HR RATING.
G. DUCT SIZE TO AIR TERMINAL SHALL BE SAME AS AIR TERMINAL INLET SIZE UNLESS OTHERWISE NOTED.
H. PROVIDE BALANCING DAMPER AT EACH DIFFUSER, GRILLE, AND BRANCH TAP-OFF IN ALL SUPPLY, RETURN AND EXHAUST DUCTWORK. LOCATE BALANCING DAMPER AS CLOSE TO BRANCH TAKE-OFF AS POSSIBLE.
I. PROVIDE CABLE OPERATED REMOTE DAMPERS TO ALL DIFFUSERS LOCATED IN DRY WALL CEILING. COORDINATE LOCATION OF REMOTE DAMPER ADJUSTMENT PANEL WITH ALL DISCIPLINES' CEILING ELEMENTS.
J. USE OF FLEXIBLE DUCTWORK IN EXHAUST AIR SYSTEM IS NOT PERMITTED.
K. USE OF FLEXIBLE DUCTWORK TO CONNECT AN AIR TERMINAL LOCATED ABOVE AN INACCESSIBLE CEILING IS NOT PERMITTED.

## SHEET KEYNOTES

M105 DIVISION 26 EQUIPMENT/FIXTURE/CONDUCTOR ONLY SHOWN FOR COORDINATION. REFER TO DIVISION 26 DOCUMENT FOR EXACT LOCATION/DIMENSION.
M401 HATCHED REGION INDICATE WALKING PATH.

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TRUE NORTH
PROJECT NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

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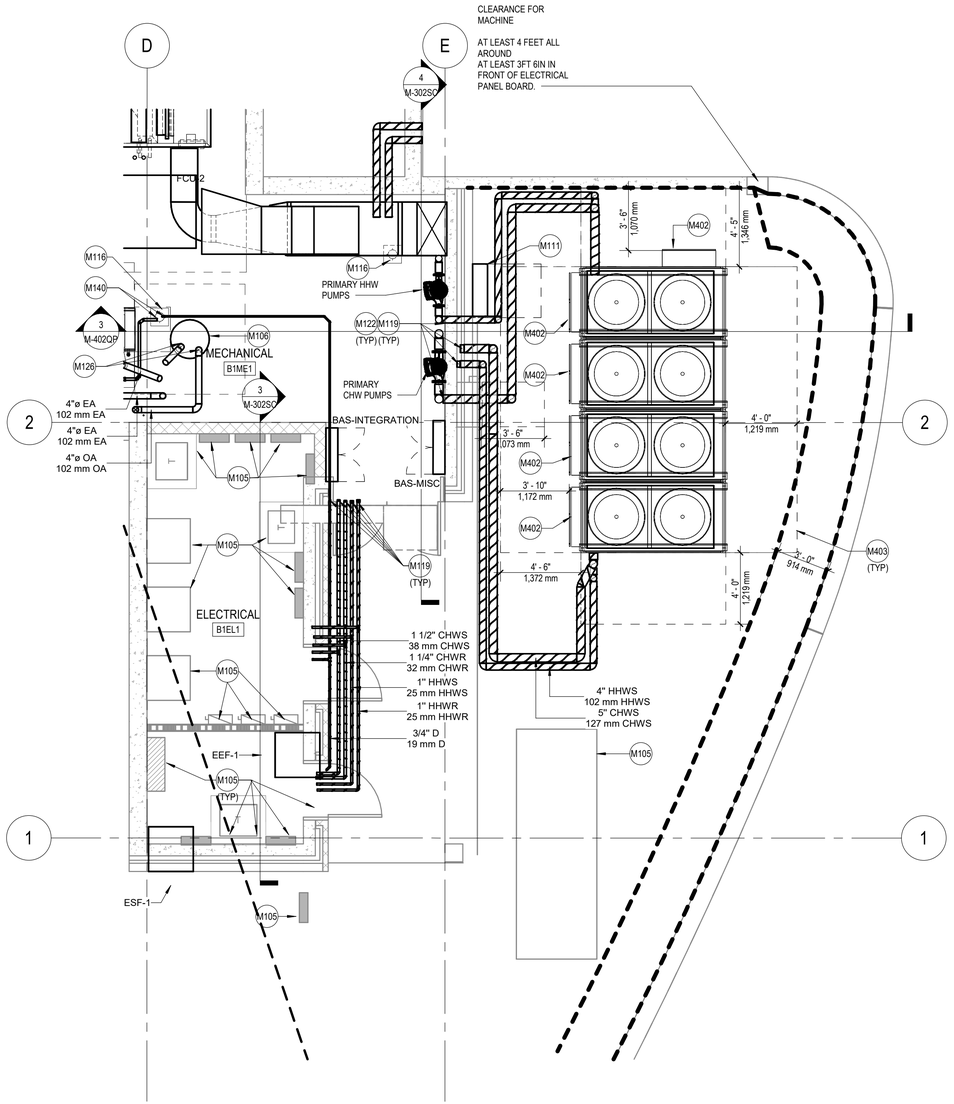
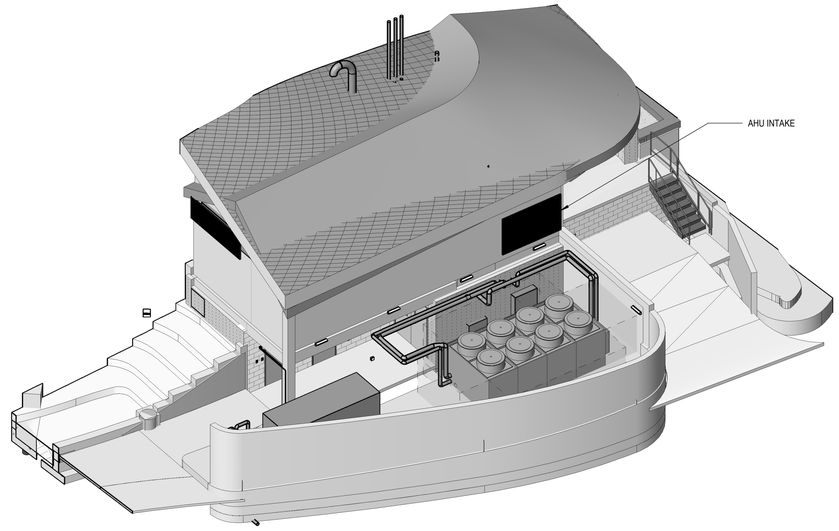
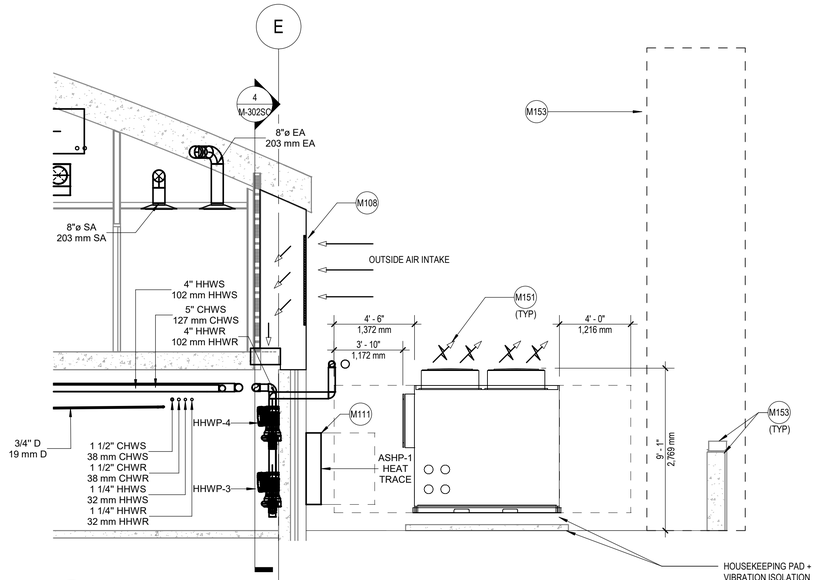
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL ROOM ACCESS WALKWAY |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |
| SHEET NO. | 266 OF 378 |
| | M-401QP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |




1 M-402QP MECHANICAL YARD (WITH BASIS OF DESIGN MACHINE)
SCALE: 1/4" = 1'-0"




2 M-402QP MECHANICAL YARD (WITH BASIS OF DESIGN MACHINE)
SCALE:




3 M-402QP INSTALLATION OF HEAT PUMP (WITH BASIS OF DESIGN MACHINE)
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. HEAT PUMP INSTALLATION MUST COMPLY WITH CONTRACT DOCUMENTS AND INSTRUCTIONS FROM SUCCESSFUL MANUFACTURER. MOST STRINGENT INSTRUCTION SHALL PREVAIL.
B. SEE MECHANICAL SCHEDULES, MECHANICAL CONTROLS AND SPECIFICATIONS FOR COMPLETE INFORMATION.
C. COORDINATE HOUSEKEEPING PAD INFORMATION WITH OTHER DISCIPLINES BASED ON DIMENSIONS OF PURCHASED HEAT PUMP ARRAY PRODUCT.

## SHEET KEYNOTES

M105 DIVISION 26 EQUIPMENT/FIXTURE/CONDUCTOR ONLY SHOWN FOR COORDINATION. REFER TO DIVISION 26 DOCUMENT FOR EXACT LOCATION/DIMENSION.
M106 SEE DOCUMENTS ISSUED BY OTHER DISCIPLINES TO VIEW EQUIPMENT AROUND THIS GENERAL REGION.
M108 SEE ARCHITECTURAL DOCUMENTS FOR MORE INFORMATION ON LOUVER/FINISH. PROVIDE AT LEAST 20SF IN FREE AREA, ACROSS THE ENTIRE INTAKE ASSEMBLY. INTAKE ASSEMBLY CONSISTS OF LOUVER, AND/OR DECORATIVE SCREENS, MESHES. PURCHASED PRODUCT MUST DOCUMENT THIS INFORMATION.
M111 MANUFACTURER FURNISHED HEAT TRACE CONTROL PANEL FOR AIR SOURCE HEAT PUMP. THE HEAT TRACE PANEL MAY BE INTEGRAL TO THE UNIT. COORDINATE LOCATION OF PANEL WITH PURCHASED PRODUCT. COORDINATE WITH ELECTRICAL CONTRACTOR ON EXACT LOCATION.
M116 UTILITY/FIXTURE/EQUIPMENT BY DIVISION 22. SEE PLUMBING DOCUMENTS FOR MORE INFORMATION.
M119 DASHED LINE PATTERN ON A PIPE INDICATES HEAT TRACED PIPING.
M122 HEAT TRACED PIPING TO AIR SOURCE HEAT PUMP. PIPING FROM MECHANICAL ROOM. SEE MP-100PP.
M126 PROVIDE A 4" (102MM) EXHAUST AND 4" (102MM) OUTSIDE AIR CONNECTION TO PLUMBING EQUIPMENT. COORDINATE WITH DIVISION 22 CONTRACTOR ON PURCHASED PRODUCT. PROVIDE END FITTINGS (REDUCERS/COUPLINGS) AS REQUIRED BY MANUFACTURER TO COMPLETE CONNECTION TO EQUIPMENT.
M140 TO DRAIN, SEE PLUMBING DRAWINGS
M151 MANUFACTURER ACCESSORY - AIRFLOW DIRECTING BAFFLES TO BE ORIENTED TO DIRECT AIRFLOW AWAY FROM OUTSIDE AIR INTAKE
M153 SEE DRAWINGS ISSUED BY OTHER DISCIPLINES FOR ADDITIONAL INFORMATION.
M402 ELECTRICAL PANEL MUST HAVE CLEAR ACCESS AS REQUIRED BY MANUFACTURER. QUOTED DIMENSION AND LOCATION OF PANEL APPLY TO BASIS OF DESIGN PRODUCT. COORDINATE WITH PURCHASED PRODUCT.
M403 WHERE MARKED CLEARANCES (EXCLUDING ELECTRICAL PANEL BOARDS) ARE NOT MET, OBTAIN AND DOCUMENT A CAPACITY DERATE FROM THE MANUFACTURER OF PURCHASED ARRAY.

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TRUE NORTH
PROJECT NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HEAT PUMP INSTALLATION AND CLEARANCE REQUIREMENTS |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |
| SHEET NO. | 267 OF 378 |
| | M-402QP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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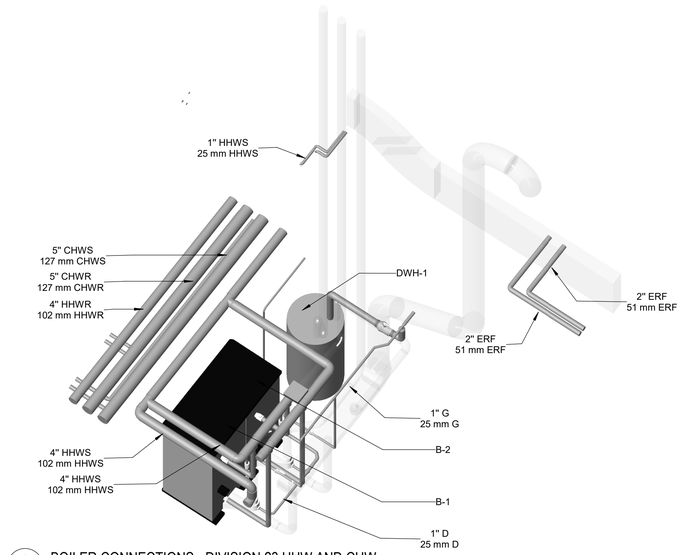
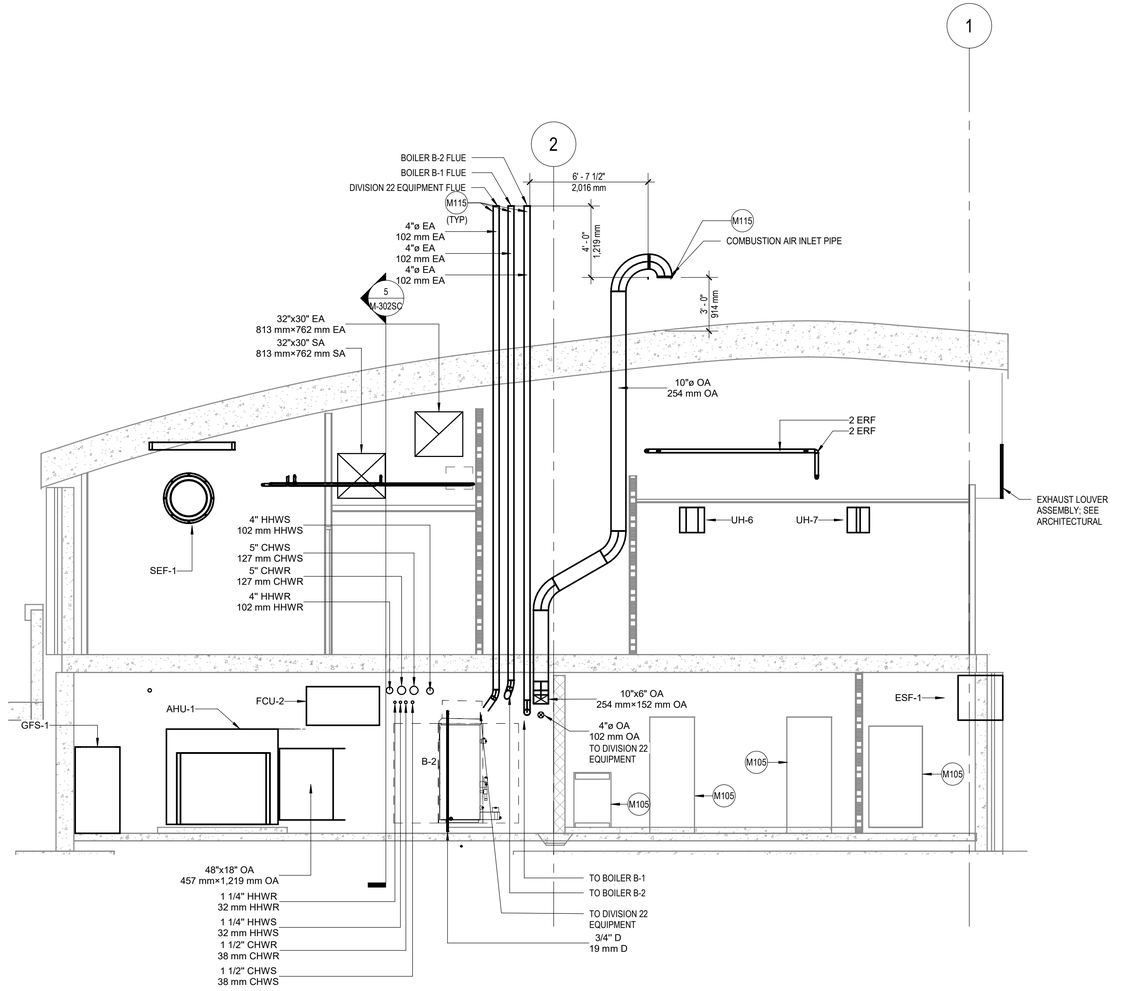



1 M-403QP BOILER CONNECTIONS - DIVISION 23 HHW AND CHW
SCALE: NOT TO SCALE




2 M-403QP BOILER CONNECTIONS - DIVISION 23 DUCTS
SCALE: NOT TO SCALE




3 M-403QP BOILER AND DIVISION 22 EQUIPMENT DUCT FLUE ROUTING
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. PROVIDE PAINTED FINISH ON ALL DUCTS EXPOSED TO EXTERIOR. PAINT COLOR SHALL BE FULLY CUSTOM AND SELECTED BY ARCHITECT.
B. PAINT MATERIAL FOR THE FLUE DUCT SHALL BE RESISTANT TO HIGH TEMPERATURE.
C. VERTICAL AND HORIZONTAL SEPARATION BETWEEN INLET AND OUTLET PIPES SHALL COMPLY WITH MANUFACTURER INSTRUCTIONS. SHOWN DIMENSIONS ON FLOOR APPLY TO BASIS OF DESIGN PRODUCT. ANY DEVIATION FROM THESE DIMENSIONS MUST BE BROUGHT TO IMMEDIATE ATTENTION OF ARCHITECT AND E.O.R.
D. NATURAL GAS PIPING IS SHOWN ONLY FOR REFERENCE. CONTRACTOR MUST REFERENCE ALL DOCUMENTS FOR COMPLETE INFORMATION.

## SHEET KEYNOTES

M105 DIVISION 26 EQUIPMENT/FIXTURE/CONDUCTOR ONLY SHOWN FOR COORDINATION. REFER TO DIVISION 26 DOCUMENT FOR EXACT LOCATION/DIMENSION.

M115 BOILER B-1, BOILER B-2 AND DIVISION 22 EQUIPMENT EXHAUST FLUE AND INTAKE OPENING. FLUE SHALL TERMINATE AT LEAST 4FT ABOVE THE INTAKE OPENING. PROVIDE WEATHER PROTECTION CAP.

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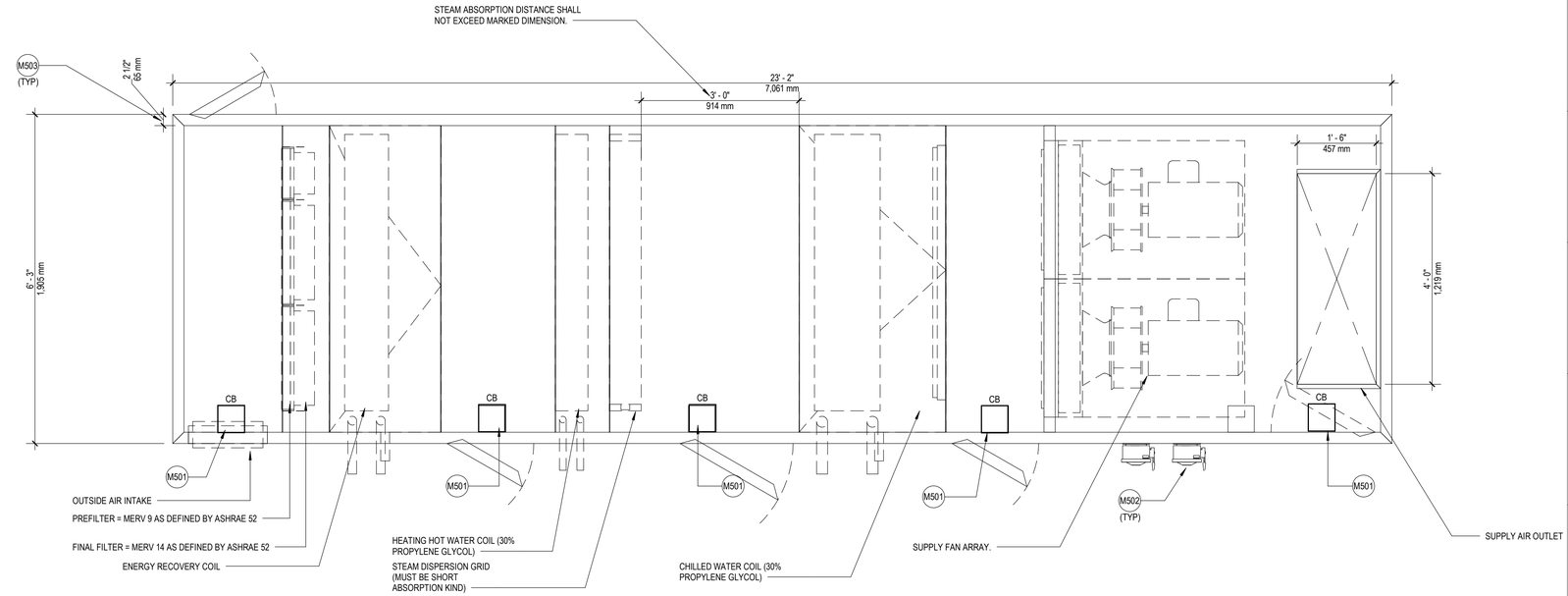
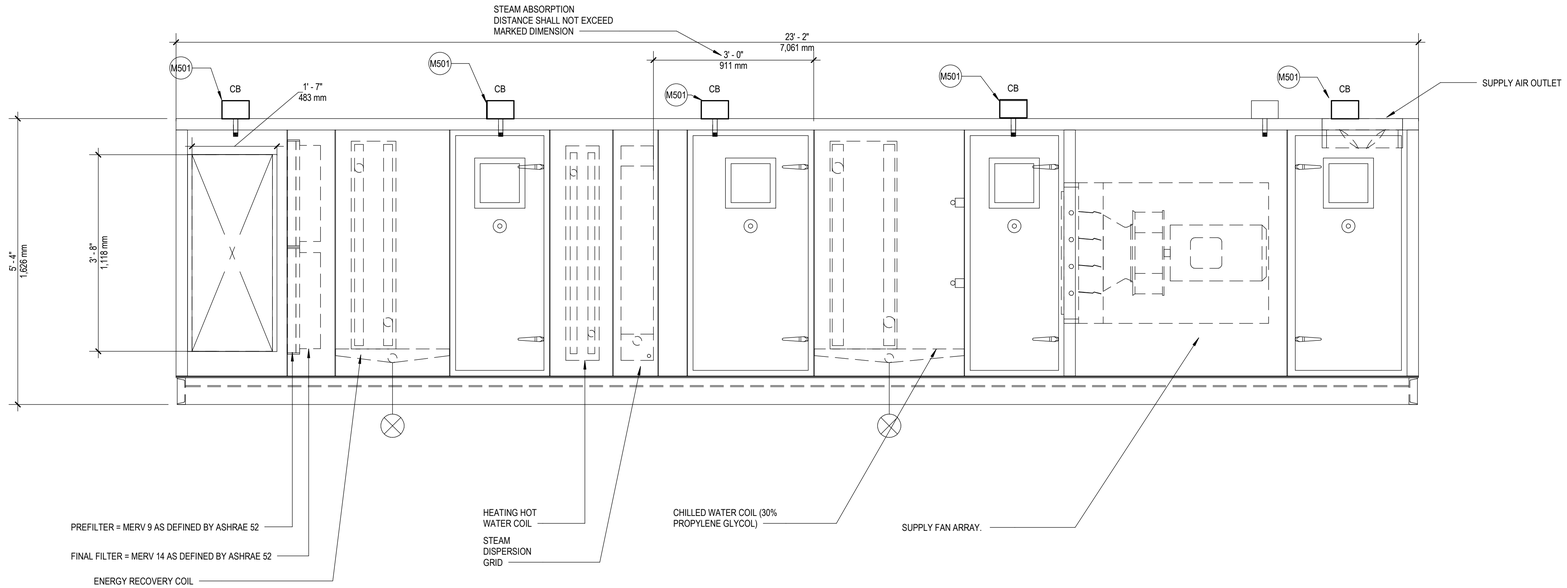
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL ROOM - BOILER SECTIONS |
| WORKING STAFF | OP / OP / IP (DESIGNED BY / DRAWN BY / CHECKED BY) |
| SHEET NO. | 268 OF 378 |
| | M-403QP (DISCIPLINE TYPE SEQUENCE DRAWING TYPE) |




1 / M-501DT FULLY CUSTOM AIR HANDLING UNIT - AHU-1

SCALE: NOT TO SCALE

## SHEET KEYNOTES

M501 PROVIDE A CONDUIT/CONDUCTOR BOX (TAGGED AS 'CB' ON THIS SHEET) AT THE INDICATED LOCATION. CONTRACTOR SHALL ROUTE ALL CONTROL AND POWER CONDUCTORS EXCLUSIVELY THROUGH THESE BOXES. FIELD PENETRATIONS OF THE AIR-HANDLING UNIT CASING ARE STRICTLY PROHIBITED. PRIOR TO SUBMISSION OF THE PRODUCT TO THE ENGINEER OF RECORD (EOR) FOR SHOP DRAWING REVIEW, THE MANUFACTURER SHALL COORDINATE WITH AND ENGAGE THE GENERAL CONTRACTOR AND ALL APPLICABLE SUBCONTRACTORS (ELECTRICAL, MECHANICAL, AND CONTROLS) TO COORDINATE AND FINALIZE THE REQUIRED OPENINGS AT EACH JUNCTION BOX.

M502 PROVIDE CASING MOUNTED FAN DISCONNECT. COORDINATE EXACT LOCATION WITH ELECTRICAL CONTRACTOR WITH PURCHASED PRODUCT'S SHOP DRAWINGS. SEE MECHANICAL SPECIFICATIONS FOR ADDITIONAL INFORMATION.

M503 THE HIGHLIGHTED CASING INSULATION THICKNESS IS SHOWN FOR THE BASIS OF DESIGN MANUFACTURER ONLY AND IS NOT INTENDED TO ESTABLISH A MINIMUM INSULATION THICKNESS FOR ALL MANUFACTURERS. ALTERNATE MANUFACTURERS MAY REQUIRE DIFFERENT INSULATION THICKNESSES TO ACHIEVE THE SPECIFIED THERMAL PERFORMANCE AND SHALL ADJUST CASING CONSTRUCTION ACCORDINGLY. THE CONTROLLING REQUIREMENT IS COMPLIANCE WITH THE SPECIFIED THERMAL PERFORMANCE; INSULATION THICKNESS IS A SELECTION VARIABLE. SEE DIVISION 23 SPECIFICATIONS FOR DETAILED AIR HANDLER CASING INSULATION AND THERMAL PERFORMANCE REQUIREMENTS..

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL DETAILS - AIR HANDLING UNIT |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

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M-501DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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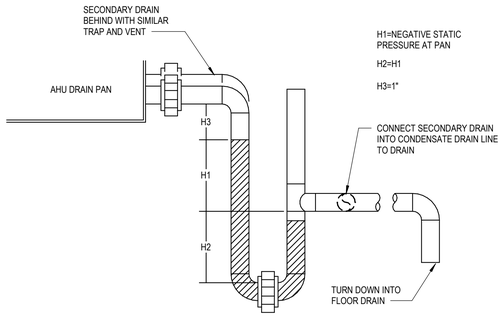
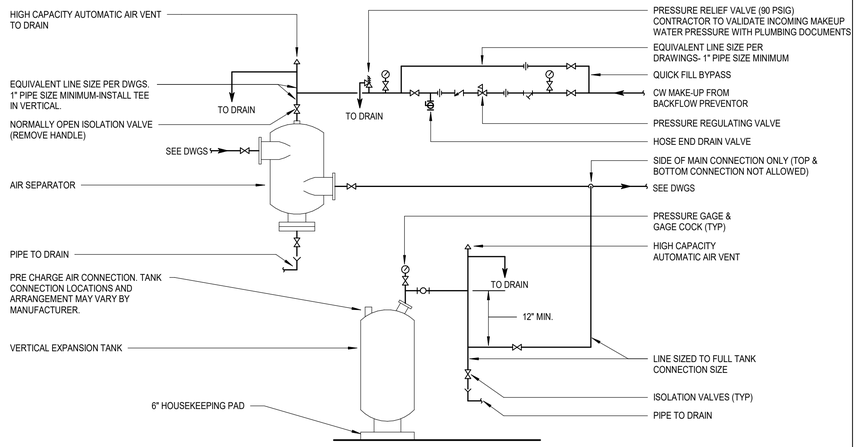
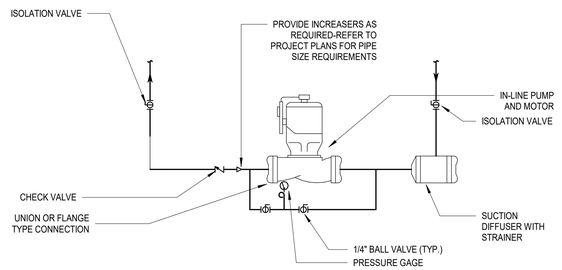



11 M-502DT **AHU DRAIN**
SCALE: NOT TO SCALE




2 M-502DT **VERTICAL EXPANSION TANK AND AIR SEPARATOR**
SCALE: NOT TO SCALE




NOTES:

SUPPORT PUMP AND PIPING FROM SUITABLE VIBRATION CONCRETE PAD ISOLATION. REFER TO SPECIFICATIONS.

PIPING SHALL BE INSTALLED SUCH THAT PUMP MAY BE COMPLETELY REMOVED WITHOUTDISMANTLING PIPE OR FITTINGS, AND WITHOUT FORCING OR SPRINGING THE PIPING.

3 M-502DT **IN-LINE PUMP DETAIL**
SCALE: NOT TO SCALE




1 M-502DT **BUFFER TANK DETAIL**
SCALE: NOT TO SCALE



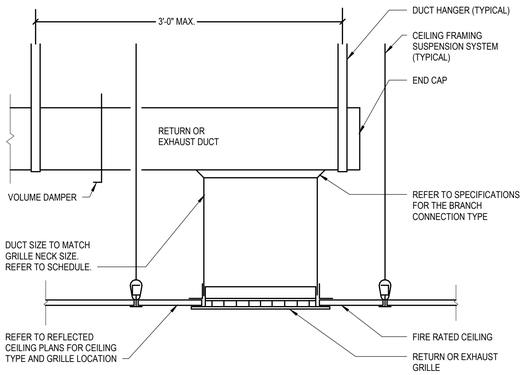

NOTES:
1. DISCHARGE STEAM LINE FROM HUMIDIFIER SHALL BE SLOPED BACK TOWARD HUMIDIFIER, AT A MINIMUM OF 1/8" PER 12".
2. INSTALL HUMIDIFIERS PER MANUFACTURERS INSTRUCTIONS AND PIPE SIZING RECOMMENDATIONS.
3. REFER TO THE TEMPERATURE CONTROL DRAWINGS FOR HUMIDIFIER CONTROL.
4. CONTRACTOR SHALL CAREFULLY CUT ACCESS IN HUMIDIFIER CASING TO GAIN ACCESS TO CONDENSATE RETURN CONNECTION. 3/4" CONDENSATE RETURN CONNECTION.

5 M-502DT **MANIFOLD STEAM HUMIDIFIER IN AHU**
SCALE: NOT TO SCALE

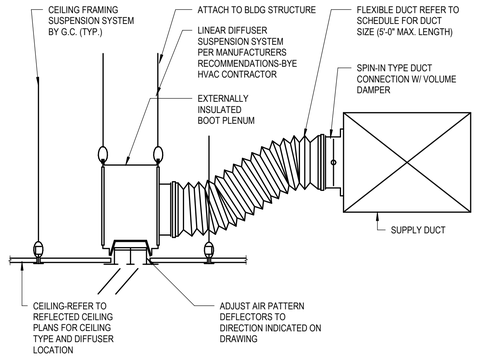



NOTES:
1. TERMINAL DEVICE HYDRONIC COIL DETAIL IS APPLICABLE TO ALL TERMINAL EQUIPMENT FOR HEATING AND COOLING SERVICE CONNECTIONS.
2. CHARACTERIZED CONTROL VALVE SHALL BE CAPABLE OF MODULATING CONTROL.
3. COMBINATION ISOLATION VALVE, STRAINER, AND PRESSURE/TEMPERATURE PORRT ON HYDRONIC SUPPLY IS ACCEPTABLE.
4. COMBINATION AUTOMATIC FLOW BALANCING VALVE AND ISOLATION VALVE ON HYDRONIC RETURN IS ACCEPTABLE.

4 M-502DT **HYDRONIC (TERMINAL DEVICE) COIL PIPING CONNECTIONS**
SCALE: NOT TO SCALE

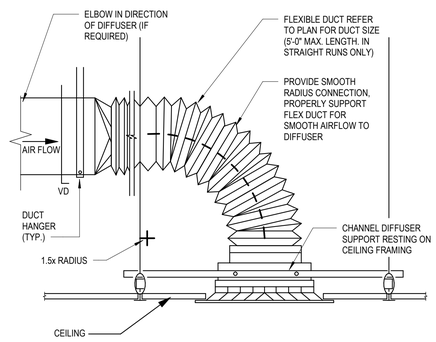



7 M-502DT **TYPICAL EXHAUST / RETURN GRILLE**
SCALE: NOT TO SCALE

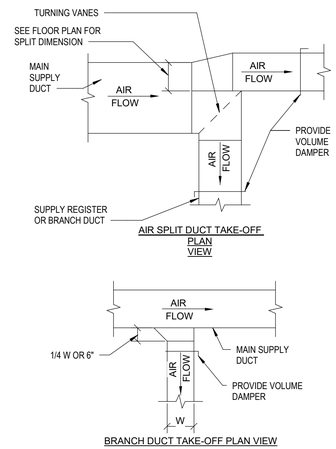



NOTES: 1. INSTALL REGISTER PER MANUFACTURERS RECOMMENDATIONS AND SMACNA DUCT CONSTRUCTION STANDARDS.
2. INSTALL FLEXIBLE DUCTWORK PER SECTION 3-SMACNA DUCT CONSTRUCTION STANDARDS.

8 M-502DT **TYPICAL LINEAR SUPPLY DIFFUSER WITH PLENUM**
SCALE: NOT TO SCALE




9 M-502DT **TYPICAL SUPPLY DIFFUSER DETAIL WITH FLEXIBLE DUCT CONNECTION**
SCALE: NOT TO SCALE




10 M-502DT **SUPPLY DUCTWORK TAKE-OFFS**
SCALE: NOT TO SCALE

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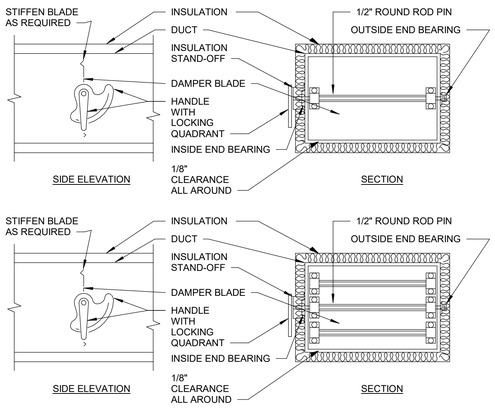
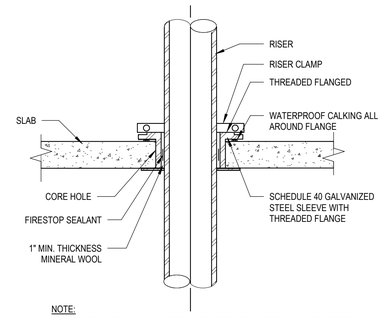
M-502DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)




NOTE:

1. DELETE INSULATION STAND-OFF ON DUCTWORK WITHOUT EXTERIOR INSULATION.
2. DUCT SIZES 8x8 INCHES (OR EQUIVALENT DIAMETER ROUND) OR SMALLER SHALL UTILIZE SINGLE BLADE DAMPER. ANY DUCTS LARGER SHALL UTILIZE OPPOSED BLADE MULTIBLADE DAMPERS.

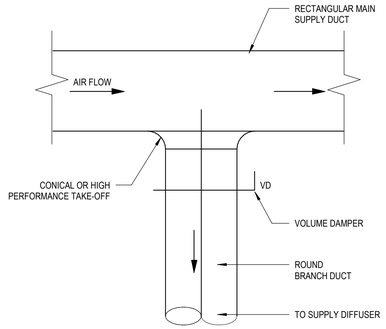
1 M-503DT — VOLUME DAMPER DETAIL
SCALE: NOT TO SCALE



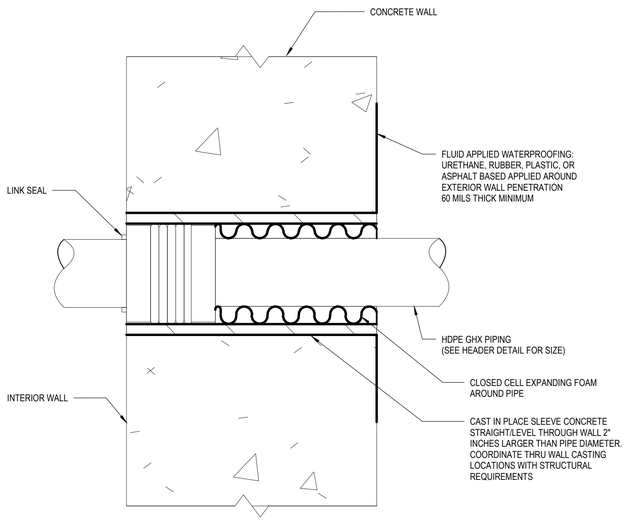

NOTE:

1. FIRESTOP SEALANTS AND PIPING PENETRATIONS THROUGH FIRE RATED ASSEMBLIES SHALL COMPLY WITH THE UL FIRE RESISTANCE DIRECTORY.

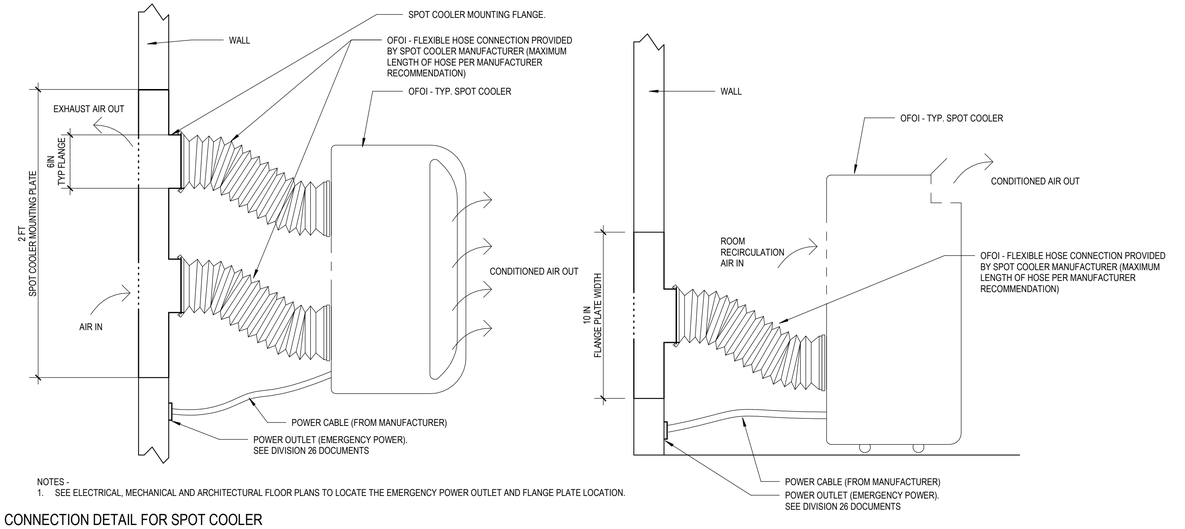
2 M-503DT — PIPE SUPPORT THROUGH SLAB
SCALE: NOT TO SCALE




3 M-503DT — TYPICAL LOW PRESSURE RECTANGULAR TO ROUND SUPPLY DUCT TAKE-OFF
SCALE: NOT TO SCALE



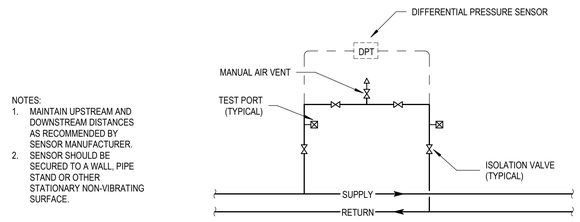

4 M-503DT — BELOW GRADE WALL PENETRATION DETAIL
SCALE: NOT TO SCALE




NOTES -

1. SEE ELECTRICAL, MECHANICAL AND ARCHITECTURAL FLOOR PLANS TO LOCATE THE EMERGENCY POWER OUTLET AND FLANGE PLATE LOCATION.

6 M-503DT — CONNECTION DETAIL FOR SPOT COOLER
SCALE: NOT TO SCALE




NOTES:

1. MAINTAIN UPSTREAM AND DOWNSTREAM DISTANCES AS RECOMMENDED BY SENSOR MANUFACTURER.
2. SENSOR SHOULD BE SECURED TO A WALL, PIPE STAND OR OTHER STATIONARY NON-VIBRATING SURFACE.

5 M-503DT — DIFFERENTIAL PRESSURE SENSOR
SCALE: NOT TO SCALE

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## CONSTRUCT ARABIAN LEOPARD FACILITY

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600 Maryland Avenue S.W. Suite 5001
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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL DETAILS |
| WORKING STAFF | OP (DESIGNED BY) · OP (DRAWN BY) · IP (CHECKED BY) |

SHEET NO. 271 OF 378

M-503DT (DISCIPLINE · TYPE · SEQUENCE · DRAWING TYPE)

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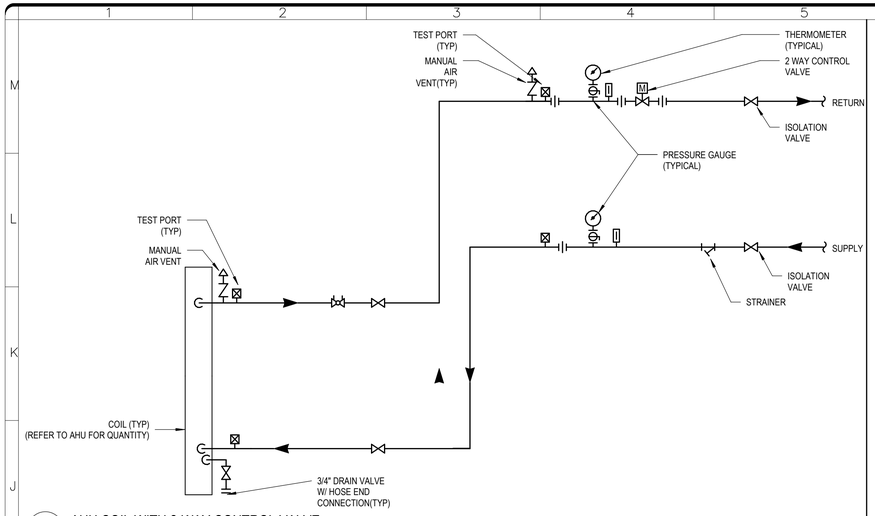
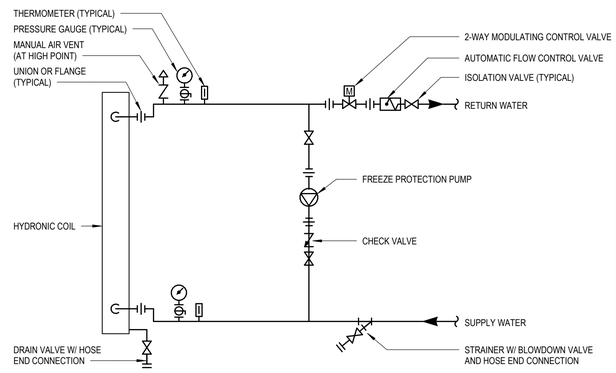
M-503DT BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

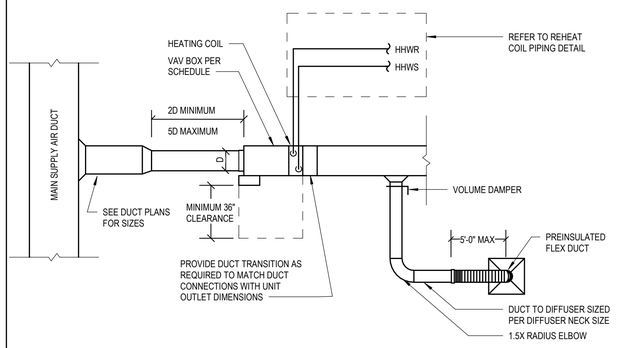



1 M-504DT AHU COIL WITH 2-WAY CONTROL VALVE
SCALE: NOT TO SCALE

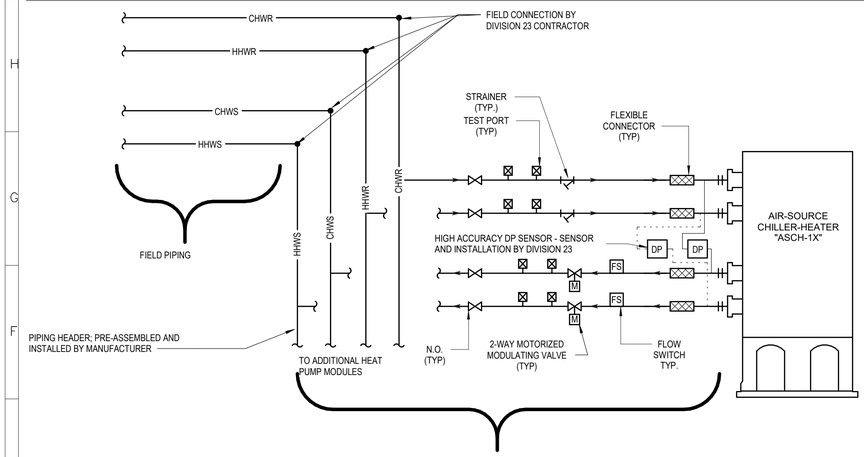



NOTES:
1. PIPING TO COIL(S) SHALL BE OFFSET AT COIL TO FACILITATE COIL REMOVAL.

6 M-504DT HYDRONIC COIL WITH 2-WAY CONTROL VALVE AND FREEZE PUMP
SCALE: NOT TO SCALE

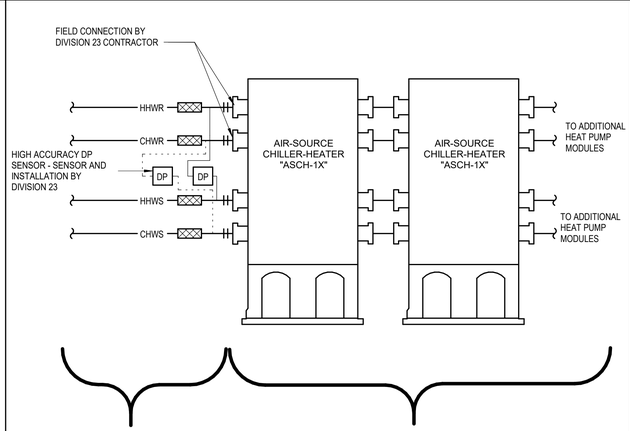



3 M-504DT TYPICAL VAV TERMINAL UNIT CONNECTION
SCALE: NOT TO SCALE



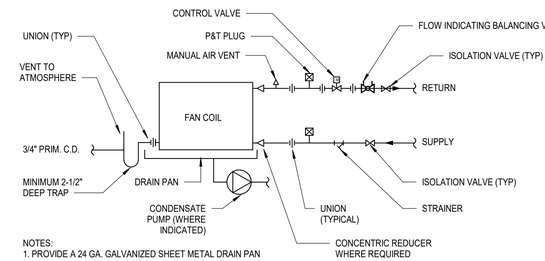

NOTES:
1. THIS DETAIL APPLY TO YORK'S YMAE 4-PIPE MODULAR HEAT PUMP.
2. YORK AIR-SOURCE HEAT PUMP IS COMPRISED OF FOUR (4) NOMINAL 35-TON MODULES, EACH WITH A SUPPLY AND RETURN CONNECTION. ENTERING AND LEAVING WATER TEMPERATURE SENSORS, INLET WATER STRAINER AND FLOW SWITCH ARE FACTORY INSTALLED ON EACH MODULE.
3. THE PIPING HEADER ALONG WITH MOTORIZED VALVES, SWITCHES AND SENSORS SHALL BE FACTORY INSTALLED, WIRED AND INTEGRATED TO THE HEAT PUMP MASTER CONTROLLER.
4. FIELD WORK ASSOCIATED FOR DIVISION 23 CONTRACTOR SHALL INCLUDE-
   A. HEAT TRACE THE HEADER AND VALVES AND APPLY INSULATION.
   B. PROVIDE CONNECTION TO FIELD PIPING.
5. REFER TO SPECIFICATIONS FOR ACCEPTABLE VALVE AND FITTING TYPES.
6. REFER TO CONTROLS SCHEMATIC FOR CONTROLS INSTRUMENTATION.
7. COORDINATE FITTINGS AND COMPONENTS SHOWN IN THIS DETAIL WITH ADDITIONAL MANUFACTURER SPECIFIC REQUIRED FITTINGS AND COMPONENTS

2 M-504DT WATER PIPING CONNECTIONS TO AIR-SOURCE HEAT PUMP
SCALE: NOT TO SCALE



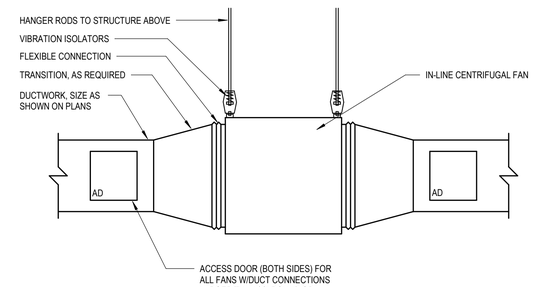

NOTES:
1. THIS DETAIL APPLY TO TRANE'S MAS 4-PIPE MODULAR HEAT PUMP.
2. MAS AIR-SOURCE HEAT PUMP IS COMPRISED OF FOUR (4) NOMINAL 30-TON MODULES.
3. SENSORS, VALVES AND SWITCHES ARE INTERNAL TO EQUIPMENT.
4. FIELD WORK ASSOCIATED FOR DIVISION 23 CONTRACTOR SHALL INCLUDE-
   A. PROVIDE CONNECTION TO HEAT PUMP.
5. REFER TO CONTROLS SCHEMATIC FOR CONTROLS INSTRUMENTATION.

4 M-504DT WATER PIPING CONNECTIONS TO AIR-SOURCE HEAT PUMP
SCALE: NOT TO SCALE




NOTES:
1. PROVIDE A 24 GA. GALVANIZED SHEET METAL DRAIN PAN UNDER UNIT WITH 3/4" SECONDARY DRAIN CONNECTION.
2. COIL CONNECTIONS SHOWN TYPICAL FOR HEATING AND COOLING PIPING.

5 M-504DT FAN COIL PIPING
SCALE: NOT TO SCALE




7 M-504DT IN-LINE FAN
SCALE: NOT TO SCALE

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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL DETAILS |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 272 OF 378

M-504DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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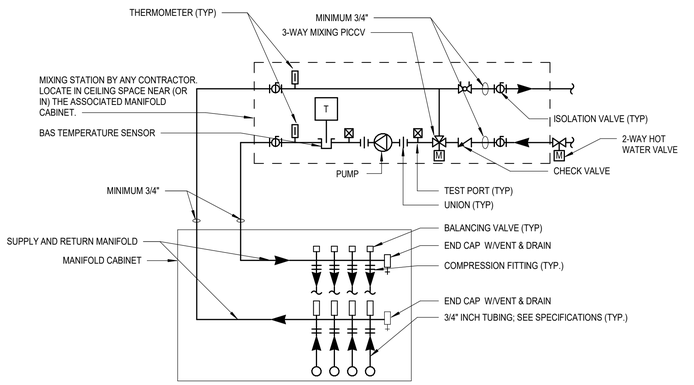
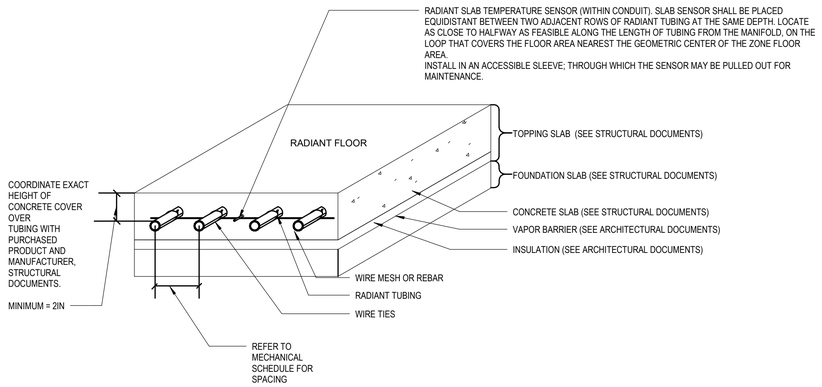
M-504DT BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS



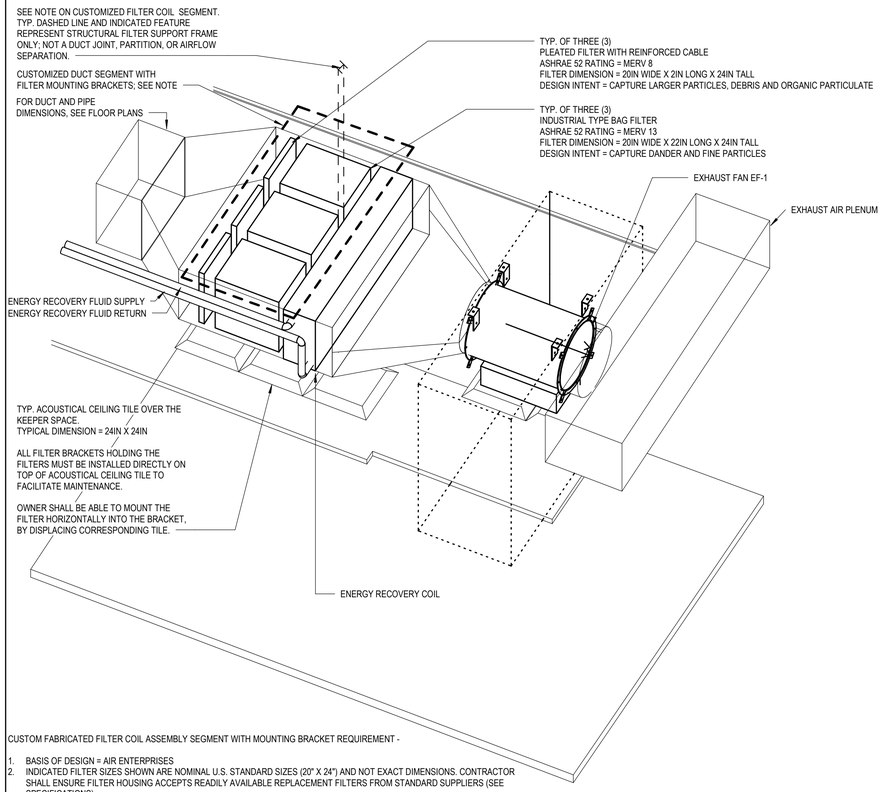

NOTES
1. PROVIDE BEND SUPPORTS FOR TUBING AT FLOOR.
2. CONTRACTOR SHALL PROVIDE MIXING STATION, SHOWN AS DASHED OUTLINE ABOVE.
3. REFER TO SCHEDULE FOR AMOUNT OF CIRCUITS PER ZONE.

1 / M-505DT — RADIANT MANIFOLD SYSTEM
SCALE: NOT TO SCALE




NOTE:
AT BUILDING EXPANSION JOINTS PROVIDE A RIGID PVC SLEEVE THAT EXTENDS A MINIMUM OF 6" BEYOND EXPANSION JOINT ON BOTH SIDES. COORDINATE WITH SUCCESSFUL RADIANT FLOOR MANUFACTURER.

2 / M-505DT — UNDERFLOOR PIPING FOR RADIANT PIPING
SCALE: NOT TO SCALE




CUSTOM FABRICATED FILTER COIL ASSEMBLY SEGMENT WITH MOUNTING BRACKET REQUIREMENT -

1. BASIS OF DESIGN = AIR ENTERPRISES
2. INDICATED FILTER SIZES SHOWN ARE NOMINAL U.S. STANDARD SIZES (20" X 24") AND NOT EXACT DIMENSIONS. CONTRACTOR SHALL ENSURE FILTER HOUSING ACCEPTS READILY AVAILABLE REPLACEMENT FILTERS FROM STANDARD SUPPLIERS (SEE SPECIFICATIONS).
   PROVIDE ADEQUATE CLEARANCE AND GASKETING FOR PROPER FIT AND NO BYPASS.
3. FABRICATE FILTER SECTION AS A RIGID, FULLY GASKETED DUCT ASSEMBLY SIZED TO ACCOMMODATE (3) PREFILTERS AND (3) BAG FILTERS IN SERIES AS SHOWN.
4. PROVIDE INTERNAL FILTER TRACK/FRAME SYSTEM TO ALLOW HORIZONTAL SLIDE-IN REMOVAL AND REPLACEMENT FROM BELOW CEILING.
5. ALL FILTER RACKS SHALL INCLUDE CONTINUOUS NEOPRENE GASKETING TO PREVENT AIR BYPASS.
6. PROVIDE REINFORCED CASING (MIN. 18 GA STAINLESS STEEL OR AS REQUIRED) TO PREVENT DEFLECTION UNDER LOADED CONDITIONS.
7. PROVIDE ACCESS VIA REMOVABLE CEILING TILE OR ACCESS PANEL; COORDINATE WITH ARCHITECTURAL CEILING SYSTEM.
8. PROVIDE STRUCTURAL SUPPORT FRAME AND MOUNTING BRACKETS; DO NOT SUPPORT FROM CEILING GRID OR DUCTWORK.
9. PROVIDE MINIMUM CLEARANCE FOR FILTER REMOVAL EQUAL TO FILTER LENGTH PLUS 2 INCHES.
10. SEAL ALL JOINTS AIRTIGHT; COMPLY WITH SMACNA SEAL CLASS A.
11. PROVIDE DIFFERENTIAL PRESSURE TAP CONNECTIONS UPSTREAM AND DOWNSTREAM OF EACH FILTER STAGE.
12. INTERIOR SURFACES SHALL BE SMOOTH AND CLEANABLE; NO PROTRUSIONS THAT CAN COLLECT DEBRIS.
13. FILTER MATERIAL SHALL BE SUITABLE FOR HIGH LINT, DANDER, AND ORGANIC PARTICULATE APPLICATIONS.
14. PROVIDE POSITIVE FILTER RETENTION (CLIPS OR HOLD-DOWNS) TO PREVENT FILTER MOVEMENT OR BYPASS.

3 / M-505DT — EXHAUST FAN AND RECOVERY COIL
SCALE: NOT TO SCALE

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REVISION DESCRIPTION

Smithsonian Institution
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF); 3001 Connecticut Avenue NW Washington, DC 20008
PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084
DRAWING TITLE: MECHANICAL DETAILS
WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)
SHEET NO. 273 OF 378
M-505DT (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

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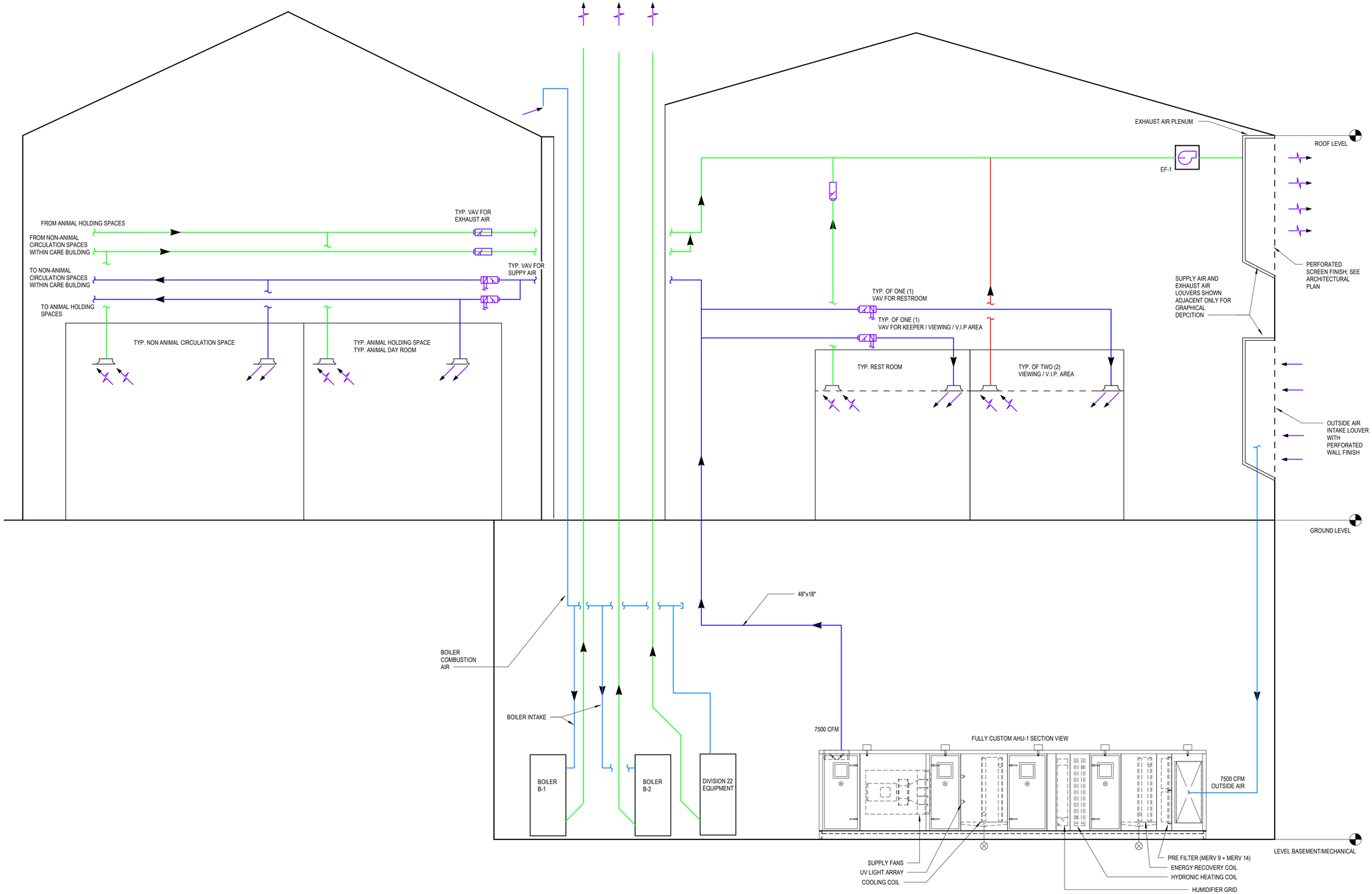


2 AIR RISER DIAGRAM
SCALE: 12" = 1'-0"

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M-601DG BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

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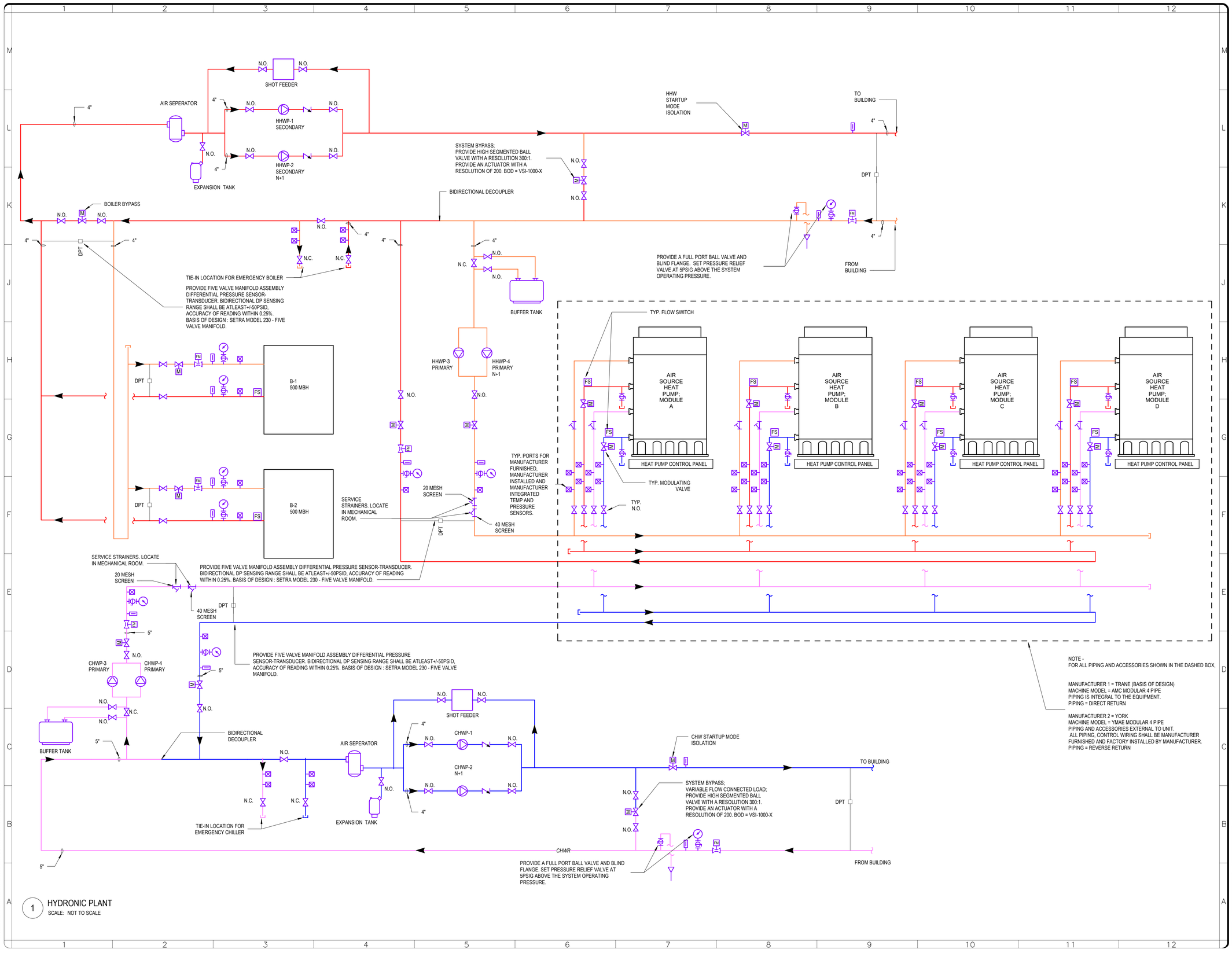
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | AIR RISER DIAGRAM |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 274 OF 378

M-601DG (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)




1 HYDRONIC PLANT
SCALE: NOT TO SCALE

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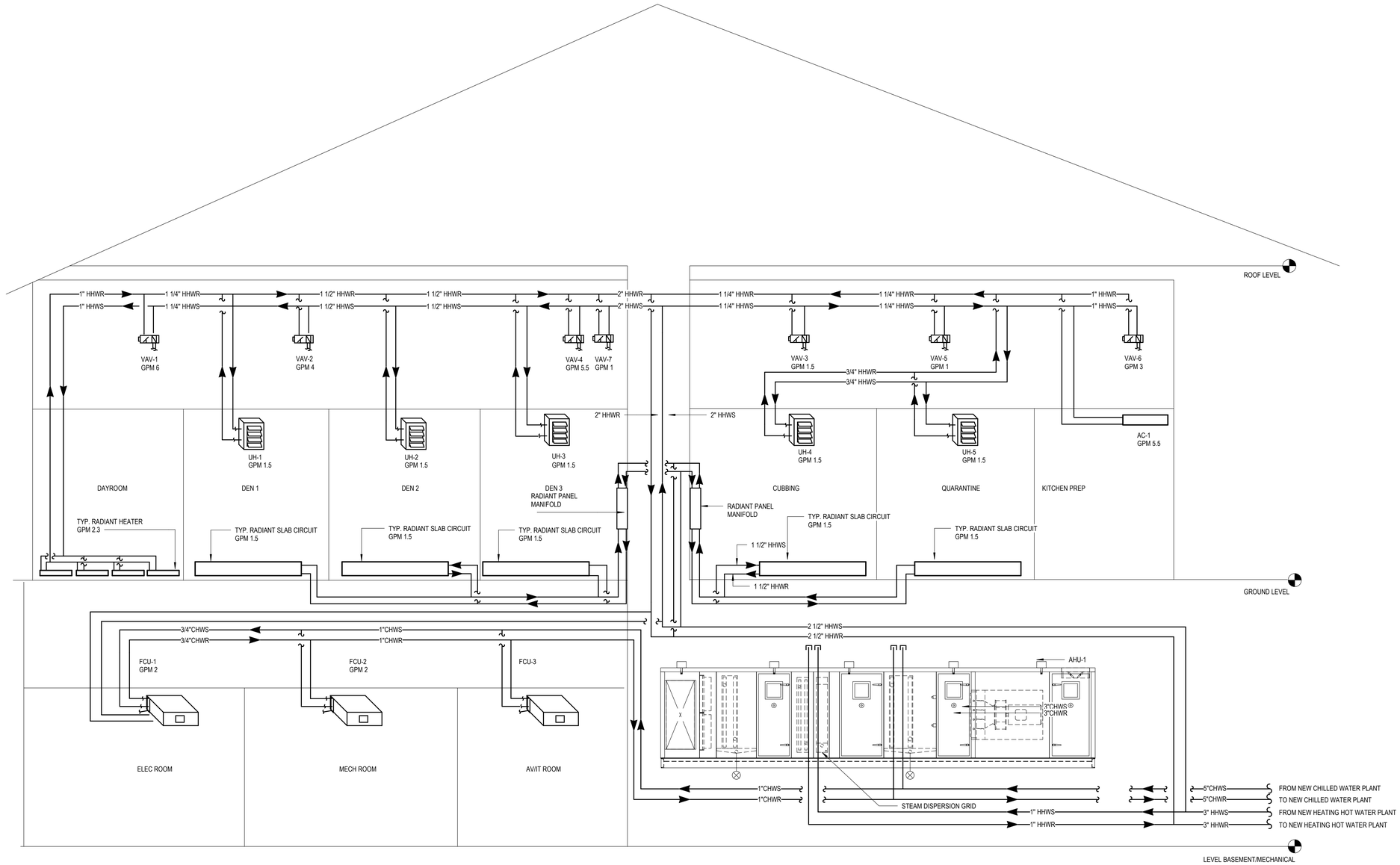
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HYDRONIC PLANT |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 275 OF 378

M-602DG (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)




1 HYDRONIC RISER DIAGRAM
SCALE: 12" = 1'-0"

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | HYDRONIC RISER |

| WORKING STAFF | DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|---|
| | OP | YW | IP |

SHEET NO. 276 OF 378

M-603DG
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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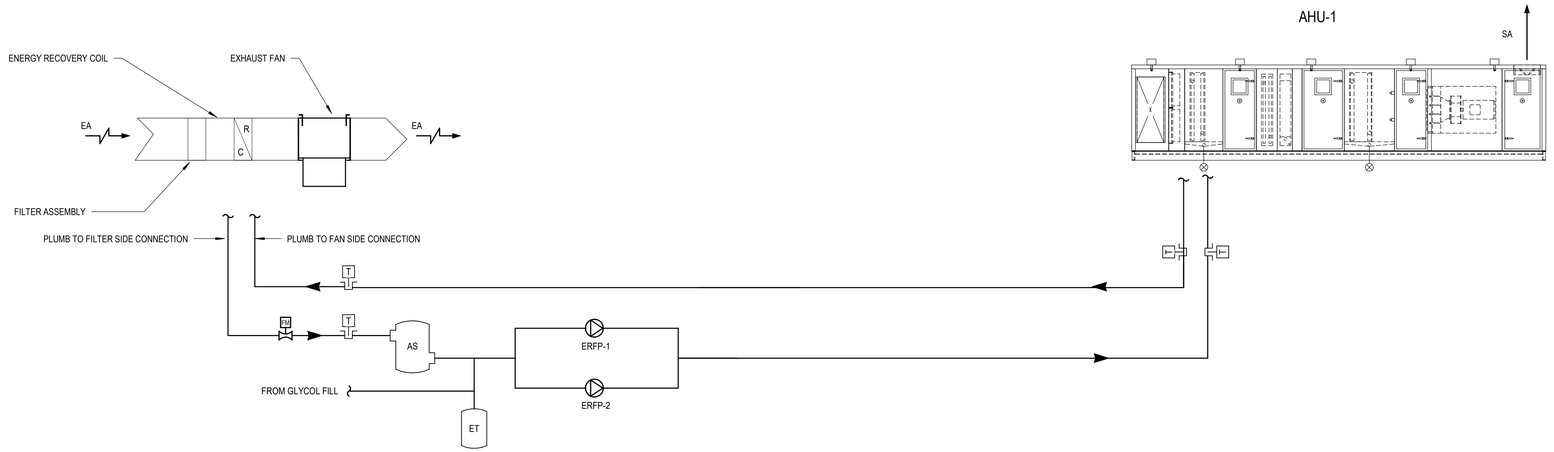
M-603DG BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




1 M-604DG ENERGY RECOVERY SYSTEM DIAGRAM
SCALE: NOT TO SCALE

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| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ENERGY RECOVERY LOOP SYSTEM DIAGRAM |
| WORKING STAFF | DESIGNED BY: OP; DRAWN BY: OP; CHECKED BY: IP |

SHEET NO. 277 OF 378

M-604DG (DISCIPLINE: M; TYPE: 6; SEQUENCE: 04; DRAWING TYPE: DG)

6/15/2026 3:01:35 PM
M-604DG BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# AIR HANDLING UNIT SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | AIRFLOW | | | SUPPLY FANS | | COILS AND GRIDS | | | | CASING | ACCESSORIES | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | HEATING | | | | | | | | | |
| | | | | SUPPLY MINIMUM | SUPPLY MAXIMUM | OUTSIDE MIN (CFM) | SF QTY | SF TAG | ENERGY RECOVERY COIL TAG | HEATING COIL TAG | STEAM DISPERSION GRID (SDG) SCHEDULE | CHW TAG | WALL THICKNESS | FINAL FILTER TAG | PREFILTER TAG | MANUFACTURER | MODEL | |
| AHU-1 | SUPPLY AIR FOR CARE BUILDING | CARE BUILDING BASEMENT MECHANICAL ROOM | 100% OUTSIDE AIR | 7,000 CFM | 7,500 CFM | 7,000 CFM | 2 | SF-1A; SF-1B | ERC-1 | HWC-1 | SDG-1 | CWC-1 | 2.5 in | AHU-FF | AHU-PF | AIR ENTERPRISES | FULLY CUSTOM | |

# AIR HANDLING UNIT AIR FILTER SCHEDULE

| TAG | SYSTEM | SYSTEM | TYPE | MAX FACE VELOCITY (FPM) | CAPACITY | HOUSING ASSEMBLY | | PRESSURE DROP | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | OVERALL | | | | |
| | | | | | | WIDTH | HEIGHT | CLEAN FILTER | DIRTY FILTER (ASHRAE 52) | |
| AHU-FF | AHU-1 | FINAL FILTER | MERV-14 | 500 ft/min | 7,500 CFM | 5' - 0" | 4' - 0" | 0.60 in-wg | 1.40 in-wg | |
| AHU-PF | AHU-1 | PRE FILTER | MERV-9 | 500 ft/min | 7,500 CFM | 5' - 0" | 4' - 0" | 0.20 in-wg | 0.60 in-wg | |

# AIR HANDLING UNIT HYDRONIC COIL SCHEDULE

| TAG | SYSTEM | LOCATION | FLUID | AIRFLOW | TOTAL CAPACITY | SENSIBLE CAPACITY | QTY OF COILS IN BANK | MIN ROWS | MAX FINS PER INCH | FACE VELOCITY | COIL SIZE DATA | | | AIRSIDE DATA | | | | | FLUID SIDE DATA | | | | BASIS OF DESIGN | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | FACE AREA | LENGTH | HEIGHT | EAT DB | EAT WB | LAT DB | LAT WB | MAX AIR PD | EWT | LWT | FLOW | WATER MAX PD | MANUFACTURER | |
| CWC-1 | AHU-1 CHILLED WATER | AHU-1 | 30% PROPYLENE GLYCOL | 7,500 CFM | 712 MBH | 283 MBH | 1 | 10 | 11 | 400 ft/min | 19 SF | 1' - 3" | 3' - 9" | 83.40 °F | 77.50 °F | 48.90 °F | 48.90 °F | 0.88 in-wg | 42 °F | 53 °F | 137 GPM | 30.0 ftH2O | HEATCRAFT COILS | |
| ERC-1 (SUMMER) | AHU-1 ENERGY RECOVERY COIL | AHU-1 | 30% PROPYLENE GLYCOL | 7,500 CFM | 31 MBH | 31 MBH | 1 | 6 | 8 | 400 ft/min | 19 SF | 10" | 3' - 9" | 83.40 °F | 77.50 °F | 79.65 °F | 76.30 °F | 0.30 in-wg | 74 °F | 79 °F | 13 GPM | 5.0 ftH2O | HEATCRAFT COILS | |
| ERC-1 (WINTER) | AHU-1 ENERGY RECOVERY COIL | AHU-1 | 30% PROPYLENE GLYCOL | 7,500 CFM | 169 MBH | 169 MBH | 1 | 6 | 8 | 400 ft/min | 19 SF | 10" | 3' - 9" | 18.00 °F | 13.86 °F | 38.74 °F | 27.28 °F | 0.30 in-wg | 60 °F | 45 °F | 24 GPM | 5.0 ftH2O | HEATCRAFT COILS | |
| PHC-1 | AHU-1 PREHEAT COIL | AHU-1 | 30% PROPYLENE GLYCOL | 7,500 CFM | 343 MBH | 343 MBH | 1 | 2 | 7 | 400 ft/min | 19 SF | 7 1/2" | 3' - 9" | 18.00 °F | 13.86 °F | 60.11 °F | 40.20 °F | 0.15 in-wg | 130 °F | 115 °F | 48 GPM | 10.0 ftH2O | HEATCRAFT COILS | |

NOTES -
1. PROVIDE SINGULAR ENERGY RECOVERY COIL. COIL PERFORMANCE IS RATED FOR A SINGULAR ERC-1 FOR SUMMER AND WINTER DESIGN DAY.

# AIR HANDLING UNIT FAN SCHEDULE

| TAG | LOCATION | SYSTEM | AIRFLOW | OPERATING POINT | | | | | | MOTOR | | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | STATIC PRESSURE | | | | | | | | | | | | | |
| | | | | INTERNAL (FOR BOD ASSEMBLY) | EXTERNAL (MINIMUM REQUIREMENT) | TOTAL | BOD STATIC EFFICIENCY (%) | BOD FAN RPM | BOD BHP | MAXIMUM PERMISSIBLE FAN SPEED | MAX RPM | HP | VOLTAGE | PHASE | OPERATING WEIGHT | MANUFACTURER | | |
| SF-1A | BASEMENT MECHANICAL ROOM | AHU-1 SUPPLY | 7,500 CFM | 4.10 in-wg | 4.00 in-wg | 8.10 in-wg | 60 | 3326 | 15.94 | 3500 | 3500 | 20 | 460 V | 3 | 325 lb | GREENHECK | |
| SF-1B | BASEMENT MECHANICAL ROOM | AHU-1 SUPPLY | 7,500 CFM | 4.10 in-wg | 4.00 in-wg | 8.10 in-wg | 60 | 3326 | 15.94' | 3500 | 3500 | 20 | 460 V | 3 | 325 lb | GREENHECK | FULLY REDUNDANT |

NOTES -
1. VFD SHALL LOCK MAXIMUM FAN SPEED AT 3450 RPM.
   A. 0 HZ AT 0 RPM AND 60HZ AT 3450 RPM.
2. PROVIDE ALL FANS WITH FACTORY MOUNTED PIEZOMETER RING AS A BACKUP AIRFLOW MEASURING DEVICE (INTEGRATION TO BAS).
3. SCHEDULED INTERNAL PRESSURE DROP INCLUDES 2IN W.G. OF TOTAL PRESSURE DROP FOR DIRTY FILTERS, SEE 'AIR HANDLING UNIT FILTER' SCHEDULE.

# AIR HANDLING UNIT STEAM DISPERSION GRID SCHEDULE

| TAG | STEAM PRODUCERS | LOCATION | TYPE | STEAM DATA | | HEATING AIRFLOW | HUMIDIFICATION | | | | | | DISPERSION GRID DIMENSIONS IN AHU | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | ENTERING AIR | | | LEAVING AIR | | | | | | | |
| | | | | STEAM FLOW RATE | GUAGE PRESSURE (PSIG) | | DRY BULB | WET BULB | GRAINS OF WATER PER POUND OF DRY AIR | DRY BULB | WET BULB | GRAINS OF WATER PER POUND OF DRY AIR | HEIGHT | WIDTH | MANUFACTURER | MODEL | |
| SDG-1 | HUM-1 AND HUM-2 | WITHIN AHU-1 | SHORT ABSORPTION MANIFOLD KIND (SAMe) | 179 lb/h | 0 | 7,000 CFM | 60.00 °F | 36.64 °F | 3.2 | 60.00 °F | 51.46 °F | 43 | 5' - 4" | 5' - 4" | CONDAIR | SAMe | |

NOTES -
1. 7,000 GRAINS OF WATER = 1 POUND (MASS) OF WATER.

# RESISTIVE ELEMENT HUMIDIFIER SCHEDULE

| TAG | LOCATION | WORKING PRINCIPLE | END USER | STEAM PRESSURE | DESIGN CAPACITY (LB/HR) | TANK VOLUME | TANK FILL RATE | TANK DRAIN RATE | INPUT POWER | VOLTAGE | PHASE | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| HUM-1 | BASEMENT MECHANICAL ROOM | ELECTRIC RESISTANCE HEAT | SDG-1 | 0 psi | 81.65 | 13.2 gal | 8 GPM | 13 GPM | 65 kW | 480 V | 3 | CONDAIR | RS 180 | |

NOTES -
1. PROVIDE BACNET INTEGRATION.
2. PROVIDE SCALE COLLECTION TANK ACCESSORY.

# FAN SCHEDULE

| TAG | LOCATION | SYSTEM | AIRFLOW | OPERATING POINT | | | | | | MOTOR | | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | STATIC PRESSURE SUMMARY | | | | | | | | | | | | | |
| | | | | ENERGY RECOVERY COIL | DIRTY FILTERS | TOTAL EXTERNAL | BOD STATIC EFFICIENCY (%) | BOD FAN RPM | BOD BHP | MAXIMUM PERMISSIBLE FAN SPEED | MAX RPM | HP | VOLTAGE | PHASE | OPERATING WEIGHT | MANUFACTURER | MODEL | |
| EEF-1 | EMERGENCY ELECTRICAL ROOM | SUPPLY AIR | 300 CFM | | | 0.50 in-wg | 31 | 1112 | 0.08 | 1750 | 1700 | 0.25 | 115 V | 1 | 130 lb | GREENHECK | AER-20-03-0605-VG | |
| EF-1 | ABOVE KEEPER'S OFFICE SPACE | EXHAUST AIR | 7,500 CFM | 1.00 in-wg | 2.00 in-wg | 4.50 in-wg | 63 | 2524 | 8.7 | 2777 | 2700 | 10 | 460 V | 3 | 500 lb | GREENHECK | QEI-18 | |
| ESF-1 | EMERGENCY ELECTRICAL ROOM | EXHAUST AIR | 300 CFM | | | 0.50 in-wg | 31 | 1112 | 0.08 | 1750 | 1700 | 0.25 | 115 V | 1 | 130 lb | GREENHECK | AER-20-03-0605-VG | |

# AIR FILTER SCHEDULE FOR EF-1

| TAG | SYSTEM | SYSTEM | FILTER TYPE | ASHRAE 52 RATING | MAX FACE VELOCITY (FPM) | CAPACITY | HOUSING ASSEMBLY | | | PRESSURE DROP | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | OVERALL | | | | | | |
| | | | | | | | CELL THICK | WIDTH | HEIGHT | CLEAN FILTER | DIRTY FILTER (ASHRAE 52) | MANUFACTURER | MODEL | |
| FILTER-E1 | EXHAUST FAN & COIL PROTECTION | EF-1 | INDUSTRIAL BAG TYPE | MERV-13 | 650 ft/min | 6,500 CFM | 1' - 10" | 1' - 8" | 2' - 1" | 0.50 in-wg | 1.40 in-wg | AAF | DRIPAK GX | |
| FILTER-E2 | EXHAUST FAN & COIL PROTECTION | EF-1 | INDUSTRIAL BAG TYPE | MERV-13 | 650 ft/min | 6,500 CFM | 1' - 10" | 1' - 8" | 2' - 1" | 0.50 in-wg | 1.40 in-wg | AAF | DRIPAK GX | |
| FILTER-E3 | EXHAUST FAN & COIL PROTECTION | EF-1 | INDUSTRIAL BAG TYPE | MERV-13 | 650 ft/min | 6,500 CFM | 1' - 10" | 1' - 8" | 2' - 1" | 0.50 in-wg | 1.40 in-wg | AAF | DRIPAK GX | |
| FILTER-E4 | EXHAUST FAN & COIL PROTECTION | EF-1 | RIGID FRAME PLEATED PANEL TYPE | MERV-8 | 650 ft/min | 6,500 CFM | 2" | 1' - 8" | 2' - 1" | 0.20 in-wg | 0.60 in-wg | AAF | PERFECTPLEAT SC M8 | |
| FILTER-E5 | EXHAUST FAN & COIL PROTECTION | EF-1 | RIGID FRAME PLEATED PANEL TYPE | MERV-8 | 650 ft/min | 6,500 CFM | 2" | 1' - 8" | 2' - 1" | 0.20 in-wg | 0.60 in-wg | AAF | PERFECTPLEAT SC M8 | |
| FILTER-E6 | EXHAUST FAN & COIL PROTECTION | EF-1 | RIGID FRAME PLEATED PANEL TYPE | MERV-8 | 650 ft/min | 6,500 CFM | 2" | 1' - 8" | 2' - 1" | 0.20 in-wg | 0.60 in-wg | AAF | PERFECTPLEAT SC M8 | |

NOTES -
1. PREFILTER E4, E5 AND E6 ARE INTENDED TO CAPTURE LARGER DEBRIS / ORGANIC PARTICLES.

ALF

CONSTRUCT ARABIAN LEOPARD FACILITY

NOT FOR CONSTRUCTION

FOR REFERENCE ONLY

STAMP

GRAPHIC SCALE(S)

DATE 06/15/2026

SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF); 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL SCHEDULES

WORKING STAFF

| OP | OP | IP |
|---|---|---|
| DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 278 OF 378

M-605SH

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

6/15/2026 3:01:35 PM
M-605SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# AIR SOURCE HEAT PUMP SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | COOLING MODE | | | | | | HEATING MODE | | | | | | SIMULTANEOUS MODE | | | | | | | | | MAX PD | OPERATING WEIGHT | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | AMBIENT TEMEPRATURE FOR HEAT REJECTION | COOLING CAPACITY | FLOW | EWT | LWT | EER (BTU/hr-kW) | AMBIENT TEMEPRATURE FOR HEAT EXTRACTION | HEATING CAPACITY | FLOW | EWT | LWT | COEFFICIENT OF PERFORMANCE (kW/kW) | HEATING CAPACITY | FLOW | HHW EWT | HHW LWT | COOLING CAPACITY | FLOW | CHW EWT | CHW LWT | COEFFICIENT OF PERFORMANCE (kW/kW) | | | MANUFACTURER | MODEL | |
| ASHP-1 | HYDRONIC PLANT FOR CARE BUILDING | OUTDOORS; HEAT PUMP YARD | ARRAY OF FOUR MODULES (ASHP-1A, ASHP-1B, ASHP-1C AND ASHP-1D) | 95 °F | 90 ton | 206 GPM | 53 °F | 42 °F | 9.6 | 18 °F | 821 MBH | 115 GPM | 115 °F | 130 °F | 2.23 | 1,205 MBH | 175 GPM | 115 °F | 130 °F | 76 ton | 169 GPM | 53 °F | 42 °F | 5.82 | 10.0 ftH2O | 12,000.00 lb | TRANE | MAS | |
| ASHP-1A | MODULE WITHIN ASHP-1 | OUTDOORS; HEAT PUMP YARD | MODULAR; VAPOR INJECTION SCROLL COMPRESSOR | 95 °F | 27 ton | 61 GPM | 53 °F | 42 °F | 9.6 | 18 °F | 274 MBH | 38 GPM | 115 °F | 130 °F | 2.23 | 402 MBH | 58 GPM | 115 °F | 130 °F | 25 ton | 56 GPM | 53 °F | 42 °F | 5.82 | 10.0 ftH2O | 3,000.00 lb | TRANE | MAS | |
| ASHP-1B | MODULE WITHIN ASHP-1 | OUTDOORS; HEAT PUMP YARD | MODULAR; VAPOR INJECTION SCROLL COMPRESSOR | 95 °F | 27 ton | 61 GPM | 53 °F | 42 °F | 9.6 | 18 °F | 274 MBH | 38 GPM | 115 °F | 130 °F | 2.23 | 402 MBH | 58 GPM | 115 °F | 130 °F | 25 ton | 56 GPM | 53 °F | 42 °F | 5.82 | 10.0 ftH2O | 3,000.00 lb | TRANE | MAS | |
| ASHP-1C | MODULE WITHIN ASHP-1 | OUTDOORS; HEAT PUMP YARD | MODULAR; VAPOR INJECTION SCROLL COMPRESSOR | 95 °F | 27 ton | 61 GPM | 53 °F | 42 °F | 9.6 | 18 °F | 274 MBH | 38 GPM | 115 °F | 130 °F | 2.23 | 402 MBH | 58 GPM | 115 °F | 130 °F | 25 ton | 56 GPM | 53 °F | 42 °F | 5.82 | 10.0 ftH2O | 3,000.00 lb | TRANE | MAS | |
| ASHP-1D | MODULE WITHIN ASHP-1 | OUTDOORS; HEAT PUMP YARD | MODULAR; VAPOR INJECTION SCROLL COMPRESSOR | 95 °F | 27 ton | 61 GPM | 53 °F | 42 °F | 9.6 | 18 °F | 274 MBH | 38 GPM | 115 °F | 130 °F | 2.23 | 402 MBH | 58 GPM | 115 °F | 130 °F | 25 ton | 56 GPM | 53 °F | 42 °F | 5.82 | 10.0 ftH2O | 3,000.00 lb | TRANE | MAS | |

NOTES -
1. PERMISSIBLE REFRIGERANT = ONLY R-454B; ASHRAE 34-2022 CLASSIFICATION = A2L.
   A. ODP (ASHRAE 34-2022) = 0
   B. GWP (ASHRAE 34-2022) = 466
2. PROVIDE A SELECTION FOR 30% PROPYLENE GLYCOL AS WORKING FLUID FOR HEATING HOT WATER AND CHILLED WATER LOOP.
3. PROVIDE AND INCLUDE FOLLOWING ACCESSORIES -
   A. INTERNAL HEAT TRACE - SINGLE POINT POWER CONNECTION - SEE ELECTRICAL DOCUMENTS FOR ADDITIONAL INFORMATION ON VOLTAGE/PHASE OF THE CONNECTION.
   B. MANUFACTURER FURNISHED, MANUFACTURER WIRED AND MANUFACTURER PROGRAMMED HUMAN MACHINE INTERFACE CONTROL PANEL ON ASHP-1A. SEE MECHANICAL CONTROLS FOR ADDITIONAL INFORMATION.
   C. SOUND ATTENUATION KIT AND SOUND ATTENUATION COMPRESSOR WRAP.
   D. FAN BAFFLE TO DEFLECT AIRFLOW FROM CONDENSER FANS.
4. SCHEDULED CAPACITIES -
   A. SCHEDULED CAPACITY FOR LINE ITEM 'ASCH-1x' INDICATE DESIGN REQUIREMENT.
   B. SCHEDULED CAPACITIES FOR INDIVIDUAL MODULES ARE SELECTION CRITERIA - CONTRACTOR MUST SELECT INDIVIDUAL ARRAYS WITHIN THE MODULE, TO MEET OR EXCEED THESE CAPACITIES.

NOTES -
4. PROVIDE A POINT OF POWER FOR EACH MODULE (4 POINTS TOTAL). ALL ELECTRICAL REQUIREMENTS OF THESE MODULES SHALL COMPLY WITH ELECTRICAL DOCUMENTS.
5. CONTRACTOR SHALL PROVIDE ALL THE RELEVANT DOCUMENTS TO MANUFACTURER FOR SELECTING THIS EQUIPMENT -
   A. PIPING FLOOR PLANS
   B. RELEVANT ARCHITECTURAL, LANDSCAPE AND ELECTRICAL FLOOR PLANS REFERENCING THE INSTALLATION LOCATION OF HEAT PUMP.
   C. SYSTEM DIAGRAM (M-602DG) AND CONTROL PLANS (M-806CP THROUGH M-809CP)
   D. EQUIPMENT SPECIFICATION 236423
   E. ENLARGED PLAN M-402QP
6. MANUFACTURER SHALL QUANTIFY THE DE-RATE OF THE MACHINE BASED ON THE LAYOUT AND PROVIDED PLANS (ARCHITECTURAL AND LANDSCAPE) AND PROVIDE THIS QUANTIFIED VALUE TO IN SUBMITTAL. SUBMITTAL DEFICIENT OF THIS VALUE WILL BE RETURNED WITHOUT REVIEW.

NOTES -
6. REDUNDANCIES -
   A. MODULE ASHP-1D IS MECHANICALLY REDUNDANT.
   B. MODULE ASHP-1D IS NOT ELECTRICALLY REDUNDANT.
7. PROVIDE COMPLETE PROGRAMMED CONTROLLER SEQUENCES (INCLUDING MANUFACTURER DEFAULTS INCLUDED WITH ATTACHED CONTROLLER) FOR OWNER'S REFERENCE.
8. COORDINATE AND ENGAGE THE MANUFACTURER TO DEVELOP PROGRAMMING REQUIREMENTS OF 230900 ASSOCIATED WITH PURCHASED PRODUCT. COORDINATE PROGRAMMING REQUIREMENTS WITH ISSUED DOCUMENTS AND MANUFACTURER PROGRAMMED SEQUENCES TO ENSURE THAT BAS OPERATES IN ACCORDANCE WITH THE PROJECT REQUIREMENTS AND DESIGN INTENT.

NOTES -
9. SOUND POWER LEVELS (AHRI 370 DEFINITIONS) SHALL NOT EXCEED FOLLOWING VALUES FOR THE ARRAY AT 100% ARRAY LOAD AND 18°F AMBIENT TEMPERATURE
   A. 63 HZ - 90DB
   B. 125 HZ - 92 DB
   C. 250 HZ - 92 DB
   D. 500 HZ - 90 DB
   E. 1000 HZ - 86 DB
   F. 2000 HZ - 83 DB
   G. 4000 HZ - 80 DB
   H. 8000 HZ - 78 DB
   I. LWA = 92 DB(A)

# BOILER SCHEDULE (HOT WATER)

| TAG | SYSTEM | LOCATION | TYPE | SELECTION CRITERIA | | | | | NATURAL GAS FLOW RATE | ELECTRICAL DATA | | VENT STACK | OPERATING WEIGHT | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | CAPACITY (SEE NOTE 1) | FLOW | EWT | LWT | MAX PD | | VOLTAGE | PHASE | | | MANUFACTURER | MODEL | |
| B-1 | HEATING HOT WATER | BASEMENT MECHANICAL ROOM | TWO PIPE; NATURAL DRAFT; CONDENSING; NATURAL GAS FIRED | 480 MBH | 70 GPM | 115 °F | 130 °F | 10.0 ftH2O | 490 ft³/h | 120 V | 1 | 4" | 675 lb | FULTON | MARK II | |
| B-2 | HEATING HOT WATER | BASEMENT MECHANICAL ROOM | TWO PIPE; NATURAL DRAFT; CONDENSING; NATURAL GAS FIRED | 480 MBH | 70 GPM | 115 °F | 130 °F | 10.0 ftH2O | 490 ft³/h | 120 V | 1 | 4" | 675 lb | FULTON | MARK II | |

NOTE -
1. OUTPUT CAPACITY AT RETURN WATER TEMPERATURE OF 115°F FOR 30% PG. PRODUCT SUBMITTAL MUST INDICATE THERMAL EFFICIENCY AT SYSTEM'S **ACTUAL** RETURN WATER TEMPERATURE. ADDITIONALLY, INCLUDE AHRI-1500 RATING OF PRODUCT.
2. PRODUCT MANUFACTURER CONTROLLER PRE-PROGRAMMED WITH SEQUENCES TO STAGE/CONTROL THE BOILERS. SEE M-850CP, M-806-CP, M-807CP AND M-808CP FOR ADDITIONAL INFORMATION.
3. MANUFACTURER MUST MAKE CONTROLLER AND REQUIRED POINTS AVAILABLE FOR INTEGRATION INTO BAS THROUGH BACNET MS/TP COMMUNICATION PROTOCOL.

# PUMP SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | FLUID | TEMPERATURE | DUTY POINT | | | | | MINIMUM NPSH REQ'D | PIPE CONNECTIONS | | MOTOR DATA | | | | | OPERATING WEIGHT | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | FLOW | DISCHARGE HEAD | MINIMUM EFFICIENCY | SPEED | BHP | | SUCTION | DISCHARGE | TYPE OF CONTROL | HP | RPM | VOLTAGE | PHASE | | MANUFACTURER | MODEL | |
| CHWP-1 | CHILLED WATER - SECONDARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 42 °F | 250 GPM | 100 ftH2O | 64 | 1588 | 10.1 | 4.07 ft | 4 | 4 | VFD | 20 | 1800 | 460 V | 3 | 610 lb | BELL&GOSSETT | e-80 | NOTE 1, 3 |
| CHWP-2 | CHILLED WATER - SECONDARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 42 °F | 250 GPM | 100 ftH2O | 64 | 1588 | 10.1 | 4.07 ft | 4 | 4 | VFD | 20 | 1800 | 460 V | 3 | 610 lb | BELL&GOSSETT | e-80 | NOTE 1, 2, 3 |
| CHWP-3 | CHILLED WATER - PRIMARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 57 °F | 250 GPM | 50 ftH2O | 67 | 1865 | 4.85 | 7.49 ft | 4 | 4 | ECM | 5.5 | 1800 | 460 V | 3 | 270 lb | BELL&GOSSETT | e-80X | |
| CHWP-4 | CHILLED WATER - PRIMARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 57 °F | 250 GPM | 50 ftH2O | 67 | 1865 | 4.85 | 7.49 ft | 4 | 4 | ECM | 5.5 | 1800 | 460 V | 3 | 270 lb | BELL&GOSSETT | e-80X | NOTE 1, 2 |
| ERFP-1 | ENERGY RECOVERY FLUID SYSTEM | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 79 °F | 25 GPM | 50 ftH2O | 34 | 1228 | 0.948 | 2.38 ft | 1 1/2 | 1 1/2 | ECM | 5.5 | 2000 | 460 V | 3 | 190 lb | BELL&GOSSETT | e-80X | |
| ERFP-2 | ENERGY RECOVERY FLUID SYSTEM | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 79 °F | 25 GPM | 50 ftH2O | 34 | 1228 | 0.948 | 2.38 ft | 1 1/2 | 1 1/2 | ECM | 5.5 | 2000 | 460 V | 3 | 190 lb | BELL&GOSSETT | e-80X | NOTE 2 |
| FPP-1 | AHU-1 FREEZE PROTECTION | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 42 °F | 25 GPM | 25 ftH2O | 59 | 1677 | 0.269 | 3.86 ft | 1 1/4 | 1 1/4 | ECM | 0.5 | 1800 | 115 V | 1 | 55 lb | BELL&GOSSETT | e-90 | |
| HHWP-1 | HEATING HOT WATER - SECONDARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 130 °F | 160 GPM | 150 ftH2O | 58 | 1664 | 10.5 | 4.34 ft | 3 | 3 | VFD | 20 | 1800 | 460 V | 3 | 575 lb | BELL&GOSSETT | e-80 | NOTE 1, 3 |
| HHWP-2 | HEATING HOT WATER - SECONDARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 130 °F | 160 GPM | 150 ftH2O | 58 | 1664 | 10.5 | 4.34 ft | 3 | 3 | VFD | 20 | 1800 | 460 V | 3 | 575 lb | BELL&GOSSETT | e-80 | NOTE 1, 2, 3 |
| HHWP-3 | HEATING HOT WATER - PRIMARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 115 °F | 160 GPM | 50 ftH2O | 60 | 1652 | 3.43 | 4.43 ft | 2 1/2 | 2 1/2 | ECM | 5.5 | 1800 | 460 V | 3 | 255 lb | BELL&GOSSETT | e-80X | |
| HHWP-4 | HEATING HOT WATER - PRIMARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 115 °F | 160 GPM | 50 ftH2O | 60 | 1652 | 3.43 | 4.43 ft | 2 1/2 | 2 1/2 | ECM | 5.5 | 1800 | 460 V | 3 | 255 lb | BELL&GOSSETT | e-80X | NOTE 2 |

NOTES -
1. PROVIDE SUCTION DIFFUSERS FOR PUMP.
2. REDUNDANT PUMP; PROVIDING N+1 REDUNDANCY ON PUMP SET.
3. PROVIDE CONCRETE VIBRATION ISOLATION INERTIA BASE.

# SUCTION DIFFUSER SCHEDULE

| TAG | TYPE | PUMP | PUMP SIZE | INLET SIZE | FLOW | PRESSURE DROP | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|
| SD-CHWP-1 | FLANGED INLET / FLANGED OUTLET | CHWP-1 | 4 | 5 | 250 GPM | 1.1 ftH2O | BELL & GOSSETT | FE-3X | |
| SD-CHWP-2 | FLANGED INLET / FLANGED OUTLET | CHWP-2 | 4 | 5 | 250 GPM | 1.1 ftH2O | BELL & GOSSETT | FE-3X | |
| SD-HHWP-1 | FLANGED INLET / FLANGED OUTLET | HHWP-1 | 3 | 4 | 160 GPM | 1.1 ftH2O | BELL & GOSSETT | ED-3X | |
| SD-HHWP-2 | FLANGED INLET / FLANGED OUTLET | HHWP-2 | 3 | 4 | 160 GPM | 1.1 ftH2O | BELL & GOSSETT | ED-3X | |

# EXPANSION TANK SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | TEMPERATURE | | SYSTEM VOLUME | TANK VOLUME | MIN ACCEPTANCE VOLUME | AIR CHARGE PRESSURE | RELIEF VALVE | OPERATING WEIGHT | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | MIN | MAX | | | | | | | MANUFACTURER | MODEL | |
| ET-CHW | CHILLED WATER (30% PG) | BASEMENT MECHANICAL ROOM | BLADDER | 40 °F | 70 °F | 1500.0 gal | 26.0 gal | 12.8 gal | 30 psi | 35 psi | 123.00 lb | BELL & GOSSETT | B-100 | |
| ET-ERF | ENERGY RECOVERY FLUID (30% PG) | BASEMENT MECHANICAL ROOM | BLADDER | 20 °F | 85 °F | 400.0 gal | 10.0 gal | 5.4 gal | 30 psi | 35 psi | 123.00 lb | BELL & GOSSETT | B-35 | |
| ET-HHW | HEATING HOT WATER (30% PG) | BASEMENT MECHANICAL ROOM | BLADDER | 50 °F | 130 °F | 1500.0 gal | 44.0 gal | 18.5 gal | 30 psi | 35 psi | 517.00 lb | BELL & GOSSETT | B-165 | |

# GLYCOL FEEDER SCHEDULE

| TAG | LOCATION | SYSTEM | TANK SIZE | CAPACITY | | PRESSURE RANGE | PUMP DATA | | ELECTRICAL DATA | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | FLOW | PRESSURE | | HP | SPEED | VOLTAGE | PHASE | MANUFACTURER | MODEL | |
| GFS-1 | BASEMENT MECHANICAL ROOM | HEATING HOT WATER AND ENERGY RECOVERY SYSTEM | 55 gal | 5 GPM | 60 psi | 10 psi - 60 psi | 0.75 | 3600 | 115 V | 1 | XYLEM | GMU-60 | |
| GFS-2 | BASEMENT MECHANICAL ROOM | CHILLED WATER SYSTEM | 55 gal | 5 GPM | 60 psi | 10 psi - 60 psi | 0.75 | 3600 | 115 V | 1 | XYLEM | GMU-60 | |

NOTES -
1. SINGLE POINT POWER CONNECTION TO UNIT.
2. PROVIDE COMPLETE WITH LOCAL DISCONNECT SWITCH, BAS INTERFACE AND ALARM, PRESSURE REDUCING VALVE BETWEEN THE UNIT AND CONNECTION TO PIPING SYSTEM.

# BUFFER TANK SCHEDULE

| TAG | LOCATION | SYSTEM | TYPE | CONNECTION SIZE | TANK MIN VOLUME | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MANUFACTURER | MODEL | |
| BT-CHW | BASEMENT MECHANICAL ROOM | CHILLED WATER | HORIZONTAL | 6 in | 850 gal | TACO | BTH-850F | |
| BT-HHW | BASEMENT MECHANICAL ROOM | HEATING HOT WATER | HORIZONTAL | 6 in | 850 gal | TACO | BTH-850F | |

NOTES -
1. ALL TANKS SHALL COMPLY WITH ASME CONSTRUCTION.
2. PROVIDE WITH HAND HOLE ON TANK FOR CLEANING.
3. PROVIDE WITH INTERNAL BAFFLE, AIR VENTS, LEGS FOR HORIZONTAL INSTALLATION AND FACTORY ELASTOMERIC INSULATION (WITH REMOVABLE SECTIONS AT SERVICE LOCATIONS).

# AIR SEPARATOR SCHEDULE

| TAG | LOCATION | TYPE | SYSTEM | CONNECTION SIZE | DESIGN FLOW RATE | MAX PRESSURE DROP | OPERATING WEIGHT | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | MANUFACTURER | MODEL | |
| AS-CHW | BASEMENT MECHANICAL ROOM | COALESCING STYLE | CHILLED WATER | 5 | 250 GPM | 1.0 ftH2O | 362 lb | BELL&GOSSETT | CRS-5F MAG | |
| AS-ERF | BASEMENT MECHANICAL ROOM | COALESCING STYLE | HEATING HOT WATER | 4 | 160 GPM | 1.0 ftH2O | 209 lb | BELL&GOSSETT | CRS-4F MAG | |
| AS-HHW | BASEMENT MECHANICAL ROOM | COALESCING STYLE | ENERGY RECOVERY FLUID | 2 | 25 GPM | 1.0 ftH2O | 110 lb | BELL&GOSSETT | CRS-2F MAG | |

NOTES -
1. ALL SEPARATORS SHALL COMPLY WITH ASME CONSTRUCTION.
2. PROVIDE FLANGED CONNECTION. GROOVED IS NOT ACCEPTABLE.
3. PROVIDE MAGNETIC INSERT ACCESSORY.
4. COALESCING MEDIA SHALL BE STAINLESS STEEL.

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REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
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DRAWING TITLE: MECHANICAL SCHEDULES
WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| OP | OP | IP |

SHEET NO. 279 OF 378

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# ENERGY RECOVERY HYDRONIC COIL SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | AIRFLOW | TOTAL CAPACITY | SENSIBLE CAPACITY | QTY OF COILS IN BANK | MIN ROWS | MAX FINS PER INCH | FACE VELOCITY | COIL SIZE DATA | | | AIRSIDE DATA | | | | | WATERSIDE DATA | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | FACE AREA | LENGTH | HEIGHT | EAT DB | EAT WB | LAT DB | LAT WB | MAX AIR PD | EWT | LWT | FLOW | WATER MAX PD | MANUFACTURER | MODEL | |
| ERC-EF (SUMMER) | EF-1 ENERGY RECOVERY COIL | KEEPER WORKSPACE | HYDRONIC | 6,500 CFM | 31 MBH | 31 MBH | 1 | 12 | 5 | 508 ft/min | 13 SF | 1' - 0" | 2' - 1" | 72.00 °F | 60.08 °F | 76.40 °F | 61.65 °F | 0.95 in-wg | 79 °F | 74 °F | 13 GPM | 10.0 ftH2O | AEROFIN | | |
| ERC-EF (WINTER) | EF-1 ENERGY RECOVERY COIL | KEEPER WORKSPACE | HYDRONIC | 6,500 CFM | 130 MBH | 130 MBH | 1 | 12 | 5 | 508 ft/min | 13 SF | 1' - 0" | 2' - 1" | 72.00 °F | 54.36 °F | 53.50 °F | 46.25 °F | 0.95 in-wg | 45 °F | 60 °F | 18 GPM | 10.0 ftH2O | AEROFIN | | |

NOTES -
1. PROVIDE SINGULAR ENERGY RECOVERY COIL. COIL PERFORMANCE IS RATED FOR A SINGULAR ERC-1 FOR SUMMER AND WINTER DESIGN DAY.

# RADIANT SLAB MANIFOLD SYSTEM SCHEDULE

| TAG | SYSTEM | LOCATION | MANIFOLD TYPE | ROOM SETPOINT | MAXIMUM PERMISSIBLE SURFACE TEMPERATURE | APPROXIMATE AREA FOR HEAT TRANSFER | MINIMUM RADIANT HEAT OUTPUT | LAYOUT | | | HYDRONIC SYSTEM | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | QUANTITY OF CIRCUITS | RADIANT PIPE SIZE | PIPING ON CENTER DISTANCE | TOTAL HHW FLOW RATE | SUPPLY WATER TEMPERATURE | MAXIMUM PERMISSIBLE PRESSURE DROP ACROSS ANY CIRCUIT | MANUFACTURER | MODEL | |
| RSS-1 | HEATING HOT WATER | DEN-1 | STAINLESS STEEL; 2 LOOP | 72 °F | 85 °F | 129 ft² | 30 Btu/(h·ft²) | 1 | 3/4 | 12 in | 1.0 GPM | 130 °F | 5 ftH2O | UPONOR | A2720202 | |
| RSS-2 | HEATING HOT WATER | DEN-2 | STAINLESS STEEL; 2 LOOP | 72 °F | 85 °F | 129 ft² | 30 Btu/(h·ft²) | 1 | 3/4 | 12 in | 1.0 GPM | 130 °F | 5 ftH2O | UPONOR | A2720202 | |
| RSS-3 | HEATING HOT WATER | DEN-3 | STAINLESS STEEL; 2 LOOP | 72 °F | 85 °F | 196 ft² | 30 Btu/(h·ft²) | 1 | 3/4 | 12 in | 1.5 GPM | 130 °F | 5 ftH2O | UPONOR | A2720202 | |
| RSS-4 | HEATING HOT WATER | CUBBING | STAINLESS STEEL; 2 LOOP | 72 °F | 85 °F | 102 ft² | 30 Btu/(h·ft²) | 1 | 3/4 | 12 in | 1.0 GPM | 130 °F | 5 ftH2O | UPONOR | A2720202 | |
| RSS-5 | HEATING HOT WATER | QUARENTINE | STAINLESS STEEL; 2 LOOP | 72 °F | 85 °F | 102 ft² | 30 Btu/(h·ft²) | 1 | 3/4 | 12 in | 1.0 GPM | 130 °F | 5 ftH2O | UPONOR | A2720202 | |

NOTES -
1. SCHEDULE INDICATE BOD FOR STAINLESS STEEL MANIFOLD. CONTRACTOR SHALL HOUSE THE MANIFOLD IN A PANEL BOARD AT INDICATED LOCATION ON DRAWINGS.
2. PANEL BOARD SHALL HOUSE ACCESSORIES AS INDICATED ON SEE M-504DT.
3. PANEL BOARD SHALL BE WALL MOUNTED (NON-RECESSED) TO FACILITATE ROUTING PEX PIPING.
4. INCLUDE A FLOATING TYPE MIXING VALVE WITH VALVE COEFFICIENT OF 1 (CV=1). BOD = BELIMO ZONE315N. INCLUDED ACTUATOR SHALL BE CAPABLE OF BAS INTEGRATION.
5. INCLUDE A RECIRCULATOR PUMP FOR EACH MANIFOLD COMPLIANT WITH DETAIL M-504DT. BOD - TACO 006 CIRCULATOR.
6. INCORPORATE MIXING STATION WITHIN THE RADIANT SLAB PANEL BOARD.
7. CONTRACTOR MAY PROVIDE ALL REQUIRED ACCESSORIES IN A SINGLE MANUFACTURER ASSEMBLED PANEL.

# HEAT TRACE SCHEDULE

| TAG | SYSTEM | PANEL LOCATION | TYPE | WATTS PER FOOT | VOLTAGE | PHASE | INPUT POWER | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| HTC-1 | DIVISION 23 HEAT TRACING | BASEMENT MECHANICAL ROOM | PARALLEL SELF REGULATING | 10 VA | 120 V | 1 | 5000 VA | CHROMALOX | SRL | |

NOTES
1. REFER TO PIPING INSULATION SPECIFICATION FOR CHILLED WATER AND HEATING HOT WATER AND FLOOR PLANS. HEAT TRACE ELEMENT SHALL BE COMPATIBLE WITH THE PURCHASED INSULATION.
2. VALIDATE PRODUCT SELECTION WITH ELECTRICAL DOCUMENTS.
3. BAS SHALL BE FULLY CAPABLE OF STARTING/STOPPING ANY/ALL HEAT TRACE CIRCUIT.
4. SEE MECHANICAL CONTROLS FOR ADDITIONAL INFORMATION. PROVIDE A DISTRIBUTION BOX ACCESSORY (BOD - MARSHALL BOX CHROMALOX) TO PROVIDE FOUR (4) LOADSIDE CIRCUITS.
5. PROVIDE ANY/ALL ACCESSORIES NEEDED TO TIE THE HEAT TRACE PANEL TO THE BAS.
6. PROVIDE HEAT TRACING ON ALL VALVES, AND PIPE ELBOWS.
7. HEAT TRACE HEATING ELEMENT MUST BE INSTALLED AS INSTRUCTED BY MANUFACTURER. ENGAGE MANUFACTURER TO REVIEW INSTALLATION OF HEATING ELEMENT ON THE PIPING ON APPROVED SHOP DRAWINGS.

# UNIT HEATER SCHEDULE (HOT WATER)

| TAG | SYSTEM | LOCATION | BOD CATALOG CAPACITY | DESIGN CAPACITY (SEE NOTE 1 AND NOTE 2) | AIRSIDE DATA | | | WATERSIDE DATA | | | | ELECTRICAL | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AIRFLOW | EAT | MIN LAT | EWT | LWT | FLOW RATE | MAX WATER PD | HP | FAN RPM | VOLTAGE | PHASE | MANUFACTURER | MODEL | |
| UH-1 | HOT WATER | LEVEL 1 | 13 MBH | 5 MBH | 400 CFM | 72 °F | 84 °F | 130 °F | 115 °F | 1.30 GPM | 1 ftH2O | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-2 | HOT WATER | LEVEL 1 | 13 MBH | 5 MBH | 400 CFM | 72 °F | 84 °F | 130 °F | 115 °F | 1.30 GPM | 1 ftH2O | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-3 | HOT WATER | LEVEL 1 | 13 MBH | 5 MBH | 400 CFM | 72 °F | 84 °F | 130 °F | 115 °F | 1.30 GPM | 1 ftH2O | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-4 | HOT WATER | LEVEL 1 | 13 MBH | 5 MBH | 400 CFM | 72 °F | 84 °F | 130 °F | 115 °F | 1.30 GPM | 1 ftH2O | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-5 | HOT WATER | LEVEL 1 | 13 MBH | 5 MBH | 400 CFM | 72 °F | 84 °F | 130 °F | 115 °F | 1.30 GPM | 1 ftH2O | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-6 | HOT WATER | LEVEL 1 | 13 MBH | 5 MBH | 400 CFM | 72 °F | 84 °F | 130 °F | 115 °F | 1.30 GPM | 1 ftH2O | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | |
| UH-7 | HOT WATER | LEVEL 1 | 13 MBH | 5 MBH | 400 CFM | 72 °F | 84 °F | 130 °F | 115 °F | 1.30 GPM | 1 ftH2O | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | |

NOTES -
1. PROVIDE SELECTION FOR SCHEDULED 'DESIGN CAPACITY'.
2. SCHEDULED 'DESIGN CAPACITY' ACCOUNTS FOR A DE-RATE FACTOR OF 0.424 BASED ON ENTERING AIR TEMPERATURE OF 72°F AND ENTERING WATER TEMPERATURE OF 130°F. ADDITIONAL DE-RATE FACTOR OF 0.912 IS APPLIED FOR WORKING FLUID (30% PROPYLENE GLYCOL), PER DAIKIN (BOD MANUFACTURER) SELECTION CATALOG. CONTRACTOR SHALL SCHEDULE AND DOCUMENT DERATE FACTOR ASSOCIATED WITH MANUFACTURER RECOMMENDED DE-RATE FACTOR FOR PURCHASED PRODUCT.
3. PROVIDE CUSTOM FABRICATED WELDED STEEL WIRE CAGE ENCLOSURE AROUND UNIT HEATER TO PROTECT EQUIPMENT FROM DAMAGE DUE TO ANIMAL. CAGE SHALL BE RIGIDLY MOUNTED TO STRUCTURE AND SHALL ALLOW ADEQUATE AIRFLOW, SERVICE ACCESS, AND CLEARANCES REQUIRED FOR PROPER OPERATION. CAGE DIMENSIONS AND SPACING SHALL BE COORDINATED WITH AND INSTALLED PER UNIT MANUFACTURER'S RECOMMENDED CLEARANCES AND APPROVED SUBMITTALS. PROVIDE REMOVABLE OR HINGED ACCESS PANEL AS REQUIRED FOR MAINTENANCE.

# FLOOR RADIANT HEATERS

| TAG | SYSTEM | LOCATION | PHYSICAL DIMENSIONS | | AIRSIDE DATA | | | | | WATERSIDE DATA | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | HEATER LENGTH | HEATER WIDTH | HEATING CAPACITY | AIRFLOW | EAT | LAT | STATIC PRESSURE | FLOW | EWT | LWT | MANUFACTURER | MODEL | |
| HTR-1 | HOT WATER | DAY ROOM | 48 in | 6 in | 5.83 MBH | 320 CFM | 72 °F | 89 °F | 0.05 in-wg | 2 GPM | 130 °F | 124 °F | PRICE | LFH | |
| HTR-2 | HOT WATER | DAY ROOM | 48 in | 6 in | 5.83 MBH | 320 CFM | 72 °F | 89 °F | 0.05 in-wg | 2 GPM | 130 °F | 124 °F | PRICE | LFH | |
| HTR-3 | HOT WATER | DAY ROOM | 48 in | 6 in | 5.83 MBH | 320 CFM | 72 °F | 89 °F | 0.05 in-wg | 2 GPM | 130 °F | 124 °F | PRICE | LFH | |
| HTR-4 | HOT WATER | DAY ROOM | 48 in | 6 in | 5.83 MBH | 320 CFM | 72 °F | 89 °F | 0.05 in-wg | 2 GPM | 130 °F | 124 °F | PRICE | LFH | |

NOTES -
1. PROVIDE CUSTOM FABRICATED WELDED STEEL WIRE CAGE ENCLOSURE AROUND HEATER TO PROTECT EQUIPMENT FROM DAMAGE DUE TO ANIMAL. CAGE SHALL BE RIGIDLY MOUNTED TO STRUCTURE AND SHALL ALLOW ADEQUATE AIRFLOW, SERVICE ACCESS, AND CLEARANCES REQUIRED FOR PROPER OPERATION. CAGE DIMENSIONS AND SPACING SHALL BE COORDINATED WITH AND INSTALLED PER UNIT MANUFACTURER'S RECOMMENDED CLEARANCES AND APPROVED SUBMITTALS. PROVIDE REMOVABLE OR HINGED ACCESS PANEL AS REQUIRED FOR MAINTENANCE.

# FAN COIL UNIT SCHEDULE

| TAG | ZONE SERVED | PHYSICAL INSTALLATION LOCATION | SYSTEM | COOLING PERFORMANCE | | | | | | | | | | HEATING PERFORMANCE | | | | | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | TOTAL CAPACITY | TOTAL SENSIBLE CAPACITY | AIRSIDE DATA | | | | | CHILLED WATER COIL | | | | TOTAL HEATING CAPACITY | AIRSIDE DATA | | | HOT WATER COIL | | | | MANUFACTURER | MODEL | |
| | | | | | | AIRFLOW | AIR EAT DB | AIR EAT WB | AIR LAT DB | AIR LAT WB | FLOW | EWT (COOLING PLANT SETBACK) | LWT | WATER PD | | AIRFLOW | EAT | LAT | EWT | LWT | FLOW | WATER PD | | | |
| FCU-1 | ELECTRICAL ROOM | BASEMENT ELECTRICAL ROOM | HEATING AND COOLING 4 PIPE; HORIZONTAL CEILING MOUNTED | 2.26 ton | 2.04 ton | 1,200 CFM | 75.00 °F | 62.50 °F | 56.30 °F | 54.80 °F | 5 GPM | 47 °F | 58 °F | 5.0 ftH2O | 12 MBH | 1,200 CFM | 75.00 °F | 84.23 °F | 130 °F | 115 °F | 2 GPM | 5.0 ftH2O | DAIKIN | FCHC | |
| FCU-2 | MECHANICAL ROOM | BASEMENT MECHANICAL ROOM | COOLING TWO PIPE; HORIZONTAL CEILING MOUNTED | 4.25 ton | 4.22 ton | 2,550 CFM | 75.00 °F | 62.50 °F | 56.90 °F | 55.70 °F | 10 GPM | 47 °F | 58 °F | 5.0 ftH2O | | | | | | | | | DAIKIN | BCHD | |
| FCU-3 | AV/IT ROOM | BASEMENT MECHANICAL ROOM | COOLING TWO PIPE;VERTICAL FLOOR MOUNTED | 4.60 ton | 4.60 ton | 3,500 CFM | 75.00 °F | 63.00 °F | 60.70 °F | 57.80 °F | 12 GPM | 48 °F | 58 °F | 5.0 ftH2O | | | | | | | | | DAIKIN | BCVD | |
| FCU-4 | EMERGENCY ELECTRICAL ROOM | BASEMENT ELECTRICAL ROOM | HEATING AND COOLING 4 PIPE; HORIZONTAL CEILING MOUNTED | 1.17 ton | 1.00 ton | 1,200 CFM | 77.13 °F | 67.25 °F | 58.00 °F | 58.00 °F | 9 GPM | 47 °F | 56 °F | 5.0 ftH2O | 36 MBH | 1,200 CFM | 60.92 °F | 88.62 °F | 130 °F | 115 °F | 5 GPM | 5.0 ftH2O | DAIKIN | FCHC | |

# AIR CURTAIN SCHEDULE

| TAG | LOCATION | HEATING DATA | | | HP OF INCLUDED FAN MOTOR | VOLTAGE | PHASE | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| | | HEATING CAPACITY | SG_EWT | FLOW | | | | | | |
| AC-1 | LEVEL 1; LOADING DOCK | 54000 Btu/h | 130 °F | 6 GPM | 0.166667 | 115 V | 1 | MARS | QP872-1VA-AL | |

NOTES -
1. PROVIDE FOLLOWING FACTORY ACCESSORIES -
   A. MOTOR CONTROL PANEL WITH BAS INTEGRATION
   B. REMOTE MOUNTED THERMOSTAT
   C. DOOR SWITCH
2. A-WEIGHT SOUND POWER LEVEL SHALL BE 51 DBA OR LESS, AT MAX FAN SPEED.

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Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL SCHEDULES
WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
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## DUCT SILENCER SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | AIRFLOW | VELOCITY | MAX AIR PD | SIZE DATA | | | | DYNAMIC INSERTION LOSS BY OCTAVE BAND (DB RE 10^(-12)) W | | | | | | | | BASIS OF DESIGN | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | DIAMETER | WIDTH | HEIGHT | LENGTH | 1 (63 HZ) | 2 (125 HZ) | 3 (250 HZ) | 4 (500 HZ) | 5 (1,000 HZ) | 6 (2,000 HZ) | 7 (4,000 HZ) | 8 (8,000 HZ) | MANUFACTURER | |
| DS-E1 | EXHAUST | LEVEL1 | EX-RDNM | 6,500 CFM | -990 ft/min | 0.10 in-wg | | 2' - 8" | 2' - 6" | 8' - 0" | 4 | 6 | 12 | 20 | 11 | 8 | 6 | 5 | PRICE | |
| DS-O1 | OUTSIDE AIR | BASEMENT MECHANICAL ROOM | EX-RDNM | 7,500 CFM | -1,920 ft/min | 0.10 in-wg | | 3' - 0" | 1' - 8" | 6' - 0" | 6 | 9 | 17 | 33 | 31 | 20 | 14 | 9 | PRICE | |
| DS-S1 | SUPPLY AIR | BASEMENT MECHANICAL ROOM | EX-RDNM | 7,500 CFM | 1,455 ft/min | 0.10 in-wg | | 3' - 8" | 1' - 6" | 5' - 0" | 3 | 5 | 10 | 23 | 24 | 16 | 10 | 7 | PRICE | |
| DS-VAVE-6 | EXHAUST VAV -6 | LEVEL1 | EX-RDNM | 900 CFM | -900 ft/min | 0.10 in-wg | | 1' - 0" | 1' - 0" | 3' - 0" | 2 | 4 | 6 | 13 | 9 | 7 | 5 | 3 | PRICE | |

NOTES:
1. TYPE:
   EX - EXTENDED CASING
   R - RECTANGULAR
   D - DISSIPATIVE
   NM - NO MEDIA
2. LENGTH SHOWN IS CENTERLINE LENGTH.
3. PRESSURE DROP AS DETERMINED PER ASTM E477-13 IN A NVLAP-ACCREDITED ACOUSTICAL LABORATORY, SHALL INCLUDE SYSTEM EFFECT FOR NEARBY DUCT ELEMENTS.
4. MANUFACTURER SHALL PROVIDE, FOR APPROVAL, PROFESSIONAL ENGINEER STAMPED ACOUSTICAL CALCULATIONS FOR ALL SYSTEMS WITH SILENCERS DEMONSTRATING:
   A. THE RESULTANT DUCTBORNE SOUND LEVELS, INCLUDING AIRBORNE AND BREAKOUT NOISE, MEET THE REQUIRED CRITERIA
   B. THE RESULTANT DROP WITH SYSTEM EFFECTS DOES NOT EXCEED SCHEDULED VALUES

## VOLUME CONTROL BOX SCHEDULE - SUPPLY VAV WITH HYDRONIC REHEAT

| TAG | LOCATION | SERVICE TO | INLET SIZE | AIRFLOWS | | | | | MAX AIR PD | HEATING COIL | | | | | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | OCCUPIED MODE | | | UNOCCUPIED MODE | | | HEATING COIL CAPACITY | HEATING AIRFLOW | ENTERING AIR TEMPERATURE | LEAVING AIR TEMPERATURE | ENTERING WATER TEMPERATURE | LEAVING WATER TEMPERATURE | FLOW | WATER PRESSURE DROP | MANUFACTURER | MODEL | |
| | | | | DESIGN | VELOCITY AT DESIGN (MAX) AIRFLOW | MINIMUM | VENTILATION MINIMUM | VAV MINIMUM | | | | | | | | | | | | |
| VAV-1 | KEEPER CORRIDOR | DAYROOM | 14 | 1,700 CFM | 1,590 ft/min | 1,400 CFM | 135 CFM | 856 CFM | 0.50 in-wg | 53 MBH | 1,400 CFM | 60 °F | 95 °F | 130 °F | 115 °F | 7.06 GPM | 5.0 ftH2O | PRICE | SDV | |
| VAV-2 | KEEPER CORRIDOR | KEEPER CORRIDOR | 10 | 800 CFM | 1,467 ft/min | 800 CFM | 30 CFM | 309 CFM | 0.50 in-wg | 30 MBH | 800 CFM | 60 °F | 95 °F | 130 °F | 115 °F | 4.03 GPM | 5.0 ftH2O | PRICE | SDV | |
| VAV-3 | KITCHEN PREP | KITCHEN PREP | 10 | 700 CFM | 1,283 ft/min | 700 CFM | 75 CFM | 309 CFM | 0.50 in-wg | 26 MBH | 700 CFM | 60 °F | 95 °F | 130 °F | 115 °F | 3.53 GPM | 5.0 ftH2O | PRICE | SDV | |
| VAV-4 | KEEPER CORRIDOR | DEN-1, DEN-2 AND DEN-3 | 10 | 1,100 CFM | 2,017 ft/min | 1,100 CFM | 105 CFM | 309 CFM | 0.50 in-wg | 42 MBH | 1,100 CFM | 60 °F | 95 °F | 130 °F | 115 °F | 5.54 GPM | 5.0 ftH2O | PRICE | SDV | |
| VAV-5 | KITCHEN PREP | CUBBING AND QUARENTINE | 10 | 600 CFM | 1,100 ft/min | 500 CFM | 50 CFM | 309 CFM | 0.50 in-wg | 19 MBH | 500 CFM | 60 °F | 95 °F | 130 °F | 115 °F | 2.52 GPM | 5.0 ftH2O | PRICE | SDV | |
| VAV-6 | KEEPER WORKSPACE | KEEPER WORKSPACE | 14 | 900 CFM | 842 ft/min | 445 CFM | 45 CFM | 445 CFM | 0.50 in-wg | 17 MBH | 450 CFM | 60 °F | 95 °F | 130 °F | 115 °F | 2.27 GPM | 5.0 ftH2O | PRICE | SDV | |
| VAV-7 | CORRIDOR ADJACENT TO DENS | ADJACENT CORRIDORS | 12 | 700 CFM | 891 ft/min | 700 CFM | 20 CFM | 198 CFM | 0.50 in-wg | 26 MBH | 700 CFM | 60 °F | 95 °F | 130 °F | 115 °F | 3.53 GPM | 5.0 ftH2O | PRICE | SDV | |

NOTES -
1. ACCOUNT FOR SPECIFIC GRAVITY OF 30% PROPYLENE GLYCOL WHEN SELECTING A PRODUCT.

## BAS PANEL SCHEDULE

| TAG | LOCATION | VOLTAGE | PHASE | REMARKS |
|---|---|---|---|---|
| BAS-MISC | BASEMENT MECHANICAL ROOM | 120 V | 1 | |
| BAS-CHW | BASEMENT MECHANICAL ROOM | 120 V | 1 | |
| BAS-HHW | BASEMENT MECHANICAL ROOM | 120 V | 1 | |
| BAS-AHU-1 | BASEMENT MECHANICAL ROOM | 120 V | 1 | |
| BAS-INTEGRATION | BASEMENT MECHANICAL ROOM | 120 V | 1 | |

GENERAL NOTES -
1. REFER TO SPECIFICATION SECTION 230900 FOR ADDITIONAL INFORMATION.
2. PROVIDE A DIVISION 23 CONTROL POWER TRANSFORMER AS REQUIRED TO POWER LOW VOLTAGE EQUIPMENT.
3. SHOWN LOCATIONS ON FLOOR PLANS ARE RECOMMENDATIONS ONLY. CONTROLS CONTRACTOR TO FIELD COORDINATE DESIRED LOCATION WITH OWNER AND OTHER TRADES AND DOCUMENT EXACT LOCATION IN SHOP DRAWINGS.
4. PROVIDE WALL RECESSED PANEL BOARD WITH DOUBLE DOOR, HINGED AT TWO EDGES.
   A. SEMI-RECESSED PANEL BOARD IS ACCEPTABLE WHERE THE PANEL DEPTH IS THICKER THAN WALL. SEE FLOOR PLANS.
5. SEE SYSTEM NETWORK ARCHITECTURE FOR ADDITIONAL INFORMATION.
6. PANELS ARE INTENDED TO
   A. PROVIDE POWER TO MOTORIZED VALVES, DAMPERS AND ACTUATORS/RELAYS.
   B. PROVIDE CENTRALIZED LOCATION TO HOUSE THE BAS CONTROLLER ASSOCIATED WITH SYSTEM. ALL POINTS SHOWN ON M8 SHALL BE INTEGRATED ON CONTROLLER(S) HOUSED IN THESE PANELS.

## VOLUME CONTROL BOX SCHEDULE - EXHAUST

| TAG | SYSTEM | INLET SIZE | AIRSIDE DATA | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|
| | | | AIRFLOW | VELOCITY AT DESIGN (MAX) AIRFLOW | MAX AIR PD | MANUFACTURER | MODEL | |
| VAVE-1 | EXHAUST | 14 in | 2,300 CFM | 2,152 ft/min | 0.50 in-wg | PRICE | SDE | |
| VAVE-2 | EXHUAST | 10 in | 800 CFM | 1,467 ft/min | 0.50 in-wg | PRICE | SDE | |
| VAVE-3 | EXHAUST | 10 in | 700 CFM | 1,283 ft/min | 0.50 in-wg | PRICE | SDE | |
| VAVE-4 | EXHAUST | 10 in | 1,100 CFM | 2,017 ft/min | 0.50 in-wg | PRICE | SDE | |
| VAVE-5 | EXHAUST | 10 in | 900 CFM | 1,650 ft/min | 0.50 in-wg | PRICE | SDE | |
| VAVE-6 | EXHAUST | 12 in | 1,200 CFM | 1,528 ft/min | 0.00 in-wg | PRICE | SDE | |
| VAVE-7 | EXHAUST | 8 in | 600 CFM | 1,719 ft/min | 0.50 in-wg | PRICE | SDE | |

## HIGH PERFORMANCE CONTROL VALVE SCHEDULE

| TAG | SYSTEM | TYPE | VALVE DIAMETER | SELECTION CRITERIA | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|
| | | | | DESIGN FLOW | DESIGN PRESSURE DROP | DESIGN COEFFICIENT (CV) | MANUFACTURER | MODEL | |
| CV-1 | BUILDING CHILLED WATER | HIGH PERFORMANCE SEGMENTED BALL | 3 in | 250 GPM | 5 psi | 116 | VSI-V | | |
| CV-2 | BUILDING HEATING HOT WATER | HIGH PERFORMANCE SEGMENTED BALL | 3 in | 160 GPM | 5 psi | 75 | VSI-V | | |

NOTES -
1. PROVIDE A HIGH PERFORMANCE SEGMENTED BALL VALVE WITH A TURNDOWN RATIO OF 300:1. PROVIDE VALVE WITH AN ELECTRIC ACTUATOR WITH A RESOLUTION OF 200 STEPS, 0.04VDC/0.5 DEGREES. BASIS OF DESIGN: VSI, SERIES 1000-X

## SMOKE EXHAUST FAN SCHEDULE

| TAG | LOCATION | SYSTEM | OPERATING POINT | | | | | MAXIMUM PERMISSIBLE PHYSICAL CONSTRUCTION SPEED LIMIT FOR FAN BODY | MOTOR | | | | OPERATING WEIGHT | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | AIRFLOW | STATIC PRESSURE SUMMARY | | BOD STATIC EFFICIENCY (%) | BOD FAN RPM | BOD BHP | | MAX RPM | HP | VOLTAGE | PHASE | | MANUFACTURER | MODEL | |
| | | | | EXTERNAL (MINIMUM REQUIREMENT) | TOTAL STATIC PRESSURE | | | | | | | | | | | | |
| SEF-1 | KITCHEN PREPARATION AREA | SMOKE EXHAUST SYSTEM | 5,000 CFM | 2.50 in-wg | 2.56 in-wg | 65 | 1770 | 3.11 | 3445 | 3600 | 5 | 460 V | 3 | 500 lb | GREENHECK | QEID-18-90 | |

NOTES -
1. VFD SHALL NOT DRIVE THE MOTOR TO A SPEED HIGHER THAN THE FAN'S PHYSICAL CONSTRUCTION LIMIT.
   A. LIMIT MAXIMUM VFD SPEED AT 3400 RPM, THAT IS, VFD AT 100% OR 60 HZ = 3400 RPM.
2. CONSTRUCTION AND CONTROLS SHALL BE UL 864 AND UUKL RATED.

## GRILLE, REGISTER, AND DIFFUSER SCHEDULE

| TAG | SYSTEM | TYPE | MINIMUM AIRLOW | MAXIMUM AIRLOW | NECK SIZE | LINEAR DIFFUSER DATA | | | FACE SIZE | MAX PD [IN-W.G] | MAX NC | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | SLOT QUANTITY | SLOT WIDTH | SLOT LENGTH | | | | MANUFACTURER | MODEL | |
| S1-1 | SUPPLY AIR | ALUMINUM SINGLE DEFLECTION | 80 | 180 | - | - | - | - | 8x6 | 0.1 | 16 | PRICE | 600 | |
| S1-2 | SUPPLY AIR | ALUMINUM SINGLE DEFLECTION | 180 | 225 | - | - | - | - | 10x6 | 0.1 | 16 | PRICE | 600 | |
| S1-3 | SUPPLY AIR | ALUMINUM SINGLE DEFLECTION | 150 | 360 | - | - | - | - | 12x8 | 0.1 | 16 | PRICE | 600 | |
| S1-4 | SUPPLY AIR | ALUMINUM SINGLE DEFLECTION | 220 | 500 | | | | | 12x12 | 0.1 | 16 | PRICE | 600 | |
| E1-1 | EXHAUST AIR | ALUMINUM SINGLE DEFLECTION | 80 | 180 | - | - | - | - | 10x8 | 0.1 | 16 | PRICE | 600 | |
| E1-2 | EXHAUST AIR | ALUMINUM SINGLE DEFLECTION | 180 | 225 | | | | | 12x8 | 0.1 | 16 | PRICE | 600 | |
| E1-3 | EXHAUST AIR | ALUMINUM SINGLE DEFLECTION | 80 | 320 | | | | | 14x10 | 0.1 | 16 | PRICE | 600 | |
| E1-4 | EXHAUST AIR | ALUMINUM SINGLE DEFLECTION | 150 | 360 | | | | | 16x10 | 0.1 | 16 | PRICE | 600 | |
| E1-5 | EXHAUST AIR | ALUMINUM SINGLE DEFLECTION | 220 | 500 | | | | | 16x14 | 0.1 | 16 | PRICE | 600 | |

NOTES:
1. REFER TO REFLECTED CEILING PLANS FOR CEILING TYPES, COORDINATE REQUIRED BORDER WITH INSTALLED LOCATION.
2. COORDINATE COLOR AND FINISH WITH ARCHITECT.
3. WHERE GRILLE IS MOUNTED DIRECTLY TO EXPOSED DUCTWORK, PROVIDE EXTRACTOR.
4. WHERE GRILLE IS MOUNTED WITH NO ACCESS TO DUCTWORK BEHIND, PROVIDE INTEGRAL DAMPER.
5. WHERE GRILLE IS MOUNTED, OPPOSED BLADE DAMPER SHALL BE INSTALLED

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# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

**NOT FOR CONSTRUCTION**

FOR REFERENCE ONLY

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**GRAPHIC SCALE(S)**

DATE: 06/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: **ARABIAN LEOPARD FACILITY (ALF)**
**3001 Connecticut Avenue NW Washington, DC 20008**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**

OPDC PROJECT NUMBER: **2533105**

A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **MECHANICAL SCHEDULES**

WORKING STAFF

| OP | OP | IP |
|---|---|---|
| DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 281 OF 378

M-608SH

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

## OVERHEAD HEATER SCHEDULE

| TAG | LOCATION | HEATING CAPACITY | INPUT POWER | VOLTAGE | PHASE | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|
| OH-1 | DAYROOM | 5 MBH | 1,450 W | 208 V | 1 | FOSTORIA | FSA | |
| OH-2 | OUTDOORS | 5 MBH | 1,450 W | 208 V | 1 | FOSTORIA | FSA | |
| OH-3 | OUTDOORS | 11 MBH | 3,150 W | 208 V | 1 | FOSTORIA | FSA | |
| OH-4 | OUTDOORS | 11 MBH | 3,150 W | 208 V | 1 | FOSTORIA | FSA | |
| OH-5 | OUTDOORS | 11 MBH | 3,150 W | 208 V | 1 | FOSTORIA | FSA | |
| OH-6 | OUTDOORS | 11 MBH | 3,150 W | 208 V | 1 | FOSTORIA | FSA | |
| OH-7 | OUTDOORS | 11 MBH | 3,150 W | 208 V | 1 | FOSTORIA | FSA | |
| OH-8 | OUTDOORS | 11 MBH | 3,150 W | 208 V | 1 | FOSTORIA | FSA | |
| OH-9 | OUTDOORS | 11 MBH | 3,150 W | 208 V | 1 | FOSTORIA | FSA | |

NOTES -
1. ALL HEATERS SHALL HAVE A PAINTED FINISH. COORDINATE WITH ARCHITECT ON INSTANCE BY INSTANCE BASIS.
2. COORDINATE WITH DIVISION 26. PROVIDE INTERMEDIATE CONTROL ENCLOSURE WITH HEAVY-DUTY CONTACTOR ADJACENT TO ALL NINE (9) HEATER RECEPTACLE. FIELD-WIRE LOW-VOLTAGE CONTROL SIDE TO BACNET RELAY MODULE. RATED FOR (AT LEAST) CONTINUOUS 5KW LOAD.

## CAGED HABITAT FAN SCHEDULE

| TAG | LOCATION | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|
| OHF-2 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |
| OHF-3 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |
| OHF-1 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |

NOTES-
1. SEE ELECTRICAL DOCUMENTS FOR ADDITIONAL INFORMATION.
2. ALL FANS SHALL HAVE A PAINTED FINISH. COORDINATE WITH ARCHITECT ON INSTANCE BY INSTANCE BASIS.

## CAGED GUEST FAN SCHEDULE

| TAG | LOCATION | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|
| OGF-1 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |
| OGF-2 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |
| OGF-3 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |

NOTES-
1. SEE ELECTRICAL DOCUMENTS FOR ADDITIONAL INFORMATION.
2. ALL FANS SHALL HAVE A PAINTED FINISH. COORDINATE WITH ARCHITECT ON INSTANCE BY INSTANCE BASIS

## OUTDOORS EXHAUST FAN SCHEDULE

| TAG | LOCATION | MAX AIRFLOW | STATIC | VOLTAGE | PHASE | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|
| OEF-1 | OUTDOOR | 500 CFM | 0.00 in-wg | 115 V | 1 | CANARM | P10 | |

NOTES-
1. PROVIDE TWO SPEED MODEL.
2. ALL FANS SHALL HAVE A PAINTED FINISH. COORDINATE WITH ARCHITECT ON INSTANCE BY INSTANCE BASIS

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WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 282 OF 378

M-609SH (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

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## FOR REFERENCE ONLY - AIR HANDLING UNIT SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | AIRFLOW | | | SUPPLY FANS | | COILS AND GRIDS | | | | CASING | ACCESSORIES | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | HEATING | | STEAM DISPERSION GRID (SDG) SCHEDULE | CHW TAG | WALL THICKNESS | FINAL FILTER TAG | PREFILTER TAG | MANUFACTURER | MODEL | |
| | | | | SUPPLY MINIMUM | SUPPLY MAXIMUM | OUTSIDE MIN (CFM) | SF QTY | SF TAG | ENERGY RECOVERY COIL TAG | HEATING COIL TAG | | | | | | | | |
| AHU-1 | SUPPLY AIR FOR CARE BUILDING | CARE BUILDING BASEMENT MECHANICAL ROOM | 100% OUTSIDE AIR | 11,893 m³/h | 12,743 m³/h | 11,893 m³/h | 2 | SF-1A; SF-1B | ERC-1 | HWC-1 | SDG-1 | CWC-1 | 6 cm | AHU-FF | AHU-PF | AIR ENTERPRISES | FULLY CUSTOM | |

## FOR REFERENCE ONLY - AIR HANDLING UNIT AIR FILTER SCHEDULE

| TAG | SYSTEM | SYSTEM | TYPE | MAX FACE VELOCITY (FPM) | CAPACITY | HOUSING ASSEMBLY | | PRESSURE DROP | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | OVERALL | | CLEAN FILTER | DIRTY FILTER (ASHRAE 52) | |
| | | | | | | WIDTH | HEIGHT | | | |
| AHU-FF | AHU-1 | FINAL FILTER | MERV-14 | 3 m/s | 12,743 m³/h | 152 cm | 122 cm | 149 Pa | 348 Pa | |
| AHU-PF | AHU-1 | PRE FILTER | MERV-9 | 3 m/s | 12,743 m³/h | 152 cm | 122 cm | 50 Pa | 149 Pa | |

## FOR REFERENCE ONLY - AIR HANDLING UNIT HYDRONIC COIL SCHEDULE

| TAG | SYSTEM | LOCATION | FLUID | AIRFLOW | TOTAL CAPACITY | SENSIBLE CAPACITY | QTY OF COILS IN BANK | MIN ROWS | MAX FINS PER INCH | FACE VELOCITY | COIL SIZE DATA | | | AIRSIDE DATA | | | | | FLUID SIDE DATA | | | | BASIS OF DESIGN | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | FACE AREA | LENGTH | HEIGHT | EAT DB | EAT WB | LAT DB | LAT WB | MAX AIR PD | EWT | LWT | FLOW | WATER MAX PD | MANUFACTURER | |
| CWC-1 | AHU-1 CHILLED WATER | AHU-1 | 30% PROPYLENE GLYCOL | 12,743 m³/h | 209 kW | 83 kW | 1 | 10 | 11 | 2 m/s | 17,419 cm² | 38 cm | 114 cm | 28.56 °C | 25.28 °C | 9.39 °C | 9.39 °C | 219 Pa | 6 °C | 12 °C | 9 L/s | 90 kPa | HEATCRAFT COILS | |
| ERC-1 (SUMMER) | AHU-1 ENERGY RECOVERY COIL | AHU-1 | 30% PROPYLENE GLYCOL | 12,743 m³/h | 9 kW | 9 kW | 1 | 6 | 8 | 2 m/s | 17,419 cm² | 25 cm | 114 cm | 28.56 °C | 25.28 °C | 26.47 °C | 24.61 °C | 74 Pa | 23 °C | 26 °C | 1 L/s | 15 kPa | HEATCRAFT COILS | |
| ERC-1 (WINTER) | AHU-1 ENERGY RECOVERY COIL | AHU-1 | 30% PROPYLENE GLYCOL | 12,743 m³/h | 49 kW | 49 kW | 1 | 6 | 8 | 2 m/s | 17,419 cm² | 25 cm | 114 cm | -7.78 °C | -10.08 °C | 3.74 °C | -2.62 °C | 74 Pa | 16 °C | 7 °C | 1 L/s | 15 kPa | HEATCRAFT COILS | |
| PHC-1 | AHU-1 PREHEAT COIL | AHU-1 | 30% PROPYLENE GLYCOL | 12,743 m³/h | 100 kW | 100 kW | 1 | 2 | 7 | 2 m/s | 17,419 cm² | 19 cm | 114 cm | -7.78 °C | -10.08 °C | 15.62 °C | 4.56 °C | 37 Pa | 54 °C | 46 °C | 3 L/s | 30 kPa | HEATCRAFT COILS | |

NOTES -
1. PROVIDE SINGULAR ENERGY RECOVERY COIL. COIL PERFORMANCE IS RATED FOR A SINGULAR ERC-1 FOR SUMMER AND WINTER DESIGN DAY.

## FOR REFERENCE ONLY - AIR HANDLING UNIT FAN SCHEDULE

| TAG | LOCATION | SYSTEM | AIRFLOW | OPERATING POINT | | | | | | MOTOR | | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | STATIC PRESSURE | | | BOD STATIC EFFICIENCY (%) | BOD FAN RPM | BOD BHP | MAXIMUM PERMISSIBLE FAN SPEED | MAX RPM | HP | VOLTAGE | PHASE | OPERATING WEIGHT | MANUFACTURER | | |
| | | | | INTERNAL (FOR BOD ASSEMBLY) | EXTERNAL (MINIMUM REQUIREMENT) | TOTAL | | | | | | | | | | | |
| SF-1A | BASEMENT MECHANICAL ROOM | AHU-1 SUPPLY | 12,743 m³/h | 1,020 Pa | 995 Pa | 2,016 Pa | 60 | 3326 | 15.94 | 3500 | 3500 | 20 | 460 V | 3 | 147 kgf | GREENHECK | |
| SF-1B | BASEMENT MECHANICAL ROOM | AHU-1 SUPPLY | 12,743 m³/h | 1,020 Pa | 995 Pa | 2,016 Pa | 60 | 3326 | 15.94' | 3500 | 3500 | 20 | 460 V | 3 | 147 kgf | GREENHECK | FULLY REDUNDANT |

NOTES -
1. VFD SHALL LOCK MAXIMUM FAN SPEED AT 3450 RPM.
   A. 0 HZ AT 0 RPM AND 60HZ AT 3450 RPM.
2. PROVIDE ALL FANS WITH FACTORY MOUNTED PIEZOMETER RING AS A BACKUP AIRFLOW MEASURING DEVICE (INTEGRATION TO BAS).
3. SCHEDULED INTERNAL PRESSURE DROP INCLUDES 510 PASCALS OF TOTAL PRESSURE DROP FOR DIRTY FILTERS, SEE 'AIR HANDLING UNIT FILTER' SCHEDULE.

## FOR REFERENCE ONLY - AIR HANDLING UNIT STEAM DISPERSION GRID SCHEDULE

| TAG | STEAM PRODUCERS | LOCATION | TYPE | STEAM DATA | | HEATING AIRFLOW | HUMIDIFICATION | | | | | | DISPERSION GRID DIMENSIONS IN AHU | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | ENTERING AIR | | | LEAVING AIR | | | | | | | |
| | | | | STEAM FLOW RATE | GUAGE PRESSURE (kPa) | | DRY BULB | WET BULB | GRAMS OF WATER PER KILOGRAM OF DRY AIR (g/kg) | DRY BULB | WET BULB | GRAMS OF WATER PER KILOGRAM OF DRY AIR (g/kg) | HEIGHT | WIDTH | MANUFACTURER | MODEL | |
| SDG-1 | HUM-1 AND HUM-2 | WITHIN AHU-1 | SHORT ABSORPTION MANIFOLD KIND (SAMe) | 81 kg/h | 0 | 11,893 m³/h | 15.56 °C | 2.58 °C | 0.45728 | 15.56 °C | 10.81 °C | 6.1447 | 163 | 163 cm | CONDAIR | SAMe | |

## FOR REFERENCE ONLY - RESISTIVE ELEMENT HUMIDIFIER SCHEDULE

| TAG | LOCATION | WORKING PRINCIPLE | END USER | STEAM PRESSURE | DESIGN CAPACITY (KG/HR) | TANK VOLUME | TANK FILL RATE | TANK DRAIN RATE | INPUT POWER | VOLTAGE | PHASE | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| HUM-1 | BASEMENT MECHANICAL ROOM | ELECTRIC RESISTANCE HEAT | SDG-1 | 0 kPa | 81.65 | 50 L | 29 L/min | 51 L/min | 65 kW | 480 V | 3 | CONDAIR | RS 180 | |

NOTES -
1. PROVIDE BACNET INTEGRATION.
2. PROVIDE SCALE COLLECTION TANK ACCESSORY.

## FOR REFERENCE ONLY - FAN SCHEDULE

| TAG | LOCATION | SYSTEM | AIRFLOW | OPERATING POINT | | | | | | | MOTOR | | | | OPERATING WEIGHT | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | STATIC PRESSURE SUMMARY | | | BOD STATIC EFFICIENCY (%) | BOD FAN RPM | BOD BHP | MAXIMUM PERMISSIBLE FAN SPEED | MAX RPM | HP | VOLTAGE | PHASE | | MANUFACTURER | MODEL | |
| | | | | ENERGY RECOVERY COIL | DIRTY FILTERS | TOTAL EXTERNAL | | | | | | | | | | | | |
| EEF-1 | EMERGENCY ELECTRICAL ROOM | SUPPLY AIR | 510 m³/h | 0 Pa | | 124 Pa | 31 | 1112 | 0.08 | 1750 | 1700 | 0.25 | 115 V | 1 | 59 kgf | GREENHECK | AER-20-03-0605-VG | |
| EF-1 | ABOVE KEEPER'S OFFICE SPACE | EXHAUST AIR | 12,743 m³/h | 249 Pa | 498 Pa | 1,120 Pa | 63 | 2524 | 8.7 | 2777 | 2700 | 10 | 460 V | 3 | 227 kgf | GREENHECK | QEI-18 | |
| ESF-1 | EMERGENCY ELECTRICAL ROOM | EXHAUST AIR | 510 m³/h | 0 Pa | | 124 Pa | 31 | 1112 | 0.08 | 1750 | 1700 | 0.25 | 115 V | 1 | 59 kgf | GREENHECK | AER-20-03-0605-VG | |

## FOR REFERENCE ONLY - AIR FILTER SCHEDULE FOR EF-1

| TAG | SYSTEM | SYSTEM | FILTER TYPE | ASHRAE 52 RATING | MAX FACE VELOCITY (FPM) | CAPACITY | HOUSING ASSEMBLY | | | PRESSURE DROP | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | CELL THICK | OVERALL | | CLEAN FILTER | DIRTY FILTER (ASHRAE 52) | MANUFACTURER | MODEL | |
| | | | | | | | | WIDTH | HEIGHT | | | | | |
| FILTER-E1 | EXHAUST FAN & COIL PROTECTION | EF-1 | INDUSTRIAL BAG TYPE | MERV-13 | 3 m/s | 11,044 m³/h | 56 cm | 51 cm | 64 cm | 124 Pa | 348 Pa | AAF | DRIPAK GX | |
| FILTER-E2 | EXHAUST FAN & COIL PROTECTION | EF-1 | INDUSTRIAL BAG TYPE | MERV-13 | 3 m/s | 11,044 m³/h | 56 cm | 51 cm | 64 cm | 124 Pa | 348 Pa | AAF | DRIPAK GX | |
| FILTER-E3 | EXHAUST FAN & COIL PROTECTION | EF-1 | INDUSTRIAL BAG TYPE | MERV-13 | 3 m/s | 11,044 m³/h | 56 cm | 51 cm | 64 cm | 124 Pa | 348 Pa | AAF | DRIPAK GX | |
| FILTER-E4 | EXHAUST FAN & COIL PROTECTION | EF-1 | RIGID FRAME PLEATED PANEL TYPE | MERV-8 | 3 m/s | 11,044 m³/h | 5 cm | 51 cm | 64 cm | 50 Pa | 149 Pa | AAF | PERFECTPLEAT SC M8 | |
| FILTER-E5 | EXHAUST FAN & COIL PROTECTION | EF-1 | RIGID FRAME PLEATED PANEL TYPE | MERV-8 | 3 m/s | 11,044 m³/h | 5 cm | 51 cm | 64 cm | 50 Pa | 149 Pa | AAF | PERFECTPLEAT SC M8 | |
| FILTER-E6 | EXHAUST FAN & COIL PROTECTION | EF-1 | RIGID FRAME PLEATED PANEL TYPE | MERV-8 | 3 m/s | 11,044 m³/h | 5 cm | 51 cm | 64 cm | 50 Pa | 149 Pa | AAF | PERFECTPLEAT SC M8 | |

NOTES -
1. PREFILTER E4, E5 AND E6 ARE INTENDED TO CAPTURE LARGER DEBRIS / ORGANIC PARTICLES.

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3001 Connecticut Avenue NW
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## FOR REFERENCE ONLY - AIR SOURCE HEAT PUMP SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | COOLING MODE: AMBIENT TEMEPRATURE FOR HEAT REJECTION | COOLING MODE: COOLING CAPACITY | COOLING MODE: FLOW | COOLING MODE: EWT | COOLING MODE: LWT | COOLING MODE: COEFFICIENT OF PERFORMANCE (kW/kW) | HEATING MODE: AMBIENT TEMEPRATURE FOR HEAT EXTRACTION | HEATING MODE: HEATING CAPACITY | HEATING MODE: FLOW | HEATING MODE: EWT | HEATING MODE: LWT | HEATING MODE: COEFFICIENT OF PERFORMANCE (kW/kW) | SIMULTANEOUS MODE: HEATING CAPACITY | SIMULTANEOUS MODE: FLOW | SIMULTANEOUS MODE: HHW EWT | SIMULTANEOUS MODE: HHW LWT | SIMULTANEOUS MODE: COOLING CAPACITY | SIMULTANEOUS MODE: FLOW | SIMULTANEOUS MODE: CHW EWT | SIMULTANEOUS MODE: CHW LWT | SIMULTANEOUS MODE: COEFFICIENT OF PERFORMANCE (kW/kW) | MAX PD | OPERATING WEIGHT | BASIS OF DESIGN: MANUFACTURER | BASIS OF DESIGN: MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ASHP-1 | HYDRONIC PLANT FOR CARE BUILDING | OUTDOORS; HEAT PUMP YARD | ARRAY OF FOUR MODULES (ASHP-1A, ASHP-1B, ASHP-1C AND ASHP-1D) | 35.00 °C | 317 kW | 13 L/s | 12 °C | 6 °C | 2.8 | -8 °C | 241 kW | 7 L/s | 46 °C | 54 °C | 2.23 | 353 kW | 11 L/s | 46 °C | 54 °C | 268 kW | 11 L/s | 12 °C | 6 °C | 5.82 | 30 kPa | 5,443 kgf | TRANE | MAS | |
| ASHP-1A | MODULE WITHIN ASHP-1 | OUTDOORS; HEAT PUMP YARD | MODULAR; VAPOR INJECTION SCROLL COMPRESSOR | 35.00 °C | 94 kW | 4 L/s | 12 °C | 6 °C | 2.8 | -8 °C | 80 kW | 2 L/s | 46 °C | 54 °C | 2.23 | 118 kW | 4 L/s | 46 °C | 54 °C | 89 kW | 4 L/s | 12 °C | 6 °C | 5.82 | 30 kPa | 1,361 kgf | TRANE | MAS | |
| ASHP-1B | MODULE WITHIN ASHP-1 | OUTDOORS; HEAT PUMP YARD | MODULAR; VAPOR INJECTION SCROLL COMPRESSOR | 35.00 °C | 94 kW | 4 L/s | 12 °C | 6 °C | 2.8 | -8 °C | 80 kW | 2 L/s | 46 °C | 54 °C | 2.23 | 118 kW | 4 L/s | 46 °C | 54 °C | 89 kW | 4 L/s | 12 °C | 6 °C | 5.82 | 30 kPa | 1,361 kgf | TRANE | MAS | |
| ASHP-1C | MODULE WITHIN ASHP-1 | OUTDOORS; HEAT PUMP YARD | MODULAR; VAPOR INJECTION SCROLL COMPRESSOR | 35.00 °C | 94 kW | 4 L/s | 12 °C | 6 °C | 2.8 | -8 °C | 80 kW | 2 L/s | 46 °C | 54 °C | 2.23 | 118 kW | 4 L/s | 46 °C | 54 °C | 89 kW | 4 L/s | 12 °C | 6 °C | 5.82 | 30 kPa | 1,361 kgf | TRANE | MAS | |
| ASHP-1D | MODULE WITHIN ASHP-1 | OUTDOORS; HEAT PUMP YARD | MODULAR; VAPOR INJECTION SCROLL COMPRESSOR | 35.00 °C | 94 kW | 4 L/s | 12 °C | 6 °C | 2.8 | -8 °C | 80 kW | 2 L/s | 46 °C | 54 °C | 2.23 | 118 kW | 4 L/s | 46 °C | 54 °C | 89 kW | 4 L/s | 12 °C | 6 °C | 5.82 | 30 kPa | 1,361 kgf | TRANE | MAS | |

NOTES -
1. REFRIGERANT = R-454B; ASHRAE 34-2022 CLASSIFICATION = A2L.
   A. ODP (ASHRAE 34-2022) = 0
   B. GWP (ASHRAE 34-2022) = 466
2. PROVIDE A SELECTION FOR 30% PROPYLENE GLYCOL AS WORKING FLUID FOR HEATING HOT WATER AND CHILLED WATER LOOP.
3. PROVIDE AND INCLUDE FOLLOWING ACCESSORIES -
   A. INTERNAL HEAT TRACE - SINGLE POINT POWER CONNECTION - SEE ELECTRICAL DOCUMENTS FOR ADDITIONAL INFORMATION ON VOLTAGE/PHASE OF THE CONNECTION.
   B. MANUFACTURER FURNISHED, MANUFACTURER WIRED AND MANUFACTURER PROGRAMMED HUMAN MACHINE INTERFACE CONTROL PANEL ON ASHP-1A. SEE MECHANICAL CONTROLS FOR ADDITIONAL INFORMATION.
   C. SOUND ATTENUATION KIT AND SOUND ATTENUATION COMPRESSOR WRAP.
4. SCHEDULED CAPACITIES -
   A. SCHEDULED CAPACITY FOR LINE ITEM 'ASCH-1x' INDICATE DESIGN REQUIREMENT.
   B. SCHEDULED CAPACITIES FOR INDIVIDUAL MODULES ARE SELECTION CRITERIA - CONTRACTOR MUST SELECT INDIVIDUAL ARRAYS WITHIN THE MODULE, TO MEET OR EXCEED THESE CAPACITIES.

NOTES -
4. PROVIDE A POINT OF POWER FOR EACH MODULE (4 POINTS TOTAL). ALL ELECTRICAL REQUIREMENTS OF THESE MODULES SHALL COMPLY WITH ELECTRICAL DOCUMENTS.
5. CONTRACTOR SHALL PROVIDE ALL THE RELEVANT DOCUMENTS TO MANUFACTURER FOR SELECTING THIS EQUIPMENT -
   A. PIPING FLOOR PLANS
   B. RELEVANT ARCHITECTURAL, LANDSCAPE AND ELECTRICAL FLOOR PLANS REFERENCING THE INSTALLATION LOCATION OF HEAT PUMP.
   C. SYSTEM DIAGRAM (M-602DG) AND CONTROL PLANS (M-806CP THROUGH M-809CP)
   D. EQUIPMENT SPECIFICATION 236423
   E. ENLARGED PLAN M-402OP
6. MANUFACTURER SHALL QUANTIFY THE DE-RATE OF THE MACHINE BASED ON THE LAYOUT AND PROVIDED PLANS (ARCHITECTURAL AND LANDSCAPE) AND PROVIDE THIS QUANTIFIED VALUE TO IN SUBMITTAL. SUBMITTAL DEFICIENT OF THIS VALUE WILL BE RETURNED WITHOUT REVIEW.

NOTES -
6. REDUNDANCIES -
   A. MODULE ASHP-1D IS MECHANICALLY REDUNDANT.
   B. MODULE ASHP-1D IS NOT ELECTRICALLY REDUNDANT.
7. PROVIDE COMPLETE PROGRAMMED CONTROLLER SEQUENCES (INCLUDING MANUFACTURER DEFAULTS INCLUDED WITH ATTACHED CONTROLLER) FOR OWNER'S REFERENCE.
8. COORDINATE AND ENGAGE THE MANUFACTURER TO DEVELOP PROGRAMMING REQUIREMENTS of 230900 ASSOCIATED WITH PURCHASED PRODUCT. COORDINATE PROGRAMMING REQUIREMENTS WITH ISSUED DOCUMENTS AND MANUFACTURER PROGRAMMED SEQUENCES TO ENSURE THAT BAS OPERATES IN ACCORDANCE WITH THE PROJECT REQUIREMENTS AND DESIGN INTENT.

NOTES -
9. SOUND POWER LEVELS (AHRI 370 DEFINITIONS) SHALL NOT EXCEED FOLLOWING VALUES FOR THE ARRAY AT 100% ARRAY LOAD AND NEGATIVE 7.78°C AMBIENT TEMPERATURE
   A. 63 HZ - 90DB
   B. 125 HZ - 92 DB
   C. 250 HZ - 92 DB
   D. 500 HZ - 90 DB
   E. 1000 HZ - 86 DB
   F. 2000 HZ - 83 DB
   G. 4000 HZ - 80 DB
   H. 8000 HZ - 78 DB
   I. LWA = 92 DB(A)

## FOR REFERENCE ONLY - BOILER SCHEDULE (HOT WATER)

| TAG | SYSTEM | LOCATION | TYPE | SELECTION CRITERIA: CAPACITY (SEE NOTE 1) | SELECTION CRITERIA: FLOW | SELECTION CRITERIA: EWT | SELECTION CRITERIA: LWT | SELECTION CRITERIA: MAX PD | NATURAL GAS FLOW RATE | ELECTRICAL DATA: VOLTAGE | ELECTRICAL DATA: PHASE | VENT STACK | OPERATING WEIGHT | BASIS OF DESIGN: MANUFACTURER | BASIS OF DESIGN: MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B-1 | HEATING HOT WATER | BASEMENT MECHANICAL ROOM | TWO PIPE; NATURAL DRAFT; CONDENSING; NATURAL GAS FIRED | 141 kW | 4 L/s | 46 °C | 54 °C | 30 kPa | 14 m³/h | 120 V | 1 | 10 cm | 306 kgf | FULTON | MARK II | |
| B-2 | HEATING HOT WATER | BASEMENT MECHANICAL ROOM | TWO PIPE; NATURAL DRAFT; CONDENSING; NATURAL GAS FIRED | 141 kW | 4 L/s | 46 °C | 54 °C | 30 kPa | 14 m³/h | 120 V | 1 | 10 cm | 306 kgf | FULTON | MARK II | |

NOTE -
1. OUTPUT CAPACITY AT RETURN WATER TEMPERATURE OF 46.11°C FOR 30% PG. PRODUCT SUBMITTAL MUST INDICATE THERMAL EFFICIENCY AT SYSTEM'S **ACTUAL** RETURN WATER TEMPERATURE. ADDITIONALLY, INCLUDE AHRI-1500 RATING OF PRODUCT.
2. PRODUCT MANUFACTURER CONTROLLER PRE-PROGRAMMED WITH SEQUENCES TO STAGE/CONTROL THE BOILERS. SEE M-850CP, M-806-CP, M-807CP AND M-808CP FOR ADDITIONAL INFORMATION.
3. MANUFACTURER MUST MAKE CONTROLLER AND REQUIRED POINTS AVAILABLE FOR INTEGRATION INTO BAS THROUGH BACNET MS/TP COMMUNICATION PROTOCOL.

## FOR REFERENCE ONLY - PUMP SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | FLUID | TEMPERATURE | DUTY POINT: FLOW | DUTY POINT: DISCHARGE HEAD | DUTY POINT: MINIMUM EFFICIENCY | DUTY POINT: SPEED | DUTY POINT: BHP | MINIMUM NPSH REQ'D | PIPE CONNECTIONS: SUCTION | PIPE CONNECTIONS: DISCHARGE | MOTOR DATA: TYPE OF CONTROL | MOTOR DATA: HP | MOTOR DATA: RPM | MOTOR DATA: VOLTAGE | MOTOR DATA: PHASE | OPERATING WEIGHT | BASIS OF DESIGN: MANUFACTURER | BASIS OF DESIGN: MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CHWP-1 | CHILLED WATER - SECONDARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 6 °C | 16 L/s | 299 kPa | 64 | 1588 | 10.1 | 1.24 m | 10 cm | 10 cm | VFD | 20 | 1800 | 460 V | 3 | 277 kgf | BELL&GOSSETT | e-80 | NOTE 1, 3 |
| CHWP-2 | CHILLED WATER - SECONDARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 6 °C | 16 L/s | 299 kPa | 64 | 1588 | 10.1` | 1.24 m | 10 cm | 10 cm | VFD | 20 | 1800 | 460 V | 3 | 277 kgf | BELL&GOSSETT | e-80 | NOTE 1, 2, 3 |
| CHWP-3 | CHILLED WATER - PRIMARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 14 °C | 16 L/s | 149 kPa | 67 | 1865 | 4.85 | 2.28 m | 10 cm | 10 cm | ECM | 5.5 | 1800 | 460 V | 3 | 122 kgf | BELL&GOSSETT | e-80X | |
| CHWP-4 | CHILLED WATER - PRIMARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 14 °C | 16 L/s | 149 kPa | 67 | 1865 | 4.85 | 2.28 m | 10 cm | 10 cm | ECM | 5.5 | 1800 | 460 V | 3 | 122 kgf | BELL&GOSSETT | e-80X | NOTE 1, 2 |
| ERFP-1 | ENERGY RECOVERY FLUID SYSTEM | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 26 °C | 2 L/s | 149 kPa | 34 | 1228 | 0.948 | 0.73 m | 4 cm | 4 cm | ECM | 5.5 | 2000 | 460 V | 3 | 86 kgf | BELL&GOSSETT | e-80X | |
| ERFP-2 | ENERGY RECOVERY FLUID SYSTEM | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 26 °C | 2 L/s | 149 kPa | 34 | 1228 | 0.948 | 0.73 m | 4 cm | 4 cm | ECM | 5.5 | 2000 | 460 V | 3 | 86 kgf | BELL&GOSSETT | e-80X | NOTE 2 |
| FPP-1 | AHU-1 FREEZE PROTECTION | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 6 °C | 2 L/s | 75 kPa | 59 | 1677 | 0.269 | 1.18 m | 3 cm | 3 cm | ECM | 0.5 | 1800 | 115 V | 1 | 25 kgf | BELL&GOSSETT | e-90 | |
| HHWP-1 | HEATING HOT WATER - SECONDARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 54 °C | 10 L/s | 448 kPa | 58 | 1684 | 10.5 | 1.32 m | 8 cm | 8 cm | VFD | 20 | 1800 | 460 V | 3 | 261 kgf | BELL&GOSSETT | e-80 | NOTE 1, 3 |
| HHWP-2 | HEATING HOT WATER - SECONDARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 54 °C | 10 L/s | 448 kPa | 58 | 1684 | 10.5 | 1.32 m | 8 cm | 8 cm | VFD | 20 | 1800 | 460 V | 3 | 261 kgf | BELL&GOSSETT | e-80 | NOTE 1, 2, 3 |
| HHWP-3 | HEATING HOT WATER - PRIMARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 46 °C | 10 L/s | 149 kPa | 60 | 1652 | 3.43 | 1.35 m | 6 cm | 6 cm | ECM | 5.5 | 1800 | 460 V | 3 | 116 kgf | BELL&GOSSETT | e-80X | |
| HHWP-4 | HEATING HOT WATER - PRIMARY PUMPS | BASEMENT MECHANICAL ROOM | INLINE | 30% PROPYLENE GLYCOL | 46 °C | 10 L/s | 149 kPa | 60 | 1652 | 3.43 | 1.35 m | 6 cm | 6 cm | ECM | 5.5 | 1800 | 460 V | 3 | 116 kgf | BELL&GOSSETT | e-80X | NOTE 2 |

NOTES -
1. PROVIDE SUCTION DIFFUSERS FOR PUMP.
2. REDUNDANT PUMP; PROVIDING N+1 REDUNDANCY ON PUMP SET.
3. PROVIDE CONCRETE VIBRATION ISOLATION INERTIA BASE.

## FOR REFERENCE ONLY - SUCTION DIFFUSER SCHEDULE

| TAG | TYPE | PUMP | PUMP SIZE | INLET SIZE | FLOW | PRESSURE DROP | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|
| SD-CHWP-1 | FLANGED INLET / FLANGED OUTLET | CHWP-1 | 10 cm | 13 cm | 16 L/s | 3 kPa | BELL & GOSSETT | FE-3X | |
| SD-CHWP-2 | FLANGED INLET / FLANGED OUTLET | CHWP-2 | 10 cm | 13 cm | 16 L/s | 3 kPa | BELL & GOSSETT | FE-3X | |
| SD-HHWP-1 | FLANGED INLET / FLANGED OUTLET | HHWP-1 | 8 cm | 10 cm | 10 L/s | 3 kPa | BELL & GOSSETT | ED-3X | |
| SD-HHWP-2 | FLANGED INLET / FLANGED OUTLET | HHWP-2 | 8 cm | 10 cm | 10 L/s | 3 kPa | BELL & GOSSETT | ED-3X | |

## FOR REFERENCE ONLY - EXPANSION TANK SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | TEMPERATURE: MIN | TEMPERATURE: MAX | SYSTEM VOLUME | TANK VOLUME | MIN ACCEPTANCE VOLUME | AIR CHARGE PRESSURE | RELIEF VALVE | OPERATING WEIGHT | BASIS OF DESIGN: MANUFACTURER | BASIS OF DESIGN: MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ET-CHW | CHILLED WATER (30% PG) | BASEMENT MECHANICAL ROOM | BLADDER | 4 °C | 21 °C | 5678 L | 98 L | 48 L | 207 kPa | 241 kPa | 56 kgf | BELL & GOSSETT | B-100 | |
| ET-ERF | ENERGY RECOVERY FLUID (30% PG) | BASEMENT MECHANICAL ROOM | BLADDER | -7 °C | 29 °C | 1514 L | 38 L | 20 L | 207 kPa | 241 kPa | 56 kgf | BELL & GOSSETT | B-35 | |
| ET-HHW | HEATING HOT WATER (30% PG) | BASEMENT MECHANICAL ROOM | BLADDER | 10 °C | 54 °C | 5678 L | 167 L | 70 L | 207 kPa | 241 kPa | 235 kgf | BELL & GOSSETT | B-165 | |

## FOR REFERENCE ONLY - GLYCOL FEEDER SCHEDULE

| TAG | LOCATION | SYSTEM | TANK SIZE | CAPACITY: FLOW | CAPACITY: PRESSURE | PRESSURE RANGE | PUMP DATA: HP | PUMP DATA: SPEED | ELECTRICAL DATA: VOLTAGE | ELECTRICAL DATA: PHASE | BASIS OF DESIGN: MANUFACTURER | BASIS OF DESIGN: MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| GFS-1 | BASEMENT MECHANICAL ROOM | HEATING HOT WATER AND ENERGY RECOVERY SYSTEM | 208 L | 19 L/min | 414 kPa | 10 psi - 60 psi | 0.75 | 3600 | 115 V | 1 | XYLEM | GMU-60 | |
| GFS-2 | BASEMENT MECHANICAL ROOM | CHILLED WATER SYSTEM | 208 L | 19 L/min | 414 kPa | 10 psi - 60 psi | 0.75 | 3600 | 115 V | 1 | XYLEM | GMU-60 | |

NOTES -
1. SINGLE POINT POWER CONNECTION TO UNIT.
2. PROVIDE COMPLETE WITH LOCAL DISCONNECT SWITCH, BAS INTERFACE AND ALARM, PRESSURE REDUCING VALVE BETWEEN THE UNIT AND CONNECTION TO PIPING SYSTEM.

## FOR REFERENCE ONLY - BUFFER TANK SCHEDULE

| TAG | LOCATION | SYSTEM | TYPE | CONNECTION SIZE | TANK MIN VOLUME | BASIS OF DESIGN: MANUFACTURER | BASIS OF DESIGN: MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|
| BT-CHW | BASEMENT MECHANICAL ROOM | CHILLED WATER | HORIZONTAL | 15 cm | 3,218 L | TACO | BTH-850F | |
| BT-HHW | BASEMENT MECHANICAL ROOM | HEATING HOT WATER | HORIZONTAL | 15 cm | 3,218 L | TACO | BTH-850F | |

NOTES -
1. ALL TANKS SHALL COMPLY WITH ASME CONSTRUCTION.
2. PROVIDE WITH HAND HOLE ON TANK FOR CLEANING.
3. PROVIDE WITH INTERNAL BAFFLE, AIR VENTS, LEGS FOR HORIZONTAL INSTALLATION AND FACTORY ELASTOMERIC INSULATION (WITH REMOVABLE SECTIONS AT SERVICE LOCATIONS).

## FOR REFERENCE ONLY - AIR SEPARATOR SCHEDULE

| TAG | LOCATION | TYPE | SYSTEM | CONNECTION SIZE | DESIGN FLOW RATE | MAX PRESSURE DROP | OPERATING WEIGHT | BASIS OF DESIGN: MANUFACTURER | BASIS OF DESIGN: MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| AS-CHW | BASEMENT MECHANICAL ROOM | COALESCING STYLE | CHILLED WATER | 13 cm | 16 L/s | 3 kPa | 164 kgf | BELL&GOSSETT | CRS-5F MAG | |
| AS-ERF | BASEMENT MECHANICAL ROOM | COALESCING STYLE | HEATING HOT WATER | 10 cm | 10 L/s | 3 kPa | 95 kgf | BELL&GOSSETT | CRS-4F MAG | |
| AS-HHW | BASEMENT MECHANICAL ROOM | COALESCING STYLE | ENERGY RECOVERY FLUID | 5 cm | 2 L/s | 3 kPa | 50 kgf | BELL&GOSSETT | CRS-2F MAG | |

NOTES -
1. ALL SEPARATORS SHALL COMPLY WITH ASME CONSTRUCTION.
2. PROVIDE FLANGED CONNECTION. GROOVED IS NOT ACCEPTABLE.
3. PROVIDE MAGNETIC INSERT ACCESSORY.
4. COALESCING MEDIA SHALL BE STAINLESS STEEL.

FOR REFERENCE ONLY

ALF

CONSTRUCT ARABIAN LEOPARD FACILITY

NOT FOR CONSTRUCTION

STAMP

GRAPHIC SCALE(S)

DATE 06/15/2026

SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF); 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL SCHEDULES

WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 284 OF 378

M-611SH (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

6/15/2026 3:01:38 PM
M-611SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# THIS SHEET IS ONLY ISSUED TO QUANTIFY PERFORMANCE OF EQUIPMENT IN METRIC UNITS FOR OWNER REFERENCE. PRODUCT DATA AND SHOP DRAWINGS MUST BE SUBMITTED IN IMPERIAL UNITS.

## FOR REFERENCE ONLY - ENERGY RECOVERY HYDRONIC COIL SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | AIRFLOW | TOTAL CAPACITY | SENSIBLE CAPACITY | QTY OF COILS IN BANK | MIN ROWS | MAX FINS PER INCH | FACE VELOCITY | COIL SIZE DATA | | | AIRSIDE DATA | | | | | WATERSIDE DATA | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | FACE AREA | LENGTH | HEIGHT | EAT DB | EAT WB | LAT DB | LAT WB | MAX AIR PD | EWT | LWT | FLOW | WATER MAX PD | MANUFACTURER | MODEL | |
| ERC-EF (SUMMER) | EF-1 ENERGY RECOVERY COIL | KEEPER WORKSPACE | HYDRONIC | 11,044 m³/h | 9 kW | 9 kW | 1 | 12 | 5 | 3 m/s | 11,892 cm² | 30 cm | 64 cm | 22.22 °C | 15.60 °C | 24.67 °C | 16.47 °C | 236.40 Pa | 26 °C | 23 °C | 1 L/s | 30 kPa | AEROFIN | | |
| ERC-EF (WINTER) | EF-1 ENERGY RECOVERY COIL | KEEPER WORKSPACE | HYDRONIC | 11,044 m³/h | 38 kW | 38 kW | 1 | 12 | 5 | 3 m/s | 11,892 cm² | 30 cm | 64 cm | 22.22 °C | 12.42 °C | 11.94 °C | 7.92 °C | 236.40 Pa | 7 °C | 16 °C | 1 L/s | 30 kPa | AEROFIN | | |

NOTES -
1. PROVIDE SINGULAR ENERGY RECOVERY COIL. COIL PERFORMANCE IS RATED FOR A SINGULAR ERC-1 FOR SUMMER AND WINTER DESIGN DAY.

## FOR REFERENCE ONLY - RADIANT SLAB MANIFOLD SYSTEM SCHEDULE

| TAG | SYSTEM | LOCATION | MANIFOLD TYPE | ROOM SETPOINT | MAXIMUM PERMISSIBLE SURFACE TEMPERATURE | APPROXIMATE AREA FOR HEAT TRANSFER | MINIMUM RADIANT HEAT OUTPUT | LAYOUT | | | HYDRONIC SYSTEM | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | QUANTITY OF CIRCUITS | RADIANT PIPE SIZE | PIPING ON CENTER DISTANCE | TOTAL HHW FLOW RATE | SUPPLY WATER TEMPERATURE | MAXIMUM PERMISSIBLE PRESSURE DROP ACROSS ANY CIRCUIT | MANUFACTURER | MODEL | |
| RSS-1 | HEATING HOT WATER | DEN-1 | STAINLESS STEEL; 2 LOOP | 22 °C | 29 °C | 12 m² | 95 W/m² | 1 | 2 cm | 30 cm | 4 L/min | 54 °C | 15 kPa | UPONOR | A2702202 | |
| RSS-2 | HEATING HOT WATER | DEN-2 | STAINLESS STEEL; 2 LOOP | 22 °C | 29 °C | 12 m² | 95 W/m² | 1 | 2 cm | 30 cm | 4 L/min | 54 °C | 15 kPa | UPONOR | A2702202 | |
| RSS-3 | HEATING HOT WATER | DEN-3 | STAINLESS STEEL; 2 LOOP | 22 °C | 29 °C | 18 m² | 95 W/m² | 1 | 2 cm | 30 cm | 6 L/min | 54 °C | 15 kPa | UPONOR | A2702202 | |
| RSS-4 | HEATING HOT WATER | CUBBING | STAINLESS STEEL; 2 LOOP | 22 °C | 29 °C | 9 m² | 95 W/m² | 1 | 2 cm | 30 cm | 4 L/min | 54 °C | 15 kPa | UPONOR | A2702202 | |
| RSS-5 | HEATING HOT WATER | QUARENTINE | STAINLESS STEEL; 2 LOOP | 22 °C | 29 °C | 9 m² | 95 W/m² | 1 | 2 cm | 30 cm | 4 L/min | 54 °C | 15 kPa | UPONOR | A2702202 | |

NOTES -
1. SCHEDULE INDICATE BOD FOR STAINLESS STEEL MANIFOLD. CONTRACTOR SHALL HOUSE THE MANIFOLD IN A PANEL BOARD AT INDICATED LOCATION ON DRAWINGS.
2. PANEL BOARD SHALL HOUSE ACCESSORIES AS INDICATED ON SEE M-504DT.
3. PANEL BOARD SHALL BE WALL MOUNTED (NON-RECESSED) TO FACILITATE ROUTING PEX PIPING.
4. INCLUDE A FLOATING TYPE MIXING VALVE WITH VALVE COEFFICIENT OF 0.865 ($K_V$=0.865). BOD = BELIMO ZONE315N. INCLUDED ACTUATOR SHALL BE CAPABLE OF BAS INTEGRATION.
5. INCLUDE A RECIRCULATOR PUMP FOR EACH MANIFOLD COMPLIANT WITH DETAIL M-504DT. BOD - TACO 006 CIRCULATOR.
6. INCORPORATE MIXING STATION WITHIN THE RADIANT SLAB PANEL BOARD.
7. CONTRACTOR MAY PROVIDE ALL REQUIRED ACCESSORIES IN A SINGLE MANUFACTURER ASSEMBLED PANEL.

## FOR REFERENCE ONLY - HEAT TRACE SCHEDULE

| TAG | SYSTEM | PANEL LOCATION | TYPE | WATTS PER FOOT | VOLTAGE | PHASE | INPUT POWER | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| HTC-1 | DIVISION 23 HEAT TRACING | BASEMENT MECHANICAL ROOM | PARALLEL SELF REGULATING | 10 VA | 120 V | 1 | 5000 VA | CHROMALOX | SRL | |

NOTES
1. REFER TO PIPING INSULATION SPECIFICATION FOR CHILLED WATER AND HEATING HOT WATER AND FLOOR PLANS. HEAT TRACE ELEMENT SHALL BE COMPATIBLE WITH THE PURCHASED INSULATION.
2. VALIDATE PRODUCT SELECTION WITH ELECTRICAL DOCUMENTS.
3. BAS SHALL BE FULLY CAPABLE OF STARTING/STOPPING ANY/ALL HEAT TRACE CIRCUIT.
4. SEE MECHANICAL CONTROLS FOR ADDITIONAL INFORMATION. PROVIDE A DISTRIBUTION BOX ACCESSORY (BOD - MARSHALL BOX CHROMALOX) TO PROVIDE FOUR (4) LOADSIDE CIRCUITS.
5. PROVIDE ANY/ALL ACCESSORIES NEEDED TO TIE THE HEAT TRACE PANEL TO THE BAS.
6. PROVIDE HEAT TRACING ON ALL VALVES, AND PIPE ELBOWS.
7. HEAT TRACE HEATING ELEMENT MUST BE INSTALLED AS INSTRUCTED BY MANUFACTURER. ENGAGE MANUFACTURER TO REVIEW INSTALLATION OF HEATING ELEMENT ON THE PIPING ON APPROVED SHOP DRAWINGS.

## FOR REFERENCE ONLY - UNIT HEATER SCHEDULE (HOT WATER)

| TAG | SYSTEM | LOCATION | BOD CATALOG CAPACITY | DESIGN CAPACITY (SEE NOTE 1 AND NOTE 2) | AIRSIDE DATA | | | WATERSIDE DATA | | | | ELECTRICAL | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AIRFLOW | EAT | MIN LAT | EWT | LWT | FLOW RATE | MAX WATER PD | HP | FAN RPM | VOLTAGE | PHASE | MANUFACTURER | MODEL | |
| UH-1 | HOT WATER | LEVEL 1 | 4 kW | 1 kW | 680 m³/h | 22 °C | 29 °C | 54 °C | 46 °C | 5 L/min | 3 kPa | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-2 | HOT WATER | LEVEL 1 | 4 kW | 1 kW | 680 m³/h | 22 °C | 29 °C | 54 °C | 46 °C | 5 L/min | 3 kPa | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-3 | HOT WATER | LEVEL 1 | 4 kW | 1 kW | 680 m³/h | 22 °C | 29 °C | 54 °C | 46 °C | 5 L/min | 3 kPa | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-4 | HOT WATER | LEVEL 1 | 4 kW | 1 kW | 680 m³/h | 22 °C | 29 °C | 54 °C | 46 °C | 5 L/min | 3 kPa | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-5 | HOT WATER | LEVEL 1 | 4 kW | 1 kW | 680 m³/h | 22 °C | 29 °C | 54 °C | 46 °C | 5 L/min | 3 kPa | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | NOTE 3 |
| UH-6 | HOT WATER | LEVEL 1 | 4 kW | 1 kW | 680 m³/h | 22 °C | 29 °C | 54 °C | 46 °C | 5 L/min | 3 kPa | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | |
| UH-7 | HOT WATER | LEVEL 1 | 4 kW | 1 kW | 680 m³/h | 22 °C | 29 °C | 54 °C | 46 °C | 5 L/min | 3 kPa | 0.033333 | 1550 | 115 V | 1 | DAIKIN | UHH-18 | |

NOTES -
1. PROVIDE SELECTION FOR SCHEDULED 'DESIGN CAPACITY'.
2. SCHEDULED 'DESIGN CAPACITY' ACCOUNTS FOR A DE-RATE FACTOR OF 0.424 BASED ON ENTERING AIR TEMPERATURE OF 22.22°C AND ENTERING WATER TEMPERATURE OF 54.44°C. ADDITIONAL DE-RATE FACTOR OF 0.912 IS APPLIED FOR WORKING FLUID (30% PROPYLENE GLYCOL), PER DAIKIN (BOD MANUFACTURER) SELECTION CATALOG. CONTRACTOR SHALL SCHEDULE AND DOCUMENT DERATE FACTOR ASSOCIATED WITH MANUFACTURER RECOMMENDED DE-RATE FACTOR FOR PURCHASED PRODUCT.
3. PROVIDE CUSTOM FABRICATED WELDED STEEL WIRE CAGE ENCLOSURE AROUND UNIT HEATER TO PROTECT EQUIPMENT FROM DAMAGE DUE TO ANIMAL. CAGE SHALL BE RIGIDLY MOUNTED TO STRUCTURE AND SHALL ALLOW ADEQUATE AIRFLOW, SERVICE ACCESS, AND CLEARANCES REQUIRED FOR PROPER OPERATION. CAGE DIMENSIONS AND SPACING SHALL BE COORDINATED WITH AND INSTALLED PER UNIT MANUFACTURER'S RECOMMENDED CLEARANCES AND APPROVED SUBMITTALS. PROVIDE REMOVABLE OR HINGED ACCESS PANEL AS REQUIRED FOR MAINTENANCE.

## FOR REFERENCE ONLY - FLOOR RADIANT HEATERS (DAYROOM)

| TAG | SYSTEM | LOCATION | PHYSICAL DIMENSIONS | | AIRSIDE DATA | | | | | WATERSIDE DATA | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | HEATER LENGTH | HEATER WIDTH | HEATING CAPACITY | AIRFLOW | EAT | LAT | STATIC PRESSURE | FLOW | EWT | LWT | MANUFACTURER | MODEL | |
| HTR-1 | HOT WATER | DAY ROOM | 122 cm | 15 cm | 1.71 kW | 544 m³/h | 22 °C | 32 °C | 12 Pa | 8 L/min | 54 °C | 51 °C | PRICE | LFH | |
| HTR-2 | HOT WATER | DAY ROOM | 122 cm | 15 cm | 1.71 kW | 544 m³/h | 22 °C | 32 °C | 12 Pa | 8 L/min | 54 °C | 51 °C | PRICE | LFH | |
| HTR-3 | HOT WATER | DAY ROOM | 122 cm | 15 cm | 1.71 kW | 544 m³/h | 22 °C | 32 °C | 12 Pa | 8 L/min | 54 °C | 51 °C | PRICE | LFH | |
| HTR-4 | HOT WATER | DAY ROOM | 122 cm | 15 cm | 1.71 kW | 544 m³/h | 22 °C | 32 °C | 12 Pa | 8 L/min | 54 °C | 51 °C | PRICE | LFH | |

NOTES -
1. PROVIDE CUSTOM FABRICATED WELDED STEEL WIRE CAGE ENCLOSURE AROUND HEATER TO PROTECT EQUIPMENT FROM DAMAGE DUE TO ANIMAL. CAGE SHALL BE RIGIDLY MOUNTED TO STRUCTURE AND SHALL ALLOW ADEQUATE AIRFLOW, SERVICE ACCESS, AND CLEARANCES REQUIRED FOR PROPER OPERATION. CAGE DIMENSIONS AND SPACING SHALL BE COORDINATED WITH AND INSTALLED PER UNIT MANUFACTURER'S RECOMMENDED CLEARANCES AND APPROVED SUBMITTALS. PROVIDE REMOVABLE OR HINGED ACCESS PANEL AS REQUIRED FOR MAINTENANCE.

## FOR REFERENCE ONLY - FAN COIL UNIT SCHEDULE

| TAG | ZONE SERVED | PHYSICAL INSTALLATION LOCATION | SYSTEM | COOLING PERFORMANCE | | | | | | | | | | HEATING PERFORMANCE | | | | | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | TOTAL CAPACITY | TOTAL SENSIBLE CAPACITY | AIRSIDE DATA | | | | | CHILLED WATER COIL | | | | TOTAL HEATING CAPACITY | AIRSIDE DATA | | | HOT WATER COIL | | | | | | |
| | | | | | | AIRFLOW | AIR EAT DB | AIR EAT WB | AIR LAT DB | AIR LAT WB | FLOW | EWT (COOLING PLANT SETBACK) | LWT | WATER PD | | AIRFLOW | AIRFLOW EAT | AIRFLOW LAT | EWT | LWT | FLOW | WATER PD | MANUFACTURER | MODEL | |
| FCU-1 | ELECTRICAL ROOM | BASEMENT ELECTRICAL ROOM | HEATING AND COOLING 4 PIPE; HORIZONTAL CEILING MOUNTED | 8 kW | 7 kW | 2,039 m³/h | 23.89 °C | 16.94 °C | 13.50 °C | 12.67 °C | 20 L/min | 8 °C | 14 °C | 15 kPa | 4 kW | 2,039 m³/h | 23.89 °C | 29.02 °C | 54 °C | 46 °C | 7 L/min | 14.9 kPa | DAIKIN | FCHC | |
| FCU-2 | MECHANICAL ROOM | BASEMENT MECHANICAL ROOM | COOLING TWO PIPE; HORIZONTAL CEILING MOUNTED | 15 kW | 15 kW | 4,332 m³/h | 23.89 °C | 16.94 °C | 13.83 °C | 13.17 °C | 38 L/min | 8 °C | 14 °C | 15 kPa | | | | | | | | | DAIKIN | BCHD | |
| FCU-3 | AV/IT ROOM | BASEMENT MECHANICAL ROOM | COOLING TWO PIPE;VERTICAL FLOOR MOUNTED | 16 kW | 16 kW | 5,947 m³/h | 23.89 °C | 17.22 °C | 15.94 °C | 14.33 °C | 45 L/min | 9 °C | 14 °C | 15 kPa | | | | | | | | | DAIKIN | BCVD | |
| FCU-4 | EMERGENCY ELECTRICAL ROOM | BASEMENT ELECTRICAL ROOM | HEATING AND COOLING 4 PIPE; HORIZONTAL CEILING MOUNTED | 4 kW | 4 kW | 2,039 m³/h | 25.07 °C | 19.58 °C | 14.44 °C | 14.44 °C | 33 L/min | 8 °C | 13 °C | 15 kPa | 11 kW | 2,039 m³/h | 16.07 °C | 31.46 °C | 54 °C | 46 °C | 20 L/min | 14.9 kPa | DAIKIN | FCHC | |

## FOR REFERENCE ONLY - AIR CURTAIN SCHEDULE

| TAG | LOCATION | HEATING DATA | | | HP OF INCLUDED FAN MOTOR | VOLTAGE | PHASE | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| | | HEATING CAPACITY | SG_EWT | FLOW | | | | | | |
| AC-1 | LEVEL 1; LOADING DOCK | 16 kW | 54 °C | 21 L/min | 0.166667 | 115 V | 1 | MARS | QP872-1VA-AL | |

NOTES -
1. PROVIDE FOLLOWING FACTORY ACCESSORIES -
   A. MOTOR CONTROL PANEL WITH BAS INTEGRATION
   B. REMOTE MOUNTED THERMOSTAT
   C. DOOR SWITCH
2. A-WEIGHT SOUND POWER LEVEL SHALL BE 51 DBA OR LESS, AT MAX FAN SPEED.

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CONSTRUCT ARABIAN LEOPARD FACILITY

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DATE 06/15/2026

SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF); 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL SCHEDULES

WORKING STAFF | DESIGNED BY: OP | DRAWN BY: OP | CHECKED BY: IP

SHEET NO. 285 of 378

M-612SH (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

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# THIS SHEET IS ONLY ISSUED TO QUANTIFY PERFORMANCE OF EQUIPMENT IN METRIC UNITS FOR OWNER REFERENCE. PRODUCT DATA AND SHOP DRAWINGS MUST BE SUBMITTED IN IMPERIAL UNITS.

## FOR REFERENCE ONLY - DUCT SILENCER SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | AIRFLOW | VELOCITY | MAX AIR PD | SIZE DATA | | | | DYNAMIC INSERTION LOSS BY OCTAVE BAND (DB RE 10^(-12)) W | | | | | | | | BASIS OF DESIGN | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | DIAMETER | WIDTH | HEIGHT | LENGTH | 1 (63 HZ) | 2 (125 HZ) | 3 (250 HZ) | 4 (500 HZ) | 5 (1,000 HZ) | 6 (2,000 HZ) | 7 (4,000 HZ) | 8 (8,000 HZ) | MANUFACTURER | |
| DS-E1 | EXHAUST | LEVEL1 | EX-RDNM | 11,044 m³/h | -5 m/s | 24.88 Pa | | 81 cm | 76 cm | 244 cm | 4 | 6 | 12 | 20 | 11 | 8 | 6 | 5 | PRICE | |
| DS-O1 | OUTSIDE AIR | BASEMENT MECHANICAL ROOM | EX-RDNM | 12,743 m³/h | -10 m/s | 24.88 Pa | | 91 cm | 51 cm | 183 cm | 6 | 9 | 17 | 33 | 31 | 20 | 14 | 9 | PRICE | |
| DS-S1 | SUPPLY AIR | BASEMENT MECHANICAL ROOM | EX-RDNM | 12,743 m³/h | 7 m/s | 24.88 Pa | | 112 cm | 46 cm | 152 cm | 3 | 5 | 10 | 23 | 24 | 16 | 10 | 7 | PRICE | |
| DS-VAVE-6 | EXHAUST VAV -6 | LEVEL1 | EX-RDNM | 1,529 m³/h | -5 m/s | 24.88 Pa | | 30 cm | 30 cm | 91 cm | 2 | 4 | 6 | 13 | 9 | 7 | 5 | 3 | PRICE | |

NOTES:
1. TYPE:
   EX - EXTENDED CASING
   R - RECTANGULAR
   D - DISSIPATIVE
   NM - NO MEDIA
2. LENGTH SHOWN IS CENTERLINE LENGTH.
3. PRESSURE DROP AS DETERMINED PER ASTM E477-13 IN A NVLAP-ACCREDITED ACOUSTICAL LABORATORY, SHALL INCLUDE SYSTEM EFFECT FOR NEARBY DUCT ELEMENTS.
4. MANUFACTURER SHALL PROVIDE, FOR APPROVAL, PROFESSIONAL ENGINEER STAMPED ACOUSTICAL CALCULATIONS FOR ALL SYSTEMS WITH SILENCERS DEMONSTRATING:
   A. THE RESULTANT DUCTBORNE SOUND LEVELS, INCLUDING AIRBORNE AND BREAKOUT NOISE, MEET THE REQUIRED CRITERIA
   B. THE RESULTANT DROP WITH SYSTEM EFFECTS DOES NOT EXCEED SCHEDULED VALUES

## FOR REFERENCE ONLY - VOLUME CONTROL BOX SCHEDULE - SUPPLY VAV WITH HYDRONIC REHEAT

| TAG | LOCATION | SERVICE TO | MINIMUM INLET SIZE | AIRFLOWS | | | | | MAX AIR PD | HEATING COIL | | | | | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | OCCUPIED MODE | | | UNOCCUPIED MODE | | | HEATING COIL CAPACITY | HEATING AIRFLOW | ENTERING AIR TEMPERATURE | LEAVING AIR TEMPERATURE | ENTERING WATER TEMPERATURE | LEAVING WATER TEMPERATURE | FLOW | WATER PRESSURE DROP | MANUFACTURER | MODEL | |
| | | | | DESIGN | VELOCITY AT DESIGN (MAX) AIRFLOW | MINIMUM | VENTILATION MINIMUM | VAV MINIMUM | | | | | | | | | | | | |
| VAV-1 | KEEPER CORRIDOR | DAYROOM | 36 cm | 2,888 m³/h | 8 m/s | 2,379 m³/h | 229 m³/h | 1,454 m³/h | 124.42 Pa | 16 kW | 2,379 m³/h | 16 °C | 35 °C | 54 °C | 46 °C | 27 L/min | 14.9 kPa | PRICE | SDV | |
| VAV-2 | KEEPER CORRIDOR | KEEPER CORRIDOR | 25 cm | 1,359 m³/h | 7 m/s | 1,359 m³/h | 51 m³/h | 525 m³/h | 124.42 Pa | 9 kW | 1,359 m³/h | 16 °C | 35 °C | 54 °C | 46 °C | 15 L/min | 14.9 kPa | PRICE | SDV | |
| VAV-3 | KITCHEN PREP | KITCHEN PREP | 25 cm | 1,189 m³/h | 7 m/s | 1,189 m³/h | 127 m³/h | 525 m³/h | 124.42 Pa | 8 kW | 1,189 m³/h | 16 °C | 35 °C | 54 °C | 46 °C | 13 L/min | 14.9 kPa | PRICE | SDV | |
| VAV-4 | KEEPER CORRIDOR | DEN-1, DEN-2 AND DEN-3 | 25 cm | 1,869 m³/h | 10 m/s | 1,869 m³/h | 178 m³/h | 525 m³/h | 124.42 Pa | 12 kW | 1,869 m³/h | 16 °C | 35 °C | 54 °C | 46 °C | 21 L/min | 14.9 kPa | PRICE | SDV | |
| VAV-5 | KITCHEN PREP | CUBBING AND QUARENTINE | 25 cm | 1,019 m³/h | 6 m/s | 850 m³/h | 85 m³/h | 525 m³/h | 124.42 Pa | 6 kW | 850 m³/h | 16 °C | 35 °C | 54 °C | 46 °C | 10 L/min | 14.9 kPa | PRICE | SDV | |
| VAV-6 | KEEPER WORKSPACE | KEEPER WORKSPACE | 36 cm | 1,529 m³/h | 4 m/s | 756 m³/h | 76 m³/h | 756 m³/h | 124.42 Pa | 5 kW | 765 m³/h | 16 °C | 35 °C | 54 °C | 46 °C | 9 L/min | 14.9 kPa | PRICE | SDV | |
| VAV-7 | CORRIDOR ADJACENT TO DENS | ADJACENT CORRIDORS | 30 cm | 1,189 m³/h | 5 m/s | 1,189 m³/h | 34 m³/h | 336 m³/h | 124.42 Pa | 8 kW | 1,189 m³/h | 16 °C | 35 °C | 54 °C | 46 °C | 13 L/min | 14.9 kPa | PRICE | SDV | |

## FOR REFERENCE ONLY - BAS PANEL SCHEDULE

| TAG | LOCATION | VOLTAGE | PHASE | REMARKS |
|---|---|---|---|---|
| BAS-MISC | BASEMENT MECHANICAL ROOM | 120 V | 1 | |
| BAS-CHW | BASEMENT MECHANICAL ROOM | 120 V | 1 | |
| BAS-HHW | BASEMENT MECHANICAL ROOM | 120 V | 1 | |
| BAS-AHU-1 | BASEMENT MECHANICAL ROOM | 120 V | 1 | |
| BAS-INTEGRATION | BASEMENT MECHANICAL ROOM | 120 V | 1 | |

GENERAL NOTES -
1. REFER TO SPECIFICATION SECTION 230900 FOR ADDITIONAL INFORMATION.
2. PROVIDE A DIVISION 23 CONTROL POWER TRANSFORMER AS REQUIRED TO POWER LOW VOLTAGE EQUIPMENT.
3. SHOWN LOCATIONS ON FLOOR PLANS ARE RECOMMENDATIONS ONLY. CONTROLS CONTRACTOR TO FIELD COORDINATE DESIRED LOCATION WITH OWNER AND OTHER TRADES AND DOCUMENT EXACT LOCATION IN SHOP DRAWINGS.
4. PROVIDE WALL RECESSED PANEL BOARD WITH DOUBLE DOOR, HINGED AT TWO EDGES.
   A. SEMI-RECESSED PANEL BOARD IS ACCEPTABLE WHERE THE PANEL DEPTH IS THICKER THAN WALL. SEE FLOOR PLANS.
5. SEE SYSTEM NETWORK ARCHITECTURE FOR ADDITIONAL INFORMATION.
6. PANELS ARE INTENDED TO
   A. PROVIDE POWER TO MOTORIZED VALVES, DAMPERS AND ACTUATORS/RELAYS.
   B. PROVIDE CENTRALIZED LOCATION TO HOUSE THE BAS CONTROLLER ASSOCIATED WITH SYSTEM. ALL POINTS SHOWN ON M8 SHALL BE INTEGRATED ON CONTROLLER(S) HOUSED IN THESE PANELS.

## FOR REFERENCE ONLY - VOLUME CONTROL BOX SCHEDULE - EXHAUST

| TAG | SYSTEM | INLET SIZE | AIRSIDE DATA | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|
| | | | AIRFLOW | VELOCITY AT DESIGN (MAX) AIRFLOW | MAX AIR PD | MANUFACTURER | MODEL | |
| VAVE-1 | EXHAUST | 36 cm | 65,129 L/min | 11 m/s | 124 Pa | PRICE | SDE | |
| VAVE-2 | EXHUAST | 25 cm | 22,653 L/min | 7 m/s | 124 Pa | PRICE | SDE | |
| VAVE-3 | EXHAUST | 25 cm | 19,822 L/min | 7 m/s | 124 Pa | PRICE | SDE | |
| VAVE-4 | EXHAUST | 25 cm | 31,149 L/min | 10 m/s | 124 Pa | PRICE | SDE | |
| VAVE-5 | EXHAUST | 25 cm | 25,485 L/min | 8 m/s | 124 Pa | PRICE | SDE | |
| VAVE-6 | EXHAUST | 30 cm | 33,980 L/min | 8 m/s | 0 Pa | PRICE | SDE | |
| VAVE-7 | EXHAUST | 20 cm | 16,990 L/min | 9 m/s | 124 Pa | PRICE | SDE | |

## FOR REFERENCE ONLY - HIGH PERFORMANCE CONTROL VALVE SCHEDULE

| TAG | SYSTEM | TYPE | VALVE DIAMETER | SELECTION CRITERIA | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|
| | | | | DESIGN FLOW | DESIGN PRESSURE DROP | DESIGN COEFFICIENT (KV) | MANUFACTURER | MODEL | |
| CV-1 | BUILDING CHILLED WATER | HIGH PERFORMANCE SEGMENTED BALL | 3 in | 16 L/s | 0.34 bar | 135 | VSI-V | | |
| CV-2 | BUILDING HEATING HOT WATER | HIGH PERFORMANCE SEGMENTED BALL | 3 in | 10 L/s | 0.34 bar | 86 | VSI-V | | |

NOTES
1. PROVIDE A HIGH PERFORMANCE SEGMENTED BALL VALVE WITH A TURNDOWN RATIO OF 300:1. PROVIDE VALVE WITH AN ELECTRIC ACTUATOR WITH A RESOLUTION OF 200 STEPS, 0.04VDC/0.5 DEGREES. BASIS OF DESIGN: VSI, SERIES 1000-X
2. VALVE COEFFICIENT ($K_V$) IN METRIC UNITS = 1.15616 * VALVE COEFFICIENT ($C_V$) IN IMPERIAL UNITS.

## FOR REFERENCE ONLY - SMOKE EXHAUST FAN SCHEDULE

| TAG | LOCATION | SYSTEM | OPERATING POINT | | | | | MAXIMUM PERMISSIBLE PHYSICAL CONSTRUCTION SPEED LIMIT FOR FAN BODY | MOTOR | | | | OPERATING WEIGHT | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | AIRFLOW | STATIC PRESSURE SUMMARY | | BOD STATIC EFFICIENCY (%) | BOD FAN RPM | BOD BHP | | MAX RPM | HP | VOLTAGE | PHASE | | MANUFACTURER | MODEL | |
| | | | | EXTERNAL (MINIMUM REQUIREMENT) | TOTAL STATIC PRESSURE | | | | | | | | | | | | |
| SEF-1 | KITCHEN PREPARATION AREA | SMOKE EXHAUST SYSTEM | 8,495 m³/h | 622 Pa | 637 Pa | 65 | 1770 | 3.11 | 3445 | 3600 | 5 | 460 V | 3 | 227 kgf | GREENHECK | QEID-18-90 | |

NOTES -
1. VFD SHALL NOT DRIVE THE MOTOR TO A SPEED HIGHER THAN THE FAN'S PHYSICAL CONSTRUCTION LIMIT.
   A. LIMIT MAXIMUM VFD SPEED AT 3400 RPM, THAT IS, VFD AT 100% OR 60 HZ = 3400 RPM.
2. CONSTRUCTION AND CONTROLS SHALL BE UL 864 AND UUKL RATED.

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DATE: 06/15/2026
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REVISION # DATE | REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL SCHEDULES
WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 286 OF 378
M-613SH (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

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# THIS SHEET IS ONLY ISSUED TO QUANTIFY PERFORMANCE OF EQUIPMENT IN METRIC UNITS FOR OWNER REFERENCE. PRODUCT DATA AND SHOP DRAWINGS MUST BE SUBMITTED IN IMPERIAL UNITS.

## FOR REFERENCE ONLY - OVERHEAD HEATER SCHEDULE

| TAG | LOCATION | HEATING CAPACITY | INPUT POWER | VOLTAGE | PHASE | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|
| OH-1 | DAYROOM | 1.45 kW | 1.45 kW | 208 V | 1 | FOSTORIA | FSA | |
| OH-2 | OUTDOORS | 1.45 kW | 1.45 kW | 208 V | 1 | FOSTORIA | FSA | |
| OH-3 | OUTDOORS | 3.16 kW | 3.15 kW | 208 V | 1 | FOSTORIA | FSA | |
| OH-4 | OUTDOORS | 3.16 kW | 3.15 kW | 208 V | 1 | FOSTORIA | FSA | |
| OH-5 | OUTDOORS | 3.16 kW | 3.15 kW | 208 V | 1 | FOSTORIA | FSA | |
| OH-6 | OUTDOORS | 3.16 kW | 3.15 kW | 208 V | 1 | FOSTORIA | FSA | |
| OH-7 | OUTDOORS | 3.16 kW | 3.15 kW | 208 V | 1 | FOSTORIA | FSA | |
| OH-8 | OUTDOORS | 3.16 kW | 3.15 kW | 208 V | 1 | FOSTORIA | FSA | |
| OH-9 | OUTDOORS | 3.16 kW | 3.15 kW | 208 V | 1 | FOSTORIA | FSA | |

NOTES -
1. ALL HEATERS SHALL HAVE A PAINTED FINISH. COORDINATE WITH ARCHITECT ON INSTANCE BY INSTANCE BASIS.
2. COORDINATE WITH DIVISION 26. PROVIDE INTERMEDIATE CONTROL ENCLOSURE WITH HEAVY-DUTY CONTACTOR ADJACENT TO ALL NINE (9) HEATER RECEPTACLE. FIELD-WIRE LOW-VOLTAGE CONTROL SIDE TO BACNET RELAY MODULE. RATED FOR (AT LEAST) CONTINUOUS 5KW LOAD.

## FOR REFERENCE ONLY - CAGED HABITAT FAN SCHEDULE

| TAG | LOCATION | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|
| OHF-2 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |
| OHF-3 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |
| OHF-1 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |

NOTES-
1. SEE ELECTRICAL DOCUMENTS FOR ADDITIONAL INFORMATION.

## FOR REFERENCE ONLY - CAGED GUEST FAN SCHEDULE

| TAG | LOCATION | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|
| OGF-1 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |
| OGF-2 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |
| OGF-3 | OUTDOORS | POLA 18 WALL FAN | BRUSHED NICKEL 339218NI | |

NOTES-
1. SEE ELECTRICAL DOCUMENTS FOR ADDITIONAL INFORMATION.

## FOR REFERENCE ONLY - OUTDOORS EXHAUST FAN SCHEDULE

| TAG | LOCATION | MAX AIRFLOW | STATIC | VOLTAGE | PHASE | MANUFACTURER | MODEL | REMARKS |
|---|---|---|---|---|---|---|---|---|
| OEF-1 | OUTDOOR | 500 CFM | 0.00 in-wg | 115 V | 1 | CANARM | P10 | |

NOTES-
1. PROVIDE TWO SPEED MODEL.

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DATE: **06/15/2026**

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REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: **ARABIAN LEOPARD FACILITY (ALF)**
**3001 Connecticut Avenue NW**
**Washington, DC 20008**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**

OPDC PROJECT NUMBER: **2533105**

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DRAWING TITLE: **MECHANICAL SCHEDULES - OUTDOOR ELEMENTS**

WORKING STAFF: **OP** (DESIGNED BY) **OP** (DRAWN BY) **IP** (CHECKED BY)

SHEET NO. **287** OF **378**

**M-614SH** (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

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# CONTROL ABBREVIATIONS

| | | | |
|---|---|---|---|
| # | | P | |
| 2-D | TWO-DIMENSIONAL | P | PUMP |
| 3-D | THREE-DIMENSIONAL | PF | POWER FACTOR |
| | | PH | PENTHOUSE |
| A | | PHC | PREHEAT COIL |
| A | AMPERES | PHDT | PREHEAT DISCHARGE TEMPERATURE |
| AAC | ADVANCED APPLICATION CONTROLLER | PMCS | POWER MONITORING & CONTROL SYSTEM |
| AC | AIR CONDITIONING UNIT | PPM | PARTS PER MILLION |
| AFMS | AIRFLOW MEASURING STATION | PS | PRESSURE SWITCH |
| AHU | AIR HANDLING UNIT | PSI | POUNDS PER SQUARE INCH |
| AI | ANALOG INPUT | | |
| AL | ALARM | R | |
| AO | ANALOG OUTPUT | RA | RETURN AIR / REFRIGERANT ALARM |
| ASC | APPLICATION SPECIFIC CONTROLLER | RAH | RETURN AIR HUMIDITY |
| AX | AUXILIARY CONTACT | RAT | RETURN AIR TEMPERATURE |
| | | RET | RETURN |
| B | | RF | RETURN FAN |
| B | BOILER | RH | RELATIVE HUMIDITY |
| BC | BUILDING CONTROLLER | RHC | REHEAT COIL |
| BD | BACKDRAFT DAMPER | RLF | RELIEF |
| BFR | BOILER FIRING RATE | ROT | ROTATION MONITORING SENSOR |
| BS | BASIN SWEEPER | RPM | ROTATIONS PER MINUTE |
| C | | S | |
| CB | CHILLED BEAM | S | SOLENOID / STATUS |
| CC | COOLING COIL | SA | SUPPLY AIR |
| CCDT | COOLING COIL DISCHARGE TEMPERATURE | SC | SELF CONTAINED UNIT |
| CH | CHILLER | SD | SMOKE DETECTOR / SMOKE DETECTOR |
| CHW | CHILLER WATER | SF | SUPPLY FAN |
| CHWR | CHILLER WATER RETURN | SP | STATIC PRESSURE |
| CHWS | CHILLER WATER SUPPLY | SPA | SETPOINT ADJUSTMENT |
| CI | CURRENT INPUT | SS | START / STOP |
| CM | CONTROL MODULE | SV | SOLENOID VALVE |
| CO | CARBON MONOXIDE | SW | SWITCH |
| CO2 | CARBON DIOXIDE | | |
| CR | CONTROL RELAY | T | |
| CSR | CURRENT SWITCH OR CURRENT SENSING UNIT | T | TEMPERATURE SENSOR |
| CT | COOLING TOWER | TEF | TOILET EXHAUST FAN |
| CU | CONDENSING UNIT | TF | TRANSFER AIR FAN |
| CW | CONDENSER WATER | TH | THERMOSTAT |
| CWR | CONDENSER WATER RETURN | | |
| CWS | CONDENSER WATER SUPPLY | V | |
| | | V | VALVE |
| D | | VAV | VARIABLE AIR VOLUME |
| D | DAMPER | VC | VALVE CONTROL |
| DAT | DISCHARGE AIR TEMPERATURE | VFD | VARIABLE FREQUENCY DRIVE |
| DC | DAMPER CONTROL | VO | VOLTAGE OUTPUT |
| DDC | DIRECT DIGITAL CONTROL | VP | VALVE POSITION |
| DI | DIGITAL INPUT | VS | VIBRATION SWITCH |
| DL | DEMAND LIMIT | | |
| DO | DIGITAL OUTPUT | W | |
| DP | PRESSURE DIFFERENCE / DAMPER POSITION | WC | WATER COLUMN |
| DPS | DIFFERENTIAL PRESSURE SWITCH | WM | WATER METER |
| DPT | DIFFERENTIAL PRESSURE SENSOR | WS | WINDOW SWITCH |
| DX | DIRECT EXPANSION | WTS | WATER TREATMENT SYSTEM |
| E | | | |
| EA | EXHAUST AIR | | |
| ECM | ELECTRONICALLY COMMUTATED MOTOR | | |
| ED | ENABLE/DISABLE | | |
| EF | EXHAUST FAN | | |
| ES | END SWITCH | | |
| F | | | |
| F | DEGREES FAHRENHEIT | | |
| FA | FIRE ALARM | | |
| FACP | FIRE ALARM CONTROL PANEL | | |
| FC | FAULT CODE | | |
| FD | FIRE DAMPER | | |
| FDAT | FAN DISCHARGE AIR TEMPERATURE | | |
| FM | FLOW METER | | |
| FS | FLOW SWITCH | | |
| FZ | FREEZESTAT | | |
| H | | | |
| H | HUMIDITY | | |
| HC | HEATING COIL | | |
| HHW | HEATING HOT WATER | | |
| HHWR | HEATING HOT WATER RETURN | | |
| HHWS | HEATING HOT WATER SUPPLY | | |
| HI | HIGH | | |
| HL | HIGH LEVEL | | |
| HM | HAND MODE | | |
| HPC | HEAD PRESSURE CONTROL | | |
| HX | HEAT EXCHANGER | | |
| HZ | HERTZ | | |
| K | | | |
| KW | KILOWATTS | | |
| L | | | |
| LL | LOW LEVEL | | |
| LO | LOW | | |
| LS/AS | LOCAL SWITCH/AUTO SWITCH | | |
| LV | LEVEL | | |
| M | | | |
| MAT | MAXIMUM | | |
| MAX | MIXED AIR TEMPERATURE | | |
| MIN | MINIMUM | | |
| MS | MOTOR STARTER | | |
| N | | | |
| NC | NORMALLY CLOSED | | |
| NO | NORMALLY OPEN | | |
| O | | | |
| OA | OUTDOOR AIR | | |
| OS | OCCUPANCY SENSOR | | |
| OVR | OVERRIDE | | |
| OWS | OPERATOR WORK STATION | | |

# CONTROL SYMBOLS

## SENSING COMPONENTS

- AIR FLOW MEASURING STATION (DUCT)
- FAN INLET BELL HOUSING AIR FLOW MEASURING STATION
- TEMPERATURE SENSOR
- TEMPERATURE SENSOR (AVERAGING)
- FREEZESTAT
- PRESSURE SWITCH
- PRESSURE SENSOR OR PRESSURESTAT
- FLOW SWITCH
- RELATIVE HUMIDITY SENSOR (DUCT)
- SMOKE DETECTOR (DUCT)
- CARBON DIOXIDE SENSOR (DUCT)
- CARBON MONOXIDE SENSOR (DUCT)
- FLOW METER
- LEVEL SENSOR
- REFRIGERANT LEAK SENSING POINT
- STATIC PRESSURE SENSOR
- DIFFERENTIAL PRESSURE SWITCH
- DIFFERENTIAL PRESSURE SENSOR
- SPACE TEMPERATURE SENSOR
- THERMOSTAT
- SPACE HUMIDITY SENSOR
- HUMIDISTAT
- HYDROGEN SENSOR (WALL)
- CARBON DIOXIDE SENSOR (WALL)
- CARBON MONOXIDE SENSOR (WALL)

## CONTROL COMPONENTS

- 2-WAY ELECTRIC CONTROL VALVE
- 3-WAY ELECTRIC CONTROL VALVE
- BUTTERFLY VALVE ELECTRIC ACTUATOR
- SEGMENTED BALL VALVE ELECTRIC ACTUATOR
- 2-WAY PNEUMATIC CONTROL VALVE
- 3-WAY PNEUMATIC CONTROL VALVE
- BUTTERFLY VALVE PNEUMATIC ACTUATOR
- SEGMENTED BALL VALVE PNEUMATIC ACTUATOR
- 2-WAY SOLENOID VALVE
- 3-WAY SOLENOID VALVE
- PARALLEL BLADE DAMPER - ELECTRIC ACTUATOR
- PARALLEL BLADE DAMPER WITH END SWITCH - ELECTRIC ACTUATOR
- PARALLEL BLADE DAMPER - PNEUMATIC ACTUATOR
- PARALLEL BLADE DAMPER WITH END SWITCH - PNEUMATIC ACTUATOR
- PARALLEL BLADE DAMPER - PNEUMATIC ACTUATOR WITH POSITIONER
- AIR FILTER WITH DIFFERENTIAL PRESSURE SENSOR AND PRESSURE GAUGE

## CONTROL ELECTRICAL COMPONENTS

- INTEGRAL FACE & BYPASS STEAM HEATING COIL
- HOT WATER OR STEAM HEATING COIL
- CHILLED WATER COOLING COIL
- DIRECT EXPANSION HEATING & COOLING COIL
- WATERSIDE ECONOMIZER COOLING COIL
- ENERGY RECOVERY COIL
- ELECTRIC DUCT HEATER
- GAS FIRED DUCT HEATER
- ELECTRIC HUMIDIFIER
- STEAM HUMIDIFIER
- GAS FIRED HUMIDIFIER
- ULTRAVIOLET LIGHT AIR PURIFIER
- FAN OR FAN ARRAY

| | |
|---|---|
| AX | AUXILIARY CONTACT |
| CI | CURRENT INPUT |
| CSR | CURRENT SENSING RELAY |
| CT | CURRENT TRANSDUCER |
| EN | ENABLE / DISABLE |
| SP | SET POINT |
| SS | START / STOP |
| VI | VOLTAGE INPUT |
| VO | VOLTAGE OUTPUT |
| VS | VIBRATION SWITCH |
| CS | CURRENT SWITCH |
| CM | CONTROL MODULE |
| TDR | TIME DELAY RELAY |
| MS | MOTOR STARTER |
| FA | FIRE ALARM INTERFACE |
| VFD | VARIABLE FREQUENCY DRIVE |
| ECM | ELECTRONICALLY COMMUTATED MOTOR |
| EPS | ELECTRIC TO PNEUMATIC SWITCH |
| EPV | ELECTRIC TO PNEUMATIC VALVE |
| ROT | ROTATION MONITORING SENSOR |
| UMCP | UNIT MANUFACTURER'S CONTROL PANEL |
| RLMS | REFRIGERANT LEAK MONITORING SYSTEM |
| FACP | FIRE ALARM CONTROL PANEL |
| EPB | EMERGENCY PUSH BUTTON |
| AFMS | AIRFLOW MEASURING STATION |
| SW | WALL MOUNTED SWITCH |
| OR | OCCUPANCY OVER RIDE SWITCH |
| OS | OCCUPANCY SENSOR |
| R | REFRIGERANT LEAK MONITORING SYSTEM |
| R | COMBINATION HORN/STROBE |
| | HORN OR SIREN |
| | BELL |
| ALT | ALTERNATOR |
| R | RELAY |
| T | POINT NAME'S INDENIFICATION (CORRESPONDS TO CONTROL ABBREVIATIONS) |
| 1 | POINT NUMBER (CONSECUTIVELY COUNTED) |

(T-1 / 1 example)

## FILTER TYPE DIAGRAMS

- FILTER
- BAG FILTER
- ROLL FILTER

# BMS POINT FUNCTION DESCRIPTIONS

HVAC CONTROLS - BMS POINT FUNCTION SCHEDULE

| S. NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | NETWORK POINT | NOTES |
|---|---|---|---|---|---|---|---|---|

POINT NAME: DESCRIPTION OF POINT.

POINT TYPE: TYPE OF INPUT OR OUTPUT CONNECTED TO A BMS CONTROL PANEL. EACH OF THE 4 TYPES (AI, AO, DI, DO) ARE DISCRIPBED BELOW:

AI (ANALOG INPUT): A SIGNAL INPUT TO THE BUILDING MANAGEMENT SYSTEM WHICH HAS A VARIABLE RANGE OF VALUES. ANALOG INPUTS PROVIDE NUMERIC DATA

RATHER THAN SIMPLE "ON/OFF" STATES. EXAMPLES OF ANALOG INPUTS FROM SENSORS INCLUDE TEMPERATURE, PRESSURE, HUMIDITY, AND CHEMICAL CONCENTRATION.

AO (ANALOG OUTPUT): A SIGNAL OUTPUT FROM THE BUILDING MANAGEMENT SYSTEM WHICH HAS A VARIABLE RANGE OF VALUES. ANALOG OUTPUTS PROVIDE

PROPORTIONAL CONTROL COMMANDS TO EQUIPMENT. EXAMPLES OF ANALOG OUTPUTS INCLUDE SET POINT ADJUSTMENT AND MOTOR SPEED.

DI (DIGITAL INPUT): A SIGNAL INPUT TO THE BUILDING MANAGEMENT SYSTEM WHICH HAS A BINARY (1 OR 0) VALUE. DIGITAL INPUTS PROVIDE DATA ABOUT THE STATUS

OF A SPACE OR EQUIPMENT. EXAMPLES OF DIGITAL INPUTS INCLUDE MOTOR STATUS ("ON/OFF"), VALVE POSITION ("OPEN/CLOSED"), OCCUPANCY

("OCCUPIED/UNOCCUPIED"), OR LIMIT ALARMS SUCH AS HIGH VOLTAGE OR LOW FUEL.

DO (DIGITAL OUTPUT): A SIGNAL OUTPUT FROM THE BUILDING MANAGEMENT SYSTEM WHICH HAS A BINARY (1 OR 0) VALUE. THESE OUTPUTS CONTROL

EQUIPMENT WHICH OPERATES IN TWO MAIN STATES (SUCH AS "ON" AND "OFF"). EXAMPLES OF THESE OUTPUTS INCLUDE MOTOR START/STOP AND DAMPER OR

VALVE POSITION CONTROL ("OPEN/CLOSE").

FAIL ON (OPEN): CONTROLLED DEVICE REVERTS TO THE OPEN POSITION OR THE ENGAGED STATE WHEN POWER OR CONTROL SIGNAL IS LOST.

FAIL OFF (CLOSED): CONTROLLED DEVICE REVERTS TO THE CLOSED POSITION OR THE DISENGAGED STATE WHEN POWER OR CONTROL SIGNAL IS LOST.

LAST COMMANDED STATE: CONTROLLED DEVICE REMAINS IN THE SAME POSITION OR STATE, WHEN POWER OR CONTROL SIGNAL IS LOST.

NETWORK POINT: POINT IS EITHER DIRECTLY OR INDIRECTLY CONNECTED TO THE BMS. IT IS COMMUNICATED TO THE BMS THROUGH A COMMUNICATION

GATEWAY FROM ANOTHER INTELLIGENT PIECE OF EQUIPMENT OR PROVIDED BY A COMMUNICATION PROTOCOL (I.E. BACNET, LONWORKS, OR MODBUS).

FOR REFERENCE ONLY

ALF

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Washington, DC 20024-2520

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
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M-801CS (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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A. **FOR EVERY SINGULAR INSTANCE OF EQUIPMENT ON THE BAS, THE CONTRACTOR SHALL COORDINATE EQUIPMENT TAGS WITH THE OWNER TO ENSURE FORMATTING IS CONSISTENT WITH THE SMITHSONIAN SERVER. CONTRACTOR SHALL INCLUDE BOTH THE SERVER TAG AND THE PHYSICAL TAG IN THE BAS AS REFERENCE INFORMATION.**

**TRIM & RESPOND SETPOINT RESET LOGIC**

A. TRIM & RESPOND SETPOINT RESET LOGIC AND ZONE/SYSTEM RESET REQUESTS WHERE REFERENCED IN SEQUENCES SHALL BE IMPLEMENTED AS DESCRIBED BELOW.

B. A "REQUEST" IS A CALL TO RESET A STATIC/DIFFERENTIAL PRESSURE OR TEMPERATURE SETPOINT, GENERATED BY DOWNSTREAM ZONES OR AIR HANDLING SYSTEMS. THESE REQUESTS ARE SENT UPSTREAM TO THE PLANT OR SYSTEM THAT SERVES THE ZONE OR AIR HANDLER WHICH GENERATED THE REQUEST.
   1. FOR EACH DOWNSTREAM ZONE OR SYSTEM, AND FOR EACH TYPE OF SETPOINT RESET REQUEST LISTED FOR THE ZONE/SYSTEM, PROVIDE THE FOLLOWING SOFTWARE POINTS:
      a. IMPORTANCE MULTIPLIER (DEFAULT = 1)
      b. REQUEST-HOURS ACCUMULATOR. PROVIDED SYSTEM-OK IS TRUE FOR THE ZONE/SYSTEM, EVERY X MINUTES (DEFAULT 5 MINUTES), ADD X DIVIDED BY 60 TIMES THE CURRENT NUMBER OF REQUESTS TO THIS REQUEST-HOURS ACCUMULATOR POINT.
      c. SYSTEM RUN-HOURS TOTAL. THIS IS THE NUMBER OF HOURS THE ZONE/SYSTEM HAS BEEN OPERATING IN ANY MODE OTHER THAN UNOCCUPIED MODE.
      d. CUMULATIVE%-REQUEST-HOURS. THIS IS THE ZONE/SYSTEM REQUEST-HOURS DIVIDED BY THE ZONE/SYSTEM RUN-HOURS (THE HOURS IN ANY MODE OTHER THAN UNOCCUPIED MODE) SINCE THE LAST RESET, EXPRESSED AS A PERCENTAGE.
      e. THE REQUEST-HOURS ACCUMULATOR AND SYSTEM RUN-HOURS TOTAL ARE RESET TO ZERO AS FOLLOWS:
         - RESET AUTOMATICALLY FOR AN INDIVIDUAL ZONE/SYSTEM WHEN THE SYSTEM RUN-HOURS TOTAL EXCEEDS 400 HOURS.
         - RESET MANUALLY BY A GLOBAL OPERATOR COMMAND. THIS COMMAND WILL SIMULTANEOUSLY RESET THE REQUEST-HOURS POINT FOR ALL ZONES SERVED BY THE SYSTEM.
      f. A LEVEL 4 ALARM IS GENERATED IF THE ZONE IMPORTANCE MULTIPLIER IS GREATER THAN ZERO, THE ZONE/SYSTEM CUMULATIVE%-REQUEST-HOURS EXCEEDS 70%, AND THE TOTAL NUMBER OF ZONE/SYSTEM RUN-HOURS EXCEEDS 40.
      g. SEE ZONE AND AIR HANDLING SYSTEM CONTROL SEQUENCES FOR LOGIC TO GENERATE REQUESTS.
      h. MULTIPLY THE NUMBER OF REQUESTS DETERMINED FROM ZONE/SYSTEM LOGIC TIMES THE IMPORTANCE MULTIPLIER AND SEND TO THE SYSTEM/PLANT THAT SERVES THE ZONE/SYSTEM. SEE SYSTEM/PLANT LOGIC TO SEE HOW REQUESTS ARE USED IN TRIM & RESPOND LOGIC.

FOR EACH UPSTREAM SYSTEM OR PLANT SETPOINT BEING CONTROLLED BY A T&R LOOP, DEFINE THE FOLLOWING VARIABLES. ALL VARIABLES BELOW SHALL BE ADJUSTABLE FROM A RESET GRAPHIC ACCESSIBLE FROM A HYPERLINK ON THE ASSOCIATED SYSTEM/PLANT GRAPHIC. INITIAL VALUES ARE DEFINED IN SYSTEM/PLANT SEQUENCES BELOW. VALUES FOR TRIM, RESPOND, TIME STEP, ETC. SHALL BE TUNED TO PROVIDE STABLE CONTROL.

TABLE B.1.C - TRIM AND RESPOND VARIABLES AND SETTINGS

| VARIABLE | DEFINITION |
|---|---|
| $SP_0$ | INITIAL SETPOINT |
| $SP_{min}$ | MINIMUM SETPOINT |
| $SP_{max}$ | MAXIMUM SETPOINT |
| $T_d$ | DELAY TIMER |
| T | TIME STEP |
| I | NUMBER OF IGNORED REQUESTS |
| R | NUMBER OF REQUESTS FROM ZONES/SYSTEMS |
| $SP_{trim}$ | TRIM AMOUNT |
| $SP_{res}$ | RESPOND AMOUNT (MUST BE OPPOSITE IN SIGN TO $SP_{trim}$ AND SHOULD GENERALLY BE HIGHER) |
| $SP_{res-max}$ | MAXIMUM RESPONSE PER TIME INTERVAL (MUST BE SAME SIGN AS $SP_{res}$) |

A. TRIM & RESPOND LOGIC SHALL RESET SETPOINT WITHIN THE RANGE SPMIN TO SPMAX. WHEN THE ASSOCIATED DEVICE (E.G. FAN, PUMP) IS OFF, THE SETPOINT SHALL BE $SP_0$. THE RESET LOGIC SHALL BE ACTIVE WHILE THE ASSOCIATED DEVICE IS PROVEN ON, STARTING $T_d$ AFTER INITIAL DEVICE START COMMAND. WHEN ACTIVE, EVERY TIME STEP T, TRIM THE SETPOINT BY $SP_{trim}$. IF THERE ARE MORE THAN I REQUESTS, RESPOND BY CHANGING THE SETPOINT BY $SP_{res}$ * (R-I), (I.E. THE NUMBER OF REQUESTS MINUS THE NUMBER OF IGNORED REQUESTS), BUT NO MORE THAN $SP_{res-max}$. IN OTHER WORDS, EVERY TIME STEP T:

CHANGE SETPOINT BY $SP_{TRIM}$
IF R>I, ALSO CHANGE SETPOINT BY (R-I)*$SP_{res}$ BUT NO LARGER THAN $SP_{res-max}$

A. FOR EACH TRIM AND RESPOND LOOP, GENERATE A GRAPHIC SIMILAR TO THE SAMPLE FIGURE. THE GRAPHIC DOES NOT NEED TO BE CUSTOMIZED TO MATCH THE FIGURE IN TERMS OF AESTHETICS; HOWEVER, IT MUST CONTAIN ALL OF THE INFORMATION AND FUNCTIONALITY SHOWN IN THE FIGURE. SUBMIT A SAMPLE OF THE INTENDED GRAPHIC FOR APPROVAL. THE FIGURE MUST BE CONSISTENT FOR EACH TRIM & RESPOND LOOP IN THE PROJECT.

**EQUIPMENT STAGING AND ROTATION**

A. PARALLEL EQUIPMENT SHALL BE LEAD/LAG OR LEAD/STANDBY ROTATED TO MAINTAIN EVEN WEAR.

B. TWO RUNTIME POINTS SHALL BE DEFINED FOR EACH EQUIPMENT:
   1. LIFETIME RUNTIME: THE CUMULATIVE RUNTIME OF THE EQUIPMENT SINCE EQUIPMENT START-UP. THIS PIONT SHALL NOT BE READILY RESETTABLE BY OPERATORS.
   2. STAGING RUNTIME: AN OPERATOR RESETTABLE RUNTIME POINT THAT STORES CUMULATIVE RUNTIME SINCE THE LAST OPERATOR RESET.

C. LEAD/LAG EQUIPMENT:
   1. UNLESS OTHERWISE NOTED, PARALLEL STAGED DEVICES (SUCH AS PUMPS, TOWERS) THAT ARE NOT REDUNDANT SHALL BE LEAD/LAG ALTERNATED WHEN MORE THAN ONE IS OFF OR MORE THAN ONE IS ON SO THAT THE EQUIPMENT WITH THE MOST OPERATING HOURS AS DETERMINED BY STAGING RUNTIME IS MADE THE LAST STAGE EQUIPMENT AND THE ONE WITH THE LEAST NUMBER OF HOURS IS MADE THE LEAD STAGE EQUIPMENT.

D. LEAD/STANDBY EQUIPMENT:
   1. UNLESS EQUIPMENT RUNS CONTINUOUSLY, PARALLEL EQUIPMENT THAT ARE 100% REDUNDANT SHALL BE LEAD/STANDBY ALTERNATED WHEN MORE THAN ONE OF THE EQUIPMENT IS OFF SO THAT THE EQUIPMENT WITH THE MOST OPERATING HOURS AS DETERMINED BY STAGING RUNTIME IS MADE THE LAST STAGE EQUIPMENT AND THE ONE WITH THE LEAST NUMBER OF HOURS IS MADE THE EARLIER STAGE EQUIPMENT.
   2. IF EQUIPMENT RUNS CONTINUOUSLY, LEAD/STANDBY POSITIONS SHALL SWITCH AT AN ADJUSTABLE DAY OF THE WEEK AND TIME (E.G. TUESDAY AT 10:00AM) BASED ON STAGING RUNTIME; STANDBY EQUIPMENT SHALL FIRST BE STARTED AND PROVEN ON BEFORE FORMER LEAD EQUIPMENT IS CHANGED TO STANDBY AND SHUT OFF.
      a. VARIABLE SPEED FANS AND PUMPS SHALL HAVE A DECELERATION RATE OF 1 HZ/SECOND OR SLOWER SET IN BAS LOGIC WHEN DISABLED TO PREVENT NUESANCE TRIPS OF OPERATING EQUIPMENT (E.G. CHILLERS).

E. EXCEPTIONS TO LEAD/LAG AND LEAD/STANDBY ROTATION
   1. OPERATORS WITH APPROPRIATE ACCESS LEVEL SHALL BE ABLE TO MANUALLY COMMAND STAGING ORDER VIA SOFTWARE POINTS, BUT NOT OVERRIDING THE IN-ALARM OR HAND-OPERATION LOGIC IN THE FOLLOWING SUBSECTIONS.
      a. STAGING ORDER CHANGES INITIATED VIA OPERATOR OVERRIDE SHALL BE INSTITUTED AS PART OF NORMAL STAGING EVENTS.
      b. STAGING ORDER SHALL REMAIN OVERRIDEN UNTIL RELEASED BY OPERATORS.
   2. FAULTED EQUIPMENT:
      a. A FAULTED EQUIPMENT IS ANY EQUIPMENT COMMANDED TO RUN THAT IS EITHER NOT RUNNING OR UNABLE TO PERFORM ITS REQUIRED DUTY. IF AN OPERATING EQUIPMENT HAS ANY FAULT CONDITION DESCRIBED SUBSEQUENTLY, A LEVEL 2 ALARM SHALL BE GENERATED AND A RESPONSE SHALL BE TRIGGERED AS DEFINED BELOW.
      b. FANS AND PUMPS
         - STATUS POINT NOT MATCHING ITS ON/OFF POINT FOR 3 SECONDS AFTER A TIME DELAY OF 15 SECONDS WHILE THE EQUIPMENT IS COMMANDED ON.
      c. HEAT PUMPS
         - SAFETY SHUTDOWN ALARM CONDITIONS EITHER THROUGH NETWORK OR HARDWIRED ALARM CONTACT.
         - HEAT PUMP MODULE IS MANUALLY SHUT OFF AS INDICATED BY THE STATUS OF THE LOCAL/AUTO SWITCH FROM CHILLER GATEWAY, OR
         - HEAT PUMP MODULE STATUS REMAINS OFF 5 MINUTES AFTER COMMAND TO START (NOTE: THIS CONDITION ONLY APPLIES WHEN A MODULE FIRST STARTS, I.E., ONCE STATUS IS PROVEN, THEN STATUS IS NO LONGER USED AS A FAULT CONDITION BECAUSE STATUS WILL COME AND GO IF A MODULE CYCLES ON LOW LOAD), OR
         - CHW AND/OR HHW ISOLATION VALVE FEEDBACK INDICATES A MISMATCH WITH THE COMMAND SIGNAL 90 SECONDS AFTER THE VALVE IS COMMANDED TO OPEN/CLOSE OR
         - FOR 10 MINUTES, CHILLED WATER RETURN TEMPERATURE HAS BEEN AT LEAST 5°F ABOVE THE CHWST SETPOINT, AND DELTA-T ACROSS THE ARRAY, AS DETERMINED BASED ON THE DIFFERENCE BETWEEN CHILLED WATER RETURN TEMPERATURE AND CHILLED WATER SUPPLY TEMPERATURE MEASURED AT THE MODULE (I.E. NOT COMMON CHWST), HAS BEEN LESS THAN 3°F.
         - FOR 10 MINUTES, HEATING HOT WATER RETURN TEMPERATURE HAS BEEN AT LEAST 5°F BELOW THE HHWST SETPOINT, AND DELTA-T ACROSS THE ARRAY, AS DETERMINED BASED ON THE DIFFERENCE BETWEEN HEATING HOT WATER RETURN TEMPERATURE AND HEATING HOT WATER SUPPLY TEMPERATURE MEASURED AT THE MODULE (I.E. NOT COMMON HHWST), HAS BEEN LESS THAN 3°F.
      d. BOILERS
         - SAFETY SHUTDOWN ALARM CONDITION EITHER THROUGH NETWORK OR HARDWIRED ALARM CONTACT, OR
         - HHW ISOLATION VALLVE FEEDBACK INDICATES VALVE IS NOT OPEN 90 SECONDS AFTER VALVE IS COMMANDED OPEN, OR
         - IF BOILER LEAVING WATER TEMPERATURE REMAINS 15°F BELOW SETPOINT FOR 15 MINUTES AND DELTA-T ACROSS THE BOILER, AS DETERMINED BASED ON THE DIFFERENCE BETWEEN HEATING HOT WATER SUPPLY TEMPERATURE AND HEATING HOT WATER RETURN TEMPERATURE MEASURED AT THE BOILER (IE., NOT COMMON HHWST), HAS BEEN LESS THAN 10°F.
   3. UPON IDENTIFICATION OF A FAULT CONDITION:
      a. FOR FANS AND PUMPS:
         - THE NEXT COMMANDED OFF EQUIPMENT IN THE STAGING ORDER, EQUIPMENT "B", SHALL BE COMMANDED ON WHILE ALARMING EQUIPMENT "A" REMAINS COMMANDED ON.
         - IF EQUIPMENT "B" FAILS TO PROVE STATUS (I.E. IT ALSO GOES INTO ALARM), IT SHALL REMAIN COMMANDED ON AND THE PRECEEDING STEP SHALL BE REPEATED UNTIL THE QUANTITY OF EQUIPMENT CALLED FOR BY THE CURRENT STAGE HAS PROVEN ON, OR THERE ARE NO MORE AVAILABLE EQUIPMENT.
         - SET ALARMING EQUIPMENT TO THE LAST POSITIONS IN THE LEAD/LAG OR LEAD/STANDBY STAGING ORDER SEQUENCED REVERSE CHRONOLOGICALLY (I.E., THE EQUIPMENT THAT ALARMED MOST RECENTLY IS SENT TO LAST POSITION).
         - STAGING ORDER OF NON-ALARMING EQUIPMENT SHALL FOLLOW THE EVEN WEAR LOGIC. EQUIPMENT IN ALARM CAN ONLY AUTOMATICALLY MOVE UP ON THE STAGING ORDER IF ANOTHER EQUIPMENT GOES INTO ALARM.
         - EQUIPMENT IN ALARM SHALL RUN IF SO CALLED FOR BY THE LEAD/LAG OR LEAD/STANDBY STAGING ORDER AND PRESENT STAGE.
      b. FOR HEAT PUMP MODULES AND BOILERS:
         - THE NEXT COMMANDED OFF EQUIPMENT IN THE STAGING ORDER, EQUIPMENT "B", SHALL BE COMMANDED ON WHILE ALARMING EQUIPMENT "A" IS COMMANDED OFF AND SET TO THE LAST POSITION IN THE LEAD/LAG STAGING ORDER.
         - IF EQUIPMENT B FAILS TO PROVE STATUS (I.E., IT ALSO GOES INTO ALARM), REPEAT THE PRECEDING STEP UNTIL THE QUANTITY OF EQUIPMENT CALLED FOR BY THE LEAD/LAG LOGIC HAVE PROVEN ON OR UNTIL ALL EQUIPMENT HAS BEEN TRIED.
         - IF ALL EQUIPMENT HAS BEEN TRIED AND THE QUANTITY OF NON-ALARMING EQUIPMENT IS LESS THAN CALLED FOR THEN THE MOST RECENTLY ALARMED EQUIPMENT WILL REMAIN COMMANDED ON.
         - STAGING ORDER OF NON-ALARMING EQUIPMENT SHALL FOLLOW THE EVEN WEAR LOGIC. EQUIPMENT IN ALARM CAN ONLY AUTOMATICALLY MOVE UP IN THE STAGING ORDER IF ANOTHER EQUIPMENT GOES INTO ALARM.
         - EQUIPMENT IN ALARM SHALL RUN IF SO CALLED FOR BY THE LEAD/LAG STAGING ORDER AND PRESENT STAGE.
      c. HAND OPERATION. IF A PIECE OF EQUIPMENT IS ON-IN-HAND (E.G., VIA AN HOA SWITCH OR LOCAL CONTROL OF VFD), THE EQUIPMENT SHALL BE SET TO THE LEAD DEVICE, AND A LEVEL 4 ALARM SHALL BE GENERATED. THE EQUIPMENT WILL REMAIN AS LEAD UNTIL HAND OPERATION IS NO LONGER DETECTED. HAND OPERATION IS DETERMINED BY THE FOLLOWING:
         - WHERE PROVIDED - HAND MODE STATUS POINT INDICATES THE DEVICE IS IN HAND MODE OR;
         - FANS AND PUMPS: STATUS POINT NOT MATCHING ITS ON/OFF POINT FOR 15 SECONDS AFTER A TIME DELAY OF 60 SECONDS WHEN THE EQUIPMENT IS COMMANDED OFF.

**HIERARCHICAL ALARM SUPPRESSION**

A. FOR EACH PIECE OF EQUIPMENT OR SPACE CONTROLLED BY THE BAS, DEFINE ITS RELATIONSHIP (IF ANY) TO OTHER EQUIPMENT IN TERMS OF "SOURCE", "LOAD", OR "SYSTEM."
   1. A COMPONENT IS A "SOURCE" IF IT PROVIDES RESOURCES TO A DOWNSTREAM COMPONENT, SUCH AS A CHILLER PROVIDING CHILLED WATER (CHW) TO AN AHU.
   2. A COMPONENT IS A "LOAD" IF IT RECEIVES RESOURCES FROM AN UPSTREAM COMPONENT, SUCH AS AN AHU THAT RECEIVES CHW FROM A CHILLER.
   3. THE SAME COMPONENT MAY BE BOTH A LOAD (RECEIVING RESOURCES FROM AN UPSTREAM SOURCE) AND A SOURCE (PROVIDING RESOURCES TO A DOWNSTREAM LOAD).
   4. A SET OF COMPONENTS IS A "SYSTEM" IF THEY SHARE A LOAD IN COMMON (I.E., COLLECTIVELY ACT AS A SOURCE TO DOWNSTREAM EQUIPMENT, SUCH AS A SET OF CHILLERS IN A LEAD/LAG RELATIONSHIP SERVING AIR HANDLERS)
      a. IF A SINGLE COMPONENT ACTS AS A SOURCE FOR DOWNSTREAM LOADS (E.G., AN AHU AS A SOURCE FOR ITS VAV BOXES), THEN THAT SINGLE-SOURCE COMPONENT SHALL BE DEFINED AS A "SYSTEM" OF ONE ELEMENT.
      b. FOR EQUIPMENT WITH ASSOCIATED PUMPS (CHILLERS, BOILERS, COOLING TOWERS):
         - IF THE PUMPS ARE IN A ONE-TO-ONE RELATIONSHIP WITH EQUIPMENT THEY SERVE, THE PUMPS SHALL BE TREATED AS PART OF THE SYSTEM TO WHICH THEY ARE ASSOCIATED (I.E. THEY ARE NOT CONSIDERED LOADS), AS A PUMP FAILURE WILL NECESSARILY DISABLE ITS ASSOCIATED EQUIPMENT.
         - IF THE PUMPS ARE HEADERED TO THE EQUIPMENT THEY SERVE, THEN THE PUMPS MAY BE TREATED AS A SYSTEM WHICH IS A LOAD RELATIVE TO THE UPSTREAM EQUIPMENT (E.G. CHILLERS) AND A SOURCE RELATIVE TO DOWNSTREAM EQUIPMENT (E.G., AIR HANDLERS).

B. FOR EACH SYSTEM, THERE SHALL BE A SystemOK FLAG, WHICH IS EITHER TRUE OR FALSE.

C. SystemOK SHALL BE TRUE WHEN ALL OF THE FOLLOWING ARE TRUE:
   1. THE SYSTEM IS PROVEN ON.
   2. THE SYSTEM IS ACHIEVING ITS TEMPERATURE AND/OR PRESSURE SETPOINT(S) FOR AT LEAST 5 MINUTES.
   3. THE SYSTEM IS READY AND ABLE TO SERVE ITS LOAD.

D. SystemOK SHALL BE FALSE WHEN THE SYSTEM IS STARTING UP (I.E., BEFORE REACHING SETPOINT) OR WHEN ENOUGH OF THE SYSTEM'S COMPONENTS ARE UNAVAILABLE (IN ALARM, DISABLED, OR TURNED OFF) TO DISRUPT THE ABILITY OF THE SYSTEM TO SERVE ITS LOAD.
   1. BY DEFAULT, LEVEL 1 THROUGH LEVEL 3 COMPONENT ALARMS (INDICATING EQUIPMENT FAILURE) SHALL INHIBIT SystemOK. LEVEL 4 COMPONENT ALARMS (MAINTENANCE AND ENERGY EFFICIENCY ALARMS) SHALL NOT AFFECT SystemOK.

E. THE BAS SHALL SELECTIVELY SUPPRESS (I.E., FAIL TO ANNOUNCE; ALARMS MAY STILL BE LOGGED TO A DATABASE) ALARMS FOR LOAD COMPONENTS IF SystemOK IS FALSE FOR THE SOURCE SYSTEM THAT SERVES THAT LOAD.
   1. IF SystemOK IS FALSE FOR A COOLING WATER SYSTEM (I.E., CHILLER, COOLING TOWER, OR ASSOCIATED PUMP), THEN ONLY HIGH-TEMPERATURE ALARMS FROM THE LOADS SHALL BE SUPPRESSED.
   2. IF SystemOK IS FALSE FOR A HEATING WATER SYSTEM (I.E., BOILER OR ASSOCIATED PUMP), THEN ONLY LOW TEMPERATURE ALARMS FROM THE LOADS SHALL BE SUPPRESSED.
   3. IF SystemOK IS FALSE FOR AN AIR-SIDE SYSTEM (AIR HANDLER, FAN COIL, VAV BOX, ETC.), THEN ALL ALARMS FROM THE LOADS SHALL BE SUPPRESSED.

F. THIS HIERARCHICAL SUPPRESSION SHALL CASCADE THROUGH MULTIPLE LEVELS OF LOAD-SOURCE RELATIONSHIPS SUCH THAT ALARMS AT DOWNSTREAM LOADS SHALL ALSO BE SUPPRESSED. REFER TO GL36 FOR ADDITIONAL EXAMPLES OF THESE RELATIONSHIPS AND LOGIC.

G. THE FOLLOWING ALARMS WILL NEVER BE SUPPRESSED BY THIS LOGIC:
   1. LIFE/SAFETY AND LEVEL 1 ALARMS
   2. FAILURE TO START ALARMS (I.E., EQUIPMENT IS COMMANDED ON, BUT STATUS POINT SHOWS EQUIPMENT TO BE OFF)
   3. FAILURE-TO-STOP/HAND ALARMS (I.E., EQUIPMENT IS COMMANDED OFF, BUT STATUS POINTS SHOWS EQUIPMENT TO BE ON)

H. TIME-BASED SUPPRESSION
   1. CALCULATE A TIME-DELAY PERIOD AFTER ANY CHANGE IN SETPOINT BASED ON THE DIFFERENCE BETWEEN THE CONTROLLED VARIABLE (E.G., ZONE TEMPERATURE) AT THE TIME OF THE CHANGE AND THE NEW SETPOINT. THE DEFAULT TIME DELAY PERIOD SHALL BE AS FOLLOWS:
      a. FOR THERMAL ZONE TEMPERATURE ALARMS: 10 MINUTES PER °F OF DIFFERENCE.
      b. FOR THERMAL ZONE TEMPERATURE COOLING REQUESTS: 5 MINUTES PER °F OF DIFFERENCE BUT NO LONGER THAN 30 MINUTES.

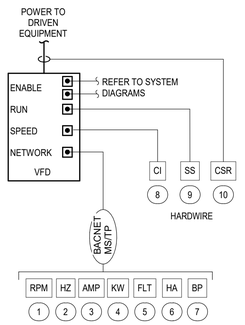
FOR THERMAL ZONE TEMPERATURE HEATING REQUESTS: 5 MINUTES PER °F OF DIFFERENCE BUT NO LONGER THAN 30 MINUTES.

VAV TEMPERATURE CONTROL LOOP OUTPUT

| TU TAG | ROOMS SERVED | ZONE COOLING | ZONE HEATING | STATUS |
|---|---|---|---|---|
| VAV-1 | RM-XXX | 85% | 85% | ON |
| VAV-2 | RM-XXX | XX% | XX% | OFF |
| VAV-3 | RM-XXX | XX% | XX% | ON |
| VAV-4 | RM-XXX | XX% | XX% | ON |
| VAV-5 | RM-XXX | XX% | XX% | ON |
| VAV-6 | RM-XXX | XX% | XX% | ON |
| VAV-7 | RM-XXX | XX% | XX% | ON |

**POLLING STATUS NOTES:**

A. THE BAS SHALL DISPLAY A POLLING STATUS SIMILAR TO TEMPLATE SHOWN ABOVE, FOR ALL TERMINAL UNITS AND ZONES FOR THE CARE BUILDING.
B. THE BAS SHALL CONSTANTLY UPDATE THE ZONE COOLING AND ZONE HEATING LOOP OUTPUT.
C. THE STATUS BUTTON SHALL BE A USER ADJUSTABLE INTERFACE POINT SUCH THAT A SINGLE (OR MULTIPLE) VAV BOX CAN QUICKLY BE REMOVED FROM THE POLLING LOGIC CONTROLLING SYSTEM RESET FUNCTIONS.
D. THE STATUS BUTTON SHALL NOT BE RESET BASED ON ANY TIMER. USER INTERFACE SHALL BE THE ONLY MEANS OF UPDATING THE POLLING STATUS.




VFD NOTES

A. UNLESS OTHERWISE NOTED, THIS DIAGRAM SHALL APPLY TO VFDs ON THE ENTIRE PROJECT SERVING PUMPS AND FANS WITH THE EXCEPTION OF VFDs THAT ARE INTEGRAL TO A PIECE OF EQUIPMENT SUCH AS A CHILLER COMPRESSOR VARIABLE SPEED DRIVE.
B. PROVIDE INTEGRATION VIA BACNET MS/TP.
C. ALL VFDs ON THE ENTIRE PROJECT SHALL BE PROVIDED BY A SINGLE MANUFACTURER UNLESS OTHERWISE INDICATED.
D. REFER TO SPECIFIC DIAGRAMS FOR SAFETY AND RUN-ONLY REQUIREMENTS FOR EACH VFD.
E. ACTUAL AMPERAGE (POINT 7) SHALL BE OBTAINED VIA BACNET MS/TP.
F. VFD SPEED POINTS
   1. THE SPEED AO SENT TO VFDs SHALL BE CONFIGURED SUCH THAT 0% SPEED CORRESPONDS TO 0 HZ, AND 100% SPEED CORRESPONDS TO MAXIMUM SPEED CONFIGURED IN THE VFD.
   2. FOR EACH PIECE OF EQUIPMENT, THE MINIMUM SPEED SHALL BE STORED IN A SINGLE SOFTWARE POINT; IN THE CASE OF A HARD-WIRED VFD INTERFACE, THE MINIMUM SPEED SHALL BE THE LOWEST SPEED COMMAND SENT TO THE DRIVE BY THE BAS. THE ACTIVE MINIMUM SPEED PARAMETER SHALL BE READ EVERY 60 MINUTES VIA THE DRIVE'S NETWORK INTERFACE. WHEN A MISMATCH BETWEEN THE DRIVE'S ACTIVE MINIMUM SPEED AND THE MINIMUM SPEED STORED IN THE SOFTWARE POINT IS DETECTED, THE MINIMUM SPEED STORED IN THE SOFTWARE POINT SHALL BE WRITTEN TO THE VFD VIA THE NETWORK INTERFACE TO RESTORE THE ACTIVE MINIMUM SPEED PARAMETER TO ITS DEFAULT VALUE, AND GENERATE A LEVEL 4 ALARM.
G. HAND-OFF-AUTO (HOA) OPERATION
   1. PROVIDE HOA SETTINGS AS PART OF THE VARIABLE FREQUENCY DRIVE (VFD) THROUGH THE DRIVES KEYPAD FOR THE SUPPLY FAN.
   2. WHEN IN HAND MODE, THE MOTOR CONTROLLED BY THE VFD WILL START.
   3. WHEN IN OFF MODE, THE MOTOR CONTROLLED BY THE VFD WILL BE OFF.
   4. WHEN IN AUTO MODE, THE MOTOR CONTROLLED BY THE VFD WILL BE STARTED AND STOPPED THROUGH THE BAS.
   5. UPON ACTIVATION, SAFTIES WILL STOP THE MOTOR IN AUTO OR HAND MODES.
H. LOCAL-REMOTE SPEED CONTROL
   1. LOCAL-REMOTE SETTINGS WILL BE PROVIDED AS PART OF THE VFD THROUGH THE DRIVES KEYPAD.
   2. IN LOCAL MODE THE MOTOR SPEED WILL BE CONTROLLED THROUGH A MANUAL SPEED CONTROL LOCATED AT THE DRIVES CONTROL PANEL.
   3. IN REMOTE MODE, THE MOTOR SPEED WILL BE CONTROLLED BY THE BAS.

## HVAC CONTROLS - BMS POINT FUNCTION SCHEDULE -...

| S. NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | NETWORK POINT | NOTES |
|---|---|---|---|---|---|---|---|---|
| VFD | | | | | | | | |
| 1 | MOTOR SPEED (RPM) | RPM | AI | | | | ■ | |
| 2 | OUTPUT FREQUENCY (HZ) | HZ | AI | | | | ■ | |
| 3 | OUTPUT CURRENT (AMPS) | AMP | AI | | | | ■ | |
| 4 | OUTPUT POWER (KW) | KW | AI | | | | ■ | |
| 5 | FAULT ALARM | FLT | DI | | | | ■ | |
| 6 | HAND MODE STATUS | HA | DI | | | | ■ | |
| 7 | BYPASS MODE STATUS | BP | DI | | | | ■ | |
| 8 | SPEED CONTROL | CI | AO | | | ■ | | |
| 9 | START/STOP | SS | DO | | | | | |
| 10 | RUN STATUS | CSR | DI | | | | | |

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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - GENERAL INFORMATION |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 289 OF 378

M-802CS

DISCIPLINE TYPE SEQUENCE DRAWING TYPE

| SYSTEM ARCHITECTURE LEGEND | |
|---|---|
| SYMBOL | DESCRIPTION |
| | OPERATOR'S WORK STATION (OWS) |
| | OPERATOR'S WORK STATION (OWS) -VIRTUAL (DEDICATED WORKSTATION CONNECTION POINT) |
| | LAPTOP - PORTIBLE PROGRAMMER'S TOOL (PPT) |
| | PRINTER |
| | BUILDING CONTROLLER (BC) |
| | ADVANCE APPLICAITON CONTROLLER (AAC) |
| | APPLICATION SPECIFIC CONTROLLER (ASC) |
| | GATEWAY |
| | BACNET IP |
| | BACNET MSTP |
| | MODBUS |
| | SWITCH / HUB |
| | CLOUD / CAMPUS NETWORK |

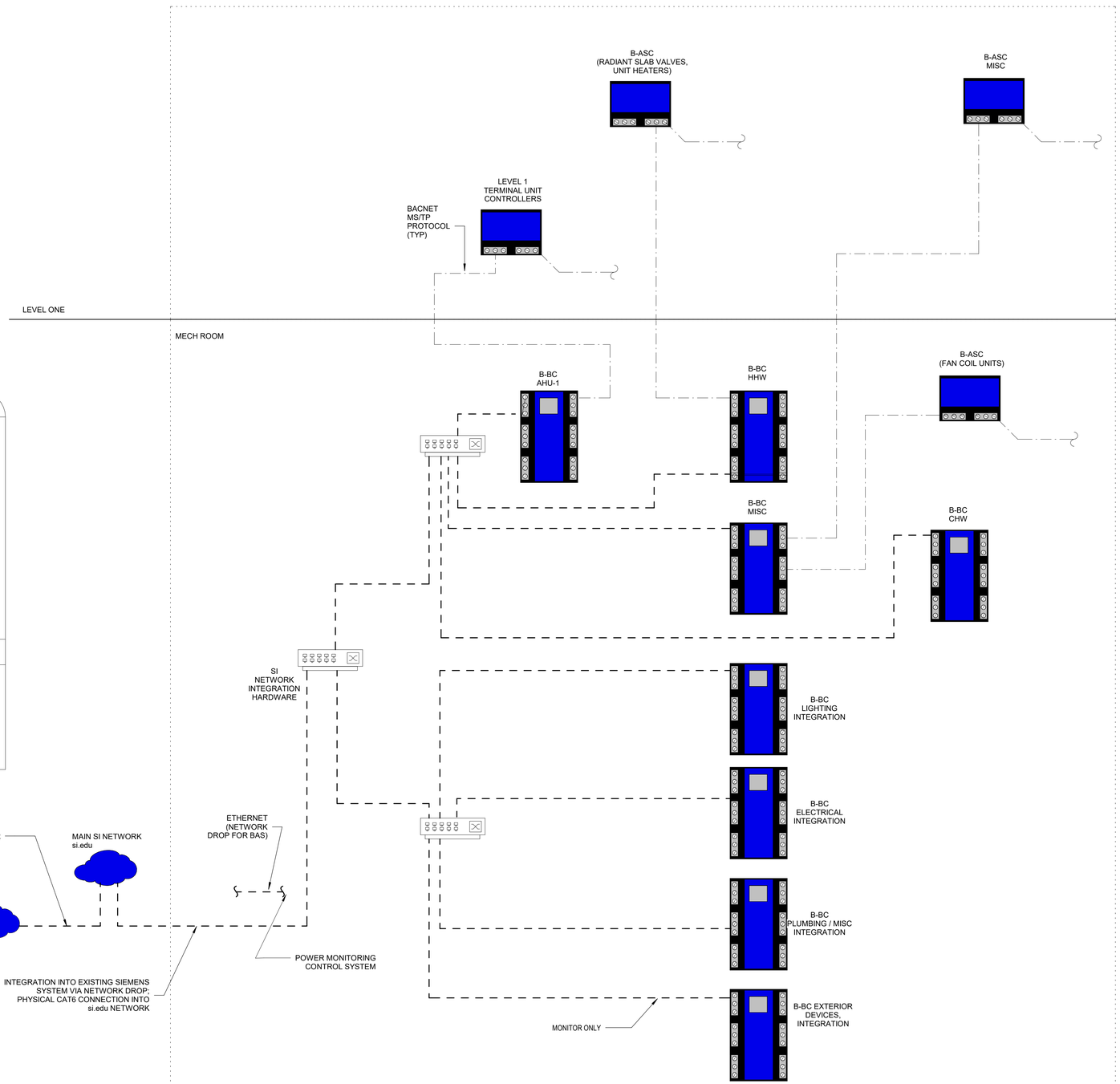
NOTE: HARDWARE AND CONTROL WIRING QUANTITIES SHOWN ON DIAGRAM ARE FOR GRAPHICAL REPRESENTATION OF OVERALL BUILDING SYSTEMS AND ARE NOT INTENDED TO REFLECT EXACT QUANTITIES. CONTROLS CONTRACTOR IS RESPONSIBLE FOR COORDINATING ALL HARDWARE AND MATERIAL QUANTITIES IN ACCORDANCE WITH ENTIRE CONTRACT DOCUMENT PACKAGE. ANY INPUT/OUTPUT EXPANSION MODULE CAPACITY SHOULD BE DETERMINED BY CONTROLS CONTRACTOR AS NECESSARY TO ACCOMPLISH THE SEQUENCE OF OPERATIONS DESCRIBED HEREIN.

| DEVICE INSTANCE CONVENTION | NETWORK NUMBER CONVENTION |
|---|---|
| C BB T N MM (Format) | S BB T N (Format) |
| 1 25 4 0 03 (Example) | 1 25 4 0 (Example) |
| MM = MAC ADDRESS (0 - 99) OR DEVICE INDEX | N = NETWORK INDEX (0 - 9) |
| N = NETWORK INDEX (0 - 9) | T = NETWORK TYPE (0 - 9)<br>0 = BACNET IP<br>1 = BACNET ETHERNET<br>2 = PTP<br>3 = ARCNET<br>4 - 7 = MESTP<br>8 - 9 = FUTURE |
| T = NETWORK TYPE (0 - 9)<br>0 = BACNET IP<br>1 = BACNET ETHERNET<br>2 = PTP<br>3 = ARCNET<br>4 - 7 = MESTP<br>8 - 9 = FUTURE | BB = BUILDING NUMBER (0 - 99) |
| BB = BUILDING NUMBER (0 - 99) | C = CAMPUS<br>1 = XXX<br>2 = XXX |
| C = CAMPUS<br>1 = XXX<br>2 = XXX | |

OBJECT NAMING CONVENTION

SELECTED CONTROLS PROVIDER TO REQUEST OWNER PROVIDED OBJECT NAMING CONVENTION DOCUMENT PRIOR TO DEVELOPMENT OF CONTROLS PACKAGE. ALL DEVICE, NETWORK AND OBJECT NAMING FOR CONTROL SYSTEM TO BE IN ACCORDANCE WITH OWNER'S STANDARDS.

NOTES -
1. CONTRACTOR SHALL COORDINATE EQUIPMENT TAGS (INCLUDING PANEL BOARDS, MODULATING VALVE ACTUATORS AND DAMPERS) WITH THE SMITHSONIAN TO ENSURE FORMATTING IS CONSISTENT WITH THE SMITHSONIAN SERVER.
2. CONTRACTOR SHALL INCLUDE BOTH THE SERVER TAG AND THE PHYSICAL TAG IN THE BAS AS REFERENCE INFORMATION.




1 M-803CP NETWORK ARCHITECTURE
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| ADDRESS | 3001 Connecticut Avenue NW<br>Washington, DC 20008 |
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| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - NETWORK ARCHITECTURE |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 290 OF 378

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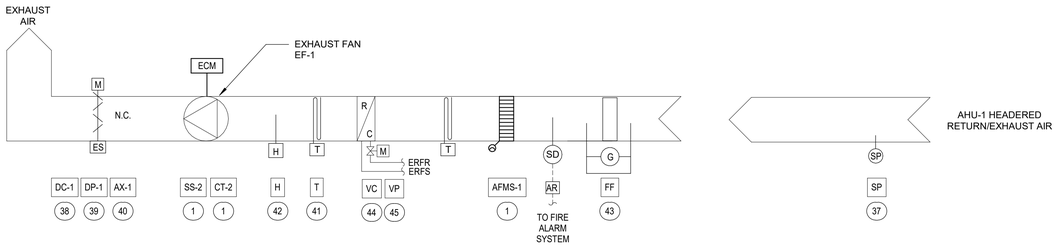
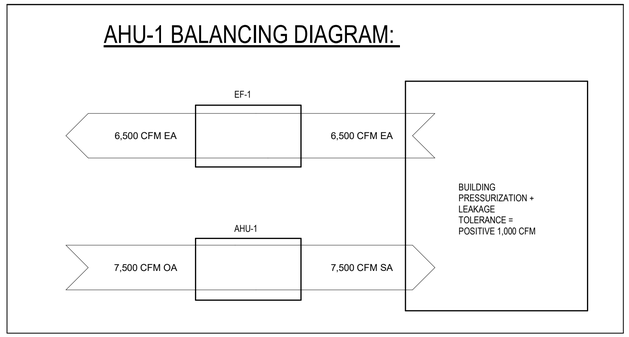
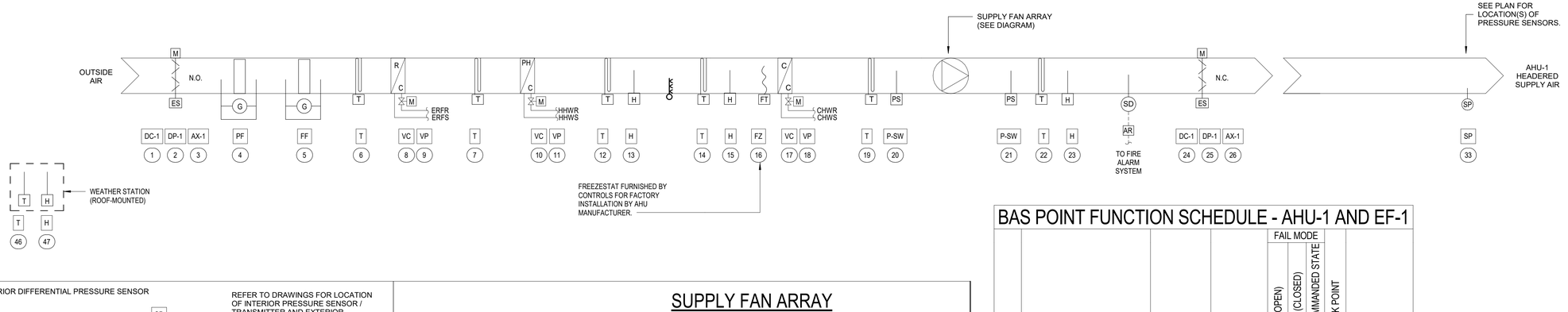
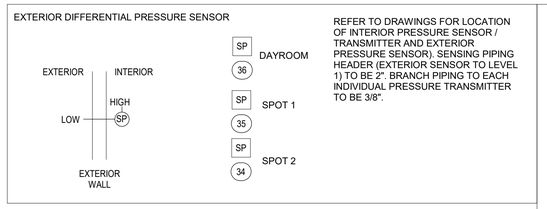



## AHU-1 BALANCING DIAGRAM:

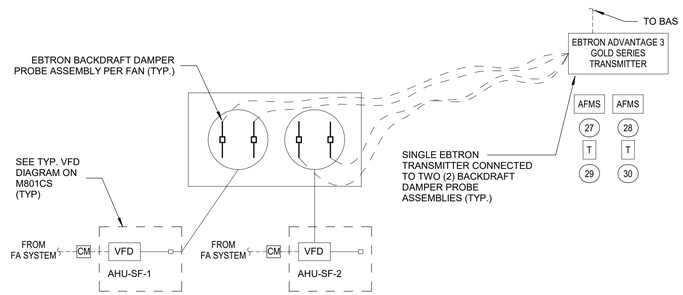


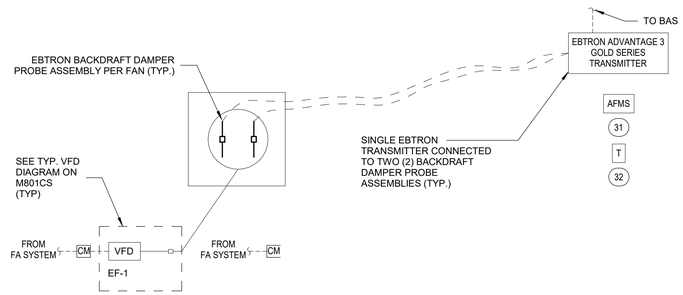




REFER TO DRAWINGS FOR LOCATION OF INTERIOR PRESSURE SENSOR / TRANSMITTER AND EXTERIOR PRESSURE SENSOR). SENSING PIPING HEADER (EXTERIOR SENSOR TO LEVEL 1) TO BE 2". BRANCH PIPING TO EACH INDIVIDUAL PRESSURE TRANSMITTER TO BE 3/8".

## SUPPLY FAN ARRAY




## EXHAUST FAN




## BAS POINT FUNCTION SCHEDULE - AHU-1 AND EF-1

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | NETWORK POINT | NOTES |
|---|---|---|---|---|---|---|---|---|
| AHU-1 | | | | | | | | |
| 1 | OA DAMPER CONTROL | DC-1 | AO | | | | | |
| 2 | OA DAMPER POSITION | DP-1 | AI | | | | | |
| 3 | DAMPER END SWITCH | AX-1 | DI | | | | | |
| 4 | PRE FILTER STATUS | PF | DI | | | | | |
| 5 | FINAL FILTER STATUS | FF | DI | | | | | |
| 6 | ERC ENTERING AIR TEMPERATURE | T | AI | | | | | |
| 7 | ERC LEAVING AIR TEMPERATURE | T | AI | | | | | |
| 8 | ERC VALVE CONTROL | VC | AO | | | | | |
| 9 | ERC VALVE POSITION | VP | AI | | | | | |
| 10 | PHC VALVE CONTROL | VC | AO | | | | | |
| 11 | PHC VALVE POSITION | VP | AI | | | | | |
| 12 | PHC LEAVING AIR TEMPERATURE | T | AI | | | | | |
| 13 | PHC LEAVING AIR HUMIDITY | H | AI | | | | | |
| 14 | SGD LEAVING AIR TEMPERATURE | T | AI | | | | | |
| 15 | SGD LEAVING AIR HUMIDITY | H | AI | | | | | |
| 16 | FREEZESTAT STATUS | FZ | DI | | | | | |
| 17 | CC VALVE CONTROL | VC | AO | | | | | |
| 18 | CC VALVE POSITION | VP | AI | | | | | |
| 19 | CC LEAVING AIR TEMPERATURE | T | AI | | | | | |
| 20 | LOW STATIC SWITCH | P-SW | DI | | | | | |
| 21 | HIGH STATIC SWITCH | P-SW | DI | | | | | |
| 22 | AHU LEAVING AIR TEMPERATURE | T | AI | | | | | |
| 23 | AHU LEAVING AIR HUMIDITY | H | AI | | | | | |
| 24 | SA DAMPER CONTROL | DC-1 | AO | | | | | |
| 25 | SA DAMPER POSITION | DP-1 | AI | | | | | |
| 26 | DAMPER END SWITCH | AX-1 | DI | | | | | |
| 27 | SF-1A AIRFLOW | AFMS | AI | | | | | |
| 28 | SF-1B FAN AIRFLOW | AFMS | AI | | | | | |
| 29 | SF-1A TEMPERATURE | T | AI | | | | | |
| 30 | SF-1B TEMPERATURE | T | AI | | | | | |
| 31 | EF-1 AIRFLOW | AFMS | AI | | | | | |
| 32 | EF-1 TEMPERATURE | T | AI | | | | | |
| BUILDING | | | | | | | | |
| 33 | AHU-1 DUCT STATIC PRESSURE SENSOR | SP | AI | | | | | |
| 34 | BLDG SP - SPOT 2 | SP | AI | | | | | |
| 35 | BLDG SP - SPOT 1 | SP | AI | | | | | |
| 36 | BLDG SP - DAYROOM | SP | AI | | | | | |
| 37 | EF-1 DUCT STATIC PRESSURE SENSOR | SP | AI | | | | | |
| EF-1 | | | | | | | | |
| 38 | EA DAMPER CONTROL | DC-1 | AO | | | | | |
| 39 | EA DAMPER POSITION | DP-1 | AI | | | | | |
| 40 | DAMPER END SWITCH | AX-1 | DI | | | | | |
| 41 | ERC LEAVING AIR TEMPERATURE | T | AI | | | | | |
| 42 | ERC LEAVING AIR HUMIDITY | H | AI | | | | | |
| 43 | FINAL FILTER STATUS | FF | DI | | | | | |
| 44 | ERC VALVE CONTROL | VC | AO | | | | | |
| 45 | ERC VALVE POSITION | VP | AI | | | | | |
| WEATHER STATION | | | | | | | | |
| 46 | OUTSIDE AIR TEMPERATURE | T | AI | | | | | |
| 47 | OUTSIDE AIR HUMIDITY | H | AI | | | | | |

1 M-804CP — CONTROLS DIAGRAM - AHU 1
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| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - AIR HANDLING UNIT CONTROL DIAGRAM |

| WORKING STAFF | OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY) |
|---|---|---|---|

SHEET NO. 291 OF 378

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**AIR HANDLING UNIT 1 DEDICATED OUTSIDE AIR SYSTEM (DOAS) SEQUENCE OF OPERATIONS**

**NOTE - COORDINATE EQUIPMENT AND POINT TAGS WITH SMITHSONIAN TO ENSURE THAT THE FORMATTING AND TAGS ARE CONSISTENT WITH THE SMITHSONIAN SERVER STANDARDS.**
**GENERAL**

THE CARE BUILDING FOR ARABIAN LEOPARD FACILITY IS SERVED BY A SINGULAR FULLY CUSTOMIZABLE AIR HANDLING UNIT. THE UNITS OPERATES AS VARIABLE VOLUME, VARYING VENTILATION AIR TO THE BUILDING TO MAINTAIN ZONE LEVEL VENTILATION, AIR CONDITIONING AND WHOLE BUILDING PRESSURIZATION. THE UNITS ALSO PROVIDE MAKE UP AIR AND LATENT SPACE CONDITIONING FOR ZONES.

THE DOAS UNIT OPERATES AT DESIGN CAPACITY DURING OCCUPIED MODE.

**ALL SETPOINTS SHALL BE USER ADJUSTABLE, UNLESS LABELED AS "NON-ADJUSTABLE".**

**SYSTEM START / STOP CONTROL**

BEFORE UNIT STARTUP, DAMPERS ARE IN THEIR CLOSED POSITIONS. WHEN THE UNIT IS ENABLED ON, THE FOLLOWING OCCURS:
1. THE OUTSIDE AIR (D-1) AND SUPPLY (D-2) AIR ISOLATION DAMPERS WILL BE OPENED. THE OUTSIDE (D-1) AIR CONTROL DAMPERS WILL MODULATE TO THE MINIMUM OPEN POSITION. SUPPLY AIR (PROOF ON END SWITCH) AND OUTSIDE AIR (PROOF ON BREAK) DAMPER ACTUATORS WILL BE HARDWIRED TO ALL SUPPLY AIR FAN VFDS.
2. EXHAUST AIR (PROOF ON END SWITCH) AND RELIEF AIR (PROOF ON BREAK) DAMPER ACTUATORS WILL BE HARDWIRED TO EXHAUST AIR FAN EF-1 VFD.
3. UPON PROOF THAT ALL DAMPERS (SUPPLY AIR, EXHAUST AIR) HAVE RESPONDED, SUPPLY FANS START WITH EXHAUST FAN EF-1. THE SUPPLY AND EXHAUST AIR FANS ARE ENABLED IN UNISON.
   A. THE BAS WILL MONITOR DAMPER ACTUATOR PROOF AND ACTIVATE AN ALARM IF THE DAMPERS DO NOT RESPOND WITHIN 60 SECONDS (ADJ.) OF THE FAN START COMMAND.
   B. FANS MODULATE TO MINIMUM SPEED FOR 30 SECONDS (ADJ.) BEFORE THE BAS BEGINS AUTOMATIC SPEED CONTROL.

WHEN COMMANDED OFF, THE SUPPLY AND EXHAUST FANS STOP AND ALL ASSOCIATED ISOLATION DAMPERS RETURN TO THEIR CLOSED POSITION. VALVES WILL BE CLOSED, ACCEPT AS REQUIRED FOR OPERATION OF FREEZE PROTECTION.

**SUPPLY / EXHAUST FAN VFD HAND-OFF-AUTO (HOA) OPERATION**

EACH FAN WILL HAVE AN INDEPENDENT VFD (WITH BYPASS). HOA SETTINGS PROVIDED AS PART OF VFD THROUGH THE DRIVE'S KEYPAD.
1. IN HAND MODE, THE ASSOCIATED FAN WILL START AND THE FAN SPEED WILL BE MANUALLY CONTROLLED THROUGH THE DRIVE'S KEYPAD.
2. IN OFF MODE, THE ASSOCIATED FAN WILL BE OFF.
3. IN AUTO MODE, THE FAN WILL BE STARTED AND STOPPED THROUGH THE BAS AND THE FAN SPEED WILL BE AUTOMATICALLY CONTROLLED BY THE BAS.
4. IN BYPASS MODE, THE ASSOCIATED FAN WILL BE STARTED AND STOPPED MANUALLY THROUGH THE DRIVE'S KEYPAD WITHOUT SPEED CONTROL.
5. AFTER A POWER FAILURE, THE BAS WILL INITIATE AN ORDERLY GLOBAL RESTART OF ALL EQUIPMENT TO PREVENT A SPIKE IN FACILITY ELECTRICAL DEMAND.
6. UPON ACTIVATION, SAFETIES STOP THE FAN IN AUTO, HAND, AND BYPASS MODES.

**SUPPLY FAN AUTOMATIC SPEED CONTROL**
1. ONCE ENABLED, THE LEAD FAN WILL OPERATE. LAG FAN WILL BE ENABLED AND DISABLED AS NECESSARY TO MAINTAIN SYSTEM STATIC PRESSURE SETPOINT(S). THE LEAD AND LAG DESIGNATION IS TO ALTERNATE ON A WEEKLY BASIS.
2. IF THE BAS IS INCREASING THE SPEED OF THE FAN, AND THE AIRFLOW THROUGH THE FAN INCREASES ABOVE 4500 CFM, THE BAS WILL ENABLE THE NEXT FAN.
3. IF THE BAS IS DECREASING THE SPEED OF THE SUPPLY FANS, AND THE AIRFLOW THROUGH EACH ACTIVE FAN IS APPROACHING 5500 CFM (ADJ.), THE BAS WILL DISABLE ONE OF THE SUPPLY FANS AND INCREASE THE SPEED OF THE REMAINING SUPPLY FAN TO MAINTAIN STATIC PRESSURE SETPOINT(S).

THE SYSTEM WILL MAINTAIN OPTIMIZED STATIC PRESSURE SETPOINT(S) THROUGH STATIC PRESSURE RESET AS FOLLOWS:
1. THE INITIAL DUCT STATIC PRESSURE SETPOINT(S) WILL BE DETERMINED BY THE TAB CONTRACTOR.
2. THE STATIC PRESSURE SETPOINT WILL RESET BASED ON OUTSIDE AIR DEMAND AS REPRESENTED BY VAV PRIMARY AIR DAMPER POSITION.
3. WHEN THE QTY OF VAV TERMINAL UNITS WITH A PRIMARY AIR DAMPER POSITION OF 90% OPEN OR LESS IS NO GREATER THAN 5 (ADJ.) UNITS, THE SYSTEM STATIC PRESSURE SETPOINT(S) WILL BE INCREMENTALLY RESET DOWN BY 0.1 WC (ADJ.) AT A FREQUENCY OF 5 MINUTES (ADJ.) TO A MINIMUM OF 0.3 WC (ADJ.) OR THE MINIMUM OPERATING SPEED OF THE SUPPLY AIR FAN SYSTEM HAS BEEN REACHED.
4. WHEN AT LEAST 2 (ADJ.) UNITS HAVE A PRIMARY AIR DAMPER POSITION OF 97% OPEN OR MORE, THE REVERSE WILL OCCUR.

**EXHAUST FAN EF-1 AUTOMATIC SPEED CONTROL**
1. ONCE ENABLED, THE EXHAUST FAN EF-1 WILL OPERATE.
2. FOR SETPOINTS AND BUILDING LEVEL CONTROL, SEE THE BUILDING PRESSURIZATION SEQUENCE.

**TEMPERATURE AND HUMIDITY CONTROL**
DISCHARGE AIR TEMPERATURE CONTROL
1. THE HEATING HOT WATER COIL SHALL MODULATE TO MAINTAIN A MINIMUM DISCHARGE AIR TEMPERATURE OF 60°F DURING HEATING MODE.
2. WHEN THE AIR HANDLING UNIT IS IN COOLING OR DEHUMIDIFICATION MODE, THE COOLING COIL SHALL MODULATE TO MAINTAIN A DISCHARGE AIR TEMPERATURE AS LOW AS 49°F.
3. THE CHILLED WATER COOLING COIL CONTROL VALVE SHALL MODULATE, IN SEQUENCE, TO MAINTAIN THE ACTIVE DISCHARGE AIR TEMPERATURE SETPOINT.
4. THE HOT WATER HEATING COIL CONTROL VALVE SHALL MODULATE AS REQUIRED TO MAINTAIN THE DISCHARGE AIR TEMPERATURE SETPOINT WHEN HEATING IS REQUIRED OR DURING REHEAT OPERATION.
5. THE CONTROL SYSTEM SHALL PREVENT SIMULTANEOUS HEATING AND COOLING EXCEPT AS REQUIRED FOR DEHUMIDIFICATION REHEAT (AT TERMINAL UNIT).

HUMIDITY / DEW POINT CONTROL
1. THE MAXIMUM DISCHARGE AIR DEW POINT TEMPERATURE SETPOINT (T-DPT) SHALL BE 49°F.
2. THE DISCHARGE AIR DEW POINT (T-DPT) SHALL BE CALCULATED USING TEMPERATURE SENSOR T-6 AND HUMIDITY SENSOR H-4.
3. IF THE MEASURED DEW POINT TEMPERATURE DOWNSTREAM OF THE ENERGY RECOVERY COIL EXCEEDS THE ACTIVE DISCHARGE AIR DEW POINT SETPOINT, THE COOLING COIL CONTROL VALVE SHALL MODULATE TO REDUCE THE DEW POINT TO SETPOINT.
4. THE INITIAL ZONE DEW POINT TEMPERATURE SETPOINT SHALL BE 53°F (ADJUSTABLE).
5. IF ANY ZONE DEW POINT EXCEEDS THE ZONE SETPOINT:
   a. THE DISCHARGE AIR DEW POINT SETPOINT SHALL BE REDUCED AT A RATE OF 0.2°F PER MINUTE.
   b. THIS SHALL CONTINUE UNTIL ALL ZONES ARE AT OR BELOW THEIR DEW POINT SETPOINT
   c. THIS RESET SHALL BE IMPLEMENTED USING TRIM AND RESPOND LOGIC

F. THE LOWEST DISCHARGE AIR DEW POINT SETPOINT SHALL BE LIMITED TO 49°F.
G. WHEN DEHUMIDIFICATION IS ACTIVE, THE COOLING COIL SHALL CONTROL TO DEW POINT (NOT DRY BULB).

HUMIDIFICATION CONTROL
1. THE HUMIDIFICATION SYSTEM SHALL UTILIZE THE ELECTRIC RESISTANCE HUMIDIFIER CONNECTED TO THE COMMON STEAM MANIFOLD.
2. SPACE HUMIDIFICATION SHALL BE MAINTAINED BETWEEN 30% RH AND 50% RH AT A SPACE TEMPERATURE OF 72°F (ADJUSTABLE).
3. HUMIDIFICATION DEMAND SHALL BE DETERMINED USING A TRIM AND RESPOND STRATEGY BASED ON ZONE RELATIVE HUMIDITY:
   A. REQUESTS ARE SENT BY ZONES.
4. THE AHU DISCHARGE AIR HUMIDITY SETPOINT SHALL BE RESET VIA TRIM AND RESPOND TO MAINTAIN ALL ZONES WITHIN THE 30–50% RH RANGE.
5. HUMIDIFIER OUTPUT SHALL MODULATE TO MAINTAIN THE ACTIVE DISCHARGE AIR HUMIDITY SETPOINT.
6. HUMIDIFICATION SAFETY AND PREHEAT CONTROL
   A. WHEN HUMIDIFICATION IS ENABLED, THE PREHEAT COIL SHALL MODULATE TO MAINTAIN A LEAVING AIR TEMPERATURE SUCH THAT THE RELATIVE HUMIDITY DOWNSTREAM OF THE HUMIDIFIER DOES NOT EXCEED 70% RH.
   B. THE CONTROL SYSTEM SHALL CALCULATE OR MONITOR LEAVING AIR TEMPERATURE AND HUMIDITY DOWNSTREAM OF THE HUMIDIFIER TO PREVENT SATURATION CONDITIONS.
   C. THE PREHEAT COIL SHALL INCREASE LEAVING AIR TEMPERATURE AS REQUIRED TO:
      a. PREVENT CONDENSATION IN DUCTWORK
      b. ENSURE THE AIR IS NOT NEAR 100% SATURATION
      c. MAINTAIN A MAXIMUM OF 70% RH AT THE HUMIDIFIER LEAVING LOCATION
   D. IF THE 70% RH LIMIT IS APPROACHED:
      a. HUMIDIFIER OUTPUT SHALL BE LIMITED OR REDUCED
      b. PREHEAT COIL OUTPUT SHALL BE INCREASED (WITHIN SAFE LIMITS)
   E. HIGH-HIGH HUMIDITY LIMIT (ADJUSTABLE, TYPICALLY ~75% RH):
      a. SHALL IMMEDIATELY DISABLE HUMIDIFIER OUTPUT
      b. GENERATE AN ALARM
   F. LOW AIRFLOW, FAN FAILURE, OR FREEZE PROTECTION CONDITIONS SHALL DISABLE HUMIDIFIER OPERATION.

SEQUENCE OF OPERATION SUMMARY
1. IN HEATING MODE:
   A. HEATING COIL MODULATES TO MAINTAIN ≥ 60°F DISCHARGE AIR TEMPERATURE
   B. COOLING COIL CLOSED
2. IN COOLING MODE:
   A. COOLING COIL MODULATES TO MAINTAIN DISCHARGE AIR TEMPERATURE (DOWN TO 49°F AS REQUIRED)
   B. HEATING COIL CLOSED
3. IN DEHUMIDIFICATION MODE:
   A. COOLING COIL MODULATES TO MAINTAIN DISCHARGE AIR DEW POINT (MAX 49°F)
   B. HEATING COIL CLOSED.
4. IN HUMIDIFICATION MODE:
   A. WHEN ZONE RH < 30%:
      a. HUMIDIFICATION ENABLED
      b. STEAM OUTPUT MODULATES TO INCREASE DISCHARGE HUMIDITY
      c. PREHEAT COIL MODULATES TO MAINTAIN ≤ 70% RH LEAVING HUMIDIFIER
   B. WHEN ZONE RH BETWEEN 30% AND 50%:
      a. HUMIDIFICATION MODULATES VIA TRIM & RESPOND TO MAINTAIN SETPOINT
   C. WHEN ZONE RH > 50%:
      a. HUMIDIFICATION DISABLED
   D. HUMIDIFICATION SHALL BE LOCKED OUT DURING MECHANICAL COOLING/DEHUMIDIFICATION MODE UNLESS SPECIFICALLY REQUIRED BY SEQUENCE OVERRIDE.

SETBACKS
1. NIGHT SETBACK TEMPERATURE CONTROL: IN THIS MODE, THE DOAS UNIT WILL PROVIDE AIR CHANGE RATE AND TEMPERATURE CONTROL TO THE ROOMS ACCORDING TO THE SETBACK TEMPERATURE SCHEDULE AND AIRFLOW MINIMUMS. THE DOAS UNIT WILL PROVIDE DEW POINT CONTROL AND THE INITIAL STEP OF TEMPERATURE CONTROL FOR ALL VAV SPACES WHERE PRIMARY AIR MINIMUM IS ZERO. NOTE, IF ADDITIONAL COOLING IS REQUIRED BEYOND THE DOAS / VAV CAPACITY TO SPACES SERVED BY CHILLED BEAM DURING UNOCCUPIED MODE, THE LOCAL TERMINAL DEVICE SHALL PROVIDE THE SECOND STEP OF SPACE CONDITIONING (REFER TO TERMINAL UNIT SEQUENCE).
2. COOLING SYSTEM OUTAGE TEMPERATURE RECOVERY: IN THE EVENT OF A POWER OUTAGE, THE DOAS COOLING LAT SHALL HAVE STAGGERED STEPPOINTS TO PREVENT CONDENSATION DUE TO TEMPERATURE AND HUMIDITY DIFFERENCE. THE COOLING LAT SHALL NEVER BE MORE THAN 18 DEGREES BELOW THE HIGHEST MEASURED SPACE DRY BULB. THE LAT SHALL DECREASE INCREMENTALLY TO MAINTAIN ZONE DEWPOINT TEMPERATURE SETPOINT.

ENERGY RECOVERY COIL (GLYCOL LOOP) CONTROL – EF-1 INTEGRATED
SYSTEM DESCRIPTION
1. THE ENERGY RECOVERY SYSTEM SHALL CONSIST OF A RUN-AROUND GLYCOL LOOP WITH:
   A. OUTSIDE AIR (OA) ENERGY RECOVERY COIL
   B. EXHAUST AIR (EA) ENERGY RECOVERY COIL
   C. GLYCOL PUMP(S)
   D. MODULATING CONTROL VALVE (IF PROVIDED)
   E. EXHAUST FAN EF-1 SERVING THE EXHAUST AIR STREAM
2. DEFINE MODES AS FOLLOWING -
   A. COOLING MODE = AHU COOLING COIL VALVE IS OPEN TO FLOW
   B. HEATING MODE = AHU HEATING COIL VALVE IS OPEN TO FLOW.
3. ENABLE CONDITIONS
   A. THE ENERGY RECOVERY SYSTEM SHALL BE ENABLED WHEN ALL THE FOLLOWING CONDITIONS ARE MET:
      a. AHU SUPPLY FAN IS PROVEN ON
      b. EXHAUST FAN EF-1 IS PROVEN ON
      c. OUTSIDE AIRFLOW AND EXHAUST AIRFLOW ARE PRESENT
      d. NO FREEZE PROTECTION CONDITION IS ACTIVE
   B. WHEN ENABLED:
      a. GLYCOL PUMP SHALL START
      b. CONTROL VALVE SHALL OPEN TO FLOW.
   C. ENTHALPY-BASED CONTROL LOGIC
      a. THE CONTROL SYSTEM SHALL CONTINUOUSLY CALCULATE:
         • OUTSIDE AIR ENTHALPY (H-OA) USING OA TEMPERATURE AND HUMIDITY
         • EXHAUST AIR ENTHALPY (H-EA) USING EA TEMPERATURE AND HUMIDITY
4. ENERGY RECOVERY SHALL BE ENABLED ONLY WHEN HEAT TRANSFER IS BENEFICIAL TO BUILDING'S HVAC SYSTEM:
   A. COOLING SEASON (SUMMER OPERATION):
      a. IF H-OA > H-EA:
         • ENERGY RECOVERY SHALL BE ENABLED
         • SYSTEM SHALL PRE-COOL AND PRE-DEHUMIDIFY OUTSIDE AIR USING EXHAUST AIR
      b. ELSEIF H-OA =< H-EA
         • DISABLE ENERGY RECOVERY SYSTEM
   B. HEATING SEASON (WINTER OPERATION):
      a. IF H-OA < H-EA:
         • ENERGY RECOVERY SHALL BE ENABLED
         • SYSTEM SHALL PRE-HEAT OUTSIDE AIR
      b. ELSEIF H-OA => H-EA
         • DISABLE ENERGY RECOVERY SYSTEM
5. MODULATION CONTROL
   A. THE ENERGY RECOVERY COIL LOOP SHALL MODULATE (VIA CONTROL VALVE AND/OR VARIABLE FLOW PUMP) TO MAINTAIN:
      a. TARGET LEAVING OUTSIDE AIR TEMPERATURE OR ENTHALPY SETPOINT, OR
      b. MAXIMIZE ENERGY TRANSFER WHILE MAINTAINING DOWNSTREAM CONTROL STABILITY
   B. CONTROL SHALL BE SMOOTH AND CONTINUOUS TO PREVENT OSCILLATION WITH COOLING/HEATING COILS.
6. DISABLE CONDITIONS
   A. THE ENERGY RECOVERY LOOP SHALL BE DISABLED UNDER ANY OF THE FOLLOWING CONDITION:
   B. ENERGY TRANSFER DOES NOT ASSIST BUILDING TEMPERATURE/HUMIDITY CONTROL:
      a. COOLING MODE: H-OA ≤ H-EA
      b. HEATING MODE: H-OA ≥ H-EA
      c. ECONOMIZER MODE IS ACTIVE AND PROVIDING FREE COOLING
      d. EXHAUST FAN EF-1 IS OFF OR FAILED
      e. SUPPLY FAN IS OFF
      f. HIGH GLYCOL LOOP TEMPERATURE OR FLOW FAILURE ALARM
      g. MAINTENANCE OVERRIDE OR MANUAL DISABLE

**OPTIMAL START CONTROL**

THE SYSTEM WILL UTILIZE AN OPTIMAL START ROUTINE TO INITIATE WARM UP / COOL DOWN PRIOR TO THE START OF EACH SCHEDULED OCCUPIED PERIOD. WARM UP MODE IS USED IF THE AVERAGE ZONE TEMPERATURE IS BELOW OCCUPIED SETPOINT. COOL DOWN MODE IS USED IF THE AVERAGE ZONE DRY BULB TEMPERATURE IS ABOVE OCCUPIED SETPOINT. THE OPTIMAL START ALGORITHM SHALL UTILIZE OUTSIDE AIR TEMPERATURE, AVERAGE DEVIATION FROM OCCUPIED MODE TEMPERATURE SETPOINT AND HISTORICAL DATA TO DETERMINE THE APPROPRIATE TIME TO INITIAL WARM UP / COOL DOWN AHEAD OF THE SCHEDULED OCCUPANCY TIME.

AT THE SCHEDULED OCCUPANCY TIME, THE SYSTEM WILL START, IF NOT ALREADY STARTED BY THE OPTIMAL START ROUTINE, AND THE UNIT WILL BE SWITCHED TO OCCUPIED MODE OPERATION.

**ALARMS AND SYSTEM DIAGNOSTICS**

THE FOLLOWING CONDITIONS WILL BE MONITORED FOR ALARM.
1. FILTER DIFFERENTIAL PRESSURE: MONITOR DP SWITCHES ACROSS ALL FILTERS AND PROVIDE MAINTENANCE ALARM ON DIRTY FILTER CONDITION.
2. FAN STATUS: MONITORING AND ALARM OF SUPPLY / RELIEF FANS
3. FAN FAILURE: COMMANDED ON, STATUS IS OFF.
4. FAN IN HAND: COMMANDED OFF, STATUS IS ON.
5. FAN VFD FAULT.
6. STATIC PRESSURE: MONITOR SUPPLY / RELIEF FAN HIGH STATIC PRESSURE AND SUPPLY / RELIEF DUCT LOW STATIC PRESSURE FOR CONDITION 25% OR GREATER FROM EFFECTIVE (RESET) SETPOINT.
7. COOLING COIL MONITORING AND ALARM
8. CONDENSATE OVERFLOW: A WATER LEVEL DETECTION DEVICE CONFORMING TO UL 508 TO ALARM IN THE EVENT THE PRIMARY DRAIN IS BLOCKED.
9. FREEZESTAT: A TEMPERATURE OF 38 DEG F (ADJ.) IS SENSED AT THE COOLING COIL INLET.
10. TEMPERATURE AND HUMIDITY MONITORING AND ALARM
11. HIGH DISCHARGE AIR TEMPERATURE IF GREATER THAN 5 DEG F (ADJ.) ABOVE EFFECTIVE DAT SETPOINT.
12. LOW DISCHARGE AIR TEMPERATURE IF LESS THAN 5 DEG F (ADJ.) BELOW EFFECTIVE DAT SETPOINT.

SAFETY DEVICES WILL BE INCORPORATED INTO A SINGLE, LOW-VOLTAGE, SERIES-CIRCUIT THAT IS HARD-WIRED TO SHUTDOWN ALL FANS, RETURN CONTROL DAMPERS AND VALVES TO THEIR NORMAL POSITION. EACH SAFETY DEVICE WILL BE WIRED AS A SUPERVISED CIRCUIT SUCH THAT REMOVAL OF THE DEVICE OR ASSOCIATED WIRING WILL RESULT IN BAS ALARM AND EQUIPMENT SHUTDOWN.
1. ALL SAFETY SHUTDOWNS WILL REQUIRE A MANUAL RESET. THE FOLLOWING CONDITIONS WILL BE SAFETY SHUTDOWNS:
   A. HIGH STATIC PRESSURE - SUPPLY / RELIEF FANS
   B. LOW STATIC PRESSURE - SUPPLY / RELIEF DUCTS
   C. SMOKE DETECTION - RELIEF AIR SMOKE DETECTOR (FROM FAS)
   D. DAMPER END SWITCH - OUTSIDE / SUPPLY / RELIEF
   E. FREEZE PROTECTION - FREEZESTAT STATUS ON ENTERING SIDE OF COOLING COIL.

**AIR HANDLING UNIT MAINTENANCE SEQUENCE OF OPERATIONS**
1. EMPLOY FOLLOWING SEQUENCES TO ENABLE MAINTENANCE MODE OF THE AHU.
2. DISABLE THE SUPPLY AND EXHAUST FANS.
3. GENERATE AN ALARM IN BAS, WHICH SHOW A CLEAR NOTIFICATION THAT THE SYSTEM IS NOT OPERATING IN NORMAL MODE. IDENTIFY THE UNIT THAT IS POWERED DOWN.
4. WHEN PROVEN THAT ALL FANS ASSCIATED WITH THE UNIT ARE OFF, CLOSE ALL THE DAMPERS (OUTSIDE AIR, SUPPLY, AND EXHAUST AIR).

TO RE-ENABLE NORMAL OPERATION MODE FROM MAINTENANCE MODE, AHU SHALL FOLLOW THE UNIT START CONTROL SEQUENCE.
1. REMOVE THE MAINTENANCE ALARM STATUS OF BAS WHEN THE AHU SYSTEM IS OPERATING BACK ON NORMAL MODE.

**SYSTEM OPERATION MONITORING AND ALARM**

THE FOLLOWING MONITORING AND ALARM FUNCTIONS WILL INITIATE A COMMISSIONING "CX" ALARM TO BE DISPLAYED AT THE BAS WORKSTATION.
1. FAN CONTROL FAULT
   A. EXCESSIVE SUPPLY FAN SPEED: IF THE SUPPLY FAN SPEED OUTPUT REMAINS ABOVE 95% FOR MORE THAN 8 HOURS (ADJ.) ACCUMULATED PER OCCUPIED PERIOD FOR AT LEAST 2 (ADJ.) CONSECUTIVE OCCUPIED PERIODS.
   B. EXCESSIVE RELIEF FAN SPEED ALARM: IF THE RELIEF FAN SPEED OUTPUT REMAINS ABOVE 95% FOR MORE THAN 8 HOURS (ADJ.) ACCUMULATED PER OCCUPIED PERIOD FOR AT LEAST 2 (ADJ.) CONSECUTIVE OCCUPIED PERIODS.
   C. CONTINUOUS AIRFLOW REQUEST ALARM: IF ONE OR MORE SUPPLY PRIMARY AIR DAMPERS (VAV) IS SENDING CONTINUOUS AIRFLOW REQUESTS AND SUPPLY DUCT STATIC PRESSURE HAS RESET TO MAXIMUM SETPOINT, AND THE COMBINED CONDITION REMAINS FOR MORE THAN 8 HOURS (ADJ.) CONTINUOUSLY.
   D. VFD LEFT IN LOCAL MANUAL OVERRIDE: NETWORK WITH VFD INTERFACE AND ALARM IF VFD IS LOCALLY LEFT IN MANUAL SPEED CONTROL MODE OR IF VFD SPEED OUTPUT SIGNAL REMAINS CONSTANT FOR MORE THAN 8 HOURS (ADJ.) CONTINUOUSLY.
2. DAMPER CONTROL FAULT
   A. DAMPER CLOSE OFF FAILURE: IF ANY OF THE DAMPER END SWITCHES SHOW THE DAMPER AS BEING OPEN WHILE THE COMMAND FROM THE BAS IS TO HAVE THE DAMPER COMPLETELY CLOSED.

**FIRE ALARM SHUTDOWN CONTROL**

THE SUPPLY AND EXHAUST FAN ARE INTERLOCKED WITH THE FIRE ALARM SYSTEM THROUGH A CONTROL MODULE (CM). THE CONTROL MODULE IS INTERLOCKED (HARD-WIRED) TO CONTROL THE FANS IN THE HAND, AUTO, AND BYPASS OPERATING MODES. THE SMOKE DETECTOR(S) LOCATED INSIDE OF THE UNIT ARE MONITORED BY THE FIRE ALARM SYSTEM.
1. WHEN THE PRODUCTS OF COMBUSTION ARE SENSED BY THE SMOKE DETECTORS, THE CONTROL MODULE WILL STOP THE FANS AND CLOSE THE SUPPLY AND RELIEF AIR ISOLATION DAMPERS. THE BAS SHOULD PROCEED WITH UNIT STOP CONTROL.
2. WHEN THE EVENT HAS ENDED AND THE FIRE ALARM SYSTEM HAS BEEN RESET, THE CONTROL MODULE WILL REQUIRE A MANUAL RESET TO RESTORE NORMAL CONTROL TO THE HVAC SYSTEM.
3. MANUAL SYSTEM SHUTDOWN SHALL ALSO BE POSSIBLE BY KEYED SWITCH AT THE AIR DISTRIBUTION SYSTEMS SHUTDOWN PANEL FOR USE BY BUILDING OR EMERGENCY RESPONSE PERSONNEL IN THE EVENT OF AN EMERGENCY.

**TERMINAL UNIT GLOBAL DAMPER POSITION CONTROL**

PROVIDE A SINGLE LOCATION WITHIN THE AHU GUI IN WHICH THE DAMPER POSITION FOR ALL TERMINAL UNITS ASSOCIATED WITH THE AIR HANDLING SYSTEM CAN BE OVERRIDDEN. THE INTENT FOR THIS FUNCTIONALITY IS TO BE ABLE TO ADJUST SYSTEM PRESSURE IN THE EVENT THAT ANY AHU FAN VFD ENTERS BYPASS MODE AND IS FORCED TO FULL SPEED.

**HEATING COIL RESET REQUESTS**
A. SEE HYDRONIC PLANT CONTROL SHEET FOR MORE INFORMATION.
B. DP RESET - HEATING COIL HEATING HOT WATER RESET REQUESTS. SEND THE PUMP SYSTEM THAT SERVES THE REHEAT COIL REQUESTS AS FOLLOWS:
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 85%.
   2. ELSEIF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.
C. SET POINT RESET - HEATING COIL REQUESTS TO RESET HEATING HOT WATER SUPPLY TEMPERATURE. SEND THE PLANT THAT GENERATES HEATING HOT WATER, REQUESTS AS FOLLOWS:
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 10%.
   2. ELSEIF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

**CHILLED WATER COIL RESET REQUESTS**
A. SEE HYDRONIC PLANT CONTROL SHEET FOR MORE INFORMATION.
B. DP RESET - COOLING COIL DP RESET REQUESTS. SEND THE PUMP SYSTEM THAT SERVES THE COOLING COIL REQUESTS AS FOLLOWS:
   1. IF THE CHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 85%.
   2. ELSEIF THE CHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.
C. SETPOINT RESET - COOLING COIL REQUESTS TO RESET CHILLED WATER SUPPLY TEMPERATURE. SEND THE PLANT THAT GENERATES CHILLED WATER, REQUESTS AS FOLLOWS:
   1. IF THE CHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 10%.
   2. ELSEIF THE CHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

**BUILDING PRESSURIZATION SEQUENCE OF OPERATION:**

1. GENERAL
   A. BUILDING PRESSURIZATION SYSTEM SHALL CONSIST OF THREE DIFFERENTIAL PRESSURE SENSORS LOCATED ON LEVEL 1. IN RESPONSE TO THE DIFFERENCE IN STATIC PRESSURE, AS MEASURED BY THESE SENSORS WITH RESPECT TO THE OUTDOOR AIR, THE BUILDING PRESSURIZATION SHALL BE MAINTAINED THROUGH EF-1 VAV CONTROL.

2. PRESSURE CONTROL EXHAUST AIR VALVE CONTROL
   A. NORMAL OPERATION
      a. THE DIFFERENTIAL PRESSURE (WITH RESPECT TO THE OUTSIDE PRESSURE) SHALL BE MEASURED BY THE THREE DIFFERENTIAL PRESSURE SENSORS (WORST CASE) ON LEVEL 1. ALL PRESSURE READINGS FROM THE SENSORS SHALL BE TRENDED BY THE BAS.
      b. THE PRESSURE SHALL BE CONTROLLED IN PARALLEL BY THE VIPKEEPER WORKSPACE EXHAUST VAV BOX (VAVE-6).
      c. MODULATE ALL OPERATING EXHAUST VAV BOXES SETPOINTS AS REQUIRED TO MAINTAIN THE AVERAGE OF ALL LEVEL STATIC PRESSURE READINGS.

3. ALARM:
   A. HIGH BUILDING PRESSURE (MORE THAN +0.10")
   B. LOW BUILDING PRESSURE (LESS THAN -0.005")

**BUILDING PRESSURE SETPOINT = 0.05" (SUMMER), 0.025" (WINTER)**

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Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
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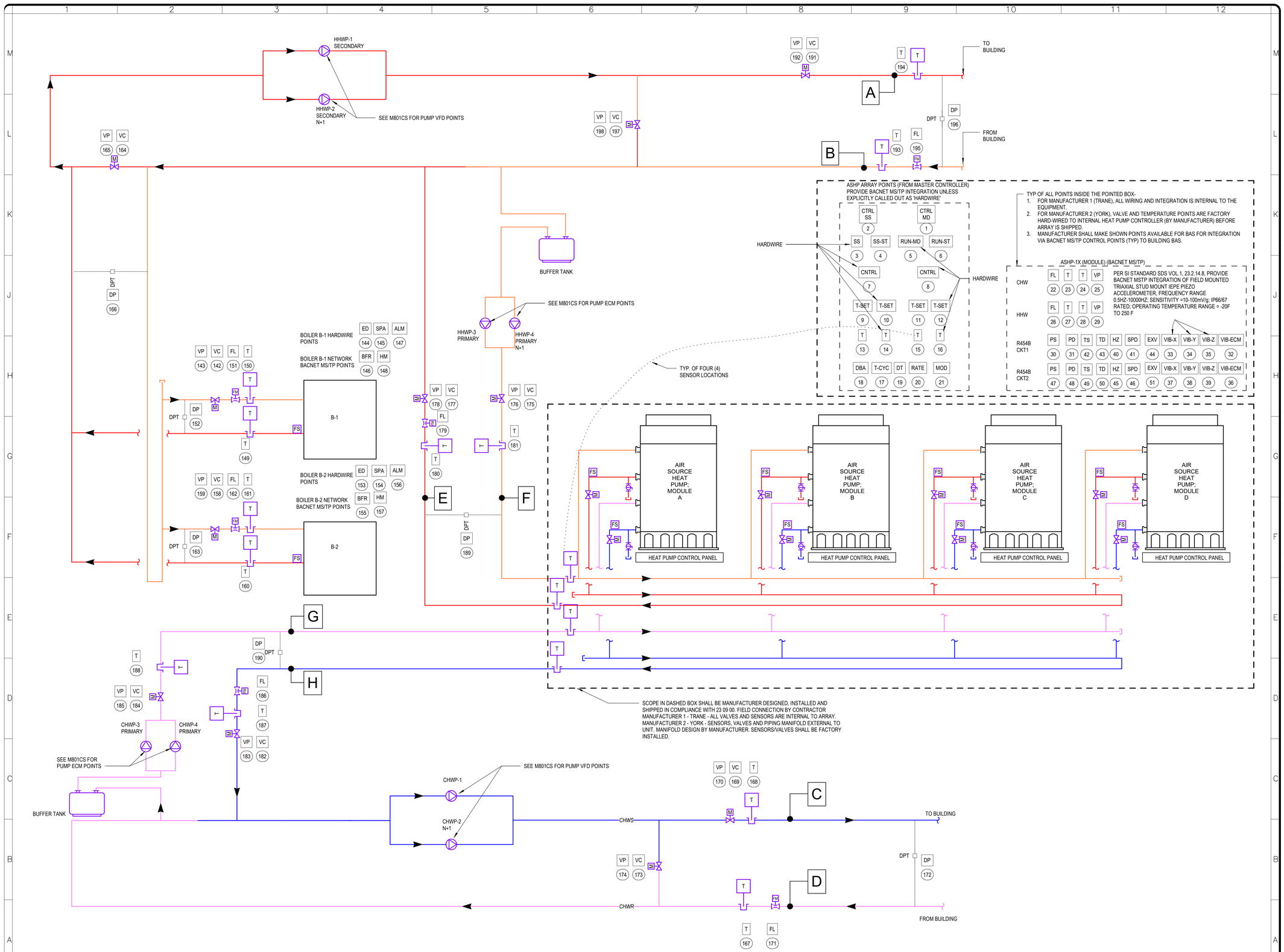
| DRAWING TITLE | MECHANICAL CONTROLS - AIR HANDLING UNIT CONTROL SEQUENCES |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 292 of 378

M-805CP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)




1 M-806CP MECHANICAL CONTROLS - HYDRONIC PLANT
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Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520


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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
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| DRAWING TITLE | MECHANICAL CONTROLS - HYDRONIC PLANT CONTROL DIAGRAM |

| WORKING STAFF | OP | OP | IP |
|---|---|---|---|
| | DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 293 OF 378

M-806CP

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

# BAS POINT FUNCTION SCHEDULE - HYDRONIC PLANT

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | NOTES |
|---|---|---|---|---|---|---|---|---|
| ASHP ARRAY | | | | | | | | |
| 1 | CONTROL MODE SELECTION AUTHORITY | CTRL MD | DO | | | | | |
| 2 | START STOP SELECTION AUTHORITY | CTRL SS | DO | | | | | |
| 3 | ARRAY START STOP | SS | DO | | | | | |
| 4 | HEAT PUMP ARRAY STATUS (ON/OFF) | SS-ST | DI | | | | | |
| 5 | RUN MODE | RUN-MD | AO | | | | | |
| 6 | RUN STATUS | RUN-ST | AI | | | | | |
| 7 | COOLING CONTROL SELECT | CNTRL | DO | | | | | |
| 8 | HEATING CONTROL SELECT | CNTRL | DO | | | | | |
| 9 | CHW ENTERING WATER TEMPERATURE SETPOINT | T-SET | AO | | | | | |
| 10 | CHW LEAVING WATER TEMPERATURE SETPOINT | T-SET | AO | | | | | |
| 11 | HHW ENTERING WATER TEMPERATURE SETPOINT | T-SET | AO | | | | | |
| 12 | HHW LEAVING WATER TEMPERATURE SETPOINT | T-SET | AO | | | | | |
| 13 | CHW ENTERING WATER TEMPERATURE | T | AI | | | | | |
| 14 | CHW LEAVING WATER TEMPERATURE | T | AI | | | | | |
| 15 | HHW ENTERING WATER TEMPERATURE | T | AI | | | | | |
| 16 | HHW LEAVING WATER TEMPERATURE | T | AI | | | | | |
| 17 | TEMPERATURE CONTROL CYCLE | T-CYC | AO | | | | | |
| 18 | NOISE REDUCTION | DBA | DO | | | | | |
| 19 | WATER TEMPERATURE CONTROL DIFFERENTIAL | DT | AO | | | | | |
| 20 | EWT PULLDOWN RATE | RATE | AO | | | | | |
| 21 | MODULE NUMBER | MOD | AO | | | | | |
| ASHP-1A | | | | | | | | |
| 22 | CHW FLOW RATE | FL | AI | | | | | |
| 23 | CHW EWT (INTO MACHINE) | T | AI | | | | | |
| 24 | CHW LWT (OUT OF MACHINE) | T | AI | | | | | |
| 25 | VALVE POSITION | VP | AI | | | | | |
| 26 | HHW FLOW RATE | FL | AI | | | | | |
| 27 | HHW EWT (INTO MACHINE) | T | AI | | | | | |
| 28 | HHW LWT (OUT OF MACHINE) | T | AI | | | | | |
| 29 | VALVE POSITION | VP | AI | | | | | |
| 30 | PRESSURE - SUCTION CKT 1 | PS | AI | | | | | |
| 31 | PRESSURE - DISCHARGE CKT 1 | PD | AI | | | | | |
| 32 | CONDENSER FAN 1 VIBRATION | VIB-ECM | AI | | | | | |
| 33 | COMPRESSOR 1 VIBRATION RMS - X (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-X | AI | | | | | |
| 34 | COMPRESSOR 1 VIBRATION RMS - Y (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Y | AI | | | | | |
| 35 | COMPRESSOR 1 VIBRATION RMS - Z (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Z | AI | | | | | |
| 36 | CONDENSER FAN 2 VIBRATION | VIB-ECM | AI | | | | | |
| 37 | COMPRESSOR 2 VIBRATION RMS - X (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-X | AI | | | | | |
| 38 | COMPRESSOR 2 VIBRATION RMS - Y (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Y | AI | | | | | |
| 39 | COMPRESSOR 2 VIBRATION RMS - Z (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Z | AI | | | | | |
| 40 | CKT 1 COMPRESSOR SPEED | HZ | AI | | | | | |
| 41 | CKT 1 FAN SPEED | SPD | AI | | | | | |
| 42 | TEMPERATURE - SUCTION CKT 1 | TS | AI | | | | | |
| 43 | TEMPERATURE - DISCHARGE CKT 1 | TD | AI | | | | | |
| 44 | CKT 1 EXPANSION VALVE | EXV | AI | | | | | |
| 45 | CKT 2 COMPRESSOR SPEED | HZ | AI | | | | | |
| 46 | CKT 2 FAN SPEED | SPD | AI | | | | | |
| 47 | PRESSURE - SUCTION CKT 2 | PS | AI | | | | | |
| 48 | PRESSURE - DISCHARGE CKT 2 | PD | AI | | | | | |
| 49 | TEMPERATURE - SUCTION CKT 2 | TS | AI | | | | | |
| 50 | TEMPERATURE - DISCHARGE CKT 2 | TD | AI | | | | | |
| 51 | CKT 2 EXPANSION VALVE | EXV | AI | | | | | |
| ASHP-1B | | | | | | | | |
| 52 | CONDENSER FAN 1 VIBRATION | VIB-ECM | AI | | | | | |
| 53 | COMPRESSOR 1 VIBRATION RMS - X (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-X | AI | | | | | |
| 54 | COMPRESSOR 1 VIBRATION RMS - Y (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Y | AI | | | | | |
| 55 | COMPRESSOR 1 VIBRATION RMS - Z (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Z | AI | | | | | |
| 56 | CONDENSER FAN 2 VIBRATION | VIB-ECM | AI | | | | | |
| 57 | COMPRESSOR 2 VIBRATION RMS - X (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-X | AI | | | | | |
| 58 | COMPRESSOR 2 VIBRATION RMS - Y (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Y | AI | | | | | |
| 59 | COMPRESSOR 2 VIBRATION RMS - Z (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Z | AI | | | | | |
| 60 | CHW FLOW RATE | FL | AI | | | | | |
| 61 | CHW EWT (INTO MACHINE) | T | AI | | | | | |
| 62 | CHW LWT (OUT OF MACHINE) | T | AI | | | | | |
| 63 | VALVE POSITION | VP | AI | | | | | |
| 64 | HHW FLOW RATE | FL | AI | | | | | |
| 65 | HHW EWT (INTO MACHINE) | T | AI | | | | | |
| 66 | HHW LWT (OUT OF MACHINE) | T | AI | | | | | |
| 67 | VALVE POSITION | VP | AI | | | | | |
| 68 | PRESSURE - SUCTION CKT 1 | PS | AI | | | | | |
| 69 | PRESSURE - DISCHARGE CKT 1 | PD | AI | | | | | |
| 70 | CKT 1 COMPRESSOR SPEED | HZ | AI | | | | | |
| 71 | CKT 1 FAN SPEED | SPD | AI | | | | | |
| 72 | TEMPERATURE - SUCTION CKT 1 | TS | AI | | | | | |
| 73 | TEMPERATURE - DISCHARGE CKT 1 | TD | AI | | | | | |
| 74 | CKT 1 EXPANSION VALVE | EXV | AI | | | | | |
| 75 | CKT 2 COMPRESSOR SPEED | HZ | AI | | | | | |
| 76 | CKT 2 FAN SPEED | SPD | AI | | | | | |
| 77 | PRESSURE - SUCTION CKT 2 | PS | AI | | | | | |
| 78 | PRESSURE - DISCHARGE CKT 2 | PD | AI | | | | | |
| 79 | TEMPERATURE - SUCTION CKT 2 | TS | AI | | | | | |
| 80 | TEMPERATURE - DISCHARGE CKT 2 | TD | AI | | | | | |
| 81 | CKT 2 EXPANSION VALVE | EXV | AI | | | | | |

# BAS POINT FUNCTION SCHEDULE - HYDRONIC PLANT

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | NOTES |
|---|---|---|---|---|---|---|---|---|
| ASHP-1C | | | | | | | | |
| 82 | CONDENSER FAN 1 VIBRATION | VIB-ECM | AI | | | | | |
| 83 | COMPRESSOR 1 VIBRATION RMS - X (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-X | AI | | | | | |
| 84 | COMPRESSOR 1 VIBRATION RMS - Y (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Y | AI | | | | | |
| 85 | COMPRESSOR 1 VIBRATION RMS - Z (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Z | AI | | | | | |
| 86 | CONDENSER FAN 2 VIBRATION | VIB-ECM | AI | | | | | |
| 87 | COMPRESSOR 2 VIBRATION RMS - X (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-X | AI | | | | | |
| 88 | COMPRESSOR 2 VIBRATION RMS - Y (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Y | AI | | | | | |
| 89 | COMPRESSOR 2 VIBRATION RMS - Z (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Z | AI | | | | | |
| 90 | CHW FLOW RATE | FL | AI | | | | | |
| 91 | CHW EWT (INTO MACHINE) | T | AI | | | | | |
| 92 | CHW LWT (OUT OF MACHINE) | T | AI | | | | | |
| 93 | VALVE POSITION | VP | AI | | | | | |
| 94 | HHW FLOW RATE | FL | AI | | | | | |
| 95 | HHW EWT (INTO MACHINE) | T | AI | | | | | |
| 96 | HHW LWT (OUT OF MACHINE) | T | AI | | | | | |
| 97 | VALVE POSITION | VP | AI | | | | | |
| 98 | PRESSURE - SUCTION CKT 1 | PS | AI | | | | | |
| 99 | PRESSURE - DISCHARGE CKT 1 | PD | AI | | | | | |
| 100 | CKT 1 COMPRESSOR SPEED | HZ | AI | | | | | |
| 101 | CKT 1 FAN SPEED | SPD | AI | | | | | |
| 102 | TEMPERATURE - SUCTION CKT 1 | TS | AI | | | | | |
| 103 | TEMPERATURE - DISCHARGE CKT 1 | TD | AI | | | | | |
| 104 | CKT 1 EXPANSION VALVE | EXV | AI | | | | | |
| 105 | CKT 2 COMPRESSOR SPEED | HZ | AI | | | | | |
| 106 | CKT 2 FAN SPEED | SPD | AI | | | | | |
| 107 | PRESSURE - SUCTION CKT 2 | PS | AI | | | | | |
| 108 | PRESSURE - DISCHARGE CKT 2 | PD | AI | | | | | |
| 109 | TEMPERATURE - SUCTION CKT 2 | TS | AI | | | | | |
| 110 | TEMPERATURE - DISCHARGE CKT 2 | TD | AI | | | | | |
| 111 | CKT 2 EXPANSION VALVE | EXV | AI | | | | | |
| ASHP-1D | | | | | | | | |
| 112 | CONDENSER FAN 1 VIBRATION | VIB-ECM | AI | | | | | |
| 113 | COMPRESSOR 1 VIBRATION RMS - X (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-X | AI | | | | | |
| 114 | COMPRESSOR 1 VIBRATION RMS - Y (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Y | AI | | | | | |
| 115 | COMPRESSOR 1 VIBRATION RMS - Z (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Z | AI | | | | | |
| 116 | CONDENSER FAN 2 VIBRATION | VIB-ECM | AI | | | | | |
| 117 | COMPRESSOR 2 VIBRATION RMS - X (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-X | AI | | | | | |
| 118 | COMPRESSOR 2 VIBRATION RMS - Y (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Y | AI | | | | | |
| 119 | COMPRESSOR 2 VIBRATION RMS - Z (DISPLACEMENT, VELOCITY OR ACCELERATION) | VIB-Z | AI | | | | | |
| 120 | CHW FLOW RATE | FL | AI | | | | | |
| 121 | CHW EWT (INTO MACHINE) | T | AI | | | | | |
| 122 | CHW LWT (OUT OF MACHINE) | T | AI | | | | | |
| 123 | VALVE POSITION | VP | AI | | | | | |
| 124 | HHW FLOW RATE | FL | AI | | | | | |
| 125 | HHW EWT (INTO MACHINE) | T | AI | | | | | |
| 126 | HHW LWT (OUT OF MACHINE) | T | AI | | | | | |
| 127 | VALVE POSITION | VP | AI | | | | | |
| 128 | PRESSURE - SUCTION CKT 1 | PS | AI | | | | | |
| 129 | PRESSURE - DISCHARGE CKT 1 | PD | AI | | | | | |
| 130 | CKT 1 COMPRESSOR SPEED | HZ | AI | | | | | |
| 131 | CKT 1 FAN SPEED | SPD | AI | | | | | |
| 132 | TEMPERATURE - SUCTION CKT 1 | TS | AI | | | | | |
| 133 | TEMPERATURE - DISCHARGE CKT 1 | TD | AI | | | | | |
| 134 | CKT 1 EXPANSION VALVE | EXV | AI | | | | | |
| 135 | CKT 2 COMPRESSOR SPEED | HZ | AI | | | | | |
| 136 | CKT 2 FAN SPEED | SPD | AI | | | | | |
| 137 | PRESSURE - SUCTION CKT 2 | PS | AI | | | | | |
| 138 | PRESSURE - DISCHARGE CKT 2 | PD | AI | | | | | |
| 139 | TEMPERATURE - SUCTION CKT 2 | TS | AI | | | | | |
| 140 | TEMPERATURE - DISCHARGE CKT 2 | TD | AI | | | | | |
| 141 | CKT 2 EXPANSION VALVE | EXV | AI | | | | | |
| BOILER B-1 | | | | | | | | |
| 142 | BOILER ISOLATION VALVE CONTROL | VC | AO | | | | | |
| 143 | BOILER ISOLATION VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 144 | BOILER ENABLE/DISABLE | ED | DO | | | | | |
| 145 | BOILER SETPOINT ADJUSTMENT | SPA | AO | | | | | |
| 146 | BOILER FIRING RATE | BFR | AI | | | | | |
| 147 | BOILER ALARM | ALM | AI | | | | | |
| 148 | BOILER HAND MODE | HM | AO | | | | | |
| 149 | BOILER B-1 SUPPLY WATER TEMPERATURE (FROM BOILER) | T | AI | | | | | |
| 150 | BOILER B-1 RETURN WATER TEMPERATURE (TO BOILER) | T | AI | | | | | |
| 151 | BOILER B-1 FLOW RATE | FL | AI | | | | | |
| 152 | BOILER DIFFERENTIAL PRESSURE | DP | AI | | | | | |
| BOILER B-2 | | | | | | | | |
| 153 | BOILER ENABLE/DISABLE | ED | AI | | | | | |
| 154 | BOILER SETPOINT ADJUSTMENT | SPA | AI | | | | | |
| 155 | BOILER FIRING RATE | BFR | AI | | | | | |
| 156 | BOILER ALARM | ALM | AI | | | | | |
| 157 | BOILER HAND MODE | HM | AI | | | | | |
| 158 | BOILER ISOLATION VALVE CONTROL | VC | AO | | | | | |
| 159 | BOILER ISOLATION VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 160 | BOILER B-1 SUPPLY WATER TEMPERATURE (FROM BOILER) | T | AI | | | | | |
| 161 | BOILER B-1 RETURN WATER TEMPERATURE (TO BOILER) | T | AI | | | | | |
| 162 | BOILER B-1 FLOW RATE | FL | AI | | | | | |
| 163 | BOILER DIFFERENTIAL PRESSURE | DP | AI | | | | | |
| BOILER SYSTEM | | | | | | | | |
| 164 | HHW BYPASS VALVE CONTROL | VC | AO | | | | | |
| 165 | HHW BYPASS VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 166 | BOILER SYSTEM DIFFERENTIAL PRESSURE | DP | AI | | | | | |

# BAS POINT FUNCTION SCHEDULE - HYDRONIC PLANT

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | NOTES |
|---|---|---|---|---|---|---|---|---|
| CHILLED WATER LOOP TO/FROM DISTRIBUTION | | | | | | | | |
| 167 | CHW RETURN WATER TEMPERATURE (FROM BUILDING) | T | AI | | | | | |
| 168 | CHW SUPPLY WATER TEMPERATURE (TO BUILDING) | T | AI | | | | | |
| 169 | PLANT ISOLATION VALVE CONTROL | VC | AO | | | | | |
| 170 | PLANT ISOLATION VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 171 | CHW FLOW RATE | FL | AI | | | | | |
| 172 | CHW SYSTEM DP | DP | AI | | | | | |
| CHW SYSTEM BYPASS | | | | | | | | |
| 173 | SYSTEM BYPASS VALVE CONTROL | VC | AO | | | | | |
| 174 | SYSTEM BYPASS VALVE POSITION FEEDBACK | VP | AI | | | | | |
| HEAT PUMP SYSTEM | | | | | | | | |
| 175 | HHW ISOLATION VALVE CONTROL | VC | AO | | | | | |
| 176 | HHW ISOLATION VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 177 | HHW ISOLATION VALVE CONTROL | VC | AO | | | | | |
| 178 | HHW ISOLATION VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 179 | HHW ARRAY FLOW RATE | FL | AI | | | | | |
| 180 | HHW SUPPLY WATER TEMPERATURE (FROM ARRAY) | T | AI | | | | | |
| 181 | HHW RETURN WATER TEMPERATURE (TO ARRAY) | T | AI | | | | | |
| 182 | CHW ISOLATION VALVE CONTROL | VC | AO | | | | | |
| 183 | CHW ISOLATION VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 184 | CHW ISOLATION VALVE CONTROL | VC | AO | | | | | |
| 185 | CHW ISOLATION VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 186 | CHW ARRAY FLOW RATE | FL | AI | | | | | |
| 187 | CHW SUPPLY WATER TEMPERATURE (FROM ARRAY) | T | AI | | | | | |
| 188 | CHW RETURN WATER TEMPERATURE (TO ARRAY) | T | AI | | | | | |
| 189 | HHW COMMON HEADER DIFFERENTIAL PRESSURE | DP | AI | | | | | |
| 190 | CHW COMMON HEADER DIFFERENTIAL PRESSURE | DP | AI | | | | | |
| HEATING HOT WATER LOOP TO/FROM DISTRIBUTION | | | | | | | | |
| 191 | PLANT ISOLATION VALVE CONTROL | VC | AO | | | | | |
| 192 | PLANT ISOLATION VALVE POSITION FEEDBACK | VP | AI | | | | | |
| 193 | HHW RETURN WATER TEMPERATURE (FROM BUILDING) | T | AI | | | | | |
| 194 | HHW SUPPLY WATER TEMPERATURE (TO BUILDING) | T | AI | | | | | |
| 195 | HHW FLOW RATE | FL | AI | | | | | |
| 196 | HHW SYSTEM DP | DP | AI | | | | | |
| HHW SYSTEM BYPASS | | | | | | | | |
| 197 | SYSTEM BYPASS VALVE CONTROL | VC | AO | | | | | |
| 198 | SYSTEM BYPASS VALVE POSITION FEEDBACK | VP | AI | | | | | |

FOR REFERENCE ONLY

ALF

CONSTRUCT ARABIAN LEOPARD FACILITY

NOT FOR CONSTRUCTION

STAMP

GRAPHIC SCALE(S)

DATE: 06/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)

ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

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DRAWING TITLE: MECHANICAL CONTROLS - HYDRONIC PLANT BAS POINT TABLE

WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 294 OF 378

M-807CP (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

6/15/2026 3:01:43 PM
M-807CP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# BUILDING HVAC CENTRAL PLANT SEQUENCE OF OPERATION

**NOTE - COORDINATE EQUIPMENT AND POINT TAGS WITH SMITHSONIAN TO ENSURE THAT THE FORMATTING AND TAGS ARE CONSISTENT WITH THE SMITHSONIAN SERVER STANDARDS.**

OVERVIEW

THE HVAC CENTRAL PLANT IS A STANDALONE CENTRAL PLANT THAT PROVIDES COOLING AND HEATING FOR THE CARE BUILDING.
ALL SET POINTS, INCLUDING BUT NOT LIMITED TO TIME DELAYS SHALL BE USER ADJUSTABLE.

THE PLANT CONTAINS PRIMARY EQUIPMENT DESIGNATED AS EITHER "SOURCE" OR "USER."
SOURCE EQUIPMENT PROVIDES HEAT PRODUCTION OR HEAT REJECTION, AND USER EQUIPMENT CONVERTS SOURCE EQUIPMENT ENERGY TO CHILLED WATER AND HEATING HOT WATER HYDRONIC LOOPS.
WORKING FLUID FOR THE CHILLED WATER AND HEATING HOT WATER LOOP IS 30% PROPYLENE GLYCOL - 70% WATER.
SOURCE EQUIPMENT AND USER EQUIPMENT TRANSFER THERMAL ENERGY WITHIN THE AIR SOURCE HEAT PUMP, THROUGH INTERNAL BRAZED PLATE HEAT EXCHANGERS.
PUMPING ARRANGEMENT IS PRIMARY-SECONDARY.
PRIMARY EQUIPMENT IS LISTED BELOW.

PRODUCERS -
AIR SOURCE HEAT PUMP ARRAY (ASHP-1) IS THE PRIMARY HEATING AND COOLING PLANT.
CHILLED WATER PRODUCERS -
1. AIR SOURCE HEAT PUMP ARRAY
   A. ASHP-1A OR MODULE A
   B. ASHP-1B OR MODULE B
   C. ASHP-1C OR MODULE C
   D. ASHP-1D OR MODULE D

HEATING HOT WATER PRODUCERS -
1. AIR SOURCE HEAT PUMP ARRAY
   A. ASHP-1A OR MODULE A
   B. ASHP-1B OR MODULE B
   C. ASHP-1C OR MODULE C
   D. ASHP-1D OR MODULE D
2. HEATING HOT WATER BOILERS
   A. BOILER B-1
   B. BOILER B-2

USERS -
CHILLED WATER USERS
1. AHU-1
2. FAN COIL UNITS

HEATING HOT WATER USERS
1. AHU-1
2. FAN COIL UNITS
3. UNIT HEATERS
4. RADIANT SLAB MIXING BOX
5. REHEAT COIL WITHIN SUPPLY AIR TERMINAL UNIT
6. AIR CURTAIN

REDUNDANCIES
1. AIR SOURCE HEAT PUMP -
   A. MODULE D IS MECHANICALLY REDUNDANT.
   B. MODULE D IS NOT ELECTRICALLY REDUNDANT.
2. BOILERS PUMP -
   A. BOTH BOILERS ARE MECHANICALLY REDUNDANT.
   B. BOTH BOILERS ARE ELECTRICALLY REDUNDANT.

**SETPOINTS**
A. TEMPERATURE SETPOINTS
   1. CHWSTminX, THE LOWEST CHILLED WATER SUPPLY TEMPERATURE SETPOINT FOR HEAT PUMP MODULE(S): **42 °F**
   2. CHWSTmax, THE MAXIMUM CHILLED WATER SUPPLY TEMPERATURE SETPOINT USED IN PLANT RESET LOGIC: **48°F**
B. HEAT PUMP ARRAY DIFFERENTIAL PRESSURE SETPOINTS
   1. HHW-DPmin, THE MINIMUM DIFFERENTIAL PRESSURE SETPOINT TO MEET MINIMUM FLOW FOR A SINGULAR MODULE:
      a. HHW-Array-DPmin SHALL BE SUFFICIENTLY HIGH AS TO ENSURE HHW-MinFlowX IS MAINTAINED THROUGH THE MODULE.
      b. THE FINAL VALUE SHALL BE CONFIRMED IN APPROVED EQUIPMENT SUBMITTAL.
   2. CHW-DPmin, THE MINIMUM DIFFERENTIAL PRESSURE SETPOINT USED IN PLANT RESET LOGIC FOR SECONDARY PUMPS: **5 PSI.**
      a. CHW-Array-DPmin SHALL BE SUFFICIENTLY HIGH AS TO ENSURE CHW-MinFlowX IS MAINTAINED THROUGH THE MODULE.
      b. THE FINAL VALUE SHALL BE CONFIRMED IN APPROVED EQUIPMENT SUBMITTAL.
C. BUILDING DIFFERENTIAL PRESSURE SETPOINTS
   1. HHW-DPmin, THE MINIMUM DIFFERENTIAL PRESSURE SETPOINT USED IN PLANT RESET LOGIC FOR SECONDARY PUMPS: **5 PSI.**
      a. HHW-DPmin SHALL BE SUFFICIENTLY HIGH AS TO ENSURE HHW-MinFlowX IS MAINTAINED THROUGH THE MODULE. THE FINAL VALUE SHALL BE DETERMINED BY THE TAB CONTRACTOR.
   2. CHW-DPmin, THE MINIMUM DIFFERENTIAL PRESSURE SETPOINT USED IN PLANT RESET LOGIC FOR SECONDARY PUMPS: **5 PSI.**
      a. CHW-DPmin SHALL BE SUFFICIENTLY HIGH AS TO ENSURE CHW-MinFlowX IS MAINTAINED THROUGH THE MODULE. THE FINAL VALUE SHALL BE DETERMINED BY THE TAB CONTRACTOR.
D. MODULE FLOW SETPOINTS
   1. CHW-MinFlowX, THE MINIMUM CHILLED WATER FLOW RATE THROUGH A SINGLE MODULE PER MANUFACTURER'S RECOMMENDATIONS : **40 GPM (FOR BASIS OF DESIGN PRODUCT)**
   2. CHW-DesFlowX, THE DESIGN CHILLED WATER FLOWRATE FOR ARRAY, IN GPM: **250 GPM**
   3. HHW-MinFlowX, THE MINIMUM CHILLED WATER FLOW RATE THROUGH A SINGLE MODULE PER MANUFACTURER'S RECOMMENDATIONS : **40 GPM (FOR BASIS OF DESIGN PRODUCT)**
   4. CHW-DesFlowX, THE DESIGN CHILLED WATER FLOWRATE FOR ARRAY, IN GPM: **160 GPM**
E. CAPACITY
   1. COOLING
      A. QchX, DESIGN CAPACITY OF SINGULAR MODULE: **30 TONS**
      B. PCHWFdesign, DESIGN PRIMARY LOOP FLOW: **250 GPM**
      C. SCHWFdesign, DESIGN SECONDARY LOOP FLOW: **250 GPM**
   2. HEATING
      A. QchX, DESIGN CAPACITY OF SINGULAR MODULE: **350 MBH**
      B. PCHWFdesign, DESIGN PRIMARY LOOP FLOW: **160 GPM**
      C. SCHWFdesign, DESIGN SECONDARY LOOP FLOW: **160 GPM**
F. HEADERED PUMP DESIGN QUANTITIES
   1. N-PCHWP, THE NUMBER OF PRIMARY CHILLED WATER PUMPS THAT OPERATE AT DESIGN CONDITIONS: **QTY = 1 (1 PUMP IS STANDBY)**
   2. M-SCHWP, THE NUMBER OF SECONDARY CHILLED WATER PUMPS THAT OPERATE AT DESIGN CONDITIONS: **QTY = 1 (1 PUMP IS STANDBY)**
   3. N-PCHWP, THE NUMBER OF PRIMARY HEATING HOT WATER PUMPS THAT OPERATE AT DESIGN CONDITIONS: **QTY = 1 (1 PUMP IS STANDBY)**
   4. M-SCHWP, THE NUMBER OF SECONDARY HEATING HOT WATER PUMPS THAT OPERATE AT DESIGN CONDITIONS: **QTY = 1 (1 PUMP IS STANDBY)**

**CONTROL STRATEGY**
**1. BUILDING ENGINEER SHALL HAVE THE ABILITY TO CHOOSE ONE OF THE FOLLOWING SEQUENCES TO RUN THE HVAC PLANT -**
   **A. OPTIMIZED FOR ENERGY EFFICIENCY (PRINCIPLES FROM ASHRAE HVAC SYSTEMS AND ASHRAE GUIDELINE 36)**
   **B. SIMPLIFIED APPROACH**
**2. OWNER SHALL HAVE THE ABILITY TO CHOOSE ONE OF THE OPTION. DISABLE SEQUENCES ASSOCIATED WITH OTHER OPTION.**

**PLANT RUN MODE -**
**1.** THE BAS DETERMINES THE RUN MODE FOR AIR SOURCE HEAT PUMP, BASED ON
**2.** BUILDING REQUIRES COOLING OR IS IN STRICT 'COOLING MODE' IF ALL OF THE FOLLOWING IS TRUE FOR 15 MIN CONTINUOUS -
   A. CHW PUMP STATUS (ANY PUMP) = ON OR CHW FLOW PROVEN
   B. CHILLED WATER RETURN TEMPERATURE ≥ CHWRTmax
   C. CHILLED WATER RETURN TEMPERATURE TREND ≥ +4°F/HR
   D. HHW PUMP STATUS = OFF OR
      a. HEATING HOT WATER RETURN TEMPERATURE ≥ HHWRTmin.
      b. HEATING HOT WATER RETURN TEMPERATURE TREND IS STABLE OR INCREASING
**3.** BUILDING REQUIRES HEATING OR IS IN STRICT 'HEATING MODE' IF ALL OF THE FOLLOWING IS TRUE FOR 15 MIN CONTINUOUS -
   A. HHW PUMP STATUS (ANY PUMP) = ON OR HHW FLOW PROVEN
   B. HEATING HOT WATER RETURN TEMPERATURE ≤ HHWRTmin
   C. HEATING HOT WATER RETURN TEMPERATURE TREND ≤ -4°F/HR
   D. CHW PUMP STATUS = OFF OR
      a. CHILLED WATER RETURN TEMPERATURE ≤ CHWRTmax
      b. CHILLED WATER RETURN TEMPERATURE TREND IS STABLE OR DECREASING
4. BUILDING REQUIRES HEATING AND COOLING OR IS IN 'SIMULTANEOUS MODE' IF ALL OF THE FOLLOWING IS TRUE FOR 15 MIN CONTINUOUS -
   A. CHW FLOW PROVEN
   B. HHW FLOW PROVEN
   C. CHILLED WATER RETURN TEMPERATURE ≥ CHWRTmax
   D. CHILLED WATER RETURN TEMPERATURE TREND ≥ +4°F/HR
   E. HEATING HOT WATER RETURN TEMPERATURE ≤ HHWRTmin
   F. HEATING HOT WATER RETURN TEMPERATURE TREND ≤ -4°F/HR

## OPTION A = PLANT CONTROL (OPTIMIZED FOR ENERGY EFFICIENCY)

**PLANT EQUIPMENT SEQUENCING**
A. COOLING
   a. BAS ENGAGES HEAT PUMP. SEE HEAT PUMP CONTROL SECTION.
   b. HEAT PUMP'S MASTER CONTROLLER SEQUENCES THE MODULES.
B. HEATING
   a. BOILERS SHALL OPERATE AS SUPPLEMENTAL HEATING ONLY. HEAT PUMP ARRAY REMAINS PRIMARY HEATING SOURCE.
   b. BAS ENGAGES HEAT PUMP. SEE HEAT PUMP CONTROL SECTION.
   c. HEAT PUMP MASTER CONTROLLER SEQUENCES THE MODULES.
   d. IF HEATING DESIGN SET-POINT IS NOT MET FOR 5 MIN CONTINUOUS (ADJ.), BAS INITIATES BOILER PLANT.
   e. BOILER MASTER CONTROLLER SEQUENCES BOILER B-1 AND BOILER B-2.

## OPTION B = PLANT CONTROL (AMBIENT BASED SOURCE SWITCH)

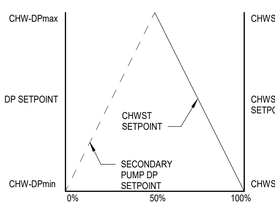
**PLANT EQUIPMENT SEQUENCING**
A. COOLING
   a. BAS ENGAGES HEAT PUMP. SEE HEAT PUMP CONTROL SECTION.
   b. HEAT PUMP'S MASTER CONTROLLER SEQUENCES THE MODULES.
B. HEATING
   a. BOILERS SHALL OPERATE AS PRIMARY HEATING SOURCE WHEN ENABLED BY AMBIENT CONDITIONS. HEAT PUMP ARRAY SHALL BE DISABLED DURING BOILER-ONLY OPERATION.
   b. BAS ENGAGES HEAT PUMP. SEE HEAT PUMP CONTROL SECTION.
   c. HEAT PUMP MASTER CONTROLLER SEQUENCES THE MODULES.
   d. BAS SWITCHES TO BOILER PLANT BASED ON AMBIENT CONDITIONS,
   e. BAS COMMANDS HEAT PUMP ARRAY TO STOP.
   f. BAS ENGAGES BOILER SYSTEM. SEE BOILER CONTROL SECTION.
   g. BOILER MASTER CONTROLLER SEQUENCES BOILER B-1 AND BOILER B-2.

**PLANT EQUIPMENT SWITCHOVER**
1. SWITCHOVER TEMPERATURE =
   A. HEAT PUMP
      a. AMBIENT DRY BULB OF 42 DEGREE F OR WARMER
      b. DEWPOINT OF 38 DEGREE F OR LOWER.
   B. BOILER
      a. AMBIENT DRY BULB OF 42 DEGREE F OR COLDER
      b. WITH AMBIENT DEW POINT OF 38 DEGREE F OR HIGHER.

**HEAT PUMP ARRAY CONTROL**
1. TO ENGAGE HEAT PUMP ARRAY, THE BAS SHALL -
   **A.** COMMAND THE NATIVE MASTER CONTROLLER OF THE ARRAY TO START.
   **B.** ONCE PROVEN ON, PROVIDE A RUN MODE TO THE MASTER CONTROLLER.
      **a.** HEATING MODE
      **b.** COOLING MODE
      **c.** SIMULTANEOUS MODE
   **C.** PROVIDE CHILLED WATER AND HEATING HOT WATER SUPPLY (TO THE BUILDING) SETPOINT TO THE MASTER CONTROLLER OF THE HEAT PUMP MODULE ARRAY.
   **D.** SEE CHILLED WATER PLANT RESET LOGIC FOR SETPOINT DETERMINATION.
   **E.** SEE HEATING HOT WATER PLANT RESET LOGIC FOR SETPOINT DETERMINATION.
2. BAS SHALL ALWAYS INITIATE THE ARRAY IN A LOW SOUND MODE.
3. MANUFACTURER PROGRAMMED LOGIC WITHIN THE MASTER CONTROLLER SHALL BE SOLELY RESPONSIBLE FOR CONTROLLING THE SLAVE MODULES -
   **A.** STAGING THE MODULES.
   **B.** SEQUENCING THE MODULES.
   **C.** COMMUNICATING SET POINTS AND CONTROL MODES
4. TO DISENGAGE HEAT PUMP ARRAY, THE BAS SHALL -
   **A.** COMMAND THE NATIVE MASTER CONTROLLER OF THE ARRAY TO STOP.

**BOILER ARRAY CONTROL**
1. UPON A SIGNAL FROM THE BAS FOR BOILER OPERATION, THE BOILER ISOLATION VALVE SHALL OPEN THE BOILER TO FLOW AND BOILER BYPASS VALVE SHALL MODULATE TO ACHIEVE DESIGN FLOW AS MEASURED BY THE BOILER INLET FLOW METER.
2. UPON VERIFICATION OF DESIGN FLOW, THE BAS SHALL INDEX THE BOILER CONTROL PANEL ON.
3. BOILER SHALL NOT BE TURNED ON UNTIL FLOW HAS BEEN ESTABLISHED VIA THE CORRESPONDENT FLOW SWITCH.
4. THE BOILER SHALL OPERATE TOGETHER TO OPTIMIZE FAN ENERGY USE.
5. THE BAS MONITORS THE OPERATION OF THE BOILER.
6. THE BOILER MANUFACTURER CONTROLLER SHALL MAINTAIN HHWST (SEE HHW PLANT RESET PARAGRAPH).
7. BOILER SHALL RUN UNTIL SIGNALED FROM THE BAS TO STOP.
8. ON VERIFICATION OF BOILER STATUS FROM THE BOILER CONTROL PANEL, THE BOILER BYPASS VALVE SHALL FULLY OPEN.
9. THE BOILER ISOLATION VALVE SHALL CLOSE.

**CHILLED WATER PLANT RESET**
A. CHWSTmin IN THE FOLLOWING LOGIC SHALL BE THE LOWEST CHWSTminX OF AIR SOURCE HEAT PUMP MODULES IN THE PLANT.
B. DIFFERENTIAL PRESSURE CONTROLLED LOOPS: CHILLED WATER SUPPLY TEMPERATURE SETPOINT CHWSTsp AND PUMP DIFFERENTIAL PRESSURE SETPOINT CHW-DPsp SHALL BE RESET BASED ON THE CURRENT VALUE OF THE LOGIC VARIABLE CALLED "CHW PLANT RESET" AS SHOWN BELOW AND DESCRIBED SUBSEQUENTLY



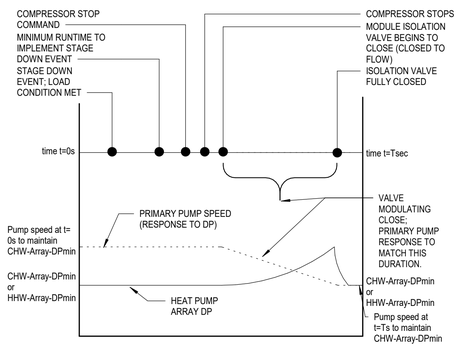

C. FROM 0% LOOP OUTPUT TO 50% LOOP OUTPUT, RESET DP SETPOINT FROM CHW-DPmin TO CHWP-DPmax.
D. FROM 50% LOOP OUTPUT TO 100% LOOP OUTPUT, RESET CHWST SETPOINT FROM CHWSTmax TO CHWSTmin.
E. CHW PLANT RESET VARIABLE SHALL BE RESET USING TRIM & RESPOND LOGIC WITH THE FOLLOWING PARAMETERS:

| VARIABLE | DEFINITION | VALUE |
|---|---|---|
| DEVICE | ASSOCIATED DEVICE (E.G., FAN, PUMP) | ANY CHW PUMP DISTRIBUTION LOOP |
| $SP_0$ | INITIAL SETPOINT | 100% |
| $SP_{min}$ | MINIMUM SETPOINT | 0% |
| $SP_{max}$ | MAXIMUM SETPOINT | 100% |
| $T_d$ | DELAY TIMER | 15 MINUTES |
| T | TIME STEP | 5 MINUTES |
| I | NUMBER OF IGNORED REQUESTS | 2 REQUESTS |
| R | NUMBER OF REQUESTS FROM ZONES/SYSTEMS | COOLING CHWST RESET REQUESTS |
| $SP_{trim}$ | TRIM AMOUNT | -2% |
| $SP_{res}$ | RESPOND AMOUNT (MUST BE OPPOSITE IN SIGN TO $SP_{trim}$ AND SHOULD GENERALLY BE HIGHER) | +3% |
| $SP_{res\text{-}max}$ | MAXIMUM RESPONSE PER TIME INTERVAL (MUST BE SAME SIGN AS $SP_{res}$) | +7% |

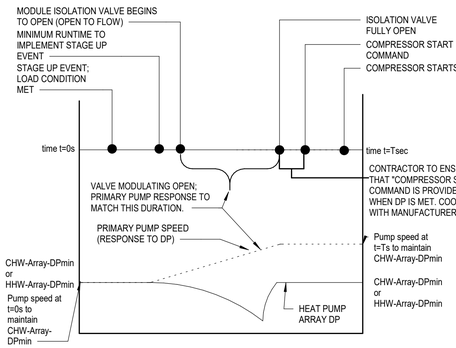
F. CHWST PLANT RESET LOOP SHALL BE ENABLED WHEN THE PLANT IS ENABLED AND DISABLED WHEN THE PLANT IS DISABLED.
G. WHEN A PLANT STAGE CHANGE IS INITIATED, CHW PLANT RESET LOGIC SHALL BE DISABLED AND VALUE FIXED AT ITS LAST VALUE FOR THE LONGER OF 15 MINUTES AND THE TIME IT TAKES FOR THE PLANT TO SUCCESSFULLY STAGE.
H. WHERE MORE THAN ONE REMOTE DP SENSOR SERVES A GIVEN SET OF PRIMARY OR SECONDARY PUMPS, REMOTE DP SETPOINTS FOR ALL REMOTE SENSORS SERVING THOSE PUMPS SHALL INCREASE IN UNISON. NOTE: IF REMOTE SENSORS HAVE DIFFERENT CHW-DPmax VALUES, THEN THE AMOUNT EACH DP SETPOINT CHANGES PER PERCENT LOOP OUTPUT WILL DIFFER.

**PRIMARY CHILLED WATER PUMPS**
A. PRIMARY PUMPS SHALL BE LEAD/STANDBY CONTROLLED PER THE GENERAL CONTROLS SHEETS.
B. ENABLE LEAD PRIMARY CHW PUMP WHEN ANY MODULE CHW ISOLATION VALVE IS COMMANDED OPEN. DISABLE THE LEAD PRIMARY CHW PUMP WHEN ALL CHW ISOLATION VALVES FOR THE MODULES ARE COMMANDED CLOSED.
C. PRIMARY PUMP SPEED SHALL BE CONTROLLED BY A REVERSE ACTING PID LOOP TO MAINTAIN THE DIFFERENTIAL PRESSURE ACROSS THE ASHP AT A SETPOINT CHW-Array-DPmin.
   a. HIGH PRECISION DIFFERENTIAL PRESSURE SENSOR WITH BIDIRECTIONAL SENSING CAPABILITY.
      • DIFFERENTIAL PRESSURE AT POINT 'E' MINUS DIFFERENTIAL PRESSURE AT POINT 'F', AT 5% OF MAXIMUM PRESSURE DROP ACROSS A HEAT PUMP MODULE

PRIMARY PUMP CONTROL INTENT - TYPICAL COMPRESSOR STAGE DOWN EVENT



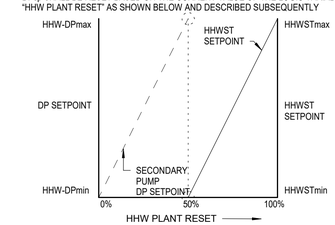

PRIMARY PUMP CONTROL INTENT - TYPICAL COMPRESSOR STAGE UP EVENT




**SECONDARY CHILLED WATER PUMPS**
A. SECONDARY PUMPS SHALL BE LEAD/STANDBY CONTROLLED PER THE GENERAL CONTROLS SHEETS.
B. ENABLE LEAD SECONDARY CHW PUMP WHEN PLANT IS ENABLED AND ANY LOAD SERVED BY THE PUMP(S) IS GENERATING A COOLING PLANT REQUEST. DISABLE THE LEAD PUMP OTHERWISE.
C. SECONDARY CHW PUMP SHALL BE STAGED AS A FUNCTION OF SCHWFR, THE RATIO OF CURRENT CHILLED WATER FLOW, $FLOW_S$, TO DESIGN FLOW.
   1. SCHWFR = $FLOW_S$ / SCHWFdesign
D. WHEN ANY SECONDARY PUMP IS PROVEN ON, PUMP SPEED WILL BE CONTROLLED BY A REVERSE ACTING PID LOOP MAINTAINING THE DIFFERENTIAL PRESSURE SIGNAL AT A SETPOINT CHW-DPsp DETERMINED BY THE RESET SCHEME DESCRIBED HEREIN. ALL PUMPS RECEIVE THE SAME SPEED SIGNAL.
E. WHERE MULTIPLE DP SENSORS EXIST, A PID LOOP SHALL RUN FOR EACH SENSOR. CHW PUMPS SHALL BE CONTROLLED TO THE HIGH SIGNAL OUTPUT OF ALL DP SENSOR LOOPS.

**CHILLED WATER SYSTEM (SECONDARY LOOP) BYPASS**
A. INTENT OF THE BYPASS TO ENSURE MINIMUM FLOW IS MAINTAINED IN THE SECONDARY LOOP WHEN SECONDARY PUMPS OR HEAT PUMP ARE ENERGIZED.
B. MINIMUM SECONDARY LOOP FLOW CONTROL
   a. PRIMARY PUMPS MAINTAIN DP ACROSS HEAT PUMP ARRAY.
   b. SYSTEM BYPASS SHALL MAINTAIN SECONDARY LOOP FLOW ABOVE 20% OF MANUFACTURER RECOMMENDED SECONDARY CHILLED WATER PUMP MINIMUM FLOW, AT ALL TIMES.
   c. CONTROL ACTION:
      • IF SECONDARY FLOW < SETPOINT, BYPASS VALVE MODULATES OPEN.
      • ELSEIF PRIMARY FLOW ≥ SETPOINT, BYPASS VALVE MODULATES CLOSED.

**HEATING HOT WATER PLANT RESET**
A. HHWSTmax IN THE FOLLOWING LOGIC SHALL BE THE HIGHEST HHWSTmaxX OF MODULES AND BOILERS IN THE PLANT.
B. DIFFERENTIAL PRESSURE CONTROLLED LOOPS: HEATING HOT WATER SUPPLY TEMPERATURE SETPOINT HHWSTsp AND PUMP DIFFERENTIAL PRESSURE SETPOINT HHW-DPsp SHALL BE RESET BASED ON THE CURRENT VALUE OF THE LOGIC VARIABLE CALLED "HHW PLANT RESET" AS SHOWN BELOW AND DESCRIBED SUBSEQUENTLY




C. FROM 0% LOOP OUTPUT TO 50% LOOP OUTPUT, RESET DP SETPOINT FROM HHW-DPmin TO HHWP-DPmax.
D. FROM 50% LOOP OUTPUT TO 100% LOOP OUTPUT, RESET HHWST SETPOINT FROM HHWSTmin TO HHWSTmax.
E. HHW PLANT RESET VARIABLE SHALL BE RESET USING TRIM & RESPOND LOGIC WITH THE FOLLOWING PARAMETERS:

| VARIABLE | DEFINITION | VALUE |
|---|---|---|
| DEVICE | ASSOCIATED DEVICE (E.G., FAN, PUMP) | ANY HHW PUMP DISTRIBUTION LOOP |
| $SP_0$ | INITIAL SETPOINT | 100% |
| $SP_{min}$ | MINIMUM SETPOINT | 0% |
| $SP_{max}$ | MAXIMUM SETPOINT | 100% |
| $T_d$ | DELAY TIMER | 15 MINUTES |
| T | TIME STEP | 5 MINUTES |
| I | NUMBER OF REQUESTS | 2 REQUESTS |
| R | NUMBER OF REQUESTS FROM ZONES/SYSTEMS | HEATING HHWST RESET REQUESTS |
| $SP_{trim}$ | TRIM AMOUNT | -2% |
| $SP_{res}$ | RESPOND AMOUNT (MUST BE OPPOSITE IN SIGN TO $SP_{trim}$ AND SHOULD GENERALLY BE HIGHER) | +3% |
| $SP_{res\text{-}max}$ | MAXIMUM RESPONSE PER TIME INTERVAL (MUST BE SAME SIGN AS $SP_{res}$) | +7% |

F. HHWST PLANT RESET LOOP SHALL BE ENABLED WHEN THE PLANT IS ENABLED AND DISABLED WHEN THE PLANT IS DISABLED.
G. WHEN A PLANT STAGE CHANGE IS INITIATED, HHW PLANT RESET LOGIC SHALL BE DISABLED AND VALUE FIXED AT ITS LAST VALUE FOR THE LONGER OF 15 MINUTES AND THE TIME IT TAKES FOR THE PLANT TO SUCCESSFULLY STAGE.
H. WHERE MORE THAN ONE REMOTE DP SENSOR SERVES A GIVEN SET OF PRIMARY OR SECONDARY PUMPS, REMOTE DP SETPOINTS FOR ALL REMOTE SENSORS SERVING THOSE PUMPS SHALL INCREASE IN UNISON. NOTE: IF REMOTE SENSORS HAVE DIFFERENT HHW-DPmax VALUES, THEN THE AMOUNT EACH DP SETPOINT CHANGES PER PERCENT LOOP OUTPUT WILL DIFFER.

**PRIMARY HEATING HOT WATER PUMPS**
A. PRIMARY PUMPS SHALL BE LEAD/STANDBY CONTROLLED PER THE GENERAL CONTROLS SHEETS.
B. ENABLE LEAD PRIMARY HHW PUMP WHEN ANY MODULE HHW ISOLATION VALVE IS COMMANDED OPEN. DISABLE THE LEAD PRIMARY HHW PUMP WHEN ALL HHW ISOLATION VALVES FOR THE MODULES ARE COMMANDED CLOSED.
C. PRIMARY PUMP SPEED SHALL BE CONTROLLED BY A REVERSE ACTING PID LOOP TO MAINTAIN THE DIFFERENTIAL PRESSURE ACROSS THE ASHP AT A SETPOINT HHW-Array-DPmin.
   a. HIGH PRECISION DIFFERENTIAL PRESSURE SENSOR WITH BIDIRECTIONAL SENSING CAPABILITY.
      • DIFFERENTIAL PRESSURE AT POINT 'G' MINUS DIFFERENTIAL PRESSURE AT POINT 'H', AT 5% OF MAXIMUM PRESSURE DROP ACROSS A HEAT PUMP MODULE

**SECONDARY HEATING HOT WATER PUMPS**
A. SECONDARY PUMPS SHALL BE LEAD/STANDBY CONTROLLED PER THE GENERAL CONTROLS SHEETS.
B. ENABLE LEAD SECONDARY HHW PUMP WHEN PLANT IS ENABLED AND ANY LOAD SERVED BY THE PUMP(S) IS GENERATING A HEATING PLANT REQUEST. DISABLE THE LEAD PUMP OTHERWISE.
C. SECONDARY HHW PUMP SHALL BE STAGED AS A FUNCTION OF SHHWFR, THE RATIO OF CURRENT HEATING WATER FLOW, $FLOW_S$, TO DESIGN FLOW.
   1. SHHWFR = $FLOW_S$ / SHHWFdesign
D. WHEN ANY SECONDARY PUMP IS PROVEN ON, PUMP SPEED WILL BE CONTROLLED BY A REVERSE ACTING PID LOOP MAINTAINING THE DIFFERENTIAL PRESSURE SIGNAL AT A SETPOINT HHW-DPsp DETERMINED BY THE RESET SCHEME DESCRIBED HEREIN. ALL PUMPS RECEIVE THE SAME SPEED SIGNAL.
E. WHERE MULTIPLE DP SENSORS EXIST, A PID LOOP SHALL RUN FOR EACH SENSOR. HHW PUMPS SHALL BE CONTROLLED TO THE HIGH SIGNAL OUTPUT OF ALL DP SENSOR LOOPS.

**HEATING HOT WATER SYSTEM (SECONDARY LOOP) BYPASS**
A. INTENT OF THE BYPASS TO ENSURE MINIMUM FLOW IS MAINTAINED IN THE SECONDARY LOOP WHEN SECONDARY PUMPS, ANYALL BOILERS OR HEAT PUMP ARE ENERGIZED.
B. MINIMUM SECONDARY LOOP FLOW CONTROL
   a. PRIMARY PUMPS MAINTAIN DP ACROSS HEAT PUMP ARRAY.
   b. SYSTEM BYPASS SHALL MAINTAIN MAXIMUM OF THE FOLLOWING -
      • SECONDARY LOOP FLOW ABOVE 20% OF MANUFACTURER RECOMMENDED SECONDARY CHILLED WATER PUMP MINIMUM FLOW, AT ALL TIMES.
      • SECONDARY LOOP FLOW ABOVE 20% OF MANUFACTURER RECOMMENDED BOILER MINIMUM FLOW, AT ALL TIMES.
   c. CONTROL ACTION:
      • IF SECONDARY FLOW < SETPOINT, BYPASS VALVE MODULATES OPEN.
      • ELSEIF PRIMARY FLOW ≥ SETPOINT, BYPASS VALVE MODULATES CLOSED.

**ALARMS**
A. MAINTENANCE INTERVAL ALARM WHEN PUMP HAS OPERATED FOR MORE THAN 3000 HOURS AS INDICATED BY THE STAGING RUNTIME: LEVEL 4. RESET THE STAGING RUNTIME INTERVAL COUNTER WHEN ALARM IS ACKNOWLEDGED.
B. MAINTENANCE INTERVAL ALARM WHEN MODULE WITHIN AN ARRAY HAS OPERATED FOR MORE THAN 1000 HOURS AS INDICATED BY THE STAGING RUNTIME: LEVEL 4. RESET THE STAGING RUNTIME INTERVAL COUNTER WHEN ALARM IS ACKNOWLEDGED.
C. HEAT PUMP MODULE/ARRAY ALARM: LEVEL 2
D. EMERGENCY OFF SWITCH: LEVEL 1
E. PUMP ALARM IS INDICATED BY THE STATUS INPUT BEING DIFFERENT FROM THE OUTPUT COMMAND FOR 15 SECONDS.
   1. COMMANDED ON, STATUS OFF: LEVEL 2. DO NOT EVALUATE ALARM UNTIL THE EQUIPMENT HAS BEEN COMMANDED ON FOR 15 SECONDS.
   2. COMMANDED OFF, STATUS ON: LEVEL 4. DO NOT EVALUATE THE ALARM UNTIL THE EQUIPMENT HAS BEEN COMMANDED OFF FOR 60 SECONDS.
F. FOR TWO-POSITION VALVES, VALVE ALARM IS INDICATED BY THE END SWITCH STATUS BEING DIFFERENT FROM THE OUTPUT COMMAND FOR 90 SECONDS.
   1. COMMANDED OPEN, STATUS NOT OPEN: LEVEL 2. DO NOT EVALUATE ALARM UNTIL THE EQUIPMENT HAS BEEN COMMANDED OPEN FOR 90 SECONDS.
   2. COMMANDED CLOSED, STATUS NOT CLOSED: LEVEL 4. DO NOT EVALUATE ALARM UNTIL THE EQUIPMENT HAS BEEN COMMANDED CLOSED FOR 90 SECONDS.
G. FOR MODULATING VALVES, VALVE ALARM IS INDICATED BY THE ANALOG POSITION FEEDBACK BEING DIFFERENT FROM THE OUTPUT COMMAND BY MORE THAN 10% FOR 90 SECONDS: LEVEL 2.
H. SENSOR FAILURE
   1. SENSOR SHALL BE DEEMED OUTSIDE OF ITS WIDEST POSSIBLE OPERATING RANGE IF ANY OF THE FOLLOWING ARE TRUE:
      a. FEEDBACK LESS THAN 2 MA FROM ANY 4 TO 20 MA TRANSDUCER; OR
      b. TEMPERATURE READING LESS THAN 0°F FROM ANY TEMPERATURE SENSOR.
   2. ANY SENSOR THAT GOES OUTSIDE OF ITS WIDEST POSSIBLE OPERATING RANGE.
      a. IF THE SENSOR IS USED FOR MONITORING ONLY: LEVEL 3
      b. IF THE SENSOR IS USED FOR CONTROL: LEVEL 2

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Washington, DC 20024-2520

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
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| SHEET NO. | 295 OF 378 |
| | M-808CP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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**PERFORMANCE MONITORING**

A. ALL CALCULATIONS AND ALL PARAMETERS SHALL BE TOGGLE-ABLE BETWEEN METRIC AND IMPERIAL UNITS. IMPERIAL UNITS SHALL BE THE DEFAULT VIEW. UNITS SHALL BE CLEARLY VISIBLE FOR ALL PARAMETERS.

B. ALL CALCULATIONS LISTED BELOW SHALL BE PERFORMED AT LEAST ONCE EVERY 30 SECONDS. TIME AVERAGED VALUES SHALL BE RECORDED AT LEAST ONCE EVERY 5 MINUTES. THE AVERAGING PERIOD SHALL EQUAL THE TRENDING INTERVAL.

C. TOTAL PLANT POWER. CALCULATE TOTAL PLANT POWER AS THE SUM OF HEAT PUMP POWER, BOILER POWER, PUMP POWER AND HEAT TRACE CIRCUIT POWER. FOR MOTORS WITH VFDS, POWER SHALL BE ACTUAL POWER AS READ THROUGH THE VFD NETWORK INTERFACE. FOR FIXED SPEED MOTORS, POWER SHALL BE ASSUMED TO BE FIXED AT BHP (FROM EQUIPMENT SCHEDULE) * 0.746 / 0.93 (APPROXIMATE MOTOR EFFICIENCY).

D. TOTAL COOLING PLANT LOAD. CALCULATE PLANT LOAD USING FLOWRATE THROUGH THE PRIMARY CIRCUIT, $FLOW_P$; CHILLED WATER RETURN TEMPERATURE ENTERING THE HEAT PUMP ARRAY, CHWRT; AND PRIMARY LOOP CHILLED WATER SUPPLY TEMPERATURE LEAVING THE PLANT, CHWST. WHERE A BTU METER IS PROVIDED, THE USE OF ITS INPUT IS ACCEPTABLE IN LIEU OF THE BELOW EQUATION PROVIDED THAT ITS LOAD INPUT IS REPORTED AND TRENDED IN TONS.
   1. $Q_{PLANT}$ [TONS] = $FLOW_P$ * ( CHWRT - CHWST ) / 24

E. TOTAL HEATING PLANT LOAD. CALCULATE PLANT LOAD USING FLOWRATE THROUGH THE PRIMARY CIRCUIT, $FLOW_P$; HEATING HOT WATER RETURN TEMPERATURE ENTERING THE HEAT PUMP ARRAY, HHWRT; AND PRIMARY LOOP HEATING WATER SUPPLY TEMPERATURE LEAVING THE PLANT, HHWST. WHERE A BTU METER IS PROVIDED, THE USE OF ITS INPUT IS ACCEPTABLE IN LIEU OF THE BELOW EQUATION PROVIDED THAT ITS LOAD INPUT IS REPORTED AND TRENDED IN TONS.
   1. $Q_{PLANT}$ [TONS] = $FLOW_P$ * ( HHWST - HHWRT ) / 24

F. HEAT PUMP INDIVIDUAL MODULE LOAD. CALCULATE LOAD FOR EACH OPERATING HEAT PUMP MODULE USING FLOWRATE THROUGH THE EQUIPMENT, $FLOW_D$; CHILLED WATER RETURN TEMPERATURE ENTERING THE EQUIPMENT, $CHWRT_D$; AND CHILLED WATER SUPPLY TEMPERATURE LEAVING THE EQUIPMENT, $CHWST_D$ INPUTS TO THE BELOW EQUATION SHALL BE DETERMINED PER THE FOLLOWING RULES.
   1. $Q_D$ [TONS] = $FLOW_D$ ( $CHWRT_D$ - $CHWST_D$ ) / 24
      a. WHERE FLOW THROUGH EACH MODULE IS INDIVIDUALLY MEASURED USING A FLOW METER, $FLOW_D$ SHALL BE THE FLOW MEASURED BY THE HEAT PUMP'S ASSOCIATED FLOW METER.
      b. $CHWRT_D$ SHALL BE THE RETURN TEMPERATURE ENTERING THE EQUIPMENT AS READ BY A HARDWIRED SENSOR IF AVAILABLE. IF A HARDWIRED SENSOR IS UNAVAILABLE FOR A MODULE, TEMPERATURE SHALL BE READ FROM A SENSOR INTERNAL TO THE INDIVIDUAL MODULE HEAT PUMP CONTROLLER THROUGH ITS NETWORK INTERFACE.
      c. $CHWST_D$ SHALL BE A HARDWIRED TEMPERATURE SENSOR AT THE OUTLET OF THE EQUIPMENT IF AVAILABLE. IF A HARDWIRED SENSOR IS UNAVAILABLE FOR A MODULE, TEMPERATURE SHALL BE READ FROM A SENSOR INTERNAL TO THE INDIVIDUAL MODULE HEAT PUMP CONTROLLER THROUGH ITS NETWORK INTERFACE. ONLY IF NEITHER OF THE ABOVE IS AVAILABLE SHALL A COMMON SUPPLY TEMPERATURE SENSOR (I.E., ONE MEASURING THE OUTPUT FROM MULTIPLE MODULES), BE USED.

G. HEAT PUMP INDIVIDUAL MODULE LOAD. CALCULATE LOAD FOR EACH OPERATING HEAT PUMP MODULE USING FLOWRATE THROUGH THE EQUIPMENT, $FLOW_D$; HEATING HOT WATER RETURN TEMPERATURE ENTERING THE EQUIPMENT, $HHWRT_D$; AND HEATING WATER SUPPLY TEMPERATURE LEAVING THE EQUIPMENT, $HHWST_D$ INPUTS TO THE BELOW EQUATION SHALL BE DETERMINED PER THE FOLLOWING RULES.
   1. $Q_D$ [MBH] = $FLOW_D$ ( $HHWST_D$ - $HHWRT_D$ ) / 24
      a. WHERE FLOW THROUGH EACH MODULE IS INDIVIDUALLY MEASURED USING A FLOW METER, $FLOW_D$ SHALL BE THE FLOW MEASURED BY THE HEAT PUMP'S ASSOCIATED FLOW METER.
      b. $HHWRT_D$ SHALL BE THE RETURN TEMPERATURE ENTERING THE EQUIPMENT AS READ BY A HARDWIRED SENSOR IF AVAILABLE. IF A HARDWIRED SENSOR IS UNAVAILABLE FOR A MODULE, TEMPERATURE SHALL BE READ FROM A SENSOR INTERNAL TO THE INDIVIDUAL MODULE HEAT PUMP CONTROLLER THROUGH ITS NETWORK INTERFACE.
      c. $HHWST_D$ SHALL BE A HARDWIRED TEMPERATURE SENSOR AT THE OUTLET OF THE EQUIPMENT IF AVAILABLE. IF A HARDWIRED SENSOR IS UNAVAILABLE FOR A MODULE, TEMPERATURE SHALL BE READ FROM A SENSOR INTERNAL TO THE INDIVIDUAL MODULE HEAT PUMP CONTROLLER THROUGH ITS NETWORK INTERFACE. ONLY IF NEITHER OF THE ABOVE IS AVAILABLE SHALL A COMMON SUPPLY TEMPERATURE SENSOR (I.E., ONE MEASURING THE OUTPUT FROM MULTIPLE MODULES), BE USED.

H. BOILER LOAD. CALCULATE LOAD FOR EACH OPERATING BOILER USING FLOWRATE THROUGH THE EQUIPMENT, $FLOW_D$; HEATING HOT WATER RETURN TEMPERATURE ENTERING THE EQUIPMENT, $HHWRT_D$; AND HEATING WATER SUPPLY TEMPERATURE LEAVING THE EQUIPMENT, $HHWST_D$ INPUTS TO THE BELOW EQUATION SHALL BE DETERMINED PER THE FOLLOWING RULES.
   1. $Q_D$ [MBH] = $FLOW_D$ ( $HHWST_D$ - $HHWRT_D$ ) / 24
      a. WHERE FLOW THROUGH EACH MODULE IS INDIVIDUALLY MEASURED USING A FLOW METER, $FLOW_D$ SHALL BE THE FLOW MEASURED BY THE HEAT PUMP'S ASSOCIATED FLOW METER.
      b. $HHWRT_D$ SHALL BE THE RETURN TEMPERATURE ENTERING THE EQUIPMENT AS READ BY A HARDWIRED SENSOR IF AVAILABLE. IF A HARDWIRED SENSOR IS UNAVAILABLE FOR A MODULE, TEMPERATURE SHALL BE READ FROM A SENSOR INTERNAL TO THE INDIVIDUAL MODULE HEAT PUMP CONTROLLER THROUGH ITS NETWORK INTERFACE.
      c. $HHWST_D$ SHALL BE A HARDWIRED TEMPERATURE SENSOR AT THE OUTLET OF THE EQUIPMENT IF AVAILABLE. IF A HARDWIRED SENSOR IS UNAVAILABLE FOR A MODULE, TEMPERATURE SHALL BE READ FROM A SENSOR INTERNAL TO THE INDIVIDUAL MODULE HEAT PUMP CONTROLLER THROUGH ITS NETWORK INTERFACE. ONLY IF NEITHER OF THE ABOVE IS AVAILABLE SHALL A COMMON SUPPLY TEMPERATURE SENSOR (I.E., ONE MEASURING THE OUTPUT FROM BOTH BOILERS), BE USED.

I. CALCULATE PLANT EFFICIENCY AS TOTAL PLANT POWER DIVIDED BY PLANT LOAD.

J. SUMMARY DATA.
   a. FOR EACH HEAT PUMP MODULE, AND FOR THE TOTAL PLANT, STATISTICS SHALL BE CALCULATED FOR RUNTIME, KWH, AVERAGE ACTUAL EFFICIENCY (KW/TON), PEAK DEMAND (TONS), AVERAGE DEMAND (TONS) AND AVERAGE LOAD (TON-HOURS), ALL ON AN INSTANTANEOUS, YEAR-TO-DATE, AND PREVIOUS-YEAR BASIS.
   b. FOR EACH BOILER, AND FOR THE TOTAL PLANT, STATISTICS SHALL BE CALCULATED FOR RUNTIME, KWH, NATURAL GAS CFH, AVERAGE ACTUAL EFFICIENCY (AFUE), PEAK DEMAND (MBH), THERMAL EFFICIENCY**, AVERAGE DEMAND (MBH) AND AVERAGE LOAD (MBH-HOURS), ALL ON AN INSTANTANEOUS, YEAR-TO-DATE, AND PREVIOUS-YEAR BASIS.
      - TO CALCULATE THERMAL EFFICIENCY = OBTAIN PRODUCT DATA FROM PURCHASED MANUFACTURER THAT SHOW THERMAL EFFICIENCY AS A FUNCTION OF RETURN WATER TEMPERATURE. THERMAL EFFICIENCY SHALL BE DETERMINED BY THE THIS DOCUMENT BASED ON INSTANTANEOUS RETURN WATER TEMPERATURE.

**AUTOMATED FAULT DETECTION AND DIAGNOSTICS**

A. PROVIDE AUTOMATIC FAULT DETECTION AND DIAGNOSTICS (AFDD) FUNCTIONALITY IN ACCORDANCE WITH ASHRAE GUIDELINE 36 - 2021 SECTION 5.20.18.

B. AFDD CONDITIONS ARE EVALUATED CONTINUOUSLY FOR THE PLANT. THE POTENTIAL FAULT CONDITIONS ARE DEPENDENT ON THE PLANT OPERATING STATE (OS) AS WELL AS VARIOUS MEASURED AND CALCULATED VARIABLES. REFER TO GL 36 SECTION 5.20.18 FOR FULL DESCRIPTION VARIABLES.

C. THE FOLLOWING POTENTIAL FAULT CONDITIONS SHALL BE EVALUATED ACCORDING TO GL 36 TABLE 5.20.18.6.
   1. DIFFERENTIAL PRESSURE IS TOO HIGH WITH THE CHILLED WATER PUMPS OFF.
   2. PRIMARY CHILLED WATER FLOW IS TOO HIGH WITH THE CHILLED WATER PUMPS OFF.
   3. SECONDARY CHILLED WATER FLOW IS TOO HIGH WITH THE CHILLED WATER PUMPS OFF.
   4. CHILLED WATER LOOP DIFFERENTIAL PRESSURE IS TOO LOW WITH CHILLED WATER PUMP(S) AT FULL SPEED.
   5. CHILLED WATER SUPPLY TEMPERATURE IS TOO HIGH.
   6. TOO MANY CHANGES IN OPERATING STATE.
   7. DIFFERENTIAL PRESSURE IS TOO HIGH WITH THE HEATING HOT WATER PUMPS OFF.
   8. PRIMARY HEATING HOT WATER FLOW IS TOO HIGH WITH THE HEATING HOT WATER PUMPS OFF.
   9. SECONDARY HEATING HOT WATER FLOW IS TOO HIGH WITH THE HEATING HOT WATER PUMPS OFF.
   10. HEATING HOT WATER LOOP DIFFERENTIAL PRESSURE IS TOO LOW WITH HEATING HOT WATER PUMP(S) AT FULL SPEED.
   11. HEATING HOT WATER SUPPLY TEMPERATURE IS TOO HIGH.
   12. TOO MANY CHANGES IN OPERATING STATE.
   13. TOO MANY HEAT PUMP STARTS.
   14. TOO MANY CHILLED WATER PLANT STAGE CHANGES.

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600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520


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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - HYDRONIC PLANT SEQUENCE OF OPERATIONS - PAGE 2 OF 2 |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 296 OF 378

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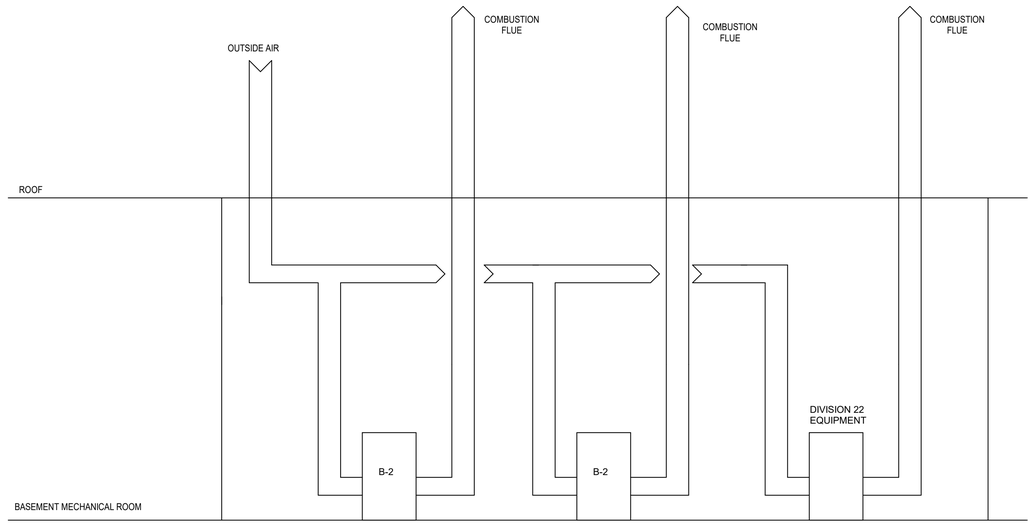


1 BOILER COMBUSTION - CONTROL DIAGRAM
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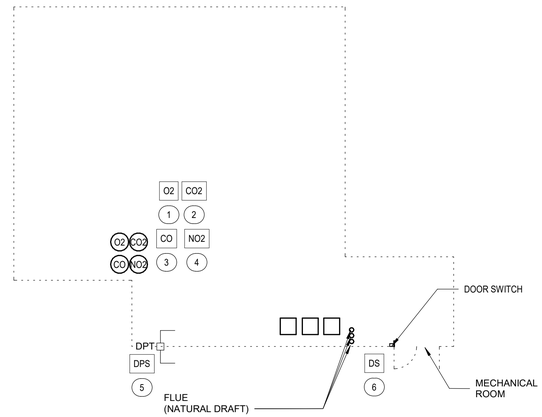


**SEQUENCE OF OPERATIONS – BOILER ROOM GAS DETECTION & SAFETY CONTROL**

1. NORMAL OPERATION
   - A. ALL SENSORS (O2, CO2, CO, NO2) SHALL CONTINUOUSLY MONITOR AIR QUALITY.
   - B. DOOR SWITCH (DS) MONITORS ACCESS STATUS (OPEN/CLOSED).
2. OXYGEN (O2) SENSOR CONTROL RESPONSE
   - A. IF O2 LEVEL DROPS BELOW 19.5%, BUT ABOVE 18%:
     - a. GENERATE LOW OXYGEN ALARM AT BAS.
   - B. INITIATE SAFETY ALERT TO OPERATORS.
   - C. ELSEIF O2 LEVEL DROPS BELOW 18%:
     - a. GENERATE CRITICAL LOW OXYGEN ALARM AT BAS.
     - b. INITIATE PRIORITY SAFETY ALERT TO OWNER.
3. CARBON DIOXIDE (CO2) CONTROL
   - A. IF CO2 EXCEEDS 1000 PPM, BUT IS BELOW 2000PPM:
     - a. GENERATE A WARNING ALARM IN BAS.
   - B. ELSEIF CO2 CONTINUES TO CONCENTRATION IS 2000 PPM OR HIGH:
     - a. GENERATE CRITICAL HIGH CARBON DIOXIDE ALARM AT BAS.
     - b. INITIATE PRIORITY SAFETY ALERT TO OWNER.
4. CARBON MONOXIDE (CO) CONTROL
5. IF CO EXCEEDS 35 PPM:
   - A. ACTIVATE HIGH PRIORITY ALARM.
   - B. SHUT DOWN BOILERS (INTERLOCK) IF LEVEL REMAINS HIGH FOR 15 MINUTES.
   - C. NOTIFY OPERATORS AND BAS.
6. NITROGEN DIOXIDE (NO2) CONTROL
   - A. IF NO2 EXCEEDS 1 PPM:
     - a. ACTIVATE ALARM CONDITION.
     - b. SHUT DOWN COMBUSTION EQUIPMENT IF LEVEL PERSISTS FOR 15 MINUTES.
7. SYSTEM FAIL CONDITIONS
   - A. IF ANY SENSOR FAILS OR IF COMMUNICATION IS LOST:
     - a. DISABLE BOILER AND DIVISION 22 EQUIPMENT OPERATION.
     - b. GENERATE AN ALARM AT THE BAS.
8. ALARM RESET
   - A. ALL ALARMS REQUIRE MANUAL ACKNOWLEDGEMENT AT BAS.
   - B. CRITICAL SAFETY LOCKOUTS REQUIRE MANUAL RESET AFTER CONDITIONS RETURN TO NORMAL.

## BAS POINT FUNCTION SCHEDULE - COMBUSTION EQUIPMENT

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | ALARM LIMITS: LOW LIMIT | ALARM LIMITS: HIGH LIMIT | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|
| 2. SENSORS | | | | | | | | | | |
| 1 | OXYGEN (DIOXYGEN) SENSOR | O2 | AI | | | | | 18% | | |
| 2 | CARBON DIOXIDE SENSOR | CO2 | AI | | | | | | 1000 PPM | |
| 3 | CARBON MONOXIDE SENSOR | CO | AI | | | | | | 80 PPM (15 MIN) | |
| 4 | NITROGEN DIOXIDE | NO2 | AI | | | | | | 1 PPM (15 MIN) | |
| 5 | ROOM DIFFERENTIAL STATIC PRESSURE SENSOR | DPS | AI | | | | | | | |
| 3. MISC | | | | | | | | | | |
| 6 | DOOR SWITCH | DS | AI | | | | | | | |

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Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL CONTROLS - COMBUSTION EQUIPMENT AND ALARMS

WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

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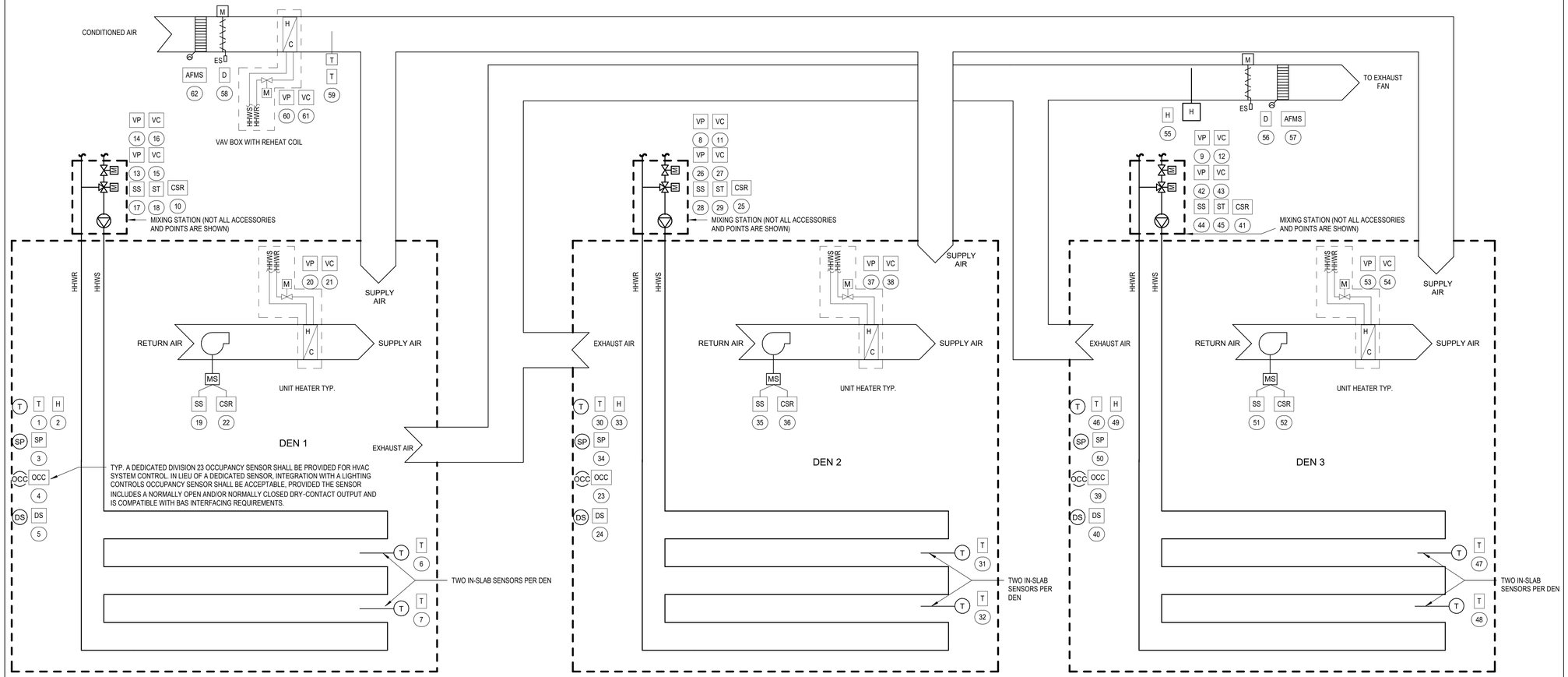


## BAS POINT FUNCTION SCHEDULE - ANIMAL SPACES (DEN 1...

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | NOTES |
|---|---|---|---|---|---|---|---|---|
| DEN 1 | | | | | | | | |
| 1 | TEMPERATURE | T | AI | | | | ■ | |
| 2 | RELATIVE HUMIDITY (MONITOR ONLY) | H | AI | | | | ■ | |
| 3 | STATIC PRESSURE SENSOR | SP | AI | | | | ■ | |
| 4 | OCCUPANCY SENSOR | OCC | DI | | | | ■ | |
| 5 | DOOR SWITCH | DS | DI | | | | ■ | |
| 6 | SLAB TEMPERATURE | T | AI | | | | ■ | |
| 7 | SLAB TEMPERATURE | T | AI | | | | ■ | |
| 10 | MIXING STATION PUMP SPEED | CSR | AI | | | | ■ | |
| 13 | MIXING STATION 3 WAY VALVE POSITION | VP | DI | | | ■ | ■ | |
| 14 | MIXING STATION VALVE POSITION | VP | AI | | | ■ | ■ | |
| 15 | MIXING STATION 3 WAY VALVE CONTROL | VC | DO | | | | ■ | |
| 16 | MIXING STATION VALVE CONTROL | VC | AO | | | | ■ | |
| 17 | MIXING STATION PUMP START STOP | SS | DO | | | | ■ | |
| 18 | MIXING STATION PUMP STATUS | ST | DI | | | | ■ | |
| 19 | UNIT HEATER FAN START STOP | SS | DO | | | | ■ | |
| 20 | UNIT HEATER VALVE POSITION | VP | AI | | | ■ | ■ | |
| 21 | UNIT HEATER VALVE CONTROL | VC | AO | | | | ■ | |
| 22 | UNIT HEATER FAN SPEED CNTRL | CSR | AI | | | | ■ | |
| DEN 2 | | | | | | | | |
| 8 | MIXING STATION VALVE POSITION | VP | AI | | | ■ | ■ | |
| 11 | MIXING STATION VALVE CONTROL | VC | AO | | | | ■ | |
| 23 | OCCUPANCY SENSOR | OCC | DI | | | | ■ | |
| 24 | DOOR SWITCH | DS | DI | | | | ■ | |
| 25 | MIXING STATION PUMP SPEED | CSR | AI | | | | ■ | |
| 26 | MIXING STATION 3 WAY VALVE POSITION | VP | DI | | | ■ | ■ | |
| 27 | MIXING STATION 3 WAY VALVE CONTROL | VC | DO | | | | ■ | |
| 28 | MIXING STATION PUMP START STOP | SS | DO | | | | ■ | |
| 29 | MIXING STATION PUMP STATUS | ST | DI | | | | ■ | |
| 30 | TEMPERATURE | T | AI | | | | ■ | |
| 31 | SLAB TEMPERATURE | T | AI | | | | ■ | |
| 32 | SLAB TEMPERATURE | T | AI | | | | ■ | |
| 33 | RELATIVE HUMIDITY (MONITOR ONLY) | H | AI | | | | ■ | |
| 34 | STATIC PRESSURE SENSOR | SP | AI | | | | ■ | |

## BAS POINT FUNCTION SCHEDULE - ANIMAL SPACES (DEN 1...

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | NOTES |
|---|---|---|---|---|---|---|---|---|
| 35 | UNIT HEATER FAN START STOP | SS | AO | | | | ■ | |
| 36 | UNIT HEATER FAN SPEED CNTRL | CSR | AI | | | | ■ | |
| 37 | UNIT HEATER VALVE POSITION | VP | AI | | | ■ | ■ | |
| 38 | UNIT HEATER VALVE CONTROL | VC | AO | | | | ■ | |
| DEN 3 | | | | | | | | |
| 9 | MIXING STATION VALVE POSITION | VP | AI | | | ■ | ■ | |
| 12 | MIXING STATION VALVE CONTROL | VC | AO | | | | ■ | |
| 39 | OCCUPANCY SENSOR | OCC | DI | | | | ■ | |
| 40 | DOOR SWITCH | DS | DI | | | | ■ | |
| 41 | MIXING STATION PUMP SPEED | CSR | AI | | | | ■ | |
| 42 | MIXING STATION 3 WAY VALVE POSITION | VP | DI | | | ■ | ■ | |
| 43 | MIXING STATION 3 WAY VALVE CONTROL | VC | DO | | | | ■ | |
| 44 | MIXING STATION PUMP START STOP | SS | DO | | | | ■ | |
| 45 | MIXING STATION PUMP STATUS | ST | DI | | | | ■ | |
| 46 | TEMPERATURE | T | AI | | | | ■ | |
| 47 | SLAB TEMPERATURE | T | AI | | | | ■ | |
| 48 | SLAB TEMPERATURE | T | AI | | | | ■ | |
| 49 | RELATIVE HUMIDITY (MONITOR ONLY) | H | AI | | | | ■ | |
| 50 | STATIC PRESSURE SENSOR | SP | AI | | | | ■ | |
| 51 | UNIT HEATER FAN START STOP | SS | DO | | | | ■ | |
| 52 | UNIT HEATER FAN SPEED CNTRL | CSR | AI | | | | ■ | |
| 53 | UNIT HEATER VALVE POSITION | VP | AI | | | ■ | ■ | |
| 54 | UNIT HEATER VALVE CONTROL | VC | AO | | | | ■ | |
| EXHAUST TERMINAL UNIT | | | | | | | | |
| 55 | HUMIDITY OF EXHAUST AIR | H | AI | | | | ■ | |
| 56 | DAMPER CONTROL | D | AO | ■ | | | | |
| 57 | EXHAUST AIR FLOW STATION | AFMS | AI | | | | | |
| SUPPLY TERMINAL UNIT | | | | | | | | |
| 58 | DAMPER CONTROL | D | AO | ■ | | | | |
| 59 | ZONE ENTERING AIR TEMPERATURE | T | AI | | | | | |
| 60 | RHEAT COIL VALVE POSITION | VP | AI | | | ■ | ■ | |
| 61 | RHEAT COIL VALVE CONTROL | VC | AO | | | | ■ | |
| 62 | SUPPLY AIR FLOW | AFMS | AI | | | | | |

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REVISION DESCRIPTION

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Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: **ARABIAN LEOPARD FACILITY (ALF)**
**3001 Connecticut Avenue NW**
**Washington, DC 20008**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**

OPDC PROJECT NUMBER: **2533105**

A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **MECHANICAL CONTROLS - ZONE CONTROL DIAGRAM - DEN 1, DEN 2 AND DEN 3**

WORKING STAFF

| OP | OP | IP |
|---|---|---|
| DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 298 OF 378

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DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

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**CONTROLS SEQUENCE OF OPERATIONS:**

**GENERAL**

A. SEQUENCES ARE DEFINED FOR FOLLOWING ANIMAL HOLDING SPACES -
   1. DEN-1
   2. DEN-2
   3. DEN-3
B. FOLLOWING SYSTEMS INTERACT AND PROVIDE TEMPERATURE AND VENTILATION CONTROL TO THE SPACES THROUGH THE BUILDING AUTOMATION SYSTEM
   1. VARIABLE AIR VOLUME (VAV) TERMINAL UNIT FOR SUPPLY AIRFLOW
   2. REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT
   3. VARIABLE AIR VOLUME (VAV) TERMINAL UNIT FOR EXHAUST AIRFLOW
   4. RADIANT SLAB SYSTEM -
      A. MIXING STATION FOR DEN 1
      B. MIXING STATION FOR DEN 2
      C. MIXING STATION FOR DEN 3
   5. FULLY REDUNDANT CONVECTIVE WALL MOUNTED UNIT HEATERS
   6. FOLLOWING INDEPENDENT SENSORS IMPLEMENT THE CONTROL LOGIC AND/OR ALARMS -
      A. TEMPERATURE
      B. HUMIDITY
      C. SLAB TEMPERATURE
      D. STATIC PRESSURE

**FORMULAS -**

A. HUMIDITY SENSOR MOUNTED WITHIN THE EXHAUST DUCTWORK SHALL BE HARD WIRED AND USED TO DETERMINE THE DEW POINT TEMPERATURE.
B. ADDITIONALLY, DEW POINT WITHIN EACH INDIVIDUAL SPACE MUST BE CALCULATED USING THE HUMIDITY SENSOR WITHIN THE SPACE.
   1. IF THE SENSOR DOES NOT HAVE THE CAPACITY TO PROVIDE DEW POINT OF THE SPACE, SPACE DEWPOINT MUST BE CALCULATED FROM THE EQUATION BELOW.
   2. ELSEIF THE SENSOR HAS THE CAPACITY TO PROVIDE DEW POINT OF THE SPACE, SPACE DEWPOINT MUST STILL BE CALCULATED FROM THE EQUATION BELOW, AND RECORDED AS 'THEORITICAL DEW POINT' WITHIN THE LOGS.
C. DEW-POINT TEMPERATURE (MAGNUS TETENS) EQUATION -
   1. WHERE T IS DRY BULB TEMPERATURE EXPRESSED **IN CELSIUS.**
   2. RH IS THE RELATIVE HUMIDITY OF THE AIR EXPRESSED AS A PERCENTAGE.
      A. HARD WIRED POINT FROM EXHAUST AIRSTREAM SHALL BE USED.
   3. 'ln(x)' IS A NATURAL LOGARITHM OF VARIABLE 'x'.
   4. Td IS THE INSTANTANEOUS DEW POINT TEMPERATURE IN FAHRENHEIT.
   5. Td = 32 + 1.8 * [ (243.04 * (ln(RH/100) + (17.625 * T) / (243.04 + T))) / (17.625 -ln(RH/100) -((17.625 * T) / (243.04 + T)))]

**SETPOINTS**

A. FOLLOWING SETPOINTS SHALL APPLY FOR ALL THE SPACES CALLED OUT ON THIS SHEET.
B. ALL SETPOINTS ARE BAS ADJUSTABLE.
C. TEMPERATURE CONTROL - COOLING
   1. DRY BULB SETPOINT = 72.0°F
      a. DEADBAND = 2°F
D. TEMPERATURE CONTROL - HEATING
   1. DRY BULB SETPOINT = 72.0°F
      a. DEADBAND = 2°F
E. RADIANT FLOOR TEMPERATURE SETPOINT = 78.0°F
F. HUMIDITY CONTROL
   1. RELATIVE HUMIDITY LOW LIMIT = 30%
   2. RELATIVE HUMIDITY HIGH LIMIT = 50%
G. ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT = POSITIVE 0.05 INCHES OF WATER (GAUGE PRESSURE)

**VENTILATION CONTROL**

A. BAS SHALL MODULATE THE TERMINAL UNITS (SUPPLY AND EXHAUST) AND THE ASSOCIATED REHEAT COIL CLOSE, IF ANY OF THE DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES (DOOR IS LEFT OPEN).
B. PARAGRAPH C THROUGH F OF THIS SECTION APPLIES WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. OCCUPIED MODE
   1. OWNER SHALL HAVE THE ABILITY TO OVERRIDE THE OCCUPANCY OF THE ZONE THROUGH THE BAS.
   2. PASSIVELY, AN OCCUPIED SIGNAL FROM ANY INDIVIDUAL WIRED SPACE OCCUPANCY SENSOR LOCATED IN VENTILATION ZONE (OR OCCUPANCY STATUS OVERRIDE BY THE BAS) SHALL ENABLE OCCUPIED MODE.
      a. NOTE: ALL OCCUPANCY SENSOR SIGNALS ARE WIRED POINTS FROM LOCAL OCCUPANCY SENSOR.
   3. THE ASSOCIATED TERMINAL UNIT PRIMARY AIR DAMPER SHALL MODULATE TO THE MIN OCCUPIED OA CFM PER TERMINAL UNIT SCHEDULES.
   4. COOLING AND HEATING:
      a. UPON A CALL FOR COOLING, INDICATED BY A RISE IN SPACE TEMPERATURE ABOVE THE SPACE TEMPERATURE SETPOINT BY ANY SPACE THERMOSTAT IN THE VENTILATION ZONE, THE BAS SHALL MODULATE THE PRIMARY AIR DAMPER OPEN TOWARD ITS PRIMARY AIR MAXIMUM CFM SETPOINT.
      b. ON BAS VERIFICATION THAT ALL TEMPERATURE CONTROL ZONES WITHIN THE VENTILATION ZONE ARE IN HEATING, THE BAS SHALL MODULATE THE PRIMARY AIR DAMPER TO ITS MIN OCCUPIED OA CFM.
D. UNOCCUPIED MODE
   1. OWNER SHALL HAVE THE ABILITY TO OVERRIDE THE OCCUPANCY OF THE ZONE THROUGH THE BAS.
   2. PASSIVELY ON BAS VERIFICATION OF 30 MINUTES CONTINUOUS (ADJ.) WITHOUT OCCUPIED SIGNAL FROM ALL OCCUPANCY SENSORS WITHIN VENTILATION CONTROL ZONE.
E. EXHAUST SYSTEM
   a. IF THE ROOM IS IN OCCUPIED MODE,
      - THE EXHAUST TERMINAL UNIT SHALL MODULATE OPEN AND EXHAUST THE DESIGN AIRFLOW (SEE MECHANICAL SCHEDULES) FROM THE SPACE.
      - THE EXHAUST TERMINAL UNIT SHALL MODULATE OPEN FURTHER TO MEET THE ZONE STATIC PRESSURE SETPOINT, WHEN SUPPLY TERMINAL UNIT IS RESPONDING TO CALL FOR COOLING.
   b. ELSEIF THE ROOM IS IN UNOCCUPIED MODE,
      - THE EXHAUST TERMINAL UNIT SHALL MODULATE PARTIALLY CLOSE TO EXHAUST THE MINIMUM UNOCCUPIED EXHAUST AIRFLOW (SEE MECHANICAL SCHEDULES) FROM THE SPACE.
F. SUPPLY SYSTEM
   a. SEE TEMPERATURE CONTROL SECTION FOR LOGIC ON REHEAT COIL MODULATION.
   b. IF THE ROOM IS IN OCCUPIED MODE,
      - STAGE 1 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN AND PROVIDE THE MINIMUM OCCUPIED AIRFLOW (SEE MECHANICAL SCHEDULES) TO THE SPACES.
      - STAGE 2 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN FURTHER, AS REQUIRED, TO MAINTAIN THE ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT.
      - STAGE 3 - SEE TEMPERATURE CONTROL SECTION.
   c. ELSEIF THE ROOM IS IN UNOCCUPIED MODE,
      - STAGE 1 - THE SUPPLY TERMINAL UNIT SHALL MODULATE TO PROVIDE THE MINIMUM UNOCCUPIED SUPPLY AIRFLOW (SEE MECHANICAL SCHEDULES) TO THE SPACE.
      - STAGE 2 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN FURTHER, AS REQUIRED, TO MAINTAIN THE ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT.
      - STAGE 3 - SEE TEMPERATURE CONTROL SECTION.

**TEMPERATURE CONTROL**

A. BAS SHALL MODULATE THE REHEAT COIL, UNIT HEATER AND SLAB HEATERS CLOSE, IF DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES (DOOR IS LEFT OPEN).
B. PARAGRAPH C THROUGH H OF THIS SECTION APPLIES WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. THE BAS SHALL MODULATE AND STAGE ZONE LEVEL HEATING SYSTEMS TO MAINTAIN THE TEMPERATURE WITHIN EACH DEN.
D. THE BAS CONTROLLER SHALL MONITOR THE ZONE TEMPERATURE, COOLING AND HEATING CONTROL VALVE(S) IN SEQUENCE TO MAINTAIN THE ZONE TEMPERATURE AT THE SETPOINTS.
E. RADIANT SLAB SYSTEM -
   1. EACH ANIMAL DEN SHALL BE SERVED BY AN INDEPENDENT HYDRONIC RADIANT SLAB HEATING SYSTEM, INCLUDING:
      a. DEDICATED SLAB (FLOOR) TEMPERATURE SENSOR
      b. DEDICATED MIXING STATION AND HEATING CONTROL VALVE
   2. EACH RADIANT SLAB SYSTEM SHALL OPERATE INDEPENDENT OF SPACE AIR TEMPERATURE, OCCUPANCY MODE, SUPPLY AND EXHAUST AIR TERMINAL OPERATION, AND REHEAT COIL OPERATION, EXCEPT AS LIMITED BY CENTRAL PLANT AVAILABILITY AND SAFETY INTERLOCKS.
   3. THE RADIANT SLAB HEATING SYSTEM SHALL BE CONTROLLED SOLELY BY MEASURED SLAB (FLOOR) TEMPERATURE RELATIVE TO THE RADIANT FLOOR TEMPERATURE SETPOINT.
   4. SPACE AIR TEMPERATURE SHALL NOT BE USED TO ENABLE, DISABLE, OR MODULATE THE RADIANT SLAB SYSTEM.
   5. RADIANT SLAB HEATING ENABLE
      a. WHEN THE MEASURED SLAB TEMPERATURE FALLS BELOW THE RADIANT FLOOR TEMPERATURE SETPOINT MINUS 2.0°F (ADJUSTABLE), THE BAS SHALL:
         - ENABLE THE RADIANT SLAB CIRCULATION PUMP WITHIN THE MIXING STATION
   6. RADIANT SLAB HEATING DISABLE
      a. WHEN THE MEASURED SLAB TEMPERATURE RISES ABOVE THE RADIANT FLOOR TEMPERATURE SETPOINT PLUS 2.0°F (ADJUSTABLE), THE BAS SHALL:
         - DISABLE THE RADIANT SLAB CIRCULATION PUMP
   7. RADIANT SLAB MIXING VALVE AND PUMP
      a. SEE MIXING STATION SEQUENCE OF OPERATIONS
   8. IF THE MEASURED SLAB TEMPERATURE EXCEEDS 85°F (ADJ.):
      a. THE BAS SHALL IMMEDIATELY DISABLE THE RADIANT SLAB CIRCULATION PUMP
      b. THE BAS SHALL DRIVE THE THREE-WAY MIXING VALVE TO FULL BYPASS
      c. THE BAS SHALL GENERATE A HIGH SLAB TEMPERATURE ALARM.
      d. HIGH-LIMIT PROTECTION SHALL BE ACTIVE REGARDLESS OF OCCUPANCY OR HEATING DEMAND.
   9. THE RADIANT SLAB HEATING SYSTEM SHALL ONLY OPERATE IN OCCUPIED MODE USING THE RADIANT FLOOR TEMPERATURE SETPOINT. OWNER SHALL HAVE THE ABILITY TO ALLOW THE SYSTEM TO OPERATE IN UNOCCUPIED MODE.
   10. SLAB SYSTEM AND PLANT RESET COORDINATION
      a. THE THREE-WAY MIXING VALVE POSITION SHALL BE USED TO GENERATE HEATING HOT WATER FOR THE LOCAL SLAB SYSTEM.
      b. THE POSITION OF THIS THREE WAY MIXING VALVE SHALL BE USED TO GENERATE PLANT-SIDE PUMP DIFFERENTIAL PRESSURE AND SUPPLY TEMPERATURE RESET REQUESTS.
      c. RADIANT SLAB OPERATION SHALL NOT BE DISABLED OR LIMITED DUE TO AIRSIDE SYSTEM RESET PRIORITIES.
F. REHEAT COIL - OCCUPIED MODE
   a. AFTER A CHANGE IS IMPLEMENTED, THE BAS SHALL WAIT FOR 3 MINUTES (ADJ.) FOR THE SYSTEM TO ADJUST.
   b. COOLING
      - ON AN INCREASE IN ZONE AIR TEMPERATURE ABOVE THE COOLING SETPOINT (CALL FOR COOLING), THE BAS SHALL MODULATE OPEN THE SUPPLY AIR DAMPER TO THE MAXIMUM POSITION.
      - THE BAS SHALL MODULATE OPEN THE REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   c. HEATING
      - ON A DECREASE IN ZONE AIR TEMPERATURE BELOW THE HEATING SETPOINT (CALL FOR HEATING), THE BAS SHALL MODULATE OPEN THE REHEAT COIL TO MAINTAIN THE SCHEDULED LEAVING AIR TEMPERATURE.
G. REHEAT COIL - UNOCCUPIED MODE
   a. AFTER A CHANGE IS IMPLEMENTED, THE BAS SHALL WAIT FOR 10 MINUTES (ADJ.) FOR THE SYSTEM TO ADJUST
   b. COOLING
      - ON AN INCREASE IN ZONE AIR TEMPERATURE ABOVE THE SETBACK COOLING SETPOINT (CALL FOR COOLING), THE BAS SHALL MODULATE OPEN THE SUPPLY AIR DAMPER BEYOND THE UNOCCUPIED MINIMUM AIRFLOW SETPOINT IN STAGES OF 5%.
      - THE BAS SHALL MODULATE OPEN THE REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   c. HEATING
      - ON A DECREASE IN ZONE AIR TEMPERATURE BELOW THE SETBACK HEATING SETPOINT (CALL FOR HEATING), THE BAS SHALL MODULATE OPEN THE REHEAT COIL TO MAINTAIN THE SCHEDULED LEAVING AIR TEMPERATURE.
H. WHEN ZONE AIR TEMPERATURE IS BETWEEN THE HEATING & COOLING SETPOINTS (DEADBAND W/O CALL FOR HEATING OR COOLING), THE REHEAT VALVES SHALL CLOSE.
I. CONVECTIVE UNIT HEATERS -
   a. THE EQUIPMENT PROVIDED WITHIN EACH DEN IS FULLY REDUNDANT.
   b. THE EQUIPMENT WILL BE STAGED ON BY THE BAS ONLY IF -
      - IF ONE OR MORE ZONES ARE OCCUPIED -
        1. SUPPLY TERMINAL UNIT DAMPER IS FULLY OPEN TO FLOW.
        2. REHEAT COIL VALVE IS FULLY OPEN TO FLOW.
        3. RADIANT SLAB SYSTEM VALVE IS FULLY OPEN TO FLOW.
   c. OWNER SHALL HAVE THE ABILITY TO ACTIVATE/DEACTIVATE THE HEATER THROUGH THE BAS.

**HUMIDITY CONTROL**

A. BAS SHALL MODULATE THE TERMINAL UNITS (SUPPLY AND EXHAUST) AND THE ASSOCIATED REHEAT COIL CLOSE, IF DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES. (DOOR IS LEFT OPEN).
B. PARAGRAPH C AND D OF THIS SECTION APPLY WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. IF THE ZONE DEW POINT EXCEEDS HIGH LIMIT
   A. THE BAS SHALL MODULATE OPEN THE REHEAT COIL CONTROL VALVE TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   B. UPON FURTHER CALL FOR DEHUMIDIFICATION, A DEW POINT RESET REQUEST SHALL BE SENT TO THE AIR HANDLING UNIT. REFER TO DEW POINT RESET LOGIC IN AHU SEQUENCE OF OPERATIONS.
D. ELSEIF ZONE DEW POINT FALLS BELOW THE LOW LIMIT
   A. THE BAS SHALL SEND A RESET REQUEST TO THE AIR HANDLING UNIT. REFER TO DEW POINT RESET LOGIC IN AHU SEQUENCE OF OPERATIONS.
   B. IF THE AIR HANDLING UNIT COOLING COIL VALVE IS ALREADY RESET AND/OR FULLY CLOSED (TO FLOW), THE BAS SHALL SEND A REQUEST TO THE CENTRAL HUMIDIFIER SYSTEM. REFER TO HUMIDIFIER CONTROL LOGIC.

**RESET REQUESTS - GENERAL**

A. THE BAS SHALL USE THE POSITION OF THE FOLLOWING TO SEND SYSTEM DIFFERENTIAL PRESSURE AND TEMPERATURE RESET REQUESTS -
   a. AIRSIDE
      - SUPPLY TERMINAL UNIT DAMPER POSITION
      - EXHAUST TERMINAL UNIT DAMPER POSITION
   b. WATERSIDE
      - REHEAT COIL CONTROL VALVE POSITION
      - RADIANT SLAB CONTROL VALVE POSITION *
      - UNIT HEATER VALVE POSITION

**PUMP DIFFERENTIAL PRESSURE RESET REQUESTS**

A. HEATING COIL HEATING HOT WATER RESET REQUESTS. SEND THE PUMP SYSTEM THAT SERVES THE REHEAT COIL REQUESTS AS FOLLOWS:
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 85%.
   2. ELSEIF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.
B. RADIANT SLAB SYSTEM HEATING HOT WATER RESET REQUESTS. SEND THE PUMP SYSTEM THAT SERVES THE RADIANT SLAB MIXING VALVE, REQUESTS AS FOLLOWS:
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 85%.
   2. ELSEIF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

**HYDRONIC PLANT SUPPLY TEMPERATURE RESET REQUESTS**

A. HEATING HOT WATER PLANT REQUESTS. SEND THE PLANT THAT GENERATES HEATING HOT WATER, REQUESTS AS FOLLOWS:
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 10%.
   2. ELSEIF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

**PRESSURE CONTROL RESET REQUESTS**

A. PRIMARY AIR STATIC PRESSURE RESET REQUESTS. SEND THE FAN SYSTEM (AHU-1 OR EF-1_ THAT SERVES THE TERMINAL UNITS, PRESSURE REQUESTS AS FOLLOWS:
   1. IF THE MEASURED AIRFLOW IS LESS THAN 85% OF THE SETPOINT WHILE THE SETPOINT IS GREATER THAN ZERO AND THE DAMPER POSITION IS GREATER THAN 95% FOR 1 MINUTE, SEND 3 REQUESTS.
   2. ELSEIF THE MEASURED AIRFLOW IS LESS THAN 90% OF THE SETPOINT WHILE THE SETPOINT IS GREATER THAN ZERO AND THE DAMPER POSITION IS GREATER THAN 95% FOR 1 MINUTE, SEND 2 REQUESTS.
   3. ELSEIF THE DAMPER POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE DAMPER POSITION IS LESS THAN 85%.
   4. ELSEIF THE DAMPER POSITION IS LESS THAN 95%, SEND 0 REQUEST(S).

**CONTROL PRIORITY**

A. THE FOLLOWING PRIORITY STRUCTURE IN ORDER OF GREATEST TO LEAST PRIORITY APPLIES DURING SYSTEM OPERATION WHEN HVAC SYSTEMS OR INDIVIDUAL COMPONENTS CANNOT MEET ALL DEMANDS PLACED UPON THE SYSTEM DURING FAILURE CONDITIONS:
   1. ROOM TEMPERATURE AND HUMIDITY SETPOINT CONTROL.
   2. ROOM PRESSURIZATION CONTROL.
   3. ROOM MINIMUM AIR EXHAUST AIRFLOW SETPOINT.

**SAFETY AND ALARM GENERATIONS**

A. THE BAS SHALL GENERATE AN ALARM WHEN ANY OF THE DOOR OF THE HABITAT IS OPEN AT REGULAR INTERVALS, AS PROVEN BY DOOR SWITCH STATUS.
   a. NOTIFY THE BUILDING ENGINEER AT FOLLOWING INTERVALS AFTER DOOR SWITCH IS TRIGGERED ACTIVE (DOOR IS LEFT OPEN).
      - 10 MINUTES
      - 20 MINUTES
      - 60 MINUTES
      - 180 MINUTES
   b. NOTIFICATION SHOULD INFORM THE ENGINEER THAT THE VENTILATION AND TEMPERATURE CONTROL IS OFF.
B. THE BAS SHALL GENERATE ALARM IN THE EVENT OF ANY OF THE FOLLOWING:
   1. MEASURED SPACE TEMPERATURE DEVIATES FROM SETPOINT BY 2°F (ADJ.) FOR 10 MINUTES CONTINUOUS.
   2. MEASURED SPACE RELATIVE HUMIDITY EXCEEDS HIGH LIMIT FOR 5 MINUTES CONTINUOUS.
   3. MEASURED EXHAUST AIRFLOW DEVIATES FROM SETPOINT BY 5% (ADJ.) OR 50 CFM (ADJ.), WHICHEVER IS HIGHER, FOR 5 MINUTES CONTINUOUS.
   4. MEASURED SUPPLY AIRFLOW DEVIATES FROM SETPOINT BY 5% (ADJ.) OR 50 CFM (ADJ.), WHICHEVER IS HIGHER, FOR 5 MINUTES CONTINUOUS.

**SOFTWARE POINTS**

A. ALL POINTS REQUIRED TO MEET THE INTENT OF THE SOO SHALL BE MONITORED AND PROVIDED WITH GRAPHICS, INCLUDING BUT NOT LIMITED TO THE FOLLOWING:
   1. TOTAL ZONE EXHAUST AIRFLOW (IN UNITS OF CUBIC FEET PER MINUTE)
   2. TOTAL ZONE SUPPLY AIRFLOW (IN UNITS OF CUBIC FEET PER MINUTE)
   3. AIRFLOW OFFSET OR PRESSURIZATION.
   4. ZONE SETPOINTS (TEMPERATURE AND HUMIDITY)
   5. AIR PRESSURIZATION STATUS

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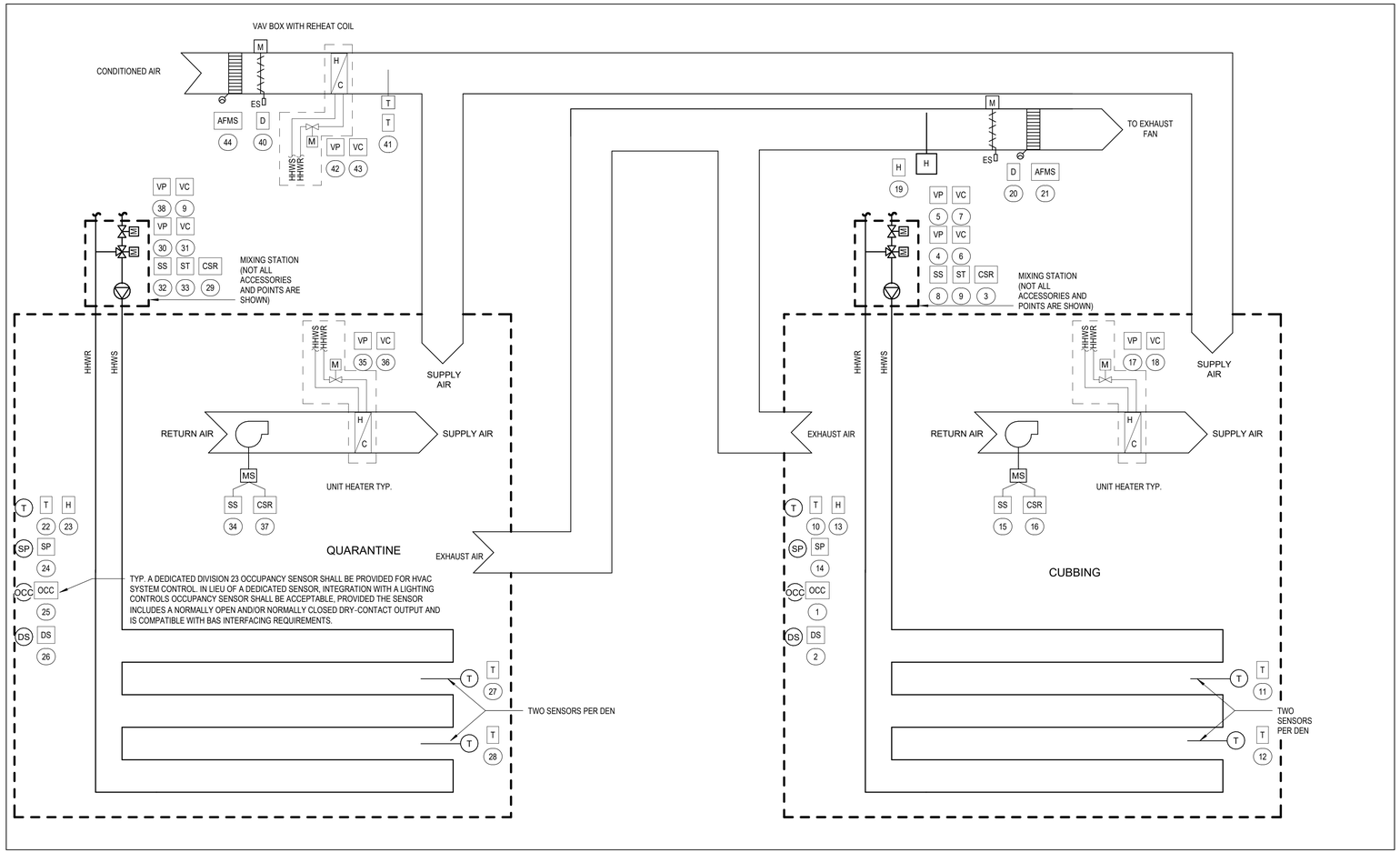
Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - ZONE CONTROL SEQUENCES - DEN 1, DEN 2 AND DEN 3 |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

SHEET NO. 299 OF 378

M-812CP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)




1 CONTROL DIAGRAM - QUARANTINE AND CUBBING
SCALE: 12" = 1'-0"

## BAS POINT FUNCTION SCHEDULE - QUARANTINE AND CUBBING

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | ALARM LIMITS: LOW LIMIT | ALARM LIMITS: HIGH LIMIT | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|
| CUBBING | | | | | | | | | | |
| 1 | OCCUPANCY SENSOR | OCC | AI | | | | ■ | | | |
| 2 | DOOR SWITCH | DS | AI | | | | ■ | | | |
| 3 | MIXING STATION PUMP SPEED | CSR | AI | | | | ■ | | | |
| 4 | MIXING STATION 3 WAY VALVE POSITION | VP | DI | | | ■ | ■ | | | |
| 5 | MIXING STATION VALVE POSITION | VP | AI | | | ■ | ■ | | | |
| 6 | MIXING STATION 3 WAY VALVE CONTROL | VC | DO | | | | ■ | | | |
| 7 | MIXING STATION VALVE CONTROL | VC | AO | | | | ■ | | | |
| 8 | MIXING STATION PUMP START STOP | SS | DO | | | | ■ | | | |
| 9 | MIXING STATION PUMP STATUS | ST | DI | | | | ■ | | | |
| 10 | TEMPERATURE | T | AI | | | | ■ | | | |
| 11 | SLAB TEMPERATURE | T | AI | | | | ■ | | | |
| 12 | SLAB TEMPERATURE | T | AI | | | | ■ | | | |
| 13 | RELATIVE HUMIDITY (MONITOR ONLY) | H | AI | | | | ■ | | | |
| 14 | STATIC PRESSURE SENSOR | SP | AI | | | | ■ | | | |
| 15 | UNIT HEATER FAN START STOP | SS | DO | | | | ■ | | | |
| 16 | UNIT HEATER FAN SPEED CNTRL | CSR | AI | | | | ■ | | | |
| 17 | UNIT HEATER VALVE POSITION | VP | AI | | | ■ | ■ | | | |
| 18 | UNIT HEATER VALVE CONTROL | VC | AO | | | | ■ | | | |
| EXHAUST TERMINAL UNIT | | | | | | | | | | |
| 19 | HUMIDITY OF EXHAUST AIR | H | AI | | | | ■ | | | |
| 20 | DAMPER CONTROL | D | AO | ■ | | | | | | |
| 21 | EXHAUST AIR FLOW STATION | AFMS | AI | | | | | | | |

## BAS POINT FUNCTION SCHEDULE - QUARANTINE AND CUBBING

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | ALARM LIMITS: LOW LIMIT | ALARM LIMITS: HIGH LIMIT | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|
| QUARANTINE | | | | | | | | | | |
| 9 | MIXING STATION VALVE CONTROL | VC | AO | | | | ■ | | | |
| 22 | TEMPERATURE | T | AI | | | | ■ | | | |
| 23 | RELATIVE HUMIDITY (MONITOR ONLY) | H | AI | | | | ■ | | | |
| 24 | STATIC PRESSURE SENSOR | SP | AI | | | | ■ | | | |
| 25 | OCCUPANCY SENSOR | OCC | AI | | | | ■ | | | |
| 26 | DOOR SWITCH | DS | AI | | | | ■ | | | |
| 27 | SLAB TEMPERATURE | T | AI | | | | ■ | | | |
| 28 | SLAB TEMPERATURE | T | AI | | | | ■ | | | |
| 29 | MIXING STATION PUMP SPEED | CSR | AI | | | | ■ | | | |
| 30 | MIXING STATION 3 WAY VALVE POSITION | VP | DI | | | ■ | ■ | | | |
| 31 | MIXING STATION 3 WAY VALVE CONTROL | VC | DO | | | | ■ | | | |
| 32 | MIXING STATION PUMP START STOP | SS | DO | | | | ■ | | | |
| 33 | MIXING STATION PUMP STATUS | ST | DI | | | | ■ | | | |
| 34 | UNIT HEATER FAN START STOP | SS | DO | | | | ■ | | | |
| 35 | UNIT HEATER VALVE POSITION | VP | AI | | | ■ | ■ | | | |
| 36 | UNIT HEATER VALVE CONTROL | VC | AO | | | | ■ | | | |
| 37 | UNIT HEATER FAN SPEED CNTRL | CSR | AI | | | | ■ | | | |
| 38 | MIXING STATION VALVE POSITION | VP | AI | | | ■ | ■ | | | |
| SUPPLY TERMINAL UNIT | | | | | | | | | | |
| 40 | DAMPER CONTROL | D | AO | ■ | | | | | | |
| 41 | ZONE ENTERING AIR TEMPERATURE | T | AI | | | | | | | |
| 42 | RHEAT COIL VALVE POSITION | VP | AI | | | ■ | ■ | | | |
| 43 | RHEAT COIL VALVE CONTROL | VC | AO | | | | ■ | | | |
| 44 | SUPPLY AIR FLOW | AFMS | AI | | | | | | | |

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REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL CONTROLS - ZONE CONTROL DIAGRAM - QUARANTINE AND CUBBING
WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 300 OF 378
M-813CP (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

**CONTROLS SEQUENCE OF OPERATIONS:**

**GENERAL**

A. SEQUENCES ARE DEFINED FOR FOLLOWING ANIMAL HOLDING SPACES -
   1. QUARANTINE DEN
   2. CUBBING DEN
B. FOLLOWING SYSTEMS INTERACT AND PROVIDE TEMPERATURE AND VENTILATION CONTROL TO THE SPACES THROUGH THE BUILDING AUTOMATION SYSTEM
   1. VARIABLE AIR VOLUME (VAV) TERMINAL UNIT FOR SUPPLY AIRFLOW
   2. REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT
   3. VARIABLE AIR VOLUME (VAV) TERMINAL UNIT FOR EXHAUST AIRFLOW
   4. RADIANT SLAB SYSTEM -
      A. MIXING STATION FOR QUARANTINE
      B. MIXING STATION FOR CUBBING
   5. FULLY REDUNDANT CONVECTIVE WALL MOUNTED UNIT HEATERS
   6. FOLLOWING INDEPENDENT SENSORS IMPLEMENT THE CONTROL LOGIC AND/OR ALARMS -
      A. TEMPERATURE
      B. HUMIDITY
      C. SLAB TEMPERATURE
      D. STATIC PRESSURE

**FORMULAS -**

A. HUMIDITY SENSOR MOUNTED WITHIN THE EXHAUST DUCTWORK SHALL BE HARD WIRED AND USED TO DETERMINE THE DEW POINT TEMPERATURE.
B. ADDITIONALLY, DEW POINT WITHIN EACH INDIVIDUAL SPACE MUST BE CALCULATED USING THE HUMIDITY SENSOR WITHIN THE SPACE.
   1. IF THE SENSOR DOES NOT HAVE THE CAPACITY TO PROVIDE DEW POINT OF THE SPACE, SPACE DEWPOINT MUST BE CALCULATED FROM THE EQUATION BELOW.
   2. ELSEIF THE SENSOR HAS THE CAPACITY TO PROVIDE DEW POINT OF THE SPACE, SPACE DEWPOINT MUST STILL BE CALCULATED FROM THE EQUATION BELOW, AND RECORDED AS 'THEORITICAL DEW POINT' WITHIN THE LOGS.
C. DEW-POINT TEMPERATURE (MAGNUS TETENS) EQUATION -
   1. WHERE T IS DRY BULB TEMPERATURE EXPRESSED **IN CELSIUS.**
   2. RH IS THE RELATIVE HUMIDITY OF THE AIR EXPRESSED AS A PERCENTAGE.
      A. HARD WIRED POINT FROM EXHAUST AIRSTREAM SHALL BE USED.
   3. 'ln(x)' IS A NATURAL LOGARITHM OF VARIABLE 'x'.
   4. Td IS THE INSTANTANEOUS DEW POINT TEMPERATURE IN FAHRENHEIT.
   5. Td = 32 + 1.8 * [ (243.04 * (ln(RH/100) + (17.625 * T) / (243.04 + T)) / (17.625 -ln(RH/100) -((17.625 * T) / (243.04 + T)))]

**SETPOINTS**

A. FOLLOWING SETPOINTS SHALL APPLY FOR ALL THE SPACES CALLED OUT ON THIS SHEET.
B. ALL SETPOINTS ARE BAS ADJUSTABLE.
C. TEMPERATURE CONTROL - COOLING
   1. DRY BULB SETPOINT = 72.0°F
      a. DEADBAND = 2°F
D. TEMPERATURE CONTROL - HEATING
   1. DRY BULB SETPOINT = 72.0°F
      a. DEADBAND = 2°F
E. RADIANT FLOOR TEMPERATURE SETPOINT = 78.0°F
F. HUMIDITY CONTROL
   1. RELATIVE HUMIDITY LOW LIMIT = 30%
   2. RELATIVE HUMIDITY HIGH LIMIT = 50%
G. ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT = POSITIVE 0.05 INCHES OF WATER (GAUGE PRESSURE)

**VENTILATION CONTROL**

A. BAS SHALL MODULATE THE TERMINAL UNITS (SUPPLY AND EXHAUST) AND THE ASSOCIATED REHEAT COIL CLOSE, IF DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES. (DOOR IS LEFT OPEN).
B. PARAGRAPH C THROUGH F OF THIS SECTION APPLY WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. OCCUPIED MODE
   1. OWNER SHALL HAVE THE ABILITY TO OVERRIDE THE OCCUPANCY OF THE ZONE THROUGH THE BAS.
   2. PASSIVELY, AN OCCUPIED SIGNAL FROM ANY INDIVIDUAL WIRED SPACE OCCUPANCY SENSOR LOCATED IN VENTILATION ZONE (OR OCCUPANCY STATUS OVERRIDE BY THE BAS) SHALL ENABLE OCCUPIED MODE.
      a. NOTE: ALL OCCUPANCY SENSOR SIGNALS ARE WIRED POINTS FROM LOCAL OCCUPANCY SENSOR.
   3. THE ASSOCIATED TERMINAL UNIT PRIMARY AIR DAMPER SHALL MODULATE TO THE MIN OCCUPIED OA POSITION PER TERMINAL UNIT SCHEDULES.
   4. COOLING AND HEATING:
      a. UPON A CALL FOR COOLING, INDICATED BY A RISE IN SPACE TEMPERATURE ABOVE THE SPACE TEMPERATURE SETPOINT BY ANY SPACE THERMOSTAT IN THE VENTILATION ZONE, THE BAS SHALL MODULATE THE PRIMARY AIR DAMPER OPEN TOWARD ITS PRIMARY AIR MAXIMUM CFM SETPOINT.
      b. ON BAS VERIFICATION THAT ALL TEMPERATURE CONTROL ZONES WITHIN THE VENTILATION ZONE ARE IN HEATING, THE BAS SHALL MODULATE THE PRIMARY AIR DAMPER TO ITS MIN OCCUPIED OA POSITION.
D. UNOCCUPIED MODE
   1. OWNER SHALL HAVE THE ABILITY TO OVERRIDE THE OCCUPANCY OF THE ZONE THROUGH THE BAS.
   2. PASSIVELY ON BAS VERIFICATION OF 30 MINUTES CONTINUOUS (ADJ.) WITHOUT OCCUPIED SIGNAL FROM ALL OCCUPANCY SENSORS WITHIN VENTILATION CONTROL ZONE.
E. EXHAUST SYSTEM
   a. IF THE ROOM IS IN OCCUPIED MODE,
      - THE EXHAUST TERMINAL UNIT SHALL MODULATE OPEN AND EXHAUST THE DESIGN AIRFLOW (SEE MECHANICAL SCHEDULES) FROM THE SPACE.
   b. ELSEIF THE ROOM IS IN UNOCCUPIED MODE,
      - THE EXHAUST TERMINAL UNIT SHALL MODULATE PARTIALLY CLOSE TO EXHAUST THE MINIMUM UNOCCUPIED EXHAUST AIRFLOW (SEE MECHANICAL SCHEDULES) FROM THE SPACE.
F. SUPPLY SYSTEM
   a. SEE TEMPERATURE CONTROL SECTION FOR LOGIC ON REHEAT COIL MODULATION.
   b. IF THE ROOM IS IN OCCUPIED MODE,
      - STAGE 1 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN AND PROVIDE THE MINIMUM OCCUPIED AIRFLOW (SEE MECHANICAL SCHEDULES) TO THE SPACES.
      - STAGE 2 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN FURTHER, AS REQUIRED, TO MAINTAIN THE ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT.
      - STAGE 3 - SEE TEMPERATURE CONTROL SECTION.
   c. ELSEIF THE ROOM IS IN UNOCCUPIED MODE,
      - STAGE 1 - THE SUPPLY TERMINAL UNIT SHALL MODULATE TO PROVIDE THE MINIMUM UNOCCUPIED SUPPLY AIRFLOW (SEE MECHANICAL SCHEDULES) TO THE SPACE.
      - STAGE 2 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN FURTHER, AS REQUIRED, TO MAINTAIN THE ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT.
      - STAGE 3 - SEE TEMPERATURE CONTROL SECTION.

**TEMPERATURE CONTROL**

A. BAS SHALL MODULATE THE REHEAT COIL, UNIT HEATER AND SLAB HEATERS CLOSE, IF DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES. (DOOR IS LEFT OPEN).
B. PARAGRAPH C THROUGH H OF THIS SECTION APPLY WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. THE BAS SHALL MODULATE AND STAGE ZONE LEVEL HEATING SYSTEMS TO MAINTAIN THE TEMPERATURE WITHIN EACH DEN.
D. THE BAS CONTROLLER SHALL MONITOR THE ZONE TEMPERATURE, COOLING AND HEATING CONTROL VALVE(S) IN SEQUENCE TO MAINTAIN THE ZONE TEMPERATURE AT THE SETPOINTS.
E. RADIANT SLAB SYSTEM -
   1. EACH ANIMAL DEN SHALL BE SERVED BY AN INDEPENDENT HYDRONIC RADIANT SLAB HEATING SYSTEM, INCLUDING:
      a. DEDICATED SLAB (FLOOR) TEMPERATURE SENSOR
      b. DEDICATED MIXING STATION AND HEATING CONTROL VALVE
   2. EACH RADIANT SLAB SYSTEM SHALL OPERATE INDEPENDENT OF SPACE AIR TEMPERATURE, OCCUPANCY MODE, SUPPLY AND EXHAUST AIR TERMINAL OPERATION, AND REHEAT COIL OPERATION, EXCEPT AS LIMITED BY CENTRAL PLANT AVAILABILITY AND SAFETY INTERLOCKS.
   3. THE RADIANT SLAB HEATING SYSTEM SHALL BE CONTROLLED SOLELY BY MEASURED SLAB (FLOOR) TEMPERATURE RELATIVE TO THE RADIANT FLOOR TEMPERATURE SETPOINT.
   4. SPACE AIR TEMPERATURE SHALL NOT BE USED TO ENABLE, DISABLE, OR MODULATE THE RADIANT SLAB SYSTEM.
   5. RADIANT SLAB HEATING ENABLE
      a. WHEN THE MEASURED SLAB TEMPERATURE FALLS BELOW THE RADIANT FLOOR TEMPERATURE SETPOINT MINUS 2.0°F (ADJUSTABLE), THE BAS SHALL:
         - ENABLE THE RADIANT SLAB CIRCULATION PUMP WITHIN THE MIXING STATION
   6. RADIANT SLAB HEATING DISABLE
      a. WHEN THE MEASURED SLAB TEMPERATURE RISES ABOVE THE RADIANT FLOOR TEMPERATURE SETPOINT PLUS 2.0°F (ADJUSTABLE), THE BAS SHALL:
         - DISABLE THE RADIANT SLAB CIRCULATION PUMP
   7. RADIANT SLAB MIXING VALVE AND PUMP
      a. SEE MIXING STATION SEQUENCE OF OPERATIONS
   8. IF THE MEASURED SLAB TEMPERATURE EXCEEDS 85°F (ADJ.):
      a. THE BAS SHALL IMMEDIATELY DISABLE THE RADIANT SLAB CIRCULATION PUMP
      b. THE BAS SHALL DRIVE THE THREE-WAY MIXING VALVE TO FULL BYPASS
      c. THE BAS SHALL GENERATE A HIGH SLAB TEMPERATURE ALARM
      d. HIGH-LIMIT PROTECTION SHALL BE ACTIVE REGARDLESS OF OCCUPANCY OR HEATING DEMAND.
   9. THE RADIANT SLAB HEATING SYSTEM SHALL ONLY OPERATE IN OCCUPIED MODE USING THE RADIANT FLOOR TEMPERATURE SETPOINT. OWNER SHALL HAVE THE ABILITY TO ALLOW THE SYSTEM TO OPERATE IN UNOCCUPIED MODE.
   10. SLAB SYSTEM AND PLANT RESET COORDINATION
      a. THE THREE-WAY MIXING VALVE POSITION SHALL BE USED TO GENERATE HEATING HOT WATER FOR THE LOCAL SLAB SYSTEM.
      b. THE POSITION OF THIS THREE WAY MIXING VALVE SHALL BE USED TO GENERATE PLANT-SIDE PUMP DIFFERENTIAL PRESSURE AND SUPPLY TEMPERATURE RESET REQUESTS.
      c. RADIANT SLAB OPERATION SHALL NOT BE DISABLED OR LIMITED DUE TO AIRSIDE SYSTEM RESET PRIORITIES.
F. REHEAT COIL - OCCUPIED MODE
   a. AFTER A CHANGE IS IMPLEMENTED, THE BAS SHALL WAIT FOR 3 MINUTES (ADJ.) FOR THE SYSTEM TO ADJUST.
   b. COOLING
      - ON AN INCREASE IN ZONE AIR TEMPERATURE ABOVE THE COOLING SETPOINT (CALL FOR COOLING), THE BAS SHALL MODULATE OPEN THE SUPPLY AIR DAMPER TO THE MAXIMUM POSITION.
      - THE BAS SHALL MODULATE OPEN THE REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   c. HEATING
      - ON A DECREASE IN ZONE AIR TEMPERATURE BELOW THE HEATING SETPOINT (CALL FOR HEATING), THE BAS SHALL MODULATE OPEN THE REHEAT COIL TO MAINTAIN THE SCHEDULED LEAVING AIR TEMPERATURE.
G. REHEAT COIL - UNOCCUPIED MODE
   a. AFTER A CHANGE IS IMPLEMENTED, THE BAS SHALL WAIT FOR 10 MINUTES (ADJ.) FOR THE SYSTEM TO ADJUST
   b. COOLING
      - ON AN INCREASE IN ZONE AIR TEMPERATURE ABOVE THE SETBACK COOLING SETPOINT (CALL FOR COOLING), THE BAS SHALL MODULATE OPEN THE SUPPLY AIR DAMPER BEYOND THE UNOCCUPIED MINIMUM AIRFLOW SETPOINT IN STAGES OF 5%.
      - THE BAS SHALL MODULATE OPEN THE REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   c. HEATING
      - ON A DECREASE IN ZONE AIR TEMPERATURE BELOW THE SETBACK HEATING SETPOINT (CALL FOR HEATING), THE BAS SHALL MODULATE OPEN THE REHEAT COIL TO MAINTAIN THE SCHEDULED LEAVING AIR TEMPERATURE.
H. WHEN ZONE AIR TEMPERATURE IS BETWEEN THE HEATING & COOLING SETPOINTS (DEADBAND W/O CALL FOR HEATING OR COOLING), THE REHEAT VALVES SHALL CLOSE.
I. CONVECTIVE UNIT HEATERS -
   a. THE EQUIPMENT PROVIDED WITHIN EACH DEN IS FULLY REDUNDANT.
   b. THE EQUIPMENT WILL BE STAGED ON BY THE BAS ONLY IF -
      - IF ONE OR MORE ZONE IS OCCUPIED -
         1. SUPPLY TERMINAL UNIT DAMPER IS FULLY OPEN TO FLOW.
         2. REHEAT COIL VALVE IS FULLY OPEN TO FLOW.
         3. RADIANT SLAB SYSTEM VALVE IS FULLY OPEN TO FLOW.
   c. OWNER SHALL HAVE THE ABILITY TO ACTIVATE/DEACTIVATE THE HEATER THROUGH THE BAS.

**HUMIDITY CONTROL**

A. BAS SHALL MODULATE THE TERMINAL UNITS (SUPPLY AND EXHAUST) AND THE ASSOCIATED REHEAT COIL CLOSE, IF DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES. (DOOR IS LEFT OPEN).
B. PARAGRAPH C AND D OF THIS SECTION APPLY WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. IF THE ZONE DEW POINT EXCEEDS HIGH LIMIT
   A. THE BAS SHALL MODULATE OPEN THE REHEAT COIL CONTROL VALVE TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   B. UPON FURTHER CALL FOR DEHUMIDIFICATION, A DEW POINT RESET REQUEST SHALL BE SENT TO THE AIR HANDLING UNIT. REFER TO DEW POINT RESET LOGIC IN AHU SEQUENCE OF OPERATIONS.
D. IF ZONE DEW POINT FALLS BELOW THE LOW LIMIT
   A. THE BAS SHALL SEND A RESET REQUEST TO THE AIR HANDLING UNIT. REFER TO DEW POINT RESET LOGIC IN AHU SEQUENCE OF OPERATIONS.
   B. IF THE AIR HANDLING UNIT COOLING COIL VALVE IS ALREADY RESET AND/OR FULLY CLOSE (TO FLOW), THE BAS SHALL SEND A REQUEST TO THE CENTRAL HUMIDIFIER SYSTEM. REFER TO HUMIDIFIER CONTROL LOGIC.

**RESET REQUESTS - GENERAL**

A. THE BAS SHALL USE THE POSITION OF FOLLOWING TO SEND SYSTEM DIFFERENTIAL PRESSURE AND TEMPERATURE RESET REQUESTS -
   a. AIRSIDE
      - SUPPLY TERMINAL UNIT DAMPER POSITION
      - EXHAUST TERMINAL UNIT DAMPER POSITION
   b. WATERSIDE
      - REHEAT COIL CONTROL VALVE POSITION
      - RADIANT SLAB CONTROL VALVE POSITION *
      - UNIT HEATER VALVE POSITION

**PUMP DIFFERENTIAL PRESSURE RESET REQUESTS**

A. HEATING COIL HEATING HOT WATER RESET REQUESTS. SEND THE PUMP SYSTEM THAT SERVES THE REHEAT COIL REQUESTS AS FOLLOWS.
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 85%.
   2. IF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.
B. RADIANT SLAB SYSTEM HEATING HOT WATER RESET REQUESTS. SEND THE PUMP SYSTEM THAT SERVES THE RADIANT SLAB MIXING VALVE, REQUESTS AS FOLLOWS.
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 85%.
   2. IF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

**HYDRONIC PLANT SUPPLY TEMPERATURE RESET REQUESTS**

A. HEATING HOT WATER PLANT REQUESTS. SEND THE PLANT THAT GENERATES HEATING HOT WATER, REQUESTS AS FOLLOWS.
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 10%.
   2. IF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

**CONTROL PRIORITY**

A. THE FOLLOWING PRIORITY STRUCTURE IN ORDER OF HIGHEST TO LEAST PRIORITY APPLIES DURING SYSTEM OPERATION WHEN HVAC SYSTEMS OR INDIVIDUAL COMPONENTS CANNOT MEET ALL DEMANDS PLACED UPON THE SYSTEM DURING FAILURE CONDITIONS.
   1. ROOM TEMPERATURE AND HUMIDITY SETPOINT CONTROL.
   2. ROOM PRESSURIZATION CONTROL.
   3. ROOM MINIMUM AIR EXHAUST AIRFLOW SETPOINT.

**SAFETY AND ALARM GENERATIONS**

A. THE BAS SHALL GENERATE AN ALARM WHEN ANY OF THE DOOR OF THE HABITAT IS OPEN AT REGULAR INTERVALS, AS PROVEN BY DOOR SWITCH STATUS.
   a. NOTIFY THE BUILDING ENGINEER AT FOLLOWING INTERVALS AFTER DOOR SWITCH IS TRIGGERED ACTIVE (DOOR IS LEFT OPEN).
      - 10 MINUTES
      - 20 MINUTES
      - 60 MINUTES
      - 180 MINUTES
   b. NOTIFICATION SHOULD INFORM THE ENGINEER THAT THE VENTILATION AND TEMPERATURE CONTROL IS OFF.
B. THE BAS SHALL GENERATE ALARM IN THE EVENT OF ANY OF THE FOLLOWING
   1. MEASURED SPACE TEMPERATURE DEVIATES FROM SETPOINT BY 2°F (ADJ.) FOR 10 MINUTES CONTINUOUS.
   2. MEASURED SPACE RELATIVE HUMIDITY EXCEEDS HIGH LIMIT FOR 5 MINUTES CONTINUOUS.
   3. MEASURED EXHAUST AIRFLOW DEVIATES FROM SETPOINT BY 5% (ADJ.) OR 50 CFM (ADJ.), WHICHEVER IS HIGHER, FOR 5 MINUTES CONTINUOUS.
   4. MEASURED SUPPLY AIRFLOW DEVIATES FROM SETPOINT BY 5% (ADJ.) OR 50 CFM (ADJ.), WHICHEVER IS HIGHER, FOR 5 MINUTES CONTINUOUS.

**SOFTWARE POINTS**

A. ALL POINTS REQUIRED TO MEET THE INTENT OF THE SOO SHALL BE MONITORED AND PROVIDED WITH GRAPHICS, INCLUDING BUT NOT LIMITED TO THE FOLLOWING
   1. TOTAL ZONE EXHAUST AIRFLOW (IN UNITS OF CUBIC FEET PER MINUTE)
   2. TOTAL ZONE SUPPLY AIRFLOW (IN UNITS OF CUBIC FEET PER MINUTE)
   3. AIRFLOW OFFSET OR PRESSURIZATION.
   4. ZONE SETPOINTS (TEMPERATURE AND HUMIDITY)
   5. AIR PRESSURIZATION STATUS

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600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF); 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

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**CONTROLS SEQUENCE OF OPERATIONS:**

**GENERAL**

A. SEQUENCES ARE DEFINED FOR FOLLOWING ANIMAL EXHIBITION SPACES -
   1. DAYROOM
B. FOLLOWING SYSTEMS INTERACT AND PROVIDE TEMPERATURE AND VENTILATION CONTROL TO THE SPACES THROUGH THE BUILDING AUTOMATION SYSTEM
   1. VARIABLE AIR VOLUME (VAV) TERMINAL UNIT FOR SUPPLY AIRFLOW
   2. REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT
   3. VARIABLE AIR VOLUME (VAV) TERMINAL UNIT FOR EXHAUST AIRFLOW
   4. FLOOR CONVECTORS
      A. HTR-1
      B. HTR-2
      C. HTR-3
      D. HTR-4
   5. FOLLOWING INDEPENDENT SENSORS IMPLEMENT THE CONTROL LOGIC AND/OR ALARMS -
      A. TEMPERATURE
      B. HUMIDITY
      C. MULLION TEMPERATURE SENSOR
      D. STATIC PRESSURE

**FORMULAS -**

A. HUMIDITY SENSOR MOUNTED WITHIN THE EXHAUST DUCTWORK SHALL BE HARD WIRED AND USED TO DETERMINE THE DEW POINT TEMPERATURE.
B. ADDITIONALLY, DEW POINT WITHIN EACH INDIVIDUAL SPACE MUST BE CALCULATED USING THE HUMIDITY SENSOR WITHIN THE SPACE.
   1. IF THE SENSOR DOES NOT HAVE THE CAPACITY TO PROVIDE DEW POINT OF THE SPACE, SPACE DEWPOINT MUST BE CALCULATED FROM THE EQUATION BELOW.
   2. ELSEIF THE SENSOR HAS THE CAPACITY TO PROVIDE DEW POINT OF THE SPACE, SPACE DEWPOINT MUST STILL BE CALCULATED FROM THE EQUATION BELOW, AND RECORDED AS 'THEORITICAL DEW POINT' WITHIN THE LOGS.
C. DEW-POINT TEMPERATURE (MAGNUS TETENS) EQUATION -
   1. WHERE T IS DRY BULB TEMPERATURE EXPRESSED **IN CELSIUS.**
   2. RH IS THE RELATIVE HUMIDITY OF THE AIR EXPRESSED AS A PERCENTAGE.
      A. HARD WIRED POINT FROM EXHAUST AIRSTREAM SHALL BE USED.
   3. 'ln(x)' IS A NATURAL LOGARITHM OF VARIABLE 'x'.
   4. Td IS THE INSTANTANEOUS DEW POINT TEMPERATURE IN FAHRENHEIT.
   5. Td = 32 + 1.8 * [ (243.04 * (ln(RH/100) + (17.625 * T) / (243.04 + T)) / (17.625 -ln(RH/100) -((17.625 * T) / (243.04 + T)))]

**SETPOINTS**

A. FOLLOWING SETPOINTS SHALL APPLY FOR ALL THE SPACES CALLED OUT ON THIS SHEET.
B. ALL SETPOINTS ARE BAS ADJUSTABLE.
C. TEMPERATURE CONTROL - COOLING
   1. DRY BULB SETPOINT = 72.0°F
      a. DEADBAND = 2°F
D. TEMPERATURE CONTROL - HEATING
   1. DRY BULB SETPOINT = 72.0°F
      a. DEADBAND = 2°F
E. MULLION TEMPERATURE SETPOINT = 72.0°F
F. HUMIDITY CONTROL
   1. RELATIVE HUMIDITY LOW LIMIT = 30%
   2. RELATIVE HUMIDITY HIGH LIMIT = 50%
G. ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT = POSITIVE 0.05 INCHES OF WATER (GAUGE PRESSURE)

**VENTILATION CONTROL**

A. BAS SHALL MODULATE THE TERMINAL UNITS (SUPPLY AND EXHAUST) AND THE ASSOCIATED REHEAT COIL CLOSE, IF ANY OF THE DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES (DOOR IS LEFT OPEN).
B. PARAGRAPH C THROUGH F OF THIS SECTION APPLIES WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. OCCUPIED MODE
   1. OWNER SHALL HAVE THE ABILITY TO OVERRIDE THE OCCUPANCY OF THE ZONE THROUGH THE BAS.
   2. PASSIVELY, AN OCCUPIED SIGNAL FROM ANY INDIVIDUAL WIRED SPACE OCCUPANCY SENSOR LOCATED IN VENTILATION ZONE (OR OCCUPANCY STATUS OVERRIDE BY THE BAS) SHALL ENABLE OCCUPIED MODE.
      a. NOTE: ALL OCCUPANCY SENSOR SIGNALS ARE WIRED POINTS FROM LOCAL OCCUPANCY SENSOR.
   3. THE ASSOCIATED TERMINAL UNIT PRIMARY AIR DAMPER SHALL MODULATE TO THE MIN OCCUPIED OA CFM PER TERMINAL UNIT SCHEDULES.
   4. COOLING AND HEATING:
      a. UPON A CALL FOR COOLING, INDICATED BY A RISE IN SPACE TEMPERATURE ABOVE THE SPACE TEMPERATURE SETPOINT BY ANY SPACE THERMOSTAT IN THE VENTILATION ZONE, THE BAS SHALL MODULATE THE PRIMARY AIR DAMPER OPEN TOWARD ITS PRIMARY AIR MAXIMUM CFM SETPOINT.
      b. ON BAS VERIFICATION THAT ALL TEMPERATURE CONTROL ZONES WITHIN THE VENTILATION ZONE ARE IN HEATING, THE BAS SHALL MODULATE THE PRIMARY AIR DAMPER TO ITS MIN OCCUPIED OA CFM.
D. UNOCCUPIED MODE
   1. OWNER SHALL HAVE THE ABILITY TO OVERRIDE THE OCCUPANCY OF THE ZONE THROUGH THE BAS.
   2. PASSIVELY ON BAS VERIFICATION OF 30 MINUTES CONTINUOUS (ADJ.) WITHOUT OCCUPIED SIGNAL FROM ALL OCCUPANCY SENSORS WITHIN VENTILATION CONTROL ZONE.
E. EXHAUST SYSTEM
   a. IF THE ROOM IS IN OCCUPIED MODE,
      - THE EXHAUST TERMINAL UNIT SHALL MODULATE OPEN AND EXHAUST THE DESIGN AIRFLOW (SEE MECHANICAL SCHEDULES) FROM THE SPACE.
      - THE EXHAUST TERMINAL UNIT SHALL MODULATE OPEN FURTHER TO MEET THE ZONE STATIC PRESSURE SETPOINT, WHEN SUPPLY TERMINAL UNIT IS RESPONDING TO CALL FOR COOLING.
   b. ELSEIF THE ROOM IS IN UNOCCUPIED MODE,
      - THE EXHAUST TERMINAL UNIT SHALL MODULATE PARTIALLY CLOSE TO EXHAUST THE MINIMUM UNOCCUPIED EXHAUST AIRFLOW (SEE MECHANICAL SCHEDULES) FROM THE SPACE.
F. SUPPLY SYSTEM
   a. SEE TEMPERATURE CONTROL SECTION FOR LOGIC ON REHEAT COIL MODULATION.
   b. IF THE ROOM IS IN OCCUPIED MODE,
      - STAGE 1 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN AND PROVIDE THE MINIMUM OCCUPIED AIRFLOW (SEE MECHANICAL SCHEDULES) TO THE SPACES.
      - STAGE 2 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN FURTHER, AS REQUIRED, TO MAINTAIN THE ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT.
      - STAGE 3 - SEE TEMPERATURE CONTROL SECTION.
   c. ELSEIF THE ROOM IS IN UNOCCUPIED MODE,
      - STAGE 1 - THE SUPPLY TERMINAL UNIT SHALL MODULATE TO PROVIDE THE MINIMUM UNOCCUPIED SUPPLY AIRFLOW (SEE MECHANICAL SCHEDULES) TO THE SPACE.
      - STAGE 2 - THE SUPPLY TERMINAL UNIT SHALL MODULATE OPEN FURTHER, AS REQUIRED, TO MAINTAIN THE ZONE STATIC PRESSURE SETPOINT WITH RESPECT TO AMBIENT.
      - STAGE 3 - SEE TEMPERATURE CONTROL SECTION.

**TEMPERATURE CONTROL**

A. BAS SHALL MODULATE THE REHEAT COIL, UNIT HEATER AND SLAB HEATERS CLOSE, IF DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES (DOOR IS LEFT OPEN).
B. PARAGRAPH C THROUGH H OF THIS SECTION APPLIES WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. THE BAS SHALL MODULATE AND STAGE ZONE LEVEL HEATING SYSTEMS TO MAINTAIN THE TEMPERATURE WITHIN EACH DEN.
D. THE BAS CONTROLLER SHALL MONITOR THE ZONE TEMPERATURE, COOLING AND HEATING CONTROL VALVE(S) IN SEQUENCE TO MAINTAIN THE ZONE TEMPERATURE AT THE SETPOINTS.
E. FLOOR CONVECTORS -
   a. FLOOR CONVECTORS ARE INTENDED TO PREVENT CONDENSATION ON MULLION.
   b. IF OUTSIDE AIR TEMPERATURE IS 45°F AND BELOW,
      - THE VALVE ASSOCIATED WITH REHEAT COIL FOR THE CONVECTORS SHALL MODULATE OPEN TO MAINTAIN MULLION TEMPERATURE SETPOINT.
   c. ELSEIF OUTSIDE AIR TEMPERATURE IS 45°F AND ABOVE,
      - THE ASSOCIATED WITH REHEAT COIL FOR THE CONVECTORS SHALL MODULATE CLOSE.
F. REHEAT COIL - OCCUPIED MODE
   a. AFTER A CHANGE IS IMPLEMENTED, THE BAS SHALL WAIT FOR 3 MINUTES (ADJ.) FOR THE SYSTEM TO ADJUST.
   b. COOLING
      - ON AN INCREASE IN ZONE AIR TEMPERATURE ABOVE THE COOLING SETPOINT (CALL FOR COOLING), THE BAS SHALL MODULATE OPEN THE SUPPLY AIR DAMPER TO THE MAXIMUM POSITION.
      - THE BAS SHALL MODULATE OPEN THE REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   c. HEATING
      - ON A DECREASE IN ZONE AIR TEMPERATURE BELOW THE HEATING SETPOINT (CALL FOR HEATING), THE BAS SHALL MODULATE OPEN THE REHEAT COIL TO MAINTAIN THE SCHEDULED LEAVING AIR TEMPERATURE.
G. REHEAT COIL - UNOCCUPIED MODE
   a. AFTER A CHANGE IS IMPLEMENTED, THE BAS SHALL WAIT FOR 10 MINUTES (ADJ.) FOR THE SYSTEM TO ADJUST
   b. COOLING
      - ON AN INCREASE IN ZONE AIR TEMPERATURE ABOVE THE SETBACK COOLING SETPOINT (CALL FOR COOLING), THE BAS SHALL MODULATE OPEN THE SUPPLY AIR DAMPER BEYOND THE UNOCCUPIED MINIMUM AIRFLOW SETPOINT IN STAGES OF 5%.
      - THE BAS SHALL MODULATE OPEN THE REHEAT COIL DOWNSTREAM OF SUPPLY TERMINAL UNIT TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   c. HEATING
      - ON A DECREASE IN ZONE AIR TEMPERATURE BELOW THE SETBACK HEATING SETPOINT (CALL FOR HEATING), THE BAS SHALL MODULATE OPEN THE REHEAT COIL TO MAINTAIN THE SCHEDULED LEAVING AIR TEMPERATURE.
H. WHEN ZONE AIR TEMPERATURE IS BETWEEN THE HEATING & COOLING SETPOINTS (DEADBAND W/O CALL FOR HEATING OR COOLING), THE REHEAT VALVES SHALL CLOSE.
I. CONVECTIVE UNIT HEATERS -
   a. THE EQUIPMENT PROVIDED WITHIN EACH DEN IS FULLY REDUNDANT.
   b. THE EQUIPMENT WILL BE STAGED ON BY THE BAS ONLY IF -
      - IF ONE OR MORE ZONES ARE OCCUPIED -
        1. SUPPLY TERMINAL UNIT DAMPER IS FULLY OPEN TO FLOW.
        2. REHEAT COIL VALVE IS FULLY OPEN TO FLOW.
        3. RADIANT SLAB SYSTEM VALVE IS FULLY OPEN TO FLOW.
   c. OWNER SHALL HAVE THE ABILITY TO ACTIVATE/DEACTIVATE THE HEATER THROUGH THE BAS.

**HUMIDITY CONTROL**

A. BAS SHALL MODULATE THE TERMINAL UNITS (SUPPLY AND EXHAUST) AND THE ASSOCIATED REHEAT COIL CLOSE, IF DOOR SWITCH STATUS IS PROVEN OPEN CONTINUOUSLY FOR 10 MINUTES. (DOOR IS LEFT OPEN).
B. PARAGRAPH C AND D OF THIS SECTION APPLY WHEN THE DOOR SWITCH STATUS IS PROVEN CLOSE.
C. IF THE ZONE DEW POINT EXCEEDS HIGH LIMIT
   A. THE BAS SHALL MODULATE OPEN THE REHEAT COIL CONTROL VALVE TO MAINTAIN LEAVING AIR TEMPERATURE AT LEAST 2°F ABOVE THE INSTANTANEOUS DEW POINT TEMPERATURE.
   B. UPON FURTHER CALL FOR DEHUMIDIFICATION, A DEW POINT RESET REQUEST SHALL BE SENT TO THE AIR HANDLING UNIT. REFER TO DEW POINT RESET LOGIC IN AHU SEQUENCE OF OPERATIONS.
D. ELSEIF ZONE DEW POINT FALLS BELOW THE LOW LIMIT
   A. THE BAS SHALL SEND A RESET REQUEST TO THE AIR HANDLING UNIT. REFER TO DEW POINT RESET LOGIC IN AHU SEQUENCE OF OPERATIONS.
   B. IF THE AIR HANDLING UNIT COOLING COIL VALVE IS ALREADY RESET AND/OR FULLY CLOSED (TO FLOW), THE BAS SHALL SEND A REQUEST TO THE CENTRAL HUMIDIFIER SYSTEM. REFER TO HUMIDIFIER CONTROL LOGIC.

**RESET REQUESTS - GENERAL**

A. THE BAS SHALL USE THE POSITION OF THE FOLLOWING TO SEND SYSTEM DIFFERENTIAL PRESSURE AND TEMPERATURE RESET REQUESTS -
   a. AIRSIDE
      - SUPPLY TERMINAL UNIT DAMPER POSITION
      - EXHAUST TERMINAL UNIT DAMPER POSITION
   b. WATERSIDE
      - REHEAT COIL CONTROL VALVE POSITION
      - RADIANT SLAB CONTROL VALVE POSITION *
      - UNIT HEATER VALVE POSITION

**PUMP DIFFERENTIAL PRESSURE RESET REQUESTS**

A. HEATING COIL HEATING HOT WATER RESET REQUESTS. SEND THE PUMP SYSTEM THAT SERVES THE REHEAT COIL REQUESTS AS FOLLOWS:
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 85%.
   2. ELSEIF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.
B. RADIANT SLAB SYSTEM HEATING HOT WATER RESET REQUESTS. SEND THE PUMP SYSTEM THAT SERVES THE RADIANT SLAB MIXING VALVE, REQUESTS AS FOLLOWS:
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 85%.
   2. ELSEIF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

**HYDRONIC PLANT SUPPLY TEMPERATURE RESET REQUESTS**

A. HEATING HOT WATER PLANT REQUESTS. SEND THE PLANT THAT GENERATES HEATING HOT WATER, REQUESTS AS FOLLOWS:
   1. IF THE HHW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HHW VALVE POSITION IS LESS THAN 10%.
   2. ELSEIF THE HHW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

**PRESSURE CONTROL RESET REQUESTS**

A. PRIMARY AIR STATIC PRESSURE RESET REQUESTS. SEND THE FAN SYSTEM (AHU-1 OR EF-1_ THAT SERVES THE TERMINAL UNITS, PRESSURE REQUESTS AS FOLLOWS:
   1. IF THE MEASURED AIRFLOW IS LESS THAN 85% OF THE SETPOINT WHILE THE SETPOINT IS GREATER THAN ZERO AND THE DAMPER POSITION IS GREATER THAN 95% FOR 1 MINUTE, SEND 3 REQUESTS.
   2. ELSEIF THE MEASURED AIRFLOW IS LESS THAN 90% OF THE SETPOINT WHILE THE SETPOINT IS GREATER THAN ZERO AND THE DAMPER POSITION IS GREATER THAN 95% FOR 1 MINUTE, SEND 2 REQUESTS.
   3. ELSEIF THE DAMPER POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE DAMPER POSITION IS LESS THAN 85%.
   4. ELSEIF THE DAMPER POSITION IS LESS THAN 95%, SEND 0 REQUEST(S).

**CONTROL PRIORITY**

A. THE FOLLOWING PRIORITY STRUCTURE IN ORDER OF GREATEST TO LEAST PRIORITY APPLIES DURING SYSTEM OPERATION WHEN HVAC SYSTEMS OR INDIVIDUAL COMPONENTS CANNOT MEET ALL DEMANDS PLACED UPON THE SYSTEM DURING FAILURE CONDITIONS:
   1. ROOM TEMPERATURE AND HUMIDITY SETPOINT CONTROL.
   2. ROOM PRESSURIZATION CONTROL.
   3. ROOM MINIMUM AIR EXHAUST AIRFLOW SETPOINT.

**SAFETY AND ALARM GENERATIONS**

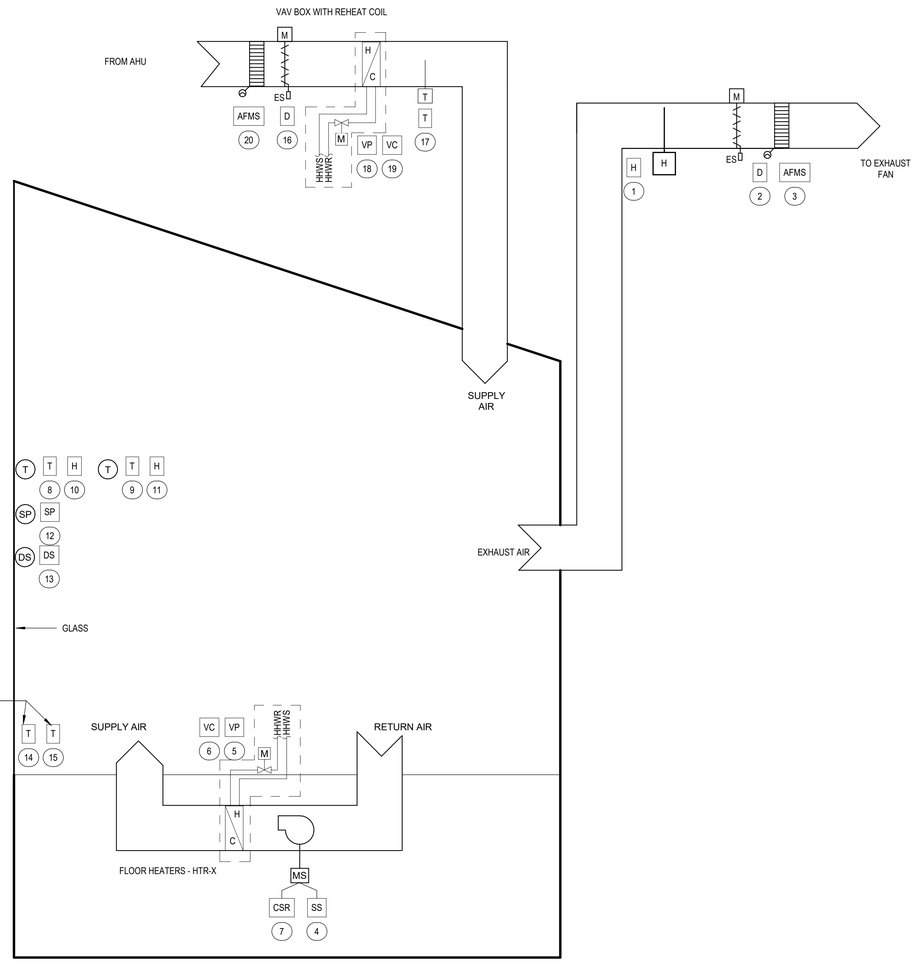
A. THE BAS SHALL GENERATE AN ALARM WHEN ANY OF THE DOOR OF THE HABITAT IS OPEN AT REGULAR INTERVALS, AS PROVEN BY DOOR SWITCH STATUS.
   a. NOTIFY THE BUILDING ENGINEER AT FOLLOWING INTERVALS AFTER DOOR SWITCH IS TRIGGERED ACTIVE (DOOR IS LEFT OPEN).
      - 10 MINUTES
      - 20 MINUTES
      - 60 MINUTES
      - 180 MINUTES
   b. NOTIFICATION SHOULD INFORM THE ENGINEER THAT THE VENTILATION AND TEMPERATURE CONTROL IS OFF.
B. THE BAS SHALL GENERATE ALARM IN THE EVENT OF ANY OF THE FOLLOWING:
   1. MEASURED SPACE TEMPERATURE DEVIATES FROM SETPOINT BY 2°F (ADJ.) FOR 10 MINUTES CONTINUOUS.
   2. MEASURED SPACE RELATIVE HUMIDITY EXCEEDS HIGH LIMIT FOR 5 MINUTES CONTINUOUS.
   3. MEASURED EXHAUST AIRFLOW DEVIATES FROM SETPOINT BY 5% (ADJ.) OR 50 CFM (ADJ.), WHICHEVER IS HIGHER, FOR 5 MINUTES CONTINUOUS.
   4. MEASURED SUPPLY AIRFLOW DEVIATES FROM SETPOINT BY 5% (ADJ.) OR 50 CFM (ADJ.), WHICHEVER IS HIGHER, FOR 5 MINUTES CONTINUOUS.

**SOFTWARE POINTS**

A. ALL POINTS REQUIRED TO MEET THE INTENT OF THE SOO SHALL BE MONITORED AND PROVIDED WITH GRAPHICS, INCLUDING BUT NOT LIMITED TO THE FOLLOWING:
   1. TOTAL ZONE EXHAUST AIRFLOW (IN UNITS OF CUBIC FEET PER MINUTE)
   2. TOTAL ZONE SUPPLY AIRFLOW (IN UNITS OF CUBIC FEET PER MINUTE)
   3. AIRFLOW OFFSET OR PRESSURIZATION.
   4. ZONE SETPOINTS (TEMPERATURE AND HUMIDITY)
   5. AIR PRESSURIZATION STATUS

## BAS POINT FUNCTION SCHEDULE - DAYROOM

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | NETWORK POINT | NOTES |
|---|---|---|---|---|---|---|---|---|
| EXHAUST TERMINAL UNIT | | | | | | | | |
| 1 | HUMIDITY OF EXHAUST AIR | H | AI | | | | | |
| 2 | DAMPER CONTROL | D | AO | ■ | | | | |
| 3 | EXHAUST AIR FLOW STATION | AFMS | AI | | | | | |
| FLOOR CONVECTORS | | | | | | | | |
| 4 | CONVECTOR FAN START STOP | SS | DO | | | | | |
| 5 | CONVECTOR VALVE POSITION | VP | DI | | | ■ | | |
| 6 | CONVECTOR VALVE CONTROL | VC | DO | | | | | |
| 7 | CONVECTOR FAN SPEED CNTRL | CSR | AI | | | | | |
| SENSORS | | | | | | | | |
| 8 | TEMPERATURE 1 | T | AI | | | | | |
| 9 | TEMPERATURE 2 | T | AI | | | | | |
| 10 | RELATIVE HUMIDITY (MONITOR ONLY) 1 | H | AI | | | | | |
| 11 | RELATIVE HUMIDITY (MONITOR ONLY) 2 | H | AI | | | | | |
| 12 | STATIC PRESSURE SENSOR | SP | AI | | | | | |
| 13 | DOOR SWITCH (AT THE BRIDGE) | DS | AI | | | | | |
| 14 | MULLION TEMPERATURE 1 | T | AI | | | | | |
| 15 | MULLION TEMPERATURE 2 | T | AI | | | | | |
| SUPPLY TERMINAL UNIT | | | | | | | | |
| 16 | DAMPER CONTROL | D | AO | ■ | | | | |
| 17 | ZONE ENTERING AIR TEMPERATURE | T | AI | | | | | |
| 18 | RHEAT COIL VALVE POSITION | VP | AI | | | ■ | | |
| 19 | RHEAT COIL VALVE CONTROL | VC | AO | | | | | |
| 20 | SUPPLY AIR FLOW | AFMS | AI | | | | | |

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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - ZONE CONTROL DIAGRAM AND SEQUENCES - DAY ROOM |

| WORKING STAFF | OP | OP | IP |
|---|---|---|---|
| | DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 302 OF 378

M-815CP

DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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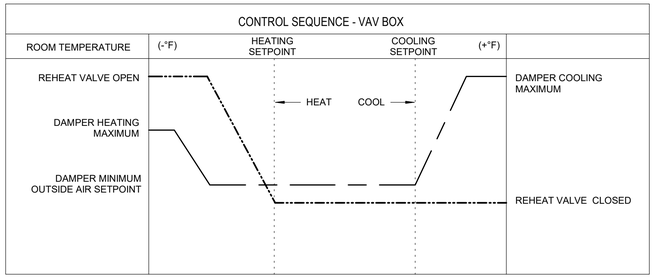


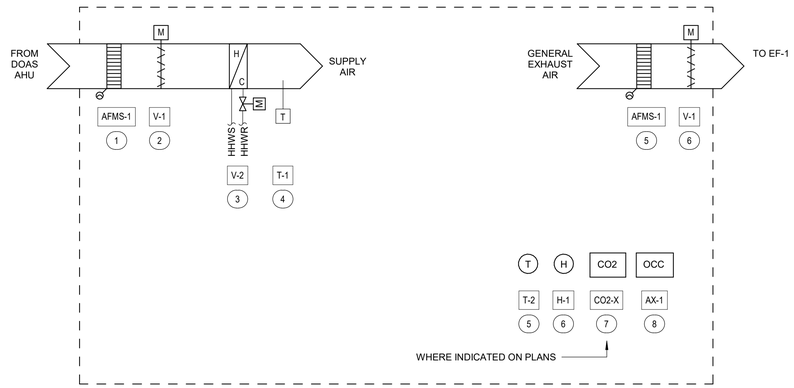


## BAS POINT FUNCTION SCHEDULE - CORRIDORS AND KEEPER OFFICE

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | ALARM LIMITS: LOW LIMIT | ALARM LIMITS: HIGH LIMIT | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|
| CORRIDORS AND KEEPER OFFICE | | | | | | | | | | |
| 1 | AIRFLOW MEASURING STATION | AFMS-1 | AI | | | | ■ | | | |
| 2 | VAV DAMPER CNTRL | V-1 | AI | ■ | | | | | | |
| 3 | REHEAT VALVE CNTRL | V-2 | AO | ■ | | | | | | |
| 4 | VAV LEAVING AIR TEMP | T-1 | AI | | | | ■ | | | |
| 5 | SPACE TEMPERATURE | T-2 | AI | | | | ■ | | | |
| 6 | SPACE HUMIDITY | H-1 | AI | | | | ■ | | | |
| 7 | SPACE CO2 | CO2-X | AI | | | | ■ | | | |
| 8 | OCCUPANCY | AX-1 | DI | | | | | | | |

**SEQUENCE OF OPERATIONS:**

**GENERAL**
THE BAS SHALL HAVE THE ABILITY TO OVERRIDE CONTROL OVER LOCAL OCCUPANCY SENSORS TO DICTATE MODE OF OPERATION. WHERE MULTIPLE TEMPERATURE SENSORS HAVE BEEN ASSIGNED TO A SINGLE ZONE, OUTSIDE AIR CONTROL TO BE BASED ON THE FURTHEST DEVIATION FROM THE SETPOINT. HEATING AND COOLING OPERATION SHALL NOT OVERLAP.

**OCCUPIED MODE / UNOCCUPIED MODE.**
HEATING:
- VAV TERMINAL UNITS WITH REHEAT COILS:
  VAV TERMINAL UNIT HEATING HOT WATER CONTROL VALVE MODULATES TO MAINTAIN SUPPLY AIR SETPOINT (85 F ADJ.).

COOLING:
- THE BAS SHALL MODULATE THE VAV BOX DAMPER TO MAINTAIN THE SPACE TEMPERATURE SETPOINT.
- AS THE SPACE TEMPERATURE INCREASES ABOVE THE SPACE TEMPERATURE SETPOINT, THE VAV BOX SHALL INCREASE THE AIRFLOW TO THE MAXIMUM DESIGN AIRFLOW.
- AS THE SPACE TEMPERATURE DECREASES BELOW THE SPACE TEMPERATURE SETPOINT, THE VAV BOX SHALL DECREASE THE AIRFLOW TO THE MINIMUM DESIGN AIRFLOW.

**NIGHT SETBACK MODE**
- HEATING: THE VAV TERMINAL UNIT IS CLOSED.
- UPON A CALL FOR HEATING, AND THE VAV SHALL OPEN TO HEATING AIRFLOW SETPOINT. REHEAT COIL VALVE SHALL MODULATE TO MAINTAIN HEATING DISCHARGE AIR TEMPERATURE SETPOINT.
- COOLING: THE VAV BOX WILL INITIALLY BE CLOSED. UPON A CALL FOR COOLING, THE VAV BOX WILL CYCLE TO PROVIDE THE FIRST STAGE OF SPACE COOLING, AS REQUIRED, TO MAINTAIN SPACE SETBACK TEMPERATURE AND DEW POINT (56 DEG F).

**ALARMS**
- DEWPOINT: IF THE SPACE DEW POINT EXCEEDS SETPOINT FOR MORE THAN 15 MINS (ADJ) AN ALARM SHALL BE SENT TO THE BAS.

**SYSTEM REQUESTS**
A. HOT-WATER RESET REQUESTS
   1. IF THE DAT IS 30°F LESS THAN SETPOINT FOR 5 MINUTES, SEND 3 REQUESTS.
   2. ELSE IF THE DAT IS 15°F LESS THAN SETPOINT FOR 5 MINUTES, SEND 2 REQUESTS.
   3. ELSE IF HW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HW VALVE POSITION IS LESS THAN 85%.
   4. ELSE IF THE HW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

B. HEATING HOT-WATER PLANT REQUESTS.
   1. SEND THE HEATING HOT-WATER PLANT THAT SERVES THE ZONE A HEATING HOT-WATER PLANT REQUEST AS FOLLOWS:
      1. IF THE HW VALVE POSITION IS GREATER THAN 95%, SEND 1 REQUEST UNTIL THE HW VALVE POSITION IS LESS THAN 10%.
      2. ELSE IF THE HW VALVE POSITION IS LESS THAN 95%, SEND 0 REQUESTS.

1 / M-816CP TERMINAL UNIT CONTROL DIAGRAM AND SEQUENCE OF OPERATIONS
SCALE: NOT TO SCALE

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| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

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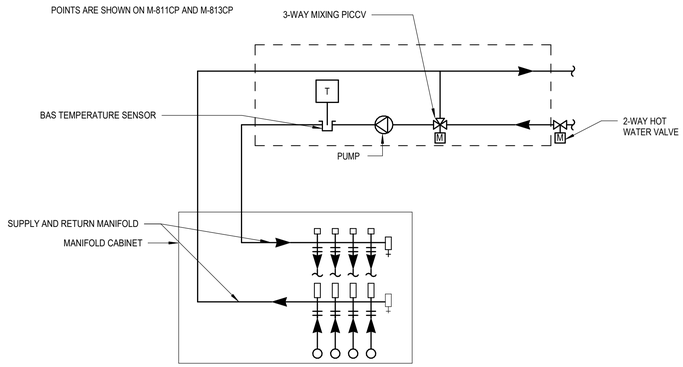


**CONTROLS SEQUENCE FOR RADIANT SLAB MIXING STATION**

1. ALL DENS (FIVE TOTAL) ARE FURNISHED WITH INDIVIDUAL RADIANT SLAB MIXING STATION.
2. THE BAS CONTROLS THE RADIANT SLAB HEATING SYSTEM THROUGH THE MIXING MANIFOLD -
   a. BAS MODULATES THE RE-CIRCULATION PUMP ON/OFF.
   b. BAS MODULATES THREE-WAY MIXING VALVE OPEN/CLOSE.
   c. BAS MODULATES HEATING HOT WATER SUPPLY ISOLATION VALVE.
3. THE MIXING STATION INCLUDES:
   1. ONE (1) MODULATING HOT WATER CONTROL VALVE
   2. ONE (1) MODULATING BYPASS/MIXING VALVE
   3. ONE (1) VARIABLE SPEED CIRCULATION PUMP
   4. ONE (1) IN-SLAB EMBEDDED TEMPERATURE SENSOR
4. THE BAS SHALL MAINTAIN A RADIANT SUPPLY WATER TEMPERATURE SETPOINT OF 110°F (ADJUSTABLE).

**SEQUENCE OF OPERATION**

1. SYSTEM ENABLE
   1. 1. THE BAS SHALL ENABLE THE RADIANT SLAB SYSTEM WHEN:
      1. a. ANY IN-SLAB EMBEDDED TEMPERATURE SENSOR READING FALLS BELOW ITS RESPECTIVE SETPOINT (ADJ)
   2. 2. WHEN ENABLED:
      1. a. THE RADIANT CIRCULATION PUMP SHALL START.
      2. b. THE MIXING VALVE CONTROL LOOP SHALL BE ENABLED.
2. HEATING MODE (DEN EMBEDDED IN-SLAB TEMPERATURE READING BELOW SETPOINT (ADJ))
3. INITIAL RESPONSE
   A. UPON A CALL FOR HEATING (SLAB TEMPERATURE BELOW SETPOINT):
      a. THE BAS SHALL TURN ON THE RADIANT SYSTEM PUMP.
      b. THE BAS SHALL TURN THE 3-WAY MIXING VALVE POSITION CLOSE FOR RECIRCULATION.
      c. THE BAS SHALL COMMAND THE HOT WATER CONTROL VALVE FULLY OPEN
   B. SUPPLY WATER TEMPERATURE CONTROL
      a. THE BAS SHALL MODULATE THE HOT WATER VALVE FULLY OPEN IF THERE IS AN INCREASED CALL FOR HEATING.
      b. IF BAS CALLS FOR MORE HEATING, THE MIXING VALVE SHALL CLOSE TO PREVENT RECIRCULATION.
   C. CONTROL SHALL BE BASED ON:
      A. FEEDBACK FROM THE EMBEDDED IN-SLAB TEMPERATURE SENSOR LOCATED IN THE FLOOR OF THE DEN(S) AND A BAS SIGNAL.
4. SATISFIED MODE (DEN SLAB IS AT OR ABOVE SETPOINT)
   A. SYSTEM RESPONSE
      a. WHEN SLAB DEN TEMPERATURE IS AT OR ABOVE SETPOINT:
         - THE BAS SHALL MODULATE THE HEATING HOT WATER SUPPLY ISOLATION VALVE CLOSE TO FLOW.
         - THE BAS SHALL OPEN 3-WAY MIXING VALVE FULLY OPEN TO RE-CIRCULATION.
         - THE BAS SHALL TURN OFF THE RE-CIRCULATION PUMP.

**PUMP CONTROL**

A. THE CIRCULATION PUMP SHALL OPERATE AS ON/OFF CONTROL.
   A. HEATING MODE -
      A. BAS SHALL START THE PUMP UPON INITIAL CALL FOR HEAT.
      B. PUMP SHALL REMAIN ON UNTIL HEATING DEMAND IS SATISFIED BASED ON:
B. RECIRCULATION MODE -
   A. THE SYSTEM SHALL RE-CIRCULATE WATER INTERNALLY THROUGH THE RADIANT SLAB SYSTEM WITHOUT INTRODUCING HEATING HOT WATER, ONCE A DAY (ADJ.)
   B. PURPOSE OF RE-CIRCULATION:
      a. PREVENT WATER STAGNATION.

**SYSTEM SAFETIES AND LIMITS**

A. THE BAS SHALL LIMIT MAXIMUM SUPPLY WATER TEMPERATURE TO 130°F (ADJ HIGH LIMIT).
B. THE BAS SHALL GENERATE AN ALARM IF:
   a. SUPPLY TEMPERATURE EXCEEDS HIGH LIMIT
   b. TEMPERATURE SENSOR FAILS
   c. PUMP STATUS DOES NOT MATCH COMMAND

**PUMP SHUTDOWN**

A. AFTER DEN(S) ARE SATISFIED FOR A CONTINUOUS PERIOD OF 10 MINUTES (ADJ):
   1. THE RESPECTIVE DEN PUMP SHALL TURN OFF UPON BAS SIGNAL
B. WHEN DEN(S) ARE UNSATISFIED FOR A CONTINUOUS PERIOD OF 10 MINUTES (ADJ):
   1. THE RESPECTIVE DEN PUMP SHALL TURN ON UPON BAS SIGNAL

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BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL CONTROLS - ZONE CONTROL DIAGRAM AND SEQUENCES - RADIANT SLAB MIXING BOX

WORKING STAFF

| OP | OP | IP |
|---|---|---|
| DESIGNED BY | DRAWN BY | CHECKED BY |

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M-817CP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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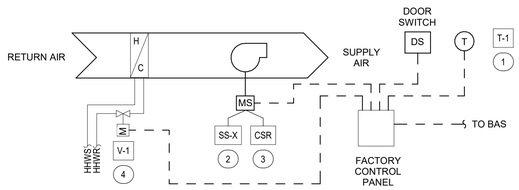
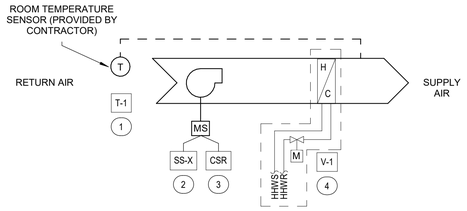



**SEQUENCE OF OPERATIONS**

A. FACTORY CONTROLLER PROVIDED WITH OPTIONAL BACNET PROTOCOL.
B. UPON ACTIVATION OF DOOR SWITCH, THE UNIT FAN(S) SHALL ENERGIZE.
C. HOT WATER VALVE SHALL BE COMMANDED OPEN UPON SPACE TEMPERATURE SENSOR FALLING BELOW 65°F (ADJ.).
   1. IF DOOR SWITCH IS NOT ACTIVATED AND TEMPERATURE SENSOR FALLS BELOW SETPOINT, UNIT FAN SHALL BE ENERGIZED.
D. ALARMS:
   1. LOW SPACE TEMPERATURE (LESS THAN 55°F (ADJ.))

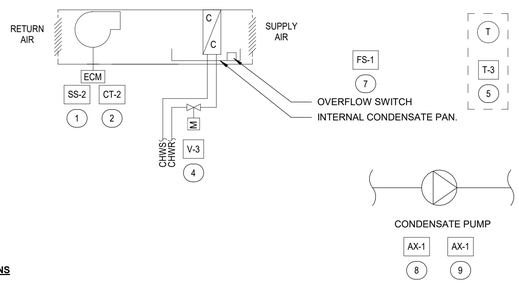
1 M-818CP AIR CURTAIN
SCALE: NONE




**SEQUENCE OF OPERATIONS**

A. THESE SEQUENCES DO NOT APPLY TO UNIT HEATERS WITHIN ANIMAL ENCLOSURES/DENS. REFER TO M-806CP. THESE SEQUENCES APPLY TO HEATERS WITHIN THE MECHANICAL ROOM
B. THE LOCAL THERMOSTAT WILL MODULATE THE HEATING HOT WATER VALVE AND CYCLE ON/OFF THE UNIT HEATER TO MAINTAIN SPACE TEMPERATURE SETPOINT 65°F (ADJ)
C. FAN SPEED:
   a. IF SPACE TEMPERATURE IS BELOW 60°F (ADJ), THE BAS SHALL TURN ON UNIT HEATER.
   b. IF SPACE TEMPERATURE IS 65°F (ADJ) AND ABOVE FOR 10 MIN CONTINUOS, BAS SHALL TURN OFF UNIT HEATER
   c. SPEED SHALL BE HIGH AS DEFAULT

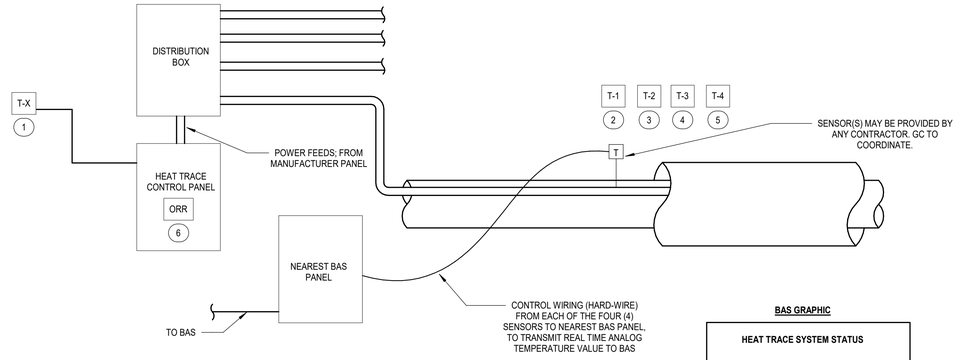
2 M-818CP UNIT HEATER
SCALE: NONE




**SEQUENCE OF OPERATIONS**

1. ENABLE/DISABLE
   A. THE BAS SHALL HAVE THE ABILITY TO ENABLE AND DISABLE THE FAN COIL UNITS AT THE OPERATORS WORKSTATION (OWS).
   B. THE BAS SHALL HAVE THE ABILITY TO ENABLE AND DISABLE THE FAN COIL UNITS BASED ON A TIME SCHEDULE SET BY THE BAS.
2. TEMPERATURE CONTROL
   A. THE FAN COIL UNIT SHALL BE PROVIDED WITH A SPACE THERMOSTAT.
   B. WHEN THERE IS NO DEMAND FROM COOLING FROM THE SPACE THERMOSTAT, THE FAN COIL UNIT FAN SHALL BE OFF AND THE CHILLED WATER VALVE SHALL BE CLOSED.
   C. WHEN THERE IS A DEMAND FOR COOLING FROM THE SPACE THERMOSTAT, THE FAN COIL UNIT FAN SHALL BE ON AND THE CHILLED WATER VALVE SHALL MODULATE OPEN TO MAINTAIN THE SPACE TEMPERATURE SETPOINT.
3. SET POINTS
   A. THE BAS SHALL BE CONFIGURED TO SEND SET POINTS REMOTELY TO A BAS CONTROLLER CONNECTED TO THE FAN COIL UNIT.
   B. THE FAN COIL UNIT SHALL BE INITIALLY SET TO MAINTAIN THE SETPOINTS IDENTIFIED IN THE SETPOINT SCHEDULE. ALL SETPOINTS SHALL BE MANUALLY ADJUSTABLE AT THE OWS.
4. LEAK DETECTION
   A. UPON SENSING A HIGH WATER LEVEL IN THE DRAIN PAN (OVERFLOW SWITCH), THE FAN SHALL SHUTDOWN, THE CONTROL VALVE SHALL CLOSE AND A BUILDING CRITICAL ALARM SHALL BE GENERATED AT THE BAS.

4 M-818CP FAN COIL UNIT (FCU) - COOLING ONLY
SCALE: NONE

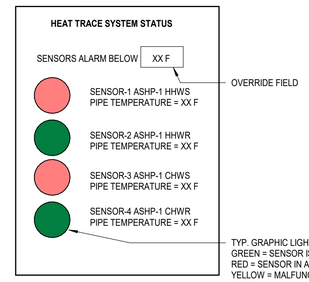



**SEQUENCE OF OPERATIONS:**

NOTE: PROVIDE FOUR (4) TEMPERATURE SENSORS TO BE MOUNTED ON PIPING.
MOUNTING INSTRUCTION - MOUNT A SENSOR AT AN ACCESSIBLE HEIGHT ON THE FOLLOWING PIPES:
1. AIR SOURCE HEAT PUMP (ASHP) = HEATING HOT WATER SUPPLY, HEATING HOT WATER RETURN, CHILLED WATER SUPPLY, AND CHILLED WATER RETURN

SEQUENCE OF OPERATIONS:
1. ALL SETPOINTS ARE USER ADJUSTABLE.
2. AN ALARM SHALL BE SENT TO THE BAS IF ANY OF THE TEMPERATURE SENSORS MOUNTED ON THE PIPE SENSE A TEMPERATURE LESS THAN 34°F DEG (ADJ) FOR MORE THAN 5 MINUTES.
3. AN ALARM SHALL BE SENT TO THE BAS WHEN AN OUTDOOR AIR TEMPERATURE DROPS BELOW 34°F DEG (ADJ) FOR MORE THAN 5 MINUTES AND HEAT TRACE PANEL IS NOT ON.

**BAS GRAPHIC**




5 M-818CP HEAT TRACE CONTROL SYSTEM
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| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - MISCELLANEOUS |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |

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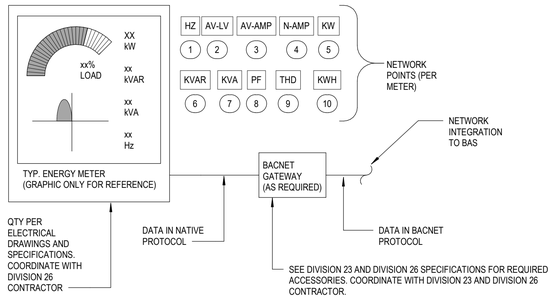
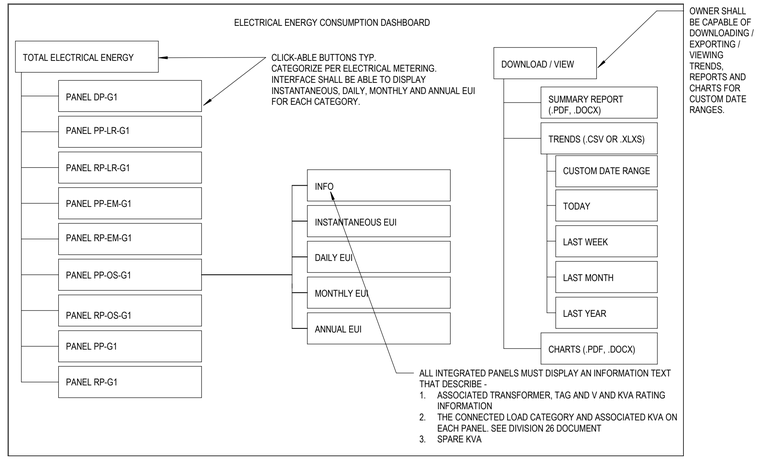


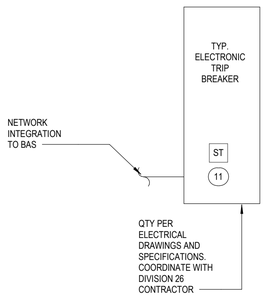




**SEQUENCE OF OPERATIONS (INTEGRATION INTO ENERGY MONITORING DASHBOARD) -**

1. SHOWN DASHBOARD IN THE GRAPHIC ABOVE IS FOR DIAGRAMMATIC REFERENCE ONLY, TO CONVEY SCOPE OF WORK AND INTEGRATION INTENT.
2. REFER TO ELECTRICAL METERING ON DIVISION 26 DRAWINGS AND SPECIFICATIONS FOR MORE INFORMATION. BAS TO MONITOR AND REPORT POWER ELECTRICAL METERS.
3. PROVIDE DISPLAY OF SHOWN POINTS IN ENERGY MONITORING DASHBOARD.
4. PROVIDE BUILDING ENERGY MONITORING DISPLAY OF TOTAL BUILDING INSTANTANEOUS KW.
5. TREND DATA EVERY 15 MINUTES (ADJ.).
6. REPORT KBTU/SF AND KWH DAILY, MONTHLY AND ANNUALLY.
7. REPORT PEAK AND AVERAGES OF ALL SHOWN ENERGY METER POINTS DAILY, MONTHLY AND ANNUALLY.
8. STORE DATA AS EXTERNAL FILES FOR A MINIMUM OF 36 MONTHS (ADJ.) ON THE SERVER, ACCESSIBLE TO BUILDING ENGINEER AND OWNER.
9. COORDINATE STORAGE LOCATION WITH BUILDING ENGINEER AND OWNER.
10. PROVIDE UPS MONITORING POINTS PER SPECIFICATION 263353.
    A. UPS ON BATTERY
    B. UPS ONLINE
    C. UPS LOAD-ON BYPASS
    D. UPS IN ALARM CONDITION
    E. UPS OFF (MAINTENANCE BYPASS CLOSED)

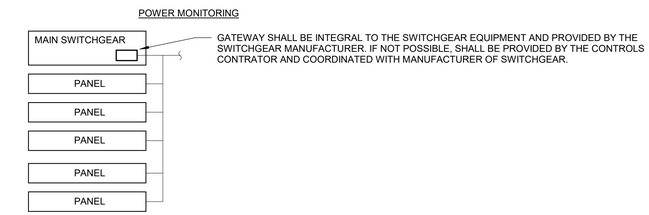
## 2 ELECTRICAL INTEGRATION - POWER MONITORING AND ENERGY METERS

SCALE: NOT TO SCALE



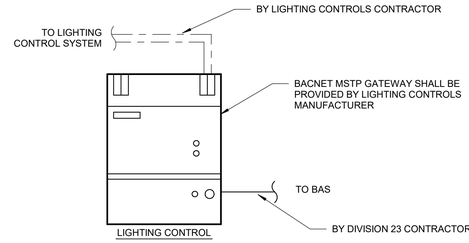

## 4 ELECTRICAL INTEGRATION ELECTRONIC TRIP BREAKER

SCALE: NOT TO SCALE




## 1 ELECTRICAL INTEGRATION - POWER MONITORING

SCALE: NOT TO SCALE




NOTES

1. BAS TO READ LIGHTING STATUS (AND OVERRIDES), SPACE OCCUPANCY, AND CLOCK/CALENDER EVENT PROGRAMMING DATA FROM LIGHTING CONTROLS SYSTEM.
   A. PROVIDE GRAPHICAL INTERFACE TO VIEW AND MONITOR ALL POINTS AT BAS.
   B. COORDINATE POINT MAPPING WITH LIGHTING CONTROL SYSTEM PROVIDER.
2. SEE SPECIFICATION SECTION 260943 NETWORK LIGHTING CONTROLS FOR RELATED REQUIREMENTS.
3. SEE DOCUMENTS ISSUED BY DIVISION 26 FOR UNIT QUANTITIES PER LIGHTING ZONE(S).

## 3 ELECTRICAL INTEGRATION - LIGHTING CONTROLS

SCALE: NOT TO SCALE

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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FOR REFERENCE ONLY

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - ELECTRICAL INTEGRATION |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |
| SHEET NO. | 306 OF 378 |
| | M-819CP (DISCIPLINE TYPE SEQUENCE DRAWING TYPE) |

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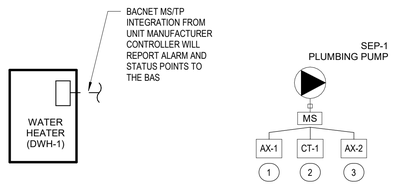
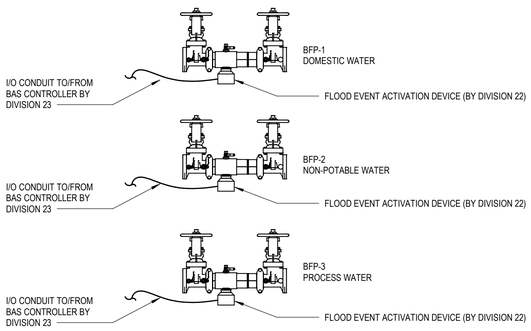
M-819CP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

WATER METERS TO PROVIDE LOCAL READOUT FOR ACTUAL FLOW. ENCODERS SHALL BE INSTALLED TO PROVIDE BAS WITH TOTALIZED FLOW AND ACTUAL FLOW VIA BACNET IP CONNECTION.

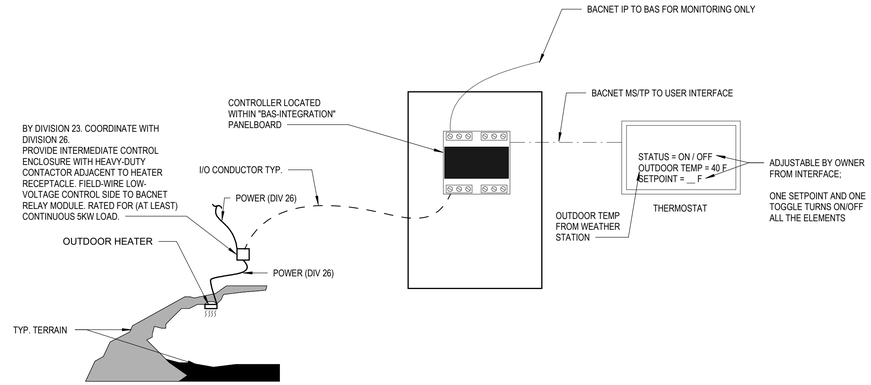



NOTES:

1. REFER TO DRAWINGS AND SPECIFICATIONS ISSUED BY DIVISION 22 FOR COMPLETE INFORMATION.
2. ALL REQUIRED BACNET GATEWAYS OR DEVICES REQUIRED TO INEGRATE CONTROLS COMMUNICATION TO THE BUILDING AUTOMATION SYSTEM SHALL BE PROVIDED BY THE EQUIPMENT MANUFACTURER.

1 / M-820CP — PLUMBING INTEGRATION
SCALE: NOT TO SCALE




2 / M-820CP — SITE HEATERS - INDEPENDENT CONTROL
SCALE: NOT TO SCALE

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DATE: 06/15/2026

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Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: MECHANICAL CONTROLS - PLUMBING AND MISC. INTEGRATION
WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 307 OF 378

M-820CP (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

FOR REFERENCE ONLY

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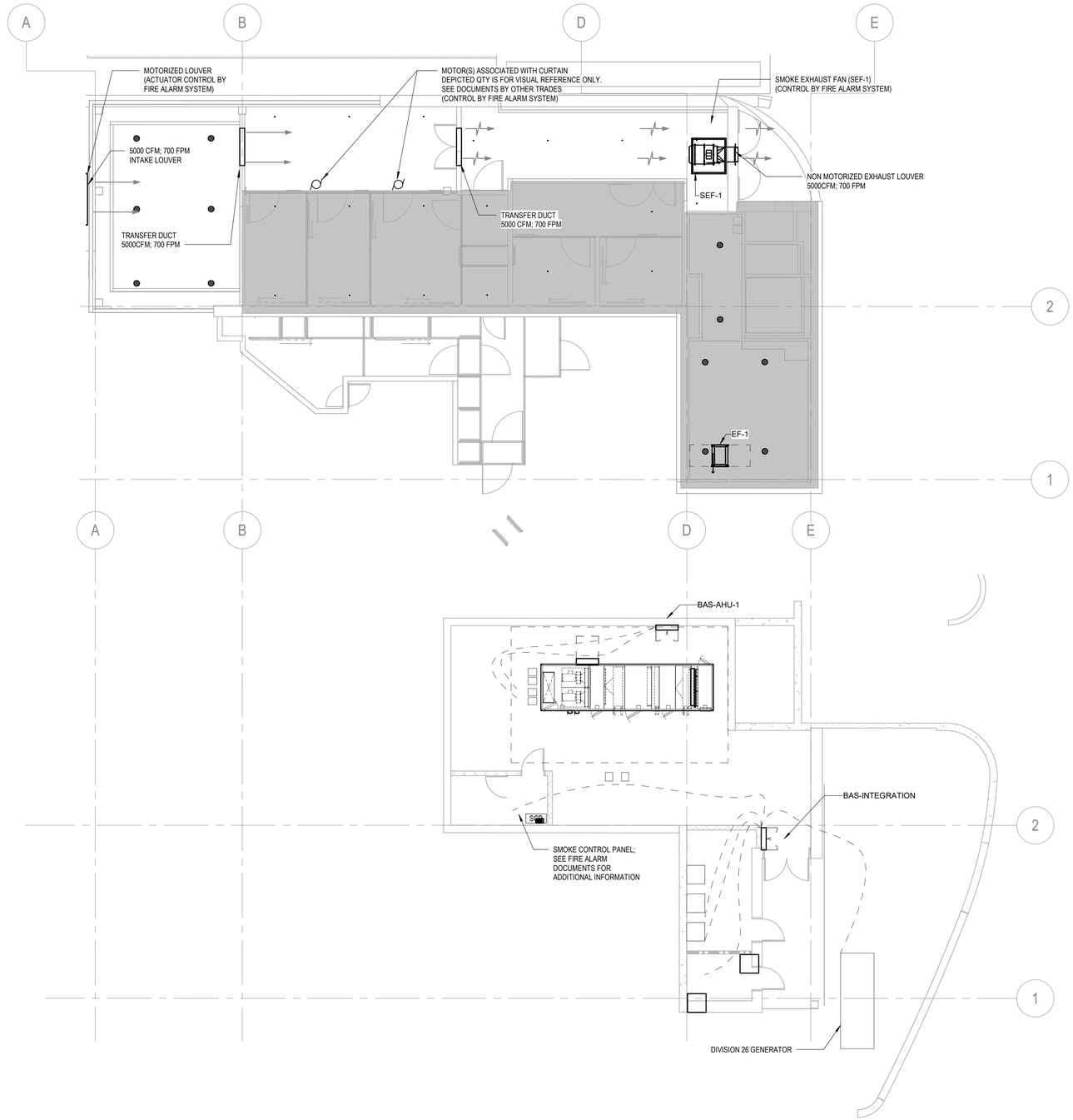
M-820CP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

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|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW<br>Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - SMOKE EXHAUST SYSTEM SCHEMATIC |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |
| SHEET NO. | 308 OF 378 |
| | M-821CP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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M-821CP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

ONLY SEQUENCE OF OPERATIONS PERTAINING THE BUILDING AUTOMATION SYSTEM IS DESCRIBED HERE.
BAS RECEIVES THE STATUS OF SMOKE EXHAUST SYSTEM THROUGH SMOKE CONTROL PANEL, THROUGH DRY CONTACT RELAY.
BAS SENDS A CONFIRMATION TO THE FIREMAN'S SCP WHEN IT EXECUTES THE SEQUENCE OF OPERATIONS STATED BELOW.
THE INTENT OF FOLLOWING SEQUENCES IS TO ENSURE THAT -

1. THE BAS DOES NOT INTERFERE WITH SMOKE REMOVAL MECHANISM
   A. ONLY EQUIPMENT COMMANDED BY FIRE ALARM SCP OR FIREMAN'S SCP SHALL OPERATE.
   B. MECHANICAL EQUIPMENT INCLUDING BUT NOT LIMITED TO HUMIDIFIER, HEATERS AND COOLING COILS SHALL NOT MANIPULATE THE ENVIRONMENT OF THE ZONE DURING THE EVENT.
   C. MECHANICAL EQUIPMENT INCLUDING BUT NOT LIMITED TO AIR HANDLING UNIT AND EXHAUST FAN SHALL NOT MANIPULATE DESIGNED AIR PATH.
2. THE BAS ACTIVELY PROHIBITS DIVISION 26 ATS TO SWITCH POWER BETWEEN SOURCES (EMERGENCY GENERATOR TO NORMAL GRID), WHEN THE SMOKE REMOVAL IS IN PROGRESS TO AVOID DAMPER/FAN/MOTOR/VFD MALFUNCTION.

FOLLOWING COMPONENTS ARE DRIVEN BY SMOKE CONTROL SYSTEM (BY OTHER TRADE) -

1. SMOKE EXHAUST FAN SEF-1 (EQUIPMENT BY DIVISION 23; CONTROL BY OTHER TRADE)
   A. DIVISION 23 CONTRACTOR MUST MAKE FAN START/STOP, FAN SPEED AND SPEED ADJUSTMENT (CONTROL AND CONFIRMATION) AVAILABLE IN NATIVE COMMUNICATION PROTOCOL OF THE FA SYSTEM. ANY EQUIPMENT/ACCESSORY SHALL BE UL 864 LISTED.
2. SMOKE EXHAUST FAN VARIABLE SPEED DRIVE (EQUIPMENT AND CONTROL BY OTHER TRADE)
3. MOTORIZED INTAKE LOUVER L-1 (EQUIPMENT BY DIVISION 23; CONTROL BY OTHER TRADE)
   A. DIVISION 23 CONTRACTOR MUST MAKE DAMPER STATUS AND DAMPER POSITION (CONTROL AND CONFIRMATION) AVAILABLE IN NATIVE COMMUNICATION PROTOCOL OF THE FA SYSTEM. ANY EQUIPMENT/ACCESSORY SHALL BE UL 864 LISTED.
4. DEN CURTAINS (EQUIPMENT AND CONTROL BY OTHER TRADE)

**SEQUENCING OF ALL COMPONENTS OF SMOKE CONTROL SYSTEM IS BY OTHER TRADE(S).**

SCENARIO A
GENERAL DESCRIPTION
1. THE BUILDING OPERATES UNDER NORMAL POWER.
2. THE SMOKE EXHAUST SYSTEM IS NOT ACTIVE.
3. EMERGENCY GENERATORS ARE OFF.

SEQUENCE OF OPERATIONS
1. THE HVAC SYSTEM OPERATES UNDER NORMAL CONDITIONS.
2. THE BAS SYSTEM OPERATES UNDER NORMAL CONDITIONS.

SCENARIO B
GENERAL DESCRIPTION
1. THE BUILDING OPERATES UNDER NORMAL POWER.
2. THE SMOKE EXHAUST SYSTEM IS ACTIVE.
3. EMERGENCY GENERATORS ARE OFF.

SEQUENCE OF OPERATIONS
   A. THE BAS COMMANDS FANS OF AHU-1 TO TURN OFF.
   B. THE BAS COMMANDS GENERAL EXHAUST FAN EF-1 TO TURN OFF.

SCENARIO C
GENERAL DESCRIPTION
1. THE BUILDING OPERATES UNDER EMERGENCY POWER, DUE TO LOSS OF NORMAL POWER.
2. THE SMOKE EXHAUST SYSTEM STATUS IS ON, AS MONITORED BY THE BAS FROM THE FIRE ALARM SYSTEM.
3. EMERGENCY GENERATORS ARE ON.
4. THE ENTIRE BAS IS OPERATING ON EMERGENCY POWER

SEQUENCE OF OPERATIONS - ORDERLY STARTUP IN EMERGENCY MODE
1. UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING IS OPERATING UNDER EMERGENCY POWER, THE BAS SHALL ISSUE A "DISABLE" COMMAND TO ALL EQUIPMENT, SHALL CLOSE ALL ASSOCIATED ISOLATION VALVES AND SHALL PERFORM AN ORDERLY STARTUP OF THE HVAC EQUIPMENT AS FOLLOWS:
   A. ENABLE THE CHILLED WATER SYSTEM PARTIALLY UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE CHILLED WATER SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION
   B. ENABLE THE HEATING HOT WATER SYSTEM UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE HEATING HOT WATER SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION
   C. KEEP THE ELECTRIC RESISTANCE HUMIDIFIER STEAM SYSTEM DISABLED UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE ELECTRIC RESISTANCE HUMIDIFIER STEAM SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION
   D. KEEP AHU-1 DISABLED.
      a. OPERATE THE AIR HANDLING UNIT UNDER THE EMERGENCY MODE OF OPERATION.
   E. KEEP EF-1 DISABLED.

SEQUENCE OF OPERATIONS - ORDERLY SHUT DOWN IN EMERGENCY MODE PRIOR TO TRANSFER TO NORMAL POWER
1. UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING HAS NORMAL POWER AVAILABLE AND THAT THE AUTOMATIC TRANSFER SWITCHES WILL SWITCH TO NORMAL POWER AND THAT THE SMOKE EXHAUST SYSTEM IS COMMANDED ON BY THE FIRE ALARM SYSTEM OR BY THE FSECP, THE BAS SHALL SEND AN INHIBIT SIGNAL TO THE ELECTRICAL SYSTEM TO PREVENT THE ELECTRICAL SYSTEM FROM SWITCHING TO NORMAL POWER.
   A. THE BAS SHALL MAINTAIN THE INHIBIT SIGNAL ACTIVE AS LONG AS REQUIRED FOR THE OPERATION OF THE SMOKE EXHAUST SYSTEM.
   B. WHEN THE SMOKE EXHAUST SYSTEM IS ACTIVE AS MONITORED BY THE BAS FROM THE FIRE ALARM SYSTEM OR THE FSECP, THE BAS SHALL IMPLEMENT THE FOLLOWING ACTIONS:
      a. DISABLE THE CHILLED WATER PLANT.
      b. DISABLE THE HEATING HOT WATER PLANT.
      c. DISABLE ALL REMAINING HVAC EQUIPMENT.
      d. AFTER ALL HVAC EQUIPMENT HAS BEEN DISABLED, AS DESCRIBED ABOVE, THE BAS SHALL WAIT 5 MINUTES (ADJ.) AND DEACTIVATE THE INHIBIT SIGNAL TO THE ELECTRICAL SYSTEM SUCH THAT THE ELECTRICAL SYSTEM CAN SWITCH TO NORMAL POWER.
2. UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING HAS NORMAL POWER AVAILABLE AND THAT THE AUTOMATIC TRANSFER SWITCHES WILL SWITCH TO NORMAL POWER WITHIN 5 MINUTES (ADJ.) AND THAT THE SMOKE EXHAUST SYSTEM IS NOT COMMANDED ON BY THE FIRE ALARM SYSTEM OR BY THE FSECP, THE BAS SHALL DISABLE ALL HVAC EQUIPMENT IN SEQUENCE.
   A. DISABLE THE CHILLED WATER PLANT.
   B. DISABLE THE HEATING HOT WATER PLANT.
   C. DISABLE ALL REMAINING HVAC EQUIPMENT.

SEQUENCE OF OPERATIONS - ORDERLY STARTUP IN NORMAL MODE
1. THE BAS SHALL PERFORM AN ORDERLY STARTUP OF THE HVAC EQUIPMENT WHEN -
   A. UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING IS OPERATING UNDER NORMAL POWER, AND THAT THE AUTOMATIC TRANSFER SWITCHES ARE IN A NORMAL MODE OF OPERATION FOR MINIMUM 5 MINUTES (ADJ.), AS TIMED BY THE BAS.
   B. THE SMOKE EXHAUST SYSTEM IS NO LONGER ACTIVE AS MONITORED BY THE BAS FROM THE FIRE ALARM SYSTEM.
2. SEQUENCES-
   A. ENABLE THE CHILLED WATER SYSTEM UNDER NORMAL MODE OF OPERATION
   B. ENABLE THE HEATING HOT WATER SYSTEM UNDER NORMAL MODE OF OPERATION
   C. ENABLE THE ELECTRIC RESISTANCE HUMIDIFIER STEAM SYSTEM UNDER NORMAL MODE OF OPERATION
   D. ENABLE AIR HANDLING UNIT.
   E. ENABLE ALL REMAINING HVAC EQUIPMENT.

SCENARIO D
GENERAL DESCRIPTION
1. THE BUILDING OPERATES UNDER EMERGENCY POWER, DUE TO LOSS OF NORMAL POWER.
2. THE SMOKE EXHAUST SYSTEM IS NOT ACTIVE.
3. EMERGENCY GENERATORS ARE ON.
4. THE ENTIRE BAS IS OPERATING ON EMERGENCY POWER

SEQUENCE OF OPERATIONS - ORDERLY STARTUP IN EMERGENCY MODE
1. UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING IS OPERATING UNDER EMERGENCY POWER, THE BAS SHALL ISSUE A "DISABLE" COMMAND TO ALL EQUIPMENT, SHALL CLOSE ALL ASSOCIATED ISOLATION VALVES AND SHALL PERFORM AN ORDERLY STARTUP OF THE HVAC EQUIPMENT AS FOLLOWS:
   A. ENABLE THE CHILLED WATER SYSTEM UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE CHILLED WATER SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION.
   B. ENABLE THE HEATING HOT WATER SYSTEM UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE HEATING HOT WATER SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION
   C. ENABLE THE ELECTRIC RESISTANCE HUMIDIFIER STEAM SYSTEM UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE ELECTRIC RESISTANCE HUMIDIFIER STEAM SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION.
2. ENABLE AHU-1 SAND EF-1
   A. OPERATE THE AIR HANDLING UNIT UNDER THE EMERGENCY MODE OF OPERATION
   B. SEE THE CORRESPONDING EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION.

SEQUENCE OF OPERATIONS - ORDERLY SHUT DOWN IN EMERGENCY MODE PRIOR TO TRANSFER TO NORMAL POWER
1. UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING HAS NORMAL POWER AVAILABLE AND THAT THE AUTOMATIC TRANSFER SWITCHES WILL SWITCH TO NORMAL MODE WITHIN 5 MINUTES (ADJ.), THE BAS SHALL DISABLE ALL HVAC EQUIPMENT IN SEQUENCE, AS FOLLOWS:
   A. DISABLE THE CHILLED WATER PLANT.
   B. DISABLE THE HEATING HOT WATER PLANT.
   C. DISABLE THE ELECTRIC RESISTANCE HUMIDIFIER STEAM SYSTEM.
   D. DISABLE ALL REMAINING HVAC EQUIPMENT.

SEQUENCE OF OPERATIONS - ORDERLY STARTUP IN NORMAL MODE
1. WITH NO HVAC EQUIPMENT RUNNING AND UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING IS OPERATING UNDER NORMAL POWER, AND THAT THE AUTOMATIC TRANSFER SWITCHES ARE IN A NORMAL MODE OF OPERATION FOR A MINIMUM 5 MINUTES (ADJ.), AS TIMED BY BAS, THE BAS SHALL PERFORM AN ORDERLY STARTUP OF THE HVAC EQUIPMENT AS FOLLOWS:
   A. ENABLE THE CHILLED WATER SYSTEM UNDER NORMAL MODE OF OPERATION
   B. ENABLE THE HEATING HOT WATER SYSTEM UNDER NORMAL MODE OF OPERATION
   C. ENABLE ELECTRIC RESISTANCE HUMIDIFIER STEAM SYSTEM UNDER NORMAL MODE OF OPERATION
   D. ENABLE ALL REMAINING HVAC EQUIPMENT.

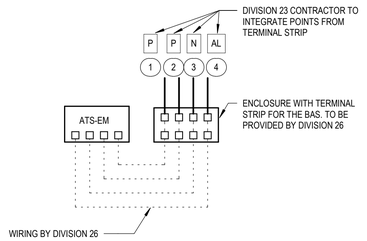
SCENARIO E
GENERAL DESCRIPTION
1. THE BUILDING OPERATES UNDER NORMAL POWER AND EMERGENCY POWER, DUE TO SCHEDULED TESTING OF THE GENERATORS.
   A. TESTING PERIOD SHALL BE NO LESS THAN __ MINUTES (SEE DOCUMENTS ISSUED BY DIVISION 26 FOR ADDITIONAL INFORMATION).
2. THE SMOKE EXHAUST SYSTEM CAN BE COMMANDED ON OR OFF BY THE FIRE ALARM SYSTEM OR BY THE FSECP.
3. EMERGENCY GENERATORS ARE ON.
4. THE ENTIRE BAS IS OPERATING ON EMERGENCY POWER.

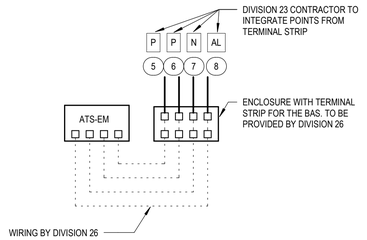
SEQUENCE OF OPERATIONS - ORDERLY SHUT DOWN IN NORMAL MODE
1. UPON RECEIVING A SIGNAL FROM THE GENERATOR TEST SWITCH TO "START", THE GENERATORS WILL ENTER TEST MODE AND THE AUTOMATIC TRANSFER SWITCHES WILL SWITCH TO EMERGENCY MODE OF OPERATION WITHIN 5 MINUTES (ADJ.), AS TIMED BY THE ELECTRICAL SYSTEM. THE BAS SHALL DISABLE ALL HVAC EQUIPMENT IN SEQUENCE, AS FOLLOWS:
   A. DISABLE THE CHILLED WATER PLANT.
   B. DISABLE THE HEATING HOT WATER PLANT.
   C. DISABLE THE ELECTRIC RESISTANCE HUMIDIFICATION SYSTEM.
   D. DISABLE ALL REMAINING HVAC EQUIPMENT THAT IS SUPPOSED TO RUN ON EMERGENCY POWER AND SHALL CLOSE ALL ASSOCIATED ISOLATION VALVES.
   E. ALL HVAC EQUIPMENT THAT IS NOT ON EMERGENCY POWER BUT STILL HAS NORMAL POWER AVAILABLE, SHALL OPERATE UNDER A "GENERATOR TEST MODE"
      a. SEE RELATED "GENERATOR TEST MODE" SEQUENCE OF OPERATIONS

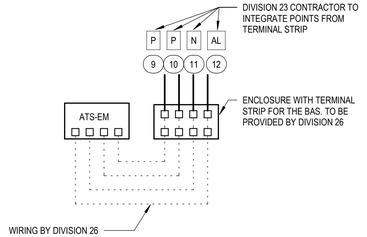
SEQUENCE OF OPERATIONS - ORDERLY STARTUP IN TEST MODE
1. UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING IS OPERATING UNDER EMERGENCY POWER, WITHIN 5 MINUTES (ADJ.) AS TIMED BY THE BAS, THE BAS SHALL PERFORM AN ORDERLY STARTUP OF THE HVAC EQUIPMENT AS FOLLOWS:
   A. ENABLE THE CHILLED WATER SYSTEM UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE CHILLED WATER SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION
   B. ENABLE THE HEATING HOT WATER SYSTEM UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE HEATING HOT WATER SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION
   C. ENABLE THE ELECTRIC RESISTANCE HUMIDIFICATION SYSTEM UNDER EMERGENCY MODE OF OPERATION
      a. SEE THE ELECTRIC RESISTANCE HUMIDIFICATION SYSTEM EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION
   D. ENABLE AHU-1.
      a. OPERATE THE AIR HANDLING UNIT UNDER THE EMERGENCY MODE OF OPERATION
      b. SEE THE CORRESPONDING EMERGENCY MODE OF OPERATION SEQUENCE OF OPERATIONS FOR ADDITIONAL INFORMATION
   E. ENABLE ALL HVAC EQUIPMENT THAT IS NOT ON EMERGENCY POWER BUT STILL HAS NORMAL POWER AVAILABLE, SHALL OPERATE UNDER A "GENERATOR TEST MODE"
      a. SEE RELATED "GENERATOR TEST MODE" SEQUENCE OF OPERATIONS

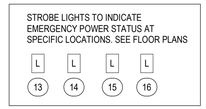
SEQUENCE OF OPERATIONS - ORDERLY SHUT DOWN IN TEST MODE
1. UPON RECEIVING A SIGNAL FROM THE GENERATOR TEST SWITCH "STOP" THAT THE GENERATOR TEST MODE WILL END AND THE AUTOMATIC TRANSFER SWITCHES WILL SWITCH TO NORMAL POWER WITHIN 5 MINUTES (ADJ.), AS TIMED BY THE ELECTRICAL SYSTEM, AND THAT THE SMOKE EXHAUST SYSTEM IS COMMANDED ON BY THE FIRE ALARM SYSTEM OR BY THE FSECP, THE BAS SHALL SEND AN INHIBIT SIGNAL TO THE ELECTRICAL SYSTEM TO PREVENT THE ELECTRICAL SYSTEM FROM SWITCHING TO NORMAL POWER.
   A. THE BAS SHALL MAINTAIN THE INHIBIT SIGNAL ACTIVE AS LONG AS REQUIRED FOR THE OPERATION OF THE SMOKE EXHAUST SYSTEM.
   B. UPON RECEIVING A COMMAND FROM THE FIRE ALARM SYSTEM OR THE FSECP TO TURN OFF THE SMOKE EXHAUST SYSTEM, THE BMS SHALL IMPLEMENT
   C. THE FOLLOWING ACTIONS:
      a. DISABLE THE CHILLED WATER PLANT.
      b. DISABLE THE HEAT HOT WATER PLANT.
      c. DISABLE THE ELECTRIC RESISTANCE HUMIDIFICATION SYSTEM.
      d. DISABLE ALL REMAINING HVAC EQUIPMENT.
      e. ALL HVAC EQUIPMENT THAT IS NOT ON EMERGENCY POWER BUT STILL HAS NORMAL POWER AVAILABLE, SHALL OPERATE UNDER A "GENERATOR TEST MODE". SEE RELATED "GENERATOR TEST MODE" SEQUENCE OF OPERATIONS.
   D. AFTER ALL HVAC EQUIPMENT HAS BEEN DISABLED, AS DESCRIBED ABOVE, THE BAS SHALL WAIT 5 MINUTES (ADJ.) AND DEACTIVATE THE INHIBIT SIGNAL TO THE ELECTRICAL SYSTEM SUCH THAT THE ELECTRICAL SYSTEM CAN SWITCH TO NORMAL POWER.
2. UPON RECEIVING A SIGNAL FROM THE GENERATOR TEST SWITCH TO "STOP". THE "GENERATOR TEST MODE WILL" END AND THE AUTOMATIC TRANSFER SWITCHES WILL SWITCH TO NORMAL MODE WITHIN 5 MINUTES (ADJ.), AS TIMED BY THE ELECTRICAL SYSTEM, AND THAT THE SMOKE EXHAUST SYSTEM IS NOT COMMANDED ON BY THE FIRE ALARM SYSTEM OR BY THE FSECP, THE BAS SHALL DISABLE ALL HVAC EQUIPMENT IN SEQUENCE.
   A. DISABLE THE CHILLED WATER PLANT.
   B. DISABLE THE HEATING HOT WATER PLANT.
   C. DISABLE THE ELECTRIC RESISTANCE HUMIDIFICATION SYSTEM.
   D. DISABLE ALL REMAINING HVAC EQUIPMENT.
   E. ALL HVAC EQUIPMENT THAT IS NOT ON EMERGENCY POWER BUT STILL HAS NORMAL POWER AVAILABLE, SHALL OPERATE UNDER A "GENERATOR TEST MODE"
      a. SEE RELATED "GENERATOR TEST MODE" SEQUENCE OF OPERATIONS

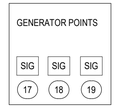
SEQUENCE OF OPERATIONS - ORDERLY STARTUP IN NORMAL MODE
1. WITH NO HVAC EQUIPMENT RUNNING AND UPON RECEIVING A SIGNAL FROM THE AUTOMATIC TRANSFER SWITCHES THAT THE BUILDING IS OPERATING UNDER NORMAL POWER, AND THAT THE AUTOMATIC TRANSFER SWITCHES ARE IN A NORMAL MODE OF OPERATION WITHIN 5 MINUTES (ADJ.), AS TIMED BY THE BAS, THE BAS SHALL PERFORM AN ORDERLY STARTUP OF THE HVAC EQUIPMENT AS FOLLOWS:
   A. ENABLE THE CHILLED WATER SYSTEM UNDER NORMAL MODE OF OPERATION
   B. ENABLE THE HEATING HOT WATER SYSTEM UNDER NORMAL MODE OF OPERATION
   C. ENABLE THE SKID MOUNTED STEAM SYSTEM UNDER NORMAL MODE OF OPERATION
   D. ENABLE THE AIR HANDLING UNIT.
   E. ENABLE ALL REMAINING HVAC EQUIPMENT.
   F. SWITCH TO A NORMAL MODE OF OPERATION ALL HVAC EQUIPMENT THAT WAS NOT ON EMERGENCY POWER BUT STILL HAD NORMAL POWER AVAILABLE AND WAS OPERATING UNDER A "GENERATOR TEST MODE"
      a. SEE RELATED "GENERATOR TEST MODE" SEQUENCE OF OPERATIONS

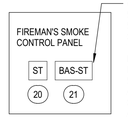


## BAS POINT FUNCTION SCHEDULE - ELECTRICAL INTEGRATION OF ATS AND GENERATOR

| S.NO | POINT NAME | TAG | POINT TYPE | FAIL MODE: FAIL ON (OPEN) | FAIL MODE: FAIL OFF (CLOSED) | FAIL MODE: LAST COMMANDED STATE | SOFTWARE: TREND | ALARM LIMITS: LOW LIMIT | ALARM LIMITS: HIGH LIMIT | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|
| ATS-EM | | | | | | | | | | |
| 1 | NORMAL POWER | P | DI | | | | ■ | | | |
| 2 | EMERGENCY POWER | P | DI | | | | ■ | | | |
| 3 | NEUTRAL POSITION | N | DI | | | | ■ | | | |
| 4 | EMERGENCY TO NORMAL POWER PRE-TRANSFER SIGNAL | AL | DI | | | | ■ | | | |
| ATS-LR | | | | | | | | | | |
| 5 | NORMAL POWER | P | DI | | | | ■ | | | |
| 6 | EMERGENCY POWER | P | DI | | | | ■ | | | |
| 7 | NEUTRAL POSITION | N | DI | | | | ■ | | | |
| 8 | EMERGENCY TO NORMAL POWER PRE-TRANSFER SIGNAL | AL | DI | | | | ■ | | | |
| ATS-OS | | | | | | | | | | |
| 9 | NORMAL POWER | P | DI | | | | ■ | | | |
| 10 | EMERGENCY POWER | P | DI | | | | ■ | | | |
| 11 | NEUTRAL POSITION | N | DI | | | | ■ | | | |
| 12 | EMERGENCY TO NORMAL POWER PRE-TRANSFER SIGNAL | AL | DI | | | | ■ | | | |
| STROBE LIGHTS | | | | | | | | | | |
| 13 | MECHANICAL ROOM EMERGENCY POWER STROBE LIGHT START/STOP | L | DO | | | | ■ | | | |
| 14 | MECHANICAL ROOM EMERGENCY POWER STROBE LIGHT START/STOP | L | DO | | | | ■ | | | |
| 15 | KEEPER SPACE EMERGENCY POWER STROBE LIGHT | L | DO | | | | ■ | | | |
| 16 | CARE BUILDING EMERGENCY POWER STROBE LIGHT | L | DO | | | | ■ | | | |
| GENERATOR RUN OVERRIDE | | | | | | | | | | |
| 19 | "SMOKE EXHAUST SYSTEM IS ACTIVE" INHIBIT SIGNAL TO GENERATOR | SIG | DI | | | | ■ | | | |
| GENERATOR TEST SWITCHES | | | | | | | | | | |
| 17 | START | SIG | DI | | | | ■ | | | |
| 18 | STOP | SIG | DI | | | | ■ | | | |
| FIREMAN'S SMOKE CONTROL PANEL | | | | | | | | | | |
| 20 | STATUS | ST | DI | | | | ■ | | | |
| 21 | STATUS OF BAS DRIVEN EQUIPMENT | BAS-ST | DO | | | | ■ | | | |
| FIRE ALARM SMOKE CONTROL PANEL SYSTEM | | | | | | | | | | |
| 21 | STATUS | ST | DI | | | | ■ | | | |
| 22 | STATUS (REDUNDANT) | ST | DI | | | | ■ | | | |

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## CONSTRUCT ARABIAN LEOPARD FACILITY

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| REVISION # | DATE | REVISION DESCRIPTION |
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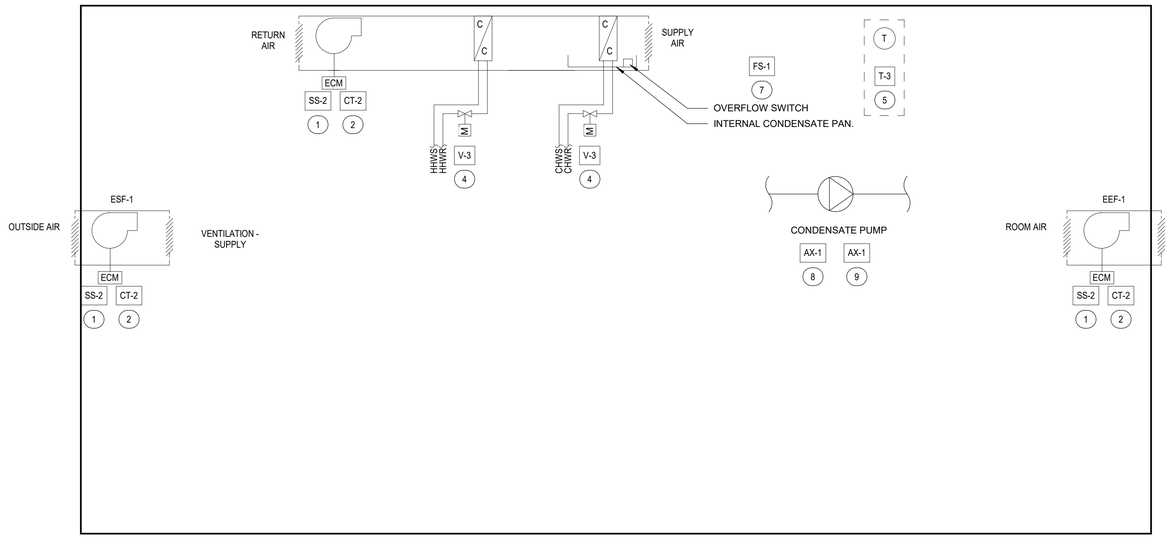
Smithsonian Institution
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008
PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084
DRAWING TITLE: MECHANICAL CONTROLS - SMOKE EXHAUST AND EMERGENCY POWER
WORKING STAFF: OP (DESIGNED BY) | OP (DRAWN BY) | IP (CHECKED BY)

SHEET NO. 309 OF 378
M-822CP (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

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DATE: 06/15/2026

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Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME / ADDRESS | ARABIAN LEOPARD FACILITY (ALF)<br>3001 Connecticut Avenue NW<br>Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | MECHANICAL CONTROLS - EMERGENCY ELECTRICAL ROOM VENTILATION AND TEMPERATURE CONTROL |
| WORKING STAFF | OP (DESIGNED BY) / OP (DRAWN BY) / IP (CHECKED BY) |
| SHEET NO. | 310 OF 378 |
| | M-823CP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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# PLUMBING ABBREVIATIONS

| ABBR | ABBREVIATION |
|---|---|
| ADA | AMERICANS WITH DISABILITIES ACT |
| AFF | ABOVE FINISHED FLOOR |
| AG | ABOVE GRADE |
| AHJ | AUTHORITY HAVING JURISDICTION |
| AMB | AMBIENT |
| ANSI | AMERICAN NATIONAL STANDARDS INSTITUTE |
| AP | ACCESS PANEL |
| APPROX | APPROXIMATE |
| BFP | BACKFLOW PREVENTOR (TYPE AS INDICATED) |
| BFV | BUTTERFLY VALVE |
| BHP | BRAKE HORSEPOWER |
| BLV | BALANCING VALVE |
| BTU/H | BRITISH THERMAL UNIT PER HOUR |
| BV | BALL VALVE |
| CA | UTILITY COMPRESSED AIR (PRESSURE AS INDICATED) |
| CAP | CAPACITY |
| CFM | CUBIC FEET PER MINUTE |
| CI | CAST IRON |
| CO | CLEANOUT |
| COMP | COMPRESSOR |
| COND | CONDENSATE |
| CONT | CONTINUATION |
| CV | CHECK VALVE |
| CW | DOMESTIC COLD WATER |
| DF | DRINKING FOUNTAIN |
| DIA | DIAMETER |
| DN | DOWN |
| DOM | DOMESTIC |
| DP | DROP |
| DWG | DRAWING |
| DWH | DOMESTIC WATER HEATER |
| DWR | DRINKING WATER RETURN |
| DWS | DRINKING WATER SUPPLY |
| (E) | EXISTING MATERIAL OR EQUIPMENT |
| EA | EACH |
| ECO | EXTERIOR CLEANOUT |
| EEW | EMERGENCY EYE WASH |
| EFF | EFFICIENCY |
| EFW | EMERGENCY FACE WASH |
| EL | ELEVATION |
| EQ | EQUIVALENT |
| ESH | EMERGENCY SHOWER |
| ET | EXPANSION TANK |
| EWC | ELECTRIC WATER COOLER |
| EWT | ENTERING WATER TEMPERATURE |
| EXP | EXPANSION |
| EXT | EXTERNAL |
| F | DEGREES FAHRENHEIT |
| FCO | FLOOR CLEANOUT |
| FD | FLOOR DRAIN |
| FL | FLOOR |
| FPM | FEET PER MINUTE |
| FPS | FEET PER SECOND |
| FS | FLOOR SINK |
| FT | FEET OR FOOT |
| FTD | FOOTER DRAINAGE |
| G | NATURAL GAS (PRESSURE AS INDICATED) |
| GAL | GALLON |
| GC | GAS COCK |
| GPD | GALLONS PER DAY |
| GPH | GALLONS PER HOUR |
| GPM | GALLONS PER MINUTE |
| GV | GATE VALVE |
| GW | GREASE WASTE |
| HB | HOSE BIBB |
| HG | MERCURY |
| HGT | HEIGHT |
| HP | HORSE POWER |
| HPS | HIGH PRESSURE STEAM |
| HR | HOUR |
| HW | DOMESTIC HOT WATER |
| HWR | DOMESTIC HOT WATER RETURN |
| ID | INSIDE DIAMETER |
| IE | INVERT ELEVATION |
| IN. | INCHES |
| K | KILOWATTS |
| LAV | LAVATORY |
| LGT | LENGTH |
| LPG | LIQUID PETROLEUM GAS |
| LPS | LOW PRESSURE STEAM |
| LWT | LEAVING WATER TEMPERATURE |
| MAX | MAXIMUM |
| MBH | 1000 BTU/HR |
| MCA | MINIMUM CURRENT AMPACITY |
| MIN | MINIMUM |
| MISC | MISCELLANEOUS |
| MOCP | MINIMUM OVER CURRENT PROTECTION |
| MPS | MEDIUM PRESSURE STEAM |
| MTG | MOUNTING |
| NA | NOT APPLICABLE |
| NC | NORMALLY CLOSED |
| NIC | NOT IN CONTRACT |
| NO | NORMALLY OPEN/ NUMBER |
| NOM | NOMINAL |
| NTS | NOT TO SCALE |
| OD | OUTSIDE DIAMETER |
| OD | OVERFLOW ROOF DRAIN |
| OFCI | OWNER FURNISHED CONTRACTOR INSTALLED |
| OFL | OVERFLOW |
| OFOI | OWNER FURNISHED OWNER INSTALLED |
| OSD | OPEN SITE DRAIN |
| P | PUMP |
| PC | PUMPED CONDENSATE |
| PCW | PROCESS COLD WATER |
| PD | PUMPED DISCHARGE (TYPE AS INDICATED) |
| PDI | PLUMBING DRAINAGE INSTITUTE |
| PH | PHASE (ELECTRICAL) |
| PPM | PARTS PER MILLION |
| PRESS | PRESSURE |
| PRV | PRESSURE REDUCING VALVE |
| PSI | POUNDS PER SQUARE INCH |
| PSIG | POUNDS PER SQUARE INCH - GAUGE |
| PV | PLUG VALVE |
| PW | PLUMBING WORK |
| RD | ROOF DRAIN |
| REF | REFERENCE |
| RL | RELOCATED MATERIAL OR EQUIPMENT |
| RPM | REVOLUTIONS PER MINUTE |
| RV | PRESSURE RELIEF VALVE |
| SAN | SANITARY OR WASTE |
| SCFM | STANDARD CUBIC FEET PER MINUTE |
| SCW | SOFTENED COLD WATER |
| SE | SEWAGE EJECTOR |
| SG | SPECIFIC GRAVITY |
| SP | SUMP PUMP |
| SPEC | SPECIFICATION |
| SQ. FT. | SQUARE FEET |
| SS | STAINLESS STEEL |
| SSD | SUB-SOIL DRAINAGE |
| ST | STORM (RAINWATER) |
| STD | STANDARD |
| STM | STEAM |
| SV | SOLENOID VALVE |
| TEMP | TEMPERATURE |
| TP | TRAP PRIMER |
| TPU | TRAP PRIMER UNIT |
| TPV | TEMPERATURE AND PRESSURE RELIEF VALVE |
| TW | TEMPERED WATER |
| TYP | TYPICAL |
| UON | UNLESS OTHERWISE NOTED |
| V | VOLTAGE/VENT |
| VAC | VACUUM |
| VEL | VELOCITY |
| VTR | VENT THRU ROOF |
| W/ | WITH |
| W/O | WITHOUT |
| WC | WATER CLOSET |
| WCO | WALL CLEAN OUT |
| WG | WATER GAUGE |
| WH | WALL HYDRANT |
| WHA | WATER HAMMER ARRESTOR |
| YD | YARD |

SEE PLUMBING SYMBOLS, LEGENDS AND EQUIPMENT SCHEDULES FOR ADDITIONAL ABBREVIATIONS. ALL ABBREVIATIONS, SYMBOLS, AND LEGENDS SHOWN ON THIS DRAWINGS ARE NOT NECESSARILY USED.

# PLUMBING FLOOR PLAN SYMBOLS

- ISOLATION VALVE
- BALANCING VALVE
- ANGLE VALVE
- CHECK VALVE
- SOLENOID VALVE
- BUTTERFLY VALVE
- BALL VALVE
- PLUG VALVE or GAS COCK
- PRESSURE REGULATOR VALVE FLANGED
- PRESSURE REGULATOR VALVE THREADED
- FLOW INDICATING BALANCE VALVE
- PRESSURE RELIEF VALVE or TEMP PRESSURE RELIEF VALVE
- STRAINER
- MECHANICAL EXPANSION JOINT
- FLEXIBLE CONNECTION
- CLEANOUT
- FLOOR CLEANOUT or EXTERIOR CLEANOUT
- FLOOR DRAIN, ROOF DRAIN
- MATERIAL OR EQUIPMENT TO BE DEMOLISHED
- MATERIAL OR EQUIPMENT EXISTING TO REMAIN
- REDUCER - CONCENTRIC
- REDUCER - ECCENTRIC
- PIPE UNION
- PIPE ANCHOR
- PIPE ALIGNMENT GUIDE/ PIPE SLEEVE
- PRESSURE GAUGE WITH COCK
- THERMOMETER
- QUICK COUPLING
- CAP OR PLUG
- PIPE UP
- PIPE DOWN
- TEE OUTLET - DOWN
- BRANCH TOP CONNECTION
- DIRECTION OF FLOW
- FLOW METER
- AIR VENT WITH VALVE
- WATER HAMMER ARRESTOR
- HOSE BIBB
- WALL HYDRANT
- POINT OF CONNECTION
- POINT OF DISCONNECT

# PLUMBING FLOOR PLAN SYMBOLS

- DCW — DOMESTIC COLD WATER
- DHW — DOMESTIC HOT WATER
- DHWR — DOMESTIC HOT WATER RETURN
- NPW — NON-POTABLE WATER
- PCW — PROCESS COLD WATER
- W — WASTE
- V — SANITARY VENT
- SAN — SANITARY DRAIN
- SAN — UNDER SLAB SANITARY DRAIN
- ST — STORM (RAINWATER) DRAIN
- OD — OVERFLOW ROOF DRAIN
- ST — UNDER SLAB STORM DRAIN
- SYS(E) — SYSTEM(EXISTING TO REMAIN)
- SYS(E) — SYSTEM(EXISTING TO BE DEMOLISHED)

# REFERENCE SYMBOLS

- 1 / E0.1 — SECTION NUMBER; DRAWING NUMBER WHERE DRAWN
- HHWS / 4 — RISER DESIGNATION (TYPE AS INDICATED)
- MXXX — KEY NOTE
- 4 — DEMO KEY NOTE

# PLUMBING GENERAL NOTES

1. INSTALL PLUMBING IN ACCORDANCE WITH THE INTERNATIONAL PLUMBING CODE 2024 AND REGULATIONS OF ALL AUTHORITIES HAVING JURISDICTION.
2. OBTAIN COMPLETE CONSTRUCTION DOCUMENTS WHICH INCLUDE BUT ARE NOT LIMITED TO ARCHITECTURAL, STRUCTURAL, MECHANICAL, PLUMBING, ELECTRICAL, AND FIRE PROTECTION DRAWINGS. ALSO OBTAIN A COMPLETE SET OF SPECIFICATIONS.
3. DURING THE COORDINATION PROCESS BRING TO THE ATTENTION OF THE ARCHITECT AND ENGINEER(S) ANY CONFLICTS. THE CONTRACTOR SHALL NOT PROCEED WITH ANY INSTALLATION EXCEPT AT HIS OWN RISK UNTIL ALL CONFLICTS ARE RESOLVED AND THE CLARIFYING INFORMATION ISSUED TO THE CONTRACTOR BY THE ARCHITECT.
4. THE CONTRACTOR SHALL VERIFY AND COORDINATE THE LOCATION OF ALL STRUCTURAL MEMBERS FOR CONFLICTS WITH PIPING. THE CONTRACTOR SHALL NOTIFY THE ARCHITECT IMMEDIATELY IF A CONFLICT IS DETERMINED.
5. PROVIDE CLEANOUTS WHERE INDICATED ON DRAWINGS AND WHERE REQUIRED BY THE APPLICABLE CODES.
6. FIRE STOP ALL OPENINGS IN COMPLIANCE WITH THE U.L. LISTING OF THE METHOD USED TO MAINTAIN THE APPROPRIATE FIRE RATING AS REQUIRED BY CODE.
7. COORDINATE ALL PIPING INSTALLED WITH ALL OTHER TRADES (I.E. STRUCTURAL, MECHANICAL, ELECTRICAL AND FIRE PROTECTION) TO ENSURE THAT ALL SYSTEMS ARE INSTALLED ABOVE THE CEILING OR IN A CONCEALED LOCATION. ALL CEILING HEIGHTS ARE INDICATED ON THE ARCHITECTURAL AND/OR INTERIOR DESIGN DRAWINGS. MAINTAIN MINIMUM CLEARANCES REQUIRED BY THE APPLICABLE CODES THROUGHOUT OUT THE BUILDING.
8. EACH FIXTURE TRAP SHALL HAVE A LIQUID TRAP SEAL, AS REQUIRED BY CODE. LIQUID TRAP SEALS SHALL BE MAINTAINED WITH A TRAP PRIMER, WHERE REQUIRED.
9. PROVIDE WATER HAMMER ARRESTORS WHERE QUICK CLOSING VALVES ARE UTILIZED. LOCATE ARRESTORS PER MANUFACTURER'S INSTRUCTIONS. ARRESTOR SHALL BE ACCESSIBLE.
10. INSULATE ALL STORM DRAIN BODIES AND HORIZONTAL PIPING. THE HORIZONTAL STORM PIPING SHALL BE INSULATED AT THE UPPER MOST LEVEL.
11. INSULATE ALL DRAINAGE PIPING RECEIVING CONDENSATE.
12. INSULATE ALL DRAINAGE AND WATER PIPING IN COMPLIANCE WITH ANSI 117.1, LOCAL CODES AND SPECIFICATIONS.
13. INSTALL VALVES AND OR ACCESSORIES SERVING EQUIPMENT TO ALLOW FOR THE REMOVAL OR SERIVICING OF THE VALVES,ACCESORIES AND EQUIPMENT.
14. WHEN PROVIDING UNIONS, VACUUM BREAKERS, STRAINERS, ETC. AS INDICATED ON DRAWINGS AND ALL EQUIPMENT, PROVIDE FOR MAINTENANCE CLEARANCE AS REQUIRED BY CODE.
15. REFER TO THE ARCHITECTURAL DRAWINGS FOR EXACT LOCATIONS OF ALL PLUMBING FIXTURES.
16. SEE THE PLUMBING FIXTURE CONNECTION SCHEDULE FOR CONNECTION SIZES.
17. ALL ABOVE CEILING WATER/LIQUID 1/4 TURN VALVES SHALL BE HORIZONTAL WITH THEIR VALVE HANDLE. PULL DOWN TOWARDS OPERATOR FOR SHUTOFF.

# DEMOLITION GENERAL NOTES

1. THE CONTRACTOR SHALL VERIFY ALL EXISTING CONDITIONS DURING THE CONTRACTOR'S PRE-BID SITE VISIT(S) AND BRING TO THE ATTENTION OF THE ARCHITECT/ENGINEER ANY DISCREPANCIES BETWEEN THE CONTRACT DOCUMENTS AND THE ACTUAL SITE CONDITIONS PRIOR TO BID FOR CLARIFICATION VIA PRE-BID RFI.
2. THE REQUIRED DEMOLITION IS NOT LIMITED TO THAT PORTION INDICATED ON THE PLANS ALONE, BUT SHALL INCLUDE ALL NECESSARY WORK INDICATED ELSEWHERE IN THE SPECIFICATIONS WHICH ARE NECESSARY TO COMPLETE THE INTENT OF THE CONTRACT DOCUMENTS.
3. THE CONTRACTOR SHALL NOT VIOLATE THE PHYSICAL SECURITY OF THE BUILDING DURING DEMOLITION OR ASSOCIATED OPERATIONS. COORDINATE DEMOLITION WITH OWNER/AGENT.
4. THE CONTRACTOR SHALL SCHEDULE WITH THE OWNER/AGENT ANY INTERRUPTIONS OF EXISITNG UTILITIES. PRIOR TO STARTING WORK THE CONTRACTOR SHALL HAVE OBTAINED WRITTEN PERMISION TO PROCEED. THE CONTRACTOR SHALL ASSUME THE BUILDING IS OCCUPIED AND OPERATIONAL AT ALL TIMES.
5. THE CONTRACTOR SHALL COORDINATE WITH ALL OTHER TRADES.
6. IT IS UNDERSTOOD THAT WORK UNDER THESE CONTRACT DOCUMENTS MAY EXTEND PAST THE LIMITS OF WORK. THE OWNER SHALL BE NOTIFIED PRIOR TO WORK STARTING IN THESE AREAS AND PROVIDE A SCHEDULE.
7. ALL FIXTURES AND EQUIIPMENT INDICATED TO BE REMOVED SHALL HAVE ALL ASSOCIATED PIPING REMOVED BACK TO THE NEAREST BRANCH, MAIN, RISER AND CAPPED IN A CONCEALED LOCATION.
8. "CONCEALED LOCATION" IS DEFINED AS BEING ABOVE, BELOW OR BEHIND A FINISHED PLANE.
9. ALL FIXTURES, EQUIPMENT, PIPING AND APPURTANCES SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND SHALL BE REMOVED IN A TIMELY MANNER. THE SALVAGE VALUE SHALL APPEAR AS A CREDIT IN THE BID.
10. ANY PIPE SHALL BE CAPPED USING A COMMERCIALLY PURCHASED CAP. A VALVE OR CRIMPING SHALL NOT CONSTITUTE A CAP AND IS UNACCEPTABLE.

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Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)

ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: PLUMBING ABBREVIATIONS /SYMBOLS/GENERAL NOTES

WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| YW | YW | PT |

SHEET NO. 311 OF 378

P-001 (DISCIPLINE: P | TYPE: - | SEQUENCE: 001 | DRAWING TYPE: -)

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1 PLUMBING UTILITY SITE PLAN
SCALE: 1 : 300

6/15/2026 3:02:42 PM
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TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

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DATE: 06/15/2026

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PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: PLUMBING UTILITY SITE PLAN

WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| YW | YW | PT |

SHEET NO. 312 OF 378

P-100SP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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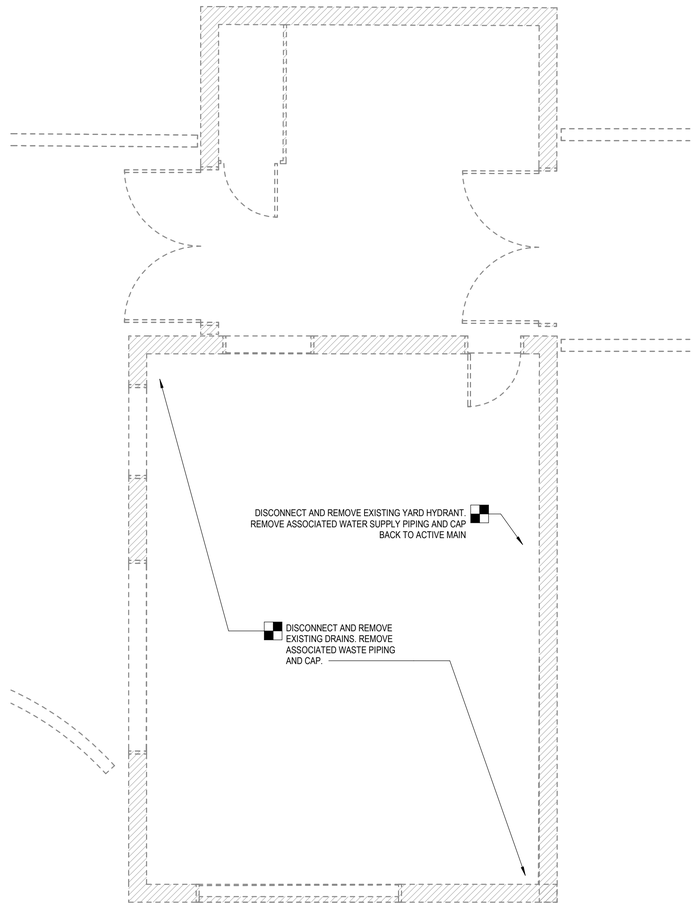


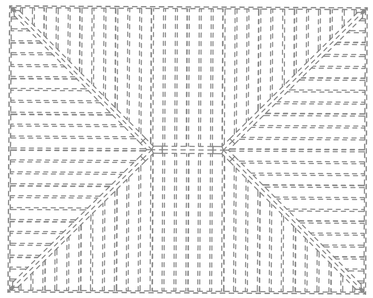

1 BISON BUILDING - PLUMBING DEMOLITION PLAN
SCALE: 1/4" = 1'-0"

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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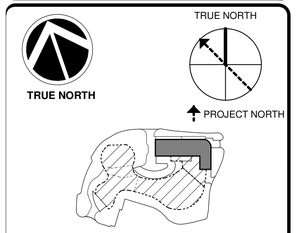
STAMP

TRUE NORTH




KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
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Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LEVEL 1 PLUMBING DEMOLITION PLAN |
| WORKING STAFF | YW (DESIGNED BY) / YW (DRAWN BY) / PT (CHECKED BY) |

SHEET NO. 313 OF 378

P-110DP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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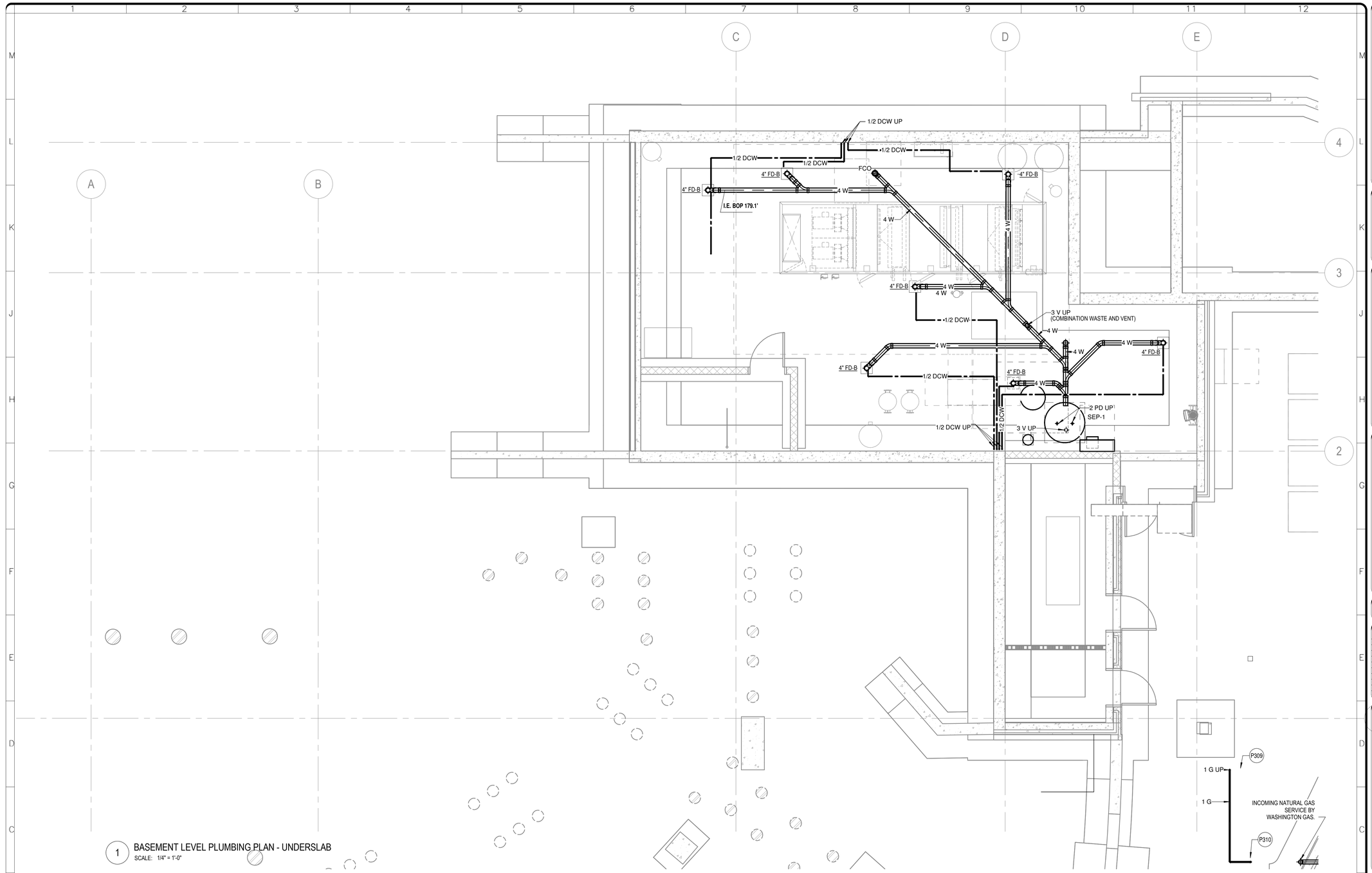
P-110DP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




1 BASEMENT LEVEL PLUMBING PLAN - UNDERSLAB
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE P-001 FOR GENERAL NOTES, SYMBOLS AND ABBREVIATIONS.

## ◯ SHEET KEYNOTES

P309 4" G UP TO EMERGENCY GENERATOR.
P310 4" G UP FROM WASHINGTON GAS METER AND REGULATOR FOR EMERGENCY GENERATOR.

## ROOM LEGEND

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BASEMENT LEVEL UNDERSLAB PLUMBING PLAN |
| WORKING STAFF | YW (DESIGNED BY) / YW (DRAWN BY) / PT (CHECKED BY) |

SHEET NO. 314 OF 378

P-100PP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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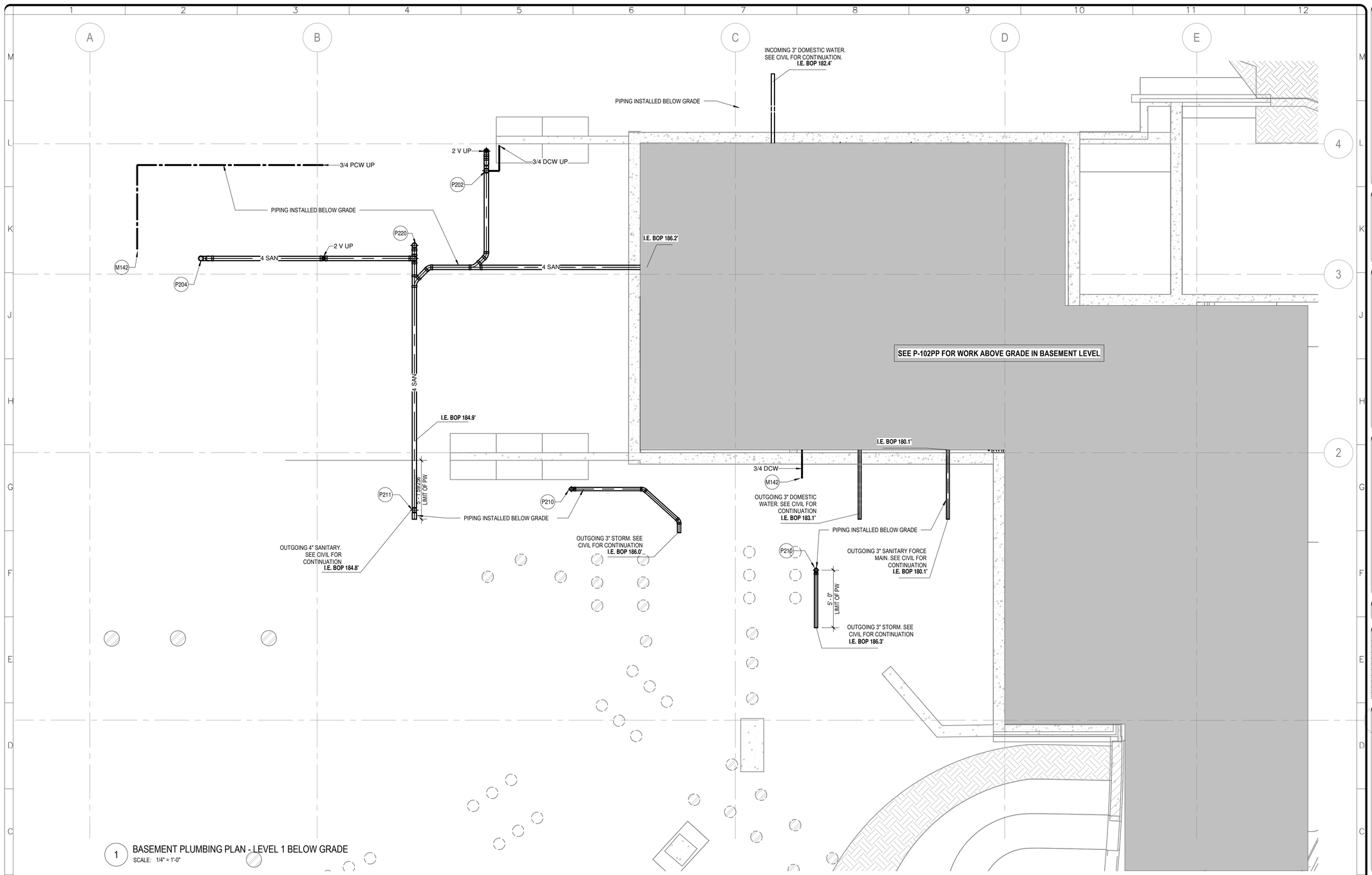


1 BASEMENT PLUMBING PLAN - LEVEL 1 BELOW GRADE
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE P-001 FOR GENERAL NOTES, SYMBOLS AND ABBREVIATIONS.

## SHEET KEYNOTES

| | |
|---|---|
| M142 | REFERENCE PLUMBING DRAWINGS FOR DRAIN LOCATION. TERMINATE OVER DRAIN PER MECHANICAL DRAWINGS AND SPECIFICATIONS. |
| P202 | 4" W&T UP, FLOOR DRAIN. |
| P204 | 4" W&T UP, TRENCH DRAIN. |
| P210 | 3" ST UP, TRENCH DRAIN. |
| P211 | 4" SAN UP, TWO-WAY CLEANOUT. |
| P220 | 4" SAN UP TO FLOOR CLEANOUT. |

## ROOM LEGEND

# ALF

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KEY PLAN

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1/4"=1'-0"

GRAPHIC SCALE(S)

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BASEMENT LEVEL PLUMBING PLAN - LEVEL 1 BELOW GRADE |
| WORKING STAFF | YW (DESIGNED BY) / YW (DRAWN BY) / PT (CHECKED BY) |
| SHEET NO. | 315 OF 378 |
| | P-101PP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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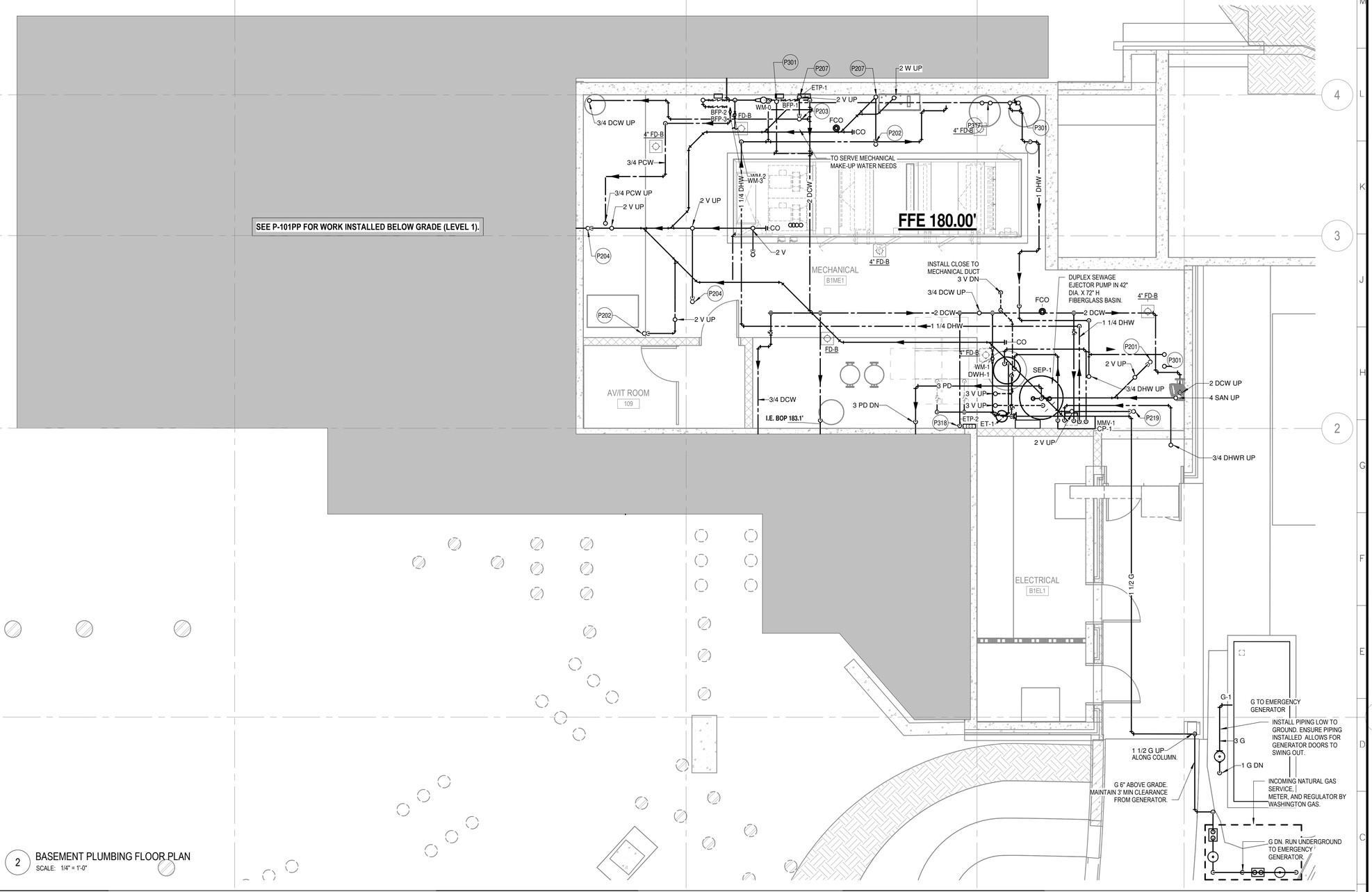


2 BASEMENT PLUMBING FLOOR PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE P-001 FOR GENERAL NOTES, SYMBOLS AND ABBREVIATIONS.

## SHEET KEYNOTES

| | |
|---|---|
| P201 | 3" W&T UP, MOP SINK. |
| P202 | 4" W&T UP, FLOOR DRAIN. |
| P203 | 4" W&T UP, FLOOR SINK. |
| P204 | 4" W&T UP, TRENCH DRAIN. |
| P207 | 4" W DN, 2" V UP |
| P219 | 2" W&T UP, FLOOR DRAIN. |
| P301 | 3/4" DCW AND DHW UP |
| P317 | 1/2" DCW & 1/2" DHW UP |
| P318 | 3/4" DN TO TRAP PRIMER. |

## ROOM LEGEND

# ALF

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TRUE NORTH

PROJECT NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

GRAPHIC SCALE(S)

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BASEMENT LEVEL PLUMBING PLAN - ABOVE GRADE |
| WORKING STAFF | YW (DESIGNED BY) / YW (DRAWN BY) / PT (CHECKED BY) |
| SHEET NO. | 316 OF 378 |
| | P-102PP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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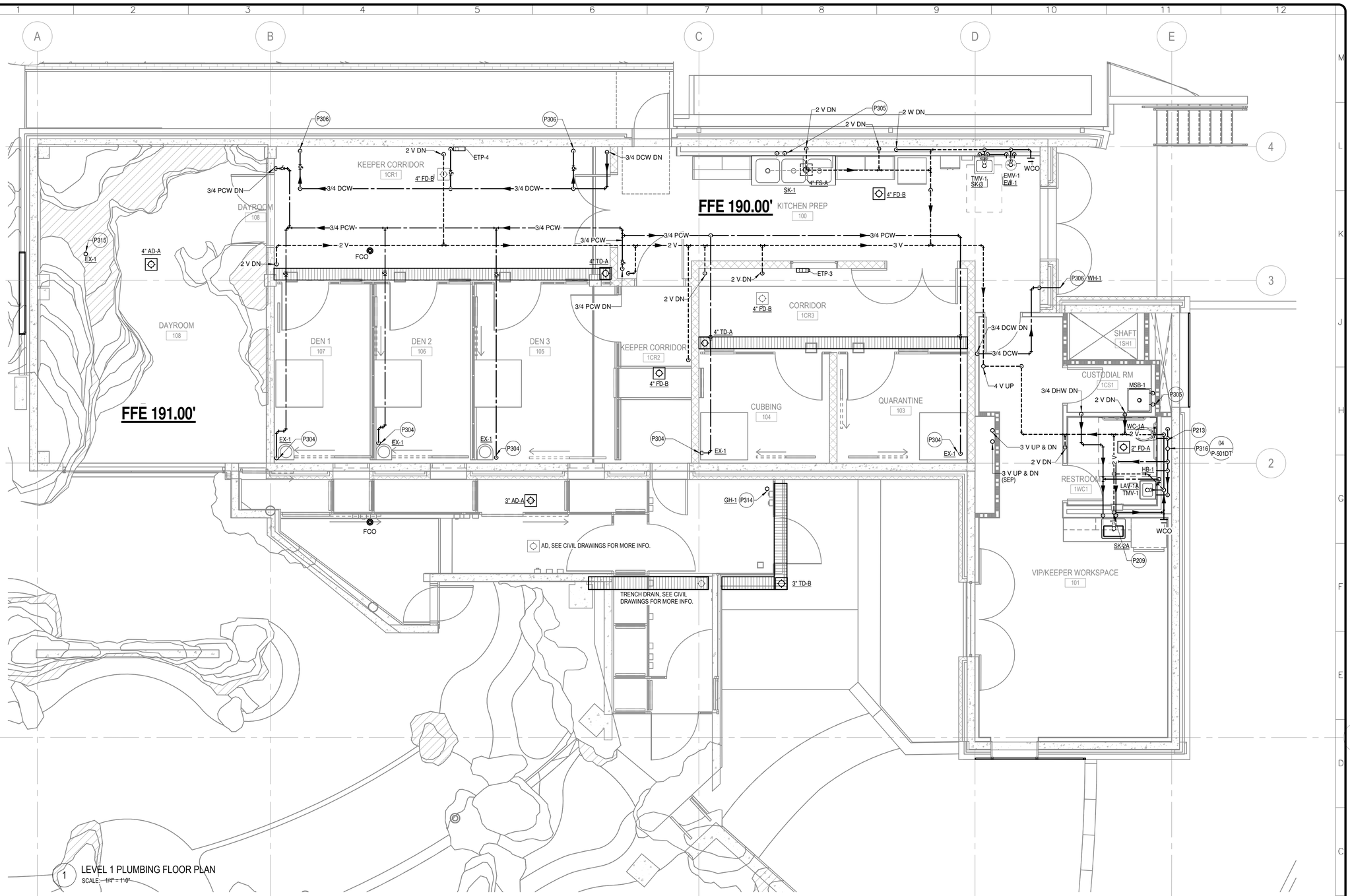
P-102PP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




1 LEVEL 1 PLUMBING FLOOR PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE P-001 FOR GENERAL NOTES, SYMBOLS AND ABBREVIATIONS.

## SHEET KEYNOTES

| | |
|---|---|
| P209 | 2" W DN, 2" V RISE |
| P213 | 4" SAN DN, 2 V RISE. |
| P304 | 3/4" PCW DP TO ANIMAL WATERER. |
| P305 | 3/4" DCW & DHW DN |
| P306 | 3/4" DCW DP TO HOSE BIBB/WALL HYDRANT. |
| P314 | 3/4" DCW DP, FROM GROUND HYDRANT. |
| P315 | 3/4" PCW DN & RISE, TO ANIMAL WATERER. |
| P316 | WATER HAMMER ARRESTOR, PROVIDE ACCESS PANEL. |

## ROOM LEGEND

# ALF

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KEY PLAN

0 2' 4' 8'
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| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LEVEL 1 PLUMBING PLAN |

| WORKING STAFF | | |
|---|---|---|
| YW | YW | PT |
| DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 317 OF 378

P-103PP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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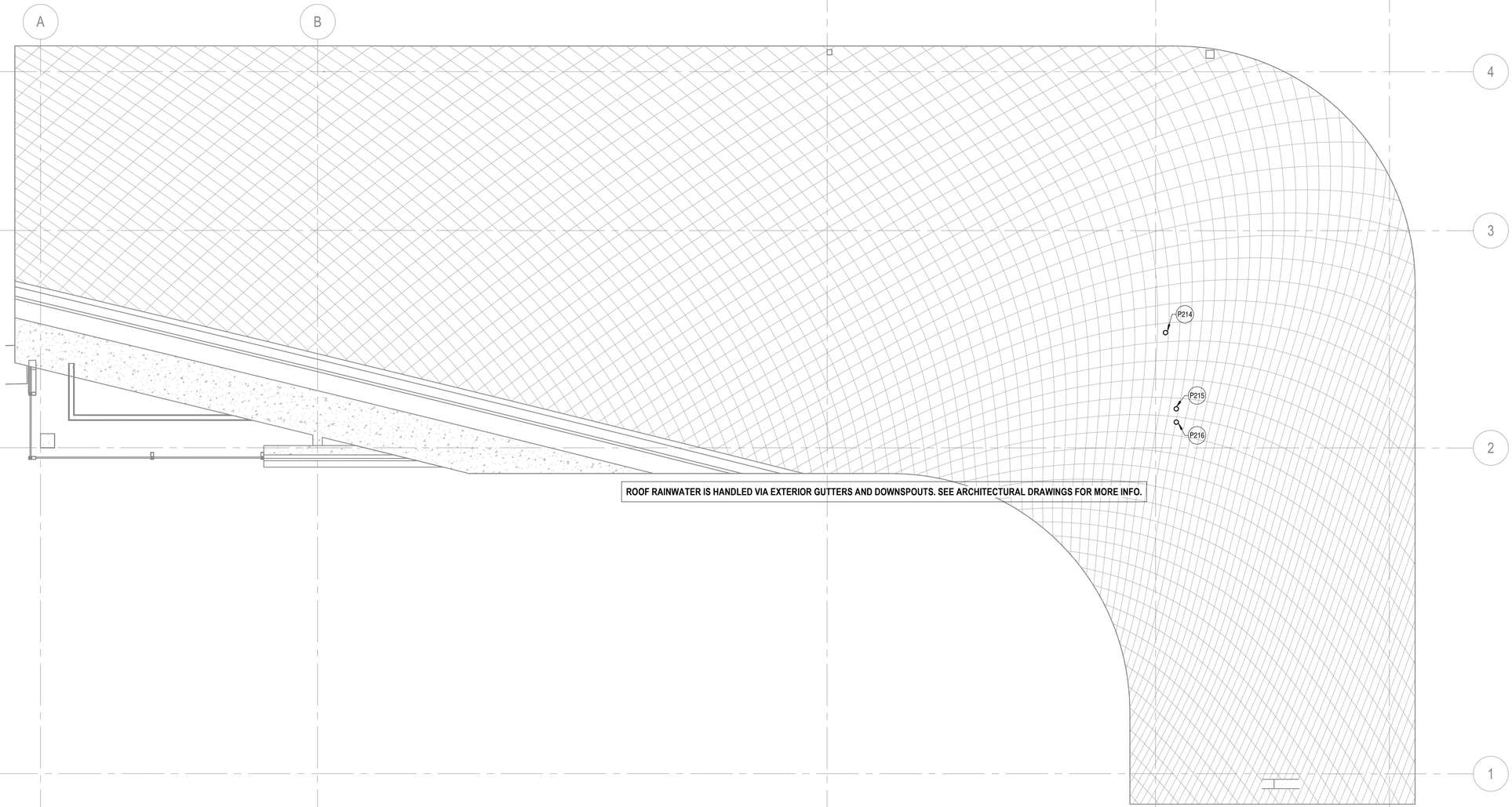
P-103PP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




ROOF RAINWATER IS HANDLED VIA EXTERIOR GUTTERS AND DOWNSPOUTS. SEE ARCHITECTURAL DRAWINGS FOR MORE INFO.

1 ROOF LEVEL PLUMBING PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE P-001 FOR GENERAL NOTES, SYMBOLS AND ABBREVIATIONS.

## SHEET KEYNOTES

P214 4" VTR.
P215 3" VTR.
P216 3" DEDICATED VTR FOR SEP.

## ROOM LEGEND

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| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ROOF PLUMBING PLAN |
| WORKING STAFF | YW (DESIGNED BY) / YW (DRAWN BY) / PT (CHECKED BY) |
| SHEET NO. | 318 OF 378 |
| | P-104PP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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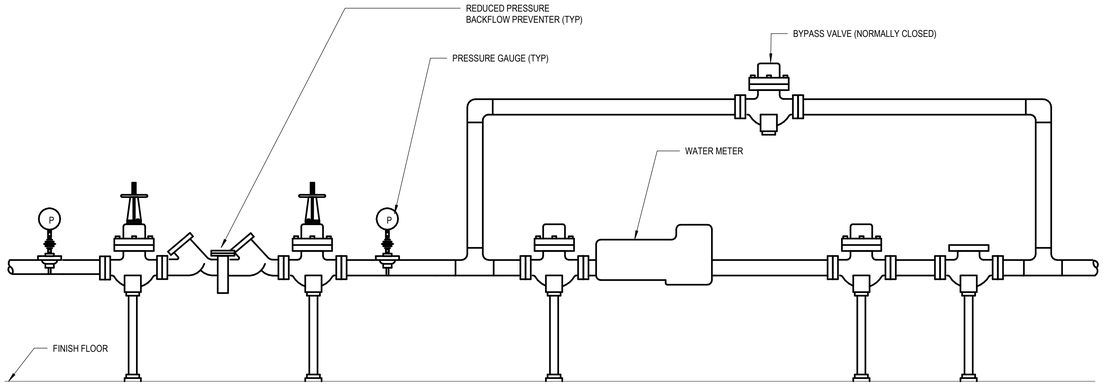
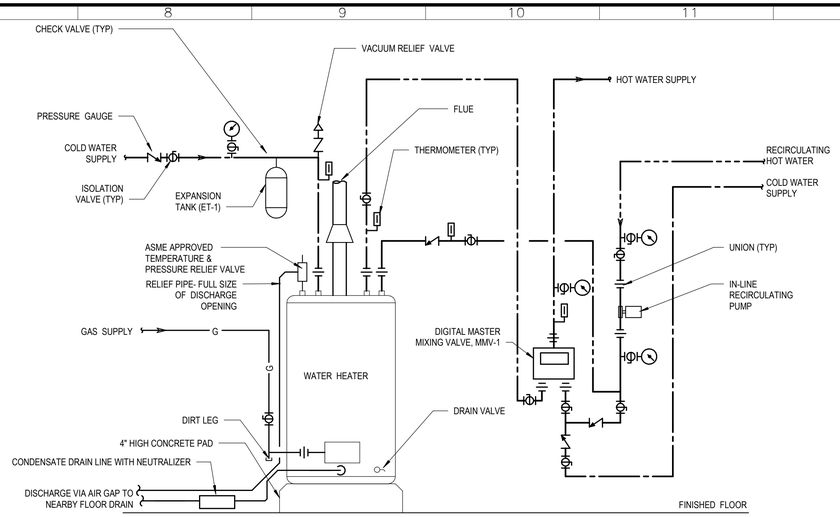
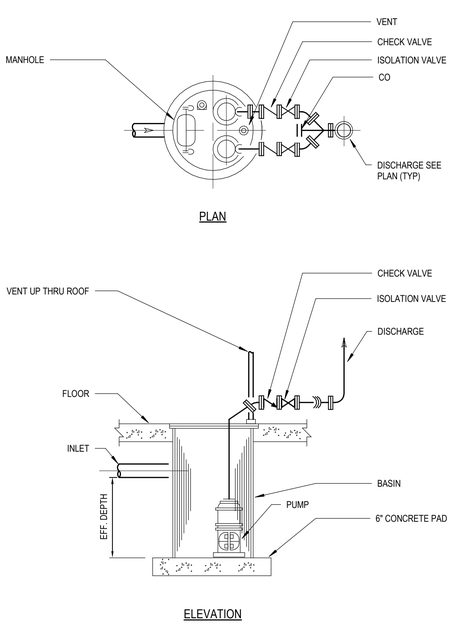



01 WATER ENTRANCE DIAGRAM
SCALE: N.T.S.



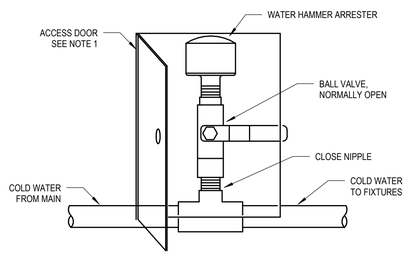

NOTE:
1. SEE FLOOR PLAN FOR CONTINUATION AND PIPE SIZES.

02 GAS FIRED STORAGE TYPE WATER HEATER
SCALE: N.T.S

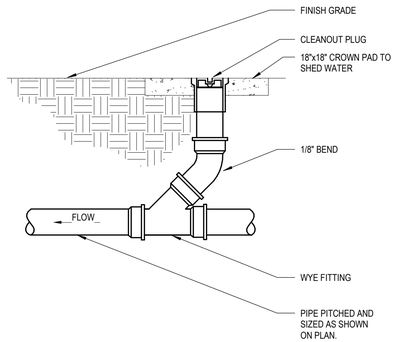



03 DUPLEX SEWAGE EJECTOR
SCALE: N.T.S

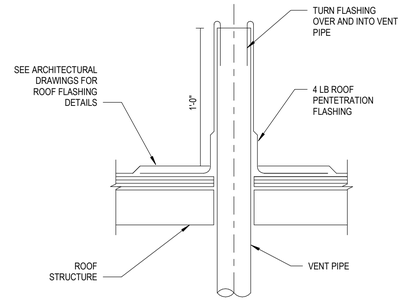
NOTES:
1. ACCESS DOOR SHALL BE 10" WIDE x 10" HIGH MINIMUM. REFER TO SPECIFICATION.
2. THIS DETAIL APPLIES TO ALL WATER CLOSET, URINAL, CLINIC SERVICE SINK AND SOLENOID QUICK-CLOSING VALVE LOCATIONS.
3. LOCATION OF ACCESS PANELS AND WATER HAMMER ARRESTER LOCATIONS SHALL BE SUBMITTED FOR ARCHITECTURE APPROVAL PRIOR TO INSTALLATION.
4. OTHER THAN THE FITTING REQUIRED BY THE ASSEMBLY SHOWN, WATER HAMMER ARRESTER SHALL BE LOCATED AS CLOSE TO MAIN AS POSSIBLE.
5. ACCESS DOORS ARE REQUIRED WHERE ARRESTORS ARE INSTALLED BEHIND SPECIALTY WALLS. CONFIRM WITH AHJ IF ACCESS DOORS ARE REQUIRED WHERE ARRESTORS ARE INSTALLED BEHIND GYPSUM BOARD.



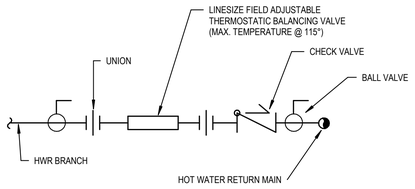

04 WATER HAMMER ARRESTER INSTALLATION
SCALE: N.T.S.



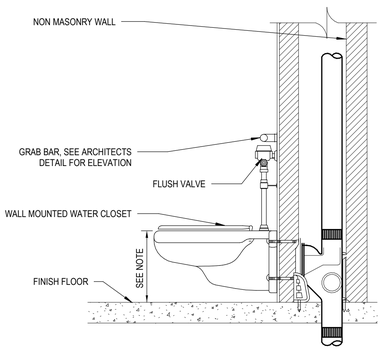

05 GRADE CLEANOUT
SCALE: N.T.S




06 VENT THRU ROOF
SCALE: N.T.S




07 BALANCING VALVE ASSEMBLY
SCALE: N.T.S.




NOTE:
ADA MOUNTING HEIGHT SHALL BE 17"-19" FROM FINISH FLOOR TO FIXTURE RIM. SEE ARCHITECTURE FOR PROJECT RIM HEIGHTS.
STANDARD USE MOUNTING HEIGHT SHALL BE 15" FROM FINISH FLOOR TO FIXTURE RIM. SEE ARCHITECTURE FOR PROJECT RIM HEIGHTS.

08 WALL MOUNTED WATER CLOSET
SCALE: N.T.S.

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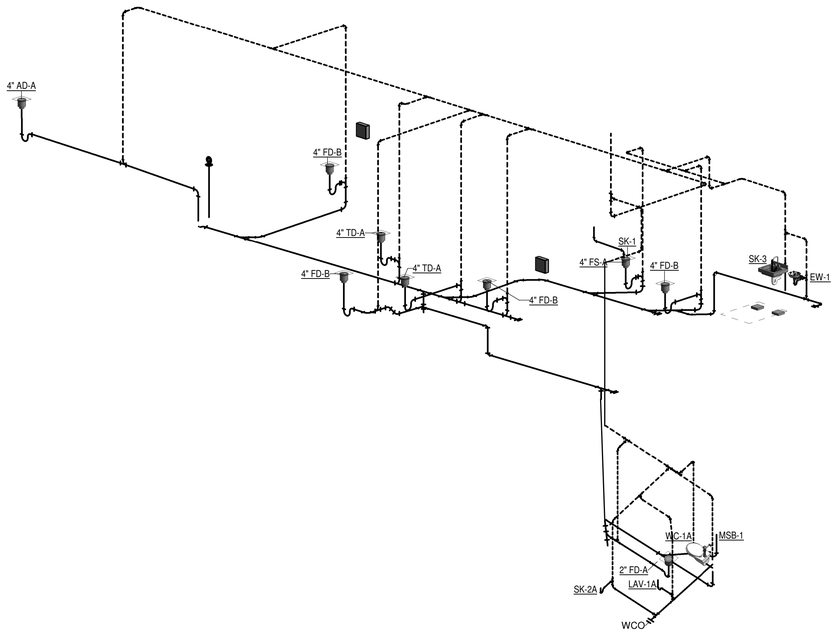
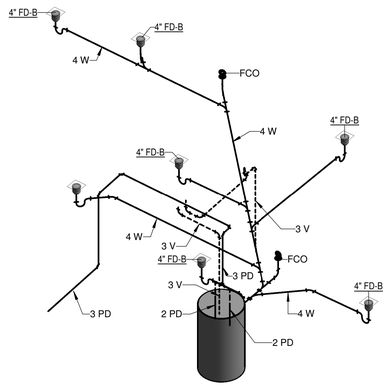
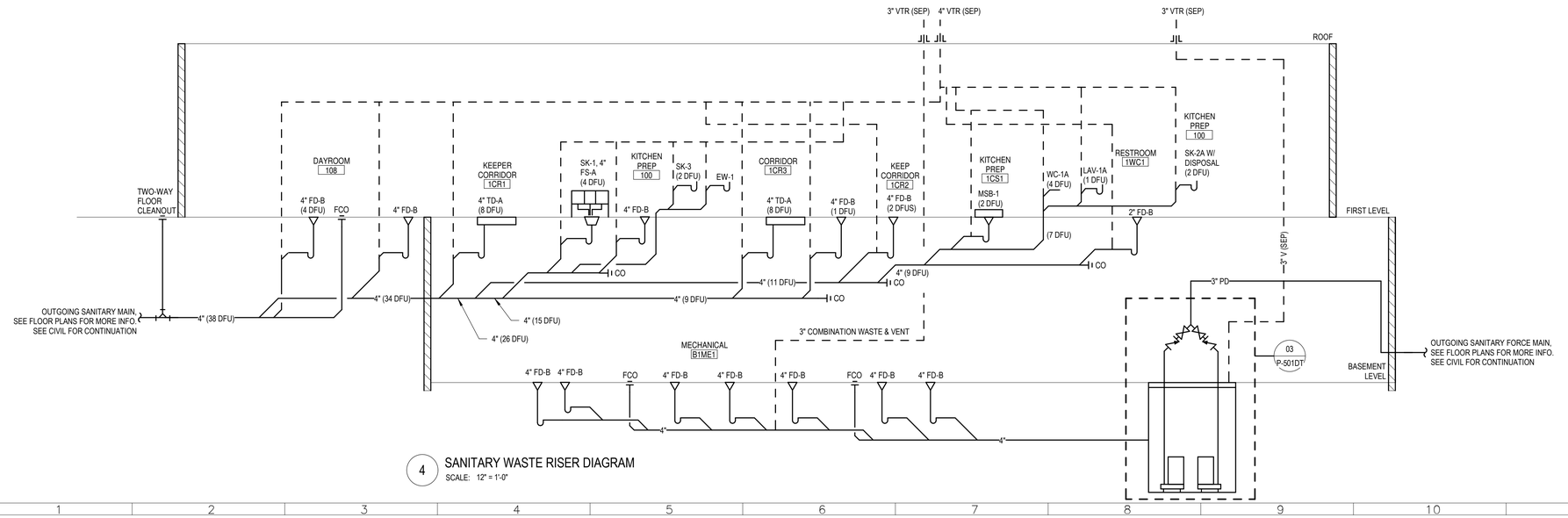
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | PLUMBING DETAILS |
| WORKING STAFF | YW / YW / PT (DESIGNED BY / DRAWN BY / CHECKED BY) |
| SHEET NO. | 319 OF 378 |
| | P-501DT (DISCIPLINE TYPE SEQUENCE DRAWING TYPE) |




1 PLBG ISO - SANITARY & VENT RISER DIAGRAM - LEVEL 1
SCALE: NOT TO SCALE




2 PLBG ISO - SANITARY & VENT RISER DIAGRAM - BASEMENT LEVEL
SCALE: NOT TO SCALE




4 SANITARY WASTE RISER DIAGRAM
SCALE: 12" = 1'-0"

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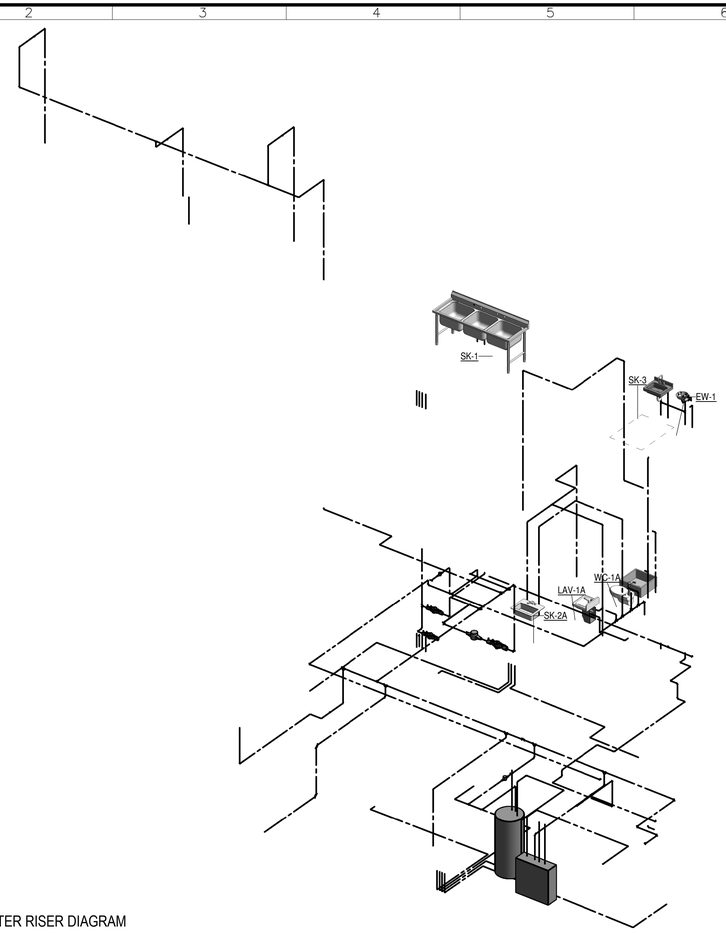
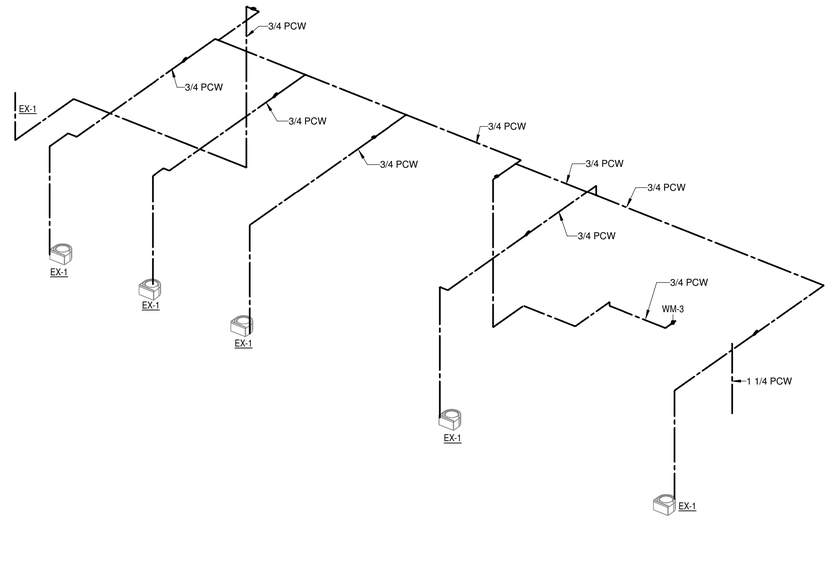
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | PLUMBING DIAGRAMS - SANITARY WASTE & VENT |
| WORKING STAFF | YW (DESIGNED BY) / YW (DRAWN BY) / PT (CHECKED BY) |

SHEET NO. 320 OF 378

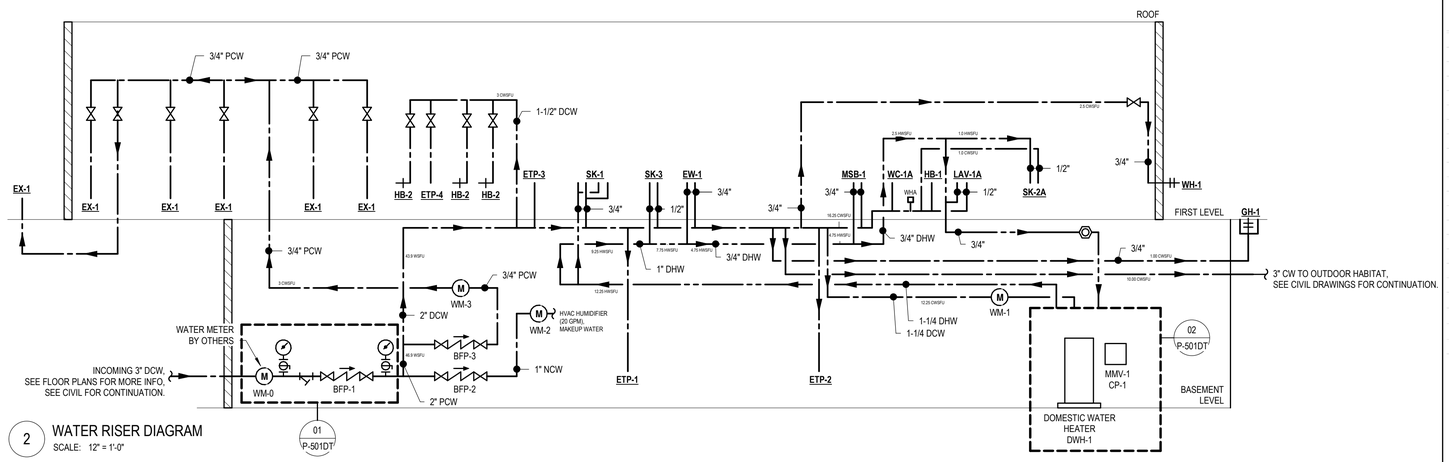
P-601DG (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)




1 PLBG ISO - DOMESTIC WATER RISER DIAGRAM
SCALE: NOT TO SCALE




3 PLBG ISO - PROCESS WATER RISER DIAGRAM
SCALE: NOT TO SCALE




2 WATER RISER DIAGRAM
SCALE: 12" = 1'-0"

| WATER CALCULATIONS - FIXTURE UNITS | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | QTY | | FU - Values | | | TOTALS | | |
| FIXTURE: | | | TOTAL | COLD | HOT | TOTAL | COLD | HOT |
| HOSE BIBB | 1 | X | 2.5 | 2.5 | | 2.5 | 2.5 | |
| HOSE BIBB, EACH ADDITIONAL | 5 | X | 1 | 1 | | 5 | 5 | |
| PANTRY SINK | 1 | X | 1.4 | 1 | 1 | 1.4 | 1 | 1 |
| 3 COMPARTMENT SINK | 1 | X | 6 | 4.5 | 4.5 | 6 | 4.5 | 4.5 |
| LAVATORY - PUBLIC | 2 | X | 2 | 1.5 | 1.5 | 4 | 3 | 3 |
| MOP SINK / SERVICE SINK | 1 | X | 3 | 2.25 | 2.25 | 3 | 2.25 | 2.25 |
| WATER CLOSET - FV - PUBLIC | 1 | X | 10 | 10 | | 10 | 10 | |
| EMERGENCY EYEWASH | 1 | X | 2 | 1.5 | 1.5 | 2 | 1.5 | 1.5 |
| ANIMAL WATERER | 6 | X | 0.5 | 0.5 | | 3 | 3 | |
| WATER TO HABITAT ( BY CIVIL) | 1 | X | 10 | 10 | | 10 | 10 | |
| BUILDING FIXTURE UNITS = | | | | | | 46.9 | 42.75 | 12.25 |

| | TOTAL | COLD | HOT |
|---|---|---|---|
| TOTAL FU = | 46.9 | 42.75 | 12.25 |

| | |
|---|---|
| HOT WATER FLOW RATE (GPM) = | 16 |
| COLD WATER FLOW RATE (GPM) = | 47 |
| FIXTURE FLOW RATE (GPM) = | 48.8 |
| FIXTURE + HVAC FLOW RATE (GPM) = | 68.8 |

# ALF

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3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: PLUMBING DIAGRAMS - WATER

WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| YW | YW | PT |

SHEET NO. 321 OF 378

P-602DG
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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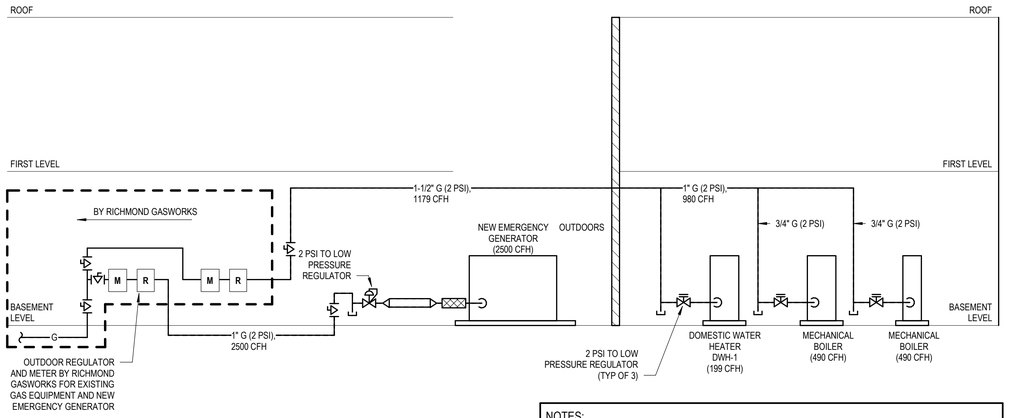



01 NATURAL GAS RISER DIAGRAM
SCALE: NOT TO SCALE

NOTES:
- PROVIDE MAIN GAS SHUTOFF VALVE TO EMERGENCY GENERATOR ON SUPPLY SIDE OF BUILDING GAS SUPPLY SYSTEM PER NFPA 110 7.9.7
- TAG BUILDING MAIN GAS SHUTOFF VALVE PER NFPA 110 7.9.8.
- GAS PIPING SIZED PER
  1. IPC 2024 HYBRID PRESSURE METHOD NOTED IN APPENDIX A A3.3.3,
  2. CHAPTER 4 TABLE 402.4(5) SCHEDULE 40 METALLIC PIPE, INLET PRESSURE = 2.0 PSI, PRESSURE DROP = 1.0 PSI,
  3. AND CHAPTER 4 TABLE 402.4(2) SCHEDULE 40 METALLIC PIPE, INLET PRESSURE = LESS THAN 2.0 PSI, PRESSURE DROP = 0.5 IN WC
  4. BUILDING 2 PSI RUN: TOTAL DEVELOPED LENGTH = 80 FT X 1.5 = 120 FT
  5. BUILDING LESS THAN 2 PSI RUN: TOTAL DEVELOPED LENGTH = 5 FT X 1.5 = 8 FT
  6. GENERATOR 2 PSI RUN: TOTAL DEVELOPED LENGTH = 15 FT X 1.5 = 23 FT
  7. GENERATOR 2 LESS THAN 2 PSI RUN: 5 FT X 1.5 = 8 FT

**TABLE 402.4(5)— SCHEDULE 40 METALLIC PIPE**

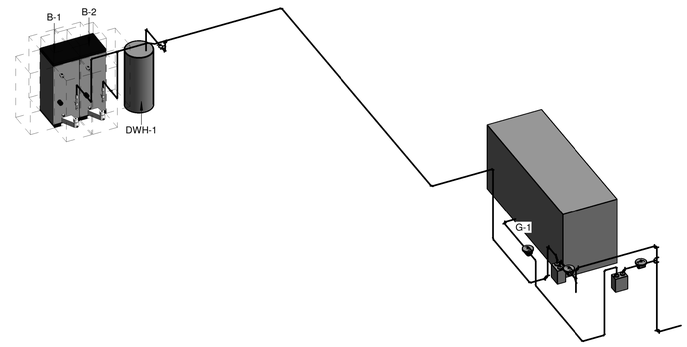
| | |
|---|---|
| **Gas** | Natural |
| **Inlet Pressure** | 2.0 psi |
| **Pressure Drop** | 1.0 psi |
| **Specific Gravity** | 0.60 |

| PIPE SIZE (inch) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Nominal** | **1/2** | **3/4** | **1** | **1 1/4** | **1 1/2** | **2** | **2 1/2** | **3** | **4** |
| **Actual ID** | **0.622** | **0.824** | **1.049** | **1.380** | **1.610** | **2.067** | **2.469** | **3.068** | **4.026** |
| **Length (ft)** | **Capacity in Cubic Feet of Gas per Hour** | | | | | | | | |
| 10 | 1,510 | 3,040 | 5,560 | 11,400 | 17,100 | 32,900 | 52,500 | 92,800 | 189,000 |
| 20 | 1,070 | 2,150 | 3,930 | 8,070 | 12,100 | 23,300 | 37,100 | 65,600 | 134,000 |
| 30 | 869 | 1,760 | 3,210 | 6,590 | 9,880 | 19,000 | 30,300 | 53,600 | 109,000 |
| 40 | 753 | 1,520 | 2,780 | 5,710 | 8,550 | 16,500 | 26,300 | 46,400 | 94,700 |
| 50 | 673 | 1,360 | 2,490 | 5,110 | 7,650 | 14,700 | 23,500 | 41,500 | 84,700 |
| 60 | 615 | 1,240 | 2,270 | 4,660 | 6,980 | 13,500 | 21,400 | 37,900 | 77,300 |
| 70 | 569 | 1,150 | 2,100 | 4,320 | 6,470 | 12,500 | 19,900 | 35,100 | 71,600 |
| 80 | 532 | 1,080 | 1,970 | 4,040 | 6,050 | 11,700 | 18,600 | 32,800 | 67,000 |
| 90 | 502 | 1,010 | 1,850 | 3,810 | 5,700 | 11,000 | 17,500 | 30,900 | 63,100 |
| 100 | 462 | 934 | 1,710 | 3,510 | 5,260 | 10,100 | 16,100 | 28,500 | 58,200 |
| 125 | 414 | 836 | 1,530 | 3,140 | 4,700 | 9,060 | 14,400 | 25,500 | 52,100 |
| 150 | 372 | 751 | 1,370 | 2,820 | 4,220 | 8,130 | 13,000 | 22,900 | 46,700 |

**TABLE 402.4(2)— SCHEDULE 40 METALLIC PIPE**

| | |
|---|---|
| **Gas** | Natural |
| **Inlet Pressure** | Less than 2 psi |
| **Pressure Drop** | 0.5 in. w.c. |
| **Specific Gravity** | 0.60 |

| PIPE SIZE (inch) | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Nominal** | **1/2** | **3/4** | **1** | **1 1/4** | **1 1/2** | **2** | **2 1/2** | **3** | **4** | **5** | **6** | **8** | **10** | **12** |
| **Actual ID** | **0.622** | **0.824** | **1.049** | **1.380** | **1.610** | **2.067** | **2.469** | **3.068** | **4.026** | **5.047** | **6.065** | **7.981** | **10.020** | **11.938** |
| **Length (ft)** | **Capacity in Cubic Feet of Gas per Hour** | | | | | | | | | | | | | |
| 10 | 172 | 360 | 678 | 1,390 | 2,090 | 4,020 | 6,400 | 11,300 | 23,100 | 41,800 | 67,600 | 139,000 | 252,000 | 399,000 |
| 20 | 118 | 247 | 466 | 957 | 1,430 | 2,760 | 4,400 | 7,780 | 15,900 | 28,700 | 46,500 | 95,500 | 173,000 | 275,000 |
| 30 | 95 | 199 | 374 | 768 | 1,150 | 2,220 | 3,530 | 6,250 | 12,700 | 23,000 | 37,300 | 76,700 | 139,000 | 220,000 |



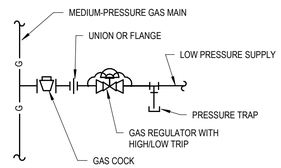

02 PLBG ISO - GAS RISER DIAGRAM
SCALE: NOT TO SCALE

## PE - NEW NATURAL GAS EQUIPMENT SCHEDULE

| TAG | DESCRIPTION | LOCATION | SUPPLY PRESSURE (IN WC) | NATURAL GAS (CFH) | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|
| | | | | | MANUFACTURER | MODEL | |
| B-1 | ENDURA XE EXE-399-500 MKII CONDENSING HYDRONIC BOILER | BASEMENT MECHANICAL ROOM | | 490 | Fulton | Fulton Thermal Corp. | |
| B-2 | ENDURA XE EXE-399-500 MKII CONDENSING HYDRONIC BOILER | BASEMENT MECHANICAL ROOM | | 490 | Fulton | Fulton Thermal Corp. | |
| DWH-1 | GAS FIRED WATER HEATER | MECHANICAL B1ME1 | | 199 | PVI INDUSTRIES | | CONNECT TO BAS. |
| TOTAL NATURAL GAS LOAD | | | | 1179 | | | |

## PE - NEW NATURAL GAS EQUIPMENT SCHEDULE - GENERATOR

| TAG | DESCRIPTION | LOCATION | SUPPLY PRESSURE (IN WC) | NATURAL GAS (CFH) | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|
| | | | | | MANUFACTURER | MODEL | |
| G-1 | EMERGENCY GENERATOR | OUTDOORS, BASEMENT LEVEL | 11 | 2500 | SEE ELEC DWGS | SEE ELEC DWGS | SEE ELEC DWGS FOR ADDITIONAL INFO. |
| TOTAL NATURAL GAS LOAD | | | | 2500 | | | |




03 MEDIUM TO LOW PRESSURE GAS REGULATOR
SCALE: N.T.S.

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

**GRAPHIC SCALE(S)**

DATE 06/15/2026
SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE
REVISION DESCRIPTION


Smithsonian Institution
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520


BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

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OPDC PROJECT NUMBER: 2533105
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WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| YW | YW | PT |

SHEET NO. 322 OF 378
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DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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## PE - PLUMBING FIXTURE SCHEDULE

| TAG | DESCRIPTION | FLOW RATE (GPM/GPF) | SERVICE SIZES | | | | BASIS OF DESIGN | | | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | CW | HW | WASTE PIPE | VENT PIPE | MANUFACTURER | MODEL | FAUCET MANUFACTURER | FAUCET MODEL | |
| EW-1 | EMERGENCY EYE WASH | 3.0 | 1/2 | 1/2 | 2 | 2 | GUARDIAN | GBF1721 | N/A | N/A | WITH EMV-1 |
| LAV-1A | LAVATORY (PUBLIC TLT) | 0.35 | 1/2 | 1/2 | 1 1/2 | 1 1/2 | AMERICAN STANDARD | LUCERNE | SLOAN | EBF-85 | HARDWIRED, AUTOMATIC SENSOR. WITH TMV-1.<br>2-HOLE SINK FOR FAUCET AND SOAP DISPENSER.<br>SOAP DISPENSER: ESD-800 (HARDWIRED) |
| MSB-1 | MOP SERVICE BASIN | - | 3/4 | 3/4 | 3 | 2 | FIAT | TSB3000 | FIAT | 830AA | TERRAZO 24" X 24" MOP BASIN. FAUCET WITH VACUUM BREAKER, INTEGRAL STOPS, PAIL HOOK, AND 3/4" HOSE THREAD |
| SK-1 | 3-COMPARTMENT SINK | - | 1/2 | 1/2 | 2 | 0 | ELKAY | | | | |
| SK-2A | PANTRY SINK | 1.5 | 1/2 | 1/2 | 1 1/2 | 1 1/2 | ELKAY | | CHICAGO FAUCETS | 434-ABCP | |
| SK-3 | HANDWASH SINK | 0.35 | 1/2 | 1/2 | 1 1/2 | 1 1/2 | ADVANCE TABCO | 7-PS-71 | CHICAGO FAUCETS | 116.114.AB.4T | STAINLESS STEEL WALL MOUNTED, WALL MOUNTED AUTOMATIC FAUCET WITH TMV-1<br>PROVIDE 0.35 GPM AERATOR. |
| WC-1A | WATER CLOSET (HANDICAP) | 1.1 | 1 1/4 | 0 | 4 | 2 | AMERICAN STANDARD | AFWALL | SLOAN | | HARDWIRED, AUTOMATIC SENSOR.WATERSENSE LABELED |

## PE - PLUMBING SPECIALTIES SCHEDULE

| TAG | DESCRIPTION | INLET SIZE | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|
| | | | MANUFACTURER | MODEL | |
| GH-1 | NONFREEZE BOXED GROUND HYDRANT | 3/4" | JR SMITH | 5812 | |
| HB-1 | HOSE BIBB | 3/4" | JR SMITH | 5618 | |
| WH-1 | NON FREEZE WALL HYDRANT | 3/4" | JR SMITH | 5515 | 1/4 TURN NON-FREEZE HYDRANT WITH AUTOMATIC DRAINING INTEGRAL VACUUM BREAKER, INTEGRAL SERVICE SHUT-OFF VALVE, DUAL CHECK VALVE, AND STAINLESS STEEL BOX |

## PE - PLUMBING FIXTURE DRAIN SCHEDULE

| TAG | DESCRIPTION | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|
| | | MANUFACTURER | MODEL | |
| AD-A | AREA DRAIN | JR SMITH | 2250 | HEAVY DUTY LARGE CAPACITY DRAIN WITH SEDIMENT BUCKET |
| FD-A | FLOOR DRAIN, FINISH FLOOR | JR SMITH | 2005 | LIGHTY DUTY |
| FD-B | FLOOR DRAIN | JR SMITH | - | |
| FS-A | FLOOR SINK | JR SMITH | 3003 | STAINLESS STEEL, 8" DEEP DRAIN BODY WITH STAINLESS STEEL DOME STRAINER. |
| TD-A | TRENCH DRAIN | BLUCHER | - | STAINLESS STEEL TRENCH DRAIN WITH CUSTOM FRP GRATE. |
| TD-B | TRENCH DRAIN, OUTDOORS | BY OTHERS | BY OTHERS | |

## PE - DOMESTIC WATER HEATER SCHEDULE

| TAG | SYSTEM | LOCATION | TYPE | STORAGE CAPACITY | RECOVERY | | EWT | LWT | GAS | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | GPH | DELTA T | | | INPUT CAPACITY | VENT DIAMETER | EFFICIENCY | MANUFACTURER | MODEL | |
| DWH-1 | DOMESTIC HOT WATER | MECHANICAL B1ME1 | CONDENSING GAS FIRED STORAGE | 25.0 gal | | 100 °F | 40 °F | 140 °F | 199000 Btu/h | 3" | | PVI INDUSTRIES | BRIGADE | CONNECT TO BAS. |

## PE - MIXING VALVE SCHEDULE

| TAG | DESCRIPTION | LOCATION | OUTLET TEMP | FLOW CAPACITY | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|
| | | | | | MANUFACTURER | MODEL | |
| EMV-1 | EMERGENCY MIXING VALVE | KITCHEN PREP 100 | 85 °F | 2.7 GPM AT 10 PSI DROP | LEONARD VALVE | 6020 LF | ASSE 1071 COMPLIANT |
| MMV-1 | MASTER MIXING VALVE | MECHANICAL B1ME1 | 120 °F | - | WATTS | INTELLISTATION 2 | ASSE 1017 COMPLIANT |
| TMV-1 | POINT OF USE THERMOSTATIC MIXING VALVE | SEE PLANS | 105 °F | 1.25 GPM AT 10 PSI DROP | LEONARD VALVE | 170D-LF | ASSE 1070 COMPLIANT |

## PE - PLUMBING PUMP SCHEDULE

| TAG | SYSTEM | LOCATION | FLOW | DISCHARGE HEAD | MINIMUM NPSH REQ'D | PIPE CONNECTIONS | | MOTOR DATA | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | SUCTION | DISCHARGE | HP | RPM | VOLTAGE | PHASE | MANUFACTURER | MODEL | |
| CP-1 | DOMESTIC HOT WATER | MECHANICAL B1ME1 | 5 GPM | 21.0 ftH2O | 0" | 0 | 0 | 0.093872 | 0 | 120 V | 1 | BELL & GOSSETT | ECOCIRC 20 18 | |
| SEP-1 | PUMPED SEWAGE | MECHANICAL B1ME1 | 30 GPM | 27.0 ftH2O | 0" | 2 | 2 | 1 | 3380 | 480 V | 3 | TSURUMI PUMP | 50C2.75-CR-63 | DUPLEX, N+1 SEWAGE EJECTOR CUTTER PUMPS.<br>WITH 42" DIAMETER X 72" H FIBERGLASS BASIN. CONNECT LEVEL ALARMS TO THE BAS. |

## PE - BACKFLOW PREVENTER SCHEDULE

| TAG | TYPE | SYSTEM | LOCATION | PIPE INLET/OUTLET | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|
| | | | | | MANUFACTURER | MODEL | |
| BFP-1 | RPZ BACKFLOW PREVENTER | DOMESTIC WATER | MECHANICAL B1ME1 | 2 | WATTS | 009-M2-QT | WITH FLOOD SENSOR AND AUTOMATIC SHUTDOWN CONTROL VALVE (BASIS OF DESIGN: WATTS LFF113FP). CONNECT TO BAS. |
| BFP-2 | RPZ BACKFLOW PREVENTER | NON-POTABLE WATER | MECHANICAL B1ME1 | 1 | WATTS | 009-M2-QT | WITH FLOOD SENSOR AND AUTOMATIC SHUTDOWN CONTROL VALVE (BASIS OF DESIGN: WATTS LFF113FP). |
| BFP-3 | RPZ BACKFLOW PREVENTER | PROCESS WATER | MECHANICAL B1ME1 | 1 | WATTS | 009-M2-QT | WITH FLOOD SENSOR AND AUTOMATIC SHUTDOWN CONTROL VALVE (BASIS OF DESIGN: WATTS LFF113FP). |

## PE - WATER METER SCHEDULE

| TAG | TYPE | LOCATION | PIPE INLET/OUTLET (MM) | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|
| | | | | MANUFACTURER | MODEL | |
| WM-0 | | | 2 | BY WATER UTILITY | BY WATER UTILITY | BUILDING WATER METER PROVIDED BY WATER UTILITY. |
| WM-1 | | | 1 | BADGER RECORDALL | MODEL 25 | DOMESTIC HOT WATER SUBMETER |
| WM-2 | | | 3/4 | BADGER RECORDALL | MODEL 25 | MECHANICAL MAKEUP WATER SUBMETER |
| WM-3 | | | 3/4 | BADGER RECORDALL | MODEL 25 | PROCESS COLD WATER SUBMETER |

NOTES: ALL WATER METERS SHALL BE CONNECTED TO THE BAS. SEE MECH DRAWINGS FOR MORE INFO.

## PE - PLUMBING ANIMAL DRINKERS SCHEDULE

| TAG | DESCRIPTION | LOCATION | INLET SIZE | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|
| | | | | MANUFACTURER | MODEL | |
| EX-1 | ANIMAL DRINKER | DEN 1, 107 | 3/4" | SEE EX SERIES DRAWINGS | SEE EX SERIES DRAWINGS | |
| EX-1 | ANIMAL DRINKER | DEN 2, 106 | 3/4" | SEE EX SERIES DRAWINGS | SEE EX SERIES DRAWINGS | |
| EX-1 | ANIMAL DRINKER | DEN 3, 105 | 3/4" | SEE EX SERIES DRAWINGS | SEE EX SERIES DRAWINGS | |
| EX-1 | ANIMAL DRINKER | CUBBING, 104 | 3/4" | SEE EX SERIES DRAWINGS | SEE EX SERIES DRAWINGS | |
| EX-1 | ANIMAL DRINKER | QUARANTINE, 103 | 3/4" | SEE EX SERIES DRAWINGS | SEE EX SERIES DRAWINGS | |
| EX-1 | ANIMAL DRINKER | DEN 1, 107 | 3/4" | SEE EX SERIES DRAWINGS | SEE EX SERIES DRAWINGS | |

NOTES: SHOWN FOR COORDINATION PURPOSES ONLY, REFER TO EX SERIES DRAWINGS FOR MORE INFO.

FOR REFERENCE ONLY

ALF

CONSTRUCT ARABIAN LEOPARD FACILITY

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DATE 06/15/2026

SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

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SHEET NO. 323 OF 378

P-601SH

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

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# FIRE ALARM ABBREVIATIONS

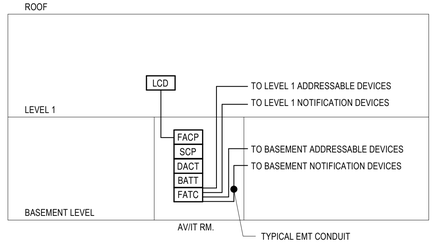
| | | |
|---|---|---|
| A | A | AMPERES |
| | AC | ALTERNATING CURRENT |
| | AF | AMPERE FRAME (BREAKER RATING) |
| | AFC | ABOVE FINISHED COUNTER |
| | AFF | ABOVE FINISHED FLOOR |
| | AFG | ABOVE FINISHED GRADE |
| | AHU | AIR HANDLING UNIT |
| | AIC | AMPERE INTERRUPTING CAPACITY |
| | ALT | ALTERNATE |
| | ARCH | ARCHITECT |
| | AT | AMPERES TRIP |
| | ATS | AUTOMATIC TRANSFER SWITCH |
| | AUTO | AUTOMATIC |
| | AUX | AUXILIARY |
| | AWG | AMERICAN WIRE GAUGE |
| B | BRKR, BKR | BREAKER |
| C | C | CONDUIT |
| | CATV | CABLE TELEVISION |
| | CB | CIRCUIT BREAKER |
| | CCTV | CLOSED CIRCUIT TELEVISION |
| | CKT | CIRCUIT |
| | CLG | CEILING |
| | COM, COMM | COMMUNICATIONS |
| | COTR | CONTRACTING OFFICER'S TECHNICAL REPRESENTATITIVE |
| | CP | CONTROL PANEL |
| | CPT | CONTROL POWER TRANSFORMER |
| | CT | CURRENT TRANSFORMER |
| | CU | COPPER |
| | BLDG | BUILDING |
| D | DC | DIRECT CURRENT |
| | DISC | DISCONNECT |
| | DIST | DISTRIBUTION |
| | DIV | DIVISION |
| | DN | DOWN |
| | DP | DISTRIBUTION PANEL |
| | DPDT | DOUBLE POLE DOUBLE THROW |
| | DPST | DOUBLE POLE SINGLE THROW |
| | DWG | DRAWING |
| | E, EX, EXIST | EXISTING |
| E | EG | EQUIPMENT GROUND |
| | ELEC | ELECTRIC, ELECTRICAL |
| | ELEV | ELEVATOR |
| | EM, EMERG | EMERGENCY |
| | EMT | ELECTRIC METALLIC CONDUIT |
| | ENCL | ENCLOSURE |
| | EQ, EQUIP | EQUIPMENT |
| | EWC | ELECTRIC WATER COOLER |
| | EWH | ELECTRIC WATER HEATER |
| | F | FUSE |
| F | FA | FIRE ALARM |
| | FAAP | FIRE ALARM ANNUNCIATOR PANEL |
| | FACP | FIRE ALARM CONTROL PANEL |
| | FC | FOOT CANDLE |
| | FDR | FEEDER |
| | FL, FLR | FLOOR |
| | FLA | FULL LOAD AMPS |
| | FLEX | FLEXIBLE |
| | FLUOR | FLUORESCENT |
| | FT | FOOT/FEET (') |
| | NAC | FIRE ALARM EXTENDER PANEL |
| G | G, GND, GRD | GROUND |
| | GEN | GENERATOR |
| | GFI | GROUND FAULT INTERRUPTER |
| H | H | HORIZONTAL MOUNTING |
| | HH | HAND HOLE |
| | HOA | HAND-OFF-AUTOMATIC |
| | HP | HORSEPOWER |
| | HPS | HIGH PRESSURE SODIUM |
| | HR | HOUR |
| | HT | HEIGHT |
| | HTR | HEATER |
| | HVAC | HEATING VENTILATION AND AIR CONDITIONING |
| | HZ | HERTZ |
| I | IG | ISOLATED GROUND |
| | IN | INCH/INCHES (") |
| | INCAND | INCANDESCENT |
| J | JB, JBOX | JUNCTION BOX |
| K | K | KEY INTERLOCK |
| | Kcmil | 1000 CIRCULAR mils |
| | KV | KILOVOLTS (THOUSAND VOLTS) |
| | KVA | KILOVOLTS-AMPERES (THOUSAND VOLT-AMPS) |
| | kW | KILOWATTS (THOUSAND WATTS) |
| | KWH | KILOWATT-HOURS |
| | L-G | LINE TO GROUND |
| | L-L | LINE TO LINE |
| | L-N | LINE TO NEUTRAL |
| | LA | LIGHTNING ARRESTOR |
| L | LAN | LOCAL AREA NETWORK |
| | LC, LCP | LIGHTING CONTROL PANEL |
| | LP | LIGHTING PANEL |
| | LSI | LONG-TIME, SHORT-TIME, INSTANTANEOUS |
| | LSIA | LONG-TIME, SHORT-TIME, INSTANTANEOUS, GROUND-FAULT ALARM ONLY |
| | LSIG | LONG-TIME, SHORT-TIME, INSTANTANEOUS, GROUND-FAULT |
| | LTG | LIGHTING |
| | LV | LOW VOLTAGE (BELOW 50 VOLTS) |
| M | MCA | MINIMUM CIRCUIT AMPACITY |
| | MCB | MAIN CIRCUIT BREAKER |
| | MCC | MOTOR CONTROL CENTER |
| | MECH | MECHANICAL |
| | MFR | MANUFACTURER |
| | MH | MANHOLE |
| | MIN | MINIMUM |
| | MISC | MISCELLANEOUS |
| | MLO | MAIN LUGS ONLY |
| | MOCP | MAXIMUM OVERCURRENT PROTECTION |
| | MTD | MOUNTED |
| | MTG | MOUNTING |
| | MTS | MANUAL TRANSFER SWITCH |
| | MV | MEDIUM VOLTAGE (OVER 600V LESS THAN 35KV) |
| N | N | NEUTRAL |
| | NC | NORMALLY CLOSED |
| | NEC | NATIONAL ELECTRICAL CODE |
| | NEMA | NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION |
| | NIC | NOT IN CONTRACT |
| | NO | NORMALLY OPEN |
| | NO., NUM, # | NUMBER |
| | NTS | NOT TO SCALE |
| O | OC | ON CENTER |
| | OCPD | OVERCURRENT PROTECTION DEVICE |
| | OFCI | OWNER FURNISHED, CONTRACTOR INSTALLED |
| | OFOI | OWNER FURNISHED, OWNER INSTALLED |
| | OH | OVERHEAD |
| | OHE/T | OVERHEAD ELECTRIC/TELEPHONE |
| | OPP | OPPOSITE |
| | % Z | PERCENT IMPEDANCE |
| P | P | POLE |
| | PA | PUBLIC ADDRESS |
| | PB | PULL BOX |
| | PDP | POWER DISTRIBUTION PANEL |
| | PE | PHOTO ELECTRIC |
| | PF | POWER FACTOR |
| | PH, Ø | PHASE |
| | PNL | PANEL |
| | PRI | PRIMARY |
| | PT | POTENTIAL TRANSFORMER |
| | PVC | POLYVINYL CHLORIDE |
| | PWR | POWER |
| Q | QTY | QUANTITY |
| R | R, RE | RELOCATE AS SHOWN |
| | RCLP | REMOTE CONTROL LIGHTING PANEL |
| | RCPT | RECEPTACLE |
| | REF | REFRIGERATOR |
| | RF | RADIO FREQUENCY |
| | RLA | RATED (RUNNING) LOAD AMPS |
| | RM | ROOM |
| | RP | RECEPTACLE PANELBOARD |
| | RSC | RIGID STEEL CONDUIT |
| S | SCH, SCHED | SCHEDULE |
| | SEC | SECONDARY |
| | SF | SQUARE FOOT |
| | SPD | SURGE PROTECTIVE DEVICE |
| | SPDT | SINGLE POLE DOUBLE THROW |
| | SPEC(S) | SPECIFICATION(S) |
| | SPKR | SPEAKER |
| | SPST | SINGLE POLE SINGLE THROW |
| | STD | STANDARD |
| | SW | SWITCH |
| | SWBD | SWITCHBOARD |
| | SWGR | SWITCHGEAR |
| | SYM | SYMMETRICAL |
| T | TB | TERMINAL BLOCK |
| | TBB | TELEPHONE BACKBOARD |
| | TC | TIME CLOCK |
| | TEL, TELE | TELEPHONE |
| | TELECOM | TELECOMMUNICATIONS |
| | TP | TAMPERPROOF |
| | TV | TELEVISION |
| | TYP | TYPICAL |
| U | UG | UNDERGROUND |
| | UGP | UNDERGROUND PRIMARY |
| | UGS | UNDERGROUND SECONDARY |
| | UGT | UNDERGROUND TELEPHONE |
| | UL | UNDERWRITER'S LABORATORY |
| | UON | UNLESS OTHERWISE NOTED |
| | UPS | UNINTERRUPTIBLE POWER SUPPLY |
| V | V | VOLTS |
| | VA | VOLT-AMPERES |
| | VFD | VARIABLE FREQUENCY DRIVE/ VARIABLE FREQUENCY MOTOR CONTROLLER |
| | W | WIRE |
| | W/ | WITH |
| | W/O | WITHOUT |
| | WHM | WATT HOUR METER |
| W | WP | WEATHERPROOF |
| X | X | REMOVE DEVICE |
| | XFMR | TRANSFORMER |

# FIRE ALARM SYMBOLS

| | |
|---|---|
| | STROBE - WALL MOUNTED |
| | STROBE - CEILING MOUNTED |
| | BELL - WALL MOUNTED |
| | HORN - WALL MOUNTED |
| | HORN - CEILING MOUNTED |
| | HORN W/STROBE - WALL MOUNTED |
| | HORN W/STROBE - CEILING MOUNTED |
| | FA SPEAKER - WALL MOUNTED |
| | FA SPEAKER - CEILING MOUNTED |
| | FA SPEAKER W/STROBE - WALL MOUNTED |
| | FA SPEAKER W/STROBE - CEILING MOUNTED |
| | FIRE FIGHTER'S TELEPHONE |
| FA | FIRE FIGHTER'S TELEPHONE JACK |
| | SMOKE DETECTOR - CEILING OR SURFACE MOUNTED |
| H | SMOKE DETECTOR - WALL MOUNTED |
| | SMOKE DETECTOR - DUCT |
| ER | SMOKE DETECTOR - ELEVATOR RECALL |
| BT/BR | SMOKE DETECTOR CEILING OR WALL MOUNTED - BEAM TYPE ("BT" DENOTES TRANSMITTER, "BR" DENOTES RECEIVER) |
| H CO | CARBON MONOXIDE DETECTOR - CEILING MOUNTED |
| CO | CARBON MONOXIDE DETECTOR - WALL MOUNTED |
| | DUCT SMOKE DETECTOR REMOTE TEST STATION |
| F | HEAT DETECTOR - FIXED TEMPERATURE |
| R/F | HEAT DETECTOR - COMBINATION OF RATE OF RISE AND FIXED TEMPERATURE |
| | JUNCTION BOX (RECESSED/SURFACE) FOR FIRE ALARM SYSTEMS PLANS, FOR CONNECTIONS ONLY |

| | |
|---|---|
| F | MANUAL FIRE ALARM PULL STATION |
| WF | WATER FLOW SWITCH |
| TS | VALVE SUPERVISORY (TAMPER) SWITCH |
| PS | WATER FLOW SWITCH - PRESSURE |
| AP | HI / LOW AIR PRESSURE SWITCH |
| DH | MAGNETIC DOOR HOLD OPEN DEVICE |
| FI | FAULT ISOLATION MODULE |
| AI | FIRE ALARM ADDRESSABLE INTERFACE MODULE (INPUT) |
| AR | FIRE ALARM ADDRESSABLE RELAY (OUTPUT) |
| EVAC | VOICE EVACUATION PANEL |
| NAC | NOTIFICATION APPLIANCE POWER EXTENDER PANEL |
| DACT | DIGITAL ALARM COMMUNICATOR TRANSMITTER (IP DENOTES INTERNET PROTOCOL AND GSM DENOTES CELLUAR DIALER) |
| SCP | FIREFIGHTER'S SMOKE CONTROL PANEL |
| BATT | BATTERY CABINET |
| FATC | FIRE ALARM TERMINAL CABINET |
| AMP | FIRE ALARM AMPLIFIER PANEL |
| FARP | FIRE ALARM REMOTE CONTROL PANEL |
| FACP | FIRE ALARM SYSTEM CONTROL PANEL |
| GAP | FIRE ALARM ANNUNCIATOR PANEL - GRAPHIC |
| LCD | FIRE ALARM ANNUNCIATOR PANEL - LCD TYPE |
| PB | ELECTRICAL PULL BOX, FOR CONNECTION ONLY |
| EN | NEMA 1 ENCLOSURE CABINET |
| ZONE # | FIRE ALARM ZONE IDENTIFICATION NUMBER |
| | FIRE ALARM ZONE BOUNDARY |

# FIRE ALARM RISER DIAGRAM




## RISER DIAGRAM NOTES

1. FIRE ALARM RISER DIAGRAM DEPICTS DESIGN INTENT. IT IS NOT THE INTENT OF THE RISER DIAGRAM TO LIST ALL FIRE ALARM DEVICES OR PANELS. SEE FLOOR PLANS FOR NEW DEVICE OR PANEL LOCATIONS.
2. PROVIDE FAULT ISOLATION MODULES ON SLC CIRCUITS PER THE MANUFACTURER'S RECOMMENDATION.
3. PROVIDE NEW SURGE PROTECTION DEVICE FOR EACH SOURCE OF POWER TO THE FIRE ALARM SYSTEM. COORDINATE WITH ELECTRICAL INSTALLER TO PROVIDE POWER TO NAC PANELS WITH ELECTRICAL JUNCTION BOX CONTAINING TERMINATION OF 120VAC/20AMP POWER CIRCUITS FROM BUILDING ELECTRICAL DISTRIBUTION PANEL.
4. PROVIDE REMOTE POWER PANEL(S) AND AMPLIFIER(S) AS NEEDED BASED ON FIRE ALARM INSTALLER'S PROPOSED LAYOUT AND PRODUCT. PROVIDE BATTERY AND VOLTAGE DROP CALCULATIONS TO DETERMINE THE QUANTITY OF REMOTE POWER SUPPLY PANELS AND AMPLIFIERS.

# GENERAL NOTES

1. THE WORK INCLUDES, BUT IS NOT LIMITED TO DESIGNING AND PROVIDING A COMPLETELY SUPERVISED, MULTIPLEXED, INTELLIGENT, ADDRESSABLE FIRE ALARM AND DETECTION SYSTEM AS INDICATED IN THE CONTRACT DOCUMENTS. THE WORK SHALL INCLUDE, BUT NOT BE LIMITED TO, ALL WIRING, RACEWAYS, PULL BOXES, TERMINAL CABINETS, OUTLET AND MOUNTING BOXES, CONTROL EQUIPMENT, ALARM AND SUPERVISORY SIGNALS, INITIATING DEVICES, ALARM NOTIFICATION APPLIANCES, NOTIFICATION ALARM PANELS, INTERFACE EQUIPMENT, AND ALL OTHER ACCESSORIES AND MISCELLANEOUS ITEMS INCLUDING SOFTWARE, SYSTEM PROGRAMMING, TEST AND CONSTRUCTION PERMITS REQUIRED FOR A COMPLETE OPERATING SYSTEM NORMALLY AND CUSTOMARILY REQUIRED, WHETHER INDICATED OR INHERENTLY NECESSARY DUE TO THE NATURE OF THE WORK. THE WORK SHALL COMPLY WITH THE SMITHSONIAN DESIGN STANDARDS, 2024 EDITIONS OF THE INTERNATIONAL BUILDING CODE AND INTERNATIONAL FIRE CODE, NFPA 70, NFPA 72, NFPA 92, AND NFPA 150. THE WORK SHALL BE LISTED FOR THE INTENDED PURPOSE BY A NATIONAL RECOGNIZED TESTING LABORATORY.
2. WHERE ANY CONFLICT AND/OR DISCREPANCY BETWEEN THESE GENERAL NOTES AND ANY OTHER INFORMATION CONTAINED IN THE CONTRACT DOCUMENTS EXISTS, THE MORE STRINGENT REQUIREMENT SHALL PREVAIL AND BE A REQUIREMENT OF THE WORK. ANY SUCH CONFLICT AND/OR DISCREPANCY SHALL BE PROMPTLY BROUGHT TO THE ATTENTION OF THE OWNER, ARCHITECT AND ENGINEER.
3. PROVIDE ALL NECESSARY PERMITS, INSPECTIONS AND FINAL SYSTEM ACCEPTANCE FROM THE AUTHORITY HAVING JURISDICTION (AHJ) AND OWNER.
4. THE TERM "PROVIDE" SHALL MEAN FURNISH, INSTALL AND CONNECT EQUIPMENT/DEVICE FOR A COMPLETE AND OPERATIONAL SYSTEM.
5. THE TERM "REMOVE" SHALL MEAN DISCONNECT EQUIPMENT/DEVICE AND CLEAR FROM SITE, AS WELL AS MAINTAINING UPSTREAM AND DOWNSTREAM EQUIPMENT/DEVICES AS REQUIRED.
6. VERIFY AND COORDINATE ALL EXISTING DIMENSIONS AND CONDITIONS PRIOR TO PROCEEDING WITH THE WORK.
7. A WORK PLAN SHALL BE PROPOSED BY THE CONTRACTOR TO THE OWNER OR OWNER'S REPRESENTATIVE TO DEMONSTRATE THAT THE WORK WILL BE PERFORMED IN SUCH MANNER TO MINIMIZE THE AMOUNT OF TIME THE EXISTING FIRE ALARM SYSTEM WILL BE OUT OF SERVICE. NOTIFY THE OWNER AND OBTAIN WRITTEN PERMISSION AT LEAST TWO BUSINESS DAYS IN ADVANCE PRIOR TO DISRUPTING ANY EXISTING FIRE ALARM SYSTEMS. PROVIDE COMPENSATING FIRE PROTECTION INCLUDING, BUT NOT LIMITED TO, FIRE WATCH SERVICE IF THE FIRE ALARM SYSTEM IS TAKEN OUT OF SERVICE FOR MORE THAN 4 HOURS IN A 24-HOUR PERIOD.
8. COORDINATE ALL WORK AND AVOID ALL CONFLICTS WITH LIGHTING FIXTURES, DIFFUSERS, GRILLS, DUCTS, STRUCTURAL MEMBERS, PIPES AND OTHER OBSTRUCTIONS.
9. COORDINATE LOCATION OF AUTOMATIC SPRINKLER FLOW AND VALVE SUPERVISORY SWITCHES PRIOR TO CONDUIT ROUGH-IN TO ENSURE ACCESSIBILITY AND MINIMIZE LENGTH OF FLEXIBLE CONDUIT RUNS NECESSARY TO MAKE CONNECTIONS.
10. SUBJECT TO ANY CODE RESTRICTION AND/OR REQUIREMENT AND UNLESS OTHERWISE NOTED, PROVIDE ELECTRICAL METALLIC TUBING (EMT). IN AREAS WITHOUT CEILINGS, EMT CONDUIT SHALL BE INSTALLED AS UNOBTRUSIVELY AS POSSIBLE, AS CLOSE AS POSSIBLE TO FLOOR/CEILING SLAB AND PARALLEL, AND AT RIGHT ANGLES TO STRUCTURAL STEEL OR CONCRETE. CONCEAL ALL CONDUITS ABOVE SUSPENDED CEILINGS AND BEHIND WALLS. IN UNFINISHED AREAS, CONDUIT MAY BE SURFACE MOUNTED UNLESS OTHERWISE NOTED. EXPOSED CONDUIT SHALL BE PAINTED TO MATCH ADJACENT FINISH.
11. LOCATE FIRE ALARM DEVICES INSTALLED IN ACOUSTICAL TILE CEILINGS IN THE CENTER OF THE TILE.
12. SUBJECT TO ANY INFORMATION CONTAINED IN THE CONTRACT DOCUMENTS, MOUNT ALL VISUAL WARNING STROBE NOTIFICATION APPLIANCES TO ACHIEVE MAXIMUM VISIBILITY IN ACCORDANCE WITH CURRENT ADAAG GUIDELINES. PROVIDE CARE IN LOCATING VISUAL WARNING STROBE APPLIANCES TO ENSURE VISIBILITY AND AVOID OBSTRUCTIONS.
13. ALL WALL AND FLOOR PENETRATIONS SHALL BE CORE-DRILLED WITH MASONRY OR CONCRETE TYPE CORE BITS. AFTER CONDUIT INSTALLATION, ALL PENETRATIONS IN WALLS, CEILINGS AND FLOORS SHALL BE SEALED TO THE FULL THICKNESS OF THE PENETRATION WITH APPROVED FIRESTOPPING MATERIAL TO PROVIDE EQUAL OR GREATER FIRE RESISTANCE. REFER TO CODE SUMMARY SHEET, LIFE SAFETY PLANS, AND ARCHITECTURAL DRAWINGS FOR FIRE RESISTANCE RATING OF FLOORS AND WALLS, IF NOT SPECIFICALLY INDICATED. ALL CORRIDOR, ROOM SEPARATION, AND SLAB-TO-SLAB WALL SHALL BE CONSIDERED TO BE OF 1-HOUR FIRE RESISTIVE RATED CONSTRUCTION, EXCEPT STAIRWAY , SHAFT ENCLOSURES, AND FLOOR SLABS SHALL BE CONSIDERED 2-HOUR FIRE RESISTIVE RATED CONSTRUCTION. FLOOR PENETRATION FIRESTOPS SHALL BE PROVIDED WITH 2-HOUR T- AND F- RATING.
14. DAMAGE TO WALLS, CEILINGS, FLOORS, AND STRUCTURAL MEMBERS FROM THE WORK SHALL BE PATCHED, REPAIRED AND PAINTED WITH NEW MATERIALS TO MATCH ADJACENT WORK, WHETHER SPECIFICALLY NOTED OR NOT.
15. ELECTRICAL BOXES SHALL BE PROVIDED AS FOLLOWS:
    I) EXPOSED WORK - CAST METAL BELL TYPE BOXES WITH THREADED HUB, EXCEPT WHERE FACTORY BOXES ARE PROVIDED; AND
    II) CONCEALED WORK - BOXES MAY BE STAMPED STEEL BOXES.
16. PROVIDE TERMINAL CABINETS OF SOLID WALL CONSTRUCTION WITH NO FACTORY CONDUIT KNOCK OUTS.
17. PROVIDE COMPRESSION TYPE STEEL FITTINGS AND CONNECTORS FOR BOTH EXPOSED AND CONCEALED CONDUITS INCLUDING RISERS UP TO 2 INCHES IN DIAMETER.
18. PROVIDE ¾-INCH MINIMUM EMT TRADE SIZE CONDUIT, UNLESS OTHERWISE NOTED. VERTICAL RISERS SHALL BE ONE INCH DIAMETER EMT MINIMUM.
19. ALL FIRE ALARM CIRCUITS SHALL BE SUPERVISED AS FOLLOWS IN ACCORDANCE WITH NFPA 72:
    I) INITIATING - CLASS B
    II) SIGNALING LINE CIRCUITS - CLASS B
    III) NOTIFICATION APPLIANCES - CLASS B
    IV) CONTROL - CLASS A
    V) NETWORK - CLASS X
20. ALL FIRE ALARM PATHWAYS SHALL BE SURVIVABILITY LEVEL 1.
21. PROVIDE WIRING CONDUCTORS AS INDICATED BELOW:
    I) INITIATING - 16 AWG, TYPE FPLR/FPLP, SOLID COPPER
    II) SIGNALING - 16 AWG, TYPE FPLR/FPLP, SOLID COPPER
    III) NOTIFICATION APPLIANCES - 14 AWG, TYPE FPLR/FPLP, SOLID COPPER, TSP
    IV) SPEAKER - 16 AWG, TYPE FPLR/FPLP, SOLID COPPER, TSP
    V) CONTROL - 16 AWG, TYPE FPLR/FPLP, SOLID COPPER
    VI) 120 VAC POWER - 12 AWG, TYPE THHN, SOLID COPPER

# SYMBOL SHEET NOTES

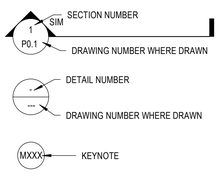
A. THE PURPOSE OF THIS SHEET IS TO ILLUSTRATE AND DEFINE TYPICAL GRAPHIC SYMBOLS WHICH MAY TYPICALLY OCCUR ON THE FIRE SUPPRESSION DRAWINGS.

B. THIS SHEET MAY ILLUSTRATE ABBREVIATIONS, SYMBOLS AND LEGENDS WHICH DO NOT OCCUR AS PART OF THIS PROJECT. REFER TO THE DRAWINGS TO DETERMINE WHICH ITEMS APPLY TO THE WORK OF THIS PROJECT.

C. ADDITIONAL SYMBOLS NOT SHOWN OR DEFINED ON THIS SHEET MAY BE USED ON THE FIRE SUPPRESSION DRAWINGS AND ARE TYPICALLY DEFINED ON OTHER SHEETS.

D. SYMBOLS ARE NOT DRAWN TO SCALE. TO DETERMINE THE ACTUAL SIZES OF ELEMENTS REPRESENTED BY SYMBOLS, REFER TO THE SPECIFICATIONS AND OTHER SHEETS, AS MAY APPLY.

# REFERENCE SYMBOLS

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# ALF

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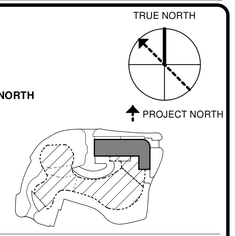


KEY PLAN

0 4' 8' 16'

1/8"=1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
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| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

Smithsonian Institution

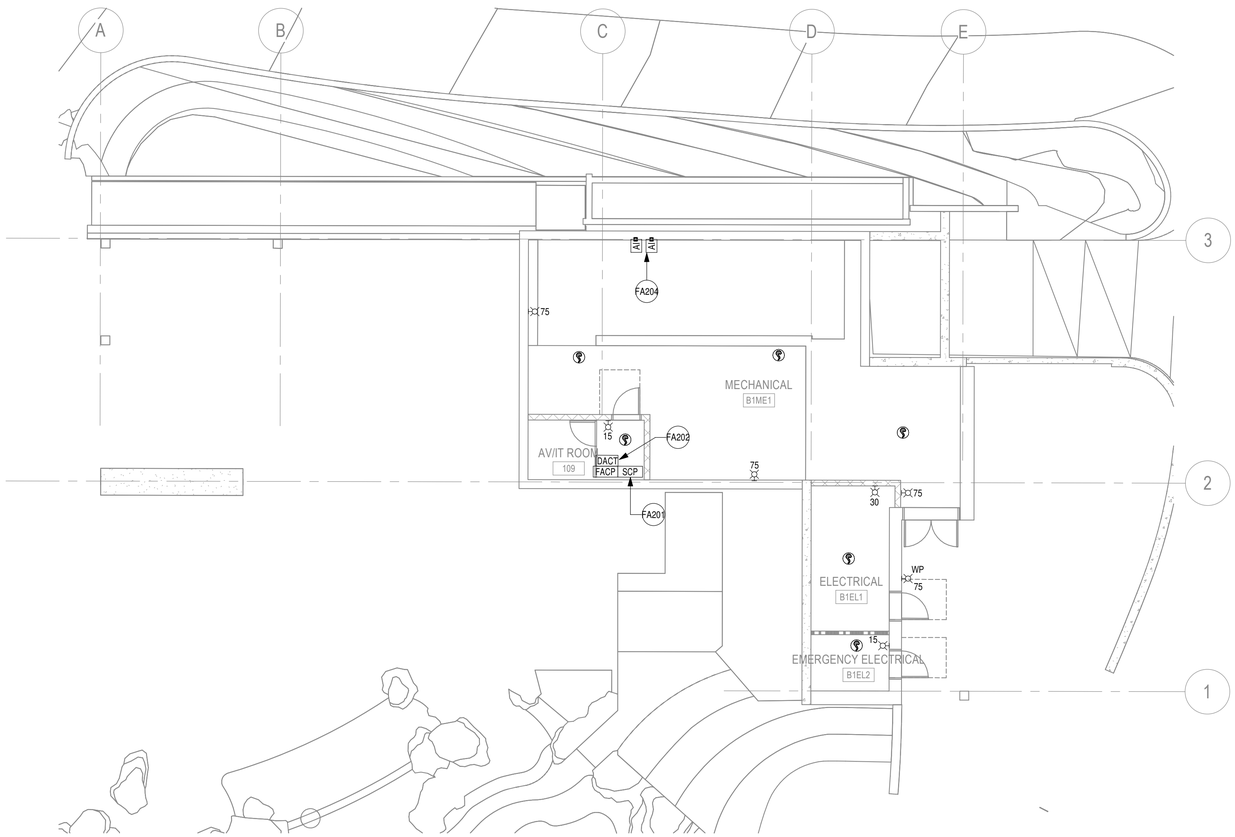
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | FIRE ALARM ABBREVIATIONS AND SYMBOLS |
| WORKING STAFF | EG (DESIGNED BY) / EG (DRAWN BY) / KL (CHECKED BY) |

SHEET NO. 324 OF 378

FA-001 (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

*FOR REFERENCE ONLY*

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FA-101CP

## BASEMENT FIRE ALARM PLAN

SCALE = 1/8" = 1'-0"

## GENERAL SHEET NOTES

A. REFER TO SHEET FA-001 FOR FIRE ALARM ABBREVIATIONS AND SYMBOLS.

B. FIRE ALARM SYSTEM SHALL BE NETWORKABLE AND ADDRESSABLE.

C. COORDINATE ALL CEILING MOUNTED FIRE ALARM DEVICES WITH FINISHED CEILING HEIGHTS ON ARCHITECTURAL REFLECTED CEILING PLANS. FOR ALL CEILING MOUNTED DEVICES MOUNTED IN OPEN CEILINGS, LOCATE AT HEIGHT SO THAT THE TOP OF THE DEVICE IS ALIGNED WITH THE LOWEST ADJACENT CEILING FOR FULL OCCUPANT VISIBILITY.

D. LOCATE FIRE ALARM DEVICES IN CENTER OF TILE WHEN INSTALLED IN LAY-IN TILE CEILING.

E. WHERE SMOKE DETECTORS ARE SHOWN IN AREAS WITHOUT HUNG CEILINGS, MOUNT DETECTORS IN EACH BEAM POCKET WHERE BEAMS EXCEED 450 mm (18 IN.) IN DEPTH AND MORE THAN 2400 mm ON CENTER. PROVIDE ADDITIONAL DETECTORS AS NECESSARY.

F. NEW FIRE ALARM DEVICES SHALL BE WHITE IN COLOR WITH A CLEAR LENS AND LABELED "ALERT".

G. ALL ADDRESSABLE INTERFACES AND RELAYS SHALL BE INSTALLED WITHIN THREE FEET OF THE CONTROLLED/MONITORED ELEMENT.

## SHEET KEYNOTES

FA201 PROVIDE FIREFIGHTER'S SMOKE CONTROL PANEL IN AV/IT ROOM.

FA202 PROVIDE NEW FACP IN AV/IT ROOM. PROVIDE KELTRON LS NET TRANSCEIVER TO CONNECT TO SMITHSONIAN CENTRAL MONITORING SYSTEM.

FA204 PROVIDE ADDRESSABLE INTERFACE MODULES, IN APPROPRIATE TYPE AND QUANTITY, TO MONITOR TAMPER AND WATERFLOW SWITCHES SERVING BACKFLOW PREVENTER.

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TRUE NORTH

PROJECT NORTH

KEY PLAN

0 4' 8' 16'

1/8"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
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| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BASEMENT FIRE ALARM PLAN |
| WORKING STAFF | EG (DESIGNED BY) / EG (DRAWN BY) / Checker (CHECKED BY) |

SHEET NO. 325 OF 378

FA-101CP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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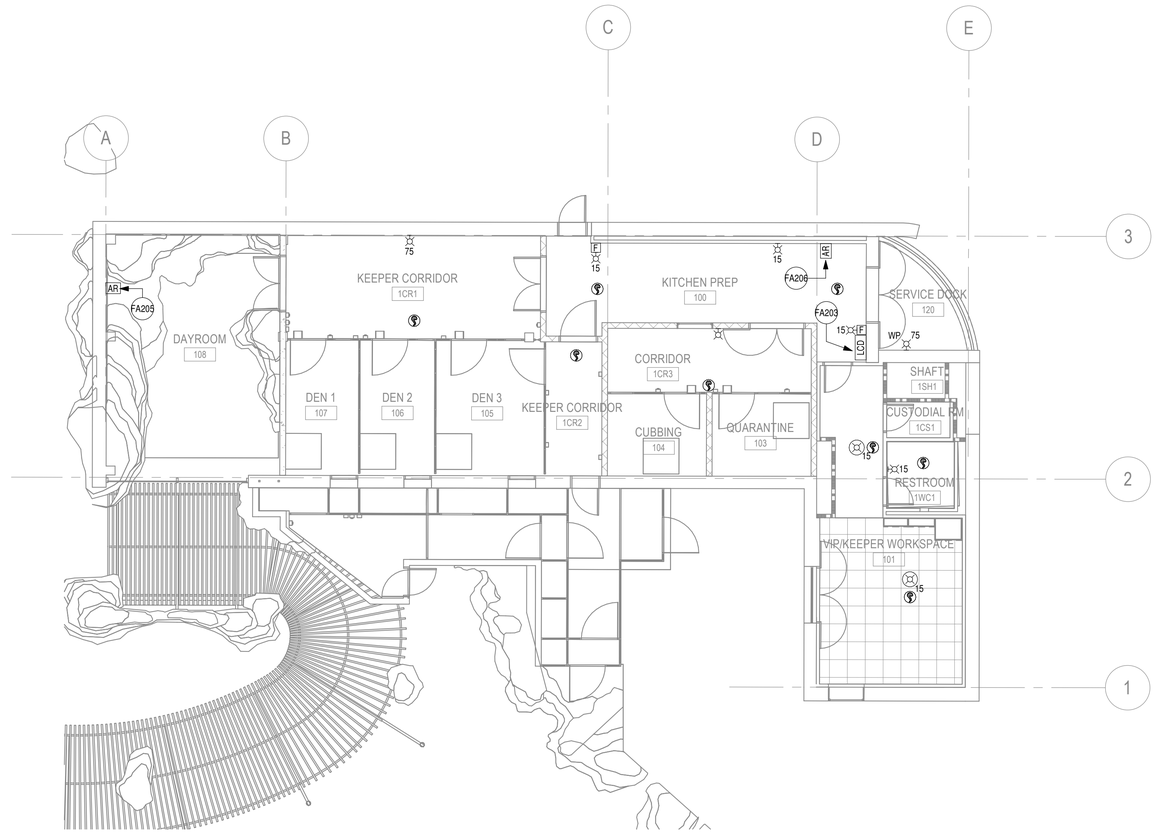


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FA-102CP

## LEVEL 1 FIRE ALARM PLAN

SCALE = 1/8" = 1'-0"

## GENERAL SHEET NOTES

A. REFER TO SHEET FA-001 FOR FIRE ALARM ABBREVIATIONS AND SYMBOLS.

B. FIRE ALARM SYSTEM SHALL BE NETWORKABLE AND ADDRESSABLE.

C. COORDINATE ALL CEILING MOUNTED FIRE ALARM DEVICES WITH FINISHED CEILING HEIGHTS ON ARCHITECTURAL REFLECTED CEILING PLANS. FOR ALL CEILING MOUNTED DEVICES MOUNTED IN OPEN CEILINGS, LOCATE AT HEIGHT SO THAT THE TOP OF THE DEVICE IS ALIGNED WITH THE LOWEST ADJACENT CEILING FOR FULL OCCUPANT VISIBILITY.

D. LOCATE FIRE ALARM DEVICES IN CENTER OF TILE WHEN INSTALLED IN LAY-IN TILE CEILING.

E. WHERE SMOKE DETECTORS ARE SHOWN IN AREAS WITHOUT HUNG CEILINGS, MOUNT DETECTORS IN EACH BEAM POCKET WHERE BEAMS EXCEED 450 mm (18 IN.) IN DEPTH AND MORE THAN 2400 mm ON CENTER. PROVIDE ADDITIONAL DETECTORS AS NECESSARY.

F. NEW FIRE ALARM DEVICES SHALL BE WHITE IN COLOR WITH A CLEAR LENS AND LABELED "ALERT".

G. ALL ADDRESSABLE INTERFACES AND RELAYS SHALL BE INSTALLED WITHIN THREE FEET OF THE CONTROLLED/MONITORED ELEMENT.

## SHEET KEYNOTES

FA203 PROVIDE NEW LCD ANNUNCIATOR.
FA205 PROVIDE ADDRESSABLE RELAY MODULE TO OPEN EXTERIOR SMOKE EXHAUST LOUVER UPON FIRE ALARM ACTIVATION.
FA206 PROVIDE ADDRESSABLE RELAY MODULE TO ACTIVATE SMOKE EXHAUST FAN UPON FIRE ALARM ACTIVATION.

# ALF

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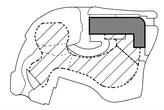

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| DRAWING TITLE | LEVEL 1 FIRE ALARM PLAN |

| WORKING STAFF | DESIGNED BY | DRAWN BY | CHECKED BY |
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| | EG | EG | KL |

SHEET NO. 326 OF 378

FA-102CP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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| SYSTEM INPUT | | SYSTEM OUTPUT | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A | B | C | D | E | F | G | H | I | J | K |
| | | ALARM ANNUNCIATION @ FACP AND REMOTE ANNUNCIATOR BY DEVICE | TROUBLE ANNUNCIATION @ FACP AND REMORE ANNUNCIATOR BY DEVICE | SUPERVISORY ANNUNCIATION @ FACP AND REMOTE ANNUNCIATOR PANEL | TRANSMIT SIGNAL TO SUPERVISING STATION | RECORD AND PRINT CHANGE OF SYSTEMS STATUS | ANNUNCIATE ASSOCIATED INFORMATION ON ALL REMOTE FAACPS | ACTIVATE CLEAR (FIRE) VISIBLE NOTIFICATION DEVICES | SHUT DOWN ONLY THE ASSOCIATED AHU AND INDICATE AT REMOTE STATION | MUTE ALL PUBLIC ADDRESS/SOUND SYSTEMS | INITIATE SECURITY LOCK RELEASE | INITIATE DOOR HOLD OPEN RELEASE |
| 1 | MANUAL PULL STATION | ■ | | | ■ | ■ | ■ | ■ | | ■ | ■ | ■ |
| 2 | SMOKE DETECTORS - SPOT TYPE | ■ | | | ■ | ■ | ■ | ■ | | ■ | ■ | ■ |
| 3 | DUCT MOUNTED SMOKE DETECTOR @ AHU (RETURN) FOR AHUS | | | ■ | ■ | ■ | ■ | | ■ | | | |
| 4 | SPRINKLER WATERFLOW SWITCH | ■ | | | ■ | ■ | ■ | ■ | | ■ | ■ | ■ |
| 5 | SPRINKLER VALVE SUPERVISORY SWITCH | | | ■ | ■ | ■ | ■ | | | | | |
| 6 | FIRE ALARM AC POWER FAIL | | ■ | | ■ | ■ | ■ | | | | | |
| 7 | FIRE ALARM SYSTEM LOW BATTERY | | ■ | | ■ | ■ | ■ | | | | | |
| 8 | OPEN CIRCUIT | | ■ | | ■ | ■ | ■ | | | | | |
| 9 | GROUND FAULT | | ■ | | ■ | ■ | ■ | | | | | |
| 10 | SLC/INITIATING DEVICE CIRCUIT FAULT | | ■ | | ■ | ■ | ■ | | | | | |
| 11 | NOTIFICATION APPLIANCE CIRCUIT SHORT | | ■ | | ■ | ■ | ■ | | | | | |
| 12 | GENERATOR RUNNING | | | ■ | ■ | ■ | ■ | | | | | |
| 13 | GENERATOR LOSS OF POWER | | | ■ | ■ | ■ | ■ | | | | | |

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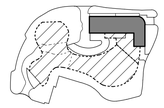

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GRAPHIC SCALE(S)

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SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE / WORKING STAFF: FIRE ALARM SYSTEM OPERATING MATRIX

| Designer | Author | Checker |
|---|---|---|
| DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 327 OF 378

FA-601SH

DISCIPLINE TYPE SEQUENCE DRAWING TYPE

*FOR REFERENCE ONLY*

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# FIRE SUPPRESSION ABBREVIATIONS

| | | |
|---|---|---|
| A | ABBR | ABBREVIATION |
| | ADA | AMERICAN WITH DISABILITIES ACT |
| | AFF | ABOVE FINISHED FLOOR |
| | AFG | ABOVE FINISHED GRADE |
| | AMB | AMBIENT |
| | ANSI | AMERICAN NATIONAL STANDARDS |
| | AP | ACCESS PANEL |
| | APPROX | APPROXIMATE |
| B | BFP | BACKFLOW PREVENTER (TYPE AS INDICATED) |
| | BHP | BRAKE HORSEPOWER |
| | BOP | BOTTOM OF PIPE |
| | BTUIH | BRITISH THERMAL UNITS PER HOUR |
| | BV | BALL VALVE |
| C | CAP | CAPACITY |
| | CFM | CUBIC FEET PER MINUTE |
| | CI | CAST IRON |
| | COMP | COMPRESSOR |
| | COND | CONDENSATE |
| | CONT | CONTINUATION |
| | COTR | CONTRACTING OFFICER'S TECHNICAL REPRESENTATITIVE |
| | CV | CHECK VALVE |
| D | DIA | DIAMETER |
| | DN | DOWN |
| | DOM | DOMESTIC |
| | DWG | DRAWING |
| E | (E) | EXISTING MATERIAL OR EQUIPMENT |
| | EA | EACH |
| | EFF | EFFICIENCY |
| | EL | ELEVATION |
| | EQ | EQUIVALENT |
| | ET | EXPANSION TANK |
| | EWT | ENTERING WATER TEMPERATURE |
| | EXP | EXPANSION |
| | EXT | EXTERNAL |
| | F | FIRE |
| | FCU | FAN COIL UNIT |
| | FHV | FIRE HOSE VALVE |
| | FL | FLOOR |
| | FPM | FEET PER MINUTE |
| | FPS | FEET PER SECOND |
| | FS | FLOW SWITCH |
| | FT | FEET OR FOOT |
| F | °F | DEGREES IN FAHRENHEIT |
| G | GAL | GALLON |
| | GPD | GALLONS PER DAY |
| | GPH | GALLONS PER HOUR |
| | GPM | GALLONS PER MNUTE |
| | GV | GATE VALVE |
| H | HG | MERCURY |
| | HGL | HYDRAULIC GRADE LINE |
| | HGT | HEIGHT |
| | HP | HORSE POWER |
| | HPS | HIGH PRESSURE STEAM |
| | HR | HOUR |
| I | ID | INSIDE DIAMETER |
| | IE | INVERT ELEVATION |
| | IN | INCHES |
| K | kW | KILOWATTS |
| L | LGT | LENGTH |
| | LWT | LEAVING WATER TEMPERATURE |
| M | MAX | MAXIMUM |
| | MBH | 1000 BTU/HR |
| | MECH | MECHANICAL |
| | MIN | MINIMUM |
| | MISC | MISCELLANEOUS |
| | MTG | MOUNTING |
| N | N.C. | NORMALLY CLOSED |
| | N.O. | NORMALLY OPEN |
| | NA | NOT APPLICABLE |
| | NIC | NOT IN CONTRACT |
| | NO | NUMBER |
| | NOM | NOMINAL |
| | NTS | NOT TO SCALE |
| O | OD | OUTSIDE DIAMETER |
| | OS&Y | OUTSIDE SCREW AND YOKE |
| P | PH | PHASE (ELECTRICAL) |
| | PPM | PARTS PER MILLION |
| | PRESS | PRESSURE |
| | PRV | PRESSURE REDUCING VALVE |
| | PSI | POUNDS PER SQUARE INCH |
| | PSIG | POUNDS PER SQUARE INCH - GAUGE |
| R | REF | REFERENCE |
| | RL | RELOCATED MATERIAL OR EQUIPMENT |
| | RPM | REVOLUTIONS PER MINUTE |
| S | SCFM | STANDARD CUBIC FEET PER MINUTE |
| | SE | SEWAGE EJECTOR |
| | SF | SQUARE FEET |
| | SG | SPECIFIC GRAVITY |
| | SPEC | SPECIFICATION |
| | SPR | SPRINKLER |
| | SPR/STP | SPRINKLER/STANDPIPE |
| | SQ. FT. | SQUARE FEET |
| | SS | STAINLESS STEEL |
| | STD | STANDARD |
| | STM | STEAM |
| T | TEMP | TEMPERATURE |
| | TS | TAMPER SWITCH |
| | TYP | TYPICAL |
| U | UON | UNLESS OTHERWISE NOTED |
| V | V | VOLT |
| | VEL | VELOCITY |
| W | W/ | WITH |
| | W/O | WITHOUT |
| | WG | WATER GAUGE |
| Y | YD | YARD |
| Z | ZCA | ZONE CONTROL ASSEMBLY |

# SPRINKLER HAZARD CLASSIFICATION CRITERIA

NFPA 13 HAZARD CLASSIFICATION

OCCUPANCY CLASSIFICATION.....ORDINARY HAZARD 2 (WET)
DESIGN DENSITY.....0.20 GPM/SF
REMOTE HYDRAULIC AREA .....1500 SF

OCCUPANCY CLASSIFICATION.....ORDINARY HAZARD 2 (DRY)
DESIGN DENSITY.....0.20 GPM/SF
REMOTE HYDRAULIC AREA .....1950 SF

# SYMBOL SHEET NOTES

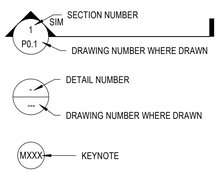
A. THE PURPOSE OF THIS SHEET IS TO ILLUSTRATE AND DEFINE TYPICAL GRAPHIC SYMBOLS WHICH MAY TYPICALLY OCCUR ON THE FIRE SUPPRESSION DRAWINGS.

B. THIS SHEET MAY ILLUSTRATE ABBREVIATIONS, SYMBOLS AND LEGENDS WHICH DO NOT OCCUR AS PART OF THIS PROJECT. REFER TO THE DRAWINGS TO DETERMINE WHICH ITEMS APPLY TO THE WORK OF THIS PROJECT.

C. ADDITIONAL SYMBOLS NOT SHOWN OR DEFINED ON THIS SHEET MAY BE USED ON THE FIRE SUPPRESSION DRAWINGS AND ARE TYPICALLY DEFINED ON OTHER SHEETS.

D. SYMBOLS ARE NOT DRAWN TO SCALE. TO DETERMINE THE ACTUAL SIZES OF ELEMENTS REPRESENTED BY SYMBOLS, REFER TO THE SPECIFICATIONS AND OTHER SHEETS, AS MAY APPLY.

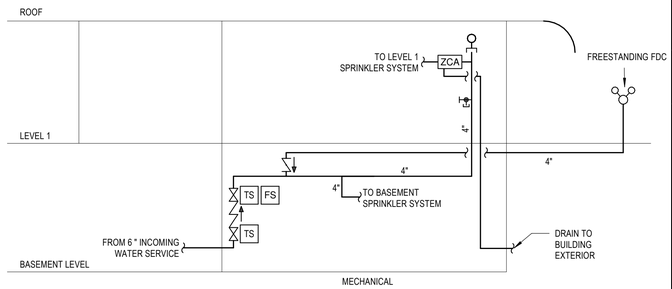
# REFERENCE SYMBOLS




# FIRE SUPPRESSION LEGENDS AND SYMBOLS

- NEW FIRE PROTECTION PIPING
- EXISTING FIRE PROTECTION PIPING TO REMAIN
- FIRE PROTECTION PIPING TO BE DEMOLISHED
- RECESSED PENDENT SPRINKLER
- CONCEALED SPRINKLER
- PENDENT SPRINKLER
- SIDEWALL SPRINKLER
- UPRIGHT SPRINKLER
- ELBOW
- CAP OR PLUG
- TEE
- TEE DOWN
- ELBOW - TURNED DOWN
- PIPE CONTINUATION
- CONCENTRIC REDUCER
- ECCENTRIC REDUCER
- OS&Y VALVE
- CHECK VALVE
- BUTTERFLY VALVE
- ALARM CHECK VALVE
- DRY PIPE VALVE
- PRE-ACTION VALVE
- DELUGE VALVE
- PRESSURE REDUCING VALVE
- PRESSURE RELIEF VALVE
- GATE VALVE - NON RISING STEM
- WALL MOUNTED POST INDICATOR VALVE
- BACKFLOW PREVENTER ASSEMBLY
- FIRE PUMP
- JOCKEY PUMP
- PUMP/BACKFLOW TEST CONNECTION, WALL MOUNTED
- PUMP/BACKFLOW TEST CONNECTION, FREESTANDING
- FIRE PUMP FLOW METER
- ZCA — ZONE CONTROL ASSEMBLY CONSISTING OF CHECK VALVE, BUTTERFLY VALVE, PRESSURE GAGE, WATER FLOW SWITCH, VALVE SUPERVISORY SWITCH, PRESSURE RELIEF VALVE, AND INSPECTORS TEST AND DRAIN.
- 2.5" FIRE HOSE VALVE CONNECTION
- 2.5" FIRE HOSE VALVE IN CABINET
- HOSE STATION
- FS — WATER FLOW SWITCH
- TS — VALVE SUPERVISORY SWITCH
- INSPECTOR'S TEST CONNECTION
- A — AIR COMPRESSOR
- N — NITROGEN GENERATOR
- A — RISER MOUNTED AIR COMPRESSOR
- HYDRAULIC CALCULATION AREA
- # — HYDRAULIC NODE
- FDC, WALL MOUNTED
- FDC, FREESTANDING
- PRESSURE GAUGE
- WATER MOTOR GONG

# FIRE SUPPRESSION RISER DIAGRAM




FIRE PROTECTION RISER DIAGRAM GENERAL NOTES:

1. RISER DIAGRAM IS SCHEMATIC IN NATURE AND INTENDED TO INDICATE GENERAL ARRANGEMENT OF SYSTEM CONNECTIONS ONLY. REFER TO FLOOR PLANS, SPECIFICATIONS, AND APPLICABLE CODES AND STANDARDS FOR COMPLETE SYSTEM REQUIREMENTS. PROVIDE ALL VALVES AND ACCESSORIES NECESSARY FOR A COMPLETE SYSTEM, WHETHER INDICATED HERE OR NOT.
2. DRAINS OMITTED FROM RISER DIAGRAM FOR CLARITY. PROVIDE DRAINS AS REQUIRED BY APPLICABLE CODES AND THE CONTRACT DOCUMENTS. ROUTE ALL DRAINS TO THE EXTERIOR, UNLESS NOTED OTHERWISE.
3. AIR RELEASE VALVE/VENT SHALL BE PROVIDED AT SYSTEM HIGH POINT TO ALLOW AIR TO BE REMOVED FROM THE STANDPIPE/SPRINKLER SYSTEM.

# FIRE SUPPRESSION GENERAL NOTES

1. PROVIDE A COMPLETE HYDRAULICALLY- CALCULATED AUTOMATIC WET SPRINKLER SYSTEM THROUGHOUT THE AREA OF WORK, INCLUDING BUT NOT LIMITED TO BACKFLOW PREVENTER ASSEMBLY, WATER FLOW ALARM SWITCHES, VALVE SUPERVISORY SWITCHES, CHECK VALVES, CONTROL VALVES, PIPING, HANGERS, SPRINKLERS, ALL ASSOCIATED EQUIPMENT, WIRING AND APPURTENANCES, NORMALLY AND CUSTOMARILY REQUIRED, WHETHER INDICATED OR INHERENTLY NECESSARY DUE TO THE NATURE OF THE WORK, FOR A COMPLETE AND OPERATIONAL SYSTEM.
2. ALL WORK SHALL CONFORM TO THE APPLICABLE PROVISIONS OF THE SMITHSONIAN DESIGN STANDARDS, 2024 EDITION OF THE IBC AND IFC, AND THE LATEST EDITIONS OF NFPA 13, 14, 70 AND 72.
3. WHERE ANY CONFLICT AND/OR DISCREPANCY BETWEEN THESE GENERAL NOTES AND ANY OTHER INFORMATION CONTAINED IN THE CONTRACT DOCUMENTS EXISTS, THE MORE STRINGENT REQUIREMENT SHALL PREVAIL AND BE A REQUIREMENT OF THE WORK. THAT NOTWITHSTANDING, ANY SUCH CONFLICT AND/OR DISCREPANCY SHALL PROMPTLY BE BROUGHT TO THE ATTENTION OF THE OWNER, ARCHITECT AND ENGINEER.
4. A WET PIPE SPRINKLER SYSTEM SHALL BE INSTALLED IN ALL HEATED SPACES. A DRY PIPE SYSTEM WILL BE INSTALLED IN ALL SPACES REQUIRING A SPRINKLER(S) WHERE EXPOSED TO FREEZING TEMPERATURES.
5. THE TERM "PROVIDE" SHALL MEAN FURNISH, INSTALL AND CONNECT EQUIPMENT/DEVICE FOR A COMPLETE AND OPERATIONAL SYSTEM. THE TERM "REMOVE" SHALL MEAN DISCONNECT AND CLEAR EQUIPMENT/DEVICE FROM SITE, AS WELL AS MAINTAINING UPSTREAM AND DOWNSTREAM EQUIPMENT/DEVICES AS REQUIRED.
6. THE CONTRACT DOCUMENTS ARE DIAGRAMMATIC AND INTENDED TO SHOW DESIGN INTENT ONLY. RUN ALL PIPING STRAIGHT AND TIGHT TO THE FACE OF STRUCTURAL MEMBERS AND WALLS, PARALLEL AND AT RIGHT ANGLES TO STRUCTURAL STEEL OR CONCRETE ELEMENTS.
7. CONTRACTOR SHALL FIELD VERIFY ALL CONDITIONS AND DIMENSIONS ON DRAWINGS PRIOR TO EXECUTION OF THIS CONTRACT. IT SHALL BE THE RESPONSIBILITY OF CONTRACTOR TO VERIFY JOB CONDITIONS AND COORDINATE ALL SYSTEM COMPONENTS. CONTRACTOR SHALL REPORT ANY DISCREPANCIES WITH THESE DRAWINGS TO THE ENGINEER IMMEDIATELY.
8. COORDINATE ALL WORK AND AVOID ALL CONFLICTS WITH LIGHTING FIXTURES, DIFFUSERS, GRILLS, DUCTS, STRUCTURAL MEMBERS, PIPES AND OTHER OBSTRUCTIONS.
9. NO STRUCTURAL MEMBERS SHALL BE CUT, DRILLED OR BURNED UNLESS PREVIOUSLY APPROVED BY THE STRUCTURAL ENGINEER. FURTHER, ANY APPROVED CUTTING OR PATCHING OF ANY STRUCTURAL WORK SHALL NOT RESULT IN A REDUCTION OF LOAD-CARRYING AND/OR LOAD-DEFLECTION RATIO. PRIOR TO SUCH WORK, OBTAIN APPROVAL FROM THE ENGINEER OF RECORD.
10. ALL WALL AND FLOOR PENETRATIONS SHALL BE CORE-DRILLED. PROVIDE PIPE SLEEVES FOR WALL AND FLOOR PENETRATIONS AS INDICATED IN THE SPECIFICATIONS. ALL PENETRATIONS IN WALLS, CEILINGS AND FLOORS SHALL BE PROVIDED WITH A UL-LISTED PENETRATION SEAL ASSEMBLY, WITH A FIRE RATING EQUIVALENT TO OR GREATER THAN THE FIRE RATING OF THE BARRIER PENETRATED STAIRWAY, SHAFT ENCLOSURES, AND FLOOR SLABS SHALL BE CONSIDERED 2-HOUR FIRE RESISTIVE RATED CONSTRUCTION UNLESS OTHERWISE NOTED.
11. PRIOR TO CORE DRILLING, CONTRACTOR SHALL VERIFY EXISTING CONDITION NOT TO DAMAGE ANY STRUCTURAL ELEMENT, CONDUIT, REBAR, AND ETC. ANY DAMAGE RESULTING FROM THE WORK, WHETHER INTENTIONAL OR NOT, SHALL BE REPAIRED AT NO ADDITIONAL COST TO, AND TO THE SATISFACTION OF THE OWNER.
12. ALL DAMAGE TO WALLS, CEILINGS, FLOORS, AND STRUCTURAL MEMBERS FROM THE WORK SHALL BE PATCHED, REPAIRED AND PAINTED WITH NEW MATERIALS TO MATCH ADJACENT WORK, WHETHER SPECIFICALLY NOTED OR NOT.
13. WHERE POSSIBLE, SPRINKLERS SHALL BE ALIGNED WITH OTHER SPRINKLERS INSTALLED IN THE SAME ROOM OR AREA.
14. AS A MINIMUM, MAINTAIN 6-FT 8-INCH HEAD ROOM CLEARANCE BETWEEN BOTTOM OF SPRINKLER PIPE AND FINISHED FLOOR, INCLUDING STAIRWELLS.
15. CONTRACTOR SHALL PAY ALL REQUIRED FEES AND OBTAIN ALL NECESSARY PERMITS AND LICENSES FOR INSTALLATION OF THE WORK.
16. COORDINATE AND SEQUENCE THE WORK PRIOR TO FABRICATION AND INSTALLATION OF EQUIPMENT. MAKE NECESSARY ACCOMMODATIONS TO MEET THE INTENT OF THE DRAWINGS AND ENSURE A COORDINATED INSTALLATION.
17. RUN ALL EXPOSED ELEMENTS NEATLY, TIGHT TO THE STRUCTURE (EXCEPT WHERE NOTED), PARALLEL TO WALLS, AND AS INCONSPICUOUSLY AS POSSIBLE.
18. INSTALL ALL CEILING SPRINKLERS IN 2 FT X 2 FT ACOUSTICAL PANEL CEILINGS IN THE CENTER OF THE PANELS; AND IN 2 FT X 4 FT PANELS, INSTALL DEVICE IN THE CENTER OF THE 2 FT X 4 FT OR 2 FT X 2 FT SQUARE AREAS.
19. PROVIDE SPRINKLER(S) UNDER ALL OBSTRUCTIONS GREATER THAN 48 INCHES IN WIDTH AND AS NEEDED TO PROVIDE PROPER COVERAGE AS REQUIRED BY NFPA 13.
20. PROVIDE LISTED GUARDS FOR SPRINKLER PROTECTION WHEN INSTALLED BELOW 7 FT AFF.
21. CONTRACTOR SHALL PROVIDE AN AUTOMATIC AIR VENTING DEVICE NEAR THE HIGHEST POINT WET PIPE SPRINKLER SYSTEM(S) PER NFPA 13.
22. CONTRACTOR SHALL PROVIDE AUXILIARY LOW POINT DRAINS WHERE REQUIRED PER NFPA 13. WHERE AUXILIARY DRAINS ARE INSTALLED BEHIND HARD CEILINGS, PROVIDE AN ACCESSIBLE PANEL DIRECTLY BENEATH THE DRAIN.
23. PER DISCUSSIONS WITH OSHEM, CONTRACTOR SHALL BE PERMITTED TO USE FLOW TEST DATA FROM 2025 FLOW TEST PROVIDED BELOW.
24. ALL EQUIPMENT SHALL BEAR UL LISTING LABEL AT MINIMUM, OR IF UL LABEL IS NOT AVAILABLE, THE ITEM SHALL BE TESTED AND LABELED BY A NATIONALLY RECOGNIZED TESTING AGENCY THAT MEETS UL AND AHJ REQUIREMENTS FOR TESTING FACILITIES. ALL EQUIPMENT SHALL BE INSTALLED PER THE LISTING AND MANUFACTURER'S INSTALLATION INSTRUCTIONS.
25. SPRINKLER SYSTEM WATERFLOW AND CONTROL VALVE SUPERVISORY SWITCHES SHALL BE MONITORED BY BUILDING FIRE ALARM SYSTEM. COORDINATE WITH FIRE ALARM CONTRACTOR SO THAT ELECTRICAL CONNECTION CAN BE MADE BETWEEN THESE DEVICES AND BUILDING FIRE ALARM SYSTEM.
26. CONTRACTOR SHALL TAKE NECESSARY MEASURES TO KEEP THE PREMISES DRY AT ALL TIMES AND PREVENT WATER DAMAGE TO SURFACES. REPAIR WATER DAMAGE RESULTING FROM THE WORK, WHETHER INTENTIONAL OR NOT, AT NO ADDITIONAL COST TO, AND TO THE SATISFACTION OF THE OWNER.
27. CONTRACTOR SHALL COORDINATE WITH OWNER TO AVOID FREEZING CONDITIONS DURING CONSTRUCTION ONCE WATER IS FILLED IN THE SPRINKLER SYSTEM.
28. HYDROSTATIC TESTING OF FIRE SUPPRESSION SYSTEMS SHALL BE WITNESSED BY ENGINEER AND/OR AHJ, BEFORE CEILING CLOSE IN. CONTRACTOR SHALL COORDINATE WITH ENGINEER AND/OR AHJ PRIOR TO TESTING.
29. CONTRACTOR SHALL CONDUCT FORWARD FLOW TESTING AND THE TESTING SHALL BE WITNESSED BY ENGINEER AND/OR AUTHORITY HAVING JURISDICTION.
30. PROVIDE DRAINS REQUIRED BY APPLICABLE CODES, WHETHER INDICATED IN THESE PLANS OR NOT. DISCHARGE ALL DRAINS TO THE EXTERIOR, UNLESS NOTED OTHERWISE.
    A. DISCHARGE DRAINS AT SIX INCHES ABOVE FINISHED GRADE. PROVIDE A CONCRETE SPLASH-BLOCK, WITH MINIMUM DIMENSIONS OF TWO FEET BY TWO FEET, AT EACH DRAIN.
    B. DRAINS LOCATED IN STORIES BELOW GRADE MAY DISCHARGE AT A FLOOR DRAIN OR SUMP CAPABLE OF RECEIVING THE MAXIMUM DISCHARGE. TEST CONNECTIONS FOR ALL FLOORS SHALL DISCHARGE TO THE EXTERIOR.
31. PROVIDE PRINTED METAL OR ENGRAVED PLASTIC NAMEPLATES TO IDENTIFY FUNCTION OF ALL ISOLATION VALVES, DRAIN VALVES, AND OTHER FIRE PROTECTION SYSTEM EQUIPMENT. PERMANENTLY AFFIX TO VALVE BODY OR HANG WITH CHAIN. CONCEALED COMPONENTS SHALL ADDITIONALLY HAVE PRINTED METAL OR ENGRAVED PLASTIC NAMEPLATES IDENTIFYING THE COMPONENT LOCATION AND FUNCTION PERMANENTLY AFFIXED TO ACCESS PANEL DOORS OR THE CEILING GRID WHERE THE COMPONENT IS ACCESSED.

# FIRE SUPPRESSION DESIGN PARAMETERS

1. FIRE HYDRANT FLOW TEST DATA
    a. DATE TEST WAS PERFORMED: 3/14/25
    b. PRESSURE HYDRANT:
        ii. STATIC PRESSURE (PSI): 82
        iii. RESIDUAL PRESSURE (PSI): 60
        iv. FLOW (GPM): 876
2. SPRINKLER SYSTEM DESIGN CRITERIA:
    a. ORDINARY HAZARD GROUP 2 (WET) -
       ALL AREAS ARE ORDINARY HAZARD GROUP 2 (WET) UNLESS OTHERWISE NOTED.
    b. ORDINARY HAZARD GROUP 2 (DRY) -
       ELEC ROOM, SERVICE DOCK, AREA BENEATH OVERHANG
3. SYSTEM PROTECTION AREA LIMITATIONS:
    a. THE MAXIMUM FLOOR AREA ON ANY ONE FLOOR TO BE PROTECTED BY SPRINKLERS SUPPLIED BY ANY ONE SPRINKLER RISER OR COMBINED SYSTEM SHALL BE 52,000 SQ FT
    b. THERE IS NO LIMIT TO THE NUMBER OF FLOORS THAT CAN BE PROTECTED BY A SINGLE RISER, BECAUSE EACH FLOOR IS CONSIDERED A SEPARATE FIRE AREA.
4. SPRINKLER DESIGN DENSITIES/DEMAND CRITERIA:
    a. ORDINARY HAZARD GROUP 2 - 0.20 GPM PER SQ FT OVER THE HYDRAULICALLY MOST REMOTE 1500 SQ FT.
    b ORDINARY HAZARD GROUP 2 (DRY) - 0.20 GPM PER SQ FT OVER THE HYDRAULICALLY MOST REMOTE 1950 SQ FT

FOR REFERENCE ONLY

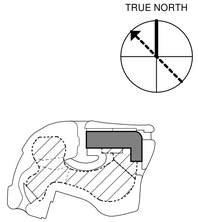
# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

TRUE NORTH



KEY PLAN

0 4' 8' 16'
1/8"=1'-0"

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
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Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | FIRE SUPPRESSION ABBREVIATIONS AND SYMBOLS |
| WORKING STAFF | EG (DESIGNED BY) / EG (DRAWN BY) / KL (CHECKED BY) |

SHEET NO. 328 OF 378

FX-001
DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

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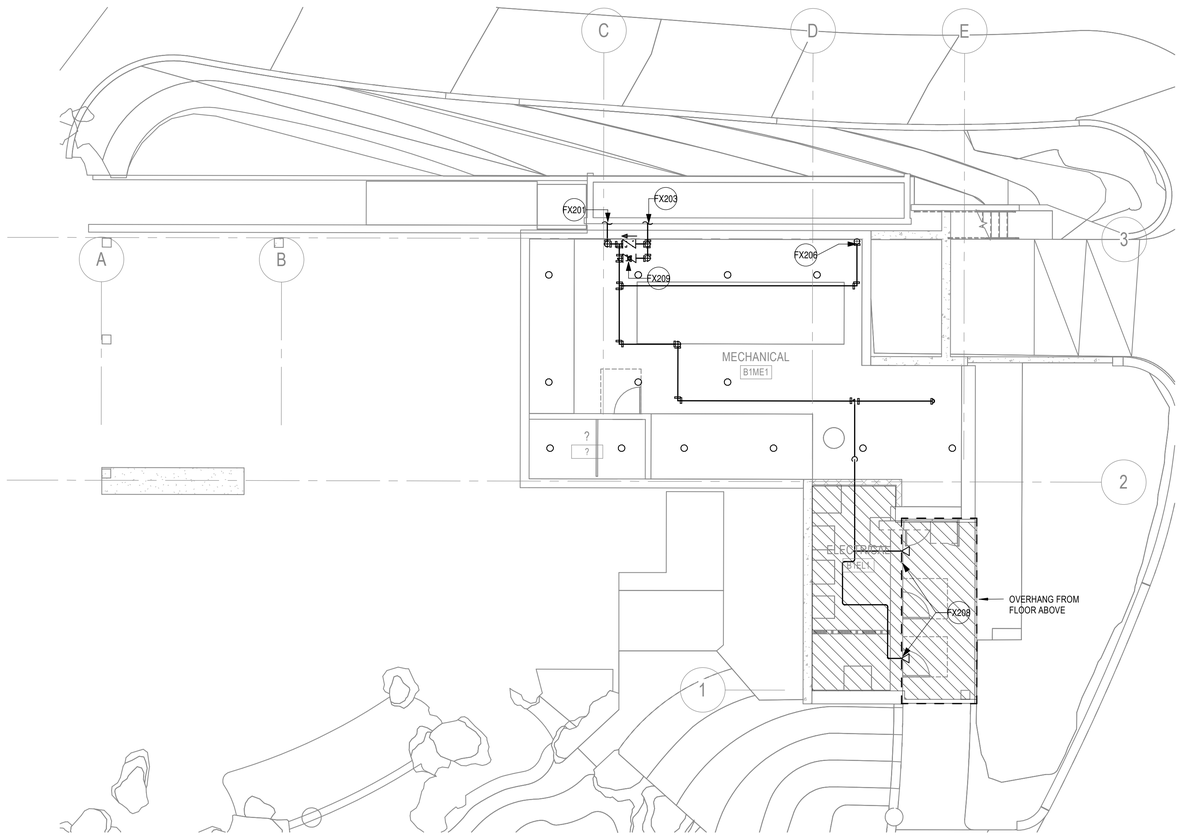


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FX-101FP

## BASEMENT - FIRE SUPPRESSION PLAN

SCALE = 1/8" = 1'-0"

## GENERAL SHEET NOTES

A. REFER TO SHEET FX-001 FOR FIRE SUPPRESSION ABBREVIATIONS AND SYMBOLS.

B. ALL SPACES SHALL BE DESIGNED PER OH-2 CRITERIA, UNLESS OTHERWISE NOTED, AS REQUIRED BY SMITHSONIAN DESIGN STANDARDS SECTION 11.5.6.1. SPRINKLER SYSTEMS SHALL MEET A REQUIRED DENSITY AND COVERAGE AREA OF 8.1 mm/min OVER 140 m² (0.20 GPM/SF OVER 1500 SF).

C. SPRINKLERS IN AREAS WITH EXPOSED CEILINGS SHALL BE UPRIGHT TYPE. SPRINKLERS IN GYPSUM AND ACT CEILINGS SHALL BE CONCEALED TYPE. COORDINATE CONCEALED SPRINKLER COVER PLATE COLOR WITH CEILING FINISH. REFER TO ARCHITECTURAL DRAWINGS FOR CEILING FINISHES.

## SHEET KEYNOTES

| | |
|---|---|
| FX201 | INCOMING 6" LINE FOR FIRE WATER SERVICE |
| FX203 | INCOMING 4" LINE FROM FREESTANDING FDC. |
| FX206 | 4" RISER UP TO LEVEL 1. |
| FX208 | PROVIDE DRY BARREL SIDEWALL SPRINKLERS TO PROTECT EXTERIOR AREA BENEATH OVERHANG. |
| FX209 | PROVIDE SUPERVISED NORMALLY CLOSED BUTTERFLY VALVE ON FDC BYPASS LINE FOR FORWARD FLOW TESTING. |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

0 4' 8' 16'

1/8"=1'-0"

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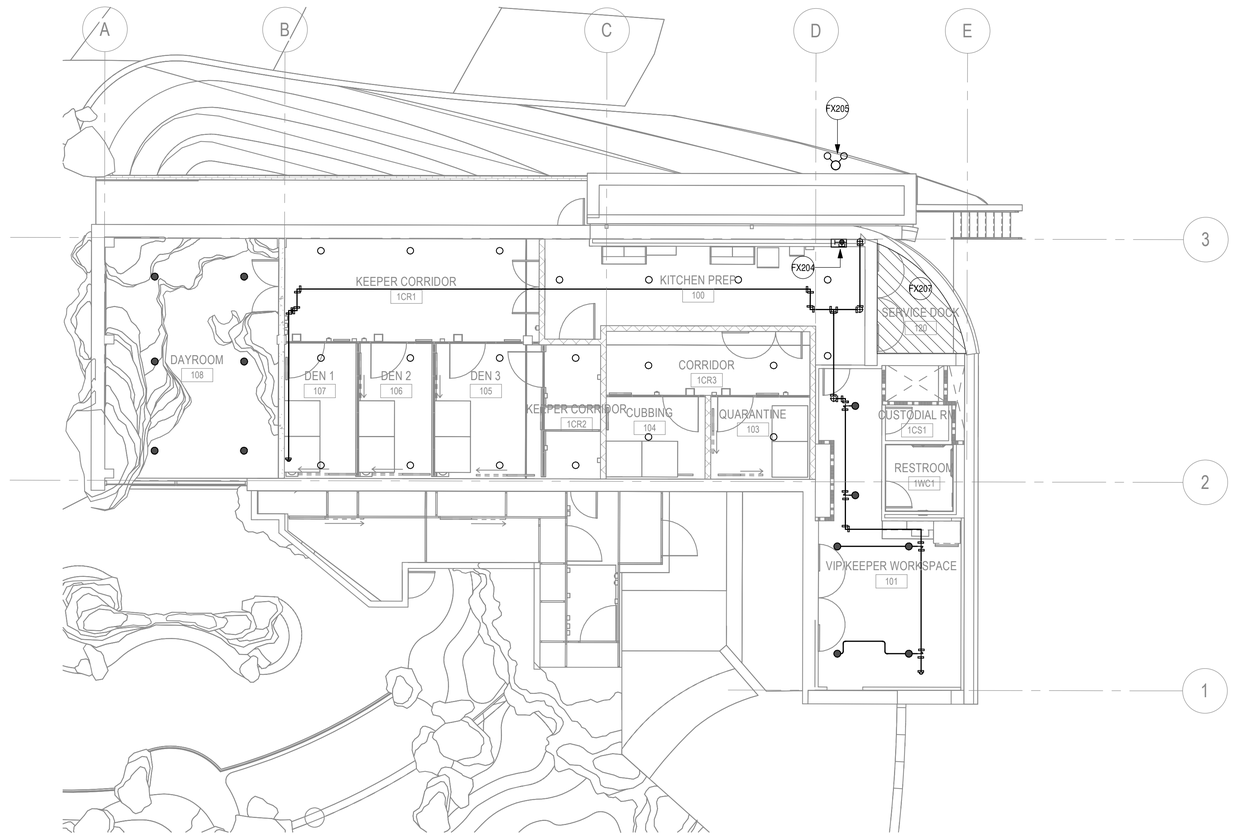


1 FX-102FP

## LEVEL 1 - FIRE SUPPRESSION PLAN

SCALE = 1/8" = 1'-0"

## GENERAL SHEET NOTES

A. REFER TO SHEET FX-001 FOR FIRE SUPPRESSION ABBREVIATIONS AND SYMBOLS.

B. ALL SPACES SHALL BE DESIGNED PER OH-2 CRITERIA, UNLESS OTHERWISE NOTED, AS REQUIRED BY SMITHSONIAN DESIGN STANDARDS SECTION 11.5.6.1. SPRINKLER SYSTEMS SHALL MEET A REQUIRED DENSITY AND COVERAGE AREA OF 8.1 mm/min OVER 140 m² (0.20 GPM/SF OVER 1500 SF).

C. SPRINKLERS IN AREAS WITH EXPOSED CEILINGS SHALL BE UPRIGHT TYPE. SPRINKLERS IN GYPSUM AND ACT CEILINGS SHALL BE CONCEALED TYPE. COORDINATE CONCEALED SPRINKLER COVER PLATE COLOR WITH CEILING FINISH. REFER TO ARCHITECTURAL DRAWINGS FOR CEILING FINISHES.

## SHEET KEYNOTES

| | |
|---|---|
| FX204 | PROVIDE HOSE VALVE CABINET FOR 2.5" HOSE VALVE. |
| FX205 | PROVIDE TWO-WAY FREESTANDING FDC. SEE CIVIL UTILITY PLAN FOR EXACT LOCATION. |
| FX207 | PROVIDE DRY BARREL SIDEWALL SPRINKLER TO PROTECT SERVICE DOCK AREA BENEATH ROOF OVERHANG. |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

0 4' 8' 16'

1/8"=1'-0"

GRAPHIC SCALE(S)

DATE: 06/15/26

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION


Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520


BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: LEVEL 1 FIRE SUPPRESSION PLAN

WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| EG | EG | KL |

SHEET NO. 330 OF 378

FX-102FP

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

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# ELECTRICAL ABBREVIATIONS

| | | |
|---|---|---|
| A | A | AMPERES |
| | AC | ALTERNATING CURRENT |
| | AF | AMPERE FRAME (BREAKER RATING) |
| | AFC | ABOVE FINISHED COUNTER |
| | AFF | ABOVE FINISHED FLOOR |
| | AFG | ABOVE FINISHED GRADE |
| | AHU | AIR HANDLING UNIT |
| | AIC | AMPERE INTERRUPTING CAPACITY |
| | ALT | ALTERNATE |
| | ARCH | ARCHITECT |
| | AT | AMPERES TRIP |
| | ATS | AUTOMATIC TRANSFER SWITCH |
| | AUTO | AUTOMATIC |
| | AUX | AUXILIARY |
| | AWG | AMERICAN WIRE GAUGE |
| B | BLDG | BUILDING |
| | BRKR, BKR | BREAKER |
| C | C | CONDUIT |
| | CATV | CABLE TELEVISION |
| | CB | CIRCUIT BREAKER |
| | CCTV | CLOSED CIRCUIT TELEVISION |
| | CE | CONTROLLED EMERGENCY |
| | CKT | CIRCUIT |
| | CLD | CURRENT LIMITING DEVICE |
| | CLG | CEILING |
| | COM, COMM | COMMUNICATIONS |
| | COPY | COPIER |
| | CP | CONTROL PANEL |
| | CPT | CONTROL POWER TRANSFORMER |
| | CT | CURRENT TRANSFORMER |
| | CU | COPPER |
| D | DC | DIRECT CURRENT |
| | DISC | DISCONNECT |
| | DIST | DISTRIBUTION |
| | DIV | DIVISION |
| | DN | DOWN |
| | DP | DISTRIBUTION PANEL |
| | DPDT | DOUBLE POLE DOUBLE THROW |
| | DPST | DOUBLE POLE SINGLE THROW |
| | DW | DISHWASHER |
| | DWG | DRAWING |
| E | E, EX, EXIST | EXISTING |
| | EG | EQUIPMENT GROUND |
| | ELEC | ELECTRIC, ELECTRICAL |
| | ELEV | ELEVATOR |
| | EM. EMEG | EMERGENCY |
| | EMT | ELECTRIC METALLIC CONDUIT |
| | ENCL | ENCLOSURE |
| | EQ, EQIP | EQUIPMENT |
| | EWC | ELECTRIC WATER COOLER |
| | EWH | ELECTRIC WATER HEATER |
| F | F | FUSE |
| | FA | FIRE ALARM |
| | FAAP | FIRE ALARM ANNUNCIATOR PANEL |
| | FACP | FIRE ALARM CONTROL PANEL |
| | FAEP | FIRE ALARM EXTENDER PANEL |
| | FC | FOOT CANDLE |
| | FDR | FEEDER |
| | FL, FLR | FLOOR |
| | FLA | FULL LOAD AMPS |
| | FLEX | FLEXIBLE |
| | FLUOR | FLUORESCENT |
| | FS | FLUSH SENSOR |
| | FT | FOOT / FEET (') |
| G | G, GND, GRD | GROUND |
| | GEN | GENERATOR |
| | GFI | GROUND FAULT INTERRUPTER |
| H | H | HORIZONTAL MOUNTING |
| | HH | HAND HOLE |
| | HOA | HAND-OFF-AUTOMATIC |
| | HP | HORSEPOWER |
| | HPS | HIGH PRESSURE SODIUM |
| | HR | HOUR |
| | HT | HEIGHT |
| | HTR | HEATER |
| | HVAC | HEATING VENTILATION AND AIR CONDITIONING |
| | HZ | HERTZ |
| I | IG | ISOLATED GROUND |
| | IN | INCH / INCHES (") |
| | INCAND | INCANDESCENT |
| | JB, JBOX | JUNCTION BOX |
| K | K | KEY INTERLOCK |
| | Kcmil | 1000 CIRCULAR mils |
| | KV | KILOVOLTS (THOUSAND VOLTS) |
| | KVA | KILOVOLTS-AMPERES (THOUSAND VOLT-AMPS) |
| | kW | KILOWATTS (THOUSAND WATTS) |
| | KWH | KILOWATT-HOURS |
| | L-G | LINE TO GROUND |
| | L-L | LINE TO LINE |
| | L-N | LINE TO NEUTRAL |
| L | LA | LIGHTNING ARRESTOR |
| | LAN | LOCAL AREA NETWORK |
| | LC, LCP | LIGHTING CONTROL PANEL |
| | LP | LIGHTING PANEL |
| | LS | LAVATORY SENSOR |
| | LSI | LONG-TIME, SHORT-TIME, INSTANTANEOUS |
| | LSIA | LONG-TIME, SHORT-TIME, INSTANTANEOUS,GROUND-FAULT ALARM ONLY |
| | LSIG | LONG-TIME, SHORT-TIME, INSTANTANEOUS,GROUND-FAULT |
| | LTG | LIGHTING |
| | LV | LOW VOLTAGE (BELOW 50 VOLTS) |
| M | MCA | MINIMUM CIRCUIT AMPACITY |
| | MCB | MAIN CIRCUIT BREAKER |
| | MCC | MOTOR CONTROL CENTER |
| | MECH | MECHANICAL |
| | MFR | MANUFACTURER |
| | MH | MANHOLE |
| | MIN | MINIMUM |
| | MISC | MISCELLANEOUS |
| | MLO | MAIN LUGS ONLY |
| | MOCP | MAXIMUM OVERCURRENT PROTECTION |
| | MTD | MOUNTED |
| | MTG | MOUNTING |
| | MTS | MANUAL TRANSFER SWITCH |
| | MV | MEDIUM VOLTAGE (OVER 600V LESS THAN 35KV) |
| N | N | NEUTRAL |
| | NC | NORMALLY CLOSED |
| | NEC | NATIONAL ELECTRICAL CODE |
| | NEMA | NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION |
| | NIC | NOT IN CONTRACT |
| | NL | NIGHTLIGHT |
| | NO | NORMALLY OPEN |
| | NO., NUM. # | NUMBER |
| | NTS | NOT TO SCALE |
| O | OC | ON CENTER |
| | OCPD | OVERCURRENT PROTECTION DEVICE |
| | OFCI | OWNER FURNISHED, CONTRACTOR INSTALLED |
| | OFOI | OWNER FURNISHED, OWNER INSTALLED |
| | OH | OVERHEAD |
| | OHD | OVERHEAD DOOR |
| | OHE/T | OVERHEAD ELECTRIC/TELEPHONE |
| | OPP | OPPOSITE |
| P | P | POLE |
| | PA | PUBLIC ADDRESS |
| | PB | PULL BOX |
| | PDP | POWER DISTRIBUTION PANEL |
| | PE | PHOTO ELECTRIC |
| | PF | POWER FACTOR |
| | PH, Ø | PHASE |
| | PNL | PANEL |
| | PRI | PRIMARY |
| | PT | POTENTIAL TRANSFORMER |
| | PVC | POLYVINYL CHLORIDE |
| | PWR | POWER |
| Q | QTY | QUANTITY |
| R | R, RE | RELOCATE AS SHOWN |
| | RCLP | REMOTE CONTROL LIGHTING PANEL |
| | RCPT | RECEPTACLE |
| | REF | REFRIGERATOR |
| | RF | RADIO FREQUENCY |
| | RLA | RATED (RUNNING) LOAD AMPS |
| | RM | ROOM |
| | RP | RECEPTACLE PANELBOARD |
| | RSC | RIGID STEEL CONDUIT |
| S | SCH, SCHED | SCHEDULE |
| | SEC | SECONDARY |
| | SF | SQUARE FOOT |
| | SPD | SURGE PROTECTIVE DEVICE |
| | SPDT | SINGLE POLE DOUBLE THROW |
| | SPEC(S) | SPECIFICATION(S) |
| | SPKR | SPEAKER |
| | SPST | SINGLE POLE SINGLE THROW |
| | STD | STANDARD |
| | SW | SWITCH |
| | SWBD | SWITCHBOARD |
| | SWGR | SWITCHGEAR |
| | SYM | SYMMETRICAL |
| T | TB | TERMINAL BLOCK |
| | TBB | TELEPHONE BACKBOARD |
| | TC | TIME CLOCK |
| | TEL, TELE | TELEPHONE |
| | TELECOM | TELECOMMUNICATIONS |
| | TP | TAMPERPROOF |
| | TV | TELEVISION |
| | TYP | TYPICAL |
| U | UG | UNDERGROUND |
| | UGP | UNDERGROUND PRIMARY |
| | UGS | UNDERGROUND SECONDARY |
| | UGT | UNDERGROUND TELEPHONE |
| | UL | UNDERWRITER'S LABORATORY |
| | UON | UNLESS OTHERWISE NOTED |
| | UPS | UNINTERRUPTIBLE POWER SUPPLY |
| V | V | VOLTS |
| | VA | VOLT-AMPERES |
| | VFD | VARIABLE FREQUENCY DRIVE/ VARIABLE FREQUENCY MOTOR CONTROLLER |
| W | W | WIRE |
| | W/ | WITH |
| | W/O | WITHOUT |
| | WHM | WATT HOUR METER |
| | WP | WEATHERPROOF |
| X | X | REMOVE DEVICE |
| | XFMR | TRANSFORMER |
| Z | %Z | PERCENT IMPEDANCE |

# DISTRIBUTION AND DIAGRAM SYMBOLS

- SWITCH - SINGLE THROW
- FUSED SWITCH
- T — TRANSFORMER - TYPE AND RATING AS INDICATED
- POTENTIAL TRANSFORMER
- CPT — CONTROL POWER TRANSFORMER
- CURRENT TRANSFORMER
- DELTA CONNECTION
- WYE CONNECTION - SOLID GROUND
- E — ELECTRIC INTERLOCK SYSTEM
- K — KEY INTERLOCKING SYSTEM
- DRAWOUT TYPE, CIRCUIT BREAKER
- STATIONARY CIRCUIT BREAKER
- FUSE
- ATS — AUTOMATIC TRANSFER SWITCH
- BUS DUCT PLUG-IN UNIT
- LIGHTNING ARRESTOR
- GROUND
- STRESS CONE
- M, EMU — ELECTRONIC METERING UNIT
- SPD — SURGE PROTECTIVE DEVICE
- DISTRIBUTION PANELBOARD
- PANELBOARD - (FLUSH/SURFACE)
- LCP — LIGHTING CONTROL PANEL, SURFACE MTD. UON.
- LRP — LOW VOLTAGE LIGHTING RELAY PANEL, SURFACE MTD. UON
- MOTOR CONTROL CENTER
- UPS — UNINTERRUPTIBLE POWER SUPPLY
- TBB — TELECOMMUNICATIONS BACKBOARD
- RP-XXXX — SYMBOL INDICATES TO WHICH PANELBOARD INDICATED CIRCUITS IN THE ROOM ARE TO BE CONNECTED.
- GAP — GENERATOR ANNUNCIATOR (DERANGEMENT) PANEL
- G — GENERATOR

# ELECTRICAL WIRING SYMBOLS

- A-1 — HOMERUN TO PANEL WITH CIRCUIT NUMBER(S) AS INDICATED
- CONDUIT RUN CONCEALED IN FINISHED AREAS, EXPOSED IN UNFINISHED AREAS OR BELOW ACCESS FLOORS
- CONDUIT CAST IN CONCRETE OR BELOW SLAB
- DEMOLITION WORK
- UG — UNDERGROUND CONDUIT/DUCTBANK
- FB — FEEDER BUSWAY
- PB — PLUG-IN BUSWAY
- G — GROUND CABLE, SIZE AS INDICATED
- GB — GROUND BUS
- CT — CABLE TRAY
- CONDUIT TURNED UP
- CONDUIT TURNED DOWN
- EMH — ELECTRIC MANHOLE
- EHH — ELECTRIC HANDHOLE
- TMH — TELEPHONE MANHOLE
- GROUND ROD

# ELECTRICAL EQUIPMENT NAMING CONVENTION KEY

XX-XXXX

- PREFIX
- SYSTEM (IF APPLICABLE)
- LETTER ID
- LEVEL

| PREFIX | DESCRIPTION |
|---|---|
| DP | DISTRIBUTION PANELBOARD |
| LP | LIGHTING PANELBOARD |
| PP | POWER PANELBOARD |
| RP | RECEPTACLE PANELBOARD |
| SWB | SWITCHBOARD |
| T | TRANSFORMER |

| SYSTEM | DESCRIPTION |
|---|---|
| EM | EMERGENCY |
| LR | LEGALLY REQUIRED |
| OS | OPTIONAL STANDBY |

| LEVEL | DESCRIPTION |
|---|---|
| G | GROUND LEVEL |
| 1 | FIRST LEVEL |
| R | ROOF LEVEL |

# ELECTRICAL EQUIPMENT SYMBOLS

- MOTOR AND CONNECTION
- DISCONNECT SWITCH, NON-FUSED. PROVIDE SWITCH AMPACITY EQUAL TO OR GREATER THAN FEEDER AMPACITY, UON
- DISCONNECT SWITCH, FUSED. PROVIDE SWITCH AMPACITY EQUAL TO OR GREATER THAN FEEDER AMPACITY, UON. FUSE AMPACITY AS INDICATED AND NO LESS THAN REQUIRED BY EQUIPMENT NAMEPLATE.
- CB — CIRCUIT BREAKER IN NEMA 1 ENCLOSURE (FLUSH/SURFACE)
- COMBINATION MOTOR STARTER WITH DISCONNECT SWITCH. PROVIDE SWITCH AMPACITY EQUAL TO OR GREATER THAN FEEDER AMPACITY, UON. FUSE AMPACITY AS INDICATED AND NO LESS THAN REQUIRED BY EQUIPMENT NAMEPLATE.
- HAND-OFF-AUTO SELECTOR SWITCH WITH PILOT LIGHT
- M — MANUAL MOTOR STARTER WITH PILOT LIGHT AS INDICATED.
- PC — PACKAGE CONTROL UNIT (PROVIDED BY EQUIPMENT SUPPLIER)
- VFD — VARIABLE FREQUENCY DRIVE/VARIABLE FREQUENCY MOTOR CONTROLLER
- PUSHBUTTON STATION
- EMERGENCY POWER OFF
- C — CONTACTOR
- GROUNDING BUS BAR

# REFERENCE SYMBOLS

- 1 / E0.1 — SECTION NUMBER; SIM; DRAWING NUMBER WHERE DRAWN
- DETAIL NUMBER; DRAWING NUMBER WHERE DRAWN
- EXXX — KEYNOTE
- EDXX — DEMO KEYNOTE
- XXX — FEEDER KEYNOTE

# GENERAL NOTES

A. PROVIDE 1#12 + 1#12N + 1#12G FOR 20A BRANCH CIRCUITING, UON; MAXIMUM OF THREE CIRCUITS PER CONDUIT; MINIMUM CONDUIT SIZE OF 3/4" C, UON.

B. PROVIDE A DEDICATED NEUTRAL WIRE FOR EACH BRANCH CIRCUIT.

C. COORDINATE DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN.

D. PROVIDE FIRE STOPPING FOR CONDUITS AND ELECTRICAL EQUIPMENT FOR FLOOR SLABS, WALLS AND CEILINGS TO MAINTAIN FIRE RATING.

E. INSTALL ELECTRICAL WORK IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE AND LOCAL BUILDING AND FIRE CODES, IN A NEAT AND WORKMANLIKE MANNER.

F. LIMIT VOLTAGE DROP TO 2% FOR FEEDERS AND 3% FOR BRANCH CIRCUITS. INCLUDE DERATING FACTOR FOR ROOF-MOUNTED CONDUITS.

G. REFER TO ARCH. REFLECTED CEILING PLANS FOR EXACT LOCATION OF CEILING MOUNTED DEVICES AND LIGHT FIXTURES, UNLESS OTHERWISE INDICATED.

H. WHERE MULTIPLE SWITCHES, RECEPTACLES, AND OTHER OUTLETS (EXCEPT WALL PHONES) ARE INDICATED PROVIDE MULTI-GANG BACK BOXES WITH GANG BARRIERS AND A COMMON FACEPLATE.

I. WHERE DIFFERENT RECESSED ELECTRICAL DEVICES WITH THE SAME MOUNTING HEIGHTS ARE INDICATED SIDE-BY-SIDE, MOUNT THE DEVICES SO THAT THERE IS FOUR INCHES BETWEEN ADJACENT VERTICAL EDGES OF THE FACEPLATES, UON.

J. WHERE ELECTRICAL DEVICES WITH DIFFERENT MOUNTING HEIGHTS ARE LOCATED IN THE SAME AREA ALIGN DEVICES VERTICALLY THROUGH THEIR CENTERLINES, UON.

K. WHERE EXIT SIGNS ARE INDICATED ABOVE DOOR MOUNT AS FOLLOWS, CENTER THE EXIT SIGN BETWEEN TOP OF DOOR FRAME AND CEILING IF DISTANCE BETWEEN TOP OF DOOR FRAME AND CEILING IS 24 INCHES OR LESS; OTHERWISE MOUNT BOTTOM OF EXIT SIGN 6 INCHES FROM TOP OF DOOR FRAME, MOUNT OTHER WALL MOUNTED EXIT SIGNS IN THE SAME AREA AT THE SAME HEIGHT. COORDINATE WITH ARCHITECTURAL DETAILS.

L. EXIT SIGNS ARE TO BE TYPE "X1", UON.

M. PROVIDE SEPARATE NEUTRALS FOR DIMMING CIRCUITS.

N. PROVIDE FEEDERS AND BRANCH CIRCUITS WHICH HAVE AN AMPACITY EQUAL TO OR GREATER THAN THE CIRCUIT OVERCURRENT PROTECTIVE DEVICE RATING, U.O.N.

O. FURNITURE LAYOUTS ARE FOR REFERENCE ONLY. COORDINATE THE FINAL LOCATION OF ELECTRICAL DEVICES AND OUTLETS WITH ARCHITECT, OWNER AND FINAL FURNITURE PLANS PRIOR TO INSTALLATION.

P. MOUNT DEVICES FLUSH WITH CONCEALED CONDUIT, EXCEPT AS NOTED ON THE DRAWINGS AND IN MECHANICAL AND ELECTRICAL EQUIPMENT ROOMS.

Q. THE DRAWINGS REPRESENT ELECTRICAL DESIGN INTENT. THEY ARE DIAGRAMMATIC. PROVIDE ALL COMPONENTS NECESSARY FOR COMPLETE OPERATING ELECTRICAL SYSTEMS.

R. WHEN MORE THAN ONE 20A., 1P., 120 VOLT CIRCUIT FEEDS A SURFACE RACEWAY CONNECT RECEPTACLES TO ALTERNATE CIRCUITS AS FOLLOWS - U.O.N.

# MOUNTING HEIGHTS

| | |
|---|---|
| ACCESS CONTROL DEVICES | 3'-6" AFF |
| CLOCK OUTLETS | 7'-0" AFF |
| DIMMERS | 3'-6" AFF |
| DISCONNECT SWITCHES | 5'-6" AFF |
| INDIVIDUAL CIRCUIT BREAKERS | 5'-6" AFF |
| INTERCOM | 3'-6" AFF |
| MOTOR STARTERS | 5'-6" AFF |
| PANELBOARDS & CABINETS | 6'-0" AFF TO TOP |
| PUSHBUTTONS | 3'-6" AFF |
| RECEPTACLES | 1'-6" AFF |
| RECEPTACLES ABOVE FINISHED COUNTER | 8" AFC |
| VANITY LIGHT IN TOILET | 6" ABOVE THE MIRROR, UON |
| VOLUME CONTROLS | 3'-6" AFF |
| WALL SWITCHES | 3'-6" AFF |
| TELECOMMUNICATIONS OUTLETS | 1'-6" AFF |
| TELECOMMUNICATIONS OUTLETS - WALL PHONE | 3'-6" AFF |
| TV OUTLETS | 5'-0" AFF |
| NURSE CALL DOME LIGHTS | 7'-6" AFF |
| NURSE CALL STATIONS | 3'-6" AFF |
| FIRE ALARM - PULL STATIONS | 3'-6" AFF |
| FIREMAN'S PHONE | 3'-6" AFF |
| FIRE ALARM DEVICES HORN / SPEAKERS / STROBES (MOUNTING HEIGHT AT THE TOP OF THE FIRE ALARM DEVICE OR 6" BELOW CEILING WHICH IS LOWER) | 7'-2" AFF TO TOP |

**MOUNTING HEIGHT NOTES:**

1. ALL ELEVATIONS ARE TO CENTER LINE OF DEVICE, UNLESS OTHERWISE NOTED.
2. REFER TO EQUIPMENT ELEVATION DRAWINGS FOR COORDINATION WITH CASEWORK.
3. MOUNT MANUAL MOTOR STARTER ADJACENT TO OR ON UNIT.
4. REFER TO ARCHITECTURAL A0 SERIES FOR ADDITIONAL ELECTRICAL DEVICE MOUNTING HEIGHTS AND DEVICE ALIGNMENT RULES.
5. TYPICAL MOUNTING HEIGHTS UNLESS OTHERWISE NOTED.

# SYMBOL SHEET NOTES

A. THE PURPOSE OF THIS SHEET IS TO ILLUSTRATE AND DEFINE TYPICAL GRAPHIC SYMBOLS WHICH MAY TYPICALLY OCCUR ON THE ELECTRICAL DRAWINGS.

B. THIS SHEET MAY ILLUSTRATE ABBREVIATIONS, SYMBOLS AND LEGENDS WHICH DO NOT OCCUR AS PART OF THIS PROJECT. REFER TO THE DRAWINGS TO DETERMINE WHICH ITEMS APPLY TO THE WORK OF THIS PROJECT.

C. ADDITIONAL SYMBOLS NOT SHOWN OR DEFINED ON THIS SHEET MAY BE USED ON THE ELECTRICAL DRAWINGS AND ARE TYPICALLY DEFINED ON OTHER SHEETS.

D. SYMBOLS ARE NOT DRAWN TO SCALE. TO DETERMINE THE ACTUAL SIZES OF ELEMENTS REPRESENTED BY SYMBOLS, REFER TO THE SPECIFICATIONS AND OTHER SHEETS, AS MAY APPLY.

ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # | DATE | REVISION DESIGNATION |
|---|---|---|
| 1 | 05/11/2026 | AMENDMENT 01 |

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ELECTRICAL ABBREVIATIONS AND SYMBOLS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 331 of 378

E-001CS

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

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# ELECTRICAL LIGHTING SYMBOLS

| | |
|---|---|
| | WALL MOUNTED LIGHTING FIXTURE- TYPE AS INDICATED |
| (#) a | LIGHTING FIXTURE - TYPE AS INDICATED (SUBSCRIPT INDICATES SWITCH CONTROL), (#) - NETWORKED LIGHTING CONTROL ZONE |
| | LIGHTING FIXTURE - TYPE AS INDICATED |
| | WALLWASH LIGHT FIXTURE |
| | SHADING INDICATES FIXTURE CONNECTED TO LIFE SAFETY BRANCH (SWITCHED) |
| | LIGHTING TRACK - TYPE AS INDICATED<br>CURRENT LIMITING DEVICE, CLD, SHOWN AS A BOX ON END OF THE TRACK. |
| | EXIT LIGHT - WALL/CEILING MOUNTED - DIRECTIONAL ARROWS AS INDICATED SHADING INDICATES ILLUMINATED FACE |
| | SITE LIGHTING FIXTURE TYPE AS INDICATED |
| XX | LIGHTING FIXTURE TYPE DESIGNATION - "XX" INDICATES SPECIFIC TYPE REFER TO LIGHT FIXTURE SCHEDULE |
| | SINGLE OR DOUBLE, REMOTE EMERGENCY LIGHTING FIXTURE |
| $S_a$ | SINGLE POLE SWITCH (SUBSCRIPT INDICATES FIXTURES CONTROLLED) |
| $S_3$ | THREE-WAY SWITCH (S3P-THREE-WAY SWITCH WITH PILOT LIGHT) |
| $S_4$ | FOUR - WAY SWITCH |
| $S_P$ | SINGLE POLE SWITCH WITH PILOT LIGHT |
| $S_K$ | KEY OPERATED SWITCH |
| $S_3S_3$ | DUAL LEVEL SWITCHING. PROVIDE BALLASTS/DRIVERS REQUIRED TO PROVIDE DUAL LEVEL SWITCHING. |
| $S_L$ | LOW VOLTAGE SWITCH |
| $S_{L1}$ | SINGLE ZONE LOW VOLTAGE ON/OFF SWITCH |
| $S_{L1D}$ | SINGLE ZONE LOW VOLTAGE DIMMER SWITCH |
| $S_{L2}$ | 2 ZONE LOW VOLTAGE ON/OFF SWITCH |
| $S_{L2D}$ | 2 ZONE LOW VOLTAGE DIMMER SWITCH |
| $S_{L4}$ | 4 ZONE LOW VOLTAGE ON/OFF SWITCH |
| $S_{L4D}$ | 4 ZONE LOW VOLTAGE DIMMER SWITCH |
| $S_D$ | DIMMER SWITCH |
| $S_{OS}$ | OCCUPANCY SENSOR WALL SWITCH |
| PI PI | PASSIVE INFRARED OCCUPANCY SENSOR (WALL/CEILING) |
| U U | ULTRASONIC OCCUPANCY SENSOR (WALL/CEILING) |
| DT DT | PASSIVE INFRARED/ULTRASONIC OCCUPANCY SENSOR (WALL/CEILING) |
| PH | PHOTO CELL |
| CS | COMBINATION PHOTOCELL/OCCUPANCY SENSOR |
| ELTD | EMERGENCY LIGHTING TRANSFER DEVICE |

# ELECTRICAL POWER SYMBOLS

| | |
|---|---|
| | SINGLE RECEPTACLE, NEMA 5-20R (SLASH INDICATES MOUNTING ABOVE FINISH COUNTER) |
| | DUPLEX RECEPTACLE, NEMA 5-20R (SLASH INDICATES MOUNTING ABOVE FINISH COUNTER) |
| | GFI, DUPLEX RECEPTACLE, NEMA 5-20R (SLASH INDICATES MOUNTING ABOVE FINISH COUNTER) |
| | QUADRUPLEX RECEPTACLE, (2) NEMA 5-20R, DUPLEX RECEPTACLES WITH A COMMON FACEPLATE. (SLASH INDICATES MOUNTING ABOVE FINISH COUNTER) |
| | GFI, QUADRUPLEX RECEPTACLE, (2) NEMA 5-20R, DUPLEX RECEPTACLES WITH A COMMON FACEPLATE. (SLASH INDICATES MOUNTING ABOVE FINISH COUNTER) |
| | SINGLE ISOLATED GROUND RECEPTACLE, NEMA 5-20R (SLASH INDICATES MOUNTING ABOVE FINISH COUNTER) |
| | DUPLEX ISOLATED GROUND RECEPTACLE, NEMA 5-20R (SLASH INDICATES MOUNTING ABOVE FINISH COUNTER) |
| | QUADRUPLEX ISOLATED GROUND RECEPTACLE, (2)NEMA 5-20R, DUPLEX ISOLATED GROUND RECEPTACLES WITH A COMMON FACEPLATE. (SLASH INDICATES MOUNTING ABOVE FINISH COUNTER) |
| | FULLY CONTROLLED DUPLEX/QUADRUPLEX RECEPTACLE, NEMA 5-20R. PROVIDE ALL WIRING, EQUIPMENT, POWER PACKS, AND OCCUPANCY SENSOR INTEGRATION REQUIRED TO TURN OFF RECEPTACLE(S) WHEN SPACE IS UNOCCUPIED. |
| | HALF CONTROLLED DUPLEX/QUADRUPLEX RECEPTACLE, NEMA 5-20R. PROVIDE ALL WIRING, EQUIPMENT, POWER PACKS, AND OCCUPANCY SENSOR INTEGRATION REQUIRED TO TURN OFF HALF OF RECEPTACLE(S) WHEN SPACE IS UNOCCUPIED. |
| | HALF SWITCHED DUPLEX RECEPTACLE, NEMA 5-20R |
| | EMERGENCY DUPLEX RECEPTACLE, NEMA 5-20R (SHADING INDICATING EMERGENCY RECEPTACLE TYPICAL FOR ALL OTHER RECEPTACLE TYPES) |
| | COMBINATION USB DUPLEX RECEPTACLE, NEMA 5-20R, WITH (2) 5VDC, COMBINATION TYPE A AND C USB CHARGING PORTS. |
| 6-30R | SPECIAL PURPOSE RECEPTACLE. TYPE AS INDICATED ON DRAWINGS. COORDINATE RECEPTACLE TYPE WITH ACTUAL EQUIPMENT PRIOR TO INSTALLATION. |
| | CEILING MOUNTED SINGLE RECEPTACLE, NEMA 5-20R |
| | CEILING MOUNTED SINGLE ISOLATED GROUND RECEPTACLE, NEMA 5-20R |
| | CEILING MOUNTED DUPLEX RECEPTACLE, NEMA 5-20R ("CR" SUBSCRIPT INDICATES CORD REEL) |
| | CEILING MOUNTED DUPLEX ISOLATED GROUND RECEPTACLE, NEMA 5-20R |
| | CEILING MOUNTED QUADRUPLEX RECEPTACLE, (2) NEMA 5-20R, DUPLEX RECEPTACLES WITH A COMMON FACEPLATE. |
| | CEILING MOUNTED QUADRUPLEX ISOLATED GROUND RECEPTACLE, (2)NEMA 5-20R, DUPLEX ISOLATED GROUND RECEPTACLES WITH A COMMON FACEPLATE. |
| | SINGLE RECEPTACLE, NEMA 5-20R, FLOOR MTD SERVICE FITTING, FLUSH TYPE UON ("S" DENOTES SURFACE TYPE) |
| | SINGLE IG RECEPTACLE, NEMA 5-20R, FLOOR MTD SERVICE FITTING, FLUSH TYPE UON ("S" DENOTES SURFACE TYPE) |
| | DUPLEX RECEPTACLE, NEMA 5-20R, FLOOR MTD SERVICE FITTING, FLUSH TYPE UON ("S" DENOTES SURFACE TYPE) |
| | DUPLEX IG RECEPTACLE, NEMA 5-20R, FLOOR MTD SERVICE FITTING, FLUSH TYPE UON ("S" DENOTES SURFACE TYPE) |
| | QUADRUPLEX RECEPTACLE, (2) NEMA 5-20R, FLOOR MTD SERVICE FITTING, FLUSH TYPE UON ("S" DENOTES SURFACE TYPE) |
| | QUADRUPLEX IG RECEPTACLE, (2) NEMA 5-20R, FLOOR MTD SERVICE FITTING, FLUSH TYPE UON ("S" DENOTES SURFACE TYPE) |
| | WHERE INDICATED IN AN EXISTING FLOOR, PROVIDE CORE DRILL PACKAGE TO MAINTAIN FIRE RATING OF FLOOR. EXTEND CONDUIT AND WIRING IN CEILING SPACE OF FLOOR BELOW TO ELECTRICAL CLOSET OR PANELBOARD. |
| FBX | FLOOR BOX ("X" DENOTES TYPE AS SPECIFIED) |
| PTX | POKE-THROUGH ASSEMBLY ("X" DENOTES TYPE AS SPECIFIED) |
| AFX | ACCESS FLOOR BOX ("X" DENOTES TYPE AS SPECIFIED) |
| | REFER TO FLOOR BOX / POKE-THROUGH SCHEDULE FOR TYPES. |
| J J J | JUNCTION BOX (CEILING, FLOOR OR WALL MOUNTED) INDICATED IN POWER, LIGHTING & FIRE ALARM SYSTEMS PLANS. |
| | DIRECT CONNECTION TO EQUIPMENT |
| P P P | POWER SERVICE FITTING (CEILING, FLOOR OR WALL MOUNTED) FOR CONNECTION TO FURNITURE SYSTEM AS SPECIFIED. |
| SR-X | SURFACE RACEWAY ("X" DENOTES TYPE AS SPECIFIED) |
| HC | ADA DOOR ACTUATOR |
| PD PD PD | COMBINATION POWER AND TELECOMMUNICATIONS BOX (CEILING, FLOOR OR WALL MOUNTED) FOR POWER AND DATA CONNECTION INDICATED IN POWER & SPECIAL SYSTEMS PLANS. |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|
| 1 | 05/11/2026 | AMENDMENT 01 |

FOR REFERENCE ONLY


Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520


| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ELECTRICAL ABBREVIATIONS AND SYMBOLS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 332 OF 378

E-002CS (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)


6/15/2026 2:48:05 PM
E-002CS BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

## FEEDER & BRANCH CIRCUIT SIZING SCHEDULE - GENERAL PURPOSE

(NOTES 1 & 2)

| OVERCURRENT DEVICE RATING (AMPERES) | WIRE SIZE - AWG OR KCMIL | | CONDUIT SIZE | | | NOTE |
|---|---|---|---|---|---|---|
| | PHASE & NEUTRAL | E.G. | 2 WIRE | 3 WIRE | 4 WIRE | |
| 15-20 | 12 | 12 | 3/4" | 3/4" | 3/4" | |
| 25-30 | 10 | 10 | 3/4" | 3/4" | 3/4" | |
| 35-40 | 8 | 10 | 3/4" | 3/4" | 3/4" | |
| 45-50 | 8(6) | 10 | 3/4" | 3/4" | 3/4"(1") | |
| 60 | 6(4) | 10 | 3/4"(1") | 3/4"(1") | 1"(1 1/4") | |
| 70 | 6(4) | 8 | 3/4"(1") | 3/4"(1") | 1"(1 1/4") | |
| 80-90 | 4(2) | 8 | 1" | 1"(1 1/4") | 1 1/4" | |
| 100 | 3(2) | 8 | 1"(1 1/4") | 1 1/4" | 1 1/4" | |
| 110 | 2(1) | 6 | 1 1/4" | 1 1/4"(1 1/2") | 1 1/4"(1 1/2") | |
| 125 | 1(1/0) | 6 | 1 1/4" | 1 1/2" | 1 1/2"(2") | |
| 150 | 1/0 | 6 | 1 1/4" | 1 1/2" | 2" | |
| 175 | 2/0 | 6 | 1 1/2" | 2" | 2" | |
| 200 | 3/0 | 6 | 1 1/2" | 2" | 2" | |
| 225 | 4/0 | 4 | 2" | 2" | 2 1/2" | |
| 250 | 250 | 4 | 2" | 2 1/2" | 2 1/2" | |
| 300 | 350 | 4 | 2 1/2" | 3" | 3" | |
| 350 | 500 | 3 | 3" | 3" | 3 1/2" | |
| 400 | 2-3/0 | 2-3 | 2-2" | 2-2" | 2-2" | |
| 450 | 2-4/0 | 2-2 | 2-2" | 2-2" | 2-2 1/2" | |
| 500 | 2-250 | 2-2 | 2-2" | 2-2 1/2" | 2-2 1/2" | |
| 600 | 2-350 | 2-1 | 2-2 1/2" | 2-3" | 2-3" | |
| 700 | 2-500 | 2-1/0 | 2-3" | 2-3" | 2-3 1/2" | |
| 800 | 3-300 | 3-1/0 | 3-2 1/2" | 3-3" | 3-3" | |
| 1000 | 3-400 | 3-2/0 | 3-2 1/2" | 3-3" | 3-3" | |
| 1200 | 4-350 | 4-3/0 | 4-2 1/2" | 4-3" | 4-3" | |
| 1600 | 5-400 | 5-4/0 | 5-2 1/2" | 5-3" | 5-3" | |
| 2000 | 6-400 | 6-250 | 6-2 1/2" | 6-3" | 6-3" | |

CIRCUIT SIZING SCHEDULES NOTES:

1. BASED ON THHN/THWN, 90°., 600V., INSULATED, COPPER WIRE APPLIED AT 75°FOR TERMINATIONS RATED AT 60°C/75°C AND 75°C. FOR TERMINATIONS RATED AT 60°C PROVIDE WIRE AND CONDUIT SIZES INDICATED IN PARENTHESIS.
2. BASED ON WIRE OUTSIDE DIAMETERS AND RIGID METALLIC CONDUIT INSIDE DIAMETERS AS PROVIDED IN THE NEC. DO NOT REDUCE CONDUIT SIZE FOR NON-RIGID METALLIC APPLICATION. REFER TO NEC FOR CONDUIT TYPES MORE RESTRICTIVE THAN RIGID METALLIC.
3. BASED ON MOTOR FULL LOAD AMPERES AS PROVIDED BY THE NEC.
4. BASED ON MOTOR RUNNING OVERLOAD PROTECTION PROVIDED BY THERMAL OVERLOAD RELAYS.
5. MOTOR STARTING TYPE BASED ON 460V., 3 PHASE, FULL VOLTAGE NON-REVERSING EXCEPT FOR MOTORS SIZED 75HP OR GREATER WHICH ARE BASED ON 460V., 3 PHASE, PART WINDING REDUCED VOLTAGE STARTING.
6. TRANSFORMER CIRCUITS BASED ON 480V TO 208/120V., 3 PHASE, 4 WIRE,DRY TYPE.
7. PROVIDE THREE PHASE WIRES AND ONE DOUBLE AMPACITY NEUTRAL FOR 110 AMPACITY CIRCUITS AND LESS. PROVIDE THREE PHASE WIRES AND TWO NEUTRAL WIRES, SIZES AS INDICATED FOR 125 AMPACITY CIRCUITS AND GREATER.

## 480V., THREE PHASE CIRCUIT LENGTH TABLE

| BREAKER AMPACITY (AMPS) | MAX. CIRCUIT LOAD (AMPS) | MAXIMUM LENGTH IN FEET | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | NO.12 | NO.10 | NO.8 | NO.6 | NO.4 | NO.3 | NO.2 | NO.1 | 1/0 | 2/0 | 3/0 | 4/0 | 250 | 350 | 500 | 2-3/0 | 2-4/0 | 2-250 | 2-350 | 2-500 | 3-300 | 3-400 | 4-350 | 5-400 | 6-400 | 6-500 |
| 20 | 16 | 253 | 403 | 642 | 1019 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 30 | 24 | 169 | 269 | 428 | 679 | 1079 | 1364 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 40 | 32 | - | 201 | 321 | 509 | 809 | 1034 | 1293 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 50 | 40 | - | - | 257 | 408 | 648 | 818 | 1034 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 60 | 48 | - | - | - | 340 | 540 | 682 | 862 | 1083 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 70 | 56 | - | - | - | 291 | 462 | 584 | 739 | 928 | 1169 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 80 | 64 | - | - | - | - | 405 | 511 | 646 | 812 | 1023 | 1286 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 90 | 72 | - | - | - | - | 360 | 455 | 574 | 722 | 909 | 1143 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 100 | 80 | - | - | - | - | - | 409 | 517 | 650 | 818 | 1029 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 125 | 100 | - | - | - | - | - | - | 414 | 520 | 655 | 823 | 1043 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 150 | 120 | - | - | - | - | - | - | - | - | 546 | 689 | 869 | 1107 | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 175 | 140 | - | - | - | - | - | - | - | - | - | 588 | 745 | 949 | 1110 | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 200 | 160 | - | - | - | - | - | - | - | - | - | - | 652 | 830 | 971 | 1360 | - | - | - | - | - | - | - | - | - | - | - | - |
| 225 | 180 | - | - | - | - | - | - | - | - | - | - | - | 738 | 863 | 1209 | 1743 | - | - | - | - | - | - | - | - | - | - | - |
| 250 | 200 | - | - | - | - | - | - | - | - | - | - | - | - | 777 | 1088 | 1569 | 1043 | - | - | - | - | - | - | - | - | - | - |
| 300 | 240 | - | - | - | - | - | - | - | - | - | - | - | - | - | 907 | 1307 | 869 | 1107 | - | - | - | - | - | - | - | - | - |
| 350 | 280 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 1120 | 745 | 949 | 1110 | - | - | - | - | - | - | - | - |
| 400 | 320 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 652 | 830 | 971 | 1360 | - | - | - | - | - | - | - |
| 450 | 360 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 738 | 863 | 1209 | - | - | - | - | - | - | - |
| 500 | 400 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 777 | 1088 | 1569 | - | - | - | - | - | - |
| 600 | 480 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 907 | 1307 | 1165 | - | - | - | - | - |
| 700 | 560 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 1120 | 999 | 1346 | - | - | - | - |
| 800 | 640 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 874 | 1177 | 1360 | - | - | - |
| 1000 | 800 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 942 | 1088 | 1569 | - | - |
| 1200 | 960 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 907 | 1307 | - | - |
| 1600 | 1200 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 980 | 1226 | 1307 |
| 1800 | 1440 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 1089 | 1177 |
| 2000 | 1600 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 980 | 1137 |

## 120V. SINGLE PHASE CIRCUIT LENGTH TABLE

| BREAKER AMPACITY (AMPS) | MAX. CIRCUIT CURRENT (AMPS) | MAX. CIRCUIT LOAD (VA) | MAXIMUM LENGTH IN FEET | | | | |
|---|---|---|---|---|---|---|---|
| | | | NO.12 | NO.10 | NO.8 | NO.6 | NO.4 |
| 20 | 4 | 480 | 220 | 349 | 556 | 882 | - |
| | 8 | 960 | 110 | 174 | 278 | 441 | 701 |
| | 12 | 1440 | 73 | 116 | 185 | 294 | 467 |
| | 16 | 1920 | 55 | 87 | 139 | 221 | 350 |
| 30 | 24 | 2880 | - | 58 | 93 | 147 | 234 |
| 40 | 32 | 3840 | - | - | 70 | 110 | 175 |
| 50 | 40 | 4800 | - | - | - | 88 | 140 |
| 60 | 48 | 5760 | - | - | - | - | 117 |

## 208V. SINGLE PHASE CIRCUIT LENGTH TABLE

| BREAKER AMPACITY (AMPS) | MAX. CIRCUIT CURRENT (AMPS) | MAX. CIRCUIT LOAD (VA) | MAXIMUM LENGTH IN FEET | | | | |
|---|---|---|---|---|---|---|---|
| | | | NO.12 | NO.10 | NO.8 | NO.6 | NO.4 |
| 20 | 4 | 832 | 380 | 605 | 964 | - | - |
| | 8 | 1664 | 190 | 302 | 482 | 765 | - |
| | 12 | 2496 | 127 | 202 | 321 | 510 | 810 |
| | 16 | 3328 | 95 | 151 | 241 | 382 | 607 |
| 30 | 24 | 4992 | - | 101 | 161 | 255 | 405 |
| 40 | 32 | 6656 | - | - | 121 | 191 | 304 |
| 50 | 40 | 8320 | - | - | - | 153 | 243 |
| 60 | 48 | 9984 | - | - | - | - | 202 |

## 208V. THREE PHASE CIRCUIT LENGTH TABLE

| BREAKER AMPACITY (AMPS) | MAX. CIRCUIT CURRENT (AMPS) | MAX. CIRCUIT LOAD (VA) | MAXIMUM LENGTH IN FEET | | | | |
|---|---|---|---|---|---|---|---|
| | | | NO.12 | NO.10 | NO.8 | NO.6 | NO.4 |
| 20 | 4 | 1440 | 439 | 698 | 1113 | - | - |
| | 8 | 2880 | 220 | 349 | 557 | 883 | - |
| | 12 | 4320 | 127 | 233 | 371 | 589 | 935 |
| | 16 | 5760 | 95 | 175 | 278 | 442 | 701 |
| 30 | 24 | 8640 | - | 116 | 186 | 294 | 468 |
| 40 | 32 | 11520 | - | - | 139 | 221 | 351 |
| 50 | 40 | 14400 | - | - | - | 177 | 281 |
| 60 | 48 | 17280 | - | - | - | - | 234 |

CIRCUIT MAXIMUM DISTANCE TABLE NOTES:

CIRCUIT MAXIMUM DISTANCE IS BASED ON NEC CHAPTER 9, TABLE 8 CONDUCTOR PROPERTIES FOR COATED COPPER CONDUCTORS AT 75 DEGREES CELSIUS.

MINIMUM WIRE SIZE BASED ON NEC TABLE 310.15 (B)(16) FOR COPPER THHN/THWN, AT 90 DEGREES CELSIUS.

## 277V. SINGLE PHASE CIRCUIT LENGTH TABLE

| BREAKER AMPACITY (AMPS) | MAXIMUM LENGTH IN FEET | | |
|---|---|---|---|
| | NO.12 | NO.10 | |
| 20 | 200 | 300 | |

## MOTOR CIRCUIT SIZING SCHEDULE

(FOR 460V, 3 PHASE MOTORS) (NOTES 3,4,5)

| MOTOR HP | SWITCH/FUSE | CIRCUIT BREAKER | STARTER | CONDUIT & WIRE | | |
|---|---|---|---|---|---|---|
| | | | SIZE/TYPE | PHASE | E.G. | CONDUIT |
| 1/2 | 30/3A. | 15A | 1 | 12 | 12 | 3/4" |
| 3/4 | 30/3A. | 15A | 1 | 12 | 12 | 3/4" |
| 1 | 30/6A. | 15A | 1 | 12 | 12 | 3/4" |
| 1 1/2 | 30/6A. | 15A | 1 | 12 | 12 | 3/4" |
| 2 | 30/6A. | 15A | 1 | 12 | 12 | 3/4" |
| 3 | 30/10A. | 15A | 1 | 12 | 12 | 3/4" |
| 5 | 30/15A. | 20A | 1 | 12 | 12 | 3/4" |
| 7 1/2 | 30/20A. | 25A | 1 | 12 | 12 | 3/4" |
| 10 | 30/25A. | 35A | 1 | 12 | 12 | 3/4" |
| 15 | 60/40A. | 50A | 2 | 10 | 10 | 3/4" |
| 20 | 60/50A. | 60A | 2 | 8 | 10 | 3/4" |
| 25 | 60/60A. | 80A | 2 | 6 | 8 | 3/4" |
| 30 | 100/70A. | 100A | 3 | 6 | 8 | 3/4" |
| 40 | 100/100A. | 125A | 3 | 4 | 6 | 1" |
| 50 | 200/125A. | 150A | 3 | 3 | 6 | 1 1/4" |
| 60 | 200/150A. | 175A | 4 | 1 | 6 | 1 1/4" |
| 75 | 200/175A. | 225A | 4 | 1/0 | 4 | 1 1/2" |
| 100 | 400/225A. | 300A | 4 | 2/0 | 4 | 2" |
| 125 | 400/300A. | 350A | 5 | 3/0 | 3 | 2" |
| 150 | 400/350A. | 450A | 5 | 4/0 | 2 | 2" |
| 200 | 600/450A. | 600A | 5 | 350 | 1 | 2 1/2" |

## MOTOR CIRCUIT SIZING SCHEDULE

(FOR 208V, 3 PHASE MOTORS NOTES 3,4,5)

| MOTOR HP | SWITCH/FUSE | CIRCUIT BREAKER | STARTER | CONDUIT & WIRE | | |
|---|---|---|---|---|---|---|
| | | | SIZE/TYPE | PHASE | E.G. | CONDUIT |
| 1/2 | 30/4A. | 15A | 1 | 12 | 12 | 3/4" |
| 3/4 | 30/6.25A. | 15A | 1 | 12 | 12 | 3/4" |
| 1 | 30/8A. | 15A | 1 | 12 | 12 | 3/4" |
| 1 1/2 | 30/15A. | 20A | 1 | 12 | 12 | 3/4" |
| 2 | 30/15A. | 20A | 1 | 12 | 12 | 3/4" |
| 3 | 30/20A. | 25A | 1 | 12 | 12 | 3/4" |
| 5 | 30/30A. | 40A | 1 | 10 | 10 | 3/4" |
| 7 1/2 | 60/45A. | 60A | 1 | 8 | 10 | 3/4" |
| 10 | 60/60A. | 70A | 2 | 8 | 8 | 3/4" |
| 15 | 100/90A. | 110A | 3 | 4 | 6 | 1" |
| 20 | 200/110A. | 125A | 3 | 3 | 6 | 1 1/4" |
| 25 | 200/150A. | 175A | 3 | 2 | 6 | 1 1/4" |
| 30 | 200/175A. | 200A | 4 | 1 | 6 | 1 1/4" |
| 40 | 200/200A. | 250A | 4 | 1/0 | 4 | 1 1/2" |
| 50 | 400/250A. | 350A | 5 | 3/0 | 3 | 2" |
| 60 | 400/300A. | 400A | 5 | 4/0 | 3 | 2" |
| 75 | 400/400A. | 500A | 5 | 300 | 2 | 2 1/2" |
| 100 | 600/500A. | 600A | 6 | 500 | 1 | 3" |
| 125 | 800/700A. | 800A | 6 | 2-4/0 | 2-1/0 | 2-2" |
| 150 | 800/700A. | 800A | 6 | 2-250 | 2-1/0 | 2-2" |
| 200 | 1200/1000A | 1200A | 7 | 2-400 | 2-3/0 | 2-2 1/2" |

## TRANSFORMER CIRCUIT SIZING SCHEDULE - GENERAL PURPOSE TYPE

(NOTE 6)

| TRANSF. KVA | PRIMARY CIRCUIT (480V.) | | SECONDARY CIRCUIT (208/120V.) | | |
|---|---|---|---|---|---|
| | SWITCH/FUSE OR CIRCUIT BREAKER | PRIMARY FEEDER | SWITCH/FUSE OR CIRCUIT BREAKER | SYSTEM/EQUIPMENT BONDING JUMPER (GROUND WIRE) | SECONDARY FEEDER |
| 9 | 30/20A. | 20A., 3W. | 30/30A. | #8 | 30A., 4W. |
| 15 | 30/25A. | 25A., 3W. | 60/60A. | #8 | 60A., 4W. |
| 30 | 60/45A. | 45A., 3W. | 100/100A. | #8 | 100A., 4W. |
| 45 | 100/70A. | 70A., 3W. | 200/175A. | #4 | 175A., 4W. |
| 75 | 200/125A. | 125A., 3W. | 400/300A. | #2 | 300A., 4W. |
| 112 1/2 | 200/175A. | 175A., 3W. | 400/400A. | #1/0 | 400A., 4W. |
| 150 | 400/225A. | 225A., 3W. | 600/600A. | #2/0 | 600A., 4W. |
| 225 | 400/350A. | 350A., 3W. | 800/800A. | #3/0 | 800A., 4W. |
| 300 | 600/500A. | 500A., 3W. | 1200/1000A. | #3/0 | 1000A., 4W. |
| | | | | | |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

**GRAPHIC SCALE(S)**

DATE: 06/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|
| 1 | 05/11/2026 | AMENDMENT 01 |

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)

ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: ELECTRICAL SCHEDULES

WORKING STAFF

| NS | NS | KM |
|---|---|---|
| DESIGNED BY | DRAWN BY | CHECKED BY |

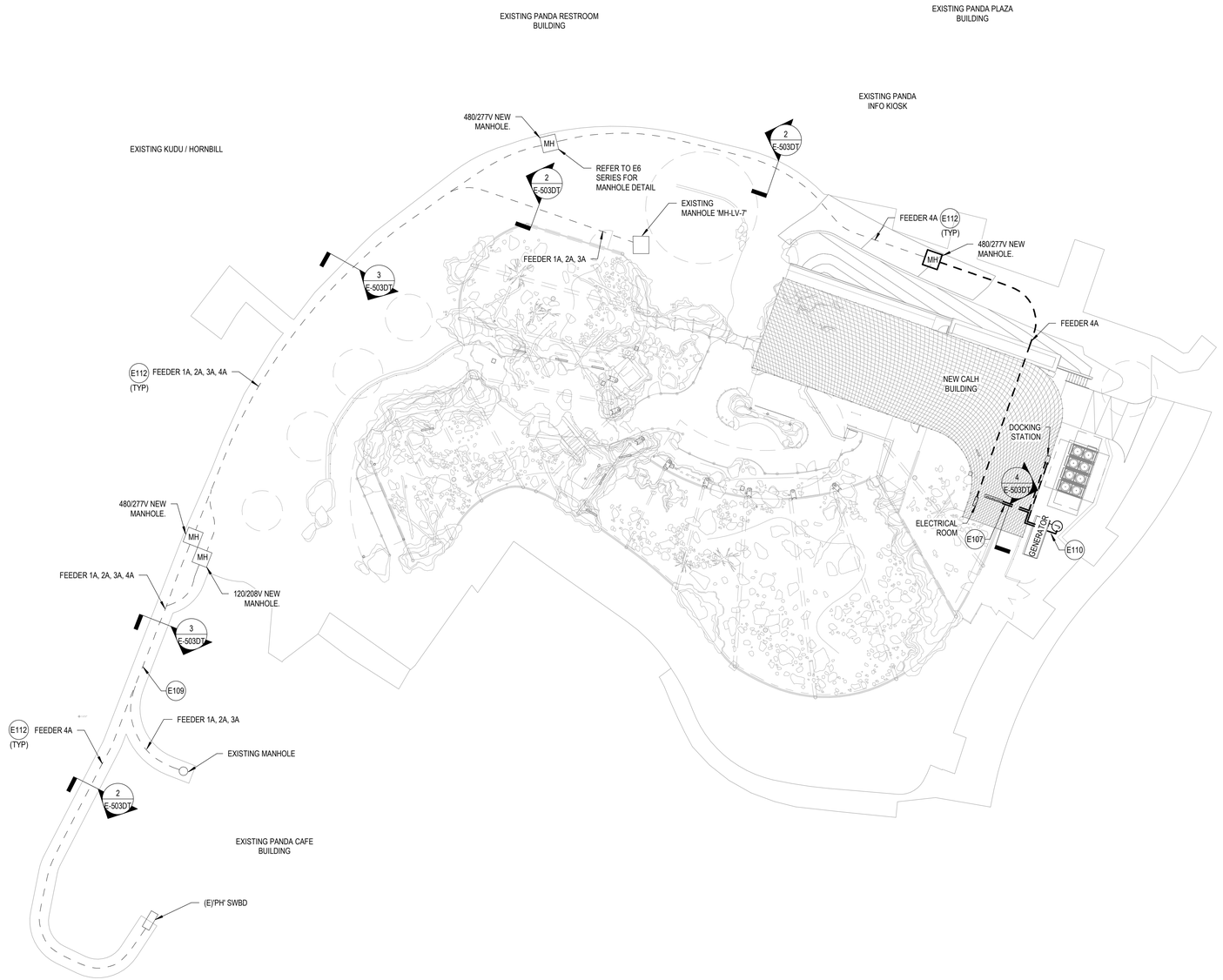
SHEET NO. 333 OF 378

E-003CS

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

FOR REFERENCE ONLY

6/15/2026 2:48:05 PM
E-003CS BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




1 ELECTRICAL SITE PLAN
SCALE: 1" = 20'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. SEE DRAWING SERIES E6 FOR POWER RISER DIAGRAM AND ALL SCHEDULES.
C. COORDINATE WITH LOCAL UTILITY FOR ALL SITE SERVICE REQUIREMENTS.
D. CONTACT LOCAL UTILITY TO IDENTIFY ALL EXISTING UNDERGROUND UTILITIES.
E. SITE WORK WILL INCLUDE ELEMENTS NOT DELINEATED ON THIS SITE PLAN INCLUDING BUT NOT LIMITED TO UNDERGROUND BRANCH CIRCUIT CONDUITS TO HABITAT HEATERS, SITE LIGHTING, RECEPTACLES, GROUNDING ELECTRODES, AND GROUNDING CONDUCTORS FOR BUILDING GROUNDING AND/OR LIGHTNING PROTECTION. REFER TO BUILDING ABOVE GRADE PACKAGE FOR MORE INFORMATION ON THESE COMPONENTS.

## SHEET KEYNOTES

E107 ROUTE CONDUIT IN CONCRETE ENCASED DUCT BANK 36" MINIMUM BELOW GRADE TO NEW DOCKING STATION. REFER TO E6 SERIES RISER DIAGRAM FOR MORE INFORMATION.
E109 THERE MAY BE ADDITIONAL SITE UTILITIES NOT DELINEATED ON THIS PLAN THAT SHALL BE RE-ROUTED AS A RESULT OF SITE/CIVIL WORK. THE SITE CONTRACTOR SHALL PERFORM A THOROUGH SURVEY AND ENSURE ALL ELECTRICAL AND MISCELLANEOUS RE-ROUTING HAVE BEEN CAPTURED IN BID.
E110 PROVIDE (2)1-1/2" CONDUIT IN CONCRETE ENCASED FOR GENERATOR LOAD CENTER.
E112 REFER TO SHEET SERIES E5 AND E6 FOR ELECTRICAL POWER DETAILS AND ONLINE DIAGRAM FOR FEEDER INFORMATION.

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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KEY PLAN

GRAPHIC SCALE(S)

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| REVISION # | DATE | REVISION DESCRIPTION |
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| 1 | 05/11/2026 | AMENDMENT 01 |

Smithsonian Institution

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|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ELECTRICAL SITE PLANS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 334 of 378

E-100SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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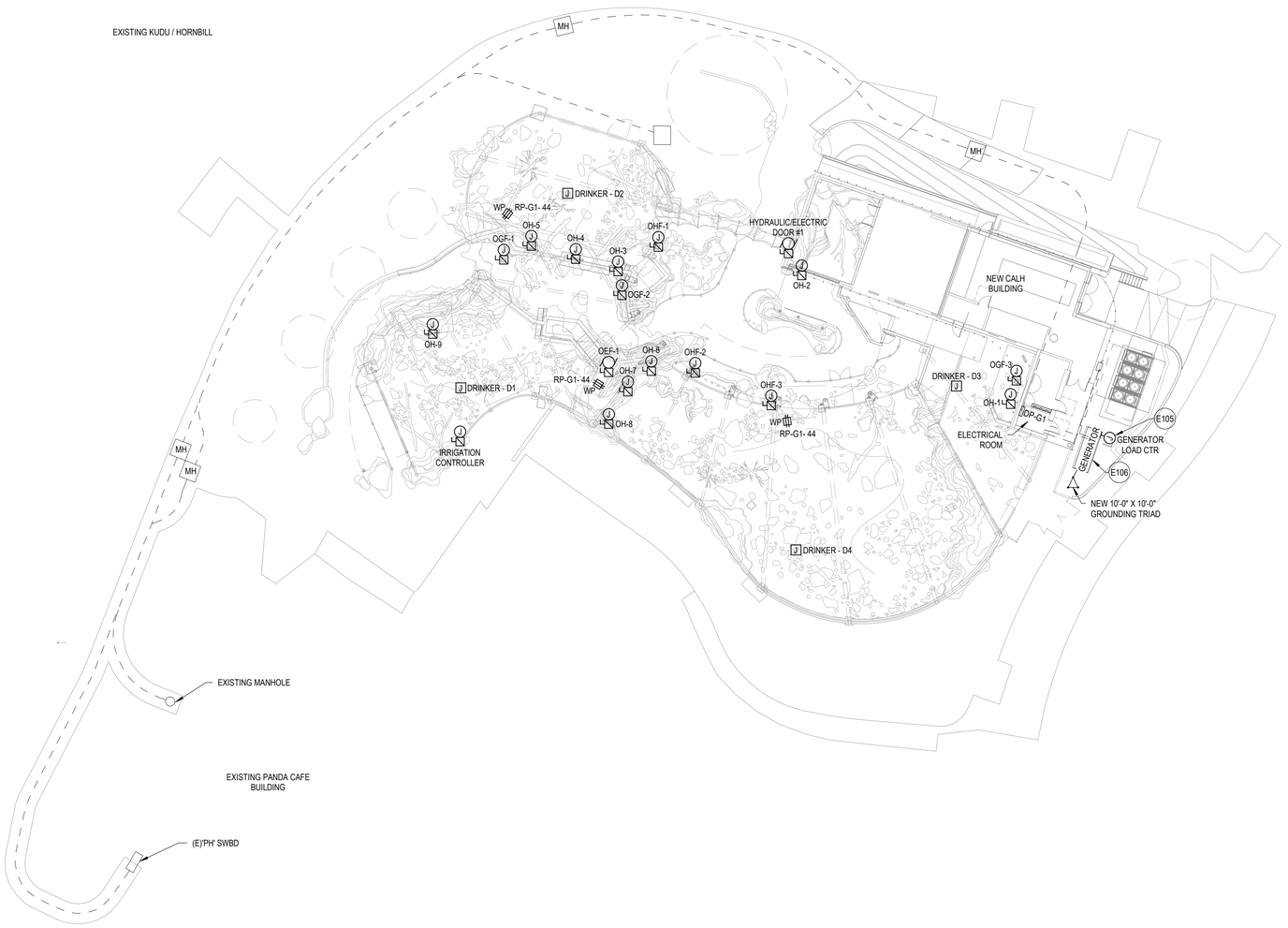


1 POWER SITE PLAN
SCALE: 1" = 20'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. SEE DRAWING SERIES E6 FOR POWER RISER DIAGRAM AND ALL SCHEDULES.
C. COORDINATE WITH LOCAL UTILITY FOR ALL SITE SERVICE REQUIREMENTS.
D. CONTACT LOCAL UTILITY TO IDENTIFY ALL EXISTING UNDERGROUND UTILITIES.
E. SITE WORK WILL INCLUDE ELEMENTS NOT DELINEATED ON THIS SITE PLAN INCLUDING BUT NOT LIMITED TO UNDERGROUND BRANCH CIRCUIT CONDUITS TO HABITAT HEATERS, SITE LIGHTING, RECEPTACLES, GROUNDING ELECTRODES, AND GROUNDING CONDUCTORS FOR BUILDING GROUNDING AND/OR LIGHTNING PROTECTION. REFER TO BUILDING ABOVE GRADE PACKAGE FOR MORE INFORMATION ON THESE COMPONENTS.

## SHEET KEYNOTES

E105 LOAD CENTER WILL BE INTEGRAL TO ENCLOSURE AND FEEDER INDICATED TO STUB-UP INTO GENERATOR.
E106 NEW NATURAL GAS LIFE SAFETY GENERATOR. REFER TO E6 SERIES RISER DIAGRAM FOR MORE INFORMATION.

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | POWER SITE PLANS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |
| SHEET NO. | 335 OF 378 |
| | ES-101SP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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1 LIGHTING SITE PLAN
SCALE: 1" = 20'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. SEE DRAWING SERIES E6 FOR POWER RISER DIAGRAM AND ALL SCHEDULES.
C. COORDINATE WITH LOCAL UTILITY FOR ALL SITE SERVICE REQUIREMENTS.
D. CONTACT LOCAL UTILITY TO IDENTIFY ALL EXISTING UNDERGROUND UTILITIES.
E. SITE WORK WILL INCLUDE ELEMENTS NOT DELINEATED ON THIS SITE PLAN INCLUDING BUT NOT LIMITED TO UNDERGROUND BRANCH CIRCUIT CONDUITS TO HABITAT HEATERS, SITE LIGHTING, RECEPTACLES, GROUNDING ELECTRODES, AND GROUNDING CONDUCTORS FOR BUILDING GROUNDING AND/OR LIGHTNING PROTECTION. REFER TO BUILDING ABOVE GRADE PACKAGE FOR MORE INFORMATION ON THESE COMPONENTS.

## SHEET KEYNOTES

E301 RECESS HANDRAIL XL1 LIGHT INTO HANDRAIL. COORDINATE AND CONCEAL POWER ROUTING BACK TO MANUFACTURER APPROVED REMOTE DRIVER LOCATED IN KEEPERS CORRIDOR 1CR1 AND/OR DIRECT BURIAL ENCLOSURE WITHIN LANDSCAPING. ALL CABLING TO AND BETWEEN HANDRAILS TO BE WITHIN CONCEALED CONDUIT. ALL HANDRAILS TO BE GROUNDED TO BUILDING GROUNDING SYSTEM SEPARATE FROM LED DRIVER EQUIPMENT GROUNDING CONDUCTOR. COORDINATE WITH ARCHITECTURE AND LANDSCAPE FOR DRIVER LOCATION. CONCEAL ALL WIRING AND DRIVER FROM VIEW.

# ALF

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| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LIGHTING SITE PLANS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 336 OF 378

ES-102SP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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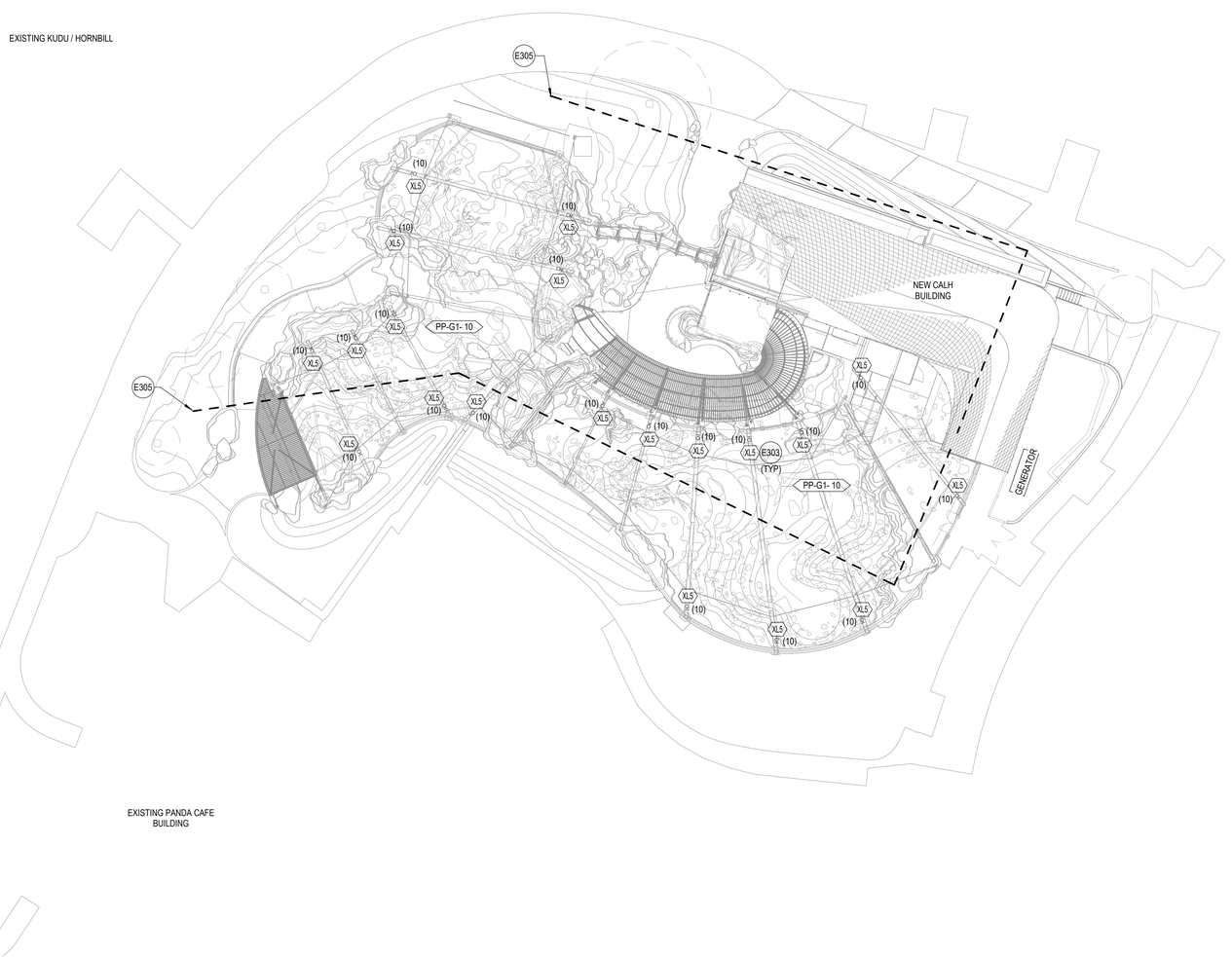


1 LIGHTING HABITAT SITE PLAN
SCALE: 1" = 20'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. SEE DRAWING SERIES E6 FOR POWER RISER DIAGRAM AND ALL SCHEDULES.
C. COORDINATE WITH LOCAL UTILITY FOR ALL SITE SERVICE REQUIREMENTS.
D. CONTACT LOCAL UTILITY TO IDENTIFY ALL EXISTING UNDERGROUND UTILITIES.
E. SITE WORK WILL INCLUDE ELEMENTS NOT DELINEATED ON THIS SITE PLAN INCLUDING BUT NOT LIMITED TO UNDERGROUND BRANCH CIRCUIT CONDUITS TO HABITAT HEATERS, SITE LIGHTING, RECEPTACLES, GROUNDING ELECTRODES, AND GROUNDING CONDUCTORS FOR BUILDING GROUNDING AND/OR LIGHTNING PROTECTION. REFER TO BUILDING ABOVE GRADE PACKAGE FOR MORE INFORMATION ON THESE COMPONENTS.

## SHEET KEYNOTES

E303 REFERENCE SHR DRAWINGS FOR XL5 MOUNTING DETAIL.
E305 PROVIDE 2"PVC UNDERGROUND CONDUIT FOR SPECIAL EVENTS LIGHTING ELECTRICAL CONNECTION. COORDINATE CONDUIT RUN AND EXACT LOCATION WITH SI. PROVIDE FISH WIRE AND END CAP.

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PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084
DRAWING TITLE: LIGHTING HABITAT SITE PLANS
WORKING STAFF: NS (DESIGNED BY) | NS (DRAWN BY) | KM (CHECKED BY)

SHEET NO. 337 OF 378
ES-103SP (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

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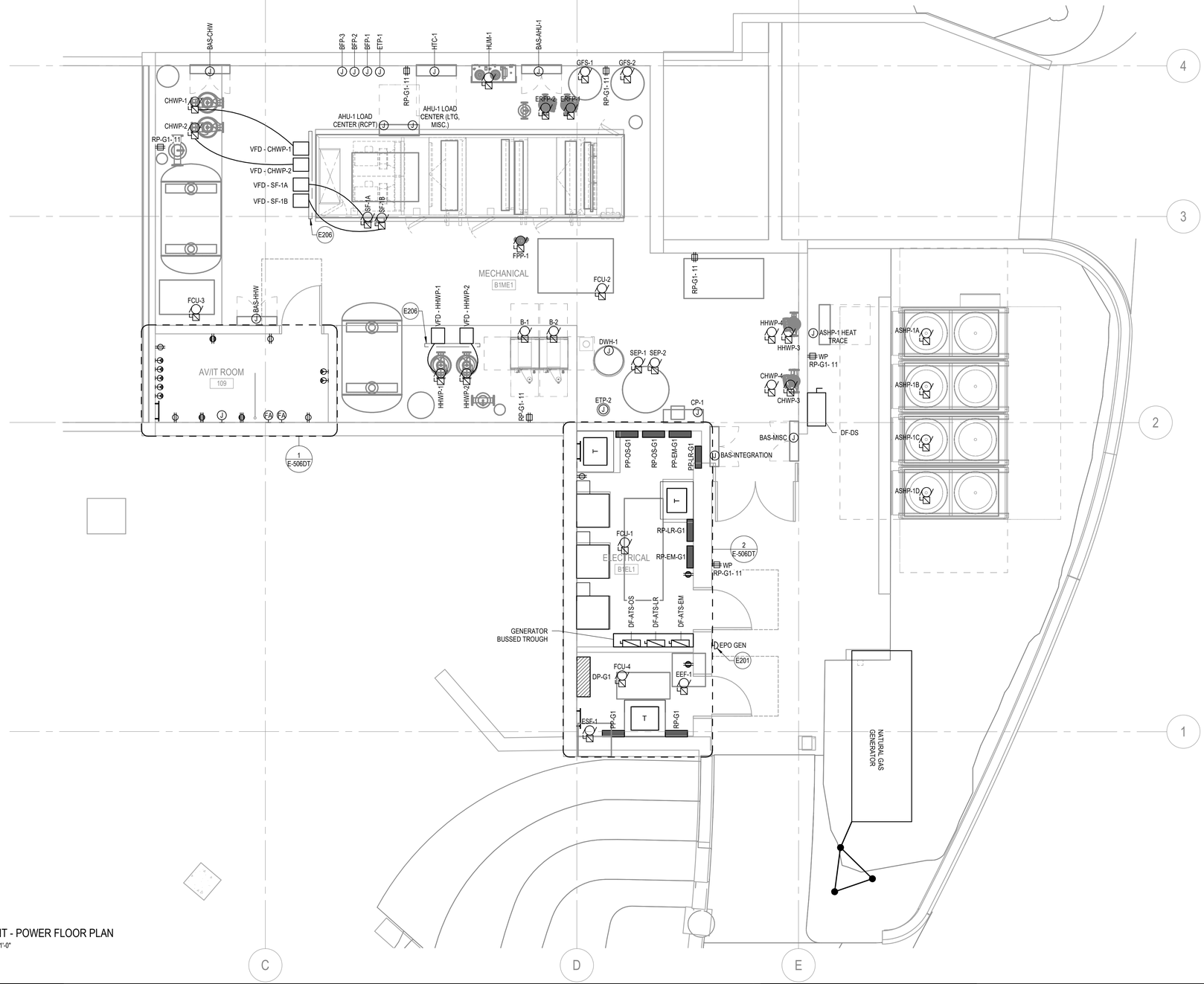


1 BASEMENT - POWER FLOOR PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. COORDINATE ALL DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN.
C. THE CONTRACTOR SHALL COORDINATE EXACT LOCATION OF EQUIPMENT POWER CONNECTIONS WITH EQUIPMENT INSTALLER PRIOR TO ROUGH-IN.
D. SEE MECHANICAL EQUIPMENT SCHEDULE FOR ASSOCIATED OVERCURRENT DEVICES, DISCONNECT SWITCHES, STARTERS AND WIRES.
E. CONTRACTOR SHALL BE RESPONSIBLE FOR SUPPLYING ANY REQUIRED TRANSFORMER/POWER SUPPLIES FOR DOOR HARDWARE. COORDINATE WITH ARCHITECTURAL AND SECURITY DRAWINGS FOR DOOR HARDWARE. PROVIDE 120V POWER AS REQUIRED.
F. CONTRACTOR SHALL COORDINATE AND FIELD VERIFY REQUIREMENT OF NEUTRAL CONDUCTOR FOR ALL EQUIPMENT AND PROVIDE NEUTRAL CONDUCTOR AS NECESSARY.
G. TRIPPING OF GFCI RECEPTACLES SHALL NOT AFFECT THE DOWNSTREAM RECEPTACLES.
H. NO MORE THAN 3 CIRCUITS MAY SHARE A HOMERUN.
I. FOR ALL MECHANICAL EQUIPMENT, PROVIDE ADDITIONAL 120V CONNECTIONS REQUIRED BY MANUFACTURER.
J. FOR EACH AIR HANDLING UNIT, PROVIDE (1)120V CONNECTION FOR LIGHTING AND RECEPTACLES. CIRCUIT TO MECHANICAL PANEL WITHIN THE SPACE.
K. PROVIDE 120V CIRCUITS FOR CONTROL PANELS AS REQUIRED BY CONTROL VENDOR. CIRCUIT TO MECHANICAL PANEL WITHIN THE SPACE.
L. ALL UNUSED WALL OUTLETS TO BE COVERED WITH BLANK PLATE. ALL BLANK PLATE COLORS TO MATCH WALL COLOR.
M. SEE ANY COMMUNICATIONS PATHWAY PLAN AND AV DRAWINGS, FOR ADDITIONAL CONDUIT TO BE PROVIDED BY ELECTRICAL CONTRACTOR.
N. ALL CONDUIT ROUTED WITHIN SLAB SHALL COMPLY WITH GUIDELINES LISTED IN STRUCTURAL DRAWINGS.
M. FOR DEVICES LABELED WITH RECTANGULAR TAG, REFER TO KITCHEN EQUIPMENT SCHEDULE ON E7 SERIES CONNECTIONS, CIRCUITS AND DEVICE TYPES.
N. UNLESS OTHERWISE NOTED, ALL SWITCHED RECEPTACLES TO BE CONTROLLED BY OCCUPANCY SENSOR OR CONTACTORS. SEE CONTROLS SCHEDULES ON AND DETAILS ON FOR ADDITIONAL CONNECTION INFORMATION.
O. COORDINATE FINAL LOCATION OF MECHANICAL AND PLUMBING EQUIPMENT AND ASSOCIATED DISCONNECT SWITCHES, STARTERS, VFDS, CONTROL POWER AND OTHER POWER REQUIREMENTS WITH DIV 22 AND 23.
P. CONTRACTOR IS RESPONSIBLE FOR ALL WIRING FROM VFDs AND DISCONNECTS SHOWN IN DRAWINGS TO THE EQUIPMENT AND IS TO COORDINATE WITH EQUIPMENT MANUFACTURER AND INSTALLATION INSTRUCTIONS.
Q. WALL MOUNTED AND CEILING MOUNTED TRANSFORMERS SHALL BE MOUNTED SUCH THAT THE TRANSFORMER ENCLOSURE AND ALL ELEMENTS OF THE HANGER SUPPORT SYSTEM ARE LOCATED ABOVE THE MINIMUM WORKING SPACE HEIGHT IN COMPLIANCE WITH NEC 110.26.
R. PROVIDE GROUND FAULT PROTECTION FOR ALL 120-VOLT 15 AND 20-AMP RECEPTACLES LOCATED IN KITCHEN AREAS AS REQUIRED BY NEC 210.8(B).

## SHEET KEYNOTES

E201 REMOTE EPO FOR GENERATOR. PROVIDE LEXAN COVER OVER SWITCH AND COORDINATE MOUNTING WITH EXISTING FACADE AND ROUTE COMMUNICATION WIRE FROM EPO TO GENERATOR CONTROL PANEL VIA DEDICATED 1" DIA EXISTING CONDUIT ROUTED TO THE GENERATOR PAD ORIGINATING THE IN THE MAIN ELECTRICAL ROOM.

E206 PROVIDE UNISTRUT SYSTEM TO INSTALL ALL ELECTRICAL EQUIPMENT.

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TRUE NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
GRAPHIC SCALE(S)

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BASEMENT POWER PLAN |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 338 OF 378

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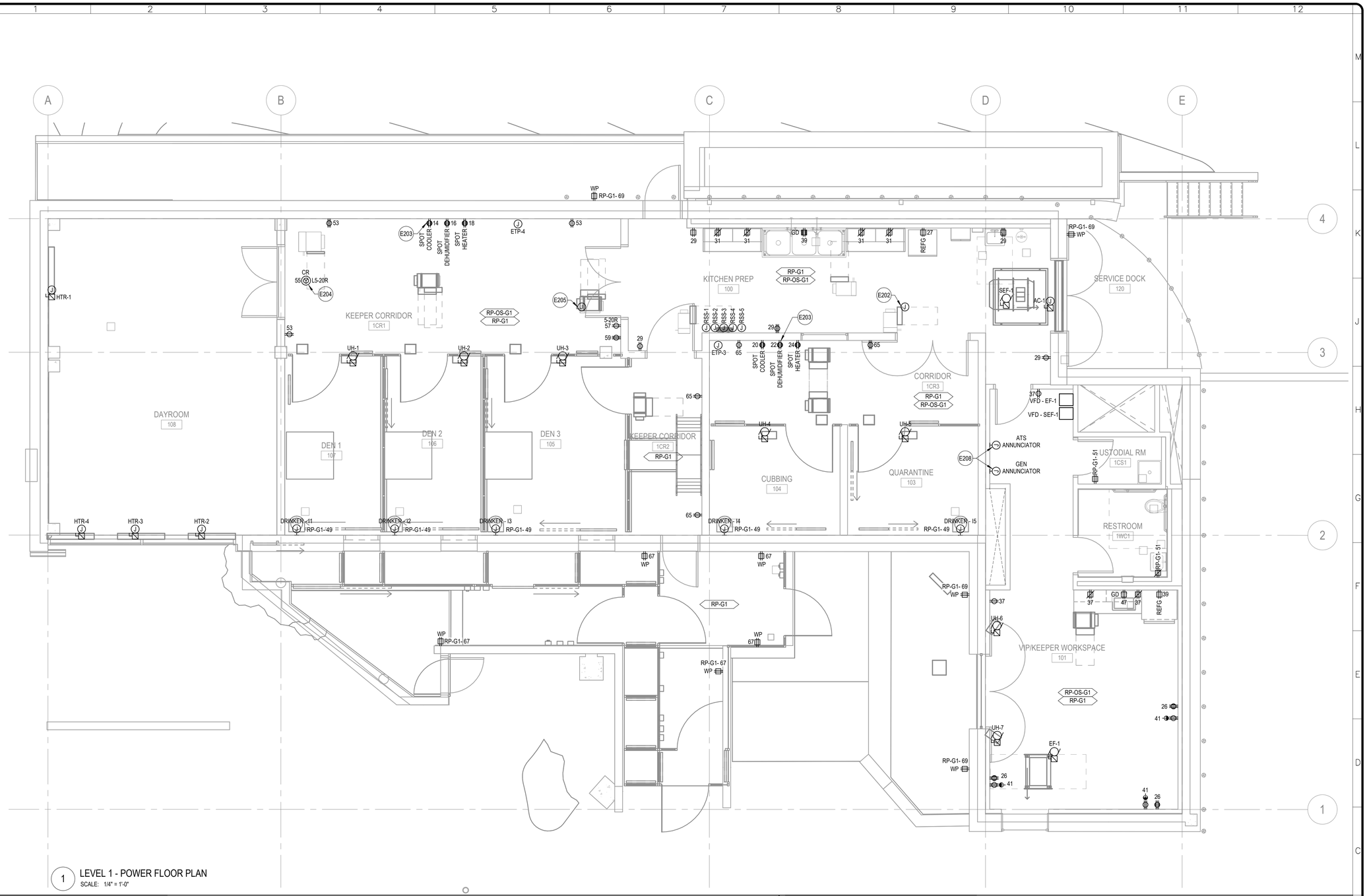
DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE




1 LEVEL 1 - POWER FLOOR PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. COORDINATE ALL DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN.
C. THE CONTRACTOR SHALL COORDINATE EXACT LOCATION OF EQUIPMENT POWER CONNECTIONS WITH EQUIPMENT INSTALLER PRIOR TO ROUGH-IN.
D. SEE MECHANICAL EQUIPMENT SCHEDULE FOR ASSOCIATED OVERCURRENT DEVICES, DISCONNECT SWITCHES, STARTERS AND WIRES.
E. CONTRACTOR SHALL BE RESPONSIBLE FOR SUPPLYING ANY REQUIRED TRANSFORMER/POWER SUPPLIES FOR DOOR HARDWARE. COORDINATE WITH ARCHITECTURAL AND SECURITY DRAWINGS FOR DOOR HARDWARE. PROVIDE 120V POWER AS REQUIRED.
F. CONTRACTOR SHALL COORDINATE AND FIELD VERIFY REQUIREMENT OF NEUTRAL CONDUCTOR FOR ALL EQUIPMENT AND PROVIDE NEUTRAL CONDUCTOR AS NECESSARY.
G. TRIPPING OF GFCI RECEPTACLES SHALL NOT AFFECT THE DOWNSTREAM RECEPTACLES.
H. NO MORE THAN 3 CIRCUITS MAY SHARE A HOMERUN.
I. FOR ALL MECHANICAL EQUIPMENT, PROVIDE ADDITIONAL 120V CONNECTIONS REQUIRED BY MANUFACTURER.
J. FOR EACH AIR HANDLING UNIT, PROVIDE (1)120V CONNECTION FOR LIGHTING AND RECEPTACLES. CIRCUIT TO MECHANICAL PANEL WITHIN THE SPACE.
K. PROVIDE 120V CIRCUITS FOR CONTROL PANELS AS REQUIRED BY CONTROL VENDOR. CIRCUIT TO MECHANICAL PANEL WITHIN THE SPACE.
L. ALL UNUSED WALL OUTLETS TO BE COVERED WITH BLANK PLATE. ALL BLANK PLATE COLORS TO MATCH WALL COLOR.
M. SEE ANY COMMUNICATIONS PATHWAY PLAN AND AV DRAWINGS, FOR ADDITIONAL CONDUIT TO BE PROVIDED BY ELECTRICAL CONTRACTOR.
N. ALL CONDUIT ROUTED WITHIN SLAB SHALL COMPLY WITH GUIDELINES LISTED IN STRUCTURAL DRAWINGS.
M. FOR DEVICES LABELED WITH RECTANGULAR TAG, REFER TO KITCHEN EQUIPMENT SCHEDULE ON E7 SERIES CONNECTIONS, CIRCUITS AND DEVICE TYPES.
N. UNLESS OTHERWISE NOTED, ALL SWITCHED RECEPTACLES TO BE CONTROLLED BY OCCUPANCY SENSOR OR CONTACTORS. SEE CONTROLS SCHEDULES ON AND DETAILS ON FOR ADDITIONAL CONNECTION INFORMATION.
O. COORDINATE FINAL LOCATION OF MECHANICAL AND PLUMBING EQUIPMENT AND ASSOCIATED DISCONNECT SWITCHES, STARTERS, VFDS, CONTROL POWER AND OTHER POWER REQUIREMENTS WITH DIV 22 AND 23.
P. CONTRACTOR IS RESPONSIBLE FOR ALL WIRING FROM VFDs AND DISCONNECTS SHOWN IN DRAWINGS TO THE EQUIPMENT AND IS TO COORDINATE WITH EQUIPMENT MANUFACTURER AND INSTALLATION INSTRUCTIONS.
Q. WALL MOUNTED AND CEILING MOUNTED TRANSFORMERS SHALL BE MOUNTED SUCH THAT THE TRANSFORMER ENCLOSURE AND ALL ELEMENTS OF THE HANGER SUPPORT SYSTEM ARE LOCATED ABOVE THE MINIMUM WORKING SPACE HEIGHT IN COMPLIANCE WITH NEC 110.26.
R. PROVIDE GROUND FAULT PROTECTION FOR ALL 120-VOLT 15 AND 20-AMP RECEPTACLES LOCATED IN KITCHEN AREAS AS REQUIRED BY NEC 210.8(B).

## SHEET KEYNOTES

E202 CONTRACTOR SHALL NOT CONNECT MORE THAN 8 VAV TO A SINGLE JUNCTION BOX.
E203 ELECTRICAL CONTRACTOR TO COORDINATE WITH ARCHITECT AND MECHANICAL DRAWINGS EXACT LOCATION OF SPOT COOLERS AND DEHUMIDIFIER.
E204 PROVIDE HUBBELL CORD REEL WITH TWIST-LOCK RECEPTACLE EQUAL TO MODEL HBL25123. COORDINATE FINAL NEMA CONFIGURATION OF RECEPTACLE WITH CORD REEL PLUG AND MOUNTING DETAILS WITH ARCHITECTURAL DRAWINGS.
E205 CONTRACTOR SHALL NOT CONNECT MORE THAN 8 VAVE TO A SINGLE JUNCTION BOX.
E208 GENERATOR AND ATS STATUS PANEL AND ANNUNCIATOR PANEL TO BE PROVIDED BY GENERATOR MANUFACTURER. PROVIDE ALL NECESSARY WIRING AND CONNECTIONS BETWEEN THESE PANELS AND THE GENERATOR AND ATS'S. REMOTE ATS MONITORING PANEL TO BE INSTALLED AND CONNECTED TO ALL ATS'S TO PROVIDE STATUS OF ALL ATS'S AND SHALL BE PROVIDED WITH ATS PACKAGE FROM MANUFACTURER. GENERATOR ANNUNCIATOR PANEL AND GENERATOR CONTROL PANEL SHALL REPORT STATUS AND ALARM FUNCTIONS TO THE BAS.

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KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
GRAPHIC SCALE(S)

| DATE | SUBMISSION |
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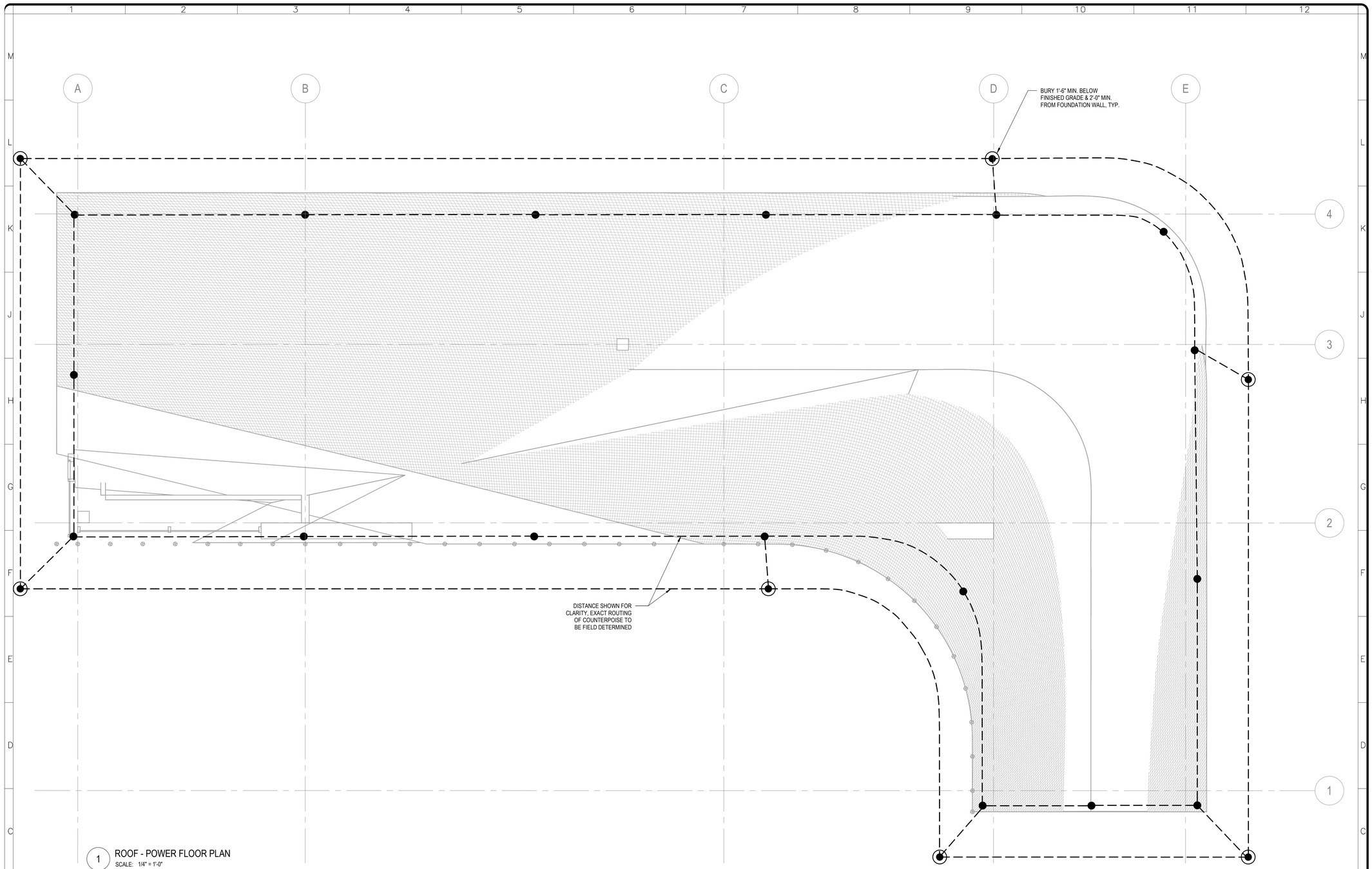
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LEVEL 1 POWER PLAN |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |
| SHEET NO. | 339 OF 378 |
| | EP-101FP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |




1 ROOF - POWER FLOOR PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING E0 SERIES FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. LOCATE AIR TEMINALS AS SHOWN. ENSURE THAT ALL POINTS ARE WITHIN 2'-0" OF OUTSIDE BUILDING EDGE, OUTSIDE CORNERS, RIDGE ENDS, AND THAT MAX SPACING DOES NOT EXCEED 20'-0", AND THAT MIN PROJECTION ABOVE OBJECT PROTECTED IS 10"; POINTS PROJECTING 24" MAY BE SPACED AT 25'-0" MAX.
C. MAINTAIN HORIZONTAL OR DOWNWARD COURSING OF MAIN CONDUCTOR. ENSURE THAT ALL BENDS HAVE AT LEAST AN 8" RADIUS AND DO NOT EXCEED 90 DEGREES.
D. ATTACH ALL EXPOSED ROOF, DOWN LEAD AND BONDING CABLES AT 3'-0" ON CENTER MAX. VERIFY COMPATIBILITY OF ADHESIVE ON MEMBRANE ROOF APPLICATION PRIOR TO INSTALLATION.
E. GROUND ROD ELECTRODES SHALL BE INSTALLED AS SHOWN, BUT IN NO INSTANCE SHALL THEY BE LESS THAN 1'-0" BELOW GRADE AND 2'-0" MIN. FROM FOUNDATION WALL. DRIVEN RODS SHALL PENETRATE THE EARTH AT LEAST 10'-0".
F. BOND TO WATER SERVICE AND OTHER PIPING SYSTEM AS SHOWN AND AS REQUIRED BY CODE.
G. MAIN SIZE LIGHTNING CONDUCTOR BONDED TO MAIN GROUND BUS FIELD VERIFY LOCATION 1-1/4" CONDUIT FOR ACCESS, INSTALLED BY OTHERS. INTERCONNECT LIGHTNING PROTECTION GROUND TO TELEPHONE AND OTHER BUILDING GROUND SYSTEM LOCATION FIELD DETERMINED OR AS REQUIRED BY CODE.
H. LB'S AND SIMILAR CONDUITS BODIES MAY NOT BE USED IN THE INSTALLATION OF DOWNLEAD CONDUITS, AS THEY DO NOT ADHERE TO THE REQUIRED 8" MIN BEND RADIUS.
I. SYSTEM SHALL BE INSTALLED AS SHOWN TO ENSURE PROPER CODE COMPLIANCE AND SYSTEM CERTIFICATION. ANY MAJOR VARIANCE SHALL BE RESUBMITTED FOR APPROVAL.
J. "AS-BUILT" DRAWINGS SHALL BE SUBMITTED IN ACCORDANCE WITH CERTIFICATION PROCEDURES.
K. ALL MATERIALS TO BE UL APPROVED WITH APPROPRIATE UL96 MARKINGS.
L. COMPLETED INSTALLATION AS SHOWN SHALL BEAR THE APPROPRIATE UL CERTIFICATION.
M. INSTALLATION SHALL COMPLY IN ALL RESPECTS TO L.P.I. CODE 175, NFPA 780 AND UL96A.

## SHEET KEYNOTES

# ALF

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TRUE NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
**GRAPHIC SCALE(S)**

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ROOF POWER PLAN |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 340 OF 378

EP-102FP

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

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1 BASEMENT - LIGHTING FLOOR PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, AND GENERAL NOTES.
B. SEE DRAWING SERIES E6 FOR ALL SCHEDULES AND CONTROL PANEL (LIGHTING & RECEPTACLE) INFORMATION.
C. COORDINATE ALL DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN.
D. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATION OF LIGHT FIXTURES AND OTHER CEILING MOUNTED DEVICES.
E. PROVIDE DEDICATED NEUTRALS FOR EACH DIMMING CIRCUIT.
F. ALL LIGHTING BRANCH CIRCUIT WIRING SHALL BE A MINIMUM SIZE OF #10 AWG. COORDINATE FINAL WIRE SIZE REQUIRED WITH SCHEDULES ON SHEET E-003.
G. CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING PROPER WIRE TYPE AND QUANTITY WITH MANUFACTURER.
H. LOWERCASE ALPHABETICAL SUBSCRIPT ON FIXTURES/DEVICES INDICATES SWITCHING ZONE WITHIN SPACE.
I. COORDINATE FIXTURES WITH ARCHITECTURAL CEILING AND WALL TYPES.
J. FIXTURES INDICATED ON PLANS WITH BOTH EMERGENCY AND NORMAL POWER CIRCUIT NUMBERS SHALL BE CONNECTED TO UL924 AUTOMATIC LOAD CONTROL RELAY DEVICE TO ALLOW FOR EMERGENCY FIXTURES TO BE CONTROLLED BY INDICATED RELAY OR SWITCH DURING NORMAL POWER, AND SWITCH TO EMERGENCY POWER CIRCUIT (FULL OUTPUT) DURING NORMAL POWER LOSS. PROVIDE # OF DEVICES AND ALL ACCESSORIES AND WIRING REQUIRED PER MANUFACTURER'S WIRING INSTRUCTIONS. FOR AUTOMATIC LOAD CONTROL RELAY DEVICES CONTROLLING MULTIPLE FIXTURES, LOCATE CLOSE TO FIRST EMERGENCY FIXTURE IN CONTROL ZONE IN NEAREST ACCESSIBLE LOCATION. COORDINATE PROPER WIRE TYPE AND QUANTITY WITH MANUFACTURER.
K. AUTOMATIC TRANSFER DEVICES SHALL BE WIRED SUCH THAT EXIT SIGNS WILL NOT BE SWITCHED WITH EMERGENCY LIGHTING TO MAINTAIN CONTINUOUS ON OPERATION. CONNECT EXIT SIGNS TO NEAREST 277V EMERGENCY CIRCUIT ON FLOOR.
L. ARCHITECT HAS FINAL APPROVAL OVER ALL FINISHES.
M. WHERE MORE THAN ONE SWITCH IS SHOWN AT ANY ONE LOCATION, GANG ALL SWITCHES UNDER ONE PLATE. PROVIDE BARRIERS BETWEEN SWITCHES CONNECTED TO DIFFERENT CIRCUITS. WHERE MORE THAN ONE DIMMER IS SHOWN AT ANY ONE LOCATION, MOUNT IN A MANNER TO GIVE THE APPEARANCE OF GANG MOUNTED UNDER ONE PLATE. FOLLOW MANUFACTURERS INSTRUCTIONS FOR DISTANCES AND DERATING.
N. COORDINATE ALL WALL MOUNTED WIRING DEVICE LOCATIONS WITH ARCHITECTURAL ELEVATIONS AND LIGHTING FIXTURE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLAN ON SHEET. REFER TO ARCH. REFLECTED CEILING PLANS FOR EXACT LOCATION OF LIGHT FIXTURES AND OTHER CEILING MOUNTED DEVICES.
O. THE LIGHTING CIRCUITING IS DIAGRAMMATIC. PROVIDE SWITCH LEGS, UNSWITCHED PHASE WIRES AND TRAVELERS AS NECESSARY.
P. FOR ALL CEILING MOUNTED OCCUPANCY SENSORS AND EXIT SIGNS MOUNTED IN OPEN CEILINGS, LOCATE AT HEIGHT SO THAT TOP OF DEVICE IS ALIGNED WITH LOWEST ADJACENT CEILING TO MAINTAIN INTENDED OPERATIONAL RANGE AND VISIBILITY.
Q. WHERE GENERAL PURPOSE RECEPTACLES ARE SHOWN WITHIN 2' OF A LIGHTING CONTROL DEVICE, ALIGN DEVICES VERTICALLY IN ELEVATION. COORDINATE LOCATIONS OF MECHANICAL THERMOSTATS AND ALIGN WITH LIGHTING CONTROL.
R. FOR ALL LIGHTING CONTROL DEVICE SYMBOLS SHOWN ON PLANS, SEE DRAWING SERIES E0 FOR LIGHTING CONTROL DESCRIPTION AND TYPE/FUNCTIONALITY OF DEVICE(S) TO BE PROVIDED WITHIN ROOM. FOR ALL LOW VOLTAGE WALL SWITCH DIMMERS, PROVIDE SEPARATE BUTTONS FOR EACH DIMMING ZONE TO ALLOW FULL RANGE DIMMING OF EACH ZONE.
S. FIELD COORDINATE LIGHT FIXTURE LOCATIONS IN ELECTRICAL, MECHANICAL AND TELECOM ROOMS WITH EQUIPMENT, DUCTS, PIPING, TELECOM, SECURITY, AND ALL UTILITIES IN SPACE.
T. EXIT SIGNS ARE REQUIRED TO BE READILY VISIBLE. ENSURE THE EXIT SIGNS ARE NOT OBSTRUCTED.

## SHEET KEYNOTES

E304 PROVIDE A DIMMING WITH ON/OFF MULTI- BUTTON SWITCH WITH PILOT LIGHT (ONE BUTTON PER NETWORK ZONE) FOR ALL NETWORK ZONES FED BY THE LIGHTING PANELBOARD WITHIN ELECTRICAL ROOM.

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

**NOT FOR CONSTRUCTION**

STAMP

TRUE NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | BASEMENT LIGHTING PLAN |

| WORKING STAFF | NS | NS | KM |
|---|---|---|---|
| | DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 341 OF 378

EL-100FP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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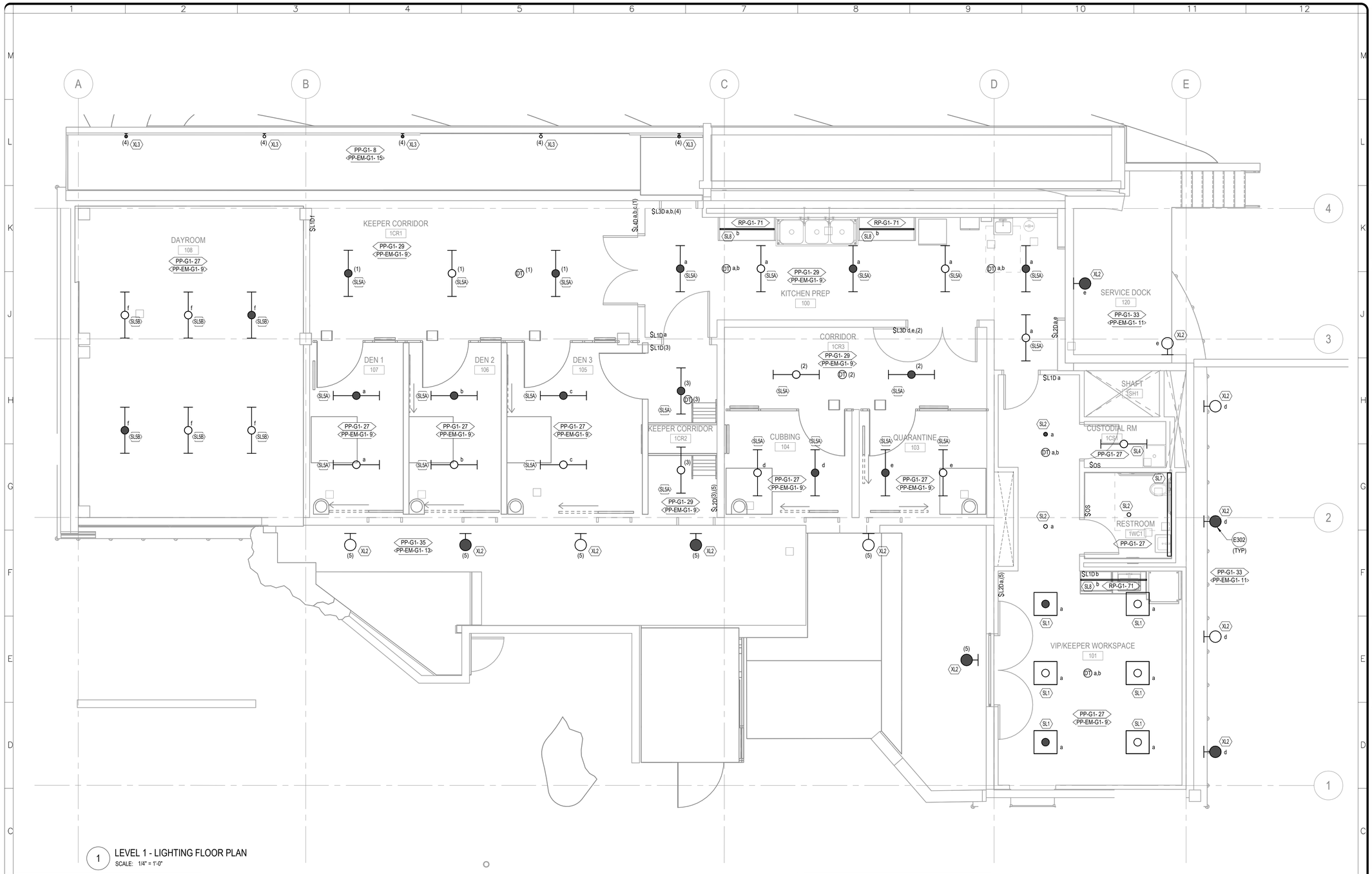


1 LEVEL 1 - LIGHTING FLOOR PLAN
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, AND GENERAL NOTES.
B. SEE DRAWING SERIES E6 FOR ALL SCHEDULES AND CONTROL PANEL (LIGHTING & RECEPTACLE) INFORMATION.
C. COORDINATE ALL DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN.
D. REFER TO ARCHITECTURAL REFLECTED CEILING PLANS FOR EXACT LOCATION OF LIGHT FIXTURES AND OTHER CEILING MOUNTED DEVICES.
E. PROVIDE DEDICATED NEUTRALS FOR EACH DIMMING CIRCUIT.
F. ALL LIGHTING BRANCH CIRCUIT WIRING SHALL BE A MINIMUM SIZE OF #10 AWG. COORDINATE FINAL WIRE SIZE REQUIRED WITH SCHEDULES ON SHEET E-003.
G. CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING PROPER WIRE TYPE AND QUANTITY WITH MANUFACTURER.
H. LOWERCASE ALPHABETICAL SUBSCRIPT ON FIXTURES/DEVICES INDICATES SWITCHING ZONE WITHIN SPACE.
I. COORDINATE FIXTURES WITH ARCHITECTURAL CEILING AND WALL TYPES.
J. FIXTURES INDICATED ON PLANS WITH BOTH EMERGENCY AND NORMAL POWER CIRCUIT NUMBERS SHALL BE CONNECTED TO UL924 AUTOMATIC LOAD CONTROL RELAY DEVICE TO ALLOW FOR EMERGENCY FIXTURES TO BE CONTROLLED BY INDICATED RELAY OR SWITCH DURING NORMAL POWER, AND SWITCH TO EMERGENCY POWER CIRCUIT (FULL OUTPUT) DURING NORMAL POWER LOSS. PROVIDE # OF DEVICES AND ALL ACCESSORIES AND WIRING REQUIRED PER MANUFACTURER'S WIRING INSTRUCTIONS. FOR AUTOMATIC LOAD CONTROL RELAY DEVICES CONTROLLING MULTIPLE FIXTURES, LOCATE CLOSE TO FIRST EMERGENCY FIXTURE IN CONTROL ZONE IN NEAREST ACCESSIBLE LOCATION. COORDINATE PROPER WIRE TYPE AND QUANTITY WITH MANUFACTURER.
K. AUTOMATIC TRANSFER DEVICES SHALL BE WIRED SUCH THAT EXIT SIGNS WILL NOT BE SWITCHED WITH EMERGENCY LIGHTING TO MAINTAIN CONTINUOUS ON OPERATION. CONNECT EXIT SIGNS TO NEAREST 277V EMERGENCY CIRCUIT ON FLOOR.
L. ARCHITECT HAS FINAL APPROVAL OVER ALL FINISHES.
M. WHERE MORE THAN ONE SWITCH IS SHOWN AT ANY ONE LOCATION, GANG ALL SWITCHES UNDER ONE PLATE. PROVIDE BARRIERS BETWEEN SWITCHES CONNECTED TO DIFFERENT CIRCUITS. WHERE MORE THAN ONE DIMMER IS SHOWN AT ANY ONE LOCATION, MOUNT IN A MANNER TO GIVE THE APPEARANCE OF GANG MOUNTED UNDER ONE PLATE. FOLLOW MANUFACTURERS INSTRUCTIONS FOR DISTANCES AND DERATING.
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S. FIELD COORDINATE LIGHT FIXTURE LOCATIONS IN ELECTRICAL, MECHANICAL AND TELECOM ROOMS WITH EQUIPMENT, DUCTS, PIPING, TELECOM, SECURITY, AND ALL UTILITIES IN SPACE.
T. EXIT SIGNS ARE REQUIRED TO BE READILY VISIBLE. ENSURE THE EXIT SIGNS ARE NOT OBSTRUCTED.

## SHEET KEYNOTES

E302 LIGHTING SWITCH TO CONTROL FACADE YARD FIXTURE TO BE LOCATED AT BASEMENT CORRIDOR.

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# ALF

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STAMP

TRUE NORTH

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
GRAPHIC SCALE(S)

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| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|

Smithsonian Institution

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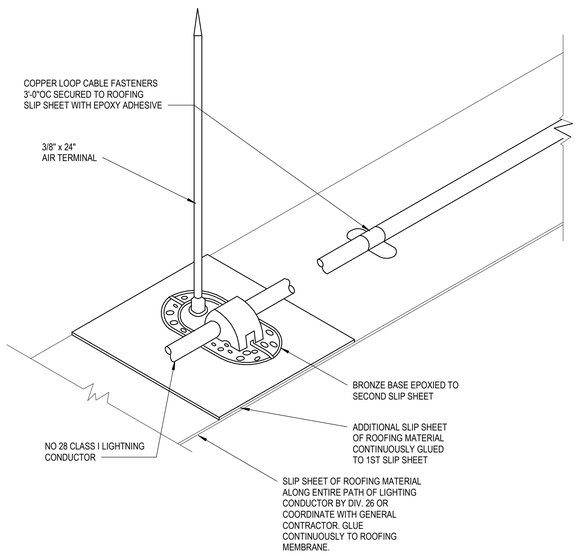
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | LEVEL 1 LIGHTING PLAN |

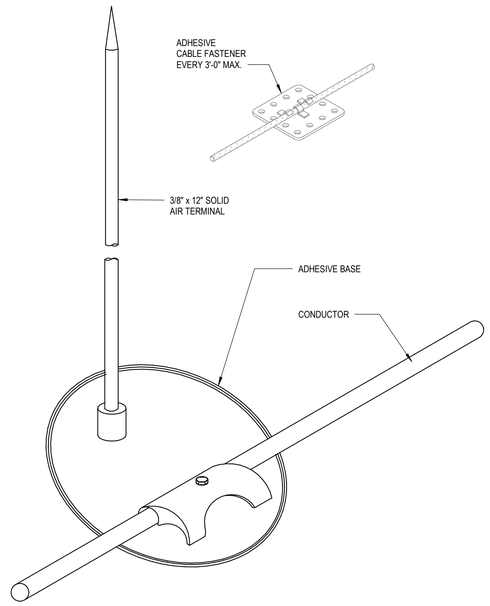
| WORKING STAFF | DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|---|
| | NS | NS | KM |

SHEET NO. 342 OF 378

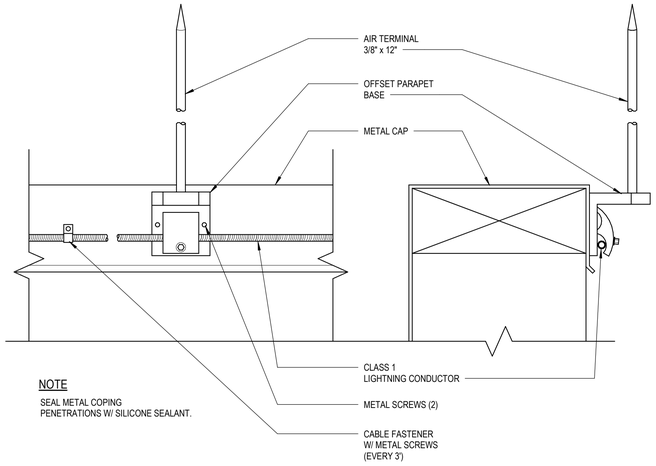
EL-101FP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE



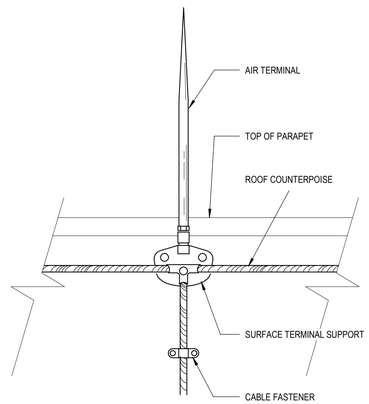

1 AIR TERMINAL CENTER ROOF
SCALE: NOT TO SCALE



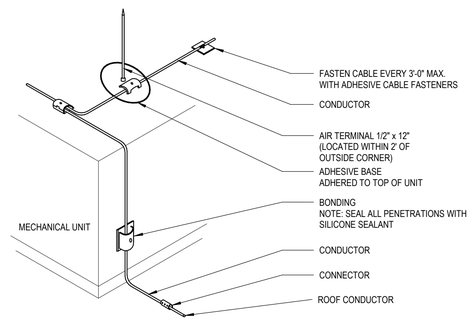

2 MID-ROOF ADHESIVE AIR TERMINAL BASE
SCALE: NOT TO SCALE




3 PARAPET AIR TERMINAL
SCALE: NOT TO SCALE




4 PARAPET MOUNTED AIR TERMINAL
SCALE: NOT TO SCALE




5 ROOF TOP UNIT
SCALE: NOT TO SCALE

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ALL SCHEDULES AND CONTROL PANEL (LIGHTING & RECEPTACLE) INFORMATION.
B. COORDINATE WITH ELECTRICAL AND LIGHTING DRAWINGS TO DETERMINE THE NUMBER OF ALL CONTROLS, MODULES, AND DEVICES. DO NOT PROVIDE QUANTITIES BASED SOLELY ON DIAGRAMS.

## SHEET KEYNOTES

ALF

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STAMP

KEY PLAN

**GRAPHIC SCALE(S)**

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600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

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|---|---|
| BUILDING NAME / ADDRESS | ARABIAN LEOPARD FACILITY (ALF)<br>3001 Connecticut Avenue NW<br>Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ELECTRICAL LIGHTNING DETAILS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 343 OF 378

E-500DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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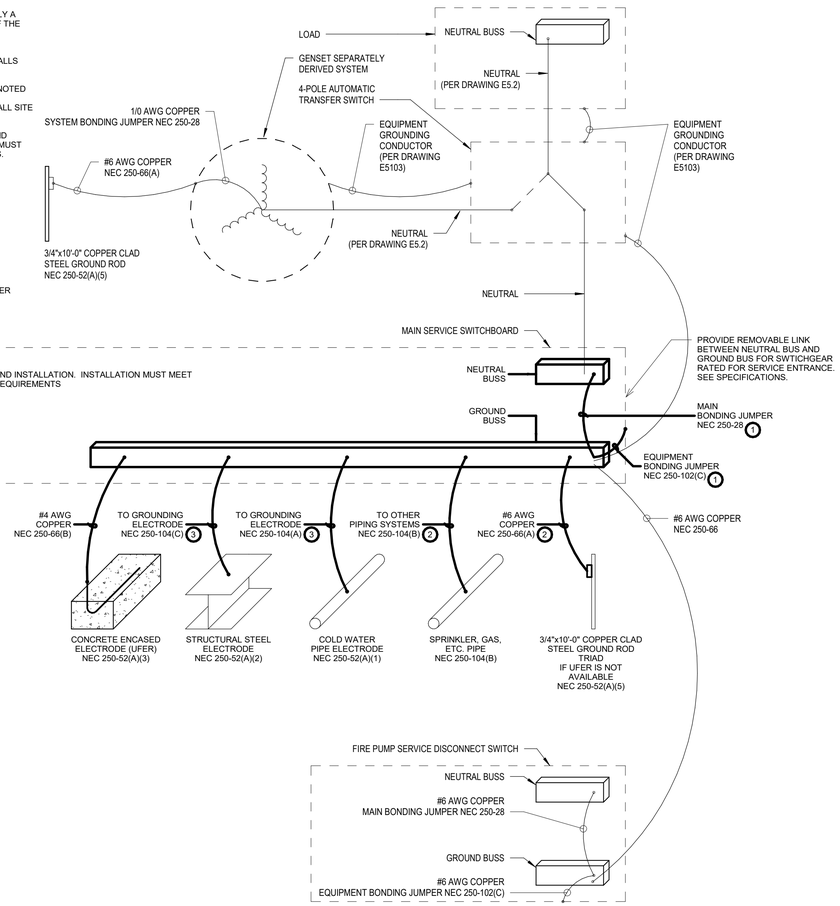
GENERAL NOTES - GROUNDING DETAIL:

1. THE GROUNDING ELECTRODE CONDUCTOR CONNECTION POINT IS NOT NECESSARILY A PHYSICAL CONNECTION. IT IS PROVIDED TO ILLUSTRATE THE INTERCONNECTION OF THE GROUNDING ELECTRODE SYSTEM. IT COULD, FOR EXAMPLE, BE THE WATER PIPE.
2. NEC REFERENCES ARE FROM 2017 NATIONAL ELECTRICAL CODE.
3. BONDS SHALL BE COMPRESSION TYPE. INTERIOR BONDS MAY BE EXOTHERMIC. UNDERGROUND CONDITIONS ARE TO BE WELDED/EXOTHERMIC EXCEPT AT TEST WALLS AND AS OTHERWISE NOTED.
4. BOND SIZE SHALL MATCH CONDUCTORS SHOWN ON FEEDER SCHEDULE.
5. GROUND CONDUCTORS SHALL BE STRANDED COPPER INSULATED CABLE, UNLESS NOTED OTHERWISE.
6. SERVICE ENTRANCE GROUND ROD TRIAD LOCATION MUST BE COORDINATED WITH ALL SITE AND UNDERGROUND ELEMENTS PRIOR TO ROUTING AND INSTALLATION.
7. UTILITY VAULT GROUND ROD LOCATIONS MUST BE COORDINATED WITH ALL SITE AND UNDERGROUND ELEMENTS PRIOR TO ROUTING AND INSTALLATION. INSTALLATION MUST MEET ALL REQUIREMENTS OF THE POWER AUTHORITY GROUNDING REQUIREMENTS.

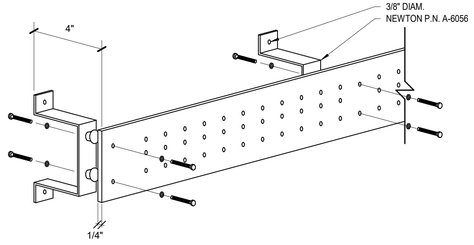
KEYED NOTES - GROUNDING DETAIL:

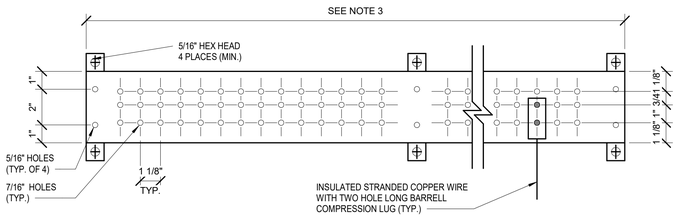
1. SIZE PER TABLE 250-102(C)(1) UP TO 1100 KCMIL. SIZE TO 12.5% OF FEEDERS WHEN OVER 1100 KCMIL. MAIN BONDING JUMPER FOR SERVICES GREATER THAN 1000A, PROVIDED WITH SES ARE ACCEPTABLE.
2. SIZE PER TABLE 250-122. ASSUMES MAIN DEVICE RATING IS EQUAL TO FEEDER SIZE.
3. SIZE PER TABLE 250-66.

| BONDING CONDUTOR SIZE | | | |
|---|---|---|---|
| SES SIZE | MBJ/EBJ (1) | PIPING (2) | GE (3) |
| 100A | 6 | 8 | 6 |
| 200A | 4 | 6 | 4 |
| 400A | 1/0 | 3 | 1/0 |
| 600A | 2/0 | 1 | 2/0 |
| 800A | 2/0 | 1/0 | 2/0 |
| 1000A | 3/0 | 2/0 | 3/0 |
| 1200A | 4/0 | 3/0 | 3/0 |
| 1600A | 250 KCMIL | 4/0 | 3/0 |
| 2000A | 300 KCMIL | 250 KCMIL | 3/0 |
| 2500A | 500 KCMIL | 350 KCMIL | 3/0 |
| 3000A | 500 KCMIL | 400 KCMIL | 3/0 |
| 3200A | 500 KCMIL | 400 KCMIL | 3/0 |
| 4000A | (2) 400 KCMIL | 500 KCMIL | 3/0 |
| 5000A | (2) 500 KCMIL | 500 KCMIL | 4/0 |

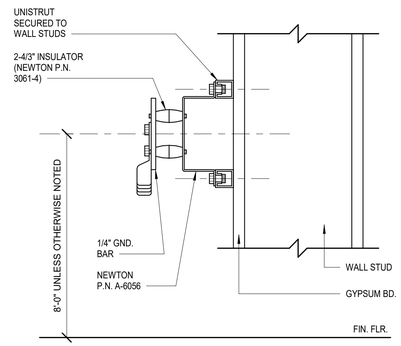



1 GROUNDING DETAIL
SCALE: NOT TO SCALE

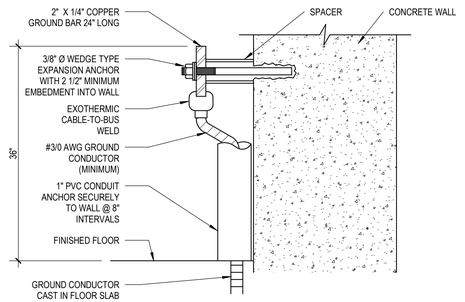



2 GROUNDING BUS BAR DETAIL
SCALE: NOT TO SCALE




NOTE:
1. ALL GROUNDING LEADS CONNECTED TO THE GROUNDING BUSBAR SHALL BE PROVIDED WITH TAGS INDICATING GROUNDING CONDUCTOR END POINT.
2. PROVIDE PLACARD TO READ "CRITICAL CONNECTIONS DO NOT REMOVE OR DISCONNECT. IF CONNECTIONS ARE LOOSE OR MUST BE REMOVED, CALL PHONE # ....."
3. TELECOMMUNICATION GROUNDING BUS (TGB) SHALL BE 1/4"D x 2"H x 12"W; MAIN ELECTRICAL ROOM GROUNDING BUS (MEGB); ELECTRICAL ROOM GROUNDING BUS (EGB); MAIN GROUNDING BUS (MGB), MAIN TELECOMMUNICATION GROUNDING BUS (TMGB), SHALL BE 1/4"D x 4"H x 24"W.




NOTES:
1. COPPER GROUND BAR SHALL BE ANCHORED TO THE WALL WITHIN 6" OF EACH END.
2. PROVIDE GROUND BUS IN MAIN ELECTRIC ROOM, MAIN EMERGENCY ELECTRICAL ROOM AND WHERE INDICATED ON DRAWINGS AND IN SPECIFICATIONS

3 ELECTRICAL ROOM GROUND BAR MOUNTING DETAIL
SCALE: NOT TO SCALE

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ALL SCHEDULES AND CONTROL PANEL (LIGHTING & RECEPTACLE) INFORMATION.
B. COORDINATE WITH ELECTRICAL AND LIGHTING DRAWINGS TO DETERMINE THE NUMBER OF ALL CONTROLS, MODULES, AND DEVICES. DO NOT PROVIDE QUANTITIES BASED SOLELY ON DIAGRAMS.

## SHEET KEYNOTES

ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

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KEY PLAN

**GRAPHIC SCALE(S)**

DATE: 06/15/2026
SUBMISSION: 65% CONSTRUCTION DOCUMENTS

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|---|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
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BUILDING NAME / ADDRESS: **ARABIAN LEOPARD FACILITY (ALF)**
**3001 Connecticut Avenue NW Washington, DC 20008**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**
OPDC PROJECT NUMBER: **2533105**
A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **ELECTRICAL GROUNDING DETAILS**
WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| NS | NS | KM |

SHEET NO. 344 OF 378

E-501DT
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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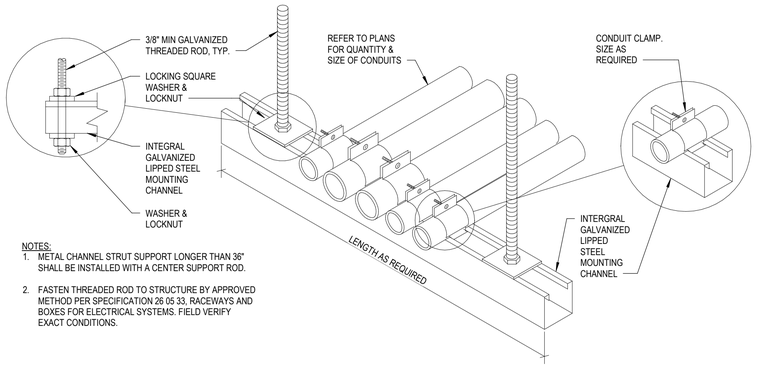
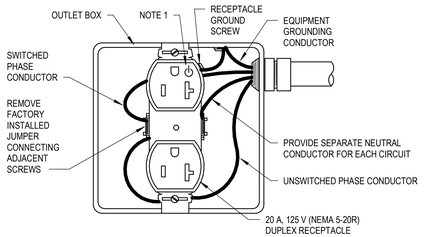
E-501DT BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS



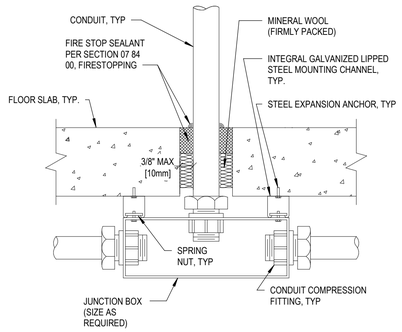

NOTES:
1. METAL CHANNEL STRUT SUPPORT LONGER THAN 36" SHALL BE INSTALLED WITH A CENTER SUPPORT ROD.
2. FASTEN THREADED ROD TO STRUCTURE BY APPROVED METHOD PER SPECIFICATION 26 05 33, RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS. FIELD VERIFY EXACT CONDITIONS.

1 MTG, CONDUIT TRAPEZE DETAIL
SCALE: NOT TO SCALE



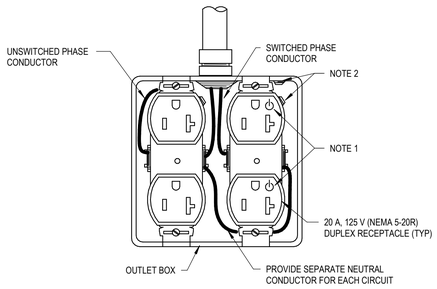

NOTES :
1. PROVIDE CONTROLLED RECEPTACLE IDENTIFICATION PER NEC 2014 ARTICLE 406.3(E).

2 CONTROLLED DUPLEX RECEPTACLE
SCALE: NOT TO SCALE



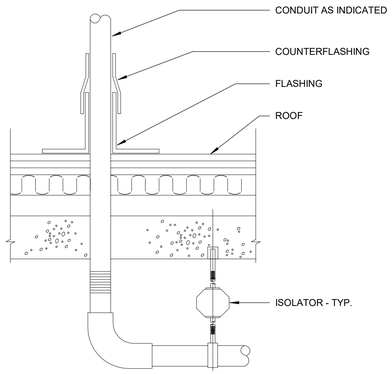

REINFORCING WITHIN THE EXISTING CONCRETE FLOOR CONSTRUCTION SHOULD BE PROPERLY LOCATED PRIOR TO ANY CORING, DRILLING, ETC. THROUGH SLAB PENETRATIONS REQUIRE BOTH TOP AND BOTTOM REINFORCING BE PROPERLY LOCATED. DO NOT DRILL, CORE, OR DAMAGE THE EXISTING REINFORCING.

4 FLOOR SLAB PENETRATION DETAIL
SCALE: NOT TO SCALE




NOTES :
1. PROVIDE CONTROLLED RECEPTACLE IDENTIFICATION PER NEC 2014 ARTICLE 406.3(E).
2. EQUIPMENT GROUNDING CONDUCTORS NOT SHOWN FOR CLARITY. PROVIDE EQUIPMENT GROUNDING FOR ALL DEVICES AND METALLIC BOXES PER SPECIFICATIONS AND NEC ARTICLE 250.

3 CONTROLLED QUAD RECEPTACLE
SCALE: NOT TO SCALE




5 E6.X - PWR - ROOF PENETRATION DETAIL
SCALE: NOT TO SCALE

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ALL SCHEDULES AND CONTROL PANEL (LIGHTING & RECEPTACLE) INFORMATION.
B. COORDINATE WITH ELECTRICAL AND LIGHTING DRAWINGS TO DETERMINE THE NUMBER OF ALL CONTROLS, MODULES, AND DEVICES. DO NOT PROVIDE QUANTITIES BASED SOLELY ON DIAGRAMS.

## SHEET KEYNOTES

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

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KEY PLAN

**GRAPHIC SCALE(S)**

| DATE | SUBMISSION |
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| REVISION # DATE | REVISION DESCRIPTION |
|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

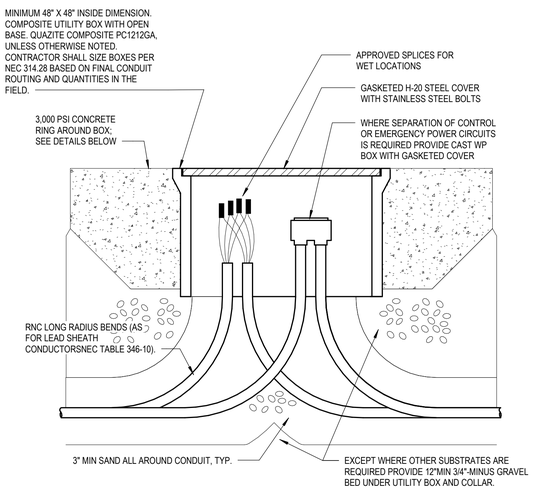
| | |
|---|---|
| BUILDING NAME / ADDRESS | ARABIAN LEOPARD FACILITY (ALF)<br>3001 Connecticut Avenue NW<br>Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ELECTRICAL POWER DETAILS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 345 OF 378

E-502DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

*FOR REFERENCE ONLY*

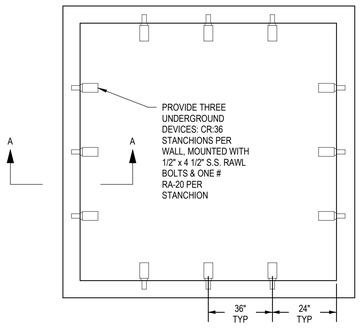
6/15/2026 2:49:20 PM
E-502DT BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS



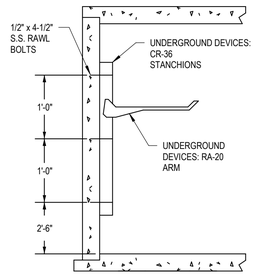

NOTE:

SIZE AS REQUIRED. PROVIDE EXTRA DEPTH BY STACKING WHERE REQUIRED WHERE REQUIRED BY CODE. SEE GENERAL NOTES ON SHEET E001.

1 HAND HOLE FINISHED GRADE
SCALE: NOT TO SCALE




PLAN VIEW

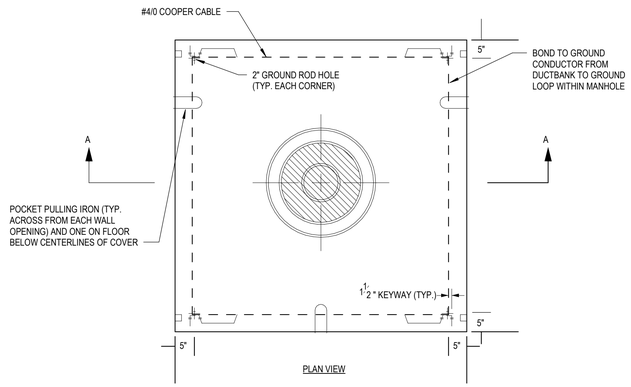



SECTION A-A

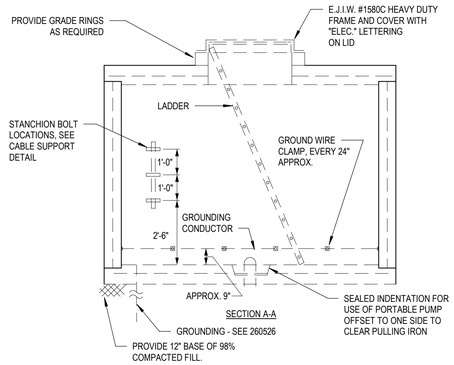
NOTES

1. REFER TO CIVIL ENGINEERING DRAWINGS FOR MANHOLE DEPTH

5 MANHOLE CABLE SUPPORT DETAIL
SCALE: NOT TO SCALE




PLAN VIEW

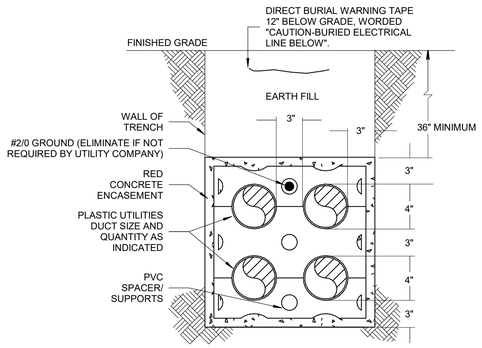



SECTION A-A

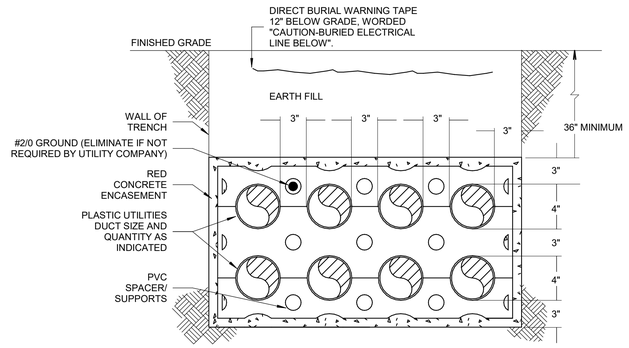
NOTES

1. REFER TO CIVIL ENGINEERING DRAWINGS FOR MANHOLE DEPTH
2. PROVIDE GALVANIZED STEEL LADDER. (CONFIRM LENGTH)

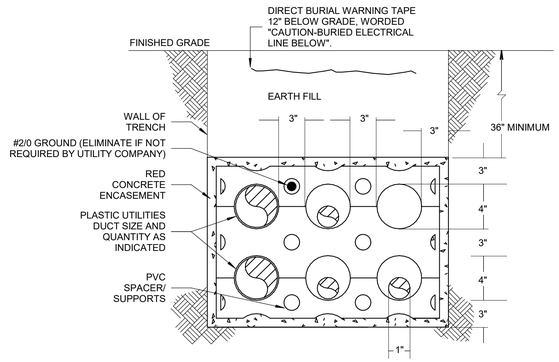
6 MANHOLE GROUNDING
SCALE: NOT TO SCALE




2 SITE-DUCT BANK 4-WAY WITH SPACERS
SCALE: NOT TO SCALE




3 SITE-DUCT BANK 8-WAY WITH SPACERS
SCALE: NOT TO SCALE




4 SITE-GENERATOR DUCT BANK 6-WAY WITH SPACERS
SCALE: NOT TO SCALE

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ALL SCHEDULES AND CONTROL PANEL (LIGHTING & RECEPTACLE) INFORMATION.
B. COORDINATE WITH ELECTRICAL AND LIGHTING DRAWINGS TO DETERMINE THE NUMBER OF ALL CONTROLS, MODULES, AND DEVICES. DO NOT PROVIDE QUANTITIES BASED SOLELY ON DIAGRAMS.

## SHEET KEYNOTES

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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KEY PLAN

**GRAPHIC SCALE(S)**

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| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|
| 1 | 05/11/2026 | AMENDMENT 01 |

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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ELECTRICAL POWER DETAILS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 346 OF 378

E-503DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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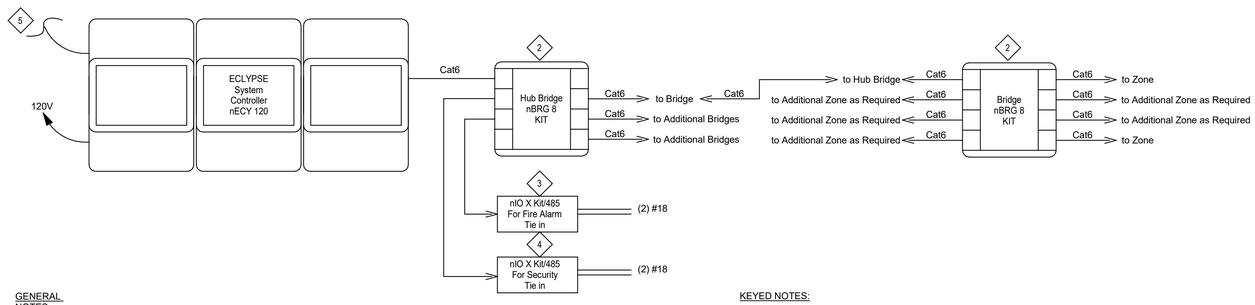
6/15/2026 2:49:20 PM
E-503DT BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS



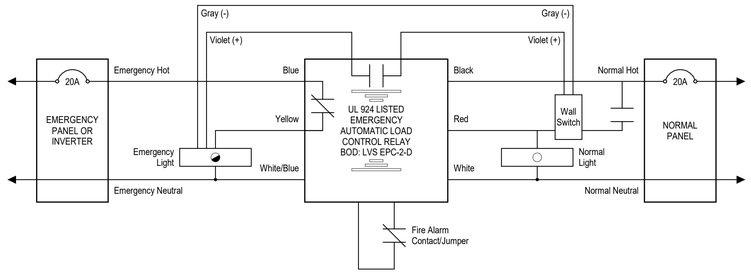

GENERAL NOTES:

1. PROVIDE NEW NEMA ENCLOSURE FOR NEW LIGHTING SYSTEM CONTROLLER, ECLYSPE nECY 120, AS SHOWN IN E3 SERIES, AND COORDINATE MOUNTING. PROVIDE SINGLE 120V DEDICATED CIRCUIT FOR SYSTEM CONTROLLER AND SYSTEMS FURNITURE RECEPTACLE RELAY PANEL; COORDINATE WITH E3 AND E7 SERIES.
2. SYSTEM SHALL HAVE HEAD END CAPABILITY OF PROVIDING TIME BASED CONTROL TO ALL AREAS REQUIRING TIME CONTROL PER SEQUENCE OF OPERATIONS. SYSTEM BACKBONE DEVICES SHALL BE PROVIDED PER MANUFACTURERS RECOMMENDATION.
3. COORDINATE FINAL LOCATION OF LOCAL BRIDGE AND HUB BRIDGE IN FIELD TO MINIMIZE CONDUIT AND CONDUCTOR RUNS.
4. PROVIDE NUMBER OF BRIDGES REQUIRED PER E3 SERIES.
5. PROVIDE NUMBER OF ZONES REQUIRED PER E3 SERIES.
6. ECLYPSE nECY BETWEEN G1 AND G2 LEVEL SHALL BE CONNECTED TOGETHER TO A LAN OR FROM THEIR ETHERNET PORTS FOR NETWORK LEVEL COMMUNICATION.

KEYED NOTES:

1. NOT USED.
2. COORDINATE NUMBER OF BRIDGES WITH NUMBER OF ZONES REQUIRED PER E3 SERIES.
3. COORDINATE WITH FIRE PROTECTION CONTRACTOR/INSTALLER TO PROVIDE TIE-IN TO FIRE ALARM SYSTEM. PROGRAM HEADEND CONTROLLER TO PROVIDE 100% FIXTURE OUTPUT FOR ALL EMERGENCY FIXTURES WHEN TRIGGERED BY THE FIRE ALARM SYSTEM.
4. COORDINATE WITH SECURITY CONSULTANT/CLIENT TO PROVIDE TIE-IN TO SECURITY SYSTEM. PROGRAM HEADEND CONTROLLER TO PROVIDE 100% FIXTURE OUTPUT FOR ALL EMERGENCY FIXTURES WHEN TRIGGERED BY THE SECURITY SYSTEM.
5. PROVIDE BACKNET IP OR BACNET MS/TP ENABLE CONNECTION TO BUILDING MANAGEMENT SYSTEM. COORDINATE WITH MECHANICAL DRAWINGS FOR CONNECTION TO BMS.

1 N-LIGHT MAIN CONTROL IN ELECTRICAL ROOM
SCALE: NOT TO SCALE




GENERAL NOTES:

1. UL924 DEVICE INDICATED IN DIGITAL LIGHTING CONTROL DETAILS IS ACCEPTABLE FOR DIGITAL LIGHTING CONTROL SYSTEMS.
2. PROVIDE QUANTITIES AS REQUIRED FOR CONTROL OF EMERGENCY FIXTURES WITH NORMAL LIGHTING FIXTURES INDICATED ON E3 SERIES.

2 EMERGENCY AUTOMATIC LOAD CONTROL RELAY WIRING DIAGRAM
SCALE: NOT TO SCALE

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ALL SCHEDULES AND CONTROL PANEL (LIGHTING & RECEPTACLE) INFORMATION.
B. COORDINATE WITH ELECTRICAL AND LIGHTING DRAWINGS TO DETERMINE THE NUMBER OF ALL CONTROLS, MODULES, AND DEVICES. DO NOT PROVIDE QUANTITIES BASED SOLELY ON DIAGRAMS.

## SHEET KEYNOTES

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600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF) — 3001 Connecticut Avenue NW Washington, DC 20008
PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084
DRAWING TITLE: ELECTRICAL LIGHTING DETAILS
WORKING STAFF: NS (DESIGNED BY) | NS (DRAWN BY) | KM (CHECKED BY)

SHEET NO. 347 OF 378

E-504DT (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

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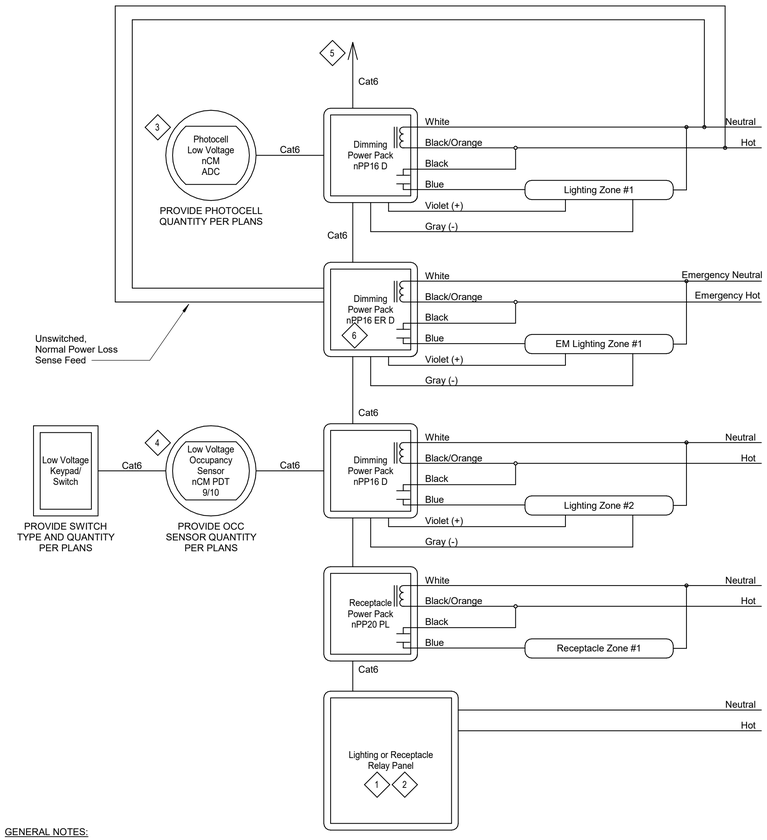


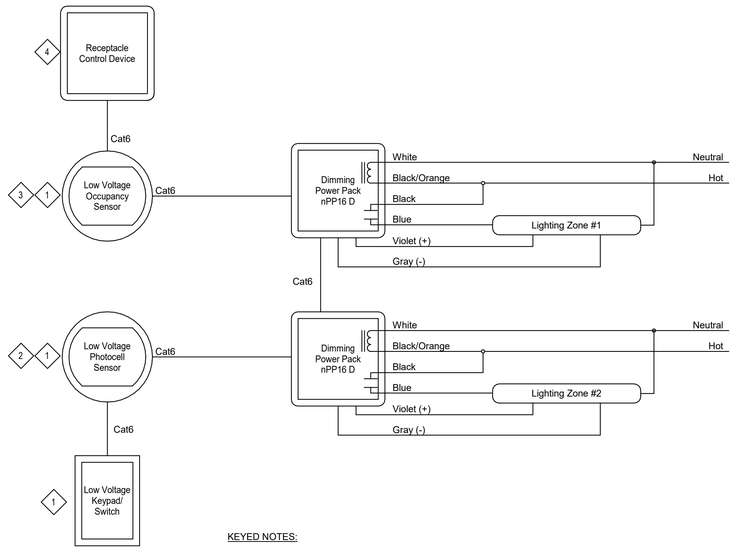

GENERAL NOTES:

1. COORDINATE WITH E0 SERIES FOR LIGHTING CONTROL DEVICE DESCRIPTIONS. COORDINATE WITH E2 SERIES FOR CONTROLLED RECEPTACLES. COORDINATE WITH E3 SERIES FOR LIGHTING CONTROL ZONES. COORDINATE WITH E6 SERIES FOR LIGHTING FIXTURE DRIVER TECHNOLOGY, LIGHTING CONTROL PROGRAMMING INTENT, AND CONTROLLED RECEPTACLE CIRCUITS. ALL DEVICE QUANTITIES AND CATALOG NUMBERS SHOWN ABOVE FOR REFERENCE ONLY. ONLY PROVIDE ADDITIONAL DEVICES AS SHOWN/INDICATED ON PLANS OR REQUIRED FOR SYSTEM TO FUNCTION AS INTENDED. COORDINATE WITH POWER AND LIGHTING DRAWINGS TO DETERMINE THE NUMBER AND TYPE OF ALL CONTROLS, MODULES, AND DEVICES REQUIRED FOR FULLY FUNCTIONING SYSTEM. MODEL NUMBERS ARE SHOWN FOR REFERENCE ONLY. COORDINATE WITH LIGHTING CONTROLS MANUFACTURER FOR RECOMMENDED CONFIGURATIONS TO ACHEIVE DESIGN INTENT INDICATED ON THE PLANS.
2. DIAGRAM SHOWN INDICATES TYPICAL LIGHTING CONTROL DEVICES FOR EACH SPACE. SEPARATELY NETWORK EACH SPACE TO LIGHTING CONTROL SYSTEM BRIDGE, UON. COORDINATE WITH CONTROL SUBSCRIPTS INDICATED IN PLAN FOR CONTROL ZONES SHARED ACROSS MULTIPLE SPACES.
3. PROVIDE POWERPACKS AND DEVICES REQUIRED TO CONTROL LIGHTING FIXTURE ZONES INDICATED IN THE E3 SERIES. PROVIDE POWER PACK QUANTITIES AND TYPE AS REQUIRED BY E3 SERIES.
   a. PROVIDE DIMMING OR NON-DIMMING POWER PACKS AS REQUIRED FOR LOCAL KEYPAD AND OCCUPANCY SENSOR CONTROL ZONES INDICATED IN E3 SERIES.
   b. PROVIDE DIMMING POWER PACKS AS REQUIRED FOR DAYLIGHT HARVESTING IN CONTROL ZONES CONTROLLED BY PHOTOCELL IN E3 SERIES.
   c. PROVIDE DIMMING OR NON-DIMMING POWER PACK THAT IS UL924 LISTED FOR ALL CONTROLLED EMERGENCY LIGHTING FIXTURES INDICATED ON E3 SERIES.
   d. PROVIDE 20A PLUG LOAD RATED SWITCHING POWER PACK FOR CONTROLLED RECEPTACLES INDICATED IN E2 SERIES.
4. CONTROL DIAGRAMS BASED ON ACUITY NLIGHT CONTROLS.
5. FOR FIXTURES INDICATED ON PLANS REQUIRED TO TURN ON TO 50% AND THEN 100%, IT IS ACCEPTABLE TO PROVIDE DIMMING POWER PACK TO DELIVER INTENDED FUNCTIONALITY. FIXTURE TYPE AND DRIVER INDICATED ON THE LIGHTING FIXTURE SCHEDULE MUST BE COORDINATED WITH CONTROL MODULE TECHNOLOGY.
6. PROVIDE POWER SUPPLIES AS REQUIRED FOR DEVICE QUANTITIES AND ZONES SHOWN ON THE PLANS. COORDINATE WITH MANUFACTURER FOR POWER SUPPLIES REQUIRED FOR LONG CABLING RUNS. POWER SUPPLY, DIMMING CONTROLLER, AND CONNECTED OCCUPANCY SENSORS TO SHARE 24 VOLT GROUND. COORDINATE POWER PACK(S) LOCATION WITH MANUFACTURER TO LIMIT VOLTAGE DROP PER MANUFACTURER RECOMMENDATIONS AND TO PROVIDE IN AN ACCESSIBLE LOCATION.
7. CONTRACTOR TO CONFIRM NECESSARY POWER PACK, ACCESSORIES, AND WIRING FOR FIXTURES WITH NON 0-10V DIMMING.
8. COORDINATE WITH MECHANICAL CONTRACTOR FOR MECHANICAL SYSTEMS TIE-IN TO LIGHTING CONTROL OCCUPANCY SENSORS AND BMS.

KEYED NOTES:

1. PROVIDE PLUG LOAD RATED POWERPACK, nPP20 PL, FOR ALL NON-SYSTEMS FURNITURE CONTROLLED RECEPTACLES. PROVIDE PLUG LOAD RATED RELAY PANEL, ARP INTENC16 NLT 16-RELAY HINGED LOCKING DOOR AND SURFACE MOUNTED ENCLOSURE, FOR ALL SYSTEMS FURNITURE CONTROLLED RECEPTACLES.
2. PROVIDE ONE RELAY FOR EACH CONTROLLED CIRCUIT; COORDINATE WITH E7 SERIES FOR SYSTEMS FURNITURE CIRCUITS TO BE CONTROLLED. CIRCUITS CONTROLLED BY RELAY PANEL ARE TO BE CONTROLLED BY LIGHTING CONTROL SYSTEM, COORDINATE WITH E3 SERIES AND E6 SERIES FOR TIMECLOCK SCHEDULE AND OCCUPANCY SENSOR OVERRIDE. RELAY PANELS ARE LOCATED IN THE MAIN ELECTRICAL ROOM.
3. PHOTOCELL TO MAINTAIN TARGET ILLUMINANCE INDICATED IN THE E7 SERIES LIGHTING CONTROL MATRIX THROUGHOUT THE ENTIRE SPACE SERVED.
4. COORDINATE WITH MECHANICAL CONTRACTOR FOR MECHANICAL SYSTEMS TIE-IN TO LIGHTING CONTROL OCCUPANCY SENSORS AND BMS.
5. PROVIDE NETWORK CONNECTION TO BRIDGE FOR EACH SPACE.
6. PROVIDE REQUIRED CONTACTS AND INTERFACES WITH FIRE ALARM SYSTEM SUCH THAT EMERGENCY LIGHTING TURNS ON TO FULL BRIGHTNESS IN THE EVENT OF FIRE ALARM ACTIVATION.

1 TYPICAL LOW VOLTAGE NETWORKED SPACE
SCALE: NOT TO SCALE




KEYED NOTES:

1. PROVIDE QUANTITIES PER E3 SERIES.
2. PHOTOCELL TO MAINTAIN TARGET ILLUMINANCE INDICATED IN THE E7 SERIES LIGHTING CONTROL MATRIX THROUGHOUT THE ENTIRE SPACE SERVED.
3. PROVIDE ISOLATED RELAY INTERFACE AS REQUIRED BY DIVISION 23. COORDINATE WITH M SERIES CONTROLS DETAILS FOR INTEGRATION AND CONTROLLABILITY.
4. PROVIDE PLUG LOAD RATED POWERPACK, nPP20 PL, FOR ALL CONTROLLED RECEPTACLES PER E2 SERIES. REFERENCE E6 SERIES FOR RECEPTACLE CONTROL WIRING DETAILS.

GENERAL NOTES:

1. COORDINATE WITH E0 SERIES FOR LIGHTING CONTROL DEVICE DESCRIPTIONS. COORDINATE WITH E2 SERIES FOR CONTROLLED RECEPTACLES. COORDINATE WITH E3 SERIES FOR LIGHTING CONTROL ZONES. COORDINATE WITH E6 SERIES FOR LIGHTING FIXTURE DRIVER TECHNOLOGY, LIGHTING CONTROL PROGRAMMING INTENT, AND CONTROLLED RECEPTACLE CIRCUITS. ALL DEVICE QUANTITIES AND CATALOG NUMBERS SHOWN ABOVE FOR REFERENCE ONLY. ONLY PROVIDE ADDITIONAL DEVICES AS SHOWN/INDICATED ON PLANS OR REQUIRED FOR SYSTEM TO FUNCTION AS INTENDED. COORDINATE WITH POWER AND LIGHTING DRAWINGS TO DETERMINE THE NUMBER AND TYPE OF ALL CONTROLS, MODULES, AND DEVICES REQUIRED FOR FULLY FUNCTIONING SYSTEM. MODEL NUMBERS ARE SHOWN FOR REFERENCE ONLY. COORDINATE WITH LIGHTING CONTROLS MANUFACTURER FOR RECOMMENDED CONFIGURATIONS TO ACHEIVE DESIGN INTENT INDICATED ON THE PLANS.
2. DIAGRAM SHOWN INDICATES TYPICAL LIGHTING CONTROL DEVICES FOR EACH SPACE. COORDINATE WITH CONTROL SUBSCRIPTS INDICATED IN PLAN FOR CONTROL ZONES SHARED ACROSS MULTIPLE SPACES.
3. PROVIDE POWERPACKS AND DEVICES REQUIRED TO CONTROL LIGHTING FIXTURE ZONES INDICATED IN THE E3 SERIES. PROVIDE POWER PACK QUANTITIES AND TYPE AS REQUIRED BY E3 SERIES.
   - PROVIDE DIMMING OR NON-DIMMING POWER PACKS AS REQUIRED FOR LOCAL KEYPAD AND OCCUPANCY SENSOR CONTROL ZONES INDICATED IN E3 SERIES.
   - PROVIDE DIMMING POWER PACKS AS REQUIRED FOR DAYLIGHT HARVESTING IN CONTROL ZONES CONTROLLED BY PHOTOCELL IN E3 SERIES.
   - PROVIDE DIMMING OR NON-DIMMING POWER PACK THAT IS UL924 LISTED FOR ALL CONTROLLED EMERGENCY LIGHTING FIXTURES INDICATED ON E3 SERIES.
   - PROVIDE 20A PLUG LOAD RATED SWITCHING POWER PACK FOR CONTROLLED RECEPTACLES INDICATED IN E2 SERIES.
   - PROVIDE DIMMING OR NON-DIMMING POWER PACKS WITH VOLTAGES AS REQUIRED FOR CONTROLLED FIXTURE VOLTAGES AS INDICATED IN LIGHTING FIXTURE SCHEDULE IN E6 SERIES.
   - PROVIDE DIMMING POWER PACKS AS REQUIRED FOR CONTROLLED FIXTURE DIMMING PROTOCOL AS INDICATED IN LIGHTING FIXTURE SCHEDULE IN E7 SERIES.
4. CONTROL DIAGRAMS BASED ON ACUITY NLIGHT CONTROLS. APPROVED EQUAL: WATTSTOPPER.
5. FOR FIXTURES INDICATED ON PLANS REQUIRED TO TURN ON TO 50% AND THEN 100%, IT IS ACCEPTABLE TO PROVIDE DIMMING POWER PACK TO DELIVER INTENDED FUNCTIONALITY. FIXTURE TYPE AND DRIVER INDICATED ON THE LIGHTING FIXTURE SCHEDULE MUST BE COORDINATED WITH CONTROL MODULE TECHNOLOGY.
6. PROVIDE POWER SUPPLIES AS REQUIRED FOR DEVICE QUANTITIES AND ZONES SHOWN ON THE PLANS. COORDINATE POWER PACK(S) LOCATION WITH MANUFACTURER TO LIMIT VOLTAGE DROP PER MANUFACTURER RECOMMENDATIONS AND TO PROVIDE IN AN ACCESSIBLE LOCATION.
7. CONTRACTOR TO CONFIRM NECESSARY POWER PACK, ACCESSORIES, AND WIRING FOR FIXTURES WITH NON 0-10V DIMMING.

2 TYPICAL LOW VOLTAGE STANDALONE SPACE
SCALE: NOT TO SCALE

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ALL SCHEDULES AND CONTROL PANEL (LIGHTING & RECEPTACLE) INFORMATION.
B. COORDINATE WITH ELECTRICAL AND LIGHTING DRAWINGS TO DETERMINE THE NUMBER OF ALL CONTROLS, MODULES, AND DEVICES. DO NOT PROVIDE QUANTITIES BASED SOLELY ON DIAGRAMS.

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| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ELECTRICAL LIGHTING DETAILS |
| WORKING STAFF | NS (DESIGNED BY) / NS (DRAWN BY) / KM (CHECKED BY) |

SHEET NO. 348 OF 378

E-505DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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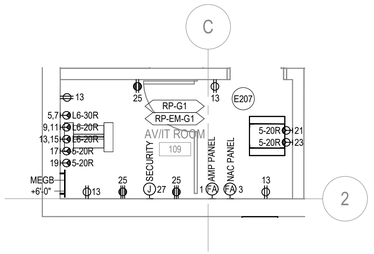
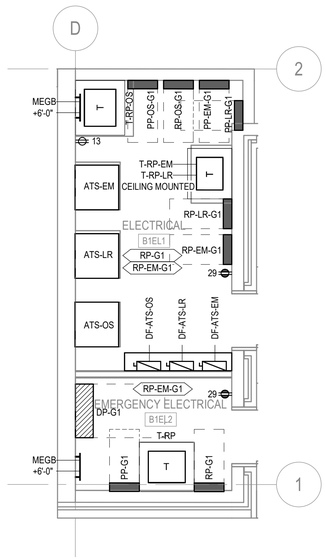



1 AV/IT ROOM 109
SCALE: 1/4" = 1'-0"




2 ELECTRICAL AND EMERGENCY ELECTRICAL ROOM - B1EL1 & B1EL2
SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. COORDINATE ALL DEVICE LOCATIONS AND MOUNTING HEIGHTS WITH ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN.
C. THE CONTRACTOR SHALL COORDINATE EXACT LOCATION OF EQUIPMENT POWER CONNECTIONS WITH EQUIPMENT INSTALLER PRIOR TO ROUGH-IN.
D. SEE MECHANICAL EQUIPMENT SCHEDULE FOR ASSOCIATED OVERCURRENT DEVICES, DISCONNECT SWITCHES, STARTERS AND WIRES.
E. CONTRACTOR SHALL BE RESPONSIBLE FOR SUPPLYING ANY REQUIRED TRANSFORMER/POWER SUPPLIES FOR DOOR HARDWARE. COORDINATE WITH ARCHITECTURAL AND SECURITY DRAWINGS FOR DOOR HARDWARE. PROVIDE 120V POWER AS REQUIRED.
F. CONTRACTOR SHALL COORDINATE AND FIELD VERIFY REQUIREMENT OF NEUTRAL CONDUCTOR FOR ALL EQUIPMENT AND PROVIDE NEUTRAL CONDUCTOR AS NECESSARY.
G. TRIPPING OF GFCI RECEPTACLES SHALL NOT AFFECT THE DOWNSTREAM RECEPTACLES.
H. NO MORE THAN 3 CIRCUITS MAY SHARE A HOMERUN.
I. FOR ALL MECHANICAL EQUIPMENT, PROVIDE ADDITIONAL 120V CONNECTIONS REQUIRED BY MANUFACTURER.
J. FOR EACH AIR HANDLING UNIT, PROVIDE (1)120V CONNECTION FOR LIGHTING AND RECEPTACLES. CIRCUIT TO MECHANICAL PANEL WITHIN THE SPACE.
K. PROVIDE 120V CIRCUITS FOR CONTROL PANELS AS REQUIRED BY CONTROL VENDOR. CIRCUIT TO MECHANICAL PANEL WITHIN THE SPACE.
L. ALL UNUSED WALL OUTLETS TO BE COVERED WITH BLANK PLATE. ALL BLANK PLATE COLORS TO MATCH WALL COLOR.
M. SEE ANY COMMUNICATIONS PATHWAY PLAN AND AV DRAWINGS, FOR ADDITIONAL CONDUIT TO BE PROVIDED BY ELECTRICAL CONTRACTOR.
N. ALL CONDUIT ROUTED WITHIN SLAB SHALL COMPLY WITH GUIDELINES LISTED IN STRUCTURAL DRAWINGS.
M. FOR DEVICES LABELED WITH RECTANGULAR TAG, REFER TO KITCHEN EQUIPMENT SCHEDULE ON E7 SERIES CONNECTIONS, CIRCUITS AND DEVICE TYPES.
N. UNLESS OTHERWISE NOTED, ALL SWITCHED RECEPTACLES TO BE CONTROLLED BY OCCUPANCY SENSOR OR CONTACTORS. SEE CONTROLS SCHEDULES ON AND DETAILS ON FOR ADDITIONAL CONNECTION INFORMATION.
O. COORDINATE FINAL LOCATION OF MECHANICAL AND PLUMBING EQUIPMENT AND ASSOCIATED DISCONNECT SWITCHES, STARTERS, VFDS, CONTROL POWER AND OTHER POWER REQUIREMENTS WITH DIV 22 AND 23.
P. CONTRACTOR IS RESPONSIBLE FOR ALL WIRING FROM VFDs AND DISCONNECTS SHOWN IN DRAWINGS TO THE EQUIPMENT AND IS TO COORDINATE WITH EQUIPMENT MANUFACTURER AND INSTALLATION INSTRUCTIONS.
Q. WALL MOUNTED AND CEILING MOUNTED TRANSFORMERS SHALL BE MOUNTED SUCH THAT THE TRANSFORMER ENCLOSURE AND ALL ELEMENTS OF THE HANGER SUPPORT SYSTEM ARE LOCATED ABOVE THE MINIMUM WORKING SPACE HEIGHT IN COMPLIANCE WITH NEC 110.26.
R. PROVIDE GROUND FAULT PROTECTION FOR ALL 120-VOLT 15 AND 20-AMP RECEPTACLES LOCATED IN KITCHEN AREAS AS REQUIRED BY NEC 210.8(B).

## SHEET KEYNOTES

E207 ALL RECEPTACLES FOR IT/AV RACK TO BE MOUNTED ON THE WALL +60" A.F.F. REFER TO TELECOM DRAWINGS FRO ADDITIONAL INFORMATION.

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
**GRAPHIC SCALE(S)**

DATE: 06/15/2026
SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE | REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

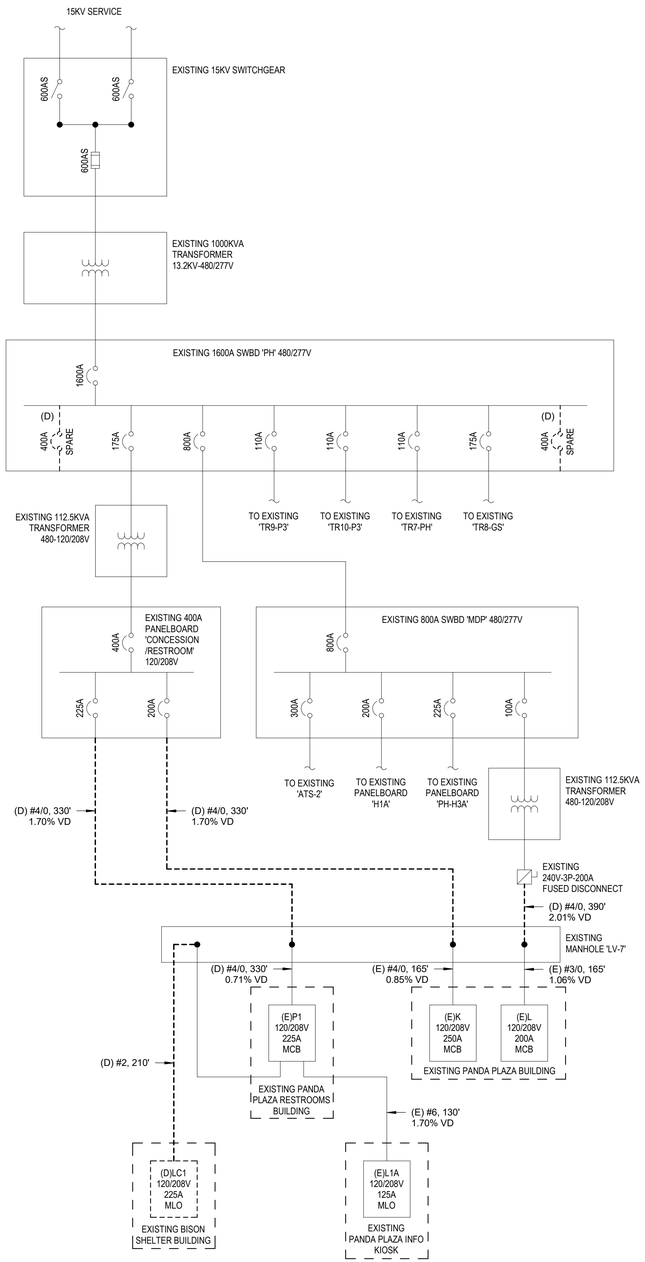
PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: ENLARGED PLAN
WORKING STAFF: NS (DESIGNED BY) | NS (DRAWN BY) | KM (CHECKED BY)

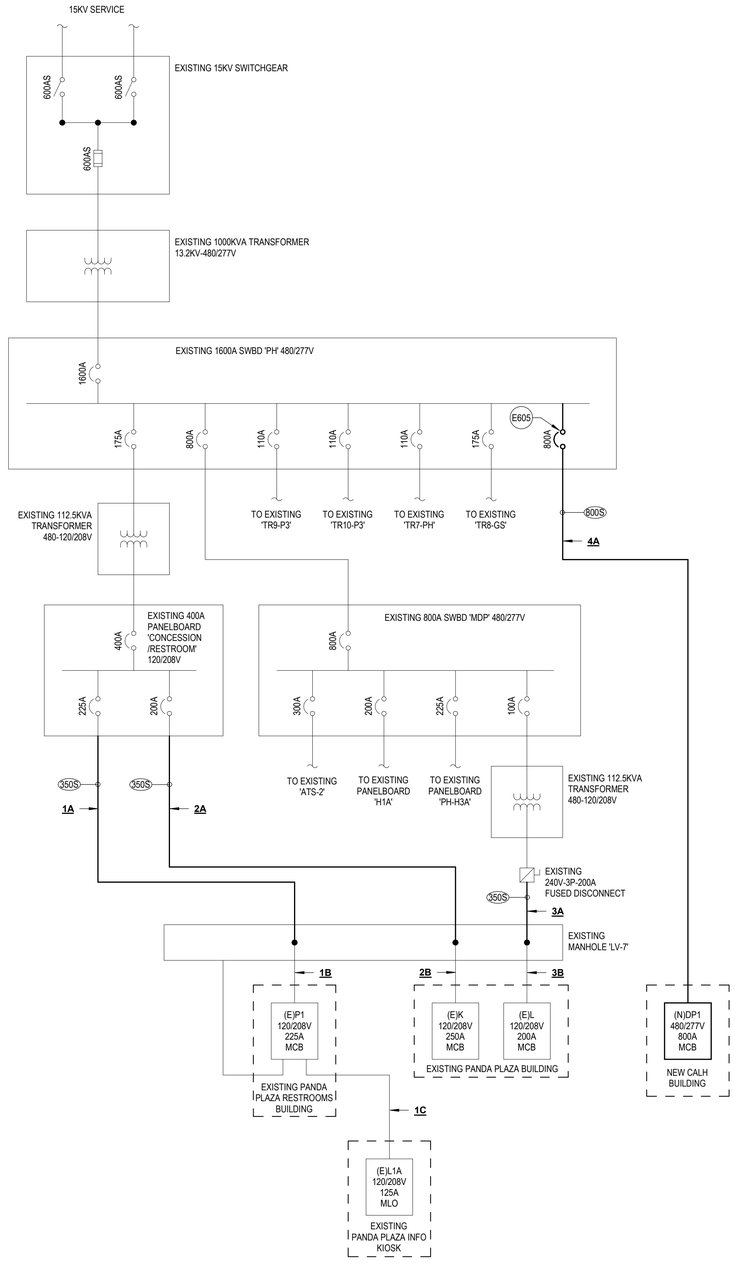
SHEET NO. 349 OF 378
E-506DT (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

FOR REFERENCE ONLY

6/15/2026 2:49:21 PM
E-506DT BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




1 EXISTING ELECTRICAL ONE LINE/RISER DIAGRAM SWBD 'PH' DEMOLITION
SCALE: NOT TO SCALE




2 EXISTING ELECTRICAL ONE LINE/RISER DIAGRAM SWBD 'PH' NEW WORK
SCALE: NOT TO SCALE

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 SERIES FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES.
B. THE CONTRACTOR SHALL COORDINATE EXACT LOCATION OF EQUIPMENT POWER CONNECTIONS WITH EQUIPMENT INSTALLER PRIOR TO ROUGH-IN.
C. COORDINATE FINAL LOCATION OF MECHANICAL EQUIPMENT AND ASSOCIATED DISCONNECT SWITCHES, STARTERS, VFDS, CONTROL POWER AND OTHER POWER REQUIREMENTS WITH DIV 23 AND MECHANICAL DRAWINGS.
D. SEE MECHANICAL EQUIPMENT SCHEDULE FOR ASSOCIATED OVERCURRENT DEVICES, DISCONNECT SWITCHES, STARTERS AND WIRES.
E. FOR ALL MECHANICAL EQUIPMENT, PROVIDE ADDITIONAL 120V CONNECTIONS REQUIRED BY MANUFACTURER.
F. VOLTAGE DROP SHALL BE LIMITED TO LESS THAN 3% FOR ALL BRANCH CIRCUITS AND 2% FOR ALL FEEDERS.
G. CONTRACTOR SHALL COORDINATE AND FIELD VERIFY REQUIREMENT OF NEUTRAL CONDUCTOR FOR ALL EQUIPMENT AND PROVIDE NEUTRAL CONDUCTOR AS NECESSARY.
H. DEMOLISH UNDERGROUND FEEDER INDICATED WITH SUBSCRIPT (D) AND HEAVY WEIGHT DASHED LINES.
I. MAINTAIN AND PRESERVE UNDERGROUND FEEDERS INDICATED WITH SUBSCRIPT (E). PROTECT FROM DAMAGE DURING CONSTRUCTION.

## SHEET KEYNOTES

E605 REMOVE THE TWO EXISTING 400A CIRCUIT BREAKERS AND PROVIDE ONE NEW TMAX 800A FRAME XT7.

## FEEDER SCHEDULE

| | |
|---|---|
| 350S | 4#500KCMIL + 1#3 G, 4" C |
| 800S | 3 SETS OF 4#500KCMIL + 1#2/0 G, (3) 4" C + (1) 4" C SPARE |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

GRAPHIC SCALE(S)

DATE 06/15/2026
SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE: 1 05/11/2026
REVISION DESCRIPTION: AMENDMENT 01

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: ELECTRICAL DIAGRAMS
WORKING STAFF

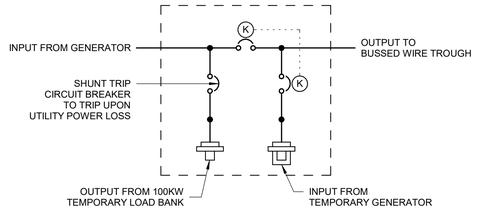
| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| NS | NS | KM |

SHEET NO. 350 OF 378

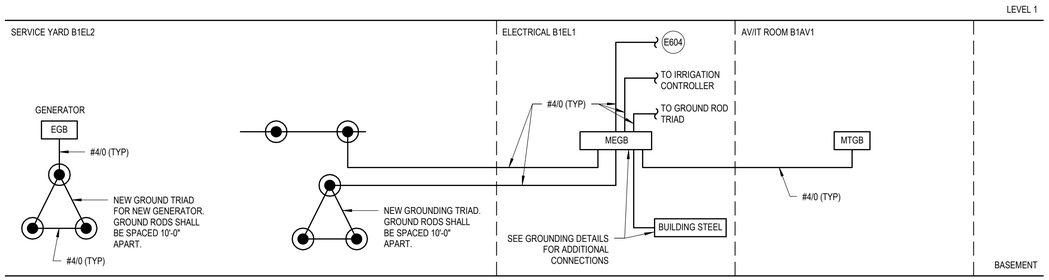
E-600SH
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

FOR REFERENCE ONLY

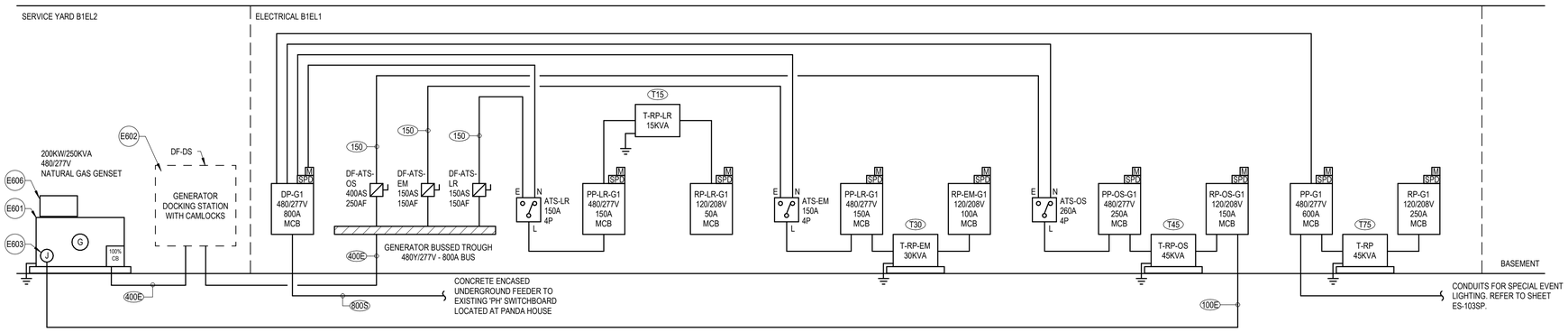
6/15/2026 2:49:21 PM
E-600SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




3 DUAL PURPOSE DOCKING STATION FOR LOAD BANK & TEMPORARY GENERATOR DETAIL
SCALE: NOT TO SCALE




2 ELECTRICAL GROUNDING RISER DIAGRAM
SCALE: NOT TO SCALE




1 ELECTRICAL ONE LINE/RISER DIAGRAM
SCALE: NOT TO SCALE

## AUTOMATIC TRANSFER SWITCH SCHEDULE

| ID | ATS FRAME/BUS AMP RATING | VOLTS | POLES (SEE NOTE 2) | TRANSITION TYPE | BYPASS/ ISOLATION SWITCH | WCR (kA) | WCR CYCLE RATING | GENERATOR START DELAY (SEC.) | UTILITY TO GENERATOR TRANSFER DELAY (SEC.) | UTILITY TO GENERATOR TRANSFER PRIORITY | GENERATOR TO UTILITY TRANSFER DELAY (SEC.) | GENERATOR TO UTILITY TRANSFER PRIORITY | GENERATOR COOL-DOWN PERIOD (MIN.) | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ATS-EM | 150 | 480 | 4 | OPEN | YES | 65 | 3 | 2 | 0 | 1 | 600 | 5 | 30 | 1, 2, 3, 4 |
| ATS-LR | 150 | 480 | 4 | OPEN | YES | 65 | 3 | 2 | 30 | 2 | 480 | 4 | 30 | 1, 2, 3, 4 |
| ATS-OS | 260 | 480 | 4 | DELAYED | YES | 65 | 3 | 2 | 120 | 4 | 360 | 2 | 30 | 1, 2, 3, 4 |

NOTES
1. FRONT ACCESS WITH SIDE ACCESS CLEARANCE
2. ALL TRANSFER SWITCHES SHALL BE CONSTRUCTED WITH SWITCHED NEUTRAL
3. PROVIDE INDICATED FRAME/BUS RATING TO ENSURE A MINIMUM 65KA FOR TIME-BASED 3-CYCLE ("ANY BREAKER") WCR.
4. PICKUP VOLTAGE: FACTORY SETTING AT 90 PERCENT; DROPOUT VOLTAGE: FACTORY SETTING AT 85 PERCENT.
5. PROVIDE SIGNAL-BEFORE-TRANSFER CONTACTS: A SET OF NORMALLY OPEN DRY CONTACTS TO OPERATE IN ADVANCE OF RETRANSFER TO NORMAL SOURCE. TIME DELAY SET TO 30 SECONDS.
6. REFER TO MECHANICAL DRAWINGS FOR MECHANICAL CONTROLS DETAILS.

## GENERAL SHEET NOTES

A. SEE DRAWING SERIES E0 SERIES FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES.
B. THE CONTRACTOR SHALL COORDINATE EXACT LOCATION OF EQUIPMENT POWER CONNECTIONS WITH EQUIPMENT INSTALLER PRIOR TO ROUGH-IN.
C. COORDINATE FINAL LOCATION OF MECHANICAL EQUIPMENT AND ASSOCIATED DISCONNECT SWITCHES, STARTERS, VFDS, CONTROL POWER AND OTHER POWER REQUIREMENTS WITH DIV 23 AND MECHANICAL DRAWINGS.
D. SEE MECHANICAL EQUIPMENT SCHEDULE FOR ASSOCIATED OVERCURRENT DEVICES, DISCONNECT SWITCHES, STARTERS AND WIRES.
E. FOR ALL MECHANICAL EQUIPMENT, PROVIDE ADDITIONAL 120V CONNECTIONS REQUIRED BY MANUFACTURER.
F. VOLTAGE DROP SHALL BE LIMITED TO LESS THAN 3% FOR ALL BRANCH CIRCUITS AND 2% FOR ALL FEEDERS.
G. CONTRACTOR SHALL COORDINATE AND FIELD VERIFY REQUIREMENT OF NEUTRAL CONDUCTOR FOR ALL EQUIPMENT AND PROVIDE NEUTRAL CONDUCTOR AS NECESSARY.
H. DEMOLISH UNDERGROUND FEEDER INDICATED WITH SUBSCRIPT (D) AND HEAVY WEIGHT DASHED LINES.
I. MAINTAIN AND PRESERVE UNDERGROUND FEEDERS INDICATED WITH SUBSCRIPT (E). PROTECT FROM DAMAGE DURING CONSTRUCTION.

## SHEET KEYNOTES

E601 PROVIDE A JUNCTION BOX AND A 1" CONDUIT WITH PULL STRING FOR GENERATOR START/STOP SIGNAL WIRING TO EACH AUTOMATIC TRANSFER SWITCH AS REQUIRED FOR REMOTE ANNUNCIATOR.
E602 DOCKING STATION WITH SHUNT TRIP TYPE CIRCUIT BREAKER WITH ENCLOSED CIRCUIT BREAKER. AMPACITY OF CIRCUIT BREAKER ON DOCKING STATION, SHOULD MATCH THE CIRCUIT BREAKER SIZE ON THE GENERATOR. PROVIDE SIGNAL WIRE FROM EMERGENCY TRANSFER SWITCH. DOCKING STATION TO DROP LOAD IF POWER LOSS IS DETECTED.
E603 GENERATOR LOAD CENTER.
E604 PROVIDE BONDING JUMPER #4/0 AWG AROUND WATER/GAS METER. REVIEW BONDING AT GAS MAIN WITH DC DEPARTMENT OF BUILDINGS. PROVIDE BONDING TO MEET BOTH WATER AND GAS UTILITY REQUIREMENTS.
E606 NEW GAS GENSET TO BE PROVIDED WITH TOP MOUNTED 100KW INTEGRATED PERMANENT LOAD BANK.

## FEEDER SCHEDULE

100E 4#1 + 1#8 G, 1 1/2" C
150 4#1/0 + 1#6 G, 2" C
400E 2 SETS OF 4#3/0 + 1#3 G, (2) 4" C + (1) 4" C SPARE. (3) 1" C FOR CONTROL BETWEEN GENERATOR AND TRANSFER SWITCHES. GENERATOR FEEDER SHALL BE CONCRETE ENCASED.
800S 3 SETS OF 4#500KCMIL + 1#2/0 G, (3) 4" C + (1) 4" C SPARE
T15 15KVA TRANSFORMER, 480-208Y/120V, 3PH PRIMARY FEEDER: 3#10 + 1#10 G, 3/4" C SECONDARY FEEDER: 4#4 + 1#8 G, 1 1/4" C BONDING JUMPERS & GROUNDING ELECTRODE CONDUCTOR: #8
T30 30KVA TRANSFORMER, 480-208Y/120V, 3PH PRIMARY FEEDER: 3#6 + 1#10 G, 3/4" C SECONDARY FEEDER: 4#1 + 1#6 G, 1 1/2" C BONDING JUMPERS & GROUNDING ELECTRODE CONDUCTOR: #6
T45 45KVA TRANSFORMER, 480-208Y/120V, 3PH PRIMARY FEEDER: 3#4 + 1#8 G, 1" C SECONDARY FEEDER: 4#2/0 + 1#4 G, 2" C BONDING JUMPERS & GROUNDING ELECTRODE CONDUCTOR: #4
T75 75KVA TRANSFORMER, 480-208Y/120V, 3PH PRIMARY FEEDER: 3#1 + 1#6 G, 1 1/2" C SECONDARY FEEDER: 4#350KCMIL + 1#2 G, 3" C BONDING JUMPERS & GROUNDING ELECTRODE CONDUCTOR: #2

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

GRAPHIC SCALE(S)

DATE: 06/15/2026
SUBMISSION: 65% CONSTRUCTION DOCUMENTS

| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|


Smithsonian Institution
Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520


BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: ELECTRICAL DIAGRAMS
WORKING STAFF

| NS | NS | KM |
|---|---|---|
| DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 351 OF 378

E-602SH
DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

FOR REFERENCE ONLY

6/15/2026 2:49:21 PM
E-602SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# MECHANICAL EQUIPMENT SCHEDULE

| TAG | PANEL | CIRCUIT NUMBER | INPUT POWER | MOTOR HP1 | MOTOR HP2 | MOTOR HP3 | MOTOR HP4 | VOLTAGE | PHASE | TOTAL POWER LOAD | MOTOR RATED SWITCH | COMBINATION STARTER NEMA SIZE | ECM | VFD | DISC. SWITCH AMPS | DISC. SWITCH POLES | DISC. SWITCH FUSED | PHASE & NEUTRAL SETS | PHASE & NEUTRAL NO. OF WIRES | PHASE & NEUTRAL AWG | GROUND NO. OF WIRES | GROUND AWG | CONDUIT SETS | CONDUIT SIZE | COMMENTS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AC-1 | RP-OS-G1 | 37 | 0 VA | 1/6 | | | | 120 V | 1 | 528 VA | No | | No | No | 30 A | 2 | 10 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| AHU-1 LOAD CENTER (LTG, MISC.) | PP-OS-G1 | 43 | 2000 VA | | | | | 277 V | 1 | 2000 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| AHU-1 LOAD CENTER (RCPT) | RP-OS-G1 | 1 | 1800 VA | | | | | 120 V | 1 | 1800 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| ASHP-1 HEAT TRACE | RP-OS-G1 | 35 | 0 VA | | | | | 120 V | 1 | 1632 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #10 | 1 | 3/4" | |
| ASHP-1A | PP-OS-G1 | 20,22,24 | 0 VA | | | | | 480 V | 3 | 55062 VA | No | | No | No | 100 A | 3 | 100 A | 1 | 3 | #3 | 1 | #8 | 1 | 1 1/4" | |
| ASHP-1B | PP-G1 | 7,9,11 | 0 VA | | | | | 480 V | 3 | 55062 VA | No | | No | No | 100 A | 3 | 100 A | 1 | 3 | #3 | 1 | #8 | 1 | 1 1/4" | |
| ASHP-1C | PP-G1 | 13,15,17 | 0 VA | | | | | 480 V | 3 | 55062 VA | No | | No | No | 100 A | 3 | 100 A | 1 | 3 | #3 | 1 | #8 | 1 | 1 1/4" | |
| ASHP-1D | PP-G1 | 19,21,23 | 0 VA | | | | | 480 V | 3 | 55062 VA | No | | No | No | 100 A | 3 | 100 A | 1 | 3 | #3 | 1 | #8 | 1 | 1 1/4" | |
| B-1 | RP-OS-G1 | 15 | 0 VA | | | | | 120 V | 1 | 960 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| B-2 | RP-OS-G1 | 17 | 0 VA | | | | | 120 V | 1 | 960 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| BAS-AHU-1 | RP-OS-G1 | 5 | 0 VA | | | | | 120 V | 1 | 1080 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| BAS-CHW | RP-OS-G1 | 11 | 0 VA | | | | | 120 V | 1 | 1080 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| BAS-HHW | RP-OS-G1 | 7 | 0 VA | | | | | 120 V | 1 | 1080 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| BAS-INTEGRATION | RP-OS-G1 | 3 | 0 VA | | | | | 120 V | 1 | 1080 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| BAS-MISC | RP-OS-G1 | 9 | 0 VA | | | | | 120 V | 1 | 1080 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| BFP-1 | RP-G1 | 5 | 0 VA | | | | | 120 V | 1 | 0 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| BFP-2 | RP-G1 | 7 | 0 VA | | | | | 120 V | 1 | 0 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| BFP-3 | RP-G1 | 9 | 0 VA | | | | | 120 V | 1 | 0 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| CHWP-1 | PP-OS-G1 | 26,28,30 | 0 VA | 20 | | | | 480 V | 3 | 22437 VA | No | | No | No | 60 A | 3 | 50 A | 1 | 3 | #8 | 1 | #10 | 1 | 3/4" | |
| CHWP-2 | PP-OS-G1 | 32,34,36 | 0 VA | 20 | | | | 480 V | 3 | 22437 VA | No | | No | No | 60 A | 3 | 50 A | 1 | 3 | #8 | 1 | #10 | 1 | 3/4" | |
| CHWP-3 | PP-OS-G1 | 38,40,42 | 0 VA | 5 | | | | 480 V | 3 | 6316 VA | No | | No | No | 30 A | 3 | 15 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| CHWP-4 | PP-OS-G1 | 44,46,48 | 0 VA | 5 | | | | 480 V | 3 | 6316 VA | No | | No | No | 30 A | 3 | 15 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| CP-1 | RP-OS-G1 | 28 | 0 VA | | | | | 120 V | 1 | 0 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| DWH-1 | RP-OS-G1 | 30 | 600 VA | | | | | 120 V | 1 | 600 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| EEF-1 | RP-G1 | 15 | 0 VA | 1/4 | | | | 120 V | 1 | 720 VA | No | | No | No | 30 A | 2 | 15 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| EF-1 | PP-OS-G1 | 14,16,18 | 0 VA | 10 | | | | 480 V | 3 | 11634 VA | No | | No | No | 30 A | 3 | 25 A | 1 | 3 | #12 | 1 | #10 | 1 | 3/4" | |
| ERPP-1 | PP-OS-G1 | 2,4,6 | 0 VA | 5 | | | | 480 V | 3 | 6316 VA | No | | No | No | 30 A | 3 | 15 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| ERPP-2 | PP-OS-G1 | 8,10,12 | 0 VA | 5 | | | | 480 V | 3 | 6316 VA | No | | No | No | 30 A | 3 | 15 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| ESF-1 | RP-G1 | 17 | 0 VA | 1/4 | | | | 120 V | 1 | 720 VA | No | | No | No | 30 A | 2 | 15 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| ETP-1 | RP-G1 | 1 | 6 VA | | | | | 120 V | 1 | 6 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| ETP-2 | RP-G1 | 3 | 6 VA | | | | | 120 V | 1 | 6 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| ETP-3 | RP-G1 | 63 | 6 VA | | | | | 120 V | 1 | 6 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| ETP-4 | RP-G1 | 61 | 6 VA | | | | | 120 V | 1 | 6 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| FCU-1 | RP-OS-G1 | 32,34 | 0 VA | 1/3 | 1/3 | | | 208 V | 1 | 1248 VA | No | | No | No | 30 A | 2 | 15 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| FCU-2 | RP-G1 | 23,25 | 0 VA | 3/4 | 3/4 | | | 208 V | 1 | 2954 VA | No | | No | No | 30 A | 2 | 25 A | 1 | 2 | #12 | 1 | #10 | 1 | 3/4" | |
| FCU-3 | RP-OS-G1 | 36,38 | 0 VA | 1 | 1 | | | 208 V | 1 | 3910 VA | No | | No | No | 30 A | 2 | 25 A | 1 | 2 | #10 | 1 | #10 | 1 | 3/4" | |
| FCU-4 | RP-OS-G1 | 40,42 | 0 VA | 3/4 | 3/4 | | | 208 V | 1 | 2954 VA | No | | No | No | 30 A | 2 | 25 A | 1 | 2 | #12 | 1 | #10 | 1 | 3/4" | |
| FPP-1 | RP-OS-G1 | 13 | 0 VA | 1/2 | | | | 120 V | 1 | 1176 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| GFS-1 | RP-G1 | 19 | 0 VA | 3/4 | | | | 120 V | 1 | 1656 VA | No | | No | No | 30 A | 2 | 25 A | 1 | 2 | #12 | 1 | #10 | 1 | 3/4" | |
| GFS-2 | RP-G1 | 21 | 0 VA | 3/4 | | | | 115 V | 1 | 1656 VA | No | | No | No | 30 A | 2 | 25 A | 1 | 3 | #12 | 1 | #10 | 1 | 3/4" | |
| HHWP-1 | PP-OS-G1 | 13,15,17 | 0 VA | 20 | | | | 480 V | 3 | 22437 VA | No | | No | No | 60 A | 3 | 50 A | 1 | 3 | #8 | 1 | #10 | 1 | 3/4" | |
| HHWP-2 | PP-OS-G1 | 19,21,23 | 0 VA | 20 | | | | 480 V | 3 | 22437 VA | No | | No | No | 60 A | 3 | 50 A | 1 | 3 | #8 | 1 | #10 | 1 | 3/4" | |
| HHWP-3 | PP-OS-G1 | 25,27,29 | 0 VA | 5 | | | | 480 V | 3 | 6316 VA | No | | No | No | 30 A | 3 | 15 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| HHWP-4 | PP-OS-G1 | 31,33,35 | 0 VA | 5 | | | | 480 V | 3 | 6316 VA | No | | No | No | 30 A | 3 | 15 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| HTC-1 | RP-OS-G1 | 19 | 5000 VA | | | | | 120 V | 1 | 5000 VA | No | | No | No | | | | 1 | 2 | #6 | 1 | #10 | 1 | 3/4" | |
| HTR-1 | RP-OS-G1 | 2 | 100 VA | | | | | 120 V | 1 | 100 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| HTR-2 | RP-OS-G1 | 4 | 100 VA | | | | | 120 V | 1 | 100 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| HTR-3 | RP-OS-G1 | 6 | 100 VA | | | | | 120 V | 1 | 100 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| HTR-4 | RP-OS-G1 | 8 | 100 VA | | | | | 120 V | 1 | 100 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| HUM-1 | PP-G1 | 1,3,5 | 65000 VA | | | | | 480 V | 3 | 65000 VA | No | | No | No | 100 A | 3 | 100 A | 1 | 3 | #1 | 1 | #8 | 1 | 1 1/4" | |
| HYDRAULIC/ELECTRIC DOOR #1 | RP-G1 | 50,52 | 0 VA | 1 | | | | 208 V | 1 | 1830 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| IRRIGATION CONTROLLER | RP-G1 | 46 | 1800 VA | | | | | 120 V | 1 | 1800 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OEF-1 | RP-G1 | 48 | 60 VA | | | | | 120 V | 1 | 60 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OGF-1 | RP-G1 | 42 | 240 VA | | | | | 120 V | 1 | 240 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OGF-2 | RP-G1 | 42 | 240 VA | | | | | 120 V | 1 | 240 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OGF-3 | RP-G1 | 42 | 240 VA | | | | | 120 V | 1 | 240 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-1 | RP-G1 | 2,4 | 3150 VA | | | | | 208 V | 1 | 3150 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-2 | RP-G1 | 6,8 | 3150 VA | | | | | 208 V | 1 | 3150 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-3 | RP-G1 | 10,12 | 3150 VA | | | | | 208 V | 1 | 3150 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-4 | RP-G1 | 14,16 | 3150 VA | | | | | 208 V | 1 | 3150 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-5 | RP-G1 | 18,20 | 3150 VA | | | | | 208 V | 1 | 3150 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-6 | RP-G1 | 22,24 | 1450 VA | | | | | 208 V | 1 | 1450 VA | No | | No | No | 30 A | 2 | 10 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-7 | RP-G1 | 26,28 | 3150 VA | | | | | 208 V | 1 | 3150 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-8 | RP-G1 | 30,32 | 3150 VA | | | | | 208 V | 1 | 3150 VA | No | | No | No | 30 A | 2 | 20 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OH-9 | RP-G1 | 34,36 | 1450 VA | | | | | 208 V | 1 | 1450 VA | No | | No | No | 30 A | 2 | 10 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OHF-1 | RP-G1 | 40 | 120 VA | | | | | 120 V | 1 | 120 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OHF-2 | RP-G1 | 40 | 120 VA | | | | | 120 V | 1 | 120 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| OHF-3 | RP-G1 | 40 | 120 VA | | | | | 120 V | 1 | 120 VA | No | | No | No | 30 A | 2 | 3 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| RSS-1 | RP-G1 | 33 | 400 VA | | | | | 120 V | 1 | 400 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| RSS-2 | RP-G1 | 33 | 400 VA | | | | | 120 V | 1 | 400 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| RSS-3 | RP-G1 | 33 | 400 VA | | | | | 120 V | 1 | 400 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| RSS-4 | RP-G1 | 35 | 400 VA | | | | | 120 V | 1 | 400 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| RSS-5 | RP-G1 | 35 | 400 VA | | | | | 120 V | 1 | 400 VA | No | | No | No | | | | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| SEF-1 | PP-LR-G1 | 7,9,11 | 0 VA | 5 | | | | 480 V | 3 | 6316 VA | No | | No | No | 30 A | 3 | 15 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| SEP-1 | PP-LR-G1 | 1,3,5 | 0 VA | 1 | 1 | | | 480 V | 3 | 3490 VA | No | | No | No | 30 A | 3 | 6 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| SEP-2 | PP-LR-G1 | 2,4,6 | 0 VA | 1 | 1 | | | 480 V | 3 | 3490 VA | No | | No | No | 30 A | 3 | 6 A | 1 | 3 | #12 | 1 | #12 | 1 | 3/4" | |
| SF-1A | PP-OS-G1 | 1,3,5 | 0 VA | 20 | | | | 480 V | 3 | 22437 VA | No | | No | No | 60 A | 3 | 50 A | 1 | 3 | #8 | 1 | #10 | 1 | 3/4" | |
| SF-1B | PP-OS-G1 | 7,9,11 | 0 VA | 20 | | | | 480 V | 3 | 22437 VA | No | | No | No | 60 A | 3 | 50 A | 1 | 3 | #8 | 1 | #10 | 1 | 3/4" | |
| UH-1 | RP-OS-G1 | 21 | 0 VA | 1/30 | | | | 120 V | 1 | 228 VA | No | | No | No | 30 A | 2 | 6 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| UH-2 | RP-OS-G1 | 23 | 0 VA | 1/30 | | | | 120 V | 1 | 228 VA | No | | No | No | 30 A | 2 | 6 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| UH-3 | RP-OS-G1 | 25 | 0 VA | 1/30 | | | | 120 V | 1 | 228 VA | No | | No | No | 30 A | 2 | 6 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| UH-4 | RP-OS-G1 | 27 | 0 VA | 1/30 | | | | 120 V | 1 | 228 VA | No | | No | No | 30 A | 2 | 6 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| UH-5 | RP-OS-G1 | 29 | 0 VA | 1/30 | | | | 120 V | 1 | 228 VA | No | | No | No | 30 A | 2 | 6 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| UH-6 | RP-OS-G1 | 31 | 0 VA | 1/30 | | | | 120 V | 1 | 228 VA | No | | No | No | 30 A | 2 | 6 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |
| UH-7 | RP-OS-G1 | 33 | 0 VA | 1/30 | | | | 120 V | 1 | 228 VA | No | | No | No | 30 A | 2 | 6 A | 1 | 2 | #12 | 1 | #12 | 1 | 3/4" | |

## GENERAL SHEET NOTES

A. COORDINATE ALL FINAL LOCATIONS, LOADS, AND CONNECTION REQUIREMENTS WITH MECHANICAL DRAWINGS AND FINAL EQUIPMENT SELECTION.
B. CONTRACTOR TO VERIFY THAT LUGS ON ALL MECHANICAL AND ELECTRICAL EQUIPMENT CAN ACCEPT THE WIRE SIZES INDICATED ON SCHEDULE AND PLANS.
C. CONTRACTOR TO VERIFY WIRE SIZES WITH FINAL EQUIPMENT LOCATIONS TO ACCOUNT FOR VOLTAGE DROP ON ALL MECHANICAL EQUIPMENT.
D. FOR ALL MECHANICAL EQUIPMENT, PROVIDE ADDITIONAL 120V CONNECTIONS AS REQUIRED BY MANUFACTURER.
E. COORDINATE ALL MECHANICAL DISCONNECT, VFD, STARTER AND CONTROLLER REQUIREMENTS WITH MANUFACTURER.

## SHEET KEYNOTES

FOR REFERENCE ONLY

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

NOT FOR CONSTRUCTION

STAMP

KEY PLAN

GRAPHIC SCALE(S)

DATE 06/15/2026
SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE
REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: EQUIPMENT SCHEDULES
WORKING STAFF: NS (DESIGNED BY) | NS (DRAWN BY) | KM (CHECKED BY)

SHEET NO. 352 OF 378
E-603SH (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

6/15/2026 2:49:21 PM
E-603SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# LIGHTING FIXTURE SCHEDULE

| TYPE | DESCRIPTION | MANUFACTURER | MODEL | LAMP | DRIVER | VOLTAGE | MOUNTING | COMMENTS |
|---|---|---|---|---|---|---|---|---|
| SL1 | RECESSED 2 FT WIDTH BY 2 FT LENGTH TROFFER; EXTRUDED ALUMINUM SIDES AND COLD ROLLED STEEL END CAPS; MICROPRISMATIC CONICAL DEGLARING LENS; LOW OUTPUT LAMBERTION DISTRIBUTION | MARK LIGHTING | WHSPR TROFFER | LED; 16 W; 2000 LM; 3500K CCT; 90+ CRI | 0-10V 10% DIMMING | 277 V | RECESSED | |
| SL2 | RECESSED DOWNLIGHT; 4" APERTURE; ALUMINUM HOUSING; FLANGED TRIM; FROSTED LENS; 50° BEAM; MATTE WHITE; IP20 | LUMENWERX | AERA 4 DOWNLIGHT | LED; 20 W; 2000 LM; 3500K CCT; 90+ CRI | 0-10V 1% DIMMING | 277 V | RECESSED | |
| SL4 | LINEAR PENDANT; 4 FT LENGTH; 4" APERTURE; EXTRUDED ALUMINUM; TEMPERED CLEAR GLASS LENS; HIGH EFFICIENCY LAMBERTION DISTRIBUTION; INTEGRAL DRIVER; 36" HANGER CHAIN TO BE INCLUDED FOR MOUNTED | LITHONIA LIGHTING | ZL1D | LED; 30 W; 750 LM/FT; 3500K CCT; 80+ CRI | 0-10V 10% DIMMING | 277 V | PENDANT | |
| SL5A | LINEAR PENDANT; 4 FT LENGTH; 4" APERTURE; EXTRUDED ALUMINUM; STEM MOUNT; REGRESSED DIFFUSE LENS; WIDE DISTRIBUTION; INTEGRAL DRIVER; MARINE FINISH IP66 RATED; IK08 RATED | LUMENWERX | V4SEALP | LED; 44 W; 1000 LM/FT (D); 3500K CCT; 90+ CRI | 0-10V 1% DIMMING | 277 V | PENDANT | |
| SL5B | LINEAR PENDANT; 4 FT LENGTH; 4" APERTURE; EXTRUDED ALUMINUM; SLOPED CEILING STEM MOUNT; REGRESSED DIFFUSE LENS; WIDE DISTRIBUTION; INTEGRAL DRIVER; MARINE FINISH IP66 RATED; IK08 RATED | LUMENWERX | V4SEALP | LED; 44 W; 1000 LM/FT (D); 3500K TO 2200K CCT DIM TO WARM; 90+ CRI | 0-10V 1% DIMMING | 277 V | PENDANT | |
| SL6 | LINEAR SURFACE MOUNT; 4 FT LENGTH; 4.5" APERTURE; EXTRUDED ALUMINUM; TEMPERED CLEAR GLASS LENS; HIGH EFFICIENCY LAMBERTION DISTRIBUTION; INTEGRAL DRIVER; WET RATED | LUMENWERX | VIAWETS | LED; 28 W; 500 LM/LF; 3000K CCT; 90+ CRI | 0-10V 1% DIMMING | 277 V | SURFACE | |
| SL7 | LINEAR RECESSED PERIMETER FIXTURE; 4" APERTURE; EXTRUDED ALUMINUM; REGRESSED DIFFUSE LENS; WIDE DISTRIBUTION; INTEGRAL DRIVER; IP66 RATED | LUMENWERX | V3SEALPERL | LED; 8 W/LF; 500 LM/LF; 3500K CCT; 90+ CRI | 0-10V 1% DIMMING | 277 V | RECESSED | |
| SL8 | LINEAR UNDERCABINET FIXTURE; 1.2" APERTURE; WIDE DISTRIBUTION; IP20 RATED | PRECISELED | LUMAMATE | LED; 5 W/LF; 500 LM/LF; 3500K CCT; 95+ CRI | 0-10V 1% DIMMING | 120 V | UNDERCABINET | |
| XL1 | RECESSED IN HANDRAIL FLUSH MOUNTING; ASYMMETRIC DISTRIBUTION; IP68 RATED; IK09 RATED | WAGNER | LUMENPOD 28 | LED; 3 W; 250 LM; 3000K CCT; 80+ CRI | 0-10V DIMMING | 120 V | HANDRAIL | |
| XL2 | WALL MOUNTED; 16" WIDTH BY 4" DEPTH BY 5" HEIGHT; TYPE I DISTRIBUTION; ALUMINUM ALLOY; MARINE GRADE FINISH; IP64 RATED | BEGA | 24 373 | LED; 21 W; 1800 LM; 3000K CCT; 80+ CRI | 0-10V DIMMING | 277 V | WALL | |
| XL3 | WALL MOUNTED; 3" DIAMETER BY 2" DEPTH; DARK BRONZE LIVING PATINA FINISH; MARINE GRADE FINISH; IP67 RATED, DARK SKY RATED | AURORA LIGHTING | MICRO CASCADE STW636203 | LED; 3 W; 100 LM; 3000K CCT; 80+ CRI | 0-10V 10% DIMMING | 277 V | WALL | |
| XL4A | RIGID TRACK MOUNTED CYLINDER DOWNLIGHT; 2.5" HEIGHT; 1.5" DIAMETER APERTURE; POWDER COATED STEEL HOUSING; FROSTED LENS; HONEYCOMB LOUVER; 35° BEAM; MATTE BLACK; IP65 | TEGAN LIGHTING | KORE-EX5 LED DOWNLIGHT | LED; 7 W; 500 LM; 3000K CCT; 80+ CRI | 0-10V DIMMING | 277 V | TRACK | |
| XL4B | RIGID TRACK MOUNTED ADJUSTABLE CYLINDER DOWNLIGHT; 2.5" HEIGHT; 1.5" DIAMETER APERTURE; POWDER COATED STEEL HOUSING; FROSTED LENS; HONEYCOMB LOUVER; 35° BEAM; MATTE BLACK; IP65 | TEGAN LIGHTING | KORE-EX5 LED ADJUSTABLE DOWNLIGHT | LED; 7 W; 500 LM; 3000K CCT; 80+ CRI | 0-10V DIMMING | 277 V | TRACK | |
| XL5 | WALL MOUNTED; 16" WIDTH BY 4" DEPTH BY 5" HEIGHT; TYPE IV DISTRIBUTION; ALUMINUM ALLOY; MARINE GRADE FINISH; IP65 RATED | BEGA | 84 098 | LED; 24 W; 2700 LM; 3000K CCT; 80+ CRI | 0-10V DIMMING | 277 V | WALL | INCLUDE CUSTOM MOD MOUNTING ARM/BACKPLATE; REFERENCE SHR DRAWINGS FOR DETAIL; CUSTOM RAL BRONZE FINISH |
| XL6 | CEILING MOUNTED CYLINDER DOWNLIGHT; 4" WIDTH BY 5" HEIGHT; ALUMINUM ALLOY; MARINE GRADE FINISH; IP65 RATED | BEGA | 24 404 | LED; 8 W; 900 3000K CCT; 80+ CRI | 0-10V 1% DIMMING | 277 V | SURFACE | |
| XL7 | LINEAR TAPELIGHT FIXTURE; 0.5" APERTURE; WIDE DISTRIBUTION; IP65 RATED | PRECISELED | BEAM WET LED TAPELIGHT | LED; 5 W/LF; 180 LM/LF; 3000K CCT; 95+ CRI | 0-10V 1% DIMMING | 277 V | UNDERCABINET | |

# LIGHTING CONTROL SCHEDULE

| SPACE TYPE | BAS / NETWORK OR STANDALONE | TIMECLOCK OVERRIDE(2) | | NON-TIMECLOCK OVERRIDE(3)(4) | | | | | RECEPTACLE CONTROL | TARGET AVERAGE ILLUMINANCE(1) (FOOTCANDLES) |
|---|---|---|---|---|---|---|---|---|---|---|
| | | SCHEDULED ON OUTPUT | SCHEDULED SHUTOFF(5) | MANUAL ON OR AUTO ON | OCCUPIED OUTPUT | MANUAL OFF OR AUTO OFF | UNOCCUPIED OUTPUT | TIME DELAY (MINUTES) | | |
| VIP/KEEPER WORKPLACE | STANDALONE | 50% | - | MO | 100% | AO | 0% | 20 | YES | 30 @ 2'-6"AFF |
| RESTROOM | STANDALONE | 100% | - | AO | 100% | AO | 0% | 20 | NO | 20 @ 3'-0"AFF |
| CUSTODIAL ROOM | STANDALONE | 100% | - | AO | 100% | AO | 0% | 20 | NO | 10 @ 0'-0"AFF |
| KEEPER CORRIDOR, CORRIDOR | NETWORKED | 50% | 8:00PM-6:00AM | AO | 100% | AO | 20% | 20 | NO | 11 @ 0'-0"AFF |
| KITCHEN PREP | STANDALONE | 50% | - | MO | 100% | AO | 0% | 20 | NO | 50 @ 2'-6"AFF |
| IT / AV | STANDALONE | - | - | MO | 100% | MO | 0% | - | NO | 10 @ 0'-0"AFF |
| ELECTRICAL / MECHANICAL | STANDALONE | - | - | MO | 100% | MO | 0% | - | NO | 30 @ 2'-6"AFF |
| CUBBING, QUARANTINE | STANDALONE | 50% | - | MO | 100% | AO | 0% | 20 | NO | 50 @ 0'-0"AFF |
| DEN'S, DAYROOM | STANDALONE | 50% | - | MO | 100% | AO | 0% | 20 | NO | 40 @ 0'-0"AFF |
| SERVICE DOCK | STANDALONE | 50% | - | MO | 100% | AO | 0% | 20 | NO | 10 @ 0'-0"AFF |
| EXTERIOR FAÇADE (TIE TO PHOTOCELL OR TIMECLOCK) | NETWORKED | 100% | 12:00AM-6:00AM | AO | 100% | AO | 0% | - | NO | 5 @ 0'-0"AFF |
| EXTERIOR (TIE TO PHOTOCELL OR TIMECLOCK) | NETWORKED | 25% | 12:00AM-6:00AM | AO | 100% | AO | 0% | - | NO | 5 @ 0'-0"AFF |

GENERAL NOTES:

A. CONTRACTOR TO REVIEW AND CONFIRM ALL CONTROL PROGRAMMING REQUIREMENTS WITH OWNER PRIOR TO PROGRAMMING.
B. CONTROL SCHEDULE IS TO SHOW PROGRAMMING DESIGN INTENT.
C. COORDINATE WITH E0 COVERSHEET AND E3 SERIES FOR MANUAL KEYPAD OVERRIDE.
D. AUTOMATIC ON SIGNAL FROM OCCUPANCY/VACANCY SENSOR TO SIGNAL SYSTEM TO ADJUST FIXTURE TO OUTPUT AS INDICATED IN "OCCUPIED OUTPUT" COLUMN.
E. OCCUPANCY/VACANCY SENSORS TO DETECT VACANCY AND SIGNAL SYSTEM TO ADJUST FIXTURES TO OUTPUT AS INDICATED IN "UNOCCUPIED OUTPUT" COLUMN.
F. OCCUPANCY SENSORS TO PROVIDE TIMECLOCK OVERRIDE DURING "NON-BUSINESS HOURS".
G. BMS TO PROVIDE SIGNAL TO ENERGIZE LIGHTING IN SPACES WITH TIME SCHEDULE INDICATED. PROVIDE STANDALONE LIGHTING AND RECEPTACLE CONTROLS TO SPACES WHERE TIMECLOCK SCHEDULE IS NOT PROVIDED.
H. FOR ANY CONTROLLED RECEPTACLES IN STANDALONE SPACES, OCCUPANCY SENSOR TO ENERGIZE RECEPTACLE DURING OCCUPANCY AND DE-ENERGIZE RECEPTACLE DURING VACANCY. FOR ANY NETWORKED SPACES, RECEPTACLE CONTROL...
I. "BUSINESS HOURS" ARE MONDAY THROUGH FRIDAY, 6:00AM-8:00PM, EXCLUDING HOLIDAYS. NON-BUSINESS HOURS ARE ALL HOURS NOT DURING BUSINESS HOURS.
J. FUNCTIONAL TESTING: LIGHTING CONTROL DEVICES AND CONTROL SYSTEMS SHALL BE TESTED TO ENSURE THAT CONTROL HARDWARE AND SOFTWARE ARE CALIBRATED, ADJUSTED, PROGRAMMED, AND IN PROPER WORKING CONDITION IN ACCORDANCE WITH THE CONSTRUCTION DOCUMENTS AND MANUFACTURER'S INSTALLATION INSTRUCTIONS.

KEYED NOTES:

(1) TARGET AVERAGE ILLUMINANCE COLUMN IS ONLY TO BE USED TO CALIBRATE PHOTOCELL ILLUMINANCE LEVELS WHERE APPLICABLE AS INDICATED ON E3 SERIES. PHOTOCELL SHALL PROVIDE NO LESS THAN ILLUMINANCE LEVEL..
(2) TIMECLOCK OVERRIDES INDICATE OUTPUT AND TIME CLOCK SCHEDULE FOR NETWORKED SPACES ONLY. SCHEDULED ON OUTPUT INDICATES THE FIXTURE OUTPUT DURING "BUSINESS HOURS" REGARDLESS OF OCCUPANCY. SCHEDULED SHUTOFF INDICATES HOURS THE OCCUPANCY/VACANCY SENSOR WILL PROVIDE CONTROL. REFERENCE NON-TIMECLOCK OVERRIDE COLUMNS FOR OUTPUT DURING TIMES INDICATED.
(3) NON-TIMECLOCK OVERRIDES INDICATE CONTROL TYPE (AUTO/MANUAL) AND FIXTURE OUTPUT DURING "NON-BUSINESS HOURS"(SEE NOTE (4)). MANUAL ON OR AUTO ON COLUMN INDICATES CONTROL TYPE FOR THE FIXTURE IN THAT SPACE, WHETHER IT REQUIRES DIRECT OR INDIRECT USER INTERACTION TO ENERGIZE FIXTURES. OCCUPIED OUTPUT INDICATES THE FIXTURE OUTPUT UPON USER INTERACTION. MANUAL OFF OR AUTO OFF COLUMN INDICATES CONTROL TYPE FOR THE FIXTURE IN THAT SPACE, WHETHER IT REQUIRES DIRECT OR INDIRECT USER INTERACTION TO DEENERGIZE FIXTURES. UNOCCUPIED OUTPUT INDICATES THE FIXTURE OUTPUT UPON USER INTERACTION. TIME DELAY INDICATES THE NUMBER OF MINUTES REQUIRED FOR VACANCY TO BE DETECTED BEFORE FIXTURES WILL AUTOMATICALLY DEENERGIZE.
(4) FOR STANDALONE SPACES, NON-TIMECLOCK OVERRIDE IS APPLICABLE FOR ALL HOURS OF OPERATION, INCLUDING "BUSINESS HOURS" AND "NON-BUSINESS HOURS".
(5) "SCHEDULED SHUTOFF" IS DURING "NON-BUSINESS HOURS".

## GENERAL SHEET NOTES

A. CONTRACTOR TO COORDINATE WITH OTHER TRADES, CEILING TYPES, ARCHITECTURAL MOUNTING DETAILS, ELEVATIONS, AND FIELD CONDITIONS. CONTRACTOR TO PROVIDE ALL EQUIPMENT AND ACCESSORIES NECESSARY TO INSTALL LIGHTING FIXTURES IN SPECIFIED CEILING CONDITION.
B. CONTRACTOR TO COORDINATE FIXTURE LOCATIONS AND MOUNTING DETAILS WITH ARCHITECTURAL PLANS.
C. CONTRACTOR TO COORDINATE ALL FIXTURE AND DRIVER COMPATIBILITY AND CONFIRM COMPATIBILITY WITH INSTALLED OCCUPANCY AND DIMMING SYSTEMS.
D. ALL FIXTURE SUBSTITUTIONS SHALL HAVE A WORKING SAMPLE PROVIDED FOR REVIEW WITHIN 10 WORKING DAYS UPON REQUEST FROM THE DESIGN TEAM.
E. JUDGMENT OF FIXTURE QUALITY AND OPTICS, AS EQUAL, SHALL BE DETERMINED BY THE ARCHITECT OR LIGHTING DESIGNER.
F. ALL FIXTURES SHALL HAVE APPROPRIATE UL LABEL, DAMP, OR WET AS REQUIRED BY LOCAL CODES.
G. ALL FIXTURES SHALL INCLUDE ALL ACCESSORIES REQUIRED FOR INSTALLATION ACCORDING TO ALL LOCAL AND NATIONAL CODES.
H. PRIOR TO ORDERING ANY LIGHTING EQUIPMENT, THE CONTRACTOR SHALL VERIFY ALL LOCATIONS AND RECESS DEPTHS.
I. ALL LAMPS SHALL BE INSTALLED ACCORDING TO THE LIGHTING FIXTURE SCHEDULE.
J. CONTRACTOR SHALL VERIFY FINAL VOLTAGES AND CEILING TRIM COMPATIBILITY PRIOR TO ORDERING FIXTURES.
K. ALL FINISHES TO BE VERIFIED BY ARCHITECT / INTERIOR DESIGNER, PRIOR TO ORDERING.
L. ANY UNSPECIFIED OR TBD FINISHES ARE TO BE ASSUMED AS STANDARD AVAILABLE FINISHES FOR PRICING, UNLESS OTHERWISE NOTED.
M. ALL FIXTURES SHALL BE ORDERED WITH THE APPROPRIATE BALLASTS THAT HAVE UL AND CBM LABELS. ALL BALLASTS MUST CONFORM TO LOCAL ENERGY CODE REQUIREMENTS FOR PERFORMANCE, SWITCHING, AND WIRING (I.E. TANDEM).
N. PROVIDE APPROVED FIRE-RATED ENCLOSURES FOR ALL LIGHTING FIXTURES LOCATED IN A FIRE-RATED CEILING OR WALL.
O. WALLS DIRECTLY ILLUMINATED SHALL BE INSTALLED AND FINISHED IN A MANNER TO ELIMINATE SHADOWS OR BLEMISHES (I.E. HANG DRY WALL VERTICALLY).
P. ALL FIXTURES LOCATED IN AN INSULATED CEILING AREA, SHOULD HAVE AN IC HOUSING (ELECTRICAL CONTRACTOR TO COORDINATE).
Q. ALL WHITE TRIM DOWNLIGHTS TO BE PAINTED TO MATCH THE COLOR OF THE CEILING. IF CEILING IS NOT WHITE. THIS APPLIES TO BOTH OPEN DOWNLIGHTS AS WELL AS PINHOLE APERTURE DOWNLIGHTS.

## SHEET KEYNOTES

ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

**GRAPHIC SCALE(S)**

DATE 06/15/2026
SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE
REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)
ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 00100084

DRAWING TITLE: LIGHTING FIXTURE SCHEDULE
WORKING STAFF: NS (DESIGNED BY) | NS (DRAWN BY) | KM (CHECKED BY)

SHEET NO. 353 OF 378
E-604SH (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

FOR REFERENCE ONLY

6/15/2026 2:49:21 PM
E-604SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# Emergency Generator Matrix for Mechanical, Plumbing, Fire Alarm and Electrical Equipment

| Discipline | Equipment Tag | Equipment | On Generator | | ON in SUMMER | ON in WINTER | Load | | QTY | Total Connected Load | | Load Connected on LR ATS | | Load Connected on EM ATS | | Load Connected on OS ATS | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | YES | NO | | | | | | | | | | | | | |
| MECH | HUM-1 | AHU-1 - Electric Humidifier Coil - 65KW - (Animal Space) | | X | | X | 65,000 | VA | 1 | 65,000 | VA | | VA | | VA | | VA |
| MECH | SF-1A | AHU-1 - Fan #1 - 20HP - (Animal Space) | X | | X | X | 18,650 | VA | 1 | 18,650 | VA | | VA | | VA | 18,650 | VA |
| MECH | SF-1B | AHU-1 - Fan #2 - 20HP (Redundant) - (Animal Space) | X | | X | X | 18,650 | VA | 1 | 0 | VA | | VA | | VA | 0 | VA |
| MECH | AHU-1 Load Center | AHU-1 - Load Center (Integral Lights, MISC) - (Animal Space) - 2000W | X | | X | X | 2,000 | VA | 1 | 2,000 | VA | | VA | | VA | 2,000 | VA |
| MECH | EF-1 | EF-1 - Exhaust Fan - 10HP - (Animal Space) | X | | X | X | 9,325 | VA | 1 | 9,325 | VA | | VA | | VA | 9,325 | VA |
| MECH - FA | SEF-1 | SEF-1 - Smoke Exhaust Fan - 7.5HP | X | | X | X | 6,994 | VA | 1 | 6,994 | VA | 6,994 | VA | | VA | | VA |
| MECH | FCU-2 | FCU - Fan Coil Units - Total Unit 14.2FLA / (2) Motors .75HP - Mech Room | | X | X | X | 699 | VA | 2 | 1,399 | VA | | VA | | VA | | VA |
| MECH | FCU-1 | FCU - Fan Coil Units - Total Unit 6FLA / (2) Motors .33HP - Elec Room | | X | X | X | 308 | VA | 2 | 615 | VA | | VA | | VA | 615 | VA |
| MECH | FCU-3 | FCU - Fan Coil Units - Total Unit 18.8FLA / (2) Motors 1HP - AV/IT Room | | X | X | X | 933 | VA | 2 | 1,865 | VA | | VA | | VA | 1,865 | VA |
| MECH | FCU-4 | FCU - Fan Coil Units - Total Unit 6FLA / (2) Motors .33HP - EM Elec Room | | X | X | X | 308 | VA | 2 | 615 | VA | | VA | | VA | 615 | VA |
| MECH | ESF-1 | ESF - Emergency Supply Fan - Total Unit Motor .25HP - EM Elec Room | | X | X | X | 233 | VA | 1 | 233 | VA | | VA | | VA | | VA |
| MECH | EEF-1 | EEF - Emergency Exhaust Fan - Total Unit Motor .25HP - EM Elec Room | | X | X | X | 233 | VA | 2 | 466 | VA | | VA | | VA | | VA |
| MECH | VAV-1 to 7, VAVE-1 to 7 | AHU-1 - VAV Boxes - 200VA | X | | X | X | 200 | VA | 14 | 2,800 | VA | | VA | | VA | 2,800 | VA |
| MECH | AC-1 | Air Curtain - 0.167HP - Loading Dock | X | | X | X | 156 | VA | 1 | 156 | VA | | VA | | VA | 156 | VA |
| MECH | BAS-AHU-1, BAS-HHW, BAS-CHW, BAS-MISC, BAS-INTEGRATION | BAS Panels - 1800VA | X | | X | X | 1,800 | VA | 5 | 9,000 | VA | | VA | | VA | 9,000 | VA |
| MECH | UH-1 to 7 | UH - Wall Mounted Unit Heaters - 1/30HP - Animal Space/Day Room/VIP (Redundant) | X | | X | X | 31 | VA | 7 | 218 | VA | | VA | | VA | 218 | VA |
| MECH | HTR-1 to 4 | UH - Wall Mounted Unit Heaters - 1A MCA MOCP 15Amp 100W - Day Room | X | | | X | 100 | VA | 4 | 400 | VA | | VA | | VA | 400 | VA |
| MECH (Four Point) | ASHP-1A, 1B, 1C and 1D | ASHP - Air Source Heat Pumps - 66.26A Four Point | | X | X | X | 55,022 | VA | 4 | 220,089 | VA | | VA | | VA | 55,022 | VA |
| MECH | B1 | B1 - Natural Gas Boilers - 8A | X | | | X | 960 | VA | 1 | 960 | VA | | VA | | VA | 960 | VA |
| MECH | B2 | B1 - Natural Gas Boilers - 8A (Redundant Normal Operation Only) | X | | | X | 960 | VA | 1 | 960 | VA | | VA | | VA | 960 | VA |
| MECH | HHWP-1 | HHWP - Heating Hot Water Pumps - 20HP | X | | X | X | 18,650 | VA | 1 | 18,650 | VA | | VA | | VA | 18,650 | VA |
| MECH | HHWP-2 | HHWP - Heating Hot Water Pumps - 20HP (Redundant) | X | | X | X | 18,650 | VA | 1 | 0 | VA | | VA | | VA | 0 | VA |
| MECH | HHWP-3 | HHWP - Heating Hot Water Pumps - 5.5HP | | X | X | X | 5,129 | VA | 1 | 5,129 | VA | | VA | | VA | 5,129 | VA |
| MECH | HHWP-4 | HHWP - Heating Hot Water Pumps - 5.5HP (Redundant) | | X | X | X | 5,129 | VA | 1 | 0 | VA | | VA | | VA | 0 | VA |
| MECH | CHWP-1 | CHWP - Chilled Water Pumps - 20HP | | X | X | X | 18,650 | VA | 1 | 18,650 | VA | | VA | | VA | 18,650 | VA |
| MECH | CHWP-2 | CHWP - Chilled Water Pumps - 20HP (Redundant) | | X | X | X | 18,650 | VA | 1 | 0 | VA | | VA | | VA | 0 | VA |
| MECH | CHWP-3 | CHWP - Chilled Water Pumps - 5.5HP | | X | X | X | 5,129 | VA | 1 | 5,129 | VA | | VA | | VA | 5,129 | VA |
| MECH | CHWP-4 | CHWP - Chilled Water Pumps - 5.5HP (Redundant) | | X | X | X | 5,129 | VA | 1 | 0 | VA | | VA | | VA | 0 | VA |
| MECH | ASHP-1 Heat Trace | HT - Heat Trace - 17A MCA - Air Source Heat Pump | X | | | X | 1,632 | VA | 4 | 6,528 | VA | | VA | | VA | 6,528 | VA |
| MECH | HTC-1 | HT - Heat Trace - 5KW - HVAC Piping | X | | | X | 5,000 | VA | 1 | 5,000 | VA | | VA | | VA | 5,000 | VA |
| MECH | ERFP-1 | Energy Recovery Fluid Pumps - 5.5HP | X | | X | X | 5,129 | VA | 1 | 5,129 | VA | | VA | | VA | 5,129 | VA |
| MECH | ERFP-2 | Energy Recovery Fluid Pumps - 5.5HP (Redundant) | X | | X | X | 5,129 | VA | 1 | 0 | VA | | VA | | VA | 0 | VA |
| MECH | FPP-1 | FPP - AHU-1 Freez Protection - .5HP | X | | | X | 466 | VA | 1 | 466 | VA | | VA | | VA | 466 | VA |
| MECH | Spot Coolers-1 & 2 | Spot Coolers - 1000W | X | | X | | 1,000 | VA | 2 | 2,000 | VA | | VA | | VA | 2,000 | VA |
| MECH | Spot Dehumidifiers-1 & 2 | Spot Dehumidifiers - 1000W | X | | X | X | 1,000 | VA | 2 | 2,000 | VA | | VA | | VA | 2,000 | VA |
| MECH | Spot Heaters-1 & 2 | Spot Heaters - 1000W | X | | | X | 1,000 | VA | 2 | 2,000 | VA | | VA | | VA | 2,000 | VA |
| MECH | RSS-1 to 5 | Radiant Slab Pumps - 400W | X | | | X | 400 | | 5 | 2,000 | VA | | VA | | VA | | VA |
| MECH | GFS-1 & 2 | Glycol Feeder - .75HP | X | | X | X | 699 | | 2 | 1,399 | VA | | VA | | VA | | VA |
| MECH | OH-1 to 2 | Outdoor Caged Radiant Heaters - 1450W | | X | | X | 1,450 | VA | 2 | 2,900 | VA | | VA | | VA | | VA |
| MECH | OH-3 to 9 | Outdoor Caged Radiant Heaters - 3150W | | X | | X | 3,150 | VA | 7 | 22,050 | VA | | VA | | VA | | VA |
| MECH | OHF-1 to 3 | Outdoor Caged Fans - 240W | | X | X | X | 240 | VA | 3 | 720 | VA | | VA | | VA | | VA |
| MECH | OGF-1 to 3 | Outdoor Caged Fans - 120W | | X | X | X | 120 | VA | 3 | 360 | VA | | VA | | VA | | VA |
| MECH | OEF-1 | Outdoor Caged Exhaust Fan - 60W | | X | X | X | 60 | VA | 1 | 60 | VA | | VA | | VA | | VA |
| * MECH - SHR | | Electrical/Hydroulic Animal Shift Door - 5HP | X | | X | X | 4,663 | VA | 1 | 4,663 | VA | | VA | | VA | 4,663 | VA |
| PLUMBING | DWH-1 | Gas Water heater - 5Amps | | X | X | X | 600 | VA | 1 | 600 | VA | | VA | | VA | 600 | VA |
| PLUMBING | CP-1 | Recirculation Pumps - 70W | | X | X | X | 70 | VA | 1 | 70 | VA | | VA | | VA | 70 | VA |
| PLUMBING | SEP-1 First Pump | Sewage Ejector Pumps - 3/4HP | X | | X | X | 699 | VA | 1 | 699 | VA | 699 | VA | | VA | | VA |
| PLUMBING | SEP-1 Second Pump | Sewage Ejector Pumps - 3/4HP (Redundant) | X | | X | X | 699 | VA | 1 | 0 | VA | 0 | VA | | VA | | VA |
| PLUMBING | ETP-1 to 4 | Electronic Trap Primers - 6W | | X | X | X | 6 | VA | 3 | 18 | VA | | VA | | VA | | VA |
| *PLUMBING | | Leak Detection Sensors - 1Amp | X | | X | X | 120 | VA | 3 | 360 | VA | | VA | | VA | 360 | VA |
| PLUMBING | GD-1 | Garbage Disposal - 2HP | X | | X | X | 1,865 | VA | 1 | 1,865 | VA | | VA | | VA | 1,865 | VA |
| PLUMBING | GD-2 | Pantry Garbage Disposal - 1/2HP | | X | X | X | 466 | VA | 1 | 466 | VA | | VA | | VA | | VA |
| PLUMBING - SHR | | Indoor Animal Water Stations - 250W | X | | X | X | 250 | VA | 6 | 1,500 | VA | | VA | | VA | | VA |
| PLUMBING - SHR | | Outdoor Caged Animal Water Stations - 250W | | X | X | X | 250 | VA | 4 | 1,000 | VA | | VA | | VA | | VA |
| *PLUMBING | | HT - Heat Trace - 5W - Generator Room Piping | X | | X | X | 5 | VA | 50 | 250 | VA | | VA | | VA | 250 | VA |
| ELEC | | 3 W/SF Receptacle Load | X | | X | X | 15,000 | VA | 1 | 15,000 | VA | | VA | 15,000 | VA | | VA |
| ELEC | | Specialty Lighting | | X | X | X | 5,000 | VA | 1 | 5,000 | VA | | VA | | VA | | VA |
| EM Lighting, Fire Alarm & Security | | 0.5 W/SF Emergency | X | | X | X | 0.5 | VA | 5000 | 2,500 | VA | | VA | 2,500 | VA | | VA |

ALF

CONSTRUCT ARABIAN LEOPARD FACILITY

NOT FOR CONSTRUCTION

FOR REFERENCE ONLY

STAMP

KEY PLAN

GRAPHIC SCALE(S)

DATE 06/15/2026

SUBMISSION 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF); 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: GENERATOR LOAD MATRIX SCHEDULE

WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| NS | NS | KM |

SHEET NO. 354 of 378

E-605SH

DISCIPLINE TYPE SEQUENCE DRAWING TYPE

6/15/2026 2:49:22 PM
E-605SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

## Switchboard: DP-G1

**Location:** EMERGENCY ELECTRICAL... **Volts:** 480Y/277 **A.I.C. Rating:** 35,000 A
**Supply From:** **Phases:** 3 **Mains Type:** MCB
**Mounting:** Surface **Wires:** 4 **Mains Rating:** 800 A
**Enclosure:** Type 1 **MCB Rating:** 800 A

| CKT | Circuit Description | # of Poles | Frame Size | Trip Rating | Load | Remarks |
|---|---|---|---|---|---|---|
| 1 | PP-G1 | 3 | 600 A | 600 A | 290945 VA | |
| 2 | ATS-EM | 3 | 150 A | 150 A | 16862 VA | |
| 3 | ATS-LR | 3 | 150 A | 150 A | 13296 VA | |
| 4 | ATS-OS | 3 | 260 A | 260 A | 279396 VA | |
| 5 | SPARE | 1 | 600 A | 600 A | 0 VA | |
| 6 | SPARE | 1 | 150 A | 150 A | 0 VA | |
| 7 | SPARE | 1 | 150 A | 150 A | 0 VA | |
| 8 | SPARE | 1 | 150 A | 150 A | 0 VA | |
| 9 | SPARE | 1 | 20 A | 20 A | 0 VA | |
| 10 | SPARE | 1 | 20 A | 20 A | 0 VA | |
| 11 | SPARE | 1 | 20 A | 20 A | 0 VA | |
| 12 | SPACE | 1 | -- | -- | -- | |
| 13 | SPACE | 1 | -- | -- | -- | |
| 14 | SPACE | 1 | -- | -- | -- | |
| | | | | **Total Conn. Load:** | 600498 VA | |
| | | | | **Total Amps:** | 722 A | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| Equipment | 18000 VA | 100.00% | 18000 VA | | |
| Lighting | 6014 VA | 100.00% | 6014 VA | **Total Conn. Load:** | 600498 VA |
| Power | 1830 VA | 100.00% | 1830 VA | **Total Est. Demand:** | 448460 VA |
| Receptacle | 18140 VA | 77.56% | 14070 VA | **Total Conn.:** | 722 A |
| Mechanical Equipment | 144418 VA | 100.00% | 144418 VA | **Total Est. Demand:** | 539 A |
| Appliance | 6036 VA | 100.00% | 6036 VA | | |
| Mechanical Equipment 50% (N+1) | 185812 VA | 50.00% | 92906 VA | | |
| Mechanical Equipment 75% (N+1) | 220248 VA | 75.00% | 165186 VA | | |

**Notes:**

## Panelboard: PP-EM-G1

**Location:** ELECTRICAL B1EL1 **Volts:** 480Y/277 **A.I.C. Rating:** 35,000 A
**Supply From:** ATS-EM **Phases:** 3 **Mains Type:** MCB
**Mounting:** Surface **Wires:** 4 **Bus Rating:** 150 A
**Enclosure:** Type 1 **MCB Rating:** 150 A

| CKT | Circuit Description | Trip | Poles | A (VA) | | B (VA) | | C (VA) | | Poles | Trip | Circuit Description | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | 5327 | 0 | | | | | 1 | 20 A | SPARE | 2 |
| 3 | RP-EM-G1 VIA T-RP-EM 30KVA | 45 A | 3 | | | 4000 | 0 | | | 1 | 20 A | SPARE | 4 |
| 5 | | | | | | | | 6110 | 0 | 1 | 20 A | SPARE | 6 |
| 7 | EM LTG - B1EL1, B1EL2, B1ME1, 109 | 20 A | 1 | 420 | 0 | | | | | 1 | 20 A | SPARE | 8 |
| 9 | EM LTG - LEVEL1 | 20 A | 1 | | | 932 | 0 | | | 1 | 20 A | SPARE | 10 |
| 11 | EM LTG - GENERATOR YARD | 20 A | 1 | | | | | 46 | 0 | 1 | 20 A | SPARE | 12 |
| 13 | EM LTG - OUTDOOR SOUTH FACADE | 20 A | 1 | 18 | 0 | | | | | 1 | 20 A | SPARE | 14 |
| 15 | EM LTG - OUTDOOR NORTH STEP LIGHT | 20 A | 1 | | | 9 | 0 | | | 1 | 20 A | SPARE | 16 |
| 17 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 18 |
| 19 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 20 |
| 21 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 22 |
| 23 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 24 |
| 25 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 26 |
| 27 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 28 |
| 29 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 30 |
| 31 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 32 |
| 33 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 34 |
| 35 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 36 |
| 37 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 38 |
| 39 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 40 |
| 41 | | | | | | | | | | | | | 42 |
| | | | | **Total Load:** | 5765 VA | | 4941 VA | | 6156 VA | | | | |
| | | | | **Total Amps:** | 21 A | | 18 A | | 23 A | | | | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| Equipment | 14000 VA | 100.00% | 14000 VA | | |
| Lighting | 1602 VA | 100.00% | 1602 VA | **Total Conn. Load:** | 16862 VA |
| Receptacle | 1260 VA | 100.00% | 1260 VA | **Total Est. Demand:** | 16862 VA |
| | | | | **Total Conn.:** | 20 A |
| | | | | **Total Est. Demand:** | 20 A |

**Notes:**

## Panelboard: PP-G1

**Location:** EMERGENCY ELECTRICAL... **Volts:** 480Y/277 **A.I.C. Rating:** 35,000 A
**Supply From:** DP-G1 **Phases:** 3 **Mains Type:** MCB
**Mounting:** Surface **Wires:** 4 **Bus Rating:** 600 A
**Enclosure:** Type 1 **MCB Rating:** 600 A

| CKT | Circuit Description | Trip | Poles | A (VA) | | B (VA) | | C (VA) | | Poles | Trip | Circuit Description | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | 21667 | 20290 | | | | | | | | 2 |
| 3 | HUM-1 - B1ME1 | 100 A | 3 | | | 21667 | 17773 | | | 3 | 125 A | RP-G1 VIA T-RP 75KVA | 4 |
| 5 | | | | | | | | 21667 | 18359 | | | | 6 |
| 7 | | | | 18354 | 6 | | | | | 1 | 20 A | LTG - OUTDOOR NORTH STEP LIGHT | 8 |
| 9 | ASHP-1B - GENERATOR YARD | 100 A | 3 | | | 18354 | 480 | | | 1 | 20 A | LTG - HABITAT STRUCTURE 'XL5' | 10 |
| 11 | | | | | | | | 18354 | 130 | 1 | 20 A | LTG - HABITAT TOE KICK 'XL7' | 12 |
| 13 | | | | 18354 | 64 | | | | | 1 | 20 A | LTG - HABITAT STAIRWELL 'XL6' | 14 |
| 15 | ASHP-1C - GENERATOR YARD | 100 A | 3 | | | 18354 | 455 | | | 1 | 20 A | LTG - HABITAT TRACKLIGHT 'XL4A' & 'XL4B' | 16 |
| 17 | | | | | | | | 18354 | 0 | 1 | 20 A | SPARE | 18 |
| 19 | | | | 18354 | 0 | | | | | 1 | 20 A | SPARE | 20 |
| 21 | ASHP-1D - GENERATOR YARD | 100 A | 3 | | | 18354 | 0 | | | 1 | 20 A | SPARE | 22 |
| 23 | | | | | | | | 18354 | 0 | 1 | 20 A | SPARE | 24 |
| 25 | LIGHTING CONTROL PANEL - B1EL2 | 20 A | 1 | 1800 | 0 | | | | | 1 | 20 A | SPARE | 26 |
| 27 | LTG - 101, 103-8, 1CS1, 1WC1 | 20 A | 1 | | | 698 | 0 | | | 1 | 20 A | SPARE | 28 |
| 29 | LTG - 1CR1-3 | 20 A | 1 | | | | | 396 | 0 | 1 | 20 A | SPARE | 30 |
| 31 | LTG - B1ME1, 109 | 20 A | 1 | 210 | 0 | | | | | 1 | 20 A | SPARE | 32 |
| 33 | LTG - GENERATOR YARD | 20 A | 1 | | | 80 | 0 | | | 1 | 20 A | SPARE | 34 |
| 35 | LTG - OUTDOOR SOUTH FACADE | 20 A | 1 | | | | | 18 | 0 | 1 | 20 A | SPARE | 36 |
| 37 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 38 |
| 39 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 40 |
| 41 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 42 |
| | | | | **Total Load:** | 99099 VA | | 96215 VA | | 95631 VA | | | | |
| | | | | **Total Amps:** | 358 A | | 348 A | | 345 A | | | | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| Equipment | 3000 VA | 100.00% | 3000 VA | | |
| Lighting | 4412 VA | 100.00% | 4412 VA | **Total Conn. Load:** | 290945 VA |
| Power | 1830 VA | 100.00% | 1830 VA | **Total Est. Demand:** | 249338 VA |
| Receptacle | 10620 VA | 97.08% | 10310 VA | **Total Conn.:** | 350 A |
| Mechanical Equipment | 102620 VA | 100.00% | 102620 VA | **Total Est. Demand:** | 300 A |
| Appliance | 3276 VA | 100.00% | 3276 VA | | |
| Mechanical Equipment 75% (N+1) | 165186 VA | 75.00% | 123890 VA | | |
| | | | | | |

**Notes:**

## Panelboard: RP-G1

**Location:** EMERGENCY ELECTRICAL... **Volts:** 208Y/120 **A.I.C. Rating:** 10,000 A
**Supply From:** T-RP **Phases:** 3 **Mains Type:** MCB
**Mounting:** Surface **Wires:** 4 **Bus Rating:** 250 A
**Enclosure:** Type 1 **MCB Rating:** 250 A

| CKT | Circuit Description | Trip | Poles | A | | B | | C | | Poles | Trip | Circuit Description | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | ETP-1 - B1ME1 | 20 A | 1 | 6 | 1575 | | | | | 2 | 20 A | HEATER - OH-1 - HABITAT | 2 |
| 3 | ETP-2 - B1ME1 | 20 A | 1 | | | 6 | 1575 | | | | | | 4 |
| 5 | BFP-1 - B1ME1 | 20 A | 1 | | | | | 0 | 1575 | 2 | 20 A | HEATER - OH-2 - HABITAT | 6 |
| 7 | BFP-2 - B1ME1 | 20 A | 1 | 0 | 1575 | | | | | | | | 8 |
| 9 | BFP-3 - B1ME1 | 20 A | 1 | | | 0 | 1575 | | | 2 | 20 A | HEATER - OH-3 - HABITAT | 10 |
| 11 | RCPT CONVENIENCE - B1ME1, OUTDOOR | 20 A | 1 | | | | | 1260 | 1575 | | | | 12 |
| 13 | RCPT CONVENIENCE - B1EL1, 109 | 20 A | 1 | 900 | 1575 | | | | | 2 | 20 A | HEATER - OH-4 - HABITAT | 14 |
| 15 | EEF-1 - B1EL2 | 20 A | 1 | | | 720 | 1575 | | | | | | 16 |
| 17 | ESF-1 - B1EL2 | 20 A | 1 | | | | | 720 | 1575 | 2 | 20 A | HEATER - OH-5 - HABITAT | 18 |
| 19 | GFS-1 - B1ME1 | 30 A | 1 | 1656 | 1575 | | | | | | | | 20 |
| 21 | GFS-2 - B1ME1 | 30 A | 1 | | | 1656 | 725 | | | 2 | 20 A | HEATER - OH-6 - HABITAT | 22 |
| 23 | FCU-2 - B1ME1 | 35 A | 2 | | | | | 1477 | 725 | | | | 24 |
| 25 | | | | 1477 | 1575 | | | | | 2 | 20 A | HEATER - OH-7 - HABITAT | 26 |
| 27 | REFREGERATOR - 100 | 20 A | 1 | | | 1050 | 1575 | | | | | | 28 |
| 29 | RCPT CONVENIENCE - 100 | 20 A | 1 | | | | | 1260 | 1575 | 2 | 20 A | HEATER - OH-8 - HABITAT | 30 |
| 31 | RCPT CONVENIENCE - 100 | 20 A | 1 | 720 | 1575 | | | | | | | | 32 |
| 33 | RSS-1 to 3 - 100 | 20 A | 1 | | | 1200 | 725 | | | 2 | 20 A | HEATER - OH-9 - HABITAT | 34 |
| 35 | RSS-4 to 5 - 100 | 20 A | 1 | | | | | 800 | 725 | | | | 36 |
| 37 | RCPT CONVENIENCE - 101 | 20 A | 1 | 720 | 1000 | | | | | 1 | 20 A | OUTDOOR DRINKERS - D1 TO D4 HABITAT | 38 |
| 39 | REFREGERATOR - 101 | 20 A | 1 | | | 1050 | 360 | | | 1 | 20 A | CAGED FANS - OHF-1 TO 3 - HABITAT | 40 |
| 41 | RCPT - 101 | 20 A | 1 | | | | | 1080 | 720 | 1 | 20 A | CAGED FANS - OGF-1 TO 3 - HABITAT | 42 |
| 43 | ATS ANNUNCIATOR - 101 | 20 A | 1 | 0 | 540 | | | | | 1 | 20 A | RCPT CONVENIENCE - HABITAT | 44 |
| 45 | GEN ANNUNCIATOR - 101 | 20 A | 1 | | | 0 | 1800 | | | 1 | 20 A | IRRIGATION CONTROLLER - HABITAT | 46 |
| 47 | GARBAGE DISPOSAL GD-2 - 101 | 20 A | 1 | | | | | 1176 | 60 | 1 | 20 A | OEF-1 - HABITAT | 48 |
| 49 | INDOOR DRIKERS - 103 TO,107 | 20 A | 1 | 0 | 915 | | | | | 2 | 20 A | HYDROELECTRIC DOOR #1 - HABITAT | 50 |
| 51 | RCPT CONVENIENCE - 1CS1, 1WC1 | 20 A | 1 | | | 360 | 915 | | | | | | 52 |
| 53 | RCPT CONVENIENCE - 1CR1 | 20 A | 1 | | | | | 720 | 0 | 1 | 20 A | SPARE | 54 |
| 55 | RCPT CORD REEL - L5-20R - 1CR1 | 20 A | 1 | 2000 | 0 | | | | | 1 | 20 A | SPARE | 56 |
| 57 | RCPT 5-20R - 1CR1 | 20 A | 1 | | | 180 | 0 | | | 1 | 20 A | SPARE | 58 |
| 59 | RCPT CONVENIENCE - 1CR1 | 20 A | 1 | | | | | 360 | 0 | 1 | 20 A | SPARE | 60 |
| 61 | ETP-4 - 1CR1 | 20 A | 1 | 6 | 0 | | | | | 1 | 20 A | SPARE | 62 |
| 63 | ETP-3 - 1CR3 | 20 A | 1 | | | 6 | 0 | | | 1 | 20 A | SPARE | 64 |
| 65 | RCPT CONVENIENCE - 1CR2, 1CR3 | 20 A | 1 | | | | | 900 | 0 | 1 | 20 A | SPARE | 66 |
| 67 | RCPT CONVENIENCE - OUTDOOR | 20 A | 1 | 900 | 0 | | | | | 1 | 20 A | SPARE | 68 |
| 69 | RCPT CONVENIENCE - OUTDOOR | 20 A | 1 | | | 720 | 0 | | | 1 | 20 A | SPARE | 70 |
| 71 | LTG - UNDERCABINET - 100-1 | 20 A | 1 | | | | | 76 | 0 | 1 | 20 A | SPARE | 72 |
| | | | | **Total Load:** | 20290 VA | | 17773 VA | | 18359 VA | | | | |
| | | | | **Total Amps:** | 170 A | | 148 A | | 154 A | | | | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| Equipment | 3000 VA | 100.00% | 3000 VA | | |
| Lighting | 76 VA | 100.00% | 76 VA | **Total Conn. Load:** | 56422 VA |
| Power | 1830 VA | 100.00% | 1830 VA | **Total Est. Demand:** | 56112 VA |
| Receptacle | 10620 VA | 97.08% | 10310 VA | **Total Conn.:** | 157 A |
| Mechanical Equipment | 37620 VA | 100.00% | 37620 VA | **Total Est. Demand:** | 156 A |
| Appliance | 3276 VA | 100.00% | 3276 VA | | |
| | | | | | |

**Notes:**

## GENERAL SHEET NOTES

A. SEE DRAWING E0.1 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. SEE DRAWING SERIES E5 FOR POWER RISER DIAGRAM.
C. PROVIDE A MINIMUM OF 20% SPARE CIRCUIT BREAKERS ON EACH NEW PANEL.
D. CONTRACTOR TO VERIFY EXISTING AVAILABLE CIRCUITS. COORDINATE LOCATION OF AVAILABLE CIRCUITS WITH THOSE INDICATED IN PANEL SCHEDULES.
E. REUSE EXISTING CIRCUIT BREAKERS WHEN APPLICABLE. PROVIDE NEW BREAKERS IN EXISTING SPACES. COORDINATE NEW BREAKER TYPE WITH EXISTING TO REMAINING PANEL.
F. CONTRACTOR TO METER AND VERIFY EXISTING PANEL LOADS AND AVAILABLE PANEL CAPACITY PRIOR TO ADDING CIRCUITS.
G. PROVIDE PERMANENT PANEL NAMEPLATES AND TYPED PANEL SCHEDULES AFTER ALL WORK IS COMPLETE. SCHEDULES AND NAMEPLATES MUST MATCH PANEL INFORMATION, CIRCUIT BREAKERS, AND CONNECTED LOADS.
H. BREAKER TIES TO BE USED FOR ALL SYSTEMS FURNITURE CONNECTIONS PER NEC.
I. CONTRACTOR TO PROVIDE 3 PHASE CIRCUIT FOR ALL NEW SURGE PROTECTION DEVICES. COORDINATE FINAL CIRCUIT BREAKER SIZE WITH DEVICE.
J. CONTRACTOR TO PROVIDE 3P, 15A BREAKER AS REQUIRED FOR ALL NEW METERS.

## SHEET KEYNOTES

| | |
|---|---|
| DP-G1 | PP-EM-G1 |
| PP-G1 | RP-G1 |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

GRAPHIC SCALE(S)

DATE 06/15/2026 | SUBMISSION 65% CONSTRUCTION DOCUMENTS

| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: **ARABIAN LEOPARD FACILITY (ALF)** — **3001 Connecticut Avenue NW Washington, DC 20008**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**
OPDC PROJECT NUMBER: **2533105**
A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **PANELBOARD SCHEDULES**
WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| NS | NS | KM |

SHEET NO. 355 OF 378

E-606SH (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

FOR REFERENCE ONLY

6/15/2026 2:49:22 PM
E-606SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

## Panelboard: PP-LR-G1

**Location:** ELECTRICAL B1EL1 | **Volts:** 480Y/277 | **A.I.C. Rating:** 35,000 A
**Supply From:** ATS-LR | **Phases:** 3 | **Mains Type:** MCB
**Mounting:** Surface | **Wires:** 4 | **Bus Rating:** 150 A
**Enclosure:** Type 1 | **MCB Rating:** 150 A

| CKT | Circuit Description | Trip | Poles | A (VA) | | B (VA) | | C (VA) | | Poles | Trip | Circuit Description | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | 1163 | 1163 | | | | | | | | 2 |
| 3 | SEP-1 - B1ME1 | 20 A | 3 | | | 1163 | 1163 | | | 3 | 20 A | SEP-2 - B1ME1 | 4 |
| 5 | | | | | | | | 1163 | 1163 | | | | 6 |
| 7 | | | | 2105 | 0 | | | | | | | | 8 |
| 9 | SEF-1 - 100 | 20 A | 3 | | | 2105 | 0 | | | 3 | 25 A | RP-LR-G1 VIA T-RP-LR 15KVA | 10 |
| 11 | | | | | | | | 2105 | 0 | | | | 12 |
| 13 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 14 |
| 15 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 16 |
| 17 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 18 |
| 19 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 20 |
| 21 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 22 |
| 23 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 24 |
| 25 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 26 |
| 27 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 28 |
| 29 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 30 |
| 31 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 32 |
| 33 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 34 |
| 35 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 36 |
| 37 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 38 |
| 39 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 40 |
| 41 | | | | | | | | | | | | | 42 |
| | | **Total Load:** | | 4432 VA | | 4432 VA | | 4432 VA | | | | | |
| | | **Total Amps:** | | 16 A | | 16 A | | 16 A | | | | | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| Mechanical Equipment 50% (N+1) | 13296 VA | 50.00% | 6648 VA | | |
| | | | | **Total Conn. Load:** | 13296 VA |
| | | | | **Total Est. Demand:** | 6648 VA |
| | | | | **Total Conn.:** | 16 A |
| | | | | **Total Est. Demand:** | 8 A |

**Notes:**

## Panelboard: RP-LR-G1

**Location:** ELECTRICAL B1EL1 | **Volts:** 208Y/120 | **A.I.C. Rating:** 10,000 A
**Supply From:** T-RP-LR | **Phases:** 3 | **Mains Type:** MCB
**Mounting:** Surface | **Wires:** 4 | **Bus Rating:** 50 A
**Enclosure:** Type 1 | **MCB Rating:** 50 A

| CKT | Circuit Description | Trip | Poles | A (VA) | | B (VA) | | C (VA) | | Poles | Trip | Circuit Description | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 2 |
| 3 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 4 |
| 5 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 6 |
| 7 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 8 |
| 9 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 10 |
| 11 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 12 |
| 13 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 14 |
| 15 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 16 |
| 17 | SPACE | -- | 1 | | | | | -- | -- | 1 | -- | SPACE | 18 |
| 19 | SPACE | -- | 1 | -- | -- | | | | | 1 | -- | SPACE | 20 |
| 21 | SPACE | -- | 1 | | | -- | -- | | | 1 | -- | SPACE | 22 |
| 23 | SPACE | -- | 1 | | | | | -- | -- | 1 | -- | SPACE | 24 |
| 25 | SPACE | -- | 1 | -- | -- | | | | | 1 | -- | SPACE | 26 |
| 27 | SPACE | -- | 1 | | | -- | -- | | | 1 | -- | SPACE | 28 |
| 29 | SPACE | -- | 1 | | | | | -- | -- | 1 | -- | SPACE | 30 |
| 31 | SPACE | -- | 1 | -- | -- | | | | | 1 | -- | SPACE | 32 |
| 33 | SPACE | -- | 1 | | | -- | -- | | | 1 | -- | SPACE | 34 |
| 35 | SPACE | -- | 1 | | | | | -- | -- | 1 | -- | SPACE | 36 |
| 37 | SPACE | -- | 1 | -- | -- | | | | | 1 | -- | SPACE | 38 |
| 39 | SPACE | -- | 1 | | | -- | -- | | | 1 | -- | SPACE | 40 |
| 41 | SPACE | -- | 1 | | | | | -- | -- | 1 | -- | SPACE | 42 |
| | | **Total Load:** | | 0 VA | | 0 VA | | 0 VA | | | | | |
| | | **Total Amps:** | | 0 A | | 0 A | | 0 A | | | | | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| | | | | **Total Conn. Load:** | 0 VA |
| | | | | **Total Est. Demand:** | 0 VA |
| | | | | **Total Conn.:** | 0 A |
| | | | | **Total Est. Demand:** | 0 A |

**Notes:**

## Panelboard: PP-OS-G1

**Location:** ELECTRICAL B1EL1 | **Volts:** 480Y/277 | **A.I.C. Rating:** 35,000 A
**Supply From:** ATS-OS | **Phases:** 3 | **Mains Type:** MCB
**Mounting:** Surface | **Wires:** 4 | **Bus Rating:** 250 A
**Enclosure:** Type 1

| CKT | Circuit Description | Trip | Poles | A | | B | | C | | Poles | Trip | Circuit Description | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | 7479 | 2105 | | | | | | | | 2 |
| 3 | SF-1A - B1ME1 | 60 A | 3 | | | 7479 | 2105 | | | 3 | 20 A | ERFP-1 - B1ME1 | 4 |
| 5 | | | | | | | | 7479 | 2105 | | | | 6 |
| 7 | | | | 7479 | 2105 | | | | | | | | 8 |
| 9 | SF-1B - B1ME1 (REDUNDANT) | 60 A | 3 | | | 7479 | 2105 | | | 3 | 20 A | ERFP-2 - B1ME1 (REDUNDANT) | 10 |
| 11 | | | | | | | | 7479 | 2105 | | | | 12 |
| 13 | | | | 7479 | 3878 | | | | | | | | 14 |
| 15 | HHWP-1 - B1ME1 | 60 A | 3 | | | 7479 | 3878 | | | 3 | 35 A | EF-1 - 101 | 16 |
| 17 | | | | | | | | 7479 | 3878 | | | | 18 |
| 19 | | | | 7479 | 18354 | | | | | | | | 20 |
| 21 | HHWP-2 - B1ME1 (REDUNDANT) | 60 A | 3 | | | 7479 | 18354 | | | 3 | 100 A | ASHP-1A - GENERATOR YARD | 22 |
| 23 | | | | | | | | 7479 | 18354 | | | | 24 |
| 25 | | | | 2105 | 7479 | | | | | | | | 26 |
| 27 | HHWP-3 - B1ME1 | 20 A | 3 | | | 2105 | 7479 | | | 3 | 60 A | CHWP-1 - B1ME1 | 28 |
| 29 | | | | | | | | 2105 | 7479 | | | | 30 |
| 31 | | | | 2105 | 7479 | | | | | | | | 32 |
| 33 | HHWP-4 - B1ME1 (REDUNDANT) | 20 A | 3 | | | 2105 | 7479 | | | 3 | 60 A | CHWP-2 - B1ME1 (REDUNDANT) | 34 |
| 35 | | | | | | | | 2105 | 7479 | | | | 36 |
| 37 | | | | 15899 | 2105 | | | | | | | | 38 |
| 39 | OSR1 VIA T-OSR1 45KVA | 70 A | 3 | | | 11265 | 2105 | | | 3 | 20 A | CHWP-3 - B1ME1 | 40 |
| 41 | | | | | | | | 11020 | 2105 | | | | 42 |
| 43 | AHU-1 LOAD CENTER (LTG, MISC) - B1ME1 | 20 A | 1 | 2000 | 2105 | | | | | | | | 44 |
| 45 | SPARE | 20 A | 1 | | | 0 | 2105 | | | 3 | 20 A | CHWP-4 - B1ME1 (REDUNDANT) | 46 |
| 47 | SPARE | 20 A | 1 | | | | | 0 | 2105 | | | | 48 |
| 49 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 50 |
| 51 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 52 |
| 53 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 54 |
| 55 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 56 |
| 57 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 58 |
| 59 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 60 |
| 61 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 62 |
| 63 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 64 |
| 65 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 66 |
| 67 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 68 |
| 69 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 70 |
| 71 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 72 |
| | | **Total Load:** | | 97636 VA | | 91002 VA | | 90757 VA | | | | | |
| | | **Total Amps:** | | 353 A | | 329 A | | 328 A | | | | | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| Equipment | 1000 VA | 100.00% | 1000 VA | | |
| Power | 0 VA | 0.00% | 0 VA | **Total Conn. Load:** | 279396 VA |
| Receptacle | 6260 VA | 100.00% | 6260 VA | **Total Est. Demand:** | 179372 VA |
| Mechanical Equipment | 41798 VA | 100.00% | 41798 VA | **Total Conn.:** | 338 A |
| Appliance | 2760 VA | 100.00% | 2760 VA | **Total Est. Demand:** | 216 A |
| Mechanical Equipment 50% (N+1) | 172516 VA | 50.00% | 86258 VA | | |
| Mechanical Equipment 75% (N+1) | 55062 VA | 75.00% | 41297 VA | | |

**Notes:**

## Panelboard: RP-OS-G1

**Location:** ELECTRICAL B1EL1 | **Volts:** 208Y/120 | **A.I.C. Rating:** 10,000 A
**Supply From:** T-RP-OS | **Phases:** 3 | **Mains Type:** MCB
**Mounting:** Surface | **Wires:** 4 | **Mains Rating:** 150 A
**Enclosure:** Type 1 | **MCB Rating:** 150 A

| CKT | Circuit Description | Trip | Poles | A | | B | | C | | Poles | Trip | Circuit Description | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | AHU-1 LOAD CENTER (RCPT) - B1ME1 | 20 A | 1 | 1800 | 100 | | | | | 1 | 20 A | HTR-1 - 108 | 2 |
| 3 | BAS-INTEGRATION - B1ME1 | 20 A | 1 | | | 1080 | 100 | | | 1 | 20 A | HTR-2 - 108 | 4 |
| 5 | BAS-AHU-1 - B1ME1 | 20 A | 1 | | | | | 1080 | 100 | 1 | 20 A | HTR-3 - 108 | 6 |
| 7 | BAS-HHW - B1ME1 | 20 A | 1 | 1080 | 100 | | | | | 1 | 20 A | HTR-4 - 108 | 8 |
| 9 | BAS-MISC - B1ME1 | 20 A | 1 | | | 1080 | 500 | | | 1 | 20 A | VAVE-1 TO 7 - LVL1 | 10 |
| 11 | BAS-CHW - B1ME1 | 20 A | 1 | | | | | 1080 | 500 | 1 | 20 A | VAV-1 TO 7 - LVL1 | 12 |
| 13 | FPP-1 - B1ME1 | 20 A | 1 | 1176 | 1000 | | | | | 1 | 20 A | SPOT COOLER - 1CR1 | 14 |
| 15 | B-1 - B1ME1 | 20 A | 1 | | | 960 | 1000 | | | 1 | 20 A | SPOT DEHUMIDIFIER - 1CR1 | 16 |
| 17 | B-2 - B1ME1 (REDUNDANT) | 20 A | 1 | | | | | 960 | 1000 | 1 | 20 A | SPOT HEATER - 1CR1 | 18 |
| 19 | HTC-1 - B1ME1 | 60 A | 1 | 5000 | 1000 | | | | | 1 | 20 A | SPOT COOLER - 1CR3 | 20 |
| 21 | UH-1 - 107 (REDUNDANT) | 20 A | 1 | | | 228 | 1000 | | | 1 | 20 A | SPOT DEHUMIDIFIER - 1CR3 | 22 |
| 23 | UH-2 - 106 (REDUNDANT) | 20 A | 1 | | | | | 228 | 180 | 1 | 20 A | SPOT HEATER - CORRIDOR 1CR3 | 24 |
| 25 | UH-3 - 105 (REDUNDANT) | 20 A | 1 | 228 | 1080 | | | | | 1 | 20 A | RCPT CONVENIENCE - 101 | 26 |
| 27 | UH-4 - 104 (REDUNDANT) | 20 A | 1 | | | 228 | 0 | | | 1 | 20 A | CP-1 - B1ME1 | 28 |
| 29 | UH-5 - 103 (REDUNDANT) | 20 A | 1 | | | | | 228 | 600 | 1 | 20 A | DWH-1 - B1ME1 | 30 |
| 31 | UH-6 - 101 (REDUNDANT) | 20 A | 1 | 228 | 624 | | | | | 2 | 20 A | FCU-1 - B1EL1 | 32 |
| 33 | UH-7 - 101 (REDUNDANT) | 20 A | 1 | | | 228 | 624 | | | | | | 34 |
| 35 | ASHP-1 HEAT TRACE - GENERATOR YARD | 30 A | 1 | | | | | 1632 | 1955 | 2 | 45 A | FCU-3 - B1ME1 | 36 |
| 37 | AC-1 - 100 | 20 A | 1 | 528 | 1955 | | | | | | | | 38 |
| 39 | GARBAGE DISPOSAL GD-1 - 100 | 20 A | 1 | | | 2760 | 1477 | | | 2 | 35 A | FCU-4 - B1EL2 | 40 |
| 41 | SPARE | 20 A | 1 | | | | | 0 | 1477 | | | | 42 |
| 43 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 44 |
| 45 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 46 |
| 47 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 48 |
| 49 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 50 |
| 51 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 52 |
| 53 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 54 |
| 55 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 56 |
| 57 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 58 |
| 59 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 60 |
| 61 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 62 |
| 63 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 64 |
| 65 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 66 |
| 67 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 68 |
| 69 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 70 |
| 71 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 72 |
| | | **Total Load:** | | 15899 VA | | 11265 VA | | 11020 VA | | | | | |
| | | **Total Amps:** | | 133 A | | 94 A | | 92 A | | | | | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| Equipment | 1000 VA | 100.00% | 1000 VA | | |
| Receptacle | 6260 VA | 100.00% | 6260 VA | **Total Conn. Load:** | 38184 VA |
| Mechanical Equipment | 28164 VA | 100.00% | 28164 VA | **Total Est. Demand:** | 38184 VA |
| Appliance | 2760 VA | 100.00% | 2760 VA | **Total Conn.:** | 106 A |
| | | | | **Total Est. Demand:** | 106 A |

**Notes:**

## GENERAL SHEET NOTES

A. SEE DRAWING E0.1 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. SEE DRAWING SERIES E5 FOR POWER RISER DIAGRAM.
C. PROVIDE A MINIMUM OF 20% SPARE CIRCUIT BREAKERS ON EACH NEW PANEL.
D. CONTRACTOR TO VERIFY EXISTING AVAILABLE CIRCUITS. COORDINATE LOCATION OF AVAILABLE CIRCUITS WITH THOSE INDICATED IN PANEL SCHEDULES.
E. REUSE EXISTING CIRCUIT BREAKERS WHEN APPLICABLE. PROVIDE NEW BREAKERS IN EXISTING SPACES. COORDINATE NEW BREAKER TYPE WITH EXISTING TO REMAINING PANEL.
F. CONTRACTOR TO METER AND VERIFY EXISTING PANEL LOADS AND AVAILABLE PANEL CAPACITY PRIOR TO ADDING CIRCUITS.
G. PROVIDE PERMANENT PANEL NAMEPLATES AND TYPED PANEL SCHEDULES AFTER ALL WORK IS COMPLETE. SCHEDULES AND NAMEPLATES MUST MATCH PANEL INFORMATION, CIRCUIT BREAKERS, AND CONNECTED LOADS.
H. BREAKER TIES TO BE USED FOR ALL SYSTEMS FURNITURE CONNECTIONS PER NEC.
I. CONTRACTOR TO PROVIDE 3 PHASE CIRCUIT FOR ALL NEW SURGE PROTECTION DEVICES. COORDINATE FINAL CIRCUIT BREAKER SIZE WITH DEVICE.
J. CONTRACTOR TO PROVIDE 3P, 15A BREAKER AS REQUIRED FOR ALL NEW METERS.

## SHEET KEYNOTES

| | |
|---|---|
| PP-LR-G1 | RP-LR-G1 |
| PP-OS-G1 | RP-OS-G1 |

ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

**GRAPHIC SCALE(S)**

DATE: 06/15/2026 | SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # / DATE | REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: **ARABIAN LEOPARD FACILITY (ALF)**
**3001 Connecticut Avenue NW Washington, DC 20008**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**
OPDC PROJECT NUMBER: **2533105**
A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **PANELBOARD SCHEDULES**
WORKING STAFF: NS (DESIGNED BY) | NS (DRAWN BY) | KM (CHECKED BY)

SHEET NO. 356 OF 378

E-607SH (DISCIPLINE E | TYPE 6 | SEQUENCE 07 | DRAWING TYPE SH)

FOR REFERENCE ONLY

6/15/2026 2:49:22 PM
E-607SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

## Panelboard: RP-EM-G1

**Location:** ELECTRICAL B1EL1
**Supply From:** T-RP-EM
**Mounting:** Surface
**Enclosure:** Type 1

**Volts:** 208Y/120
**Phases:** 3
**Wires:** 4

**A.I.C. Rating:** 10,000 A
**Mains Type:** MCB
**Bus Rating:** 100 A
**MCB Rating:** 100 A

| CKT | Circuit Description | Trip | Poles | A (VA) | | B (VA) | | C (VA) | | Poles | Trip | Circuit Description | CKT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | AMP PANEL - 109 | 20 A | 1 | 500 | 0 | | | | | 1 | 20 A | SPARE | 2 |
| 3 | NAC PANEL - 109 | 20 A | 1 | | | 500 | 0 | | | 1 | 20 A | SPARE | 4 |
| 5 | RCPT - L6-30R - 109 | 20 A | 2 | | | | | 1000 | 0 | 1 | 20 A | SPARE | 6 |
| 7 | | | | 1000 | 0 | | | | | 1 | 20 A | SPARE | 8 |
| 9 | RCPT - L6-20R - 109 | 20 A | 2 | | | 750 | 0 | | | 1 | 20 A | SPARE | 10 |
| 11 | | | | | | | | 750 | 0 | 1 | 20 A | SPARE | 12 |
| 13 | RCPT - L6-20R - 109 | 20 A | 2 | 750 | 0 | | | | | 1 | 20 A | SPARE | 14 |
| 15 | | | | | | 750 | 0 | | | 1 | 20 A | SPARE | 16 |
| 17 | RCPT - 5-20P - 109 | 20 A | 1 | | | | | 2000 | 0 | 1 | 20 A | SPARE | 18 |
| 19 | RCPT - 5-20P - 109 | 20 A | 1 | 2000 | 0 | | | | | 1 | 20 A | SPARE | 20 |
| 21 | RCPT - 5-20P - 109 | 20 A | 1 | | | 2000 | 0 | | | 1 | 20 A | SPARE | 22 |
| 23 | RCPT - 5-20P - 109 | 20 A | 1 | | | | | 2000 | 0 | 1 | 20 A | SPARE | 24 |
| 25 | RCPT - 109 | 20 A | 1 | 900 | 0 | | | | | 1 | 20 A | SPARE | 26 |
| 27 | SECURITY J BOX - 109 | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 28 |
| 29 | RCPT - B1EL1, B1EL2 | 20 A | 1 | | | | | 360 | 0 | 1 | 20 A | SPARE | 30 |
| 31 | EM LTG - HABITAT HANDRAIL 'XL1' | 20 A | 1 | 177 | 0 | | | | | 1 | 20 A | SPARE | 32 |
| 33 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 34 |
| 35 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 36 |
| 37 | SPARE | 20 A | 1 | 0 | 0 | | | | | 1 | 20 A | SPARE | 38 |
| 39 | SPARE | 20 A | 1 | | | 0 | 0 | | | 1 | 20 A | SPARE | 40 |
| 41 | SPARE | 20 A | 1 | | | | | 0 | 0 | 1 | 20 A | SPARE | 42 |
| | | | **Total Load:** | 5327 VA | | 4000 VA | | 6110 VA | | | | | |
| | | | **Total Amps:** | 46 A | | 33 A | | 53 A | | | | | |

| Load Classification | Connected Load | Demand Factor | Estimated Demand | Panel Totals | |
|---|---|---|---|---|---|
| Equipment | 14000 VA | 100.00% | 14000 VA | | |
| Lighting | 177 VA | 100.00% | 177 VA | **Total Conn. Load:** | 15437 VA |
| Receptacle | 1260 VA | 100.00% | 1260 VA | **Total Est. Demand:** | 15437 VA |
| | | | | **Total Conn.:** | 43 A |
| | | | | **Total Est. Demand:** | 43 A |

**Notes:**

## GENERAL SHEET NOTES

A. SEE DRAWING E0.1 FOR ABBREVIATIONS, SYMBOLS, GENERAL NOTES AND DEVICE MOUNTING HEIGHT OF WALL MOUNTED DEVICES, UON.
B. SEE DRAWING SERIES E5 FOR POWER RISER DIAGRAM.
C. PROVIDE A MINIMUM OF 20% SPARE CIRCUIT BREAKERS ON EACH NEW PANEL.
D. CONTRACTOR TO VERIFY EXISTING AVAILABLE CIRCUITS. COORDINATE LOCATION OF AVAILABLE CIRCUITS WITH THOSE INDICATED IN PANEL SCHEDULES.
E. REUSE EXISTING CIRCUIT BREAKERS WHEN APPLICABLE. PROVIDE NEW BREAKERS IN EXISTING SPACES. COORDINATE NEW BREAKER TYPE WITH EXISTING TO REMAINING PANEL.
F. CONTRACTOR TO METER AND VERIFY EXISTING PANEL LOADS AND AVAILABLE PANEL CAPACITY PRIOR TO ADDING CIRCUITS.
G. PROVIDE PERMANENT PANEL NAMEPLATES AND TYPED PANEL SCHEDULES AFTER ALL WORK IS COMPLETE. SCHEDULES AND NAMEPLATES MUST MATCH PANEL INFORMATION, CIRCUIT BREAKERS, AND CONNECTED LOADS.
H. BREAKER TIES TO BE USED FOR ALL SYSTEMS FURNITURE CONNECTIONS PER NEC.
I. CONTRACTOR TO PROVIDE 3 PHASE CIRCUIT FOR ALL NEW SURGE PROTECTION DEVICES. COORDINATE FINAL CIRCUIT BREAKER SIZE WITH DEVICE.
J. CONTRACTOR TO PROVIDE 3P, 15A BREAKER AS REQUIRED FOR ALL NEW METERS.

## ○ SHEET KEYNOTES

| | |
|---|---|
| RP-EM-G1 | - |
| - | - |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

STAMP

KEY PLAN

**GRAPHIC SCALE(S)**

DATE: 06/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: **ARABIAN LEOPARD FACILITY (ALF)**
**3001 Connecticut Avenue NW**
**Washington, DC 20008**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**
OPDC PROJECT NUMBER: **2533105**
A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **PANELBOARD SCHEDULES**
WORKING STAFF: **NS** (DESIGNED BY) **NS** (DRAWN BY) **KM** (CHECKED BY)

SHEET NO. 357 OF 378

E-608SH (DISCIPLINE E, TYPE 6, SEQUENCE 08, DRAWING TYPE SH)

*FOR REFERENCE ONLY*

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E-608SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# ABBREVIATIONS

| | |
|---|---|
| A/E | ARCHITECT/ENGINEER |
| AAP | ALARM ANNUNCIATOR PANEL |
| ADA | AMERICANS WITH DISABILITIES ACT |
| AFC | ABOVE FINISHED CEILING |
| AFF | ABOVE FINISHED FLOOR |
| AFG | ABOVE FINISHED GRADE |
| AHJ | AUTHORITY HAVING JURISDICTION |
| ARCH | ARCHITECT |
| AV | AUDIOVISUAL |
| AVC | AUDIOVISUAL CONTRACTOR |
| AWG | AMERICAN WIRE GAUGE |
| BAS | BUILDING AUTOMATION SYSTEM |
| BCT | BONDING CONDUCTOR FOR TELECOMMUNICATIONS |
| BFF | BELOW FINISH FLOOR |
| BICSI | BUILDING INDUSTRY CONSULTING SERVICE INTERNATIONAL |
| C | CAMERA |
| CAT | CATEGORY |
| CATV | CABLE TELEVISION |
| CCTV | CLOSED CIRCUIT TELEVISION |
| CD | CONSTRUCTION DOCUMENTS |
| CL | CENTERLINE |
| CLG | CEILING |
| CONTR | CONTRACTOR |
| CT | CABLE TRAY |
| DEMO | DEMOLITION |
| DIA | DIAGRAM |
| DPDT | DOUBLE POLE DOUBLE THROW |
| EC | ELECTRICAL CONTRACTOR |
| ELEC | ELECTRICAL |
| EMER | EMERGENCY |
| EMT | ELECTRICAL METAL CONDUIT |
| ENCL | ENCLOSURE |
| ENT | ELECTRICAL NON-METALIC CONDUIT |
| EQ | EQUAL DIMENSION |
| FP | FIRE PROTECTION |
| FT | FEET |
| GC | GENERAL CONTRACTOR |
| GND | GROUND |
| HT | HEIGHT |
| HVAC | HEATING, VENTILATION, AND AIR CONDITIONING |
| I/O | AUDIOVISUAL I/O PANEL |
| IDS | INTRUSION DETECTION SYSTEM |
| IMC | INTERMEDIATE METAL CONDUIT |
| IN | INCHES |
| J-BOX | JUNCTION BOX |
| LAN | LOCAL AREA NETWORK |
| LV | LOW VOLTAGE |
| KP | KEYPAD |
| M | MICROPHONE |
| MAX | MAXIMUM |
| MECH | MECHANICAL |
| MIN | MINIMUM |
| MTR | MAIN TELECOMMUNICATION ROOM |
| NEC | NATIONAL ELECTRICAL CODE |
| NTS | NOT TO SCALE |
| PB | PULL BOX |
| PBB | PRIMARY BONDING BUSBAR |
| PET | PROTECTED ENTRANCE TERMINAL |
| PIRS | PASSIVE INFARED SENSOR |
| PS | PROJECTION SCREEN SWITCH |
| PST | PROJECTION SCREEN TYPE |
| PT | POKE THRU |
| PTZ | PAN / TILT / ZOOM |
| PVC | POLYVINYL CHLORIDE CONDUIT |
| PWR | POWER |
| QTY | QUANTITY |
| RCDD | REGISTERED COMMUNICATIONS DISTRIBUTION DESIGNER |
| RCP | REFLECTED CEILING PLAN |
| RECPT | RECEPTACLE |
| REQD | REQUIRED |
| REX | REQUEST TO EXIT |
| RGS | RIGID GALVANIZED STEEL CONDUIT |
| RM | ROOM |
| RMC | RIGID METAL CONDUIT |
| RS | ROOM SCHEDULER |
| RU | RACK UNIT |
| S | SPEAKER |
| SB | SPECIALTY BACK BOX |
| SBB | SECONDARY BONDING BUSBAR |
| SH | SHIELD |
| SMG | SOUND MASKING SYSTEM GENERATOR |
| SMP | SOUND MASKING SYSTEM PAGER |
| T | TIP |
| TB | TELECOMMUNICATIONS BACKBOARD |
| TBB | TELECOMMUNICATIONS BONDING BACKBONE |
| TC | TELECOMMUNICATIONS CONTRACTOR |
| TE | TELECOMMUNICATIONS ENCLOSURE |
| TP | TOUCH PANEL |
| TV | TELEVISION |
| UL | UNDERWRITERS LABORATORY |
| UON | UNLESS OTHERWISE NOTED |
| UPS | UNINTERRUPTIBLE POWER SUPPLY |
| UTIL | UTILITY |
| V | VIDEO |
| WP | WEATHERPROOF |
| WAP | WIRELESS ACCESS POINT |

# TELECOMMUNICATIONS SYMBOLS

(ALL SYMBOLS SHOWN ARE NOT NECESSARILY USED ON DRAWINGS)

## WALL MOUNTED DEVICES

- TELECOMMUNICATIONS GROUNDING BAR
- # (X) — DATA OUTLET
- # (X) — PATCHCORD OUTLET FROM DATA OUTLET (THRU FURNITURE)
- # (X) — DATA BACKBOX FOR CABLE PASS-THRU FROM TELECOM ROOM
- Y, # (X) — BACK BOX WITH POWER, DATA, AND AUDIOVISUAL
- # (X) — WIRELESS ACCESS POINT (WAP)

## VIEW REFERENCE CALLOUTS

- X, X, TTX.X, X, X — ELEVATION CALLOUT
- X, TTX.X — ROOM SECTION CALLOUT
- X, TTX.X — ENLARGED PLAN CALLOUT

# TELECOM DEVICE NUMBERING

1. "X" REFERS TO DATA TYPE. REFER TO CEILING DATA OUTLET SCHEDULE FOR FURTHER INFORMATION.
2. "#" INDICATES NUMBER OF CAT 6A CABLES. REFER TO EQUIPMENT TYPE SCHEDULES FOR ADDITIONAL INFORMATION.
3. "Y" REFERS TO BOX TYPE, REFER TO TT MANUFACTURERS LIST FOR FURTHER INFORMATION.

CEILING — # (X)

FLOOR — Y, # (X)

WALL — Y, # (X)

# TELECOMMUNICATIONS SYMBOLS

(ALL SYMBOLS SHOWN ARE NOT NECESSARILY USED ON DRAWINGS)

## CEILING MOUNTED DEVICES

- CEILING CONDUIT PENETRATION
- HORIZONTAL CONDUIT
- J — JUNCTION BOX MOUNTED 18" AFC
- CABLE TRAY
- LADDER TYPE CABLE TRAY
- # (X) — DATA OUTLET
- # (X) — WIRELESS ACCESS POINT (WAP)

## FLOOR MOUNTED DEVICES

- FLOOR CONDUIT PENETRATION
- RACK # — TELECOMMUNICATIONS 2-POST EQUIPMENT RACK WITH VERTICAL CABLE MANAGERS. LABEL INDICATES FRONT OF RACK.
- RACK # — TELECOMMUNICATIONS 4-POST EQUIPMENT RACK WITH VERTICAL CABLE MANAGERS. LABEL INDICATES FRONT OF RACK.
- CAB # — TELECOMMUNICATIONS EQUIPMENT CABINET. LABEL INDICATES FRONT OF CABINET.
- UNDERFLOOR CABLE TRAY
- FB1, # (X) — FLOOR BOX WITH DATA
- PT, # (X) — POKE-THRU WITH DATA
- CS, # (X) — CONDUIT STUB FOR DATA
- FF — FURNITURE FEED PASS-THRU
- FB1, # (X) — FLOOR BOX WITH POWER, DATA, AND AUDIOVISUAL
- PT, # (X) — POKE-THRU WITH POWER, DATA, AND AUDIOVISUAL
- CS, # (X) — (3) CONDUIT STUBS FOR POWER, DATA, AND AUDIOVISUAL

# GENERAL NOTES

1. IT IS THE INTENT OF THE DRAWINGS AND SPECIFICATIONS TO PROVIDE COMPLETE AND FULLY FUNCTIONAL LOW VOLTAGE INFRASTRUCTURE. THE CONTRACTOR SHALL PROVIDE ALL WORK AND MATERIALS NECESSARY TO DELIVER SUCH A SYSTEM. ANY DISCREPANCIES SHALL BE BROUGHT TO THE OWNER'S ATTENTION.
2. ALL WALL MOUNTED OUTLETS ARE TO BE MOUNTED 18" AFF UNLESS OTHERWISE NOTED. SEE ARCHITECTURAL DRAWINGS FOR BACK BOX MOUNTING STANDARDS.
3. WAPs NOT LOCATED UNDER CEILING TILE NEED TO BE MOUNTED LEVEL WITH ADJACENT FIXTURES.
4. WAPs MOUNTED TO UNDERSIDE OF CEILING TILE WILL HAVE BACK BOX MOUNTED ABOVE ACT. PROVIDE PLENUM RATED PATCH CABLE FROM WAP TO BACK BOX.
5. MOUNT DEVICES AND EQUIPMENT PER SPECIFICATIONS UNLESS OTHERWISE NOTED.
6. MOUNTING HEIGHTS, UNLESS OTHERWISE NOTED, ARE TO CENTER LINE OF DEVICE.
7. THE CONTRACTOR SHALL DETERMINE THE EXACT ROUTING OF RACEWAYS.
8. THE CONTRACTOR SHALL COORDINATE ALL RACEWAY PENETRATIONS AND PROVIDE FIREPROOFING AS APPLICABLE.
9. THE MAXIMUM PERMANENT HORIZONTAL LINK DISTANCE SHALL NOT EXCEED 295' AND THE MAXIMUM HORIZONTAL CHANNEL (LINK PLUS WORKSTATION CABLES AND PATCH CORDS) SHALL BE 328'.
10. ALL HORIZONTAL AND BACKBONE CABLING SHALL BE PLENUM RATED WITH THE EXCEPTION OF CABLING INSTALLED IN UNDERGROUND CONDUITS.
11. IN UNDERGROUND CONDUITS, PROVIDE CABLE LISTED FOR USE IN WET LOCATIONS. PLENUM-RATED CABLE SHALL NOT BE INSTALLED IN ANY UNDERGROUND CONDUIT.
12. CATEGORY CABLING SHALL BE TERMINATED ON 8-PIN, 8- CONDUCTOR (RJ-45) CONNECTORS USING PIN/PAIR ASSIGNMENTS IN ACCORDANCE WITH T568B PIN-OUT SEQUENCE.
13. PROVIDE CATEGORY CABLING, 8-PIN, 8-CONDUCTOR MODULAR TO 8-PIN, 8-CONDUCTOR MODULAR, PATCH CABLES IN QUANTITIES AND LENGTHS SUCH THAT ALL WORKSTATION CABLES ARE CONNECTED.
14. THE USE OF J-HOOKS SHALL BE LIMITED. WHEN USED, J-HOOKS SHALL BE CATEGORY 6 RATED AND SPACED AT INTERVALS NOT TO EXCEED 5'. J-HOOKS SHALL NOT EXCEED 24 CABLES OR 50% FILL, WHICHEVER IS LESS.
15. ALL TELECOMMUNICATIONS RACKS AND SUPPORTING STRUCTURES (CABLE TRAYS, LADDER TYPE TRAY, CONDUITS, AND BASKETS) WITHIN A TELECOMMUNICATIONS ROOM MUST BE BONDED TO THE TELECOMMUNICATIONS ROOM SIGNAL GROUND RING OR BUS BAR (SBB OR PBB) AS DEFINED IN ANSI/TIA-569-E AND IN ANSI/TIA-607-D.
16. USE ONLY COPPER CONDUCTORS FOR GROUND CONNECTIONS. THE USE OF ALUMINUM CONDUCTORS IS NOT ALLOWED.
17. CABLE TRAY SHALL BE WALL MOUNTED (PREFERABLE) OR TRAPEZE MOUNTED. CENTER HUNG CABLE TRAY IS NOT ALLOWED. CABLE TRAY SHALL NOT BE USED TO SUPPORT OTHER ITEMS, CONDUIT, CABLE TRAY, PIPING OF UTILITIES. CABLE TRAY SUPPORTS SHALL BE PER NEMA STANDARDS PUBLICATION VE 2 - METAL CABLE TRAY INSTALLATION GUIDELINES.
18. PROVIDE ACCESS PANELS IN NON LAY-IN CEILINGS SUCH THAT NO POINT IN THE CABLE TRAY SYSTEM IS MORE THAN 20' FROM AN ACCESS POINT.
19. PROVIDE A MINIMUM OF 12" OF CLEARANCE ABOVE THE TOP OF CABLE TRAYS AND A MINIMUM OF 3" CLEARANCE BELOW CABLE TRAYS.
20. REFER TO DIVISIONS 26, 27, 28 AND 33 SPECIFICATIONS FOR TELECOMMUNICATION SYSTEM ITEMS.
21. LABEL ALL CABLE TERMINATIONS AND BOTH ENDS OF ALL CABLES IN ACCORDANCE WITH ANSI/TIA-606-B.
22. ANY MOTOR, TRANSFORMER, OR OTHER ELECTRICAL DEVICE GREATER THAN 5 KVA SHALL HAVE A MINIMUM OF A 4'-0" BUFFER FROM ANY WALL OF A TELECOMMUNICATIONS ROOM.
23. ALL PENETRATIONS SHALL BE SEALED AS REQUIRED TO MAINTAIN FIRE, SMOKE, AND STC SOUND RATINGS.

# TELECOMMUNCATION DRAWINGS LIST

| SHEET NUMBER | SHEET TITLE | REVISION |
|---|---|---|
| TN-001CS | TELECOMMUNICATION SYMBOLS & NOTES | |
| TN-020SP | TELECOMMUNICATION SITE PLAN | |
| TN-100FP | TELECOMMUNICATION FLOOR PLAN - BASEMENT | |
| TN-101FP | TELECOMMUNICATION FLOOR PLAN - LEVEL 1 | |
| TN-401EP | TELECOMMUNICATION ENLARGED PLANS | |
| TN-501DT | TELECOMMUNICATION DETAILS | |
| TN-502DT | TELECOMMUNICATION DETAILS | |
| TN-601DG | TELECOMMUNICATION RISER DIAGRAMS | |

# INDUSTRY STANDARDS

TELECOMMUNICATIONS INSTALLATION SHALL COMPLAY WITH THE MOST CURRENT VERSION OF THE FOLLOWING STANDARDS:

AMERICANS WITH DISABILITIES ACT, 2010

ANSI/TIA-568.0-E - GENERIC TELECOMMUNICATIONS CABLING FOR CUSTOMER PREMISES, 2020

ANSI/TIA-568.2-D - BALANCED TWISTED-PAIR TELECOMMUNICATIONS CABLING AND COMPONENTS STANDARD, 2018

ANSI/TIA-569-D TELECOMMUNICATION PATHWAYS AND SPACES

ANSI/TIA-569.1-E - COMMERCIAL BUILDING TELECOMMUNICATIONS CABLING STANDARD, 2020

ANSI/TIA/EIA-598-D - OPTICAL FIBER CABLE COLOR CODING, 2014

ANSI/TIA-606-D - ADMINISTRATION STANDARD FOR COMMERICAL TELECOMMUNICATIONS INFRASTRUCTURE, 2021

ANSI/TIA-607-D - COMMERCIAL BUILDING GROUNDING (EARTHING AND BONDING REQUIREMENTS FOR TELECOMMUNICATIONS, 2019

ANSI/TIA-758-B - CUSTOMER-OWNED OUTSIDE PLANT TELECOMMUNICATIONS INFRASTRUCTURE STANDARD, 2012

NFPA 70 - NATIONAL ELECTRICAL CODE (NEC)

NFPA 75 - STANDARD FOR THE PROTECTION OF INFORMATION TECHNOLOGY EQUIPMENT

NFPA 76 - STANDARD FOR THE FIRE PROTECTION OF TELECOMMUNICATIONS FACILITIES

BICSI - TELECOMMUNICATIONS DISTRIBUTION METHODS MANUAL (TDMM), 15TH EDITION, 2024

# TELECOMMUNICATIONS SCHEDULES

## DATA DEVICE TYPES

| OUTLET TYPE | DESCRIPTION | MOUNTING | CABLE TYPE | DETAIL |
|---|---|---|---|---|
| (D) | DATA DEVICE | 18" AFF (UON) | CAT6A | |
| (D-AW) | ANIMAL WELFARE CAMERA | 18" AFF (UON) | CAT6A | |
| (D-PS) | PHYSICAL SECURITY CAMERA | 18" AFF (UON) | CAT6A | |
| (E) | ELECTRICAL CONTROL/ METERING | COORDINATE WITH ELEC DRAWINGS | CAT6A | |
| (M) | MECHANICAL CONTROLS | COORDINATE WITH MECH DRAWINGS | CAT6A | |
| (S) | SECURITY DEVICE | RE: TY SERIES | CAT6A | |

## CEILING DATA DEVICE TYPES

| OUTLET TYPE | DESCRIPTION | MOUNTING | CABLE TYPE | DETAIL |
|---|---|---|---|---|
| (D-AW) | SECURITY DEVICE | CEILING | | |
| (D-PS) | SECURITY DEVICE | CEILING | | |
| (WA) | WIRELESS ACCESS POINT | SURFACE MOUNTED | CAT6A | |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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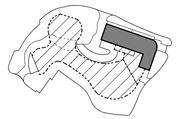

0 1" 2"
1"=1"

GRAPHIC SCALE(S)

DATE: 03/16/2026

SUBMISSION: Above Grade 35% Design Development

| REVISION # | DATE | REVISION DESCRIPTION |
|---|---|---|

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)

ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 2254

DRAWING TITLE: TELECOMMUNICATION SYMBOLS & NOTES

WORKING STAFF: JB (DESIGNED BY) | JB (DRAWN BY) | DV (CHECKED BY)

SHEET NO. 358 OF 378

TN-001CS (DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE)

FOR REFERENCE ONLY

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TN-001CS BY Approver 03/16/2026 Above Grade 35% Design Development

| KEYNOTES | |
|---|---|
| **NUMBER** | **KEYNOTE** |
| 1 | EXISTING TELECOMUNICATIONS PULL BOX. COORDINATE EXACT LOCATION OF SERVICE ENTRANCE WITH CIVIL PLANS. |
| 2 | ROUTE (2) 4" CONDUITS FROM EXISTING PULL BOX TO AV/IT ROOM B1AV1. COORDINATE ROUTE OF SERVICE CONDUITS WITH MECHANICAL SERIES DRAWINGS. |
| 3 | ROUTE 1" CONDUIT TO LOCATION AND PROVIDE EXTERIOR RATED 2-GANG BACK BOX FOR INTERPRETIVE DISPLAY. |
| 4 | PROVIDE PULL BOX OUTSIDE OF AV/IT ROOM FOR SECUIRITY CAMERA CABLING. COORDINATE FINAL PATHWAY ROUTING, PENETRATIONS, SUPPORTS, AND TERMINATION REQUIREMENTS WITH THE SECURITY CONSULTANT, ENCLOSURE ENGINEERS, AND OTHER AFFECTED TRADES. REFER TO SECURITY DRAWINGS FOR NETWORK SEPARATIONS, CAMERA LOCATIONS, MOUNTING DETAILS, AND DEVICE-SPECIFIC REQUIREMENTS. |

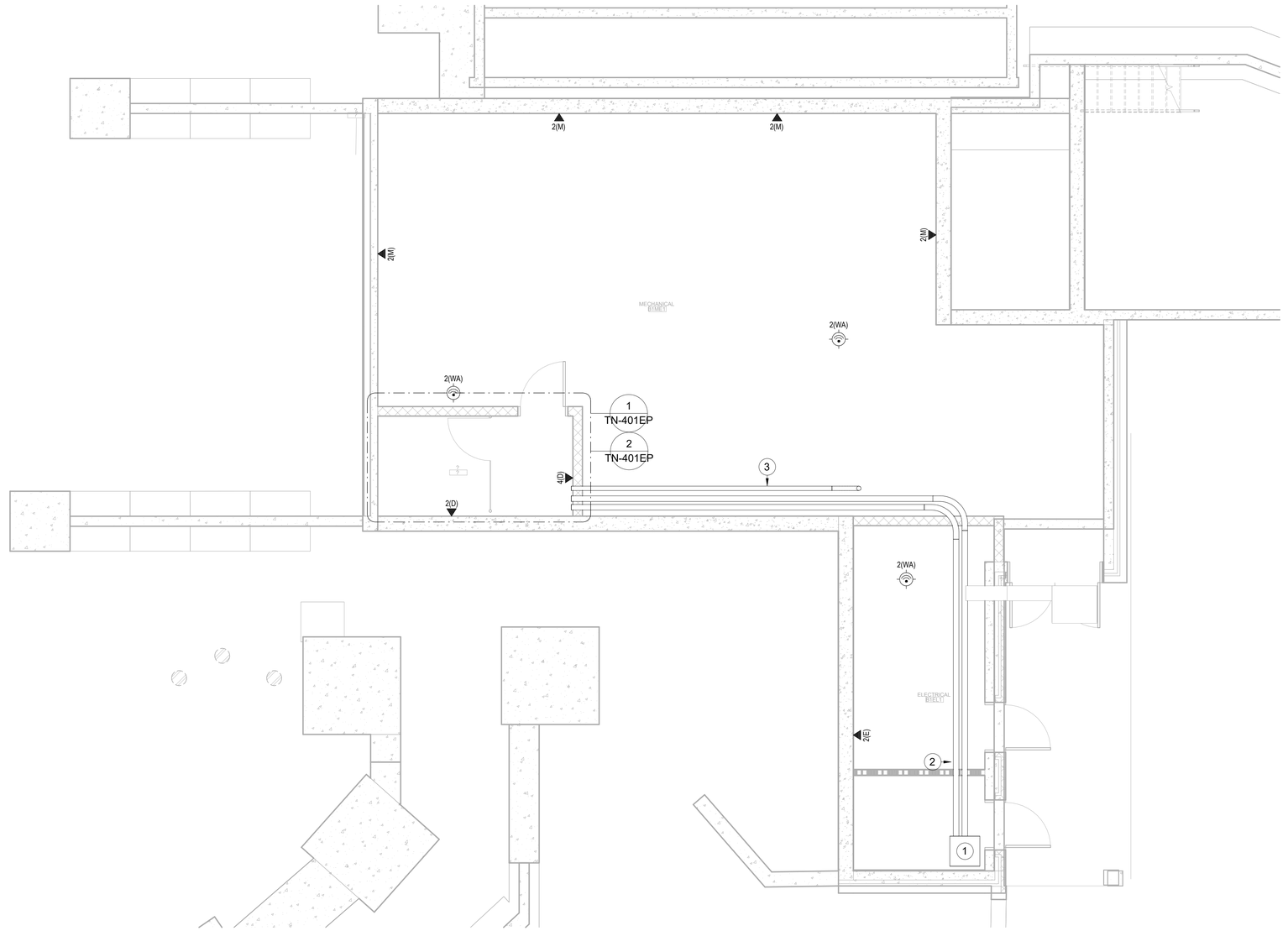
2(D-PS)
2(D-PS)
2(D-PS)
2(D-AW)
2(D-PS)
2(D-PS)
2(D-PS

## KEYNOTES

| NUMBER | KEYNOTE |
|---|---|
| 1 | PROVIDE PULL BOX FOR CONDUITS AS REQUIRED BY DIVISION 27 SPECIFICATIONS AND BICSI STANDARDS. |
| 2 | ROUTE (2) 4" CONDUITS FROM EXISTING PULL BOX TO AV/IT ROOM B1AV1. COORDINATE ROUTE OF SERVICE CONDUITS WITH MECHANICAL SERIES DRAWINGS. |
| 3 | ROUTE 1" CONDUIT TO VIP/KEEPER WORKSPACE 101 ACESSIBLE CEILING FOR NETWORK CABLING. ALL CABLING SHALL BE ROUTED IN CONDUIT EXCEPT WITHIN THE VIP WORKSPACE. CAP WITH NYLON BUSHINGS, FIRE STOP PER NEC. COORDINATE FINAL CONDUIT ROUTE WITH MECHANICAL EQUIPMENT AND DUCTWORK. |

## SHEET NOTES - FLOOR PLAN

1. REFER TO TT-001 FOR GENERAL NOTES, SYMBOL DEFINITIONS, AND ABBREVIATIONS.
2. TELECOM PLANS REPRESENT CABLE QUANTITIES TO BE PULLED AND ARE DIAGRAMMATIC IN NATURE.
3. DEVICE HEIGHTS PROVIDED ARE FOR ESTIMATION PURPOSES. REFER TO ARCHITECTURAL (A-SERIES) PLANS FOR COORDINATION OF DEVICE HEIGHTS.
4. SECURITY DROP LOCATIONS ARE FOR COORDINATION PURPOSES ONLY. VERIFY FINAL CAMERA LAYOUT WITH SECURITY DRAWINGS PRIOR TO INSTALLATION OF ANY SECURITY INFRASTRUCTURE. REFER TO TY-SERIES PLANS FOR SECURITY NETWORK DEVICE DROP REQUIREMENTS.
5. ROUTE 1" CONDUIT FROM DEVICE LOCATION TO TELECOMMUNICATION PULL BOX IN BASEMENT LEVEL. ALL CABLING IN BASEMENT LEVEL SHALL BE ROUTED IN CONDUIT. CAP WITH NYLON BUSHINGS, FIRE STOP PER NEC.
6. ROUTE 1" CONDUIT TO VIP/KEEPER WORKSPACE 101 ACCESSIBLE CEILING FOR NETWORK CABLING ON LEVEL 1. ALL CABLING SHALL BE ROUTED IN CONDUIT EXCEPT WITHIN THE VIP WORKSPACE. CAP WITH NYLON BUSHINGS, FIRE STOP PER NEC.




1 TN-100FP — TELECOMMUNICATION PLAN - BASEMENT

SCALE = 1/4" = 1'-0"

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TN-100FP BY Approver 03/16/2026 Above Grade 35% Design Development

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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STAMP

TRUE NORTH

0 1" 2"
1"=1"

GRAPHIC SCALE(S)

DATE: 03/16/2026

SUBMISSION: Above Grade 35% Design Development

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 2254

DRAWING TITLE: TELECOMMUNICATION FLOOR PLAN - BASEMENT

WORKING STAFF: JB (DESIGNED BY) | JB (DRAWN BY) | DV (CHECKED BY)

SHEET NO. 360 OF 378

TN-100FP (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

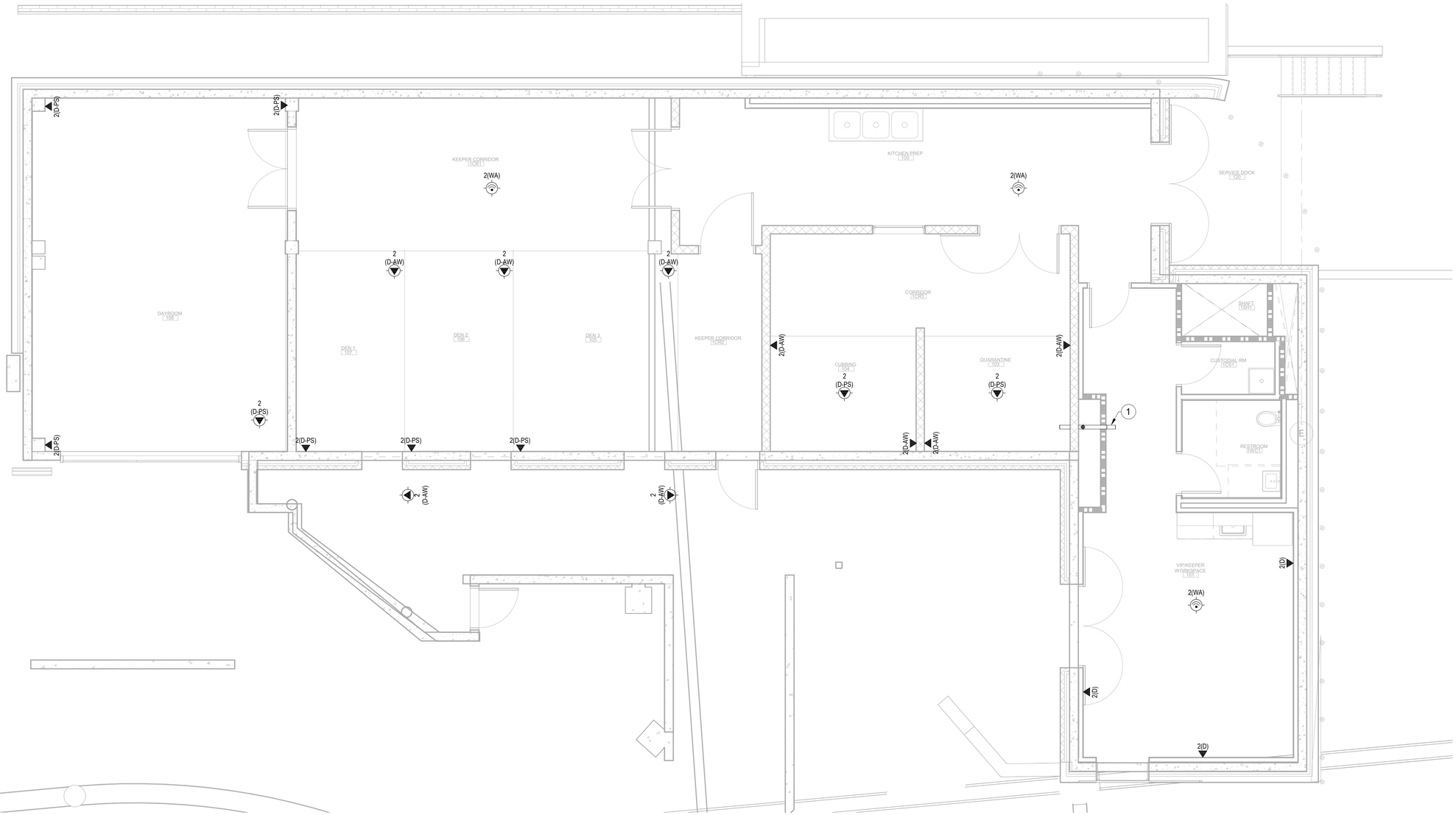
FOR REFERENCE ONLY

## KEYNOTES

| NUMBER | KEYNOTE |
|---|---|
| 1 | ROUTE 1" CONDUIT TO VIP/KEEPER WORKSPACE 101 ACESSIBLE CEILING FOR NETWORK CABLING. ALL CABLING SHALL BE ROUTED IN CONDUIT EXCEPT WITHIN THE VIP WORKSPACE. CAP WITH NYLON BUSHINGS, FIRE STOP PER NEC. COORDINATE FINAL CONDUIT ROUTE WITH MECHANICAL EQUIPMENT AND DUCTWORK. |

## SHEET NOTES - FLOOR PLAN

1. REFER TO TT-001 FOR GENERAL NOTES, SYMBOL DEFINITIONS, AND ABBREVIATIONS.
2. TELECOM PLANS REPRESENT CABLE QUANTITIES TO BE PULLED AND ARE DIAGRAMMATIC IN NATURE.
3. DEVICE HEIGHTS PROVIDED ARE FOR ESTIMATION PURPOSES. REFER TO ARCHITECTURAL (A-SERIES) PLANS FOR COORDINATION OF DEVICE HEIGHTS.
4. SECURITY DROP LOCATIONS ARE FOR COORDINATION PURPOSES ONLY. VERIFY FINAL CAMERA LAYOUT WITH SECURITY DRAWINGS PRIOR TO INSTALLATION OF ANY SECURITY INFRASTRUCTURE. REFER TO TY-SERIES PLANS FOR SECURITY NETWORK DEVICE DROP REQUIREMENTS.
5. ROUTE 1" CONDUIT FROM DEVICE LOCATION TO TELECOMMUNICATION PULL BOX IN BASEMENT LEVEL. ALL CABLING IN BASEMENT LEVEL SHALL BE ROUTED IN CONDUIT. CAP WITH NYLON BUSHINGS, FIRE STOP PER NEC.
6. ROUTE 1" CONDUIT TO VIP/KEEPER WORKSPACE 101 ACCESSIBLE CEILING FOR NETWORK CABLING ON LEVEL 1. ALL CABLING SHALL BE ROUTED IN CONDUIT EXCEPT WITHIN THE VIP WORKSPACE. CAP WITH NYLON BUSHINGS, FIRE STOP PER NEC.




1 TN-101FP — TELECOMMUNICATION PLAN - LEVEL 1

SCALE = 1/4" = 1'-0"

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TN-101FP BY Approver 03/16/2026 Above Grade 35% Design Development

# ALF

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STAMP

TRUE NORTH

0 1" 2"
1"=1"
GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
| 03/16/2026 | Above Grade 35% Design Development |

| REVISION # DATE | REVISION DESCRIPTION |
|---|---|
| | |

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 2254 |
| DRAWING TITLE | TELECOMMUNICATION FLOOR PLAN - LEVEL 1 |
| WORKING STAFF | JB (DESIGNED BY) / JB (DRAWN BY) / DV (CHECKED BY) |
| SHEET NO. | 361 OF 378 |
| | TN-101FP (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE) |

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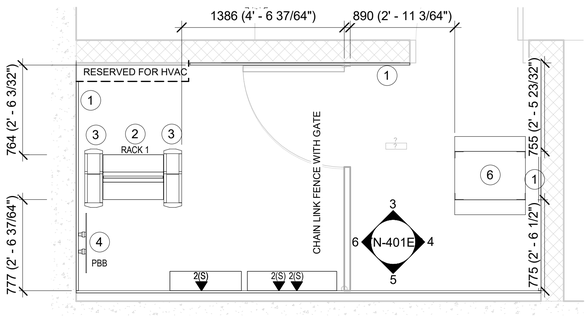
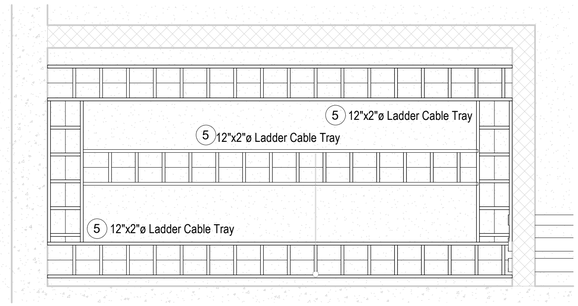


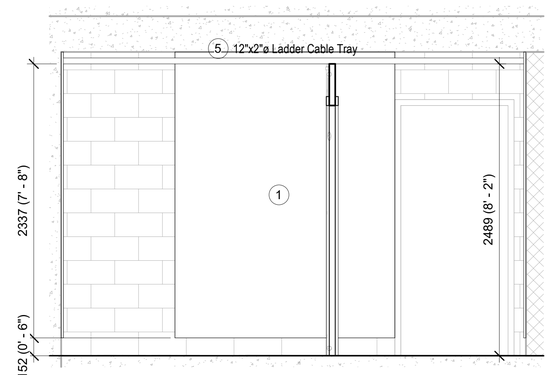

1 TN-401EP TELECOM ENLARGED PLAN - AV/IT ROOM B1AV1

SCALE = 1/2" = 1'-0"



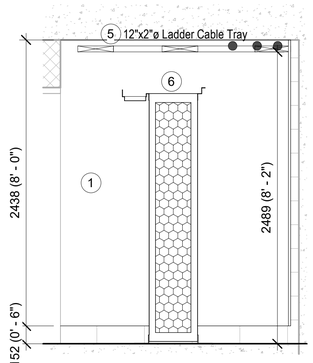

2 TN-401EP TELECOM ABOVE RACK PLAN - AV/IT ROOM B1AV1

SCALE = 1/2" = 1'-0"



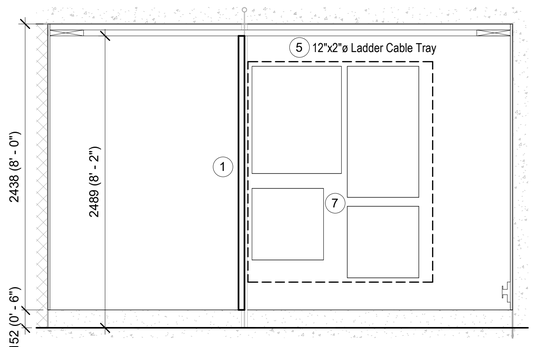

3 TN-401EP AV/IT ROOM B1AV1

SCALE = 1/2" = 1'-0"



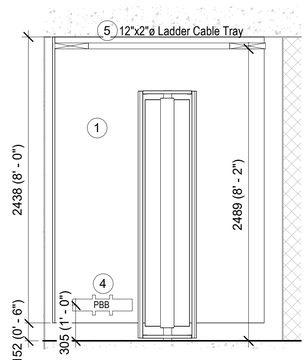

4 TN-401EP AV/IT ROOM B1AV1

SCALE = 1/2" = 1'-0"




5 TN-401EP AV/IT ROOM B1AV1

SCALE = 1/2" = 1'-0"




6 TN-401EP AV/IT ROOM B1AV1

SCALE = 1/2" = 1'-0"

## KEYNOTES

| NUMBER | KEYNOTE |
|---|---|
| 1 | PROVIDE 3/4" AC GRADE PLYWOOD, VOID-FREE WITH THE A-SIDE OUT. PAINT WITH OFF-WHITE FIRE-RETARDANT PAINT ON FACE AND EDGES. DO NOT PAINT OVER FIRE-RATED STAMP. MOUNT AT 6" AFF TO 8'-6" AFF. |
| 2 | PROVIDE 19" X 7'H TIA 2-POST TELECOMMUNICATIONS EQUIPMENT RACK. |
| 3 | PROVIDE 6" WIDE HIGH-DENSITY VERTICAL CABLE MANAGER |
| 4 | PROVIDE PRIMARY BONDING BUSBAR, MOUNT AT 72" AFF. |
| 5 | PROVIDE LADDER TYPE CABLE TRAY WITHIN TELECOMMUNICATIONS SPACE AS DIMENSIONED. BOND TO LOCAL BONDING BUSBAR W/ GREEN INSULATED #6-AWG GROUND WIRE. |
| 6 | 4-POST AUDIOVISUAL EQUIPMENT RACK. |
| 7 | ACCESS CONTROL DOOR CONTROLLER PANEL LOCATION. ROUTE ALL SECURITY CABLING FROM ACCESS CONTROL DOOR TO THIS LOCATION AND TERMINATE TO DOOR CONTROLLERS. COORDINATE FINAL SECURITY PANEL LOCATION WITH THE SECURITY DESIGNER SO THE INSTALLED PANEL AND ALL ASSOCIATED POWER EQUIPMENT MAINTAIN REQUIRED NEC WORKING CLEARANCES. |

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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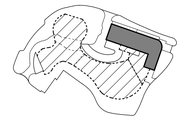

0 1" 2"

1"=1"

GRAPHIC SCALE(S)

DATE 03/16/2026

SUBMISSION Above Grade 35% Design Development

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME: ARABIAN LEOPARD FACILITY (ALF)

ADDRESS: 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 2254

DRAWING TITLE: TELECOMMUNICATION ENLARGED PLANS

WORKING STAFF: JB (DESIGNED BY) JB (DRAWN BY) DV (CHECKED BY)

SHEET NO. 362 OF 378

TN-401EP (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

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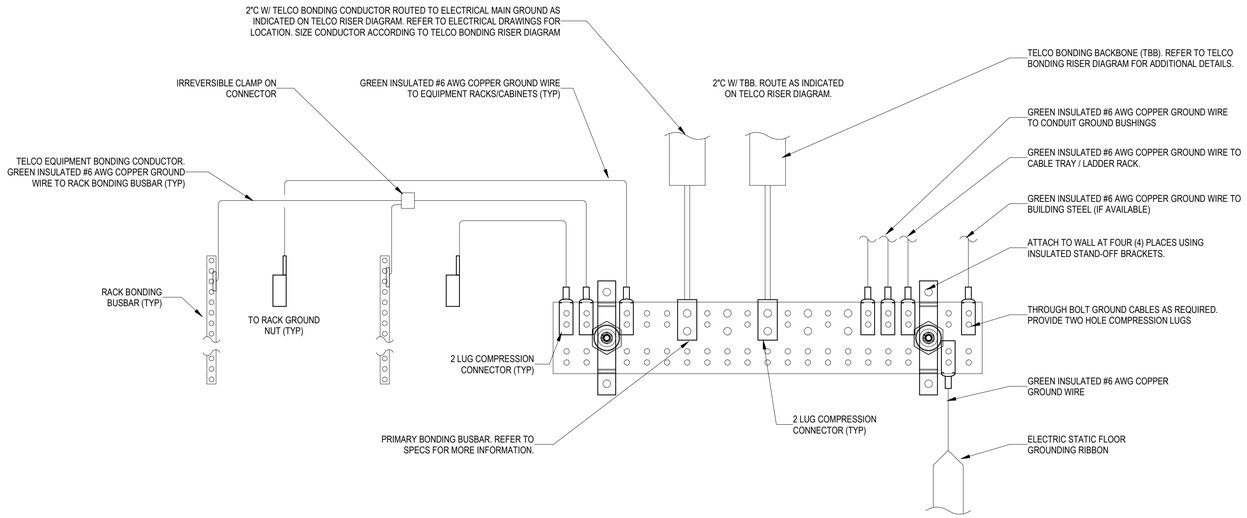
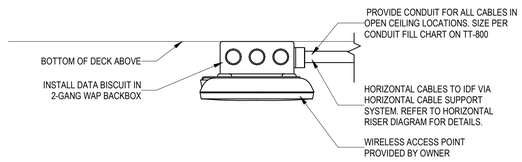
6/15/26 1:05:20 PM
TN-401EP BY Approver 03/16/2026 Above Grade 35% Design Development



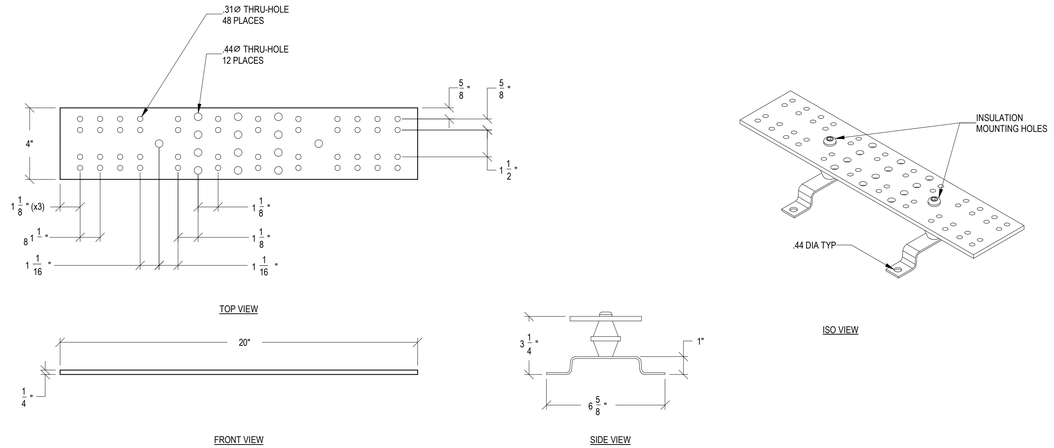

## 1 TN-501DT TT-BTS-D-GND-ATTACHMENTS PBB

SCALE =NTS



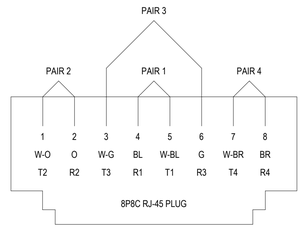

## 2 TN-501DT WAP DETAIL - SURFACE MOUNTED

SCALE =NTS




## 3 TN-501DT TELECOMMUNICATIONS MAIN GROUNDING BUSBAR - 20 INCHES

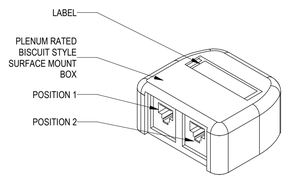
SCALE =NTS




**T568B**

## 4 TN-501DT T568B JACK WIRING DIAGRAM

SCALE =NTS




## 5 TN-501DT BISCUIT STYLE JACK

SCALE =NTS

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0 1" 2"
1"=1"

**GRAPHIC SCALE(S)**

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|---|---|
| 03/16/2026 | Above Grade 35% Design Development |

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Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 2254 |
| DRAWING TITLE | TELECOMMUNICATION DETAILS |
| WORKING STAFF | JB (DESIGNED BY) / JB (DRAWN BY) / DV (CHECKED BY) |

SHEET NO. 363 OF 378

TN-501DT (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

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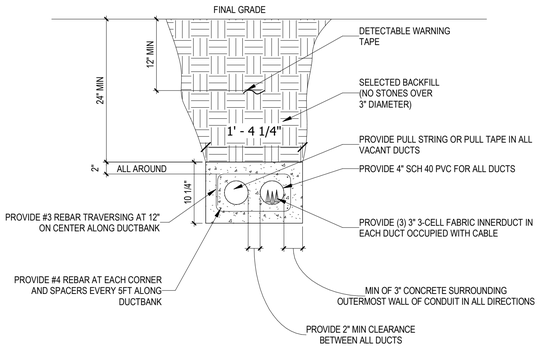
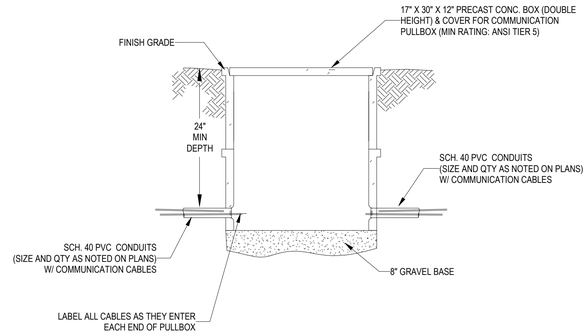


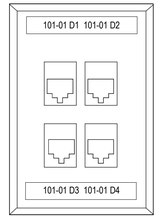

1 / TN-502DT — 2-WAY 4"C CONCRETE ENCASED DUCT BANK

SCALE =NTS




2 / TN-502DT — PULLBOX (ANSI TIER 5)

SCALE =NTS




XXX - YY DZ

XXX - ROOM NUMBER
YY - OUTLET NUMBER IN SEQUENTIAL ORDER AT DOOR AND MOVING TO THE LEFT
D - MEDIA DESIGNATION
D - DATA
F - OPTICAL FIBER
RS - ROOM SCHEDULING PANEL OUTLET
Z - SEQUENTIAL NUMBER OF JACK IN FACEPLATE

(EXAMPLE LABELS SHOWN FOR FIRST OUTLET IN ROOM 101)

3 / TN-502DT — FACEPLATE LABELING

SCALE =NTS

# ALF

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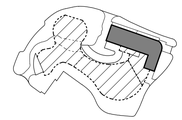

0 1" 2"
1"=1"

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PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)
OPDC PROJECT NUMBER: 2533105
A/E PROJECT NUMBER: 2254

DRAWING TITLE: TELECOMMUNICATION DETAILS
WORKING STAFF: JB (DESIGNED BY) JB (DRAWN BY) DV (CHECKED BY)

SHEET NO. 364 OF 378

TN-502DT (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

FOR REFERENCE ONLY

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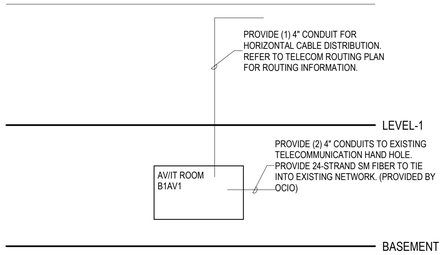
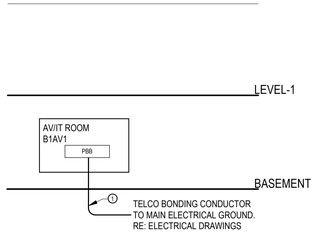
TN-502DT BY Approver 03/16/2026 Above Grade 35% Design Development




1 TN-601DG TELECOMMUNICATIONS BACKBONE RISER

SCALE =NTS




3 TN-601DG TELECOMMUNICATIONS BONDING RISER

SCALE =NTS

**TELCO BONDING SYSTEM NOTES:**

1. ALL TELCO BONDING SYSTEM CONDUCTORS TO BE CONTAINED IN 2"C WHEN NOT WITHIN A TELECOMMUNICATION SPACE.

2. CONNECT ALL MID-SPAN SBBs WITH A CONDUCTOR EQUAL TO THE SIZE OF THE TELCO BONDING BACKBONE AND IRREVERSIBLE BONDING CLAMP. DO NOT DAISY-CHAIN SBBs.

**LINETYPE LEGEND**

- ——— TELCO BONDING CONDUCTOR - SIZE ACCORDING TO GROUND CONDUCTOR SIZING CHART.
- —·—·— TELCO BONDING BACKBONE - SIZE ACCORDING TO GROUND CONDUCTOR SIZING CHART.
- – – – – BACKBONE BONDING CONDUCTOR- SIZE EQUAL TO THE TELCO BONDING BACKBONE.

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

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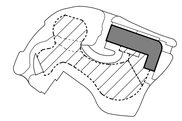

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BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF); 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 2254

DRAWING TITLE: TELECOMMUNICATION RISER DIAGRAMS

WORKING STAFF: JB (DESIGNED BY) | JB (DRAWN BY) | DV (CHECKED BY)

SHEET NO. 365 OF 378

TN-601DG (DISCIPLINE TYPE SEQUENCE DRAWING TYPE)

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TN-601DG BY Approver 03/16/2026 Above Grade 35% Design Development

# SYMBOLS LEGEND

### ACCESS CONTROL

| SYMBOL ETR | SYMBOL DEMO | SYMBOL NEW | SECTION | DESCRIPTOR |
|---|---|---|---|---|
| | | | 28 10 10 | CARD READER, 4 = RM4 ARM/ DISARM READER |
| RM | RM | RM | 28 10 10 | READER MODULE |
| | | | 28 15 15 | ELECTRIFIED LOCK, IN = ELECTRIFIED LOCKSET WITH INTEGRAL REQUEST TO EXIT, M=MAGNETIC LOCK, D=DELAYED EGRESS |
| | | | 28 13 00 | DOOR CONTACT, S= SURFACE, W=WIDE GAP |
| REX | REX | REX | 28 10 00 | REQUEST TO EXIT SENSOR |
| | | | 28 10 00 | LOCAL SOUNDER |
| ISTAR | ISTAR | ISTAR | 28 10 00 | ISTAR PANEL, GCM AND TWO (2) ACM |
| I/O | I/O | I/O | 28 10 00 | INPUT/ OUTPUT MODULES |
| | | | 28 05 07 | POWER SUPPLY |
| BAT | BAT | BAT | 28 05 07 | BATTERY |

### INTRUSION DETECTION

| SYMBOL ETR | SYMBOL DEMO | SYMBOL NEW | SECTION | DESCRIPTOR |
|---|---|---|---|---|
| | | | 28 13 00 | 360 DEGREE MOTION SENSOR |
| | | | 28 13 00 | DIRECTIONAL MOTION SENSOR, 90 DEGREE, L=LONG RANGE |
| | | | 28 13 00 | GLASS BREAK SENSOR, S= SEISMIC, A=ACOUSTIC |
| | | | 28 13 00 | SEISMIC SENSOR |

### VIDEO SURVEILANCE

| SYMBOL ETR | SYMBOL DEMO | SYMBOL NEW | SECTION | DESCRIPTOR |
|---|---|---|---|---|
| | | | 28 20 00 | SINGLE IMAGER DOME IP CAMERA |
| | | | 28 20 00 | MULTI-IMAGER DOME IP CAMERA |
| | | | 28 20 00 | PAN-TILT-ZOOM IP CAMERA |
| | | | 28 20 00 | DETENTION GRADE IP CAMERA |
| NVR | NVR | NVR | 28 20 00 | NETWORK VIDEO RECORDER |

### INTERCOMMUNICATIONS

| SYMBOL ETR | SYMBOL DEMO | SYMBOL NEW | SECTION | DESCRIPTOR |
|---|---|---|---|---|
| M | M | M | TBD | INTERCOM MASTER |
| | | | TBD | INTERCOM |
| V | V | V | TBD | VIDEO INTERCOM |
| ECB | ECB | ECB | TBD | EMERGENCY CALLBOX |

### MISCELLANIOUS

| SYMBOL ETR | SYMBOL DEMO | SYMBOL NEW | SECTION | DESCRIPTOR |
|---|---|---|---|---|
| EN | EN | EN | 28 05 31.10 | ETHERNET SWITCH |
| SPL | SPL | SPL | XXX | SECURITY PANEL LOCATION |
| CPP | CPP | CPP | XXX | COPPER PATCH PANEL |
| FPP | FPP | FPP | XXX | FIBER PATCH PANEL |
| DGP | DGP | DGP | XXX | DATA GATHERING PANEL |
| SEC | SEC | SEC | XXX | SECURITY ENCLOSURE |
| GICPE | GICPE | GICPE | XXX | GALLERY INFRASTRUCTURE CONSOLIDATION POINT ENCLOSURE, CEILING GRID MOUNTED |
| GJB | GJB | GJB | XXX | GALLERY JUNCTION BOX, 12 X 12 X 4, 4 INPUT/4 OUTPUT TERMINAL BLOCKS, CEILING GRID MOUNTED |
| TVSS | TVSS | TVSS | 28 05 09.10 | TRANSIENT VOLTAGE SURGE SUPPRESSION |
| WR | WR | WR | XXX | WIRELESS RECEIVER, ALARM |

# CAMERA TYPES AND PLACEMENT NOTES

1. CAMERA ARE PLACED TO SUPPORT SPECIFIC USES. CAMERA FIELDS OF VIEWS HAVE BEEN PROVIDED TO ILLUSTRATE THE DESIRED COVERAGE.
2. THE FOLLOWING CAMERAS ARE USED AS FOLLOWS:
   - FIELD DOME CAMERAS ARE PLANS FOR PORTAL MONITORING OR VIDEO TRAPS TO CAPTURE GENERAL SURVEILLANCE OF THE DOORWAY OR EXHIBIT OR GALLERY ENTRANCE. CAMERAS MAY ALSO BE USED FOR FIXED HIGH VALUE OBJECT COVERAGE INCLUDING ANIMALS.
   - MULTI-IMAGER CAMERAS ARE USED TO PROVIDE GENERAL VIDEO SURVEILLANCE OF LARGE CORRIDORS, LOBBIES OR GATHERING AREAS (GALLERIES, LOUNGES, ETC.).
   - PAN-TILT-ZOOM CAMERAS ARE PRIMARILY USED FOR OBJECT OR LARGE PERIMETER WINDOW ALARM ZONE PIN-POINT CALL UPS. ANIMAL AREAS MAY USE PTZ CAMERAS
   - ANTI-GRIP HIGH SECURITY CAMERAS IN DENS AND OTHER ANIMAL AREAS. CAMERAS WILL BE EQUIPPED WITH MICROPHONES.
3. ALL ZOO SECURITY CAMERAS WILL BE SUPPORTED BY INFRARED ILLUMINATORS AND MICROPHONES.
4. ZOO SECURITY CAMERA ANNOTATIONS ARE AS FOLLOWS:
   CR = CORNER MOUNT
   CM = CEILING MOUNT
   CW = CORNER WALL MOUNT
   PM = POLE MOUNT
   BM = BUILDING MOUNTED, PENDANT WALL MOUNT
   SM = STRUCTURE MOUNTED, STRUCTURAL BEAM MOUNTED
   IR = INFRARED ILLUMINATOR, 360 DEGREE FLOOD
   TM = TUNNEL MOUNT

# WIRE AND CABLE NOTES

1. FOLLOW CABLE SCHEDULE ON THIS SHEET FOR ALL WIRE AND CABLE REQUIREMENTS. ALL WIRING AND CABLING IS BY THE CONTRACTOR. SECURITY AND TELECOMMUNICATIONS SYSTEM COPPER SIGNAL CONDUCTORS SHALL NOT BE INSTALLED IN THE SAME CONDUIT AS 120 VAC LINES.
2. CABLE FILL, CONDUCTORS AND CONDUIT SHALL BE SIZED BY THE CONTRACTOR AND INSTALLED IN ACCORDANCE WITH THE NATIONAL ELECTRIC CODE (NEC) AND ALL LOCAL CODES HAVING JURISDICTION. NUMBERS OF CONDUCTORS REQUIRED MAY VARY ON THE BASIS OF THE MANUFACTURERS EQUIPMENT SELECTED BY THE CONTRACTOR. MINIMUM CONDUIT SIZE SHALL BE 3/4 INCH, SIZE UP TO PROVIDE 40% CAPACITY. CONDUITS SUPPORTING BACKBONE OR RIZERS WILL BE SIZED TO PROVIDE 50% CAPACITY.
3. ALL SECURITY WIRING SYSTEMS SHALL BE CONSISTENTLY COLOR-CODED AND TAGGED. COORDINATION OF EXACT WIRE CODING AND TAGGING IS MANDATORY. THE CONTRACTOR IS RESPONSIBLE FOR COORDINATING WIRING FROM EQUIPMENT HARDWARE MANUFACTURERS TERMINAL STRIPS TO FINAL CONNECTION POINTS OF EQUIPMENT WITHIN THE SCOPE OF THE PROJECT.
4. ALL TELECOMMUNICATIONS SECURITY CAMERA AND INTERCOM CABLING WILL BE CATEGORY 6 CABLING MEETING TIA SPECIFICATIONS.
5. SECURITY CABLING INCLUDING ETHENET CABLING WILL NOT SHARE CONDUITS DESIGNATED FOR TELECOM OR OTHER NON-SECURITY CONDUITS.

# GENERAL NOTES

1. THE CONTRACTOR SHALL PROVIDE, INSTALL, DOCUMENT AND TEST A COMPLETE TURNKEY SECURITY SYSTEM. THE SECURITY SYSTEM WILL INCLUDE A VIDEO MANAGEMENT SYSTEMS, ACCESS AND CONTROLS SYSTEM, INTRUSION DETECTION SYSTEM, AND INTERCOM.
2. ALL DRAWINGS AND SPECIFICATIONS MUST BE USED IN COMBINATION TO OBTAIN COMPLETE CONSTRUCTION INFORMATION. ALL FLOOR AND WALL CONDUIT PENETRATIONS SHALL BE SLEEVED AND SEALED TO THE FULL THICKNESS OF THE PENETRATION WITH AN APPROVED FIRE RESISTIVE SEALANT. FOLLOW PROJECT FIRE STOPPING REQUIREMENTS.
3. THE CONTRACTOR WILL BE RESPONSIBLE FOR COORDINATING FINAL SECURITY EQUIPMENT LAYOUT INCLUDING MOCKUPS OF DEVICES AND EQUIPMENT MOUNTED IN VIEW OF THE PUBLIC. CONFLICTS WILL IMMEDIATELY BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER, WHO WILL APPROVE CHANGES PRIOR TO THE INITIATION OF WORK.
4. CONTRACTOR SHALL PROVIDE ALL REQUIRED EQUIPMENT AND HARDWARE IN ORDER TO MEET THE INTENT OF THE DESIGN AND TO PROVIDE A COMPLETE AND FULLY OPERATIONAL SYSTEM.
5. ALL PRIMARY 120 VAC INPUT POWER FOR POWER SUPPLIES AND OTHER SYSTEM COMPONENTS SHALL BE PROVIDED BY THE CONTRACTOR EXCEPT AS NOTED. THE CONTRACTOR WILL BE RESPONSIBLE FOR COORDINATION OF SECURITY SYSTEM POWER REQUIREMENTS. ALL SECURITY EQUIPMENT POWER SHALL BE SUPPLIED ON DEDICATED EMERGENCY CIRCUITS.
6. ALL POWER DISTRIBUTION AND SUPPLY PANELS, WIRING, CONDUCTORS, CONDUIT, RACEWAY SYSTEMS AND TERMINATION'S FOR THE SUPPLY OF 120 VAC POWER TO THE SECURITY SYSTEM COMPONENTS SHALL BE PROVIDED BY THE CONTRACTOR. IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO COORDINATE WITH THE ELECTRICAL CONTRACTOR AND PROVIDE COMPLETE INFORMATION REGARDING ELECTRICAL REQUIREMENTS. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE INTERCONNECTION OF ELECTRONIC SECURITY SYSTEM COMPONENTS REQUIRING 120 VAC TO THE PROVIDED SUPPLIES.
7. THE CONTRACTOR SHALL BE RESPONSIBLE FOR VERIFYING SMOKE AD FIRE PARTITION LOCATIONS, AND AVOIDING CONFLICTS WITH LIGHTS, DIFFUSERS, GRILLS, AND OTHER OBSTRUCTIONS. LOCATIONS OF ALL VISIBLE DEVICES AND EQUIPMENT SHALL BE COORDINATED WITH DRAWINGS.
8. SECURITY CONTRACTOR TO PROVIDE ALL COORDINATION WITH RESPECT TO DOOR HARDWARE INCLUDING COORDINATING CONTROL REQUIREMENT SEQUENCING WITH POWER ACTUATORS, FIRE ALARM AND OTHER POWER ASSISTED CONTROLS.

# PATHWAY NOTES

1. CONTRACTOR WILL PROVIDE A COMPLETE CONDUIT SYSTEM FOR ALL SECURITY WIRE AND CABLE. COMPLETE CONDUIT SYSTEM INCLUDES BUT NOT LIMITED TO COMPONENTS, MATERIALS, OFFSETS, BENDS, ELBOWS, BACK BOXES, JUNCTION BOXES, ENCLOSURES AND OTHER SPECIFIC ELEMENTS THAT MAY BE REQUIRED FOR A COMPLETE. ALL SECURITY CABLING WILL BE IN CONDUIT. THE CONTRACTOR WILL BE REQUIRED FOR COORDINATING ALL CONDUIT REQUIREMENTS.
2. PRIOR TO INSTALLATION OF DEVICES, FIELD-VERIFY AND COODINATE FINAL DEVICE LOCATIONS WITH DUE REGARD TO AS-BUILT CONDITIONS, COORDINATION WITH THE WORK OF OTHER TRADES AND MANUFACTURERS' RECOMMENDATIONS.
3. CONDUIT SHALL BE SIZED BY THE CONTRACTOR AND INSTALLED IN ACCORDANCE WITH THE NATIONAL ELECTRIC CODE (NEC) AND ALL LOCAL CODES HAVING JURISDICTION. NUMBERS OF CONDUCTORS REQUIRED MAY VARY ON THE BASIS OF THE MANUFACTURERS EQUIPMENT SELECTED BY THE CONTRACTOR. MINIMUM CONDUIT SIZE SHALL BE 3/4 INCH, SIZE UP TO PROVIDE 40% CAPACITY. CONDUITS SUPPORTING BACKBONE OR RIZERS WILL BE SIZED TO PROVIDE 50% CAPACITY.
4. THE CONTRACTOR SHALL CONCEAL CONDUITS TO THE EXTENT POSSIBLE AND UNOBTRUSIVELY BY FOLLOWING THE STRUCTURE OF THE BUILDING. ALSO, A MOCK UP CONSTRUCTION SAMPLE IS TO BE PROVIDED IN ALL PUBLIC AREAS TO DETERMINE THE INSTALLATION MEANS AND METHODS FOR APPROVAL BY THE AUTHORIZED CONTRACT REPRESENTATIVE.
5. CONDUIT BENDS WILL BE SWEEPS TO MAINTAIN PROPER CABLE BEND RADIUS. ALL CONDUITS SHALL BE DEBURRED, HAVE 200LB PULLSTRINGS AND BUSHINGS INSTALLED. CABLE WILL NOT BE PLACED INTO CONDUITS OR SLEEVES WITHOUT FIRST DEBURRING AND INSTALLING BUSHINGS. ANY CABLE DAMAGED WILL BE REPLACED ENTIRELY.
6. PULL BOXES AND JUNCTION BOXES SHALL BE PROVIDED PER LOCAL CODES AND STANDARDS TO FACILITATE THE EASE IN PULLING OF FUTURE CABLING. AT A MINIMUM, PROVIDE ONE (1) PULL BOX FOR EVERY 100 FEET (30480 MM) OF STRAIGHT CONDUIT RUN, AND ONE (1) PULL BOX AFTER TWO (2) 90-DEGREE BENDS. ALL CONDUITS WILL BE PULLED WITH PULL STRING.
7. CABLE TRAY WITHIN THE SECURITY CLOSET ONLY WILL PERMIT ETHERNET CABLES WITHIN CABLE TRAY. CABLES WILL BE PLACED IN A RANDOM FASHION WITHOUT BUNDLING THE CABLES. CABLE TIES SHALL NOT BE USED.
8. CABLES SHALL EXIT CABLE TRAY BY MEANS OF CABLE TRAY RADIUS DROP (I.E., WATERFALL) OR CABLES TRANSITIONING FROM CABLE TRAY TO CONDUIT SPANNING NOT MORE THAN 3 INCHES (75 MM).
9. WORK AREA OUTLET CONDUIT SHALL BE A MINIMUM OF 1 INCH (25 MM) CONDUIT, UNLESS OTHERWISE NOTED.
10. SINGLE GANG PLASTER REDUCING RINGS SHALL BE PROVIDED WITH DOUBLE-GANG BACKBOXES, UNLESS OTHERWISE NOTED.
11. HORIZONTAL CABLE SHALL BE PLENUM RATED.
12. PRIOR TO ORDER, SUBMIT FACEPLATE, JACK, AND DEVICE COLOR SAMPLES TO CONTRACTING OFFICER FOR REVIEW.
13. REFER TO ARCHITECTURAL DRAWINGS AND SPECIFICATIONS FOR PROPER OUTLET MOUNTING HEIGHTS AND LOCATIONS PRIOR TO ROUGH-IN.
14. REFER TO STRUCTURAL DRAWINGS AND SPECIFICATIONS FOR DIMENSIONED CORE HOLE PENETRATIONS.
15. REFER TO LIFE SAFETY DRAWINGS FOR FIRE & SMOKE WALL LOCATIONS. CONDUIT PENETRATIONS SHALL BE PROPERLY FIRESTOPPED PER CODE. CABLE TRAY SHALL NOT PENETRATE FIRE & SMOKE WALLS. LISTED FIRESTOP ASSEMBLIES SHALL BE USED TO PENETRATE FIRE & SMOKE WALLS FOR CABLE TRAY PATHWAYS.
16. ALL CONDUITS FROM DEVICE TO TRAY SYSTEM SHALL HAVE A PULL LINE INSTALLED.
17. ALL SECURITY CONDUITS WILL BE MARKED IN ACCORDANCE WITH SI OPS MARKING STANDARDS. AT MINIMUM, MARKING CONDUITS EVERY 10 METERS (32 FEET). ALL SECURITY JUNCTION BOX COVERS WILL BE MARKED IN ACCORDANCE WITH SI OPS MARKING STANDARDS.

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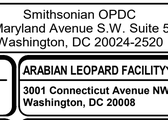

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BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITYY (ALF) — 3001 Connecticut Avenue NW Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: SECURITY REFERENCE

WORKING STAFF

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| MMH | CW | MMH |

SHEET NO. 366 OF 378

TY 000 CS

DISCIPLINE | TYPE | SEQUENCE | DRAWING TYPE

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## ABBREVIATIONS

| | |
|---|---|
| AC | ALTERNATING CURRENT |
| AFF | ABOVE FINISHED FLOOR |
| ANSI | AMERICAN NATIONAL STANDARDS INSTITUTE |
| ASTM | AMERICAN SOCIETY FOR TESTING AND MATERIALS |
| C | CONDUIT |
| CAT | CATEGORY |
| CM | CEILING MOUNTED |
| DC | DIRECT CURRENT |
| EMI | ELECTRICAL MAGNETIC INTERFERENCE |
| EMT | ELECTRICAL METALLIC TUBING |
| ESS | ELECTRONIC SECURITY SYSTEM |
| FO | FIBER OPTIC |
| IMT | INTERMEDIATE METALLIC TUBING |
| ITS | INFORMATION TECHNOLOGY SYSTEMS |
| LAN | LOCAL AREA NETWORK |
| MAX | MAXIMUM |
| MIN | MINIMUM |
| MISC | MISCELLANEOUS |
| MM | MULTIMODE |
| MTD | MOUNTED |
| NEC | NATIONAL ELECTRICAL CODE |
| NEMA | NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION |
| NTS | NOT TO SCALE |
| OC | ON CENTER |
| OPE | OWNER PROVIDED EQUIPMENT |
| OPS | OFFICE OF PROTECTION SERVICES |
| OSP | OUTSIDE PLANT |
| PC | PERSONAL COMPUTER |
| PP | PATCH PANEL |
| PR | PAIR |
| PTZ | PAN-TILT-ZOOM CAMERA |
| RM | ROOM |
| RMU | RACK MOUNTING UNIT (1.75") |
| SI | SMITHSONIAN INSTITUTION |
| SM | SINGLEMODE |
| STP | SHIELDED TWISTED PAIR |
| TBD | TO BE DETERMINED |
| TGB | TELECOMMUNICATIONS GROUNDING BUSBAR |
| TMGB | TELECOMMUNICATIONS MAIN GROUNDING BUSBAR |
| TR | TELECOMMUNICATIONS ROOM |
| TVSS | TRANSIENT VOLTAGE SURGE SUPPRESSION |
| TYP | TYPICAL |
| UL | UNDERWRITERS LABORATORIES |
| UON | UNLESS OTHERWISE NOTED |
| UPS | UNINTERRUPTED POWER SUPPLY |
| UTP | UNSHIELDED TWISTED PAIR |
| VAC | VOLTS ALTERNATING CURRENT |
| VDC | VOLTS DIRECT CURRENT |
| VP | VANDAL PROOF |
| WM | WALL MOUNTED |
| WPE | WEATHER PROOF ENCLOSURE |

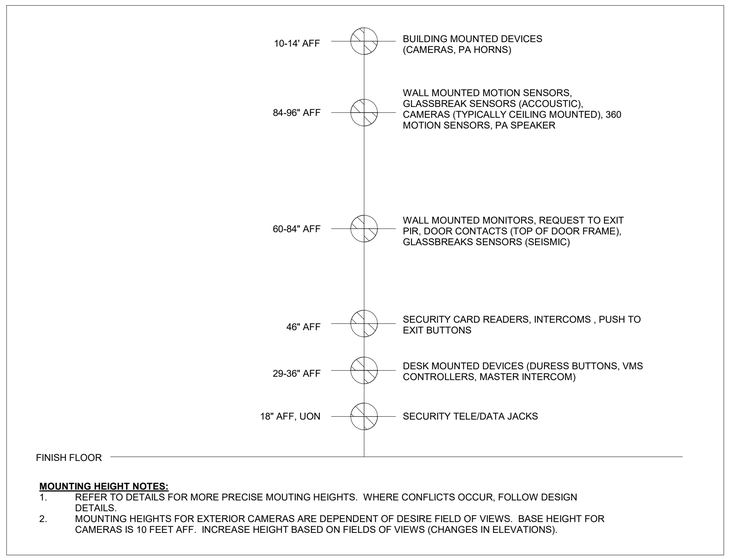
## REMOVAL NOTES

1. PRIOR TO INITIATING ANY REMOVAL WORK THE CONTRACTOR WILL PERFORM A "FUNCTIONAL CHECK" OF ALL ELECTRONIC SECURITY SYSTEMS (ESS) EQUIPMENT INCLUDING BUT NOT LIMITED TO ACCESS CONTROL PANELS, READER INTERFACE MODULES, CARD READERS, VIDEO SURVEILLANCE CAMERAS, ELECTRIFIED LOCKS, INTERCOMS, DOOR CONTACTS, MOTION SENSORS, SIEZMIC SENSORS, ASSOCIATED CONDUIT SYSTEMS (INCLUDING BACKBOXES AND ENCLOSURES) AND CABLING, AND ASSOCIATED DATA TRANSPORT SYSTEMS "FUNCTIONAL CHECK" RESULTS WILL BE COMPILED AND DELIVERED TO THE OWNER. CONTRACTOR WILL NOTIFY THE OWNER OF ANY EQUIPMENT THAT DOES NOT FUNCTION PROPERLY.
2. THE TERM "REMOVE" MEANS DISCONNECT, DE-PROGRAM, REMOVE FROM SITE AND LEGALLY DISPOSE. DISPOSAL OF ELECTRONICS AND METALS WILL BE THROUGH RECYCLING WHERE DETERMINE PRACTICAL BY THE OWNER.
3. SECURITY EQUIPMENT, COMPONENTS, AND DEVICES SCHEDULED FOR DEMOLITIOBN WILL BE REMIOVED IN SUCH A MANNER TO PRESERVE THE APPEARANCE AND OPERABILITY TO THE GREATEST EXTENT POSSIBLE.
4. CONTRACTOR WILL INVENTORY AND TURNOVER TO OWNER ALL EQUIPMENT DEEMED REUSABLE BY THE OWNER. INVENTORY WILL BE ON DIGITAL HAND RECEIPT AND WILL INCLUDE MANUFACTURER, MODEL, SERIAL NUMBER, AS APPLICABLE MAC AND IP ADDRESS AND ESTIMATED DATE MANUFACTURED. EXISTING SECURITY SYSTEMS AND ASSOCIATED DEVICES WILL REMAIN OPERATIONAL AS LONG AS POSSIBLE. COORDINATE SECURITY SECURITY SYSTEM OUTAGE DATE AND TIME AS LEAST TWO WEEKS PRIOR WITH COTR, AND OFFICE OF PROTECTION SERVICES.
5. IF SYSTEM OUTAGE EXTENDS BEYOND NORMAL BUSINESS HOURS (MONDAY - FRIDAY, 8AM - 5PM EXCEPT FEDERAL HOLIDAYS), THE CONTRACTOR WILL INCUR THE COST FOR UTILIZING THE OWNERS UNIFORMED GUARD SERIVCE TO PROVIDE ADDITIONAL SECURUITY NECESSARY TO SECURE TH AFFECTED AREA.
6. ALL EXISTING ELECTRICAL WORK WHICH WILL NOT BE RENDERED OBSOLETE AND WHICH MAY BE DISTURBED DUE TO CHANGES REQUIRED UNDER CONTRACT WILL BE RESTORED TO ITS ORIGINAL OPERATIONAL CONDITION. OTHER ELECTRICAL MATERIAL RENDERED OBSOLETE WILL BE DISCONNECTED AND REMOVED BACK TO SOURCE. REMOVAL WILL STOP WHERE DAMAGE WOULD OCCUR IF FUIRTHER REMOVAL OCCURRED, CUT CONDUIT OFF AND REMOVE WIRE CONDUCTORS UNDER THESE CONDITIONS. ABONDONED OUTLETS IN WALL WILL BE REMOVED AND WALL PATCHED AND PAINTED TO MATCH ADJACENT FINISH. ABANDONED OPUTLETS ABOVE THE FINSHED CEILING WILL BE COVERED WITH BRUSHED STAINLESS STEEL COVERPLATES OR PAINTYED TO WHICH ADJACENT FINISH. ALL UNUSED WIRING AND DEVICES WILL BE REMOVED UON.
7. PROVIDE MOCK-UP OF TEMPORARY SECURITY SYSTEM THROUGH THE COTR TO THE OFFICE OF PROTECTION SERVICES FOR APPROVAL.

## DEMOLITION NOTES

1. REFER TO DEMOLITION PLANS FOR ALL DEVICES SCHEDULED FOR DEMOLITION.
2. THE FOLLOWING APPLIES FOR EACH DEVICE SCHEDULED FOR DEMOLITION.
   A. PROVIDE LISTING OF DEVICES TO BE DEMOLISHED TO AUTHOIRZED CONTRACTING REPRESENTATIVE. LISTING WILL INCLUDE DEVICE DESCRIPTION, LOCATION, SERIAL NUMBER AND MAC ADDRESS. CONTRACTING REPRESENTATIVE WILL COORDINATE WHICH DEVICES WILL BE RETAINED BY SI OPS AS SPARES.
   B. REMOVE DEVICE, MOUNTING HARDWARE, CONDUITS AND CABLING BACK TO SOURCE. FOR DEVICES IDENTIFIED BY OPS TO BE RETAINED, PROVIDE DEVICE ON RECEIPT TO SI OPS REPRESETATIVE.
3. SPARE DEVICES WILL NOT BE RE-USED ON THE PROJECT.

## MOUNTING HEIGHTS




**MOUNTING HEIGHT NOTES:**
1. REFER TO DETAILS FOR MORE PRECISE MOUTING HEIGHTS. WHERE CONFLICTS OCCUR, FOLLOW DESIGN DETAILS.
2. MOUNTING HEIGHTS FOR EXTERIOR CAMERAS ARE DEPENDENT OF DESIRE FIELD OF VIEWS. BASE HEIGHT FOR CAMERAS IS 10 FEET AFF. INCREASE HEIGHT BASED ON FIELDS OF VIEWS (CHANGES IN ELEVATIONS).

## SECURITY DEVICE TAG LEGEND

| TAG DEVICE DESCRIPTION | TAG BREAKDOWN |
|---|---|
| ALARM INPUT TERMINAL: CONNECTED TO I-8 MODULE BOARD | AH### ACM:# 18:#:## XX<br>BUILDING ABBREVIATION<br>ISTAR NUMBER<br>ACM BOARD NUMBER<br>DEVICE SIGNAL TYPE<br>I-8 INPUT BOARD INPUT NUMBER<br>I-8 INPUT BOARD NUMBER |
| ALARM INPUT TERMINAL: CONNECTED TO ISTAR ULTRA BOARD | AH### ACM:# M:# XX<br>BUILDING ABBREVIATION<br>ISTAR NUMBER<br>ACM BOARD NUMBER<br>DEVICE SIGNAL TYPE<br>ACM BOARD INPUT NUMBER |
| RS485 TERMINAL: CONNECTED TO ISTAR ULTRA BOARD | AH### ACM:# RDR:#<br>BUILDING ABBREVIATION<br>ISTAR NUMBER<br>ACM BOARD NUMBER<br>READER TERMINAL NUMBER |
| OUTPUT TERMINAL: CONNECTED TO R-8 RELAY MODULE BOARD | AH### ACM:# R8:#:## XX<br>BUILDING ABBREVIATION<br>ISTAR NUMBER<br>ACM BOARD NUMBER<br>DEVICE SIGNAL TYPE<br>R-8 OUTPUT BOARD TERMINAL NUMBER<br>R-8 OUTPUT BOARD NUMBER |

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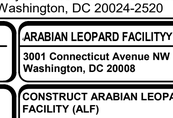

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A/E PROJECT NUMBER: 00100084

DRAWING TITLE / WORKING STAFF: SECURITY REFERENCE

| DESIGNED BY | DRAWN BY | CHECKED BY |
|---|---|---|
| MMH | CW | MMH |

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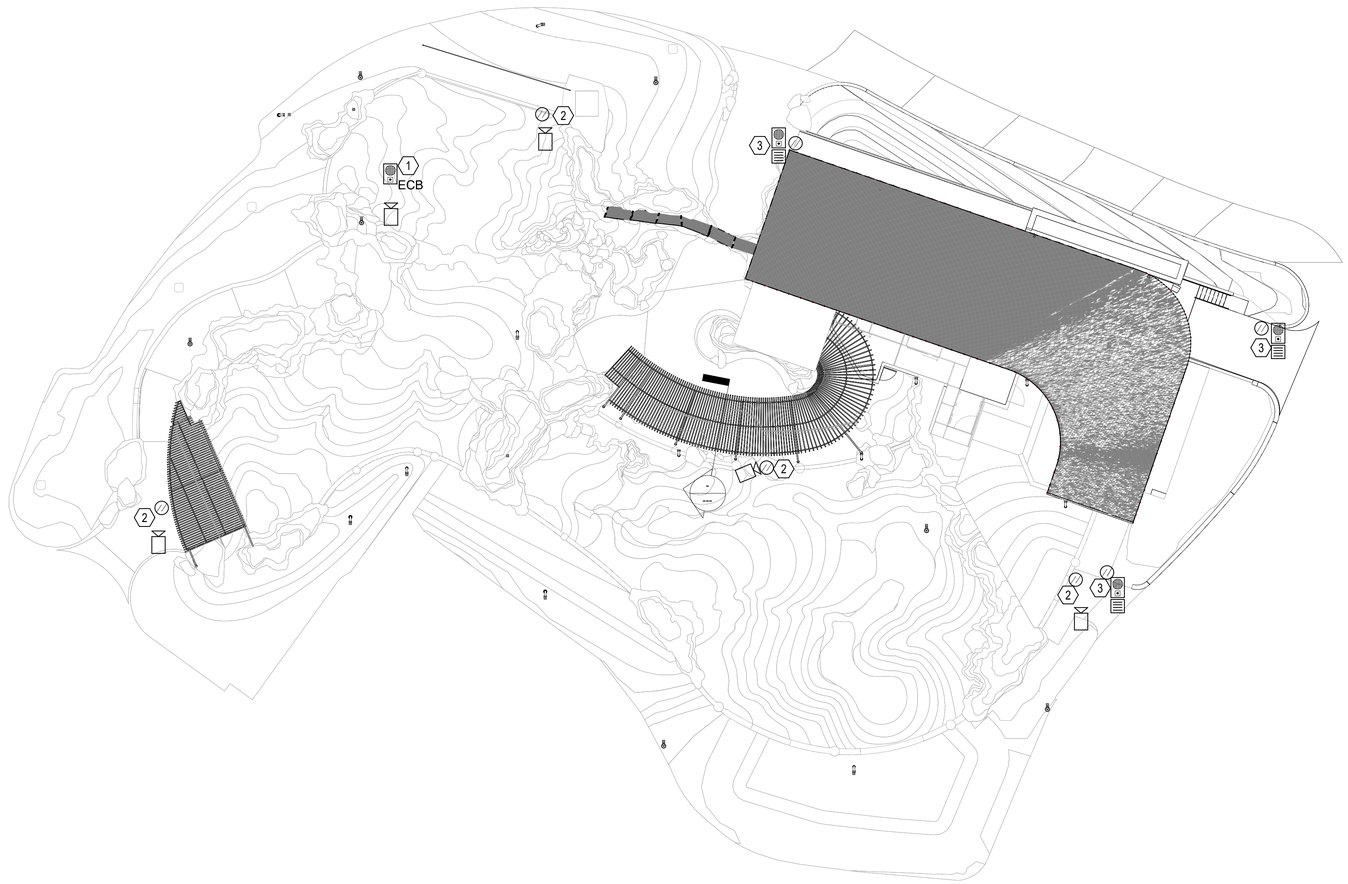


## SITE PLAN - SECURITY

1 / TY 100 SP

SCALE = 1/16" = 1'-0"

## GENERAL NOTES

1. COORDINATE ALL SITE RELATED WORK WITH SITE CIVIL WORK. A SEPARATE CONDUIT FOR SECURITY WILL BE REQUIRED.
2. SITE RELATED CAMERAS ARE CURRENTLY MOUNTED TO BUILDING EXTERIORS OR STRUCTURES.
3. DESIGN INTENT IS TO PROVIDE 90% COVERAGE OF THE PEDESTRIAN WALKING PATHS, BUILDING PERIMETERS AND GENERAL VICINITY.

## GENERAL SHEET NOTES

1 ROUTE ALL ZOO CAMERA CABLING TO THE TELECOM RACK. ALL OPS CAMERA CABLING WILL BE ROUTED TO THE OPS EQUIPMENT RACK.

## KEYNOTES

1. EMERGENCY CALL BOX LOCATION IN PEDESTRIAN TUNNEL. CAMERA SUPPORTS VIEWS OF THE TUNNEL AND CALL BOX.
2. DOOR LOCATION WILL REQUIRE DOOR CONTACT AND CAMERA COVERAGE.
3. CARD READER AND INTERCOM SUPPORTS GATE OPERATION.

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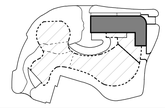
PROJECT NORTH



KEY PLAN

0 50' 100'

1"=50'-0"

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| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SECURITY SITE PLAN |
| WORKING STAFF | MMH (DESIGNED BY) / CW (DRAWN BY) / MMH (CHECKED BY) |

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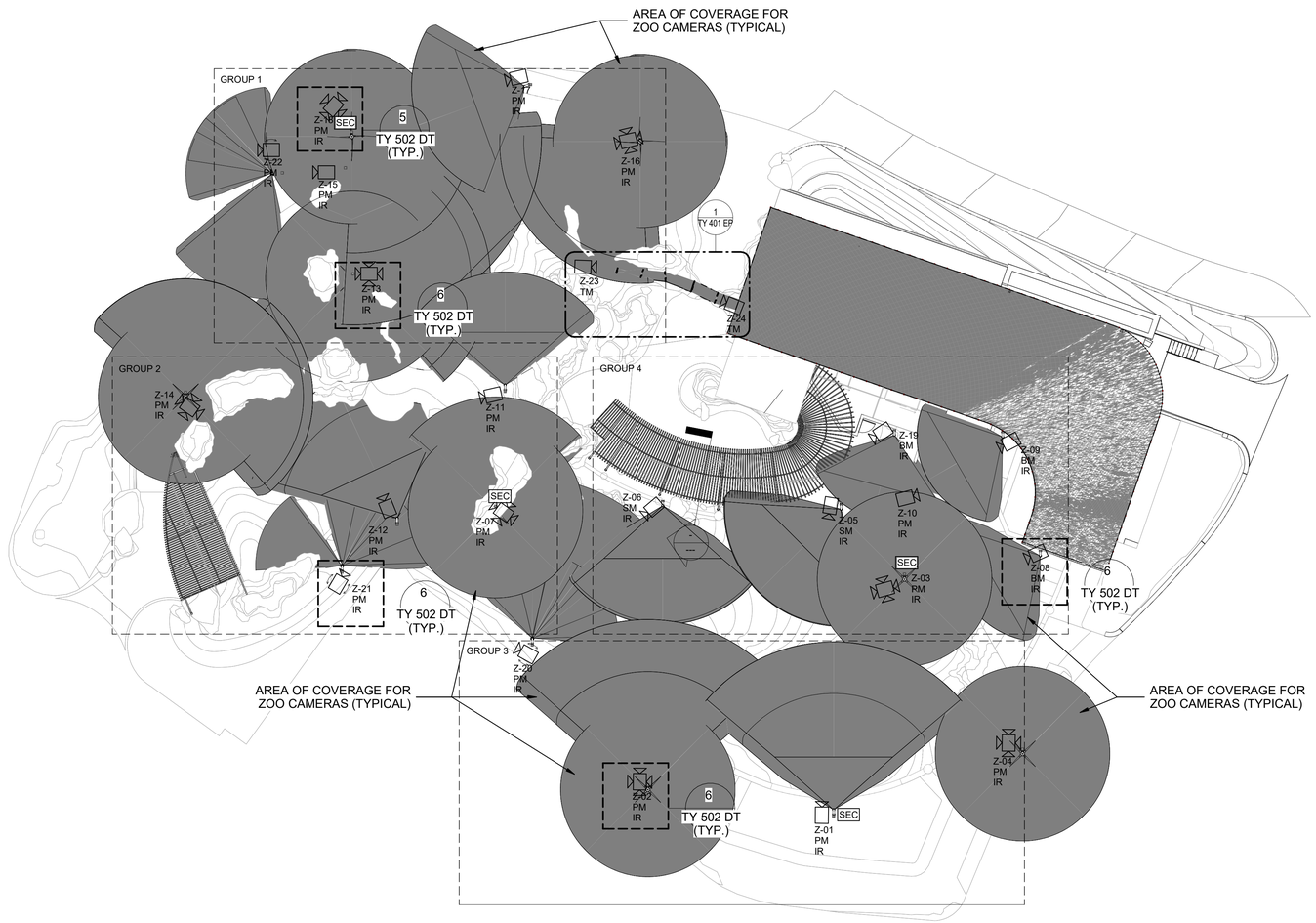


1 / TY 101 SP

## SITE PLAN - ZOO SECURITY

SCALE = 1/16" = 1'-0"

## GENERAL NOTES

1. COORDINATE ALL SITE RELATED WORK WITH SITE CIVIL WORK. A SEPARATE CONDUIT FOR SECURITY WILL BE REQUIRED.
2. SITE RELATED CAMERAS ARE CURRENTLY MOUNTED TO BUILDING EXTERIORS OR STRUCTURES.
3. DESIGN INTENT IS TO PROVIDE 90% COVERAGE OF THE PEDESTRIAN WALKING PATHS, BUILDING PERIMETERS AND GENERAL VICINITY.

## GENERAL SHEET NOTES

1 ROUTE ALL ZOO CAMERA CABLING TO THE TELECOM RACK. ALL OPS CAMERA CABLING WILL BE ROUTED TO THE OPS EQUIPMENT RACK.

## KEYNOTES

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OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: ZOO SECURITY SITE PLAN

WORKING STAFF: MMH (DESIGNED BY) | CW (DRAWN BY) | MMH (CHECKED BY)

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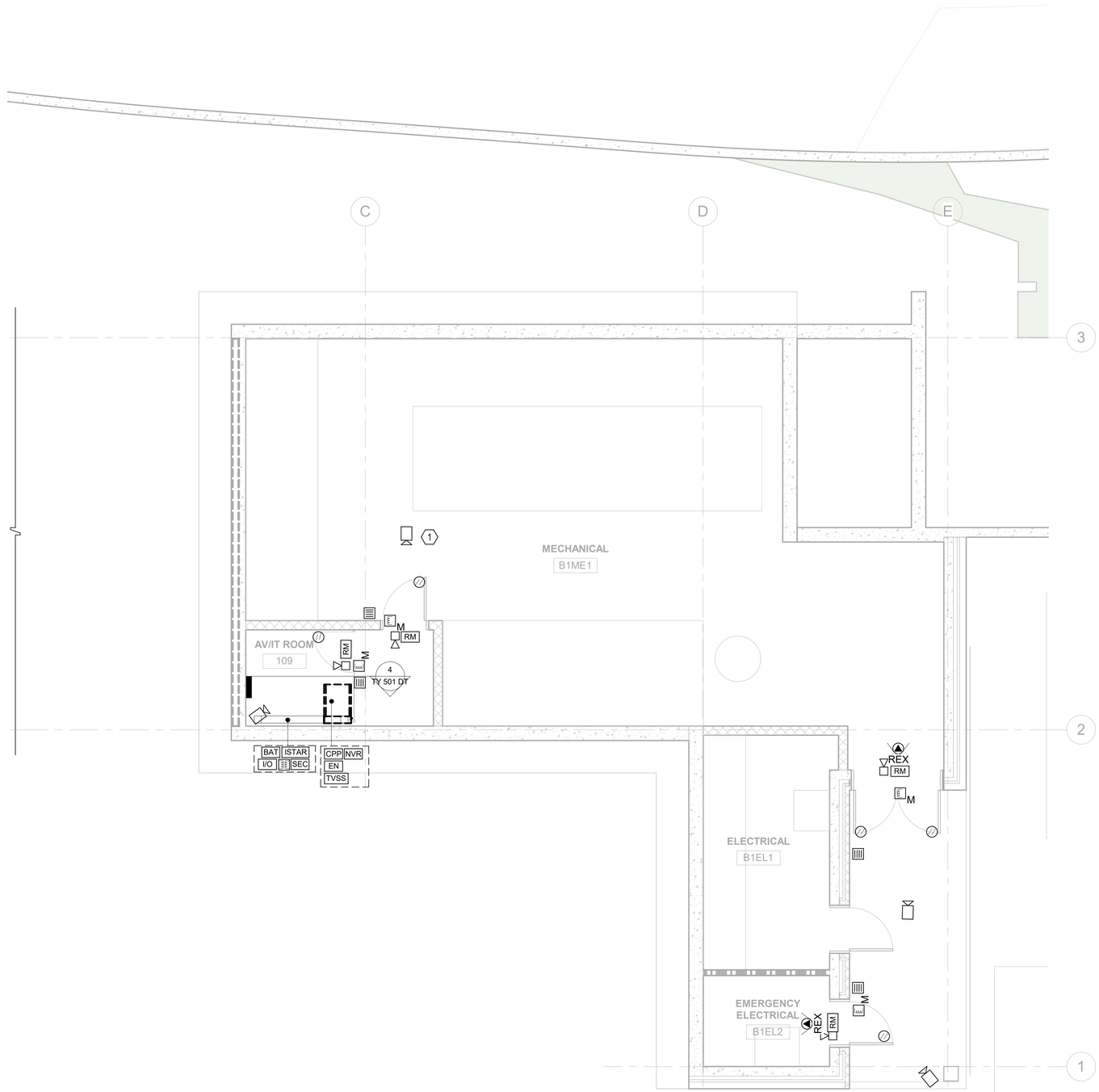


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TY 102 FP

## FINISHED FLOOR - BASMENT LEVEL - SECURITY

SCALE = 1/4" = 1'-0"

## GENERAL NOTES

1 INTERIOR CAMERAS ARE CEILING MOUNTED.

## GENERAL SHEET NOTES

1 ROUTE ALL ZOO CAMERA CABLING TO THE TELECOM RACK. ALL OPS CAMERA CABLING WILL BE ROUTED TO THE OPS EQUIPMENT RACK.

## KEYNOTES

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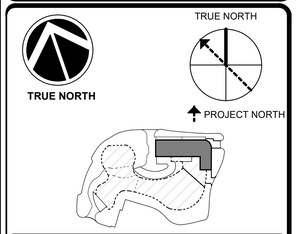


KEY PLAN

0 2' 4' 8'
1/4"=1'-0"

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| OPDC PROJECT NUMBER | 2533105 |
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| DRAWING TITLE | SECURITY BASEMENT LEVEL PLAN |
| WORKING STAFF | DESIGNED BY: MMH; DRAWN BY: CW; CHECKED BY: MMH |
| SHEET NO. | 370 OF 378 |
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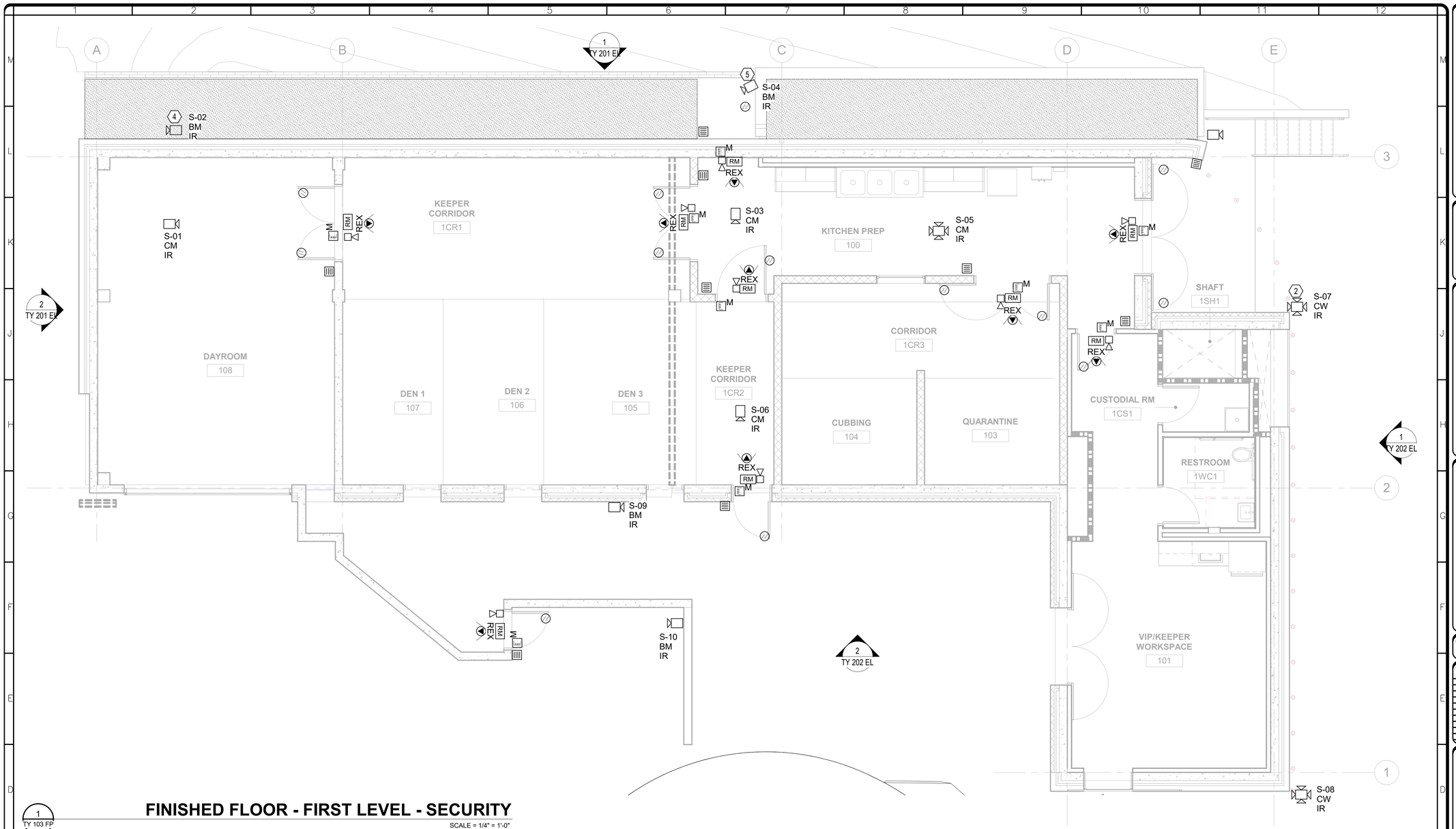


## FINISHED FLOOR - FIRST LEVEL - SECURITY

1 TY 103 FP

SCALE = 1/4" = 1'-0"

## GENERAL NOTES

1 ROUTE ALL ZOO CAMERA CABLING TO THE TELECOM RACK. ALL OPS CAMERA CABLING WILL BE ROUTED TO THE OPS EQUIPMENT RACK.

## GENERAL SHEET NOTES

1 INTERIOR CAMERAS ARE CEILING MOUNTED.

## KEYNOTES

1 SOFFIT MOUNTED CAMERA
2 SOFFIT MOUNTED CAMERA
4 WALL MOUNTED
5 WALL MOUNTED

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TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

0 2' 4' 8'

1/4"=1'-0"

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| DRAWING TITLE | SECURITY FIRST LEVEL PLAN |
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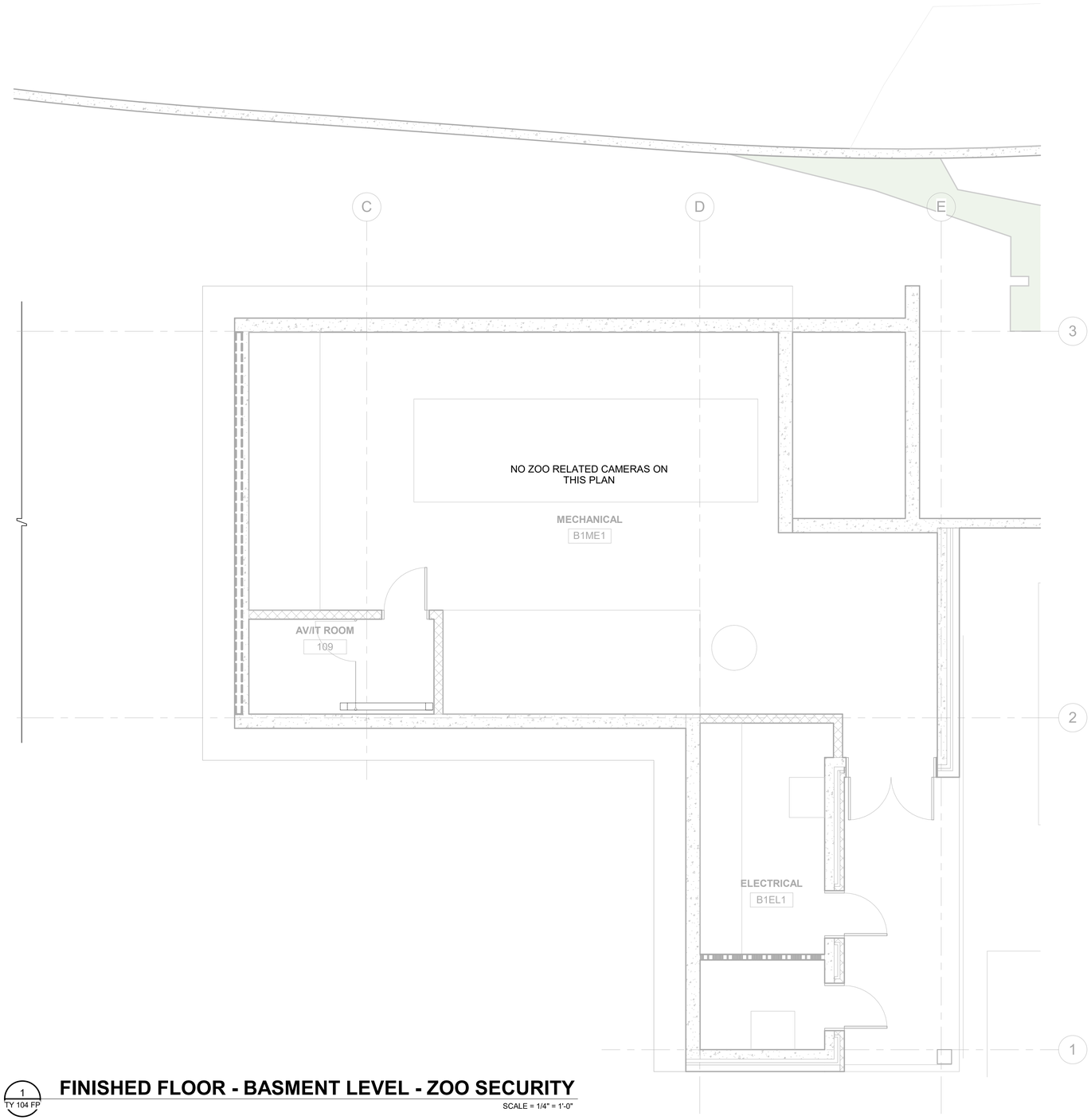


1 / TY 104 FP

## FINISHED FLOOR - BASMENT LEVEL - ZOO SECURITY

SCALE = 1/4" = 1'-0"

## GENERAL NOTES

1 INTERIOR CAMERAS ARE CEILING MOUNTED.

## KEYNOTES

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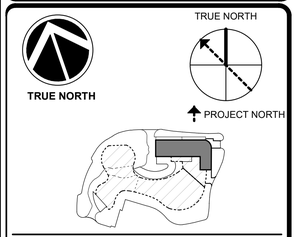


KEY PLAN

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
| 06/15/2026 | 65% CONSTRUCTION DOCUMENTS |

| REVISION # DATE | REVISION DESCRIPTION |
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600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

| | |
|---|---|
| BUILDING NAME | ARABIAN LEOPARD FACILITYY (ALF) |
| ADDRESS | 3001 Connecticut Avenue NW Washington, DC 20008 |
| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ZOO SECURITY BASEMENT LEVEL PLAN |
| WORKING STAFF | MMH / CW / MMH (DESIGNED BY / DRAWN BY / CHECKED BY) |
| SHEET NO. | 372 OF 378 |
| | TY 104 FP (DISCIPLINE TYPE SEQUENCE DRAWING TYPE) |

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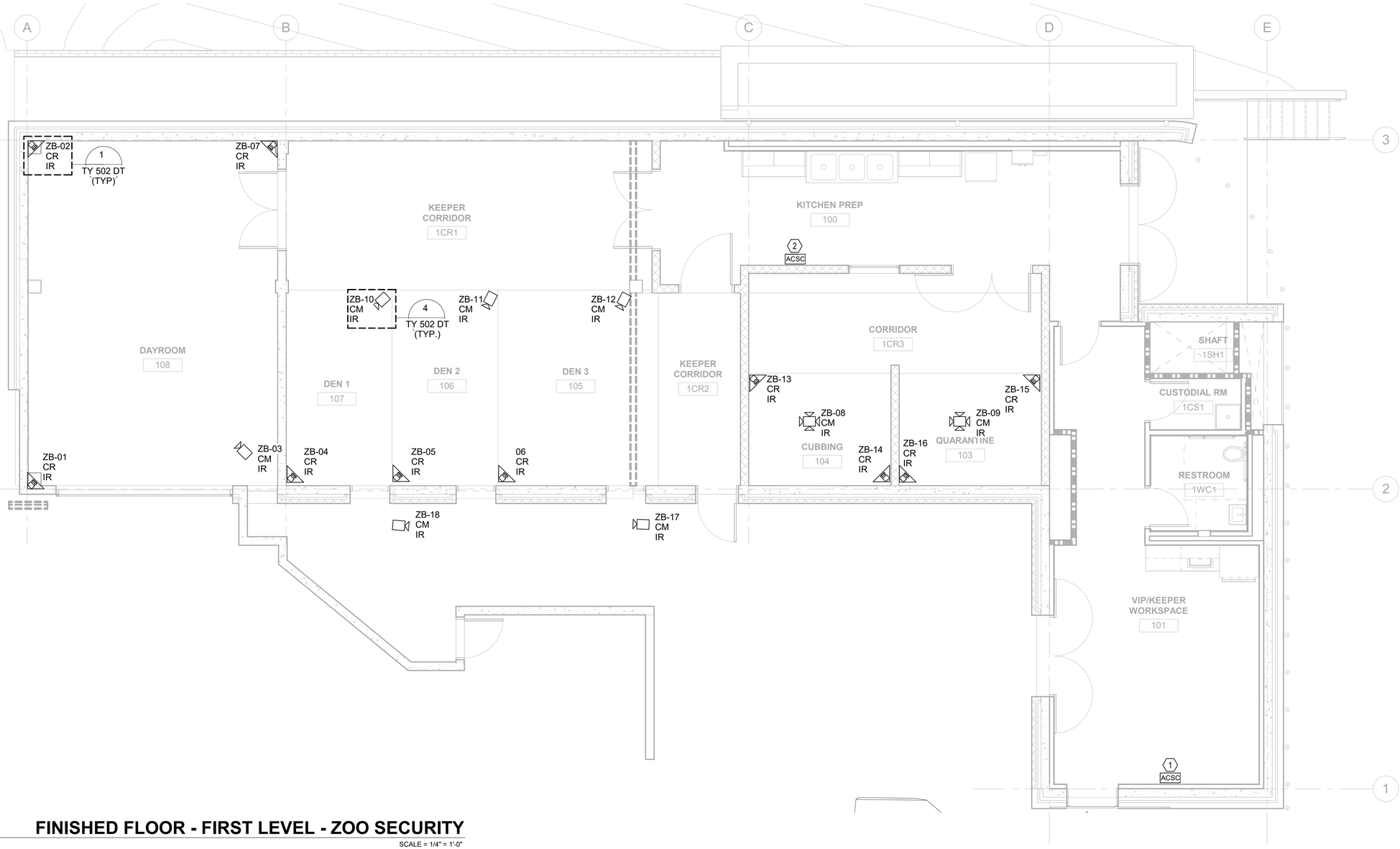
TY 104 FP BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS




1 TY 105 FP

## FINISHED FLOOR - FIRST LEVEL - ZOO SECURITY

SCALE = 1/4" = 1'-0"

## GENERAL NOTES

1 ROUTE ALL ZOO CAMERA CABLING TO THE TELECOM RACK. ALL OPS CAMERA CABLING WILL BE ROUTED TO THE OPS EQUIPMENT RACK.

## GENERAL SHEET NOTES

1 INTERIOR CAMERAS ARE CEILING MOUNTED.

## KEYNOTES

1. ZOO CAMERA VIDEO DISPLAY & COMPUTER 1:
   - A. 65" FLAT SCREEN VIDEO MONITOR
   - B. PEERLESS MODEL IM760PU RECESSED BACKBOX WITH 32-70" TV MOUNT
   - C. SMALL FORM FACTOR PC FOR GENETIC VIDEO MANAGEMENT SOFTWARE LINKED TO THE ZOO SECURITY SYSTEMS
2. ZOO CAMERA VIDEO DISPLAY & COMPUTER 2:
   - A. 65" FLAT SCREEN VIDEO MONITOR
   - B. PEERLESS MODEL SURFACE TV MOUNT
   - C. SMALL FORM FACTOR PC FOR GENETIC VIDEO MANAGEMENT SOFTWARE LINKED TO THE ZOO SECURITY SYSTEMS

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

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STAMP

TRUE NORTH

TRUE NORTH

PROJECT NORTH

KEY PLAN

0 2' 4' 8'

1/4"=1'-0"

GRAPHIC SCALE(S)

| DATE | SUBMISSION |
|---|---|
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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | ZOO SECURITY FIRST LEVEL PLAN |
| WORKING STAFF | DESIGNED BY: MMH; DRAWN BY: CW; CHECKED BY: MMH |
| SHEET NO. | 373 OF 378 |
| | TY 105 FP (DISCIPLINE TYPE SEQUENCE DRAWING TYPE) |

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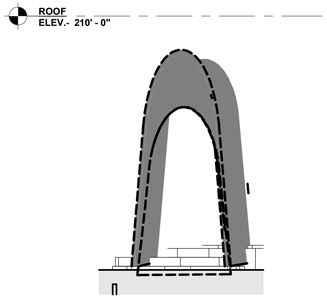
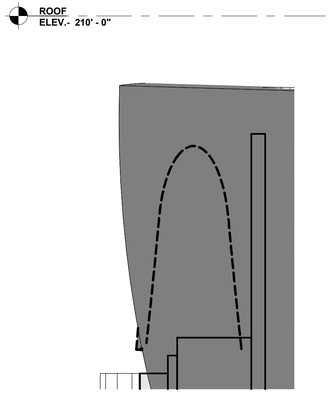
1 TY 401 EP — **ENLARGED SECURITY PLAN - TUNNEL**
SCALE = 1/4" = 1'-0"




3 TY 401 EP — **ENLARGED SECTION @ TUNNEL**
SCALE = 1/2" = 1'-0"




2 TY 401 EP — **ENLARGED SECTION @ TUNNEL**
SCALE = 1/2" = 1'-0"

# ALF

**CONSTRUCT ARABIAN LEOPARD FACILITY**

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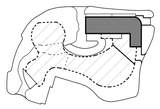
TRUE NORTH

TRUE NORTH

PROJECT NORTH



KEY PLAN

0 2' 4' 8'
1/4"=1'-0"
0 1' 2' 4'
1/2"=1'-0"

**GRAPHIC SCALE(S)**

DATE: 06/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

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BUILDING NAME / ADDRESS: **ARABIAN LEOPARD FACILITYY (ALF)**
**3001 Connecticut Avenue NW**
**Washington, DC 20008**

PROJECT TITLE: **CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)**
OPDC PROJECT NUMBER: **2533105**
A/E PROJECT NUMBER: **00100084**

DRAWING TITLE: **SECURITY ENLARGED PLANS & SECTIONS**
WORKING STAFF: MMH (DESIGNED BY) | CW (DRAWN BY) | MMH (CHECKED BY)

SHEET NO. 374 OF 378

TY 401 EP
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

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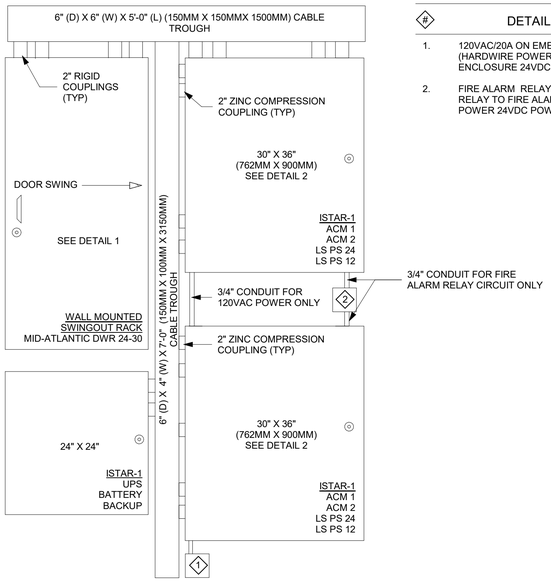
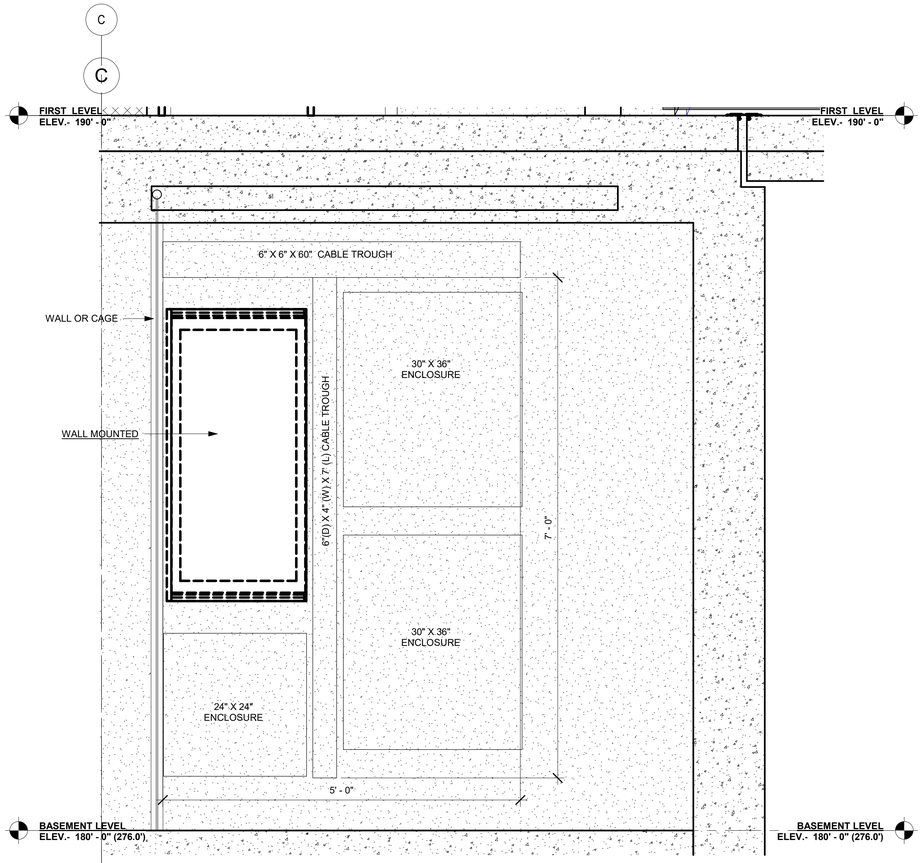



| # | DETAIL KEYNOTES |
|---|---|
| 1. | 120VAC/20A ON EMERGENCY POWER CIRCUIT (HARDWIRE POWER INTO LIFE SAFETY POWER ENCLOSURE 24VDC POWER SUPPLIES ONLY) |
| 2. | FIRE ALARM RELAY (HARDWIRE FIRE ALARM RELAY TO FIRE ALARM INPUT ON LIFE SAFETY POWER 24VDC POWER SUPPLIES ONLY) |

## SECURITY WALL ASSEMBLY (DETAILED)

3 / TY 501 DT

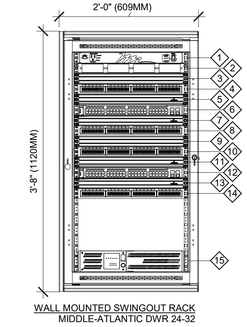
SCALE = 1" = 1'-0"




## SECURITY WALL ASSEMBLY (DEMENTIONAL MODEL)

4 / TY 501 DT

SCALE = 1" = 1'-0"

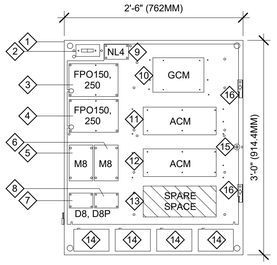



| # | DETAIL KEYNOTES |
|---|---|
| 1. | MIDDLE ATLANTIC DUAL SURGE, 8 OUTLET POWER SUPPLY, MODEL PD-915RV-RN, 1 RU |
| 2. | FIBER OPTIC PATCH PANEL, 1 RU |
| 3. | HORIZONTAL CABLE MANAGER, 1 RU |
| 4. | 24 PORT CATEGORY 6 PATCH PANEL, 1 RU |
| 5. | HORIZONTAL CABLE MANAGER, 1 RU |
| 6. | CISCO CATALYST 48 PORT POE ETHERNET SWITCH, MODEL 2960 |
| 7. | HORIZONTAL CABLE MANAGER, 1 RU |
| 8. | 24 PORT CATEGORY 6 PATCH PANEL, 1 RU |
| 9. | HORIZONTAL CABLE MANAGER, 1 RU |
| 10. | 24 PORT CATEGORY 6 PATCH PANEL, 1 RU |
| 11. | HORIZONTAL CABLE MANAGER, 1 RU |
| 12. | CISCO CATALYST 48 PORT POE ETHERNET SWITCH, MODEL 2960 |
| 13. | HORIZONTAL CABLE MANAGER, 1 RU |
| 14. | 24 PORT CATEGORY 6 PATCH PANEL, 1 RU |
| 15. | MIDDLE ATLANTIC UNITERRUPTABLE POWER SUPPLY, UPS-S2200R |

## WALL RACK LAYOUT

1 / TY 501 DT

SCALE = 1" = 1'-0"




| # | DETAIL KEYNOTES |
|---|---|
| 1. | FPO150/250-2D82M8NL4E8S UL RATED ENCLOSURE 36" H X 30" W X 4.5" D (910MM X 760MM X115MM) ENCLOSURE |
| 2. | 120VAC POWER CIRCUIT SWITCH |
| 3. | LIFE SAFETY POWER FPO 150, 250 POWER SUPPLY MODULE 1 |
| 4. | LIFE SAFETY POWER FPO 150, 250 POWER SUPPLY MODULE 2 |
| 5. | LIFE SAFETY POWER M8, SMART DISTRIBUTION MODULE 1 (INPUTS 1-8, OUTPUTS 1-8) |
| 6. | LIFE SAFETY POWER M8, SMART DISTRIBUTION MODULE 2 (INPUTS 9-16, OUTPUTS 9-16) |
| 7. | LIFE SAFETY POWER D8, DBP POWER DISTRIBUTION MODULE 1 (OUTPUTS 1-8) |
| 8. | LIFE SAFETY POWER D8, DBP POWER DISTRIBUTION MODULE 2 (OUTPUTS 9-16) |
| 9. | LIFE SAFETY POWER NL4, NETLINK REMOTED MONITORING MODULE (4 PORT) |
| 10. | SOFTWAREHOUSE GCM MODULE |
| 11. | SOFTWARE HOUSE ACM MODULE 1 |
| 12. | SOFTWARE HOUSE ACM MODULE 2 |
| 13. | SPACE FOR ADDITIONAL D8'S |
| 14. | 12VDC BACKUP BATTERY SIZE AS REQUIRED FOR 8 HOUR BOACKUP |
| 15. | ENCLOSURE LOCK |
| 16. | TAMPER SWITCH |

| DETAIL NOTES |
|---|
| 1. INSTALL THE ENCLOSURE IN ACCORDANCE WITH MANUFACTURERS RECOMMENDATIONS. |
| 2. PRIOR TO WORKING ON THE PANEL, GROUND THE PANEL TO LOCAL TGB. |
| 3. BOND THE DOOR TO THE ENCLOSURE GROUND. |
| 4. CONNECTION THE DOOR TO THE ENCLOSURE. |
| 5. CONNECT THE SHEILD DRAIN OF ALL RISERS COMMUNICATIONS LINES (CARD READERS, INPUT AND OUTPUT PANELS, AND CARD READER MODULES) TO THE PANELS. |
| 6. TERMINATE GCM TAMPER INPUT TO THE ENCLOSURE TAMPERS. |
| 7. TERMINATE GCM LOW BATTERY INPUT TO THE POWER SUPPLY LOW BATTERY RELAY INPUT. |

## LIFESAFETY POWER W/ ISTAR PANEL

2 / TY 501 DT

SCALE = 1" = 1'-0"

# ALF

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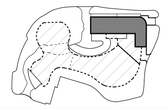

0 1" 2"

1" = 1"

GRAPHIC SCALE(S)

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|---|---|
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| PROJECT TITLE | CONSTRUCT ARABIAN LEOPARD FACILITY (ALF) |
| OPDC PROJECT NUMBER | 2533105 |
| A/E PROJECT NUMBER | 00100084 |
| DRAWING TITLE | SECURITY DETAILS |
| WORKING STAFF | Designer / Author / Checker |

SHEET NO. 375 OF 378

TY 501 DT (DISCIPLINE / TYPE / SEQUENCE / DRAWING TYPE)

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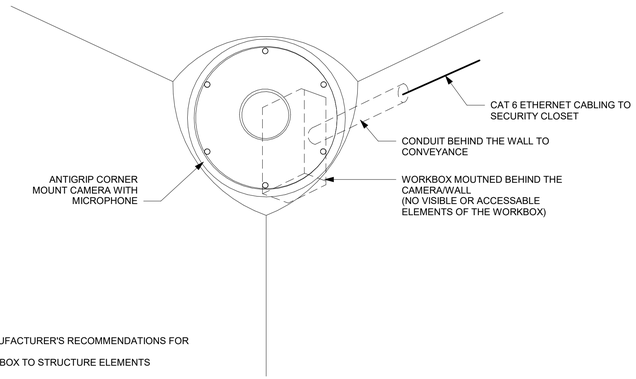
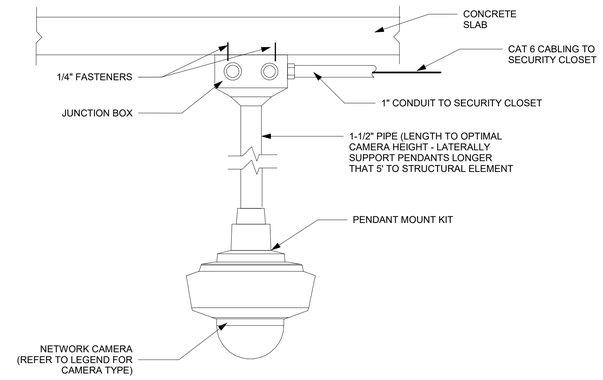



NOTES:
1. FOLLOW MANAUFACTURER'S RECOMMENDATIONS FOR INSTALLATION.
2. SECURE WORKBOX TO STRUCTURE ELEMENTS

## SECURITY - CAMERA - ANIMAL DENS, CUBBING AND QUARANTINE AREAS

1 / TY 502 DT

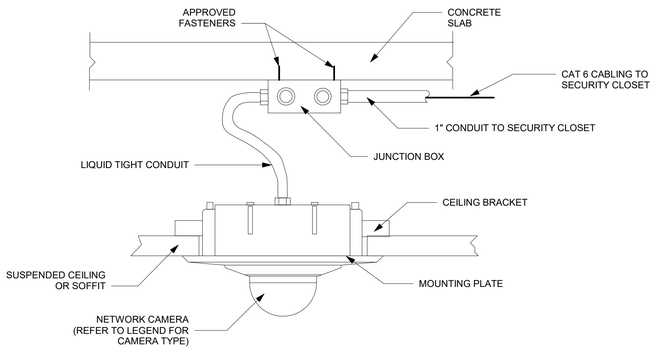
SCALE = 12" = 1'-0"




## SECURITY - INTERIOR- EXTERIOR CAMERA - PENDANT MOUNTED

2 / TY 502 DT

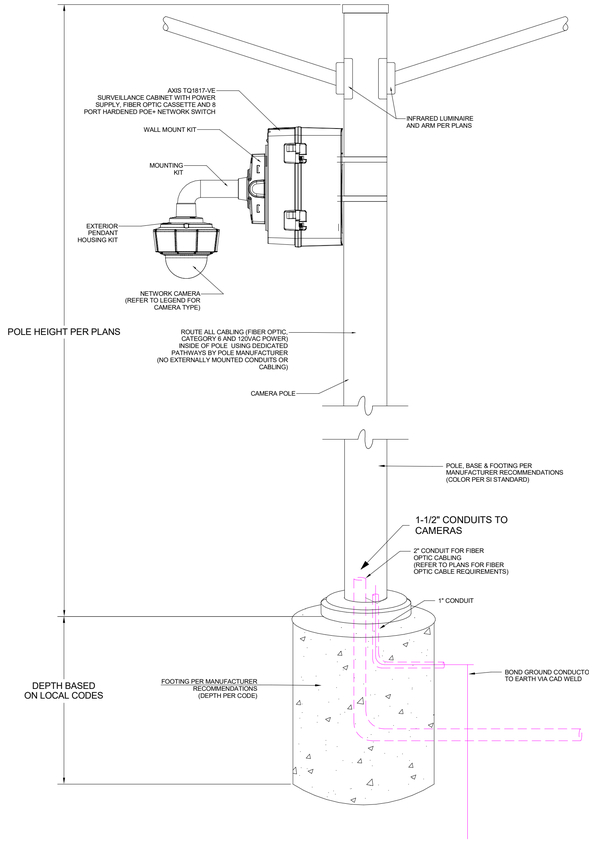
SCALE = 12" = 1'-0"




## SECURITY - INTERIOR- EXTERIOR CAMERA - SOFFIT MOUNTED

3 / TY 502 DT

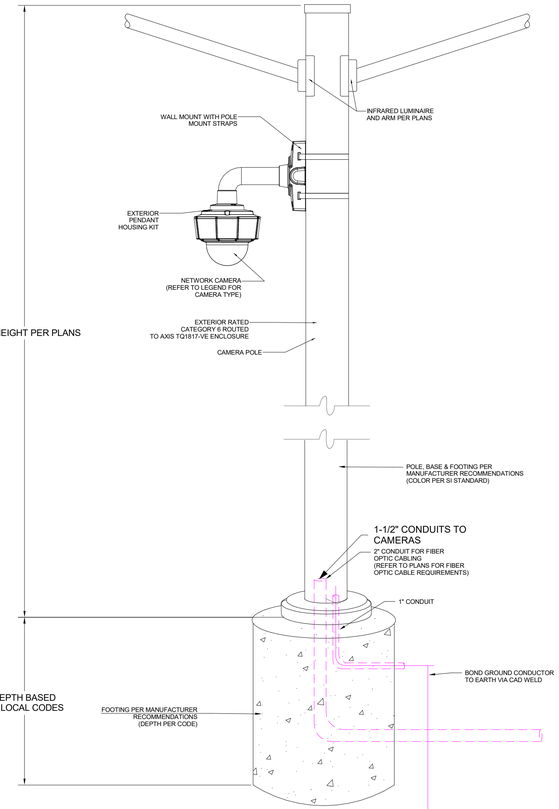
SCALE = 12" = 1'-0"




## POLE MOUNTED CAMERA WITH SECURITY ENCLOSURE

5 / TY 502 DT

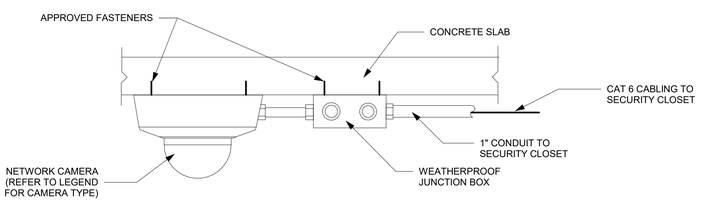
SCALE = 6" = 1'-0"




## POLE MOUNTED CAMERA WITHOUT SECURITY ENCLOSURE

6 / TY 502 DT

SCALE = 6" = 1'-0"




## SECURITY - INTERIOR-EXTERIOR CAMERA - SURFACE MOUNTED

4 / TY 502 DT

SCALE = 12" = 1'-0"

# ALF

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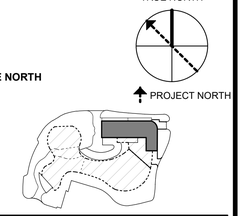


0 1" 2"
1" = 1"

GRAPHIC SCALE(S)

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| DRAWING TITLE | SECURITY DETAILS |
| WORKING STAFF | Designer / Author / Checker |
| SHEET NO. | 376 OF 378 |
| | TY 502 DT (DISCIPLINE TYPE SEQUENCE DRAWING TYPE) |

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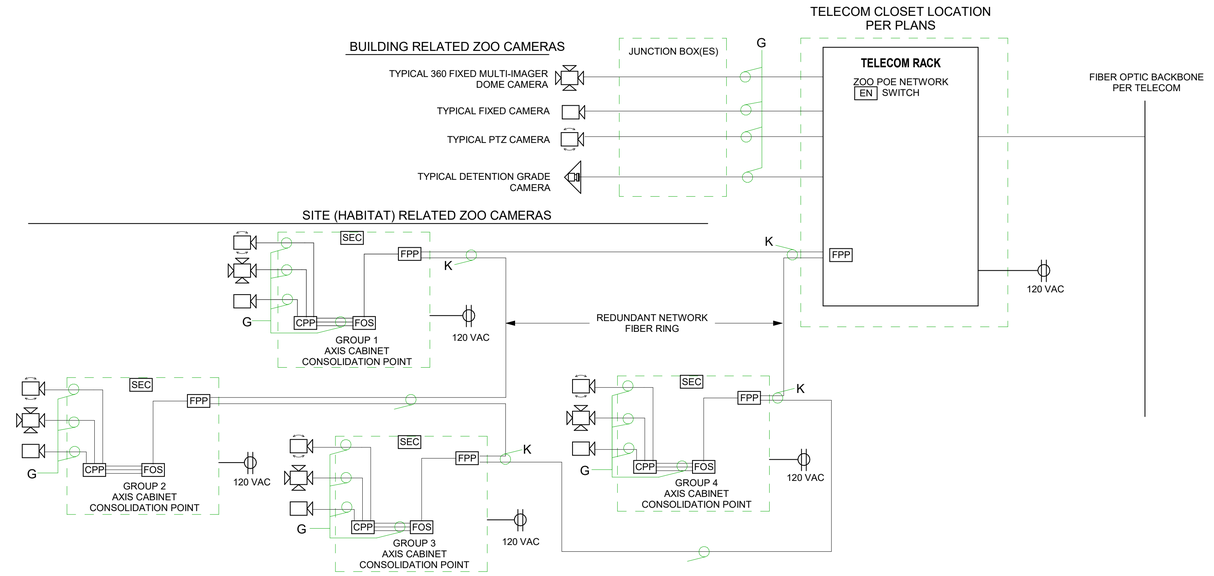
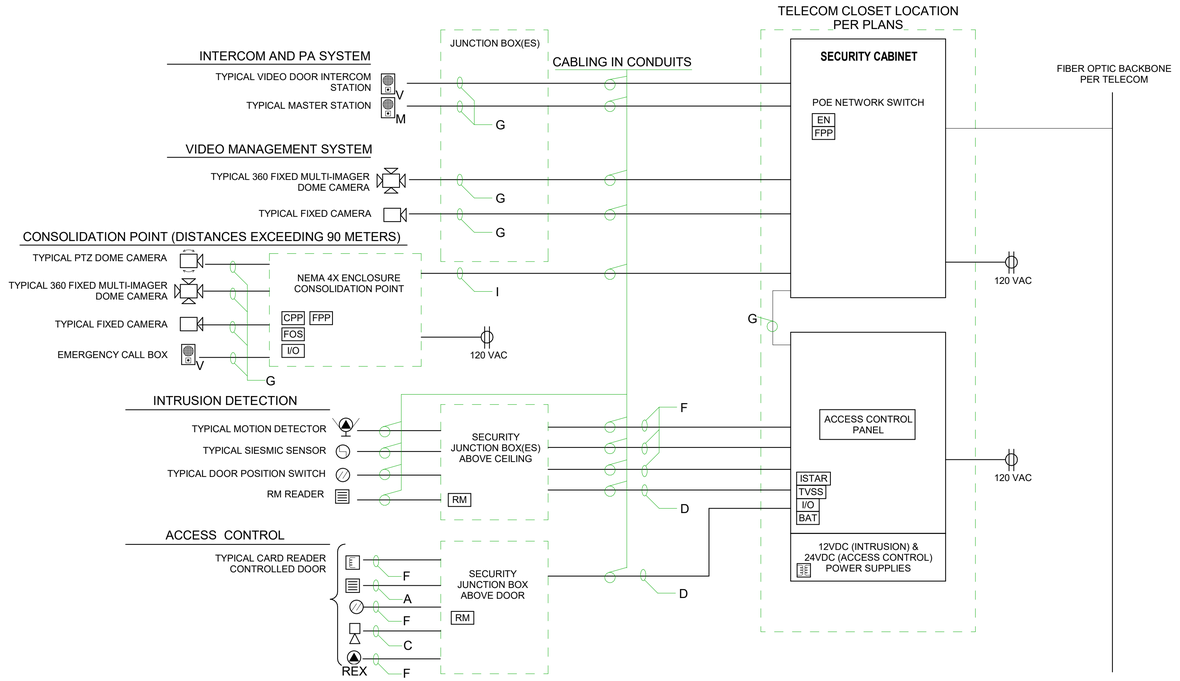



2 TY 601 DG

## ONELINE DIAGRAM - ZOO CAMERAS

SCALE = 1" = 1'-0"




1 TY 601 DG

## ONELINE DIAGRAM - OPS SYSTEM

SCALE = 1" = 1'-0"

## CABLE LEGEND

| TYPE | SIZE | # OF CONDUCTORS | DESCRIPTION | BELDEN (OR EQUIVELENT) PART NUMBER | SIGNAL TYPE |
|---|---|---|---|---|---|
| A | 22AWG | 6 | STRANDED, PVC INSULATION, PVC JACKET, OVERALL FOIL, COPPER DRAIN (SHIELDED) | 9942 | CARD READER |
| B | 18AWG | 2 | STRANDED, PVC INSULATION, PVC JACKET | 8461 | POWER |
| C | 22AWG | 2 | STRANDED, PVC INSULATION, PVC JACKET | 8442 | CONTACT |
| D | 24AWG | 4 (2 PAIR) | STRANDED, PVC INSULATION, PVC JACKET, OVERALL FOIL, COPPER DRAIN (SHIELDED) | 9842 | CONTROL (RS-485, 4-WIRE) |
| E | 24AWG | 2 (1 PAIR) | STRANDED, PVC INSULATION, PVC JACKET, OVERALL FOIL, COPPER DRAIN (SHIELDED) | 9841 | CONTROL (RS-485, 2-WIRE) |
| F | 22AWG | 4 | STRANDED, PVC INSULATION, PVC JACKET, OVERALL FOIL, COPPER DRAIN (SHIELDED) | 8729 | DEVICE WIRING |
| G | 23AWG | 6 | CATEGORY 6 OR 6+ ENHANCED CABLE, BONDED PAIRS, UNSHIELDED TWISTED PAIR, PLENUM | 7852A | COMMUNICATIONS |
| H | OM4 | 6 | FIBER OPTIC CABLE, TIGHT BUFFER TUBE, 50 MICRON | FD4L006P9 | COMMUNICATIONS |
| I | SM/MM | 12SM/24MM | FIBER OPTIC CABLE, TIGHT BUFFER, HYBRID OS2/OM4, PLENUM | FIEDH12PK | COMMUNICATIONS |
| J | SM/MM | 24SM/24MM | FIBER OPTIC CABLE, TIGHT BUFFER, HYBRID OS2/OM4, PLENUM | FIEDH22PK | COMMUNICATIONS |
| K | SM | 6SM | FIBER OPTIC CABLE, TIGHT BUFFER, HYBRID OS2, PLENUM | FEILD___PK | COMMUNICATIONS |

# ALF

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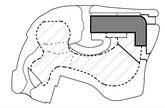

0 1" 2"
1" = 1"

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| DRAWING TITLE | SECURITY DIAGRAMS |
| WORKING STAFF | MMH (DESIGNED BY) / CW (DRAWN BY) / MMH (CHECKED BY) |

SHEET NO. 377 OF 378

TY 601 DG
DISCIPLINE TYPE SEQUENCE DRAWING TYPE

FOR REFERENCE ONLY

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TY 601 DG BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

## CAMERA SCHEDULE

| ITEM NUMBER | CAMERA NUMBER | CAMERA TYPE | INTERIOR /EXTERIOR | POLE # | MOUNT TYPE | LOCATION | GROUP | ROUTE CAT 6 TO POLE # OR BUILDING | ROUTE FIBER RING FROM / TO | POLE SECURITY CABINET POLE/CABINET # | RACK | -CAMERA INFORMATION- | | | | | | | NETWORK SWITCH INFORMATION | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | MANUFACTURER | MODEL-PART NUMBER | MAC ADDRESS | IP ADDRESS | SUBNET | GATEWAY | CABLE TYPE | ROOM/ CLOSET | RACK | SWITCH / BLADE | PORT |
| 1 | Z-01 | FIXED | EXTERIOR | 1 | POLE (6/TY502DT) | SITE | 3 | 2 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | TBD | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 2 | Z-02 | MULTIIMAGER | EXTERIOR | 2 | POLE (5/TY502DT) | SITE | 3 | 2 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 3 | Z-04 | MULTIIMAGER | EXTERIOR | 3 | POLE (6/TY502DT) | SITE | 3 | 2 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 4 | Z-20 | FIXED | EXTERIOR | 20 | POLE (6/TY502DT) | SITE | 3 | 2 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 5 | Z-07 | MULTIIMAGER | EXTERIOR | 7 | POLE (5/TY502DT) | SITE | 4 | 7 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 2 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 5 | Z-11 | FIXED | EXTERIOR | 11 | POLE (6/TY502DT) | SITE | 4 | 7 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 2 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 6 | Z-12 | MULTIIMAGER | EXTERIOR | 12 | POLE (6/TY502DT) | SITE | 4 | 7 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 2 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 7 | Z-14 | MULTIIMAGER | EXTERIOR | 14 | POLE (6/TY502DT) | SITE | 4 | 7 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 2 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 8 | Z-21 | PAN-TILT-ZOOM | EXTERIOR | 21 | POLE (6/TY502DT) | SITE | 4 | 7 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 2 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 9 | Z-13 | MULTIIMAGER | EXTERIOR | 13 | POLE (6/TY502DT) | SITE | 1 | 18 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 3 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 10 | Z-16 | MULTIIMAGER | EXTERIOR | 16 | POLE (6/TY502DT) | SITE | 1 | 18 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 3 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 11 | Z-17 | FIXED | EXTERIOR | 17 | POLE (6/TY502DT) | SITE | 1 | 18 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 3 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 12 | Z-18 | MULTIIMAGER | EXTERIOR | 18 | POLE (5/TY502DT) | SITE | 1 | 18 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 3 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 13 | Z-22 | PAN-TILT-ZOOM | EXTERIOR | 22 | POLE (6/TY502DT) | SITE | 1 | 18 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 3 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 14 | Z-23 | FIXED | EXTERIOR | N/A | TUNNEL (4/TY502DT) | SITE | 1 | 18 | SEE DETAIL 2/SHEET TY601DG | POLE 7 / CAB 3 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 15 | Z-03 | MULTIIMAGER | EXTERIOR | 3 | STRUCTURE BEAM (TBD) | SITE | 4 | 3 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | P-3748 PLVE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 16 | Z-05 | FIXED | EXTERIOR | N/A | POLE (5/TY502DT) | SITE | 4 | 3 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 17 | Z-06 | FIXED | EXTERIOR | N/A | STRUCTURE BEAM (TBD) | SITE | 4 | 3 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 18 | Z-08 | FIXED | EXTERIOR | N/A | WALL (TBD) | SITE | 4 | 3 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 19 | Z-09 | FIXED | EXTERIOR | N/A | WALL (TBD) | SITE | 4 | 3 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 20 | Z-10 | FIXED | EXTERIOR | N/A | STRUCTURE BEAM (TBD) | SITE | 4 | 3 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 21 | Z-19 | FIXED | EXTERIOR | N/A | STRUCTURE BEAM (TBD) | SITE | 4 | 3 | SEE DETAIL 2/SHEET TY601DG | POLE 2 / CAB 1 | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | FO SFP |
| 22 | Z-24 | FIXED | EXTERIOR | N/A | TUNNEL (4/TY502DT) | SITE | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA FIBER SFP FROM POLE MOUNTED CABINET | AXIS | Q3628-VE | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 23 | ZB-01 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | DAYROOM | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 24 | ZB-02 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | DAYROOM | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 25 | ZB-03 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | DAYROOM | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 26 | ZB-07 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | DAYROOM | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 27 | ZB-04 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | DEN 1 | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 28 | ZB-05 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | DEN 2 | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 29 | ZB-06 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | DEN 3 | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 30 | ZB-10 | FIXED | INTERIOR | N/A | CEILING (4/TY502DT) | KEEPER CORRIDOR | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | TBD | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 31 | ZB-11 | FIXED | INTERIOR | N/A | CEILING (4/TY502DT) | KEEPER CORRIDOR | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | TBD | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 32 | ZB-12 | FIXED | INTERIOR | N/A | CEILING (4/TY502DT) | KEEPER CORRIDOR | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | TBD | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 33 | ZB-13 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | CUBBING | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 34 | ZB-08 | 360 HEMISPHERICAL | INTERIOR | N/A | CEILING (4/TY502DT) | CUBBING | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | M3077 | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 35 | ZB-14 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | CUBBING | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 36 | ZB-09 | 360 HEMISPHERICAL | INTERIOR | N/A | CEILING (4/TY502DT) | QUARANTINE | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | M3077 | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 37 | ZB-15 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | QUARANTINE | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 38 | ZB-16 | ANTIGRIP CORNER | INTERIOR | N/A | CORNER (1/TY502DT) | QUARANTINE | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | P9117-PV | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 39 | ZB-17 | 180 HEMIPHERICAL | INTERIOR | N/A | TBD | EXTERIOR OF DENS | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | M3077 | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |
| 40 | ZB-18 | 180 HEMIPHERICAL | INTERIOR | N/A | TBD | EXTERIOR OF DENS | BUILDING | BUILDING | SEE DETAIL 2/SHEET TY601DG | N/A | IT RACK VIA CAT 6 | AXIS | M3077 | BY CONTRACTOR | BY OWNER | BY OWNER | BY OWNER | CAT 6 | AV-IT RM 109 | TELECOM | BY CONTRACTOR | BY OWNER |

**NOTE: THIS SCHEDULE IS FOR ZOO CAMERAS ONLY. OPS CAMERA SCHEDULE PENDING CAMERA MODEL VALIDATION**

1 / TY 701 SH

## ZOO ONLY CAMERA SCHEDULE

SCALE = 12" = 1'-0"

6/15/2026 9:04:05 AM
TY 701 SH BY Approver 06/15/2026 65% CONSTRUCTION DOCUMENTS

# ALF

## CONSTRUCT ARABIAN LEOPARD FACILITY

**NOT FOR CONSTRUCTION**

FOR REFERENCE ONLY

STAMP

TRUE NORTH

KEY PLAN

GRAPHIC SCALE(S)

DATE: 06/15/2026

SUBMISSION: 65% CONSTRUCTION DOCUMENTS

REVISION # DATE

REVISION DESCRIPTION

Smithsonian Institution

Smithsonian OPDC
600 Maryland Avenue S.W. Suite 5001
Washington, DC 20024-2520

BUILDING NAME / ADDRESS: ARABIAN LEOPARD FACILITY (ALF)
3001 Connecticut Avenue NW
Washington, DC 20008

PROJECT TITLE: CONSTRUCT ARABIAN LEOPARD FACILITY (ALF)

OPDC PROJECT NUMBER: 2533105

A/E PROJECT NUMBER: 00100084

DRAWING TITLE: SCHEDULES

WORKING STAFF

| Designer | Author | Checker |
|---|---|---|
| DESIGNED BY | DRAWN BY | CHECKED BY |

SHEET NO. 378 OF 378

TY 701 SH

DISCIPLINE TYPE SEQUENCE DRAWING TYPE